The SAGE Handbook of

Applied Social Research Methods

2 EDITION

Leonard Bickman
Debra J. Rog
EDITORS

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Introduction

Why a Handbook of Applied Social Research Methods?

Leonard Bickman

Debra J. Rog

This second edition of the *Handbook of Applied Social Research Methods* includes 14 chapters revised and updated from the first edition as well as 4 new chapters. We selected the combination of chapters in this second edition to represent the cutting edge of applied social research methods and important changes that have occurred in the field in the decade since the first edition was published.

One area that continues to gain prominence is the focus on qualitative research. In the first edition, 4 of the 18 chapters were focused on the qualitative approach; in this edition, a third of the *Handbook* now focuses on that approach. Moreover, research that combines quantitative and qualitative research methods, called *mixed methods*, has become a much more common requirement for studies. In Chapter 9, Abbas Tashakorri and Charles Teddlie present an approach to integrating qualitative and quantitative methods with an underlying belief that qualitative and quantitative methods are not dichotomous or discrete but are on a continuum of approaches.

Another change that is reflected in many of the revised chapters as well as in two of the new chapters is the increasing use of technology in research. The use of the Internet and computer-assisted methods is discussed in several of the chapters and is the focus of Samuel Best and Chase Harrison's chapter (Chapter 13) on Internet survey methods. In addition, Mary Kane and Bill Trochim's contribution on concept mapping in Chapter 14 offers a cutting-edge technique involving both qualitative and quantitative methods in designing research.

Finally, Michael Harrison's chapter on organizational diagnosis is a new contribution to this *Handbook* edition. Harrison's approach focuses on using methods and models from the behavioral and organization sciences to help identify what is going on in an organization and to help guide decisions based on this information.

In addition to reflecting any new developments that have occurred (such as the technological changes noted above), other changes that have been made in this edition respond to comments made about the first edition, with an emphasis on increasing the pedagogical quality of each of the chapters and the book as a whole. In particular, the text has been made more "classroom friendly" with the inclusion of discussion questions and exercises. The chapters also are current with new research cited and improved examples of those methods. Overall, however, research methods are not an area that is subject to rapid changes.

This version of the *Handbook*, like the first edition, presents the major methodological approaches to conducting applied social research that we believe need to be in a researcher's repertoire. It serves as a "handy" reference guide, covering key yet often diverse themes and developments in applied social research. Each chapter summarizes and synthesizes major topics and issues of the method and is designed with a broad perspective but provides information on additional resources for more in-depth treatment of any one topic or issue.

Applied social research methods span several substantive arenas, and the boundaries of application are not well-defined. The methods can be applied in educational settings, environmental settings, health settings, business settings, and so forth. In addition, researchers conducting applied social research come from several disciplinary backgrounds and orientations, including sociology, psychology, business, political science, education, geography, and social work, to name a few. Consequently, a range of research philosophies, designs, data collection methods, analysis techniques, and reporting methods can be considered to be "applied social research." Applied research, because it consists of a diverse set of research strategies, is difficult to define precisely and inclusively. It is probably most easily defined by what it is *not,* thus distinguishing it from basic research. Therefore, we begin by highlighting several differences between applied and basic research; we then present some specific principles relevant to most of the approaches to applied social research discussed in this *Handbook.*

Distinguishing Applied From Basic Social Research

Social scientists are frequently involved in tackling real-world social problems. The research topics are exceptionally varied. They include studying physicians' efforts to improve patients' compliance with medical regimens, determining whether drug use is decreasing at a local high school, providing up-to-date information on the operations of new educational programs and policies, evaluating the impacts of environmental disasters, and analyzing the likely effects of yet-to-be-tried programs to reduce teenage pregnancy. Researchers are asked to estimate the costs of everything from shopping center proposals to weapons systems and to speak to the relative effectiveness of alternative programs and policies. Increasingly, applied researchers are contributing to major public policy debates and decisions.

Applied research uses scientific methodology to develop information to help solve an immediate, yet usually persistent, societal problem. The applied research environment is often complex, chaotic, and highly political, with pressures for quick and conclusive answers yet little or no experimental control. Basic research, in comparison, also is firmly grounded in the scientific method but has as its goal the creation of new knowledge about how fundamental processes work. Control is often provided through a laboratory environment.

These differences between applied and basic research contexts can sometimes seem artificial to some observers, and highlighting them may create the impression that researchers in the applied community are "willing to settle" for something less than rigorous science. In practice, applied research and basic research have many more commonalities than differences; however, it is critical that applied researchers (and research consumers) understand the differences. Basic research and applied research differ in purposes, context, and methods. For ease of presentation, we discuss the differences as dichotomies; in reality, however, they fall on continua.

Differences in Purpose

Knowledge Use Versus Knowledge Production. Applied research strives to improve our understanding of a "problem," with the intent of contributing to the solution of that problem. The distinguishing feature of basic research, in contrast, is that it is intended to expand knowledge (i.e., to identify universal principles that contribute to our understanding of how the world operates). Thus, it is knowledge, as an end in itself, that motivates basic research. Applied research also may result in new knowledge, but often on a more limited basis defined by the nature of an immediate problem. Although it may be hoped that basic research findings will eventually be helpful in solving particular problems, such problem solving is not the immediate or major goal of basic research.

Broad Versus Narrow Questions. The applied researcher is often faced with "fuzzy" issues that have multiple, often broad research questions, and addresses them in a "messy" or uncontrolled environment. For example, what is the effect of the provision of mental health services to people living with AIDS? What are the causes of homelessness?

Even when the questions are well-defined, the applied environment is complex, making it difficult for the researcher to eliminate competing explanations (e.g., events other than an intervention could be likely causes for changes in attitudes or behavior). Obviously, in the example above, aspects of an individual's life other than mental health services received will affect that person's well-being. The number and complexity of measurement tasks and dynamic real-world research settings pose major challenges for applied researchers. They also often require that researchers make conscious choices (trade-offs) about the relative importance of answering various questions and the degree of confidence necessary for each answer.

In contrast, basic research investigations are usually narrow in scope. Typically, the basic researcher is investigating a very specific topic and a very tightly focused question. For example, what is the effect of white noise on the short-term recall of nonsense syllables? Or what is the effect of cocaine use on fine motor coordination? The limited focus enables the researcher to concentrate on a single measurement task and to use rigorous design approaches that allow for maximum control of potentially confounding variables. In an experiment on the effects of white noise, the laboratory setting enables the researcher to eliminate all other noise variables from the environment, so that the focus can be exclusively on the effects of the variable of interest, the white noise.

Practical Versus Statistical Significance. There are differences also between the analytic goals of applied research and those of basic research. Basic researchers generally are most concerned with determining whether or not an effect or causal relationship exists, whether or not it is in the direction predicted, and whether or not it is statistically significant. In applied research, both practical significance and statistical significance are essential. Besides determining whether or not a causal relationship exists and is statistically significant, applied researchers are interested in knowing if the effects are of sufficient size to be meaningful in a particular context. It is critical, therefore, that the applied researcher understands the level of outcome that will be considered "significant" by key audiences and interest groups. For example, what level of reduced drug use is considered a practically significant outcome of a drug program? Is a 2% drop meaningful? Thus, besides establishing whether the intervention has produced statistically significant results, applied research has the added task of determining whether the level of outcome attained is important or trivial.

Theoretical "Opportunism" Versus Theoretical "Purity." Applied researchers are more likely than basic researchers to use theory instrumentally. Related to the earlier concept of practical significance, the applied researcher is interested in applying and using a theory only if it identifies variables and concepts that will likely produce important, practical results. Purity of theory is not as much a driving force as is utility. Does the theory help solve the problem? Moreover, if several theories appear useful, then the applied researcher will combine them, it is hoped, in a creative and useful way. For those involved in evaluation research, they are most often trying to understand the "underlying theory" or logic of the program or policy they are studying and using that to guide the research.

For the basic researcher, on the other hand, it is the underlying formal theory that is of prime importance. Thus, the researcher will strive to have variables in the study that are flawless representations of the underlying theoretical constructs. In a study examining the relationships between frustration and aggression, for example, the investigator would try to be certain that the study deals with aggression and not another related construct, such as anger, and that frustration is actually manipulated, and not boredom.

Differences in Context

Open Versus Controlled Environment. The context of the research is a major factor in accounting for the differences between applied research and basic research. As noted earlier, applied research can be conducted in many diverse contexts, including business settings, hospitals, schools, prisons, and communities. These settings, and their corresponding characteristics, can pose quite different demands on applied researchers. The applied researcher is more concerned about generalizability of findings. Since application is a goal, it is important to know how dependent the results of the study are on the particular environment in which it was tested. In addition, lengthy negotiations are sometimes necessary for a researcher even to obtain permission to access the data.

Basic research, in contrast, is typically conducted in universities or similar academic environments and is relatively isolated from the government or business worlds. The environment is within the researcher's control and is subject to close monitoring.

Client Initiated Versus Researcher Initiated. The applied researcher often receives research questions from a client or research sponsor, and sometimes these questions are poorly framed and incompletely understood. Clients of applied social research can include federal government agencies, state governments and legislatures, local governments, government oversight agencies, professional or advocacy groups, private research institutions, foundations, business corporations and organizations, and service delivery agencies, among others. The client is often in control, whether through a contractual relationship or by virtue of holding a higher position within the researcher's place of employment (if the research is being conducted internally). Typically, the applied researcher needs to negotiate with the client about the project scope, cost, and deadlines. Based on these parameters, the researcher may need to make conscious trade-offs in selecting a research approach that affects what questions will be addressed and how conclusively they will be addressed.

University basic research, in contrast, is usually self-initiated, even when funding is obtained from sources outside the university environment, such as through government grants. The idea for the study, the approach to executing it, and even the timeline are generally determined by the researcher. The reality is that the basic researcher, in comparison with the applied researcher, operates in an environment with a great deal more flexibility, less need to let the research agenda be shaped by project costs, and less time pressure to deliver results by a specified deadline. Basic researchers sometimes can undertake multiyear incremental programs of research intended to build theory systematically, often with supplemental funding and support from their universities.

Research Team Versus Solo Scientist. Applied research is typically conducted by research teams. These teams are likely to be multidisciplinary, sometimes as a result of competitive positioning to win grants or contracts. Moreover, the substance of applied research often demands multidisciplinary teams, particularly for studies that address multiple questions involving different areas of inquiry (e.g., economic, political, sociological). These teams must often comprise individuals who are familiar with the substantive issue (e.g., health care) and others who have expertise in specific methodological or statistical areas (e.g., economic forecasting).

Basic research is typically conducted by an individual researcher who behaves autonomously, setting the study scope and approach. If there is a research team, it generally comprises the researcher's students or other persons that the researcher chooses from the same or similar disciplines.

Differences in Methods

External Versus Internal Validity. A key difference between applied research and basic research is the relative emphasis on internal and external validity. Whereas internal validity is essential to both types of research, external validity is much more important to applied research. Indeed, the likelihood that applied research findings will be used often depends on the researchers' ability to convince policymakers that the results are applicable to their particular setting or problem. For example, the results from a laboratory study of aggression using a bogus shock generator are not as likely to be as convincing or as useful to policymakers who are confronting the problem of violent crime as are the results of a well-designed survey describing the types and incidence of crime experienced by inner-city residents.

The Construct of Effect Versus the Construct of Cause. Applied research concentrates on the construct of effect. It is especially critical that the outcome measures are valid—that they accurately measure the variables of interest. Often, it is important for researchers to measure multiple outcomes and to use multiple measures to assess each construct fully. Mental health outcomes, for example, may include measures of daily functioning, psychiatric status, and use of hospitalization. Moreover, measures of real-world outcomes often require more than self-report and simple paper-and-pencil measures (e.g., self-report satisfaction with participation in a program). If attempts are being made to address a social problem, then real-world measures directly related to that problem are desirable. For example, if one is studying the effects of a program designed to reduce intergroup conflict and tension, then observations of the interactions among group members will have more credibility than group members' responses to questions about their attitudes toward other groups. In fact, there is much research evidence in social psychology that demonstrates that attitudes and behavior often do not relate.

Basic research, on the other hand, concentrates on the construct of cause. In laboratory studies, the independent variable (cause) must be clearly explicated and not confounded with any other variables. It is rare in applied research settings that control over an independent variable is so clear-cut. For example, in a study of the effects of a treatment program for drug abusers, it is unlikely that the researcher can isolate the aspects of the program that are responsible for the outcomes that result. This is due to both the complexity of many social programs and the researcher's inability in most circumstances to manipulate different program features to discern different effects.

Multiple Versus Single Levels of Analysis. The applied researcher, in contrast to the basic researcher, usually needs to examine a specific problem at more than one level of analysis, not only studying the individual, but often larger groups, such as organizations or even societies. For example, in one evaluation of a community crime prevention project, the researcher not only examined individual attitudes and perspectives but also measured the reactions of groups of neighbors and neighborhoods to problems of crime. These added levels of analysis may require that the researcher be conversant with concepts and research approaches found in several disciplines, such as psychology, sociology, and political science, and that he or she develop a multidisciplinary research team that can conduct the multi-level inquiry.

Similarly, because applied researchers are often given multiple questions to answer, because they must work in real-world settings, and because they often use multiple measures of effects, they are more likely to use multiple research methods, often including both quantitative and qualitative approaches. Although using multiple methods may be necessary to address multiple questions, it may also be a strategy used to triangulate on a difficult problem from several directions, thus lending additional confidence to the study results. Although it is desirable for researchers to use experimental designs whenever possible, often the applied researcher is called in after a program or intervention is in place, and consequently is precluded from building random assignment into the allocation of program resources. Thus, applied researchers often use quasi-experimental studies. The obverse, however, is rarer; quasi-experimental designs are generally not found in the studies published in basic research journals.

The Orientation of This *Handbook*

This second edition is designed to be a resource for professionals and students alike. It can be used in tandem with the *Applied Social Research Methods Series* that is coedited by the present editors. The series has more than 50 volumes related to the design of applied research, the collection of both quantitative and qualitative data, and the management and presentation of these data. Almost all the authors in the *Handbook* also authored a book in that series on the same topic.

Similar to our goal as editors of the book series, our goal in this *Handbook* is to offer a hands-on, how-to approach to research that is sensitive to the constraints and opportunities in the practical and policy environments, yet is rooted in rigorous and sound research principles. Abundant examples and illustrations, often based on the authors' own experience and work, enhance the potential usefulness of the material to students and others who may have limited experience in conducting research in applied arenas. In addition, discussion questions and exercises in each chapter are designed to increase the usefulness of the *Handbook* in the classroom environment.

The contributors to the *Handbook* represent various disciplines (sociology, business, psychology, political science, education, economics) and work in diverse settings (academic departments, research institutes, government, the private sector). Through a concise collection of their work, we hope to provide in one place a diversity of perspectives and methodologies that others can use in planning and conducting applied social research. Despite this diversity of perspectives, methods, and approaches, several central themes are stressed across the chapters. We describe these themes in turn below.

The Iterative Nature of Applied Research. In most applied research endeavors, the research question—the focus of the effort—is rarely static. Rather, to maintain the credibility, responsiveness, and quality of the research project, the researcher must typically make a series of iterations within the research design. The iteration is necessary not because of methodological inadequacies, but because of successive redefinitions of the applied problem as the project is being planned and implemented. New knowledge is gained, unanticipated obstacles are encountered, and contextual shifts take place that change the overall research situation and in turn have effects on the research. The first chapter in this *Handbook*, by Bickman and Rog, describes an iterative approach to planning applied research that continually revisits the research question as trade-offs in the design are made. In Chapter 7, Maxwell also discusses the iterative, interactive nature of qualitative research design, highlighting the unique relationships that occur in qualitative research among the purposes of the research, the conceptual context, the questions, the methods, and validity.

Multiple Stakeholders. As noted earlier, applied research involves the efforts and interests of multiple parties. Those interested in how a study gets conducted and its results can include the research sponsor, individuals involved in the intervention or program under study, the potential beneficiaries of the research (e.g., those who could be affected by the results of the research), and potential users of the research results (such as policymakers and business leaders). In some situations, the cooperation of these parties is critical to the successful implementation of the project. Usually, the involvement of these stakeholders ensures that the results of the research will be relevant, useful, and hopefully used to address the problem that the research was intended to study.

Many of the contributors to this volume stress the importance of consulting and involving stakeholders in various aspects of the research process. Bickman and Rog describe the role of stakeholders throughout the planning of a study, from the specification of research questions to the choice of designs and design trade-offs. Similarly, in Chapter 4, on planning ethically responsible research, Sieber emphasizes the importance of researchers' attending to the interests and concerns of all parties in the design stage of a study. Kane and Trochim, in Chapter 14, offer concept mapping as a structured technique for engaging stakeholders in the decision making and planning of research.

Ethical Concerns. Research ethics are important in all types of research, basic or applied. When the research involves or affects human beings, the researcher must attend to a set of ethical and legal principles and requirements that can ensure the protection of the interests of all those involved. Ethical issues, as Boruch and colleagues note in Chapter 5, commonly arise in experimental studies when individuals are asked to be randomly assigned into either a treatment condition or a control condition. However, ethical concerns are also raised in most studies in the development of strategies for obtaining informed consent, protecting privacy, guaranteeing anonymity, and/or ensuring confidentiality, and in developing research procedures that are sensitive to and respectful of the specific needs of the population involved in the research (see Sieber, Chapter 4; Fetterman, Chapter 17). As Sieber notes, although attention to ethics is important to the conduct of all studies, the need for ethical problem solving is particularly heightened when the researcher is dealing with highly political and controversial social problems, in research that involves vulnerable populations (e.g., individuals with AIDS), and in situations where stakeholders have high stakes in the outcomes of the research.

Enhancing Validity. Applied research faces challenges that threaten the validity of studies' results. Difficulties in mounting the most rigorous designs, in collecting data from objective sources, and in designing studies that have universal generalizability require innovative strategies to ensure that the research continues to produce valid results. Lipsey and Hurley, in Chapter 2, describe the link between internal validity and statistical power and how good research practice can increase the statistical power of a study. In Chapter 6, Mark and Reichardt outline the threats to validity that challenge experiments and quasi-experiments and various design strategies for controlling these threats. Henry, in his discussion of sampling in Chapter 3, focuses on external validity and the construction of samples that can provide valid information about a broader population. Other contributors in Part III (Fowler & Cosenza, Chapter 12; Lavrakas, Chapter 16; Mangione & Van Ness, Chapter 15) focus on increasing construct validity through the improvement of the design of individual questions and overall data collection tools, the training of data collectors, and the review and analysis of data.

Triangulation of Methods and Measures. One method of enhancing validity is to develop converging lines of evidence. As noted earlier, a clear hallmark of applied research is the triangulation of methods and measures to compensate for the fallibility of any single method or measure. The validity of both qualitative and quantitative applied research is bolstered by triangulation in data collection. Yin (Chapter 8), Maxwell (Chapter 7), and Fetterman (Chapter 17) stress the importance of triangulation in qualitative research design, ethnography, and case study research. Similarly, Bickman and Rog support the use of multiple data collection methods in all types of applied research.

Qualitative and Quantitative. Unlike traditional books on research methods, this volume does not have separate sections for quantitative and qualitative methods. Rather, both types of research are presented together as approaches to consider in research design, data collection, analysis, and reporting. Our emphasis is to find the tools that best fit the research question, context, and resources at hand. Often, multiple tools are needed, cutting across qualitative and quantitative boundaries, to research a topic thoroughly and provide results that can be used. Chapter 9 by Tashakkori and Teddlie specifically focuses on the use of mixed methods designs.

Several tools are described in this *Handbook*. Experimental and quasi-experimental approaches are discussed (Boruch et al., Chapter 5; Mark & Reichardt, Chapter 6; Lipsey & Hurley, Chapter 2) alongside qualitative approaches to design (Maxwell, Chapter 7), including case studies (Yin, Chapter 8) and ethnographies (Fetterman, Chapter 17) and approaches that are influenced by their setting (Harrison, Chapter 10). Data collection tools provided also include surveys (in person, mail, Internet, and telephone), focus groups (Stewart, Shamdasani, & Rook, Chapter 18), and newer approaches such as concept mapping (Kane & Trochim, Chapter 14).

Technological Advances. Recent technological advances can help applied researchers conduct their research more efficiently, with greater precision, and with greater insight than in the past. Clearly, advancements in computers have improved the quality, timeliness, and power of research. Analyses of large databases with multiple levels of data would not be possible without high-speed computers. Statistical syntheses of research studies, called *meta-analyses* (Cooper, Patall, & Lindsay, Chapter 11), have become more common in a variety of areas, in part due to the accessibility of computers. Computers are required if the Internet is going to be used for data collection as described by Best and Harrison in Chapter 13. Qualitative studies can now benefit from computer technology, with software programs that allow for the identification and analysis of themes in narratives (Tashakkori & Teddlie, Chapter 9), programs that simply allow the researcher to organize and manage the voluminous amounts of qualitative data typically collected in a study (Maxwell, Chapter 7; Yin, Chapter 8), and laptops that can be used in the field to provide for efficient data collection (Fetterman, Chapter 17). In addition to computers, other new technology provides for innovative ways of collecting data, such as through videoconferencing (Fetterman, Chapter 17) and the Internet.

However, the researcher has to be careful not to get caught up in using technology that only gives the appearance of advancement. Lavrakas points out that the use of computerized telephone interviews has not been shown to save time or money over traditional paper-and-pencil surveys.

Research Management. The nature of the context in which applied researchers work highlights the need for extensive expertise in research planning. Applied researchers must take deadlines seriously, and then design research that can deliver useful information within the constraints of budget, time, and staff available. The key to quality work is to use the most rigorous methods possible, making intelligent and conscious trade-offs in scope and conclusiveness. This does not mean that any information is better than none, but that decisions about what information to pursue must be made very deliberately with realistic assessments of the feasibility of executing the proposed research within the required time frame. Bickman and Rog (Chapter 1), and Boruch et al. (Chapter 5) describe the importance of research management from the early planning stages through the communication and reporting of results.

Conclusion

We hope that the contributions to this *Handbook* will help guide readers in selecting appropriate questions and procedures to use in applied research. Consistent with a handbook approach, the chapters are not intended to provide the details necessary for readers to use each method or to design comprehensive research; rather, they are intended to provide the general guidance readers will need to address each topic more fully. This *Handbook* should serve as an intelligent guide, helping readers select the approaches, specific designs, and data collection procedures that they can best use in applied social research.

PART I

Approaches to Applied Research

The four chapters in this section describe the key elements and approaches to designing and planning applied social research. The first chapter by Bickman and Rog presents an overview of the design process. It stresses the iterative nature of planning research as well as the multimethod approach. Planning an applied research project usually requires a great deal of learning about the context in which the study will take place as well as different stakeholder perspectives. It took one of the authors (L.B.) almost 2 years of a 6-year study to decide on the final design. The authors stress the trade-offs that are involved in the design phase as the investigator balances the needs for the research to be timely, credible, within budget, and of high quality. The authors note that as researchers make trade-offs in their research designs, they must continue to revisit the original research questions to ensure either that they can still be answered given the changes in the design or that they are revised to reflect what can be answered.

One of the aspects of planning applied research covered in Chapter 1, often overlooked in teaching and in practice, is the need for researchers to make certain that the resources necessary for implementing the research design are in place. These include both human and material resources as well as other elements that can make or break a study, such as site cooperation. Many applied research studies fail because the assumed community resources never materialize. This chapter describes how to develop both financial and time budgets and modify the study design as needed based on what resources can be made available.

The next three chapters outline the principles of three major areas of design: experimental designs, descriptive designs, and making sure that the design meets ethical standards. In Chapter 2, Lipsey and Hurley highlight the importance of planning experiments with design sensitivity in mind. Design sensitivity, also referred to as statistical power, is the ability to detect a difference between the treatment and control conditions on an outcome if that difference is really there. In a review of previous studies, they report that almost half were underpowered and, thus, lacked the ability to detect reasonable-sized effects even if they were present. The low statistical power of many projects has been recognized by editors and grant reviewers to the extent that a power analysis has increasingly become a required component of a research design. The major contribution of this chapter is that the authors illustrate how statistical power is affected by many components of a study, and they offer several approaches for increasing power other than just increasing sample size. In highlighting the components that affect statistical power, the authors illustrate several ways in which the sensitivity of the research design can be strengthened to increase the design's overall statistical power. Most important, they demonstrate how the researcher does not have to rely only on increasing the sample size to increase the power but how good research practice (e.g., the use of valid and reliable measurement, maintaining the integrity and completeness of both the treatment and control groups) can increase the effect size and, in turn, increase the statistical power of the study. The addition of the new section of multilevel designs is especially appropriate for an increasing number of studies where the unit of analysis is not an individual, such as a student, but a group such as a class or a school.

As Henry points out in Chapter 3, sampling is a critical component of almost every applied research study, but it is most critical to the conduct of descriptive studies involving surveys of particular populations (e.g., surveys of homeless individuals). Henry describes both probability and nonprobability sampling, also sometimes referred to as convenience sampling. When a random or representative sample cannot be drawn, knowing how to select the most appropriate nonprobability sample is critical. Henry provides a practical sampling design framework to help researchers structure their thinking about making sampling decisions in the context of how those decisions affect total error. Total error, defined as the difference between the true population value and the estimate based on the sample data, involves three types of error: error due to differences in the population definition, error due to the sampling approach used, and error involved in the random selection process. Henry's framework outlines the decisions that effect total error in the presampling, sampling, and postsampling phases of the research. In his chapter, however, he focuses on the implications of the researcher's answers to the questions on sampling choices. In particular, Henry illustrates the challenges in making tradeoffs to reduce total error, keeping the study goals and resources in mind.

Planning applied social research is not just application of methods; it also involves attention to ethics and the rights of research participants. In Chapter 4, Sieber discusses three major areas of ethics that need to be considered in the design of research: strategies for obtaining informed consent; issues related to, and techniques for ensuring privacy and confidentiality; and strategies for investigators to recognize research risk and, in turn, maximize the benefits of research. Sieber places special emphasis on these areas in the conduct of research with vulnerable populations (e.g., individuals with AIDS) and with children. We know that getting research approved by an institutional review board can sometimes be a long and tortuous process. This chapter, through its many examples and vignettes, will be of great help in obtaining that approval.

CHAPTER 1

Applied Research Design

A Practical Approach

Leonard Bickman

Debra J. Rog

Planning Applied Social Research

The chapters in this *Handbook* describe several approaches to conducting applied social research, including experimental studies (Boruch, Weisburd, Turner, Karpyn, & Littell, Chapter 5), qualitative research (Maxwell, Chapter 7; Fetterman, Chapter 17), and mixed methods studies (Tashakkori & Teddlie, Chapter 9). Regardless of the approach, all forms of applied research have two major phases—planning and execution—and four stages embedded within them (see Figure 1.1). In the planning phase, the researcher defines the scope of the research and develops a comprehensive research plan. During the second phase the researcher implements and monitors the plan (design, data collection and analysis, and management procedures), followed by reporting and follow-up activities.

In this chapter, we focus on the first phase of applied research, the planning phase. Figure 1.2 summarizes the research planning approach advocated here, highlighting the iterative nature of the design process. Although our chapter applies to many different types of applied social research (e.g., epidemiological, survey research, and ethnographies), our examples are largely program evaluation examples, the area in which we have the most research experience. Focusing on program evaluation also permits us to cover many different planning issues, especially the interactions with the sponsor of the research and other stakeholders.

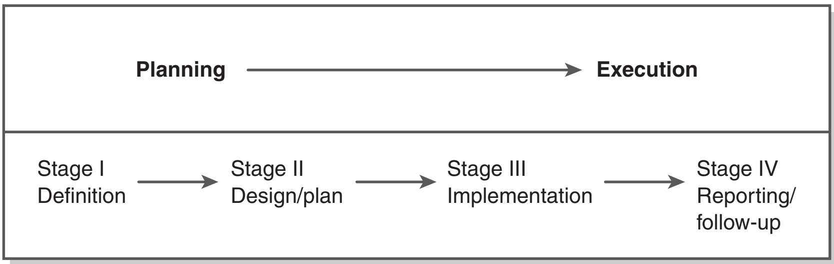

Figure 1.1 The Conduct of Applied Research

Other types of applied research need to consider the interests and needs of the research sponsor, but no other area has the variety of participants (e.g., program staff, beneficiaries, and community stakeholders) involved in the planning stage like program evaluation.

Stage I of the research process starts with the researcher's development of an understanding of the relevant problem or societal issue. This process involves working with stakeholders to refine and revise study questions to make sure that the questions can be addressed given the research conditions (e.g., time frame, resources, and context) and can provide useful information. After developing potentially researchable questions, the investigator then moves to Stage II—developing the research design and plan. This phase involves several decisions and assessments, including selecting a design and proposed data collection strategies.

As noted, the researcher needs to determine the resources necessary to conduct the study, both in the consideration of which questions are researchable as well as in making design and data collection decisions. This is an area where social science academic education and experience is most often deficient and is one reason why academically oriented researchers may at times fail to deliver research products on time and on budget.

Assessing the feasibility of conducting the study within the requisite time frame and with available resources involves analyzing a series of trade-offs in the type of design that can be employed, the data collection methods that can be implemented, the size and nature of the sample that can be considered, and other planning decisions. The researcher should discuss the full plan and analysis of any necessary trade-offs with the research client or sponsor, and agreement should be reached on its appropriateness.

As Figure 1.2 illustrates, the planning activities in Stage II often occur simultaneously, until a final research plan is developed. At any point in the Stage II process, the researcher may find it necessary to revisit and revise earlier decisions, perhaps even finding it necessary to return to Stage I and renegotiate the study questions or timeline with the research client or funder. In fact, the researcher may find that the design that has been developed does not, or cannot, answer the original questions. The researcher needs to review and correct this discrepancy before moving on to Stage III, either revising the questions to bring them in line with what can be done

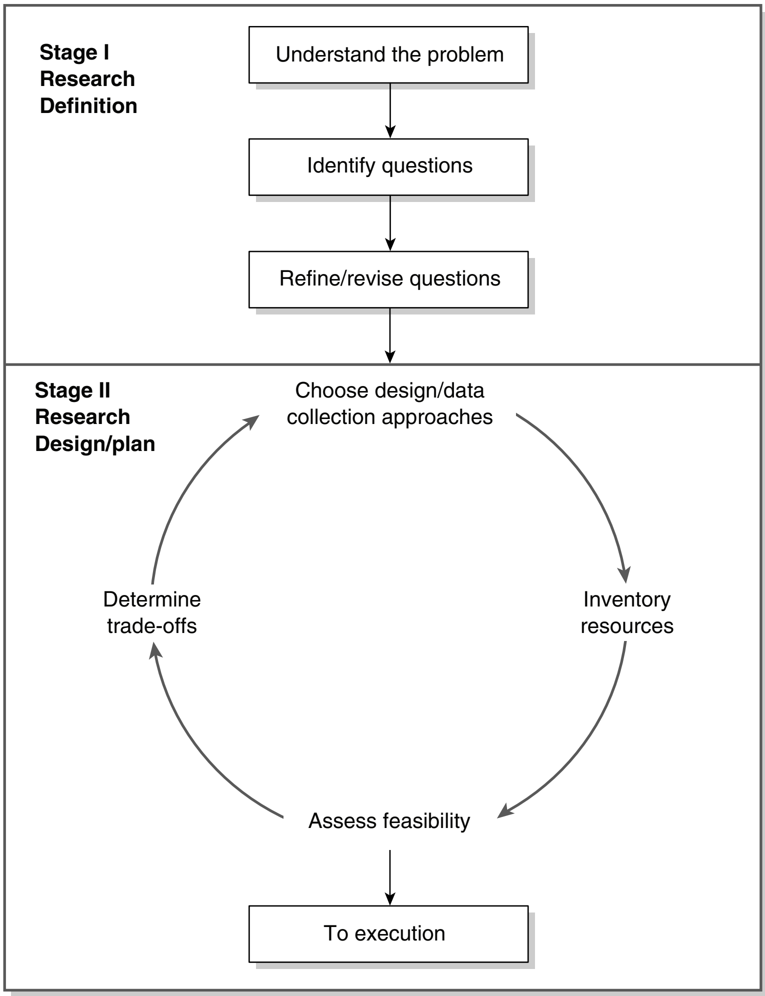

Figure 1.2 Applied Research Planning

with the design that has been developed or reconsidering the design trade-offs that were made and whether they can be revised to be in line with the questions of interest. At times, this may mean increasing the resources available, changing the sample being considered, and other decisions that can increase the plausibility of the design to address the questions of interest.

Depending on the type of applied research effort, these decisions can either be made in tandem with a client or by the research investigator alone. Clearly, involving stakeholders in the process can lengthen the planning process and at some point, may not yield the optimal design from a research perspective. There typically needs to be a balance in determining who needs to be consulted, for what decisions, and when in the process. As described later in the chapter, the researcher needs to have a clear plan and rationale for involving stakeholders in various decisions. Strategies such as concept mapping (Kane & Trochim, Chapter 14) provide a structured mechanism for obtaining input that can help in designing a study. For some research efforts, such as program evaluation, collaboration, and consultation with key stakeholders can help improve the feasibility of a study and may be important to improving the usefulness of the information (Rog, 1985). For other research situations, however, there may be need for minimal involvement of others to conduct an appropriate study. For example, if access or "buy in" is highly dependent on some of the stakeholders, then including them in all major decisions may be wise. However, technical issues, such as which statistical techniques to use, generally do not benefit from, or need stakeholder involvement. In addition, there may be situations in which the science collides with the preferences of a stakeholder. For example, a stakeholder may want to do the research quicker or with fewer participants. In cases such as these, it is critical for the researcher to provide persuasive information about the possible trade-offs of following the stakeholder advice, such as reducing the ability to find an effect if one is actually present—that is, lowering statistical power. Applied researchers often find themselves educating stakeholders about the possible trade-offs that could be made. The researcher will sometimes need to persuade stakeholders to think about the problem in a new way or demonstrate the difficulties in implementing the original design.

The culmination of Stage II is a comprehensively planned applied research project, ready for full-scale implementation. With sufficient planning completed at this point, the odds of a successful study are significantly improved, but far from guaranteed. As discussed later in this chapter, conducting pilot and feasibility studies continues to increase the odds that a study can be successfully mounted.

In the sections to follow, we outline the key activities that need to be conducted in Stage I of the planning process, followed by highlighting the key features that need to be considered in choosing a design (Stage II), and the variety of designs available for different applied research situations. We then go into greater depth on various aspects of the design process, including selecting the data collection methods and approach, determining the resources needed, and assessing the research focus.

Developing a Consensus on the Nature of the Research Problem

Before an applied research study can even begin to be designed, there has to be a clear and comprehensive understanding of the nature of the problem being addressed. For example, if the study is focused on evaluating a program for homeless families being conducted in Georgia, the researcher should know what research and other available information has been developed about the needs and characteristics of homeless families in general and specifically in Georgia; what evidence base exists, if any for the type of program being tested in this study; and so forth. In addition, if the study is being requested by an outside sponsor, it is important to have an understanding of the impetus of the study and what information is desired to inform decision making.

Strategies that can be used in gathering the needed information include the following:

- review relevant literature (research articles and reports, transcripts of legislative hearings, program descriptions, administrative reports, agency statistics, media articles, and policy/position papers by all major interested parties);
- gather current information from experts on the issue (all sides and perspectives) and major interested parties;
- conduct information-gathering visits and observations to obtain a real-world sense of the context and to talk with persons actively involved in the issue;
- initiate discussions with the research clients or sponsors (legislative members; foundation, business, organization, or agency personnel; and so on) to obtain the clearest possible picture of their concerns; and
- if it is a program evaluation, informally visit the program and talk with the staff, clients, and others who may be able to provide information on the program and/or overall research context.

Developing the Conceptual Framework

Every study, whether explicitly or implicitly, is based on a conceptual framework or model that specifies the variables of interest and the expected relationships between them. In some studies, social and behavioral science theory may serve as the basis for the conceptual framework. For example, social psychological theories such as cognitive dissonance may guide investigations of behavior change. Other studies, such as program and policy evaluations, may be based not on formal academic theory but on statements of expectations of how policies or programs are purported to work. Bickman (1987, 1990) and others (e.g., Chen, 1990) have written extensively about the need for and usefulness of program theory to guide evaluations. The framework may be relatively straightforward or it may be complex, as in the case of evaluations of comprehensive community reforms, for example, that are concerned with multiple effects and have a variety of competing explanations for the effects (e.g., Rog & Knickman, 2004).

In evaluation research, logic models have increased in popularity as a mechanism for outlining and refining the focus of a study (Frechtling, 2007; McLaughlin & Jordan, 2004; Rog, 1994; Rog & Huebner, 1992; Yin, Chapter 8, this volume). A logic model, as the name implies, displays the underlying logic of the program (i.e., how the program goals, resources, activities, and outcomes link together). In several instances, a program is designed without explicit attention to the evidence base available on the topic and/or without explicit attention to what immediate and intermediate outcomes each program component and activity needs to accomplish to ultimately reach the desired longer-term outcomes. The model helps display these gaps in logic and provides a guide for either refining the program and/or outlining more of the expectations for the program. For example, community coalitions funded to prevent community violence need to have an explicit logic that details the activities they are intended to conduct that should lead to a set of outcomes that chain logically to the prevention of violence.

The use of logic modeling in program evaluation is an outgrowth of the evaluability assessment work of Wholey and others (e.g., Wholey, 2004), which advocates describing and displaying the underlying theory of a program as it is designed and implemented prior to conducting a study of its outcomes. Evaluators have since discovered the usefulness of logic models in assisting program developers in the program design phase, guiding the evaluation of a program's effectiveness, and communicating the nature of a program as well as changes in its structure over time to a variety of audiences. A program logic model is dynamic and changes not only as the program matures but also may change as the researcher learns more about the program. In addition, a researcher may develop different levels of models for different purposes; for example, a global model may be useful for communicating to outside audiences about the nature and flow of a program, but a detailed model may be needed to help guide the measurement phase of a study.

In the design phase of a study (Stage II), the logic model will become important in guiding both the measurement and analysis of a study. For these tasks, the logic model needs to not only display the main features of a program and its outcomes but also the variables that are believed to mediate the outcomes as well as those that could moderate an intervention's impact (Baron & Kenny, 1986). Mediating variables, often referred to as intervening or process variables, are those variables through which an independent variable (or program variable) influences an outcome. For example, the underlying theory of a therapeutic program designed to improve the overall well-being of families may indicate that the effect of the program is mediated by the therapeutic alliance developed between the families and the program staff. In other words, without the development of a therapeutic alliance, it is not expected that the program can have an effect. Often, mediators are short-term outcomes that are believed to be logically necessary for a program to first accomplish in order to achieve the longer-term outcomes.

Moderators are those variables that explain differences in outcomes due to preexisting conditions. For example, demographic variables, such as gender, age, income, and others are often tested as moderators of a program's effects. Contextual variables also can act as moderators of the effects of a program; for example, a housing program for homeless families is expected to have greater effect on housing stability in communities that have higher housing vacancy rates than those with lower rates (i.e., less available housing).

Identifying the Research Questions

As noted in the introduction to this *Handbook*, one of the major differences between basic research and applied research is that the basic researcher is more autonomous than the applied researcher. Basic research, when externally funded, is typically conducted through a relatively unrestricted grant mechanism; applied research is more frequently funded through contracts and cooperative agreements. Even when applied research is funded through grant mechanisms, such as with foundations, there is usually a "client" or sponsor who specifies (or at least guides) the research agenda and requests the research results. Most often, studies have multiple stakeholders: sponsors, interested beneficiaries, and potential users (Bickman

& Rog, 1986). The questions to be addressed by an applied study tend to be posed by individuals other than the researcher, often by nontechnical persons in nontechnical language.

Therefore, one of the first activities in applied research is working with the study clients to develop a common understanding of the research agenda—the research questions. Phrasing study objectives as questions is desirable in that it leads to more clearly focused discussion of the type of information needed. It also makes it more likely that key terms (e.g., welfare dependency, drug use) will be operationalized and clearly defined. Using the logic models also helps focus the questions on what is expected from the program and to move to measurable variables to both study the process of an intervention or program as well as its expected outcomes. Later, after additional information has been gathered and reviewed, the parties will need to reconsider whether these questions are the "right" questions and whether it is possible, with a reasonable degree of confidence, to obtain answers for these questions within the available resource and time constraints.

Clarifying the Research Questions

In discussing the research agenda with clients, the researcher will usually identify several types of questions. For example, in a program evaluation, researchers are frequently asked to produce comprehensive information on both the implementation ("what actually is taking or took place") and the effects ("what caused what") of an intervention. When the research agendas are broad such as those in the example, they pose significant challenges for planning in terms of allocating data collection resources among the various study objectives. It is helpful to continue to work with the sponsors to further refine the questions to both more realistically plan the scope of the research and to also ensure that they are specific enough to be answered in a meaningful way and one that is agreed on by the clients.

The researcher should guard against biasing the scope of the research. The questions left unaddressed by a study can be as or more important than the questions answered. If the research addresses only questions likely to support only one position in a controversy and fails to develop information relevant to the concerns voiced by other interested parties, it will be seen as biased, even if the results produced are judged to be sound and conclusive. For example, an evaluation that is limited to measuring just the stated goals of a program may be biased if any possible unintended negative side effects of the program are not considered. Thus, the research agenda should be as comprehensive as is necessary to address the concerns of all parties. Resource constraints will limit the number and scope of questions that may be addressed, but at minimum the researcher should state explicitly what would be necessary for a comprehensive study and how the research meets or does not meet those requirements. Resources will also determine the degree of certainty one can have in an answer. Thus, a representative survey is much more expensive to conduct than sampling by convenience, but the generalizability of the results will be much stronger in the representative sample.

Ideally, the development of the conceptual framework/logic model will occur simultaneously with the identification of the research questions. Once the conceptual framework has been agreed on, the researcher can further refine the study questions—grouping questions and identifying which are primary and secondary questions. Areas that need clarification include the time frame of the data collection (i.e., "Will it be a cross-sectional study or one that will track individuals or cohorts over time; how long will the follow-up period be?"); how much the client wants to generalize (e.g., "Is the study interested in providing outcome information on all homeless families that could be served in the program or only those families with disabilities?"); how certain the client wants the answers to be (i.e., "How precise and definitive should the data collected be to inform the decisions?"); and what subgroups the client wants to know about (e.g., "Is the study to provide findings on homeless families in general only or is there interest in outcomes for subgroups of families, such as those who are homeless for the first time, those who are homeless more than once but for short durations, and those who are 'chronically homeless'?"). The levels of specificity should be very high at this point, enabling a clear agreement on what information will be produced. As the next section suggests, these discussions between researcher and research clients oftentimes take on the flavor of a negotiation.

Negotiating the Scope of a Study

Communication between the researcher and stakeholders (the sponsor and all other interested parties) is important in all stages of the research process. To foster maximum and accurate utilization of results, it is recommended that the researcher regularly interact with the research clients—from the initial discussions of the "problem" to recommendations and follow-up. In the planning phase, we suggest several specific communication strategies. As soon as the study is sponsored, the researcher should connect with the client to develop a common understanding of the research questions, the client's time frame for study results, and anticipated uses for the information. The parties can also discuss preliminary ideas regarding a conceptual model for the study. Even in this initial stage, it is important for the researcher to begin the discussion of the contents and appearance of the final report. This is an opportunity for the researcher to explore whether the client expects only to be provided information on study results or whether the client anticipates that the researcher will offer recommendations for action. It is also an opportunity for the researcher to determine whether he or she will be expected to provide interim findings to the client as the study progresses.

At this juncture, the researcher also needs to have an understanding of the amount of funds or resources that will be available to support the research. Cost considerations will determine the scope and nature of the project, and the investigator needs to consider the resources while identifying and reviewing the research questions. In some studies, the budget is set prior to any direct personal contact with the research client. In others, researchers may help to shape the scope and the resources needed simultaneously or there may be a pilot effort that helps design the larger study.

Based on a comprehensive review of the literature and other inputs (e.g., from experts) and an initial assessment of resources, the researcher should decide if the research questions need to be refined. The researcher and client then typically discuss the research approaches under consideration to answer these questions as well as the study limitations. This gives the researcher an opportunity to introduce constraints into the discussion regarding available resources, time frames, and any trade-offs contemplated regarding the likely precision and conclusiveness of answers to the questions.

In most cases, clients want sound, well-executed research and are sympathetic to researchers' need to preserve the integrity of the research. Some clients, however, have clear political, organizational, or personal agendas, and will push researchers to provide results in unrealistically short time frames or to produce results supporting particular positions. Other times, the subject of the study itself may generate controversy, a situation that requires the researcher to take extreme care to preserve the neutrality and credibility of the study. Several of the strategies discussed later attempt to balance client and researcher needs in a responsible fashion; others concentrate on opening research discussions up to other parties (e.g., advisory groups). In the earliest stages of research planning, it is possible to initiate many of these kinds of activities, thereby bolstering the study's credibility, and often its feasibility.

Stage II: The Research Design

Having developed a preliminary study scope during Stage I, the researcher moves to Stage II, developing a research design and plan. During this stage, the applied researcher needs to perform five activities almost simultaneously: selecting a design, choosing data collection approaches, inventorying resources, assessing the feasibility of executing the proposed approach, and determining trade-offs. These activities and decisions greatly influence one another. For example, a researcher may revisit preliminary design selections after conducting a practical assessment of the resources available to do the study, and may change data collection plans after discovering weaknesses in the data sources during planning.

The design serves as the architectural blueprint of a research project, linking design, data collection, and analysis activities to the research questions and ensuring that the complete research agenda will be addressed. A research study's credibility, usefulness, and feasibility rest with the design that is implemented. *Credibility* refers to the validity of a study and whether the design is sufficiently rigorous to provide support for definitive conclusions and desired recommendations. Credibility is also, in part, determined by who is making the judgment. To some sponsors, a credible project need only use a pre-post design. Others may require a randomized experimental design to consider the findings credible. Credibility is also determined by the research question. A representative sample will make a descriptive study more credible than a sample of convenience or one with known biases. In contrast, representativeness is not as important in a study designed to determine the causal link between a program and outcomes. The planner needs to be sure that the design matches the types of information needed. For example, under most circumstances, the simple pre-post design should not be used if the purpose of the study is to draw causal conclusions.

Usefulness refers to whether the design is appropriately targeted to answer the specific questions of interest. A sound study is of little use if it provides definitive answers to the wrong questions. *Feasibility* refers to whether the research design can be executed, given the requisite time and other resource constraints. All three factors—credibility, usefulness, and feasibility—must be considered to conduct high-quality applied research.

Design Dimensions

Maximizing Validity

In most instances, a credible research design is one that maximizes validity—it provides a clear explanation of the phenomenon under study and controls all plausible biases or confounds that could cloud or distort the research findings. Four types of validity are typically considered in the design of applied research (Bickman, 1989; Shadish, Cook, & Campbell, 2002).

- *Internal validity:* the extent to which causal conclusions can be drawn or the degree of certainty that "A" caused "B," where A is the independent variable (or program) and B is the dependent variable (or outcome).
- *External validity:* the extent to which it is possible to generalize from the data and context of the research study to other populations, times, and settings (especially those specified in the statement of the original problem/issue).
- *Construct validity:* the extent to which the constructs in the conceptual framework are successfully operationalized (e.g., measured or implemented) in the research study. For example, does the program as actually implemented accurately represent the program concept and do the outcome measures accurately represent the outcome? Programs change over time, especially if fidelity to the program model or theory is not monitored.
- *Statistical conclusion validity:* the extent to which the study has used appropriate sample size, measures, and statistical methods to enable it to detect the effects if they are present. This is also related to the statistical power.

All types of validity are important in applied research, but the relative emphases may vary, depending on the type of question under study. With questions dealing with the effectiveness of an intervention or impact, for example, more emphasis should be placed on internal and statistical conclusion validity than on external validity. The researcher of such a study is primarily concerned with finding any evidence that a causal relationship exists and is typically less concerned (at least initially) about the transferability of that effect to other locations or populations. For descriptive questions, external and construct validity may receive greater emphasis. Here, the researcher may consider the first priority to be developing a comprehensive and rich picture of a phenomenon. The need to make cause-effect attributions is not relevant. Construct validity, however, is almost always relevant.

Operationalizing the Key Variables and Concepts

The process of refining and revising the research questions undertaken in Stage I should have yielded a clear understanding of the key research variables and concepts. For example, if the researcher is charged with determining the extent of high school drug use (a descriptive task), key outcome variables might include drug type, frequency and duration of drug use, and drug sales behavior. Attention should be given at this point to reassessing whether the researcher is studying the right variables—that is, whether these are "useful" variables.

Outlining Comparisons

An integral part of design is identifying whether and what comparisons can be made—that is, which variables must be measured and compared with other variables or with themselves over time. In simple descriptive studies, there are decisions to be made regarding the time frame of an observation and how many observations are needed. Typically, there is no explicit comparison in simple descriptive studies. Normative studies are an extension of descriptive studies in that the interest is in comparing the descriptive information to some appropriate "standard." The decision for the researcher is to determine where that standard will be drawn from or how it will be developed. In correlative studies, the design is again an extension of simple descriptive work, with the difference that two or more descriptive measures are arrayed against each other to determine whether they covary. Impact or outcome studies, by far, demand the most judgment and background work. To make causal attributions (*X* causes *Y*), we must be able to compare the condition of *Y* when *X* occurred with what the condition of *Y* would have been without *X*. For example, to know if a drug treatment program reduced drug use, we need to compare drug use among those who were in the program with those who did not participate in the program.

Level of Analysis

Knowing what level of analysis is necessary is also critical to answering the "right" question. For example, if we are conducting a study of drug use among high school students in Toledo, "Are we interested in drug use by individual students, aggregate survey totals at the school level, aggregate totals at the school district, or for the city as a whole?"

Correct identification of the proper level or unit of analysis has important implications for both data collection and analysis. The Stage I client discussions should clarify the desired level of analysis. It is likely that the researcher will have to help the client think through the implications of these decisions, providing information about research options and the types of findings that would result. In addition, this is an area that is likely to be revisited if initial plans to obtain data at one level (e.g., the individual student level) prove to be prohibitively expensive or unavailable. A design fallback position may be to change to an aggregate analysis level (e.g., the school), particularly if administrative data at this level are more readily available and less costly to access.

In an experiment, the level of analysis is typically determined by the level that the intervention is introduced. For example, if the intervention was targeted at individual students, then that should usually be the level of analysis. Similarly, a classroom intervention should use classroom as the level and a schoolwide intervention should use the school. It is tempting to use the lowest level with the largest sample size because that provides the most statistical power—that is, ability to find an effect if one is there. For example, if an intervention is at the school level and there is only a treatment and control school then the sample size is two, not the total number of students. Statistical programs that take into account multilevel designs are easily accessible (Graham, Singer, & Willett, 2008). However, the real challenge with multilevel designs is finding enough units (e.g., schools) to cooperate as well as enough resources to pay for the study.

Population, Geographic, and Time Boundaries

Population, geographic, and time boundaries are related to external validity issues. Each can affect the generalizability of the research results—for instance, whether the results will be representative of all high school students, all high school students graduating within the past 3 years, all students in urban areas, and so on. Population generalizability and geographic generalizability are probably the most commonly discussed types of generalizability, and researchers frequently have heated debates concerning whether the persons or organizations that they have studied and the locations where they conducted their studies will allow them to use their findings in different locations and with different populations. In basic research, generalizability or external validity is usually not considered but in applied research some may rate it more important than internal validity (Cronbach et al., 1980).

Time boundaries also can be crucial to the generalizability of results, especially if the study involves extant data that may be more than a few years old. With the fast pace of change, questions can easily arise about whether survey data on teenagers from even just 2 years prior are reflective of current teens' attitudes and behaviors.

The researcher cannot study all people, all locations, or all time periods relevant to the problem/program under scrutiny. One of the great "inventions" for applied social research is sampling. Sampling allows the researcher to study only a subset of the units of interest and then generalize to all these units with a specifiable degree of error. It offers benefits in terms of reducing the resources necessary to do a study; it also sometimes permits more intensive scrutiny by allowing a researcher to concentrate on fewer cases. More details on sampling can be found in Henry (1990; see also Sieber, Chapter 4, this volume).

Level of Precision

Knowing how precise an answer must be is also crucial to design decisions. The level of desired precision may affect the rigor of the design. When sampling is used, the level of desired precision also has important ramifications for how the sample is drawn and the size of the sample used. In initial discussions, the researcher and the client should reach an understanding regarding the precision desired or necessary overall and with respect to conclusions that can be drawn about the findings for specific subgroups. The cost of a study is very heavily influenced by the degree of precision or certainty required. In sampling, more certainty usually requires a bigger sample size, with diminishing returns when samples approach 1,000. However, if the study is focused on subgroups, such as gender or ethnicity, then the sample at those levels of analysis must also be larger.

Another example of precision is the breadth and depth of a construct that need to be measured in a study. More breadth usually requires more questions, and greater depth often requires the use of in-depth interviewing, both likely increasing the costs of data collection especially if administered in person or with a telephone interview. The level of precision is discussed later in the section dealing with trade-offs as level of precision is often a trade-off decision that must be made within the budget of a study.

Choosing a Design

There are three main categories of applied research designs: descriptive, experimental, and quasi-experimental. In our experience, developing an applied research design rarely allows for implementing a design straight from a textbook; rather, the process more typically involves the development of a hybrid, reflecting combinations of designs and other features that can respond to multiple study questions, resource limitations, dynamics in the research context, and other constraints of the research situation (e.g., time deadlines). Thus, our intent here is to provide the reader with the tools to shape the research approach to the unique aspects of each situation. Those interested in more detailed discussion should consult Mark and Reichardt's work on quasi-experimentation (Chapter 6) and Boruch and colleagues' chapter on randomized experiments (Chapter 5). In addition, our emphasis here is on quantitative designs; for more on qualitative designs, readers should consult Maxwell (Chapter 7), Yin (Chapter 8), and Fetterman (Chapter 17).

Descriptive Research Designs

Description and Purpose. The overall purpose of descriptive research is to provide a "picture" of a phenomenon as it naturally occurs, as opposed to studying the effects of the phenomenon or intervention. Descriptive research can be designed to answer questions of a univariate, normative, or correlative nature—that is, describing only one variable, comparing the variable to a particular standard, or summarizing the relationship between two or more variables.

Key Features. Because the category of descriptive research is broad and encompasses several different types of designs, one of the easiest ways to distinguish this class of research from others is to identify what it is not: It is not designed to provide information on cause-effect relationships.

Variations. There are only a few features of descriptive research that vary. These are the representativeness of the study data sources (e.g., the subjects/entities)—that is, the manner in which the sources are selected (e.g., universe, random sample, stratified sample, nonprobability sample); the time frame of measurement—that is, whether the study is a one-shot, cross-sectional study, or a longitudinal study; whether the study involves some basis for comparison (e.g., with a standard, another group or population, data from a previous time period); and whether the design is focused on a simple descriptive question, on a normative question, or on a correlative question.

When to Use. A descriptive approach is appropriate when the researcher is attempting to answer "what is," or "what was," or "how much" questions.

Strengths. Exploratory descriptive studies can be low cost, relatively easy to implement, and able to yield results in a fairly short period of time. Some efforts, however, such as those involving major surveys, may sometimes require extensive resources and intensive measurement efforts. The costs depend on factors such as the size of the sample, the nature of the data sources, and the complexity of the data collection methods employed. Several chapters in this volume outline approaches to surveys, including mail surveys (Mangione & Van Ness, Chapter 15), internet surveys (Best & Harrison, Chapter 13), and telephone surveys (Lavrakas, Chapter 16).

Limitations. Descriptive research is not intended to answer questions of a causal nature. Major problems can arise when the results from descriptive studies are inappropriately used to make causal inferences—a temptation for consumers of correlational data.

Experimental Research Designs

Description and Purpose. The primary purpose in conducting an experimental study is to test the existence of a causal relationship between two or more variables. In an experimental study, one variable, the independent variable, is systematically varied or manipulated so that its effects on another variable, the dependent variable, can be measured. In applied research, such as in program evaluation, the "independent variable" is typically a program or intervention (e.g., a drug education program) and the "dependent variables" are the desired outcomes or effects of the program on its participants (e.g., drug use, attitudes toward drug use).

Key Features. The distinguishing characteristic of an experimental study is the random assignment of individuals or entities to the levels or conditions of the study. Random assignment is used to control most biases at the time of assignment and to help ensure that only one variable—the independent (experimental) variable—differs between conditions. With well-implemented random assignment, all individuals have an equal likelihood of being assigned either to the treatment group or to the control group. If the total number of individuals or entities assigned to the treatment and control groups is sufficiently large, then any differences between the groups should be small and due to chance.

Variations. The most basic experimental study is called a *post-only design,* in which individuals are randomly assigned either to a treatment group or to a control group, and the measurement of the effects of the treatment is conducted at a given period following the administration of the treatment. There are several variations to this simple experimental design that can respond to specific information needs as well as provide control over possible confounds or influences that may exist. Among the features that can be varied are the number and scheduling of posttest measurement or observation periods, whether a preobservation is conducted, and the number of treatment and control groups used. The post-only design is rarely used because faulty random assignment may result in the control and treatment groups not being equivalent at the start of the study. Few researchers are that (over) confident in the implementation of a field randomized design to take the chance that the results could be interpreted as being caused by faulty implementation of the design.

When to Use. An experimental study is the most appropriate approach to study cause-effect relationships. There are certain situations that are especially conductive to randomized experiments (Boruch et al., Chapter 5, this volume; Shadish et al., 2002) when random assignment is expected (i.e., certain scarce resources may already be provided on a "lottery" or random basis), when demand outstrips supply for an intervention, and when there are multiple entry groups over a period of time.

Strengths. The overwhelming strength of a randomized experiment is its control over threats to internal validity—that is, its ability to rule out potential alternative explanations for apparent treatment or program effects. This strength applies to both the variables that are measured and, more important, the variables that are not measured and, thus, are unknown to the researcher but continue to be controlled by the design.

Limitations. Randomized experiments can be difficult to implement with integrity, particularly in settings where the individuals responsible for random assignment procedures lack research training or understanding of the importance of maintaining compliance with the research protocol (Bickman, 1985; Cook, 2002; Riccio & Bloom, 2002). In addition, random assignment does not control for all biases such as participant preference for one condition over the other (Macias, Hargreaves, Bickman, Fisher, & Aronson, 2005) or local history where some external event occurs for one group but not for the other.

Quasi-Experimental Designs

Description and Purpose. Quasi-experimental designs have the same primary purpose as experimental studies—to test the existence of a causal relationship between two or more variables. They are used when random assignment is not feasible or desired.

Key Features. Quasi-experiments attempt to approximate randomized experiments by substituting other design features for the randomization process. There are generally two ways to create a quasi-experimental comparison base—through the addition of nonequivalent comparison groups or through the addition of pre- and posttreatment observations on the treated group; preferably, both methods should be used.

If comparison groups are used, they are generally referred to as *nonequivalent* comparison groups based on the fact that they cannot be equivalent with the treatment group as in a randomized experiment. The researcher, however, strives to develop procedures to make these groups as equivalent as possible to provide necessary information and control so that competing explanations for their results can be ruled out.

Variations. Quasi-experiments vary along several of the same dimensions that are relevant for experiments. Overall, there are two main types of quasi-experiments: those involving data collection from two or more nonequivalent groups and those involving multiple observations over time. More specifically, quasi-experimental designs can vary along the following dimensions: the number and scheduling of pre- or postobservation periods; the nature of the observations—whether the preobservation uses the same measurement procedure as the postobservation, or whether both are using measures that are proxies for the real concept; the manner in which the treatment and comparison groups are determined; and whether the treatment group serves as its own comparison group or a separate comparison group or groups are used.

Some of the strongest time-series designs supplement a time series for the treatment group with comparison time series for another group (or time period). Another powerful variation occurs when the researcher is able to study the effects of an intervention over time under circumstances where that intervention is both initiated and later withdrawn. A third strong design is the regression discontinuity design in which participants are assigned to a treatment or comparison group based on a clearly designated pretest score. Although this design has been used in clinical screening (e.g., CATS Consortium, 2007), it is rarely used as most studies do not involve the use of a pretest score as a cutoff.

When to Use. A quasi-experimental design is not the method of choice but rather a fallback strategy for situations in which random assignment is not feasible. Situations such as these include when the nature of the independent variable precludes the use of random assignment (e.g., exposure or involvement in a natural disaster); retrospective studies (e.g., the program is already well under way or over); studies focused on economic or social conditions, such as unemployment; when randomization is too expensive, not feasible to initiate, or impossible to monitor closely; when there are obstacles to withholding the treatment or when it seems unethical to withhold it; and when the timeline is tight and a quick decision is mandated.

Strengths. The major strength of the quasi-experimental design is that it provides an approximation to the experimental design and supports causal inferences. Although often open to several types of threats to internal validity (see Mark & Reichardt, Chapter 6), the quasi-experiment does provide a mechanism for chipping away at the uncertainty surrounding the existence of a specific causal relationship. Additional nonequivalent comparison groups also can bolster an experimental design, particularly if it is narrowly focused.

Limitations. The greatest vulnerability of quasi-experimental designs is the possibility that the comparison group created is biased and that it does not give an accurate estimate of what the situation would have been in the absence of the treatment or program. This is especially a concern when participants self-select into treatment or control groups. Although not a perfect remedy, propensity score matching is increasingly used as a technique for helping to correct for selection bias between treatment and comparison groups (Foster, 2003; Rosenbaum & Rubin, 1983, 1984; Rubin, 1997). A propensity score is a composite of variables that controls on known differences between two groups by creating matches or subgroups of cases that are similar on this score.

Selecting Data Collection Approaches

Concurrent with deciding on a design, the researcher should investigate possible data collection approaches. Most applied research studies, particularly those investigating multiple research questions, often encompass several data collection efforts. We begin this section with a discussion of the data collection issues that the researcher must consider during the planning stage, including the sources of data available, the form in which the data are available, the amount of data needed, the accuracy and reliability of the data, and whether the data fit the parameters of the design. We then review the major methods of data collection that are used in applied research and discuss the need for an analysis plan.

Sources of Data

The researcher should identify the likely sources of data to address the research questions. Data sources typically fall into one of two broad categories: primary and secondary. Among the potential primary data sources that exist for the applied researcher are people (e.g., community leaders, program participants, service providers, the general public), independent descriptive observations of events and activities, physical documents, and test results. These data are most often collected by the investigator as part of the study through one or more methods (e.g., questionnaires, interviews, observations). Secondary sources can include administrative records, management information systems, economic and social indicators, and various types of documents (e.g., prior research studies, fugitive unpublished research literature) (Gorard, 2002; Hofferth, 2005; Stewart & Kamins, 1993). Typically the investigator does not collect these data but uses already existing sources such as census data, program administrative records, and others. In recent years, there has been an increasing emphasis on performance-monitoring systems and the implementation of management information systems, especially in agencies and organizations that receive government funding. These systems can be often considered potential sources to tap in applied research projects depending on the quality and completeness of the data collected (as discussed below).

Form of the Data

The form in which the data are found is a very important factor for any applied research project and may even determine the overall feasibility of the study. Some projects are easy to conduct—the data sources are obvious and the data are already gathered, archived, and computerized. The researcher need only request access to the files and have the ability to transfer them. However, even these data may not be easy to use if the data have problems such as missing or duplicated cases or are composed of different files that require matching clients across files. Other projects are extremely difficult—identifying appropriate sources for the needed information may be confusing, and it may turn out that the procedures necessary for obtaining the information are expensive and time-consuming. Gathering data may sometimes be so difficult that the study is not feasible—at least not within the available level of resources and other constraints. For example, a study of several school systems required that the researchers have access to the student achievement data. Obtaining these data sets actually took several years because the researchers' needs were not a high priority in the school systems relative to other priorities. Moreover, one of the school districts was changing computer software, further delaying the process. The lesson here is what seems like a simple request is usually not that simple.

Possible forms of data include self-reports (e.g., attitudes, behaviors and behavioral intentions, opinions, memories, characteristics, and circumstances of individuals), computerized or manual (i.e., hard copy) research databases or administrative records, observations (e.g., events, actions, or circumstances that need to be described or recorded), biobehavioral measures (e.g., urinalysis to measure drug use), and various kinds of documentary evidence (e.g., letters, invoices, receipts, meeting minutes, memoranda, plans, reports).

Self-Report Data

When dealing with self-reported data, the researcher may ask individual research participants to provide, to the best of their ability, information on the areas of interest. These inquiries may be made through individual interviews, through telephone or mail surveys, Web-based surveys, or through written corroboration or affirmation. Self-report data may be biased if the questions deal with socially desirable behavior, thoughts, or attitudes. In general, people like to present themselves in a positive way. Making the data collection anonymous may improve the accuracy of these data, especially about sensitive topics. However, anonymous data can be difficult, but not impossible, to use in the conduct of longitudinal studies.

Extant Databases

When dealing with extant data from archival sources, the researcher is generally using the information for a purpose other than that for which they were originally collected. There are several secondary data sources that are commonly used, such as those developed by university consortia, federal sources such as the Bureau of the

Census, state and local sources such as Medicaid databases, and commercial sources such as Inform, a database of 550 business journals.

Given the enormous amount of information routinely collected on individuals in U.S. society, administrative databases are a potential bonanza for applied researchers. More and more organizations, for example, are computerizing their administrative data and archiving their full databases at least monthly. Management information systems, in particular, are becoming more common in service settings for programmatic and evaluation purposes as well as for financial disbursement purposes.

Administrative data sets, however, have one drawback in common with databases of past research—they were originally constructed for operational purposes, not to meet the specific objectives of the researcher's task. When the data are to be drawn from administrative databases, the researcher should ask the following questions: Are the records complete? Why were the data originally collected? Did the database serve some hidden political purpose that could induce systematic distortions? What procedures have been used to deal with missing data? Do the computerized records bear a close resemblance to the original records? Are some data items periodically updated or purged from the computer file? How were the data collected and entered, and by whom?

Biobehavioral Data

Biobehavioral measures are becoming increasingly important, especially in health and health-related research. Body mass index, for example, is often used in research on obesity as a measure of fitness (Flegal, Carroll, Ogden, & Johnson, 2002). Increasingly, in studies of illegal behavior, such as drug use, biobehavioral measures using urinalysis are viewed as more valid than self-reports due to the stigma associated with the behavior (e.g., Kim & Hill, 2003). Many of the measures, however, require the use of advanced technology and can increase the expense of data collection.

Observational Data

Observational procedures become necessary when events, actions, or circumstances are the major form of the data. If the events, actions, or circumstances are repetitive or numerous, this form of data can be easier to collect than data composed of rare events that are difficult to observe. Because the subject of the data collection is often complex, the researcher may need to create detailed guidelines to structure the data collection, coding, and analysis (see Maxwell, Chapter 7, for more detail on qualitative data categorization and analysis).

Documents

Documentary evidence may also serve as the basis for an applied researcher's data collection. Particular kinds of documents may allow the researcher to track what happened, when it happened, and who was involved. Examples of documentary data include meeting minutes, journals, and program reports. Investigative research may rely on documentary evidence, often in combination with data from interviews.

Amount of Data

The research planner must anticipate the amount of data that will be needed to conduct the study. Planning for the appropriate amount involves decisions regarding the number and variety of data sources, the time periods of interest, and the number of units (e.g., study participants), as well as the precision desired. As noted earlier, statistical conclusion validity concerns primarily those factors that might make it appear that there were no statistically significant effects when, in fact, there were effects.

Effect size is defined as the proportion of variance accounted for by the treatment, or as the difference between a treatment and control group measured in standard deviation units. The purpose of using standard deviation units is to produce a measure that is independent of the metric used in the original variable. Thus, we can discuss universal effect sizes regardless of whether we are measuring school grades, days absent, or self-esteem scores. This makes possible the comparison of different studies and different measures in the same study. Conversion to standard deviation units can be obtained by subtracting the mean of the control group from the mean of the treatment group and then dividing this difference by the pooled or combined standard deviations of the two groups.

There are several factors that could account for not finding an effect when there actually is one. As Lipsey and Hurley (Chapter 2) indicate, there are four factors that govern statistical power: the statistical test, the alpha level, the sample size, and the effect size. Many researchers, when aware of power concerns, mistakenly believe that increasing sample size is the only way to increase statistical power. Increasing the amount of data collected (the sample size) is clearly one route to increasing power; however, given the costs of additional data collection, the researcher should consider an increase in sample size only after he or she has thoroughly explored the alternatives of increasing the sensitivity of the measures, improving the delivery of treatment to obtain a bigger effect, selecting other statistical tests, and raising the alpha level. If planning indicates that power still may not be sufficient, then the researcher faces the choice of not conducting the study, changing the study to address more qualitative questions, or proceeding with the study but informing the clients of the risk of "missing" effects below a certain size. (More information on how to improve the statistical power of a design can be found in Lipsey & Hurley, Chapter 2.)

With qualitative studies, the same set of trade-offs are made in planning how much data to collect—that is, consideration of the number and variety of data sources available, the time periods of interest, and the number of units, as well as the precision desired (see Harrison, Chapter 10). Precision in qualitative studies, however, does not refer to statistical power as much as the need for triangulation to establish the validity of conclusions. Triangulation refers to the use of multiple data sources and/or methods to measure a construct or a phenomenon in order to see if they converge and support the same conclusions. The more diverse the sources and methods, the greater confidence there is in the convergence of the findings. Maxwell (Chapter 7) describes a number of strategies, including triangulation, for ensuring and assessing the validity of conclusions from qualitative data.

Accuracy and Reliability of Data

Data are not useful if they are not accurate, valid, and reliable. The concept of construct validity (i.e., Are we measuring what we intend to measure?) is relevant whether one is using extant data or collecting primary data. The researcher is concerned that the variables used in the study are good operationalizations of key variables in the study's conceptual framework.

The researcher must also be concerned with the possibility of large measurement errors. Whenever there is measurement of a phenomenon, there is some level of error. The error may be random or systematic. It is important for the researcher to remember that just about all measures contain some degree of error; the challenge is to minimize the error or understand it sufficiently to adjust the study. If the error is systematic (i.e., not random), the researcher may be able to correct statistically for the bias that is introduced. However, it is often difficult for the researcher to discover that any systematic error exists, let alone its magnitude. Random error can best be controlled through the use of uniform procedures in data collection.

Researchers should be cautious in the development of their own measures. As noted in other chapters in the *Handbook* (Fowler & Cosenza, Chapter 12), developing a good questionnaire requires more than writing some questions. In one of our projects, we needed to use instruments that were short, valid, reliable, and free. Unfortunately, such measures are rare in the child and adolescent mental therapeutic alliance and session impact. Developing these measures was a yearlong activity that consumed a great deal of time and money. Creating the questions was the easy part. We needed to conduct cognitive testing to determine if the respondents were interpreting the instructions and questions as expected, piloting for length, and then intensive psychometric testing that included collecting data from more than 1,000 respondents and analyzing the data using both classical and item response theory approaches. The test battery is available free at http://peabody.vanderbilt.edu/ptpb.

Design Fit

Even when accurate and reliable data exist or can be collected, the researcher must ask whether the data fit the necessary parameters of the design. Are they available on all necessary subgroups? Are they available for the appropriate time periods? Is it possible to obtain data at the right level of analysis (e.g., individual student vs. school)? Do different databases feeding into the study contain comparable variables? Are they coded the same way?

If extant databases are used, the researcher may need to ask if the database is sufficiently complete to support the research. Are all variables of interest present? If an interrupted time-series design is contemplated, the researcher may need to make sure that it is possible to obtain enough observations prior to the intervention in question and that there has been consistency in data reporting throughout the analytic time frame.

Types of Data Collection Instruments

Observational Recording Forms

Observational recording forms are guides to be used in the requesting and documenting of information. The subjects may be events, actions, or circumstances, whether live or re-created through discussions or review of written documentation. Observational recording forms are needed when there is substantial information to be collected through observational means or when there are multiple data collectors. When a study employs multiple data collectors, creating a recording guide can help the researcher make sure that all areas have been covered and can eliminate the need for recontacting research participants. Also, when there are multiple data collectors, the use of a recording form provides necessary structure to the data collection process, thereby ensuring that all collectors are following similar procedures and employing similar criteria in choosing to include or exclude information. There are several programs available that increase the ease of data collection through the use of laptops or personal digital assistants (Eid & Diener, 2006; Felce & Emerson, 2000).

Tests

In applied studies, researchers are more likely to make use of existing instruments to measure knowledge or performance than to develop new ones. Whether choosing to use a test "off the shelf" or to capitalize on an existing database that includes such data, it is very important that the researcher be thoroughly familiar with the content of the instrument, its scoring, the literature on its creation and norming, and any ongoing controversies about its accuracy. There are several compendiums of tests available that describe their characteristics (e.g., Robinson, Shaver, & Wrightsman, 1999).

Data Extraction Forms/Formats

Frequent reliance on administrative records and documents is a major factor underlying the use of this type of data collection. Whether obtaining information from manual case records or computerized data tapes, the researcher needs to screen the data source for the key variables and record them into the research database.

A data extraction form may be a manual coding sheet for recording information from a paper file folder (e.g., medical chart) or the data collector may use a portable computer to enter information directly into a preformatted research database.

Even when the original source is computerized, the researcher will still likely need to create a data extraction format. The format should identify the relevant variables on the computerized file and include a program to extract the appropriate information into the research file. In circumstances where there are multiple sources of data (e.g., monthly welfare caseload data tapes), it may be necessary to apply these procedures to multiple data sources, using another program to merge the information into the appropriate format for analysis.

Structured Interview Guides

Whenever a research project requires that the same information items be obtained from multiple individuals, it is desirable for the researcher to create a structured interview guide. The need for structured data collection processes becomes even greater when multiple data collectors are being used (see Fowler & Cosenza, Chapter 12, on standardized survey interviewing). Computer-assisted personal interviewing (CAPI) has become increasingly popular for more structured personal interviewing. With CAPI, interviewers use portable computers rather than paper questionnaires to collect and enter the data. CAPI is particularly useful for large-scale surveys and especially those with complex question patterns.

A structured interview guide may begin with an explanation of the purpose of the interview and then proceed to a set of sequenced inquiries designed to collect information about attitudes, opinions, memories of events, characteristics, and circumstances. The questions may be about the respondents themselves or about activities occurring in their environment (e.g., individual dietary habits, housing history, program activities, world events). The guide itself is typically structured to interact with the individual's responses branching from one area to the next based on the individual's previous answer. There are also instances in which semistructured or even unstructured interviews (or parts of the interview) may be appropriate. These approaches are generally appropriate for the conduct of descriptive, exploratory research in a new area of inquiry or when the construct is difficult to measure in a close-ended, structured format. For example, in collecting data on homeless families' history of residential arrangements, a semistructured residential follow-back tool (New Hampshire-Dartmouth Psychiatric Research Center, 1995; Tsemberis, McHugo, Williams, Hanrahan, & Stefancic, 2006) is commonly used to walk a person through a calendar, keying on dates that will spark the person's memory of where the person may have been living at different points in time. Some people respond better to walking backward in remembering their residential arrangements, others are more comfortable beginning at a selected starting point and progressing to the present time. Flexibility in administration is important, therefore, to obtain complete data from a variety of individuals.

Mail and Telephone Surveys

Mail and telephone surveys are used when the researcher needs to obtain the same information from large numbers of respondents. There are many parallels between these methods and structured in-person interview data collection, with the key difference being the mode of data collection. In Chapter 16, Lavrakas describes telephone survey methods, including issues of sampling and selection of respondents and supervision of interviewers. Computer-assisted telephone interviewing (CATI), the oldest form of computer-assisted interviewing, allows interviewers to ask questions over the telephone and key the data directly into the computer system. As with CAPI, CATI has a strong advantage in situations where the interview has a complex structure (e.g., complicated skip patterns) and also provides the ability to reconcile data inconsistencies at the point of data collection (e.g., Fowler, 2002). In Chapter 15, Mangione and Van Ness provide more detail on the use of mail surveys.

Web-Based or Online Surveys

Web-based surveys are becoming more popular with the advent of inexpensive software and Web storage space. This approach is excellent when surveying a specific group such as employees of a company or college students. It is typical that these groups will have access to computers and feel comfortable in their use. There are several advantages to this approach. First, the data can be collected very rapidly, clearly more so through mail and phone surveys. Second, there are no data entry costs since the respondent enters his or her own data. Third, the data are almost immediately available to the researcher. With the development of sophisticated software, the survey can be programmed with skipping and branching where questions are given to the respondent based on their previous responses. This ability is also available in CATI and computerized surveys but not in written questionnaires. Finally, the researcher can track the completion rate and respond while the survey is still in the field to increase that rate.

Audio Computer-Assisted Self-Interview

Another approach to automating the data collection process is the use of audio computer-assisted self-interview (ACASI) software. With this approach, people with lower literacy are able to participate in such interviews, since the entire interview and instructions are heard instead of just read. The research participant listens to digitally recorded question items over a headset and, if desired, can simultaneously read the questions on the computer screen. The participant responds by pressing a number key or using a touch sensitive screen.

As Dillman (2006) notes, there are often situations in which we have the need to change data collection modes or mix modes (e.g., enhancing response rates of telephone surveys by contacting individuals by Internet or in person). He cautions that the accuracy of data collection from mixed mode efforts cannot be assumed, due to, for example, unintentional differences in the question stimulus presented to respondents and differences in social desirability. Attention to potential differences in the nature of responses due to data collection mode should be considered in the design stage and checked in analysis.

Resource Planning

Before making final decisions about the specific design to use and the type of data collection procedures to employ, the investigator must take into account the resources available and the limitations of those resources. Resource planning is an integral part of the iterative Stage II planning activities (see Figure 1.2). Resources important to consider are the following:

- *Data:* What are the sources of information needed and how will they be obtained?
- *Time:* How much time is required to conduct the entire research project, including final analyses and reporting?
- *Personnel:* How many researchers are needed and what are their skills?
- *Money:* How much money is needed to implement the research and in what categories?

Data as a Resource

The most important resource for any research project consists of the data needed to answer the research question. As noted, data can be obtained primarily in two ways: from original data collected by the investigator and from existing data. We discuss below the issues associated with primary data collection and the issues involved in the use of secondary data.

Primary Data Collection

There are five major issues that the researcher needs to consider in planning for primary data collection: site selection, authorization, the data collection process, accessibility, and other support needed.

Site Selection. Applied research and basic research differ on several dimensions, as discussed earlier, but probably the most salient difference is in the location of the research. The setting has a clear impact on the research, not only in defining the population studied, but also in the researcher's formulation of the research question, the research design, the measures, and the inferences that can be drawn from the study. The setting can also determine whether there are enough research participants available.

Deciding on the appropriate number and selection of sites is an integral part of the design/data collection decision, and often there is no single correct answer. Is it best to choose "typical" sites, a "range" of sites, "representative" sites, the "best" site, or the "worst" site? There are always more salient variables for site selection than resources for study execution, and no matter what criteria are used, some critics will claim that other more important site characteristics were omitted. For this reason, we recommend that the researcher make decisions regarding site selection in close coordination with the research client and/or advisory group. In general, it is also better to concentrate on as few sites as are required, rather than stretching the time and management efforts of the research team across too many locations.

There is another major implication connected with site selection. As noted earlier, multilevel designs have implications for the number and type of sites selected. In hierarchical designs, if the research intervention is at the site level (as in the earlier school example), then the investigator needs to have a sufficient number of sites in each experimental condition to maintain enough statistical power to detect a meaningful effect. For example, if a drug prevention program is instituted at the school level, then the number of schools, not classes or students, is what is important. One of the problems of using units lower in the hierarchy, such as classes, is that there may be concern about contamination from one condition to another. In the case where teachers are delivering the intervention and they teach in more than one classroom, then it should be obvious that classroom is not a suitable unit of analysis. Even if there is little or no chance of contamination, the observations still may be correlated and not independent of each other. This correlation, sometimes called the design effect, reduces the statistical power by reducing in effect the number of participants or units. Proper design and analysis requires multiple units, with the implication that enough units have to exist in the environment to do the study. In the case of schools, there may be a sufficient number in a given city. The same may not be true for hospital emergency rooms, public housing units, or mental health centers. Studies with these organizations will typically require the participation of multiple cities. More about designing and analyzing these site-based hierarchical designs can be found in Raudenbush and Bryk (2002) and Graham et al. (2008).

The distinction between "frontstage" and "backstage" made by Goffman (1959) also helps assess the openness of the setting to research. Frontstage activities are available to anyone, whereas backstage entrance is limited. Thus in a trial, the actions that take place in the courtroom constitute frontstage activity, open to anyone who can obtain a seat. Entrance to the judge's chambers is more limited, presence during lawyer-client conferences is even more restricted, and the observation of jury deliberations is not permitted. The researcher needs to assess the openness of the setting before taking the next step—seeking authorization for the research.

Authorization. Even totally open and visible settings usually require some degree of authorization for data collection. Public space may not be as totally available to the researcher as it may seem. For example, it is a good idea to notify authorities if a research team is going to be present in some public setting for an extended period of time. Although the team members' presence may not be illegal and no permission is required for them to conduct observations or interviews, residents of the area may become suspicious and call the police.

If the setting is a closed one, the researcher will be required to obtain the permission of the individuals who control or believe they control access. If there are several sites that are eligible for participation and they are within one organization, then it behooves the researcher to explore the independence of these sites from the parent organization. For example, in doing research in school systems, it might also be advisable to approach a principal to obtain preliminary approval that then can be presented to central administration for formal approval. Most school systems have written procedures that investigators must follow if they are going to gain access to the schools.

The planner needs to know not only at which level of the organization to negotiate but also which individuals to approach. Again, this will take some intelligence gathering. Personal contacts help a great deal, because authorities are usually more likely to meet and be cooperative with the researcher if he or she is recommended by someone they know and trust. Thus, the investigator should search for some connection to the organization. If the researcher is at a university, then it is possible that someone on the board of trustees is an officer of the organization. If so, contact with the university's development office is advisable. In sum, it is best for the researcher to obtain advance recommendations from credible sources and, hence, to avoid approaching an organization cold.

Permission from a central authority, however, does not necessarily imply cooperation from the sites needed for data collection. Nowhere is this more evident than in state/county working relationships. Often, central approval will be required just for the researcher to approach local sites. However, the investigator should not assume that central approval guarantees cooperation from those lower down on the organization's hierarchy; this belief can lead the investigator to behave in an insensitive manner. Those at the upper levels of an organization tend to believe that they have more power than they actually wield. A wise investigator will put a great deal of effort into obtaining cooperation at the local level, where he or she will find the individuals who feel they control that environment and with whom he or she will be interacting during the data collection phase. A good example is the school superintendent saying that he or she strongly supports the research but in reality, each principal will have to decide to participate.

Some closed organizations have procedures that must be followed before they can issue permission to conduct research in their settings (e.g., prisons and schools). Confidentiality and informed consent are usually significant issues for any organization. Will participants be identified or identifiable? How will the data be protected from unauthorized access? Will competitors learn something about the organization from this research that will put it at a disadvantage? Will individuals in the organization be put in any jeopardy by the project? The researcher needs to resolve such issues before approaching an organization for permission.

Organizations that have experience with research usually have standard procedures for working with researchers. For example, school systems typically have standard forms for researchers to complete and deadlines by which these forms must be submitted. These organizations understand the importance of research and are accustomed to dealing with investigators. In contrast, other organizations may not be familiar with applied research. Most for-profit corporations fall into this category, as do many small nonprofit organizations. In dealing with such groups, the investigator will first have to convince the authorities that research, in general, is a good idea and that their organization will gain something from their participation. In some cases, the researcher may also have to obtain the support of staff within the participating organizations, if they are needed to collect data or to obtain access to research participants. In conducting research on programs for homeless families, for example, researchers often have to convince program staff that the research will be worthwhile, will not place the families in the position of "guinea pigs," and will treat the families with respect and dignity. Most important, an organization's decision makers must be convinced that the organization will not be taking a significant risk or taking up valuable time in participating in the study. The planner must be prepared to present a strong case for why a nonresearch-oriented organization should want to involve itself in a research project.

Finally, any agreement between the researcher and the organization should be in writing. This may take the form of a letter addressed to the organization's project liaison officer (there may be one) for the research. The letter should describe the procedures that will take place and indicate the dates that the investigator will be on-site. The agreement should be detailed and should include how the organization will cooperate with the research.

The importance of site cooperation cannot be stressed too much. Lack of cooperation or dropping out of the study are some of the major factors that cause studies to fail. It is better to recruit more sites than you think you will need because invariably some will drop out before the study starts, and others will not have the client flow that they assured you that they had. This is discussed more in the next section.

Data Collection Process. The primary purpose of obtaining access to a site is to be able to collect data from or about people. The researcher should not assume that having access ensures that the target study participants will agree to participate in the study. Moreover, the researcher should be skeptical regarding assurances from management concerning others' availability and willingness to participate in a study. In a review of 30 randomized studies in drug abuse, Dennis (1990) found that 54% seriously underestimated the client flow by an average of 37%. Realistic and accurate participant estimates are necessary for the researcher to allocate resources and to ensure sufficient statistical power. Many funding agencies require power analyses as part of submitted grant proposals. These power analyses should be supported by evidence that the number of cases in these analyses are valid estimates. Dillman's (1978, 2000) total design method has been used successfully to improve recruitment rate (Records & Rice, 2006).

A planner can try to avoid shortfalls in the number of cases or subjects needed by conducting a small pilot study. In a pilot study, the researcher can verify client flow, enrollment and attendance data, program or service capacity, and willingness to participate. In cases where potential subjects enter into some program or institution, it will be important to verify the actual subject flow (e.g., number per week). This type of study is often called a pipeline study. In some circumstances, the flow into the program is affected by seasonal issues, contextual factors, organizational changes, and other factors. In addition, program capacity also can change and affect the size of the potential study participant pool. For example, in an evaluation of a newly developed service program for homeless families, initial sample size estimates were derived by program estimates that each of 6 case managers would be working with an average of 15 families at a time for an average of 9 months. Therefore, over an 18-month period, it was expected that there would be approximately 180 families in the participant pool. However, this estimate did not account for delays in hiring the full set of case managers as well as other times when one or more positions were unfilled, delays in enrolling families, and difficulties in both having full caseloads and moving families out of service in the 9-month time period due to the problems that families faced. Therefore, with the slippage of each part of the equation, the number of potential families for the study (before even considering eligibility criteria and refusal rates) was considerably smaller than initial expectations.

Care must be especially used in defining exactly who is eligible to participate in the study. For example, a pipeline study found that there were more than enough potential participants. However, the participant sample was limited to one child per family. It was not known until the study was underway that 30% of the potential participants had a sibling receiving treatment from the same organization.

Related to the number of participants is the assurance that the research design can be successfully implemented. Randomized designs are especially vulnerable to implementation problems. It is easy to promise that there will be no new taxes, that the check is in the mail, and that a randomized experiment will be conducted—but it is often difficult to deliver on these promises. In an applied setting, the investigator should obtain agreement from authorities in writing that they will cooperate in the conduct of the study. This agreement must be detailed and procedurally oriented and should clearly specify the responsibilities of the researcher and those who control the setting. While a written document may be helpful, it is not a legal contract that can be enforced. The organization leadership can change and with it the permission to conduct the study.

The ability to implement the research depends on the ability of the investigator to carry out the planned data collection procedures. A written plan for data collection is critical to success, but it does not assure effective implementation. A pilot study or walk-through of the procedure is necessary to determine if it is feasible. In this procedure, the investigator needs to consider both accessibility and other support. Written plans agreed to before the start of the study are helpful but not the final word. The researcher needs to monitor the implementation of the research. Studies can be sabotaged by resentful employees. For example, children eligible for services were recruited from a mental health center by the staff person who determined the severity of each case on a 10-point scale. The staff person was instructed that the mild cases, rated 4 or less, or the emergency cases, rated 10, were not eligible for the study. That left us cases rated in the range of 5 to 9, which would supply the needed number of participants. In the first month, much fewer children entered the study than expected. It was discovered that the person answering the phone was rating much fewer cases in the range than needed because she didn't think the study should be done. Once the director of the center talked to her, the situation was resolved.

Accessibility. There are a large number of seemingly unimportant details that can damage a research project, if they are ignored. Will the research participants have the means to travel to the site? Is there sufficient public transportation? If not, will the investigator arrange for transportation? Will families need child care to participate? If the study is going to use an organization's space for data collection, will the investigator need a key? Is there anyone else who may use the space? Who controls scheduling and room assignments? Has this person been notified? For example, a researcher about to collect posttest data in a classroom should ensure that he or she will not be asked to vacate the space before data collection is completed.

Other Support. Are the lighting and sound sufficient for the study? If the study requires the use of electrical equipment, will there be sufficient electrical outlets? Will the equipment's cords reach the outlets or should the researcher bring extension cords? Do the participants need food or drink? Space is a precious commodity in many institutions; the researcher should never assume that the research project will have sufficient space.

Secondary Data Analysis

The use of existing data, compared with collecting primary data, has the advantage of lower costs and time savings, but it may also entail managing a large amount of flawed and/or inappropriate data. In some cases, these data exist in formats designed for research purposes; for example, there are a number of secondary data sources developed by university consortia or by federal agencies such as the Bureau of the Census. Other kinds of data exist as administrative records (e.g., mental health agency records) that were not designed to answer research questions.

In the planning process, the investigator must establish with some confidence that the records to be used contain the information required for the study. Sampling the records will not only provide the researcher with an indication of their content, it will give an idea of their quality. It is frequently the case that clinical or administrative records are not suitable for research purposes. The planner must also have some confidence in the quality of the records. Are the records complete? Why were the data originally collected? The database may serve some hidden political purpose that could induce systematic distortions. What procedures are used to deal with missing data? Are the same procedures used for all variables or only selected variables? Do the computerized records bear a close resemblance to the original records (if available)? Are some data items periodically updated or purged from the computer file? How were the data collected and entered, and by whom? What quality control and verification checks are used? To assess the quality of the database, the planner should try to interview the data collectors and others experienced with the data, observe the data entry process, and compare written records to the computerized version. Conducting an analysis of administrative records seems easy only if it is not done carefully.

The investigator should not assume that the level of effort needed to process extant data will be small or even moderate. Data sets may be exceedingly complex, with changes occurring in data fields and documentation over time. In many cases, there may be very poor documentation, making interpretation of the data difficult. Moreover, if the researcher is interested in matching cases across existing data sets (as in tracking service used across multiple county databases), he or she will need to ensure that identification fields are available in each data set to match individuals' records. Often, matching alone can take a considerable amount of time and resources.

Finally, once the researcher has judged the administrative records or other database to be of sufficient quality for the study, he or she must then go through the necessary procedures to obtain the data. In addition to determining the procedures for extracting and physically transferring the data, the investigator also must demonstrate how the confidentiality of the records will be protected. For example, school systems may want a formal contractual agreement between the university and the school system before they would release identifiable student achievement data. Knowledge of relevant laws and regulations are important. In this example, the researchers had legitimate right to the identifiable data under federal regulations, namely, the Family Educational Rights and Privacy Act (FERPA) and the Protection of Pupil Rights Amendment (PPRA). While it may seem to be a simple request, it took over a year to obtain the data.

Time as a Resource

Time takes on two important dimensions in the planning of applied research: calendar time and clock time. Calendar time is the total amount of time available for a project, and it varies across projects.

Time and the Research Question

The calendar time allotted for a study should be related to the research questions. Is the phenomenon under study something that lasts for a long period or does it exist only briefly? Does the phenomenon under study occur in cycles? Is the time allocated to data collection sufficient?

Time and Data Collection

The second way in which the researcher needs to consider time is in terms of the actual or real clock time needed to accomplish particular tasks. For example, the event that is being studied might exist infrequently and only for a short period of time; thus, a long period of calendar time might need to be devoted to the project, but only a short period of clock time for data collection. Having established the time estimates, the investigator needs to estimate how long actual data collection will take. In computing this estimate, the researcher should consider how long it will take to recruit study participants and to gain both cooperation and access. The researcher should also attempt to estimate attrition or dropout from the study. If high attrition is predicted, then more recruitment time may be needed for data collection for the study to have sufficient statistical power. Thus, in computing the time needed, the investigator should have an accurate and comprehensive picture of the environment in which the study will be conducted.

Time Budget

In planning to use any resource, the researcher should create a budget that describes how the resource will be allocated. Both calendar and clock time need to be budgeted.

To budget calendar time, the researcher must know the duration of the entire project. In applied research, the duration typically is set at the start of the project, and the investigator then tailors the research to fit the length of time available. There may be little flexibility in total calendar time on some projects. Funded research projects usually operate on a calendar basis; that is, projects are funded for specific periods of time. Investigators must plan what can be accomplished within the time available.

The second time budget a researcher must create concerns clock time. How much actual time will it take to develop a questionnaire or to interview all the participants? It is important for the investigator to decide what units of time (e.g., hours, days, months) will be used in the budget. That is, what is the smallest unit of analysis of the research process that will be useful in calculating how much time it will take to complete the research project? To answer this question, we now turn to the concepts of tasks.

Tasks and Time

To "task out" a research project, the planner must list all the significant activities (tasks) that must be performed to complete the project. The tasks in a project budget serve a purpose similar to that of the expense categories—rent, utilities, food, and so on—used in planning a personal financial budget. When listing all these expense items, one makes implicit decisions concerning the level of refinement that will be used. Major categories (such as utilities) are usually divided into finer subcategories. The degree of refinement in a research project task budget depends on how carefully the investigator needs to manage resources.

To construct a time budget, the investigator first needs to consider the time required to manage the overall process; keep various stakeholders informed as needed either through meetings, monthly reports, update telephone calls, and/or other mechanisms; maintain connections with other members of the team in team meetings, conference calls (especially if the team is in more than one location); and other activities that maintain the integrity of the project over the entire study time frame. Second, the researcher should list all the tasks that must be accomplished during the research project. Typically, these tasks can be grouped into a number of major categories. The first category usually encompasses conceptual development. This includes literature reviews and thinking and talking about the problem to be investigated. Time needs to be allocated also for consulting with experts in areas where investigators need additional advice. The literature reviews could be categorized into a number of steps, ranging from conducting computerized searches to writing a summary of the findings.

The second phase found in most projects is instrument development and refinement. Regardless of whether the investigator plans to do intensive face-to-face interviewing, self-administered questionnaires, or observations, he or she needs to allocate time to search for, adapt, or develop relevant instruments used to collect data. The researcher also needs to allocate time for pilot testing of the instruments. Pilot testing should never be left out of any project. Typically, a pilot test will reveal "new" flaws that were not noted by members of the research team in previous applications of the instrument. If multiple data collection sites are involved, it is often important to pilot the procedures in all the sites or at least a sample that represents the range of sites involved. If the data collection approach involves extracting information from administrative records, the researcher should pilot test the training planned for data extractors as well as the data coding process. Checks should be included for accuracy and consistency across coders.

When external validity or generalizability is a major concern, the researcher will need to take special care in planning the construction of the sample. The sampling procedure describes the potential subjects and how they will be selected to participate in the study. This procedure may be very complex, depending on the type of sampling plan adopted.

The next phase of research is usually the data collection. The investigator needs to determine how long it will take to gain access to the records as well as how long it will take to extract the data from the records. It is important that the researcher not only ascertains how long it will take to collect the data from the records but also discovers whether information assumed to be found in those records is there. If the researcher is planning to conduct a survey, the procedure for estimating the length of time needed for this process could be extensive. Fowler and Cosenza (Chapter 12) describe the steps involved in conducting a survey. These include developing the instrument, recruiting and training interviewers, sampling, and the actual collection of the data. Telephone interviews require some special techniques that are described in detail by Lavrakas (Chapter 16). Time must also be allotted to obtain institutional review board's approval of the project if it involves human subjects. If a project is involved in federal data collection, review may also be required by the Office of Management and Budget (OMB), which, depending on the size of the project, can involve a considerable effort to develop the OMB review package and up to 4 months for the review to occur.

The next phase usually associated with any research project is data analysis. Whether the investigator is using qualitative or quantitative methods, time must be allocated for the analysis of data. Analysis includes not only statistical testing using a computer but also the preparation of the data for computer analysis. Steps in this process include "cleaning" the data (i.e., making certain that the responses are readable and unambiguous for data entry personnel), physically entering the data, and checking for the internal consistency of the data (Smith, Breda, Simmons, Vides de Andrade, & Bickman, 2008). Once the data are clean, the first step in quantitative analysis is the production of descriptive statistics such as frequencies, means, standard deviations, and measures of skewness. More complex studies may require researchers to conduct inferential statistical tests. As part of the design, a clear and comprehensive analysis plan should be developed that includes the steps for cleaning the data as well as the sequence of analyses that will take place, including analyses that may be needed to test for possible artifacts (e.g., attrition).

Finally, time needs to be allocated for communicating the results. An applied research project almost always requires a final report, usually a lengthy, detailed analysis as well as one or more verbal briefings. Within the report itself, the researcher should take the time needed to communicate the data to the audience at the right level. In particular, visual displays can often communicate even the most complex findings in a more straightforward manner than prose.

Because most people will not read the entire report, it is critical that the researcher include a two- or three-page executive summary that succinctly and clearly summarizes the main findings. The executive summary should focus on the findings, presenting them as the highlights of the study. No matter how much effort and innovation went into data collection, these procedures are of interest primarily to other researchers, not to typical sponsors of applied research or other stakeholders. The best the researcher can hope to accomplish with these latter audiences is to educate them about the limitations of the findings based on the specific methods used.

The investigator should allocate time not only for producing a report but also for verbally communicating study findings to sponsors and perhaps to other key audiences. Moreover, if the investigator desires to have the results of the study used, it is likely that he or she needs to allocate time to work with the sponsor and other organizations in interpreting and applying the findings of the study. This last utilization-oriented perspective is often not included by researchers planning their time budgets.

Time Estimates

Once the researcher has described all the tasks and subtasks, the next part of the planning process is to estimate how long it will take to complete each task. One way to approach this problem is to reduce each task to its smallest unit. For example, in the data collection phase, an estimate of the total amount of interviewing time is needed. The simplest way to estimate this total is to calculate how long each interview should take. Pilot data are critical for helping the researcher to develop accurate estimates.

The clock-time budget indicates only how long it will take to complete each task. What this budget does not tell the researcher is the sequencing and the real calendar time needed for conducting the research. Calendar time can be calculated from clock-time estimates, but the investigator needs to make certain other assumptions as well. For example, calendar conflicts need to be considered in the budgeting. Schools, for example, have a restricted window of time for data collection, usually avoiding the month around school entry and any testing. As another example, some service programs have almost no time for researchers around the busy holiday times, making December a difficult time to schedule any onsite data collection.

Another set of assumptions is based on the time needed for data collection. For example, if the study uses interviewers to collect data and 200 hours of interviewing time are required, the length of calendar time needed will depend on several factors. Most clearly, the number of interviewers will be a critical factor. One interviewer will take a minimum of 200 hours to complete this task, whereas 200 interviewers could theoretically do it in 1 hour. However, the larger number of interviewers may create a need for other mechanisms to be put into place (e.g., interviewer supervision and monitoring) as well as create concerns regarding the quality of the data. Thus the researcher needs to specify the staffing levels and research team skills required for the project. This is the next kind of budget that needs to be developed.

Each research project has unique characteristics that make it difficult to generalize from one project to another. Estimating time and expenses is an inexact art. In most cases the research underestimates the time and cost of a project. Unexpected events that disrupt the research should be expected. Since research budgets typically do not permit funds to be reserved for unforeseen events the planner is advised to build in some aspect of the project that could be sacrificed without affecting the central features of the research. The time and funds allocated to that task can usually be used to provide the additional support needed to complete the research.

Personnel as a Resource

Skills Budget

Once the investigator has described the tasks that need to be accomplished, the next step is to decide what kinds of people are needed to carry out those tasks. What characteristics are needed for a trained observer or an interviewer? What are the requirements for a supervisor? To answer these questions, the investigator should complete a skills matrix that describes the requisite skills needed for the tasks and attaches names or positions of the research team to each cluster of skills. Typically, a single individual does not possess all the requisite skills, so a team will need to be developed for the research project. As noted earlier, in addition to specific research tasks, the investigator needs to consider management of the project. This function should be allocated to every research project. Someone will have to manage the various parts of the project to make sure that they are working together and that the schedule is being met.

Person Loading

Once the tasks are specified and the amount of time required to complete each task is estimated, the investigator must assign these tasks to individuals. The assignment plan is described by a person-loading table that shows how much time each person is supposed to work on each task.

At some point in the planning process, the researcher needs to return to real, or calendar, time, because the project will be conducted under real-time constraints. Thus the tasking chart, or Gantt chart, needs to be superimposed on a calendar. This chart simply shows the tasks on the left-hand side and the months of the study period at the top. Bars show the length of calendar time allocated for the completion of specific subtasks. The Gantt chart shows not only how long each task takes, but also the approximate relationship in calendar time between tasks. Although inexact, this chart can show the precedence of research tasks and the extent to which some tasks will overlap and require greater staff time. One of the key relationships and assumptions made in producing a plan is that no individual will work more than 40 hours a week. Thus the person-loading chart needs to be checked against the Gantt chart to make sure that tasks can be completed by those individuals assigned to them within the periods specified in the Gantt chart. Very reasonably priced computer programs are available to help the planner do these calculations and draw the appropriate charts.

Financial Resources

Usually, the biggest part of any research project's financial budget is consumed by personnel—research staff. Social science research, especially applied social science, is very labor-intensive. Moreover, the labor of some individuals can be very costly. To produce a budget based on predicted costs, the investigator needs to follow a few simple steps.

Based on the person-loading chart, the investigator can compute total personnel costs for the project by multiplying the hours allocated to various individuals by their hourly costs.

The investigator should compute personnel costs for each task. In addition, if the project will take place over a period of years, the planner will need to provide for salary increases in the estimates. Hourly cost typically includes salary and fringe benefits and may also include facilities and administration (F&A) or overhead costs. (In some instances, personnel costs need to be calculated by some other time dimensions, such as daily or yearly rates; similarly, project costs may need to be categorized by month or some time frame other than year.)

After the budget has been calculated, the investigator may be faced with a total cost that is not reasonable for the project, either because the sponsor does not have those funds available or because the bidding for the project is very competitive. If this occurs, the investigator has several alternatives. Possible alternatives are to eliminate some tasks, reduce the scope of others, and/or shift the time from more expensive to less expensive staff for certain tasks where it is reasonable. The investigator needs to use ingenuity to try to devise not only a valid, reliable, and sensitive project, but one that is efficient as well. For example, in some cases this may mean recommending streamlining data collection or streamlining the reporting requirements.

The financial budget, as well as the time budget, should force the investigator to realize the trade-offs that are involved in applied research. Should the investigator use a longer instrument, at a higher cost, or collect fewer data from more subjects? Should the subscales on an instrument be longer, and thus more reliable, or should more domains be covered, with each domain composed of fewer items and thus less reliable? Should emphasis be placed on representative sampling as opposed to a purposive sampling procedure? Should the researcher use multiple data collection techniques, such as observation and interviewing, or should the research plan include only one technique, with more data collected by that procedure? These and other such questions are ones that all research planners face. However, when a researcher is under strict time and cost limitations, the salience of these alternatives is very high.

Making Trade-Offs and Testing Feasibility

Before making a firm go/no-go decision, it is worthwhile for the researcher to take the time to assess the strengths and weaknesses of the proposed approach and decide whether it is logistically feasible. This section returns to a discussion of the iterative process that researchers typically use as they assess and refine the initial design approach. Two major activities take place: (a) identifying and deciding on design trade-offs and (b) testing the feasibility of the proposed design. These activities almost always occur simultaneously. The results may require the researcher to reconsider the potential design approach or even to return to the client to renegotiate the study questions.

Making Design Trade-Offs

Examples of areas where design trade-offs often occur include external generalizability of study results, conclusiveness of findings, precision of estimates, and comprehensiveness of measurement. Trade-offs are often forced by external limitations in dollar and staff resources, staff skills, time, and the quality of available data.

Generalizability

Generalizability refers to the extent to which research findings can be credibly applied to a wider setting than the research setting. For example, if one wants to describe the methods used in vocational computer training programs, one might decide to study a local high school, an entire community (including both high schools and vocational education agencies and institutions), or schools across the nation. These choices vary widely with respect to the resources required and the effort that must be devoted to constructing sampling frames. The trade-offs here are ones of both resources and time. Local information can be obtained much more inexpensively and quickly than can information about a larger area; however, one will not know whether the results obtained are representative of the methods used in other high schools or used nationally.

Generalizability can also involve time dimensions, as well as geographic and population dimensions. Moreover, generalizability decisions need to have a clear understanding of the generalizability boundaries at the initiation of the study.

Conclusiveness of Findings

One of the key questions the researcher must address is how conclusive the study must be. Research can be categorized as to whether it is exploratory or confirmatory in nature. An exploratory study might seek only to identify the dimensions of a problem—for example, the types of drug abuse commonly found in a high school population. More is demanded from a confirmatory study. In this case, the researcher and client have a hypothesis to test—for example, among high school students use of marijuana is twice as likely as abuse of cocaine or heroin. In this example, it would be necessary to measure with confidence the rates of drug abuse for a variety of drugs and to test the observed differences in rate of use.

Precision of Estimates

In choosing design approaches, it is essential that the researcher have an idea of how small a difference or effect it is important to be able to detect for an outcome evaluation or how precise a sample to draw for a survey. This decision drives the choice of sample sizes and sensitivity of instrumentation, and thus affects the resources that must be allocated to the study.

Sampling error in survey research poses a similar issue. The more precise the estimate required, the greater the amount of resources needed to conduct a survey. If a political candidate feels that he or she will win by a landslide, then fewer resources are required to conduct a political poll than if the race is going to be close and the candidate requires more precision or certainty concerning the outcome as predicted by a survey.

Comprehensiveness of Measurement

The last area of choice involves the comprehensiveness of measurement used in the study. It is usually desirable to use multiple methods or multiple measures in a study (especially in qualitative studies, as noted earlier) for this allows the researcher to look for consistency in results, thereby increasing confidence in findings. However, multiple measures and methods can sometimes be very expensive and potentially prohibitive. Thus researchers frequently make trade-offs between resources and comprehensiveness in designing measurement and data collection approaches.

Choosing the most appropriate strategy involves making trade-offs between the level of detail that can be obtained and the resources available. Calendar time to execute the study also may be relevant. Within the measurement area, the researcher often will have to make a decision about breadth of measurement versus depth of measurement. Here the choice is whether to cover a larger number of constructs, each with a brief instrument, or to study fewer constructs with longer and usually more sensitive instrumentation. Some trade-off between comprehensiveness (breadth) and depth is almost always made in research. Thus, within fixed resources, a decision to increase external validity by broadening the sample frame may require a reduction in resources in other aspects of the design. The researcher needs to consider which aspects of the research process require the most resources, often in consultation with the research sponsor or other possible users of the study findings.

Feasibility Testing of the Research Design/Plan

Once researchers have tentatively selected a research design, they must determine whether the design is feasible. Areas to be tested for feasibility include the assessment of any secondary data, pilot tests of data collection procedures and instruments, and pilot tests of the design itself (e.g., construction of sampling frames, data collection procedures, and other study procedures). Additionally, efforts may be needed to explore the likelihood of potential confounding factors—that is, whether external events are likely to distort study results or whether the study procedures themselves may create unintended effects. The process of feasibility testing may take as little as a few hours or may involve a trial run of all study procedures in a real-world setting and could last several weeks or months.

The premise of feasibility testing is that, although sometimes time-consuming, it can greatly improve the likelihood of success or, alternatively, can prevent resources from being wasted on research that has no chance of answering the posed questions. A no-go decision does not represent a failure on the part of the researcher but rather an opportunity to improve on the design or research procedures, and it ultimately results in better research and hopefully better research utilization. A go decision reinforces the confidence of the researcher and others in the utility of expending resources to conduct the study.

Once the researcher has appropriately balanced any design trade-offs and determined the feasibility of the research plan, he or she should hold final discussions with the research client to confirm the proposed approach. If the client's agreement is obtained, the research planning phase is complete. If agreement is not forthcoming, the process may start again, with a change in research scope (questions) or methods.

Conclusion

The key to conducting a sound applied research study is planning. In this chapter, we have described several steps that can be taken in the planning stage to bolster a study and increase its potential for successful implementation. We hope that these steps will help you to conduct applied research that is credible, feasible, and useful.

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CHAPTER 2

Design Sensitivity

Statistical Power for Applied Experimental Research

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Sean M. Hurley

Applied experimental research investigates the effects of deliberate intervention in situations of practical importance. A psychotherapist, for instance, might study the efficacy of systematic desensitization for reducing the symptoms of snake phobia, a school might evaluate the success of a drug education program, or a policymaker might ask for evidence that increasing the tax rate on gasoline will discourage consumption. The basic elements of experimental research are well-known: selection of participants and assignment of them to treatment and control conditions, preferably using a random procedure; application of the intervention of interest to the treatment group but not to the control group; monitoring the research situation to ensure that there are no differences between the treatment and control conditions other than the intervention; measurement of selected outcomes for both groups; and statistical analysis to determine if the groups differ on those dependent variable measures. To ensure that the conclusions about intervention effects drawn from experimental design are correct, the design must have both sensitivity and validity. *Sensitivity* refers to the likelihood that an effect, if present, will be detected. *Validity* refers to the likelihood that what is detected is, in fact, the effect of interest. This chapter is about the problem of sensitivity.

Sensitivity in intervention research is thus the ability to detect a difference between the treatment and control conditions on some outcome of interest. If the research design has high internal validity, that difference will represent the effect of the intervention under investigation. What, then, determines our ability to detect it? Answering this question requires that we specify what is meant by *detecting a difference* in experimental research. Following current convention, we will take this to mean that statistical criteria are used to reject the null hypothesis of no difference between the mean on the outcome measure for the persons in the treatment condition and the mean for those in the control condition. In particular, we conclude that there is an effect if an appropriate statistical test indicates a statistically significant difference between the treatment and control means.

Our goal in this chapter is to help researchers "tune" experimental design to maximize sensitivity. However, before we can offer a close examination of the practical issues related to design sensitivity, we need to present a refined framework for describing and assessing the desired result—a high probability of detecting a given magnitude of effect if it exists. This brings us to the topic of *statistical power,* the concept that will provide the idiom for this discussion of design sensitivity.

The Statistical Power Framework

In the final analysis, applied experimental research comes down to just that: analysis (data analysis, that is). After all the planning, implementation, and data collection, the researcher is left with a set of numbers on which the crucial tests of statistical significance are conducted. There are four possible scenarios for this testing. There either *is* or *is not* a real treatment versus control difference that would be apparent if we had complete data for the entire population from which our sample was drawn (but we don't). And, for each of these situations, the statistical test on the sample data either *is* or *is not* significant. The various combinations can be depicted in a 2×2 table along with the associated probabilities, as shown in Table 2.1.

Finding statistical significance when, in fact, there is no effect is known as Type I error; the Greek letter α is used to represent the probability of that happening. Failure to find statistical significance when, in fact, there is an effect is known as Type II error; the Greek letter β is used to represent that probability. Most important, statistical power is the probability $(1 - \beta)$ that statistical significance will be attained

Table 2.1 The Possibilities of Error in Statistical Significance Testing of Treatment (T) Versus Control (C) Group Differences

Conclusion From Statistical Test on Sample Data	*Population Circumstances:* *T and C Differ*	*Population Circumstances:* *T and C Do Not Differ*
Significant difference (reject null hypothesis)	Correct conclusion Probability = $1 - \beta$ (power)	Type I error Probability = α
No significant difference (fail to reject null hypothesis)	Type II error Probability = β	Correct conclusion Probability = $1 - \alpha$

given that there really is an intervention effect. This is the probability that must be maximized for a research design to be sensitive to actual intervention effects.

Note that α and β in Table 2.1 are statements of *conditional* probabilities. They are of the following form: *If* the null hypothesis is true (false), *then* the probability of an erroneous statistical conclusion is α (β). When the null hypothesis is true, the probability of a statistical conclusion error is held to 5% by the convention of setting $\alpha = .05$. When the null hypothesis is false (i.e., there is a real effect), however, the probability of error is β, and β can be quite large. If we want to design experimental research in which statistical significance is found when the intervention has a real effect, then we must design for a low β error, that is, for high statistical power ($1 - \beta$).

An important question at this juncture concerns what criterion level of statistical power the researcher should strive for—that is, what level of risk for Type II error is acceptable? By convention, researchers generally set $\alpha = .05$ as the maximum acceptable probability of a Type I error. There is no analogous convention for beta. Cohen (1977, 1988) suggested $\beta = .20$ as a reasonable value for general use (more specifically, he suggested that power, equal to $1 - \beta$, be at least .80). This suggestion represents a judgment that Type I error is four times as serious as Type II error. This position may not be defensible for many areas of applied research where a null statistical result for a genuinely effective intervention may represent a great loss of valuable practical knowledge.

A more reasoned approach would be to analyze explicitly the cost-risk issues that apply to the particular research circumstances at hand (more on this later). At the first level of analysis, the researcher might compare the relative seriousness of Type I and Type II errors. If they are judged to be equally serious, the risk of each should be kept comparable; that is, alpha should equal beta. Alternatively, if one is judged to be more serious than the other, it should be held to a stricter standard even at the expense of relaxing the other. If a convention must be adopted, it may be wise to assume that, for intervention research of potential practical value, Type II error is at least as important as Type I error. In this case, we would set $\beta = .05$, as is usually done for α, and thus attempt to design research with power ($1 - \beta$) equal to .95.

Determinants of Statistical Power

There are four factors that determine statistical power: sample size, alpha level, statistical test, and effect size.

Sample Size. Statistical significance testing is concerned with sampling error, the expectable discrepancies between sample values and the corresponding population value for a given sample statistic such as a difference between means. Because sampling error is smaller for large samples, it is less likely to obscure real differences between means and statistical power is greater.

Alpha Level. The level set for alpha influences the likelihood of statistical significance—larger alpha makes significance easier to attain than does smaller alpha. When the null hypothesis is false, therefore, statistical power increases as alpha increases.

Statistical Test. Because investigation of statistical significance is made within the framework of a particular statistical test, the test itself is one of the factors determining statistical power.

Effect Size. If there is a real difference between the treatment and control conditions, the size of that difference will influence the likelihood of attaining statistical significance. The larger the effect, the more probable is statistical significance and the greater the statistical power. For a given dependent measure, effect size can be thought of simply as the difference between the means of the treatment versus control *populations.* In this form, however, its magnitude is partly a function of how the dependent measure is scaled. For most purposes, therefore, it is preferable to use an effect size formulation that standardizes differences between means by dividing by the standard deviation to adjust for arbitrary units of measurement. The effect size (*ES*) for a given difference between means, therefore, can be represented as follows:

$$ES = \frac{\mu_t - \mu_c}{\sigma}$$

where μ_t and μ_c are the respective means for the treatment and control populations and σ is their common standard deviation. This version of the effect size index was popularized by Cohen (1977, 1988) for purposes of statistical power analysis and is widely used in meta-analysis to represent the magnitude of intervention effects (Lipsey & Wilson, 2000). By convention, effect sizes are computed so that positive values indicate a "better" outcome for the treatment group than for the control group, and negative values indicate a "better" outcome for the control group.

For all but very esoteric applications, the most practical way actually to estimate the numerical values for statistical power is to use precomputed tables or a computer program. Particularly complete and usable reference works of statistical power tables have been published by Cohen (1977, 1988). Other general reference works along similar lines include those of Kraemer and Thiemann (1987), Lipsey (1990), and Murphy and Myors (2004). Among the computer programs available for conducting statistical power calculations are Power and Precision (from Biostat), nQuery Advisor (from Statistical Solutions), and SamplePower (from SPSS). In addition, there are open access power calculators on many statistical Web sites. The reader should turn to sources such as these for information on determining statistical power beyond the few illustrative cases presented in this chapter.

Figure 2.1 presents a statistical power chart for one of the more common situations. This chart assumes (a) that the statistical test used is a *t* test, one-way ANOVA, or other parametric test in this same family (more on this later) and (b) that the conventional $\alpha = .05$ level is used as the criterion for statistical significance. Given these circumstances, the chart shows the relationships among power $(1 - \beta)$, effect size (*ES*), and sample size (*n* for each group) plotted on sideways log-log paper, which makes it easier to read values for the upper power levels and the lower

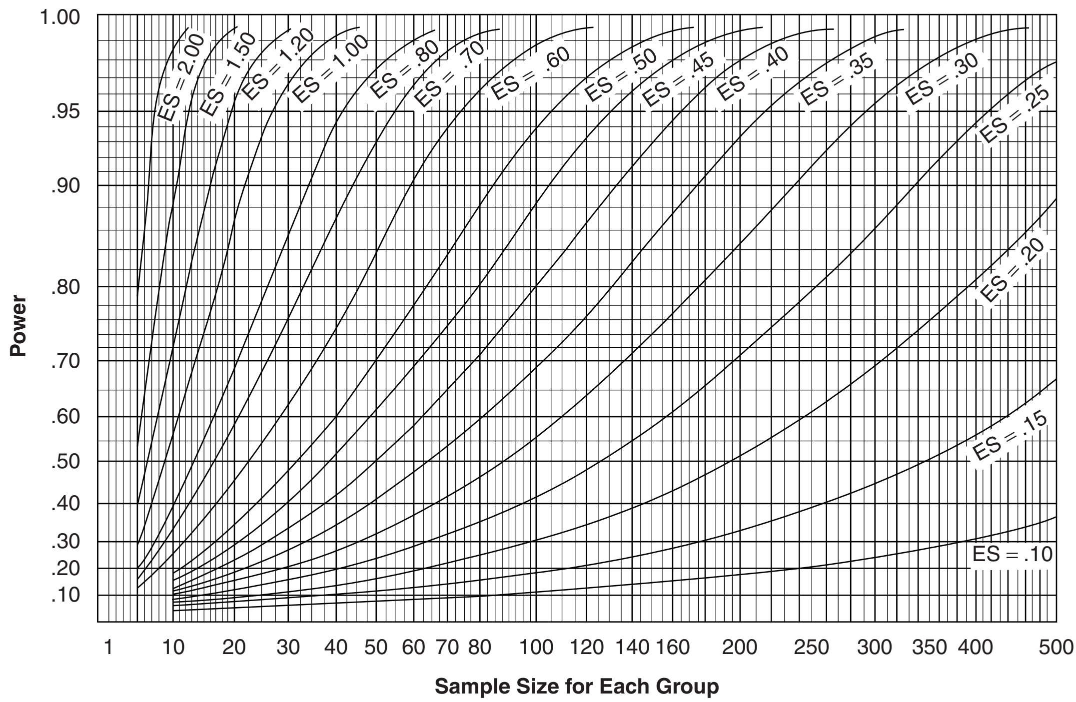

Figure 2.1 Power Chart for α = .05, Two-Tailed, or α = .025, One-Tailed

sample sizes. This chart shows, for instance, that if we have an experiment with 40 participants in each of the treatment and control groups (80 total), the power to detect an effect size of .80 (.8 standard deviations difference between the treatment and control group means) is about .94 (i.e., given a population $ES = .80$ and group $n = 40$, statistical significance would be expected 94% of the time at the $\alpha = .05$ level with a t test or one-way ANOVA).

Optimizing Statistical Power

To maximize the sensitivity of experimental research for detecting intervention effects using conventional criteria of statistical significance, the researcher must maximize statistical power. In the remainder of this chapter, we examine each of the determinants of statistical power and discuss how it can be manipulated to enhance power. The objective of this discussion is to provide the researcher with the conceptual tools to design experimental research with the greatest possible sensitivity to intervention effects given the resources available. Moreover, in those cases where an appropriately high level of statistical power cannot be attained, these same concepts can be used to analyze the limitations of the research design and guard against misinterpretation.

Sample Size

The relationship between sample size and statistical power is so close that many textbooks discuss power only in terms of determining the sample size necessary to attain a desired power level. A look at Figure 2.1 makes clear why sample size warrants so much attention. Virtually any desired level of power for detecting any given effect size can be attained by making the samples large enough.

The difficulty that the relationship between sample size and statistical power poses for intervention research is that the availability of participants is often limited. Although a researcher can increase power considerably by parading a larger number of participants through the study, there must be individuals ready to march before this becomes a practical strategy. In practical intervention situations, relatively few persons may be appropriate for the intervention or, if there are enough appropriate persons, there may be limits on the facilities for treating them. If facilities are adequate, there may be few who volunteer or whom program personnel are willing to assign; or, if assigned, few may sustain their participation until the study is complete. The challenge for the intervention researcher, therefore, is often one of keeping power at an adequate level with modest sample sizes. If modest sample sizes in fact generally provided adequate power, this particular challenge would not be very demanding. Unfortunately, they do not.

Suppose, for instance, that we decide that *ES* = .20 is the minimal effect size that we would want our intervention study to be able to detect reliably. An *ES* of .20 is equivalent to a 22% improvement in the success rate for the treatment group (more on this later). It is also the level representing the first quintile in the effect size distribution derived from meta-analyses of psychological, behavioral, and education intervention research (Lipsey & Wilson, 1993). Absent other considerations, therefore, *ES* = .20 is a reasonable minimal effect size to ask research to detect—it is not so large that it requires heroic assumptions to think it might actually be produced by an intervention and not so small that it would clearly lack practical significance.

If we calculate the sample size needed to yield a power level of .95 ($\beta = \alpha = .05$), we find that the treatment and control group must each have a minimum of about 650 participants for a total of about 1,300 in both groups (see Figure 2.1). The sample sizes in social intervention research are typically much smaller than that, often less than 100 in each group. If we want to attain a power level for *ES* = .20 that makes Type II error as small as the conventional limit on Type I error through sample size alone, then we must increase the number of participants quite substantially over the average in present practice. Even attaining the more modest .80 power level suggested as a minimum by Cohen (1988) would require a sample size of about 400 per treatment group, larger than many studies can obtain.

Increased sample size is thus an effective way to boost statistical power and should be employed whenever feasible, but its costs and limited availability of participants may restrict the researcher's ability to use this approach. It is important, therefore, that the researcher be aware of other routes to increasing statistical power. The remainder of this chapter discusses some of these alternate routes.

Alpha Level

Alpha is conventionally set at .05 for statistical significance testing and, on the surface, may seem to be the one straightforward and unproblematic element of statistical power for the intervention researcher. That impression is misleading. An α of .05 corresponds to a .95 probability of a correct statistical conclusion only when the null hypothesis is true. However, a relatively conservative alpha makes statistical significance harder to attain when the null hypothesis is false and, therefore, decreases the statistical power. Conversely, relaxing the alpha level required for statistical significance increases power. The problem is that this reduction in the probability of a Type II error comes at the expense of an increased probability of a Type I error. This means that the researcher cannot simply raise alpha until adequate power is attained but, rather, must find some appropriate balance between alpha and beta. Both Type I error (α) and Type II error (β) generally have important implications in the investigation of intervention effects. Type I error can mean that an ineffective or innocuous intervention is judged beneficial or, possibly, harmful, whereas Type II error can permit a truly effective intervention (or a truly harmful one) to go undiscovered. Though little has been written in recent years about how to think about this balancing act, useful perspectives can be found in Brown (1983), Cascio and Zedeck (1983), Nagel and Neef (1977), and Schneider and Darcy (1984). In summary form, the advice of these authors is to consider the following points in setting error risk levels.

Prior Probability. Because the null hypothesis is either true or false, only one type of inferential error is possible in a given study—Type I for a true null hypothesis and Type II for a false null hypothesis. The problem, of course, is that we do not know if the null hypothesis is true or false and, thus, do not know which type of error is relevant to our situation. However, when there is evidence that makes one alternative more likely, the associated error should be given more importance. If, for example, prior research tends to show an intervention effect, the researcher should be especially concerned about protection against Type II error and should set beta accordingly.

Directionality of Significance Testing. A significance test of a one-tailed hypothesis (e.g., that the treatment group mean is superior to the control group) conducted at a given α level has higher power (smaller beta) than a two-tailed test at the same alpha (e.g., that the treatment group is either superior *or* inferior to control). In applied intervention research, concern often centers on one direction of effects, for instance, whether a new intervention is better than an existing one. In these situations, it may be reasonable to argue that one-tailed tests are justified and that using two-tailed tests amounts to inappropriate restriction of the alpha level. Such an argument implies that a negative intervention effect, should it occur, is of no interest, however—a rather strong claim for many kinds of intervention.

Relative Costs and Benefits. Perhaps the most important aspect of error risk in intervention research has to do with the consequences of an error. Rarely will the costs of each type of error be the same, nor will the benefits of each type of correct inference. Sometimes, intervention effects and their absence can be interpreted directly in terms of dollars saved or spent, lives saved or lost, and the like. In such cases, the optimal relationship between alpha and beta error risk should be worked out according to their relative costs and benefits. When the consequences of Type I and Type II errors cannot be specified in such definite terms, the researcher may still be able to rely on some judgment about the relative seriousness of the risks. Such judgment might be obtained by asking those familiar with the intervention circumstances to rate the error risk and the degree of certainty that they feel is minimal for the conclusions of the research. This questioning, for instance, may reveal that knowledgeable persons believe, on average, that a 95% probability of detecting a meaningful effect is minimal and that Type II error is three times as serious as Type I error. This indicates that β should be set at .05 and α at .15. Nagel and Neef (1977) provided a useful decision theory approach to this judgment process that has the advantage of requiring relatively simple judgments from those whose views are relevant to the research context.

If some rational analysis of the consequences of error is not feasible, it may be necessary to resort to a convention (such as $\alpha = .05$) as a default alternative. For practical intervention research, the situation is generally one in which *both* types of errors are serious. Under these circumstances, the most straightforward approach is to set alpha risk and beta risk equal unless there is a clear reason to do otherwise. If we hold to the usual convention that α should be .05, then we should design research so that β will also be .05. If such high standards are not practical, then both alpha and beta could be relaxed to some less stringent level—for example, .10 or even .20.

To provide some framework for consideration of the design issues related to the criterion levels of alpha and beta set by the researcher, Table 2.2 shows the required sample size per group for the basic two-group experimental design at various effect sizes under various equal levels of alpha (two-tailed) and beta. It is noteworthy that maintaining relatively low levels of alpha and beta risk (e.g., .05 or below) requires either rather large effect sizes or rather large sample sizes. Moreover, relaxing alpha levels does not generally yield dramatic increases in statistical power for the most difficult to detect effect sizes. Manipulation of other aspects of the power function, such as those described later, will usually be more productive for the researcher seeking to detect potentially modest effects with modest samples sizes.

Statistical Test

Consider the prototypical experimental design in which one treatment group is compared with one control group. The basic statistical tests for analyzing this design are the familiar *t* test and one-way analysis of variance (ANOVA). These tests use an "error term" based on the within-group variability in the sample data to assess the likelihood that the mean difference between the groups could result from sampling error. To the extent that within-group variability can be eliminated, minimized, or somehow offset, intervention research will be more powerful—that is, more sensitive to true effects if they are present.

Two aspects of the statistical test are paramount in this regard. First, for a given set of treatment versus control group data, different tests may have different formulations

Table 2.2 Approximate Sample Size for Each Group Needed to Attain Various Equal Levels of Alpha and Beta for a Range of Effect Sizes

	Level of Alpha and Beta ($\alpha = \beta$)			
Effect Size	*.20*	*.10*	*.05*	*.01*
.10	900	1,715	2,600	4,810
.20	225	430	650	1,200
.30	100	190	290	535
.40	60	110	165	300
.50	35	70	105	195
.60	25	50	75	135
.70	20	35	55	100
.80	15	30	45	75
.90	10	25	35	60
1.00	10	20	30	50

of the sampling error estimate and the critical test values needed for significance. For instance, nonparametric tests—those that use only rank order or categorical information from dependent variable scores—generally have less inherent power than do parametric tests, which use scores representing degrees of the variable along some continuum.

The second and most important aspect of a statistical test that is relevant to power is the way it partitions sampling error and which components of that error variance are used in the significance test. It is often the case in intervention research that some of the variability on a given dependent measure is associated with participant characteristics that are not likely to change as a result of intervention. If certain factors extraneous to the intervention effect of interest contribute to the population variability on the dependent measure, the variability associated with those factors can be removed from the estimate of sampling error against which differences between treatment and control means are tested with corresponding increases in power.

A simple example might best illustrate the issue. Suppose that men and women, on average, differ in the amount of weight they can lift. Suppose further that we want to assess the effects of an exercise regimen that is expected to increase muscular strength. Forming treatment and control groups by simple random sampling of the undifferentiated population would mean that part of the within-group variability that is presumed to reflect the luck of the draw (sampling error) would be the natural differences between men and women. This source of variability may well be judged irrelevant to an assessment of the intervention effect—the intervention may rightfully be judged effective if it increases the strength of women relative to the natural variability in women's strength and that of men relative to the natural variability in men's strength. The corresponding sampling procedure is not simple random sampling but stratified random sampling, drawing women and men separately so that the experimental sample contains identified subgroups of women and men. The estimate of sampling error in this case comes from the within-group variance—within experimental condition within gender—and omits the between-gender variance, which has now been identified as having a source other than the luck of the draw.

All statistical significance tests assess effects relative to an estimate of sampling error but they may make different assumptions about the nature of the sampling and, hence, the magnitude of the sampling error. The challenge to the intervention researcher is to identify the measurable extraneous factors that contribute to population variability and then use (or assume) a sampling strategy and corresponding statistical test that assesses intervention effects against an appropriate estimate of sampling error. Where there are important extraneous factors that correlate with the dependent variable (and there almost always are), using a statistical significance test that partitions them out of the error term can greatly increase statistical power. With this in mind, we review below some of the more useful of the variance control statistical designs with regard to their influence on power.

Analysis of Covariance

One of the most useful of the variance control designs for intervention research is the one-way analysis of covariance (ANCOVA). Functionally, the ANCOVA is like the simple one-way ANOVA, except that the dependent variable variance that is correlated with a covariate variable (or linear combination of covariate variables) is removed from the error term used for significance testing. For example, a researcher with a reading achievement test as a dependent variable may wish to remove the component of performance associated with IQ before comparing the treatment and control groups. IQ differences may well be viewed as nuisance variance that is correlated with reading scores but is not especially relevant to the impact of the program on those scores. That is, irrespective of a student's IQ score, we would still expect an effective reading program to boost the reading score.

It is convenient to think of the influence of variance control statistical designs on statistical power as a matter of adjusting the effect size in the power relationship. Recall that *ES*, as it is used in statistical power determination, is defined as $(\mu_t - \mu_c)/\sigma$ where σ is the pooled within-groups standard deviation. For assessing the power of variance control designs, we adjust this *ES* to create a new value that is the one that is operative for statistical power determination. For the ANCOVA statistical design, the operative *ES* for power determination is as follows:

$$ES_{ac} = \frac{\mu_t - \mu_c}{\sigma\sqrt{1 - r_{dc}^2}},$$

where ES_{ac} is the effect size formulation for the one-way ANCOVA; μ_t and μ_c are the means for the treatment and control populations, respectively; σ is the common standard deviation; and r_{dc} is the correlation between the dependent variable and the covariate. As this formula shows, the operative effect size for power determination using ANCOVA is inflated by a factor of $1/\sqrt{1-r^2}$, which multiples *ES* by 1.15 when $r = .50$, and 2.29 when $r = .90$. Thus, when the correlation of the covariate(s) with the dependent variable is substantial, the effect of ANCOVA on statistical power can be equivalent to more than doubling the operative effect size. Examination on Figure 2.1 reveals that such an increase in the operative effect size can greatly enhance power at any given sample size.

An especially useful application of ANCOVA in intervention research is when both pretest and posttest values on the dependent measure are available. In many cases of experimental research, preexisting individual differences on the characteristic that intervention is intended to change will not constitute an appropriate standard for judging intervention effects. Of more relevance will be the size of the intervention effect relative to the dispersion of scores for respondents that began at the same initial or baseline level on that characteristic. In such situations, a pretest measure is an obvious candidate for use as a covariate in ANCOVA. Because pretest-posttest correlations are generally high, often approaching the test-retest reliability of the measure, the pretest as a covariate can dramatically increase the operative effect size in statistical power. Indeed, ANCOVA with the pretest as the covariate is so powerful and so readily attainable in most instances of intervention research that it should be taken as the standard to be used routinely unless there are good reasons to the contrary.

ANOVA With a Blocking Factor

In the blocked ANOVA design, participants are first categorized into blocks, that is, groups of participants who are similar to each other on some characteristic related to the dependent variable. For example, to use gender as a blocking variable, one would first divide participants into males and females, then assign some males to the treatment group and the rest to the control group and, separately, assign some females to treatment and the rest to control.

In the blocked design, the overall variance on the dependent measure can be viewed as the sum of two components: the within-blocks variance and the between-blocks variance. Enhanced statistical power is gained in this design because it removes the contribution of the between-blocks variance from the error term against which effects are tested. As in the ANCOVA case, this influence on power can be represented in terms of an adjusted effect size. If we let PV_b equal the proportion of the total dependent variable variance associated with the difference between blocks, the operative *ES* for this case is as follows:

$$ES_{ab} = \frac{\mu_t - \mu_c}{\sigma\sqrt{1 - PV_b}},$$

where ES_{ab} is the effect size formulation for the blocked one-way ANOVA, σ is the pooled within-groups standard deviation (as in the unadjusted *ES*), and PV_b is

σ_b^2/σ^2 with σ_b^2 the between-blocks variance and σ^2 the common variance of the treatment and control populations.

The researcher, therefore, can estimate PV_b, the between-blocks variance, as a proportion of the common (or pooled) variance within experimental groups and use it to adjust the effect size estimate in such a way as to yield the operative effect size associated with the statistical power of this design. If, for instance, the blocking factor accounts for as much as half of the common variance, the operative *ES* increases by more than 40%, with a correspondingly large increase in power.

Power Advantages of Variance Control Designs

The variance control statistical designs described above all have the effect of reducing the denominator of the effect size index and, hence, increasing the operative effect size that determines statistical power. Depending on the amount of variance controlled in these designs, the multiplier effect on the effect size can be quite considerable. Table 2.3 summarizes that multiplier effect for different proportions of the within-groups variance associated with the control variable. Although the effects are modest when the control variable accounts for a small proportion of the dependent variable variance, they are quite considerable for higher proportions. For instance, when the control variable accounts for as much as 75% of the variance, the operative effect size is double what it would be without the control variable. Reference back to Figure 2.1, the statistical power chart, will reveal that a doubling of the effect size has a major effect on statistical power. Careful use of variance control designs, therefore, is one of the most important tactics that the intervention researcher can use to increase statistical power without requiring additional participants in the samples.

Effect Size

The effect size parameter in statistical power can be thought of as a signal-to-noise ratio. The signal is the difference between treatment and control population means on the dependent measure (the *ES* numerator, $\mu_t - \mu_c$). The noise is the within-groups variability on that dependent measure (the *ES* denominator, σ). Effect size and, hence, statistical power is large when the signal-to-noise ratio is high—that is, when the *ES* numerator is large relative to the *ES* denominator. In the preceding section, we saw that variance control statistical designs increase statistical power by removing some portion of nuisance variance from the *ES* denominator and making the operative *ES* for statistical power purposes proportionately larger. Here, we will look at some other approaches to increasing the signal-to-noise ratio represented by the effect size.

Dependent Measures

The dependent measures in intervention research yield the set of numerical values on which statistical significance testing is performed. Each such measure chosen

Table 2.3 Multiplier by Which *ES* Increases When a Covariate or Blocking Variable Is Used to Reduce Within-Groups Variance

Proportion of Variance Associated With Control Variable[a]	*Multiplier for ES Increase*
.05	1.03
.10	1.05
.15	1.08
.20	1.12
.25	1.15
.30	1.20
.35	1.24
.40	1.29
.45	1.35
.50	1.41
.55	1.49
.60	1.58
.65	1.69
.70	1.83
.75	2.00
.80	2.24
.85	2.58
.90	3.16
.95	4.47
.99	10.00

a. r^2 for ANCOVA, PV_b for blocked ANOVA.

for a study constitutes a sort of listening station for certain effects expected to result from the intervention. If the listening station is in the wrong place or is unresponsive to effects when they are actually present, nothing will be heard. To optimize the signal-to-noise ratio represented in the effect size, the ideal measure for intervention effects is one that is maximally responsive to any change that the intervention brings about (making a large *ES* numerator) and minimally responsive to anything else (making a small *ES* denominator). In particular, three aspects of outcome measurement have direct consequences for the magnitude of the effect size parameter and, therefore, statistical power: (a) validity for measuring change, (b) reliability, and (c) discrimination of individual differences among respondents.

Validity for Change. For a measure to respond to the signal, that is, to intervention effects, it must, of course, be a valid measure of the characteristic that the intervention is expected to change. But validity alone is not sufficient to make a measure responsive to intervention effects. What is required is validity for *change.* A measure can be a valid indicator of a characteristic but still not be a valid indicator of change on that characteristic. Validity for change means that the measure shows an observable *difference* when there is, in fact, a change on the characteristic measured that is of sufficient magnitude to be interesting in the context of application.

There are various ways in which a measure can lack validity for change. For one, it may be scaled in units that are too gross to detect the change. A measure of mortality (death rate), for instance, is a valid indicator of health status but is insensitive to variations in how sick people are. Graduated measures, those that range over some continuum, are generally more sensitive to change than categorical measures, because the latter record changes only between categories, not within them. The number of readmissions to a mental hospital, for example, constitutes a continuum that can differentiate one readmission from many. This continuum is often represented categorically as "readmitted" versus "not readmitted," however, with a consequent loss of sensitivity to change and statistical power.

Another way in which a measure may lack validity for measuring change is by having a floor or ceiling that limits downward or upward response. A high school-level mathematics achievement test might be quite unresponsive to improvements in Albert Einstein's understanding of mathematics—he would most likely score at the top of the scale with or without such improvements. Also, a measure may be specifically designed to cancel out certain types of change, as when scores on IQ tests are scaled by age norms to adjust away age differences in ability to answer the items correctly.

In short, measures that are valid for change will respond when intervention alters the characteristic of interest and, therefore, will differentiate a treatment group from a control group. The stronger this differentiation, the greater the contrast between the group means will be and, correspondingly, the larger the effect size.

Reliability. Turning now to the noise in the signal detection analogy, we must consider variance in the dependent measure scores that may obscure any signal due to intervention effects. Random error variance—that is, unreliability in the measure—is obviously such a noise. Unreliability represents fluctuations in the measure that are unrelated to the characteristic being measured, including intervention effects on that characteristic. Measures with lower measurement error will yield less variation in the distribution of scores for participants within experimental groups. Because within-groups variance is the basis for the denominator of the *ES* ratio, less measurement error makes that denominator smaller and the overall *ES* larger.

Some measurement error is intrinsic—it follows from the properties of the measure. Self-administered questionnaires, for instance, are influenced by fluctuations in respondents' attention, motivation, comprehension, and so forth. Some measurement error is procedural—it results from inconsistent or inappropriate application of the measure. Raters who must report on an observed characteristic, for instance, may not be trained to use the same standards for their judgment, or the conditions of observation may vary for different study participants in ways that influence their ratings.

Also included in measurement error is systematic but irrelevant variation—response of the measure to characteristics other than the one of interest. When these other characteristics vary differently than the one being measured, they introduce noise into a measure. For example, frequency of arrest, which may be used to assess the effects of intervention for juvenile delinquency, indexes police behavior (e.g., patrol and arrest practices) as well as the criminal behavior of the juveniles. If the irrelevant characteristic to which the measure is also responding can be identified and separately measured, its influence can be removed by including it as a covariate in an ANCOVA, as discussed above. For instance, if we knew the police precinct in which each arrest was made, we could include that information as control variables (dummy coding each precinct as involved vs. not involved in a given arrest) that would eliminate variation in police behavior across precincts from the effect size for a delinquency intervention.

Discrimination of Individual Differences. Another source of systematic but often irrelevant variation that is especially important in intervention effectiveness research has to do with relatively stable individual differences on the characteristic measured. When a measure is able to discriminate strongly among respondents, the variance of its distribution of scores is increased. This variation does not represent error, as respondents may truly differ, but it nonetheless contributes to the noise variance that can obscure intervention effects. In a reading improvement program, for example, the primary interest is whether each participant shows improvement in reading level, irrespective of his or her initial reading level, reading aptitude, and so forth. If the measure selected is responsive to such other differences, the variability may be so great as to overshadow any gains from the program.

Where psychological and educational effects of intervention are at issue, an important distinction is between "psychometric" measures, designed primarily to discriminate individual differences, and "edumetric" measures, designed primarily to detect change (Carver, 1974). Psychometric measures are those developed using techniques that spread out the scores of respondents; IQ tests, aptitude tests, personality tests, and other such standardized tests would generally be psychometric measures. By comparison, edumetric measures are those developed through the sampling of some defined content domain that represents the new responses participants are expected to acquire as a result of intervention. Mastery tests, such as those an elementary school teacher would give students to determine whether they have learned to do long division, are examples of edumetric tests.

Because they are keyed specifically to the sets of responses expected to result from intervention, edumetric tests, or measures constructed along similar lines, are more sensitive than psychometric tests to the changes induced by intervention and less sensitive to preexisting individual differences. To the extent that any measure reflects less heterogeneity among participants, within-group variability on that measure is smaller. That, in turn, results in a smaller denominator for the *ES* ratio and a corresponding increase in statistical power.

The Independent Variable

The independent variable in intervention research is defined by the contrast between the experimental conditions (e.g., treatment and control) to which participants are exposed. When more contrast is designed into the study, the effect size can be correspondingly larger if the intervention is effective.

Dose Response. Experimental design is based on the premise that intervention levels can be made to vary and that different levels might result in different responses. Generally speaking, the "stronger" the intervention, the larger the response should be. One way to attain a large effect size, therefore, is to design intervention research with the strongest possible dose of the intervention represented in the treatment condition. In testing a new math curriculum, for instance, the researcher might want the teachers to be very well-trained to deliver it and to spend a significant amount of class time doing so. If the intervention is effective, the larger effect size resulting from a stronger dose will increase statistical power for detecting the effect.

Optimizing the strength of the intervention operationalized in research requires some basis for judging what might constitute the optimal configuration for producing the expected effects. There may be insufficient research directly on the intervention under study (else why do the research), but there may be other sources of information that can be used to configure the intervention so that it is sufficiently strong to potentially show detectable effects. One source, for example, is the experience and intuition of practitioners in the domain where the intervention, or variants, is applied.

Variable Delivery of the Intervention. The integrity or fidelity of an intervention is the degree to which it is delivered as planned and, in particular, the degree to which it is delivered in a uniform manner in the right amounts to the right participants at the right time. At one end of the continuum, we might consider the case of intervention research conducted under tightly controlled clinical or laboratory conditions in which delivery can be regulated very closely. Under these conditions, we would expect a high degree of intervention integrity, that is, delivery of a constant, appropriate dose to each participant.

Intervention research, however, cannot always be conducted under such carefully regulated circumstances. It must often be done in the field with volunteer participants whose compliance with the intervention regimen is difficult to ensure. Moreover, the interventions of interest are often not those for which dosage is easily determined and monitored, nor are they necessarily delivered uniformly. The result is that the participants in a treatment group may receive widely different amounts and even kinds of intervention (e.g., different mixes of components). If participants' responses to intervention vary with its amount and kind, then it follows that variation in the intervention will generate additional variation in the outcome measures.

When treatment and control groups are compared in a statistical analysis, all that usually registers as an intervention effect is the difference between the treatment group's mean score and the control group's mean score on the dependent variable. If there is variation around those means, it goes into the within-groups variance of the effect size denominator, making the overall *ES* smaller. Maintaining a uniform application of treatment and control conditions is the best way to prevent this problem. One useful safeguard is for the researcher to actually measure the amount of intervention received by each participant in the treatment and control conditions (presumably little or none in the control). This technique yields information about how much variability there actually was and generates a covariate that may permit statistical adjustment of any unwanted variability.

Control Group Contrast. Not all aspects of the relationship between the independent variable and the effect size have to do primarily with the intervention. The choice of a control condition also plays an important role. The contrast between the treatment and control means can be heightened or diminished by the choice of a control that is more or less different from the treatment condition in its expected effects on the dependent measure.

Generally, the sharpest contrast can be expected when what the control group receives involves no aspects of the intervention or any other attention—that is, a "no treatment" control. For some situations, however, this type of control may be unrepresentative of participants' experiences in nonexperimental conditions or may be unethical. This occurs particularly for interventions that address problems that do not normally go unattended—severe illness, for example. In such situations, other forms of control groups are often used. The "treatment as usual" control group, for instance, receives the usual services in comparison to a treatment group that receives innovative services. Or a placebo control might be used in which the control group receives attention similar to that received by the treatment group but without the specific active ingredient that is presumed to be the basis of the intervention's efficacy. Finally, the intervention of interest may simply be compared with some alternative intervention, for example, traditional psychotherapy compared with behavior modification as treatment for anxiety.

The types of control conditions described above are listed in approximate order according to the magnitude of the contrast they would generally be expected to show when compared with an effective intervention. The researcher's choice of a control group, therefore, will influence the size of the potential contrast and hence of the potential effect size that appears in a study. Selection of the control group likely to show the greatest contrast from among those appropriate to the research issues can thus have an important bearing on the statistical power of the design.

Statistical Power for Multilevel Designs

For the experimental designs discussed in the previous sections, we have assumed that the units on which the dependent variables were measured are the same units that were randomly assigned to treatment and control conditions. In social science intervention studies, those units are typically individual people. Research designs for some intervention situations, however, involve assignment of clusters of units to experimental conditions or delivery of treatment at the cluster level, but measurement of the outcomes on the individual units within those clusters. Such designs are especially common in education research where classrooms or entire schools may be assigned to treatment and control conditions with student grades or achievement test scores as the dependent variable. Similarly, patients whose outcomes are of interest might be clustered within hospitals assigned to treatment and control conditions, energy use might be examined for apartments clustered within housing projects assigned to receive a weatherization program or not, and so forth. Even when individuals are randomly assigned to conditions, if the treatment and control conditions are implemented on clusters, for example, classrooms, there are still multiple levels in the design. These types of designs may also have other levels or groupings in between the units of measurement and the units of randomization. For example, students (whose achievement scores are the outcomes of interest) might be clustered within classrooms that are clustered within schools that are clustered within school districts that are assigned to intervention and control conditions. For simplicity, the discussion here will be limited to two-level models, but the general principles can be extended to designs with more than two levels.

These cluster or multilevel designs have distinct characteristics that affect statistical power. One way to think about them is in terms of the sample size for the experiment—a critical factor for power discussed earlier. Is the pertinent sample size the number of clusters assigned to the experimental conditions or is it the number of units within all those clusters on which the outcomes are measured? The answer, and the main source of complexity for power analysis, is that it could be either or something in between. The operative sample size is the number of *statistically independent* units represented in the study. Participants within a cluster (e.g., students within a classroom) are likely to have dependent measure scores that are more similar to each other than to participants in different clusters either because of the natural sorting processes that have put them in that cluster or because of similar influences that they share as members of it. If so, their scores are not statistically independent—there is some degree of predictability from one to another within a classroom. When there is statistical dependence among the scores within clusters, the operative sample size is no longer the number of units measured but, instead, shrinks toward the number of clusters assigned, which is always a smaller number (Snijders & Bosker, 1999).

Statistical analysis for multilevel designs and, correspondingly, statistical power considerations must, therefore, take into account the within- and between-cluster variance structure of the data. If there is relative homogeneity within clusters and heterogeneity between clusters, the results will be quite different than if it is the other way around. Specialized statistical programs are available for analyzing multilevel data, for example, HLM (Raudenbush, Bryk, & Congdon, 2004), MLwiN (Rasbash, Steele, Browne, & Prosser, 2004), and, more generally, mixed models analysis routines in the major computer programs such as SPSS, SAS, and Stata. In the sections that follow, we identify the distinctive issues associated with statistical power in multilevel designs and describe ways in which it can be optimized and estimated.

Determinants of Statistical Power for Multilevel Designs

Basically, the same four factors that influence power in single-level designs apply to multilevel designs—sample size, alpha level, the statistical test (especially whether variance controls are included), and effect size. The alpha level at which the intervention effect is tested and the effect size are defined virtually the same way in multilevel designs as in single-level ones and function the same way in power analysis. It should be particularly noted that despite the greater complexity of the structure of the variance within treatment and control groups in multilevel designs, the effect size parameter remains the same. It is still defined as the difference between the mean score on the dependent variable for all the individuals in the treatment group and the mean for all the individuals in the control group divided by the common standard deviation of all the scores within the treatment and control groups. In a multilevel design, the variance represented in that standard deviation could, in turn, be decomposed into between- and within-cluster components or built up from them. It is, nonetheless, the same treatment or control population variance (estimated from sample values) irrespective of whether the participants providing scores have been sampled individually or clusterwise.

The statistical analysis on the other hand will be different—it will involve a multilevel statistical model that represents participant scores at the lowest level and the clusters that were randomized at the highest level. One important implication of this multilevel structure is that variance control techniques, such as use of selected covariates, can be applied at both the participant and cluster levels of the analysis. Similarly, sample size applies at both levels and involves the number of clusters assigned to experimental conditions and the number of participants within clusters who provide scores on the dependent measures.

One additional factor distinctive to multilevel designs also plays an important role in statistical power: the intracluster correlation (ICC; Hox, 2002; Raudenbush & Bryk, 2002; Snijders & Bosker, 1999). The ICC is a measure of the proportion of the total variance of the dependent variable scores that occurs between clusters. It can be represented as follows:

$$\rho = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}},$$

where the numerator is the variance between the clusters and the denominator is the total variance in the model (between-cluster plus within-cluster variance).

If none of the variability in the data is accounted for by between-cluster differences, then the ICC will be 0 and the effective sample size for the study will simply be the total number of participants in the study. If, on the other hand, all the variability is accounted for by between-cluster differences, then the ICC will be 1 and the effective *N* for the study will be the number of clusters. In practice, the ICC will be somewhere between these two extremes, and the effective *N* of the study will be somewhere in between the number of participants and the number of clusters.

Figure 2.2 contains a graph that depicts the effect of the magnitude of the ICC on the power to detect an effect size of .40 at α = .05 with 50 clusters total (evenly divided between treatment and control) and 15 participants per cluster. As the figure shows, even small increases in the ICC can substantially reduce the power.

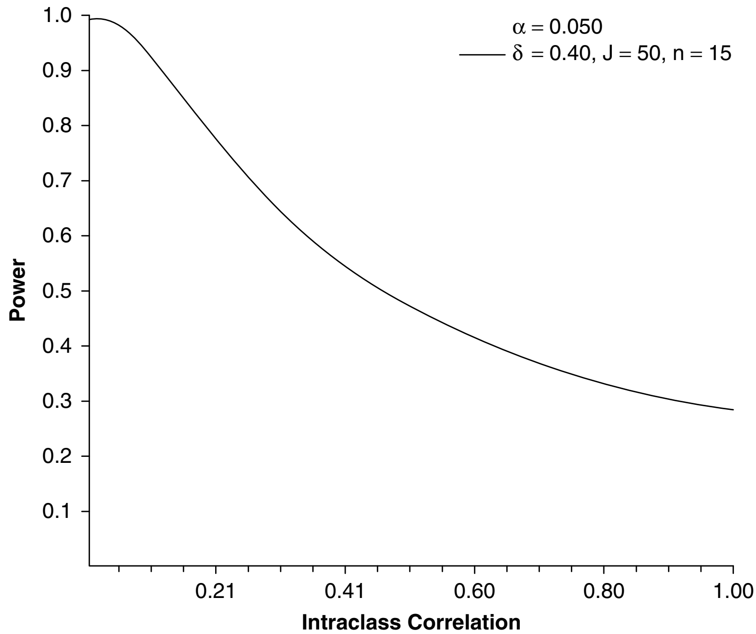

Figure 2.2 The Relationship Between ICC and Power to Detect an Effect Size of .40, With 50 Clusters Total, 15 Participants per Cluster, and α = .05 (graph generated using optimal design software)

Clearly, the ICC is crucial for determining statistical power when planning a study. Unfortunately, the researcher has no control over what the ICC will be for a particular study. Thus, when estimating the statistical power of a planned study, the researcher should consider the ICC values that have been reported for similar research designs. For example, the ICCs for the educational achievement outcomes of students clustered within classroom or schools typically range from approximately .15 to .25 (Hedges & Hedberg, 2006).

Unlike the ICC, the number of clusters and the number of participants within each cluster are usually within the researcher's control, at least to the extent that resources allow. Unfortunately, in multilevel analyses the total number of participants (which are usually more plentiful) has less of an effect on power than the number of clusters (which are often available only in limited numbers). This is in contrast to single-level designs in which the sample size at the participant level plays a large role in determining power. See Figure 2.3 for a graph depicting the relationship between sample size at the participant level and power to detect an effect size of .40 at α = .05 for a study with 50 clusters total and an ICC of .20. Once clusters have around 15 participants each, adding additional participants yields only modest gains in power.

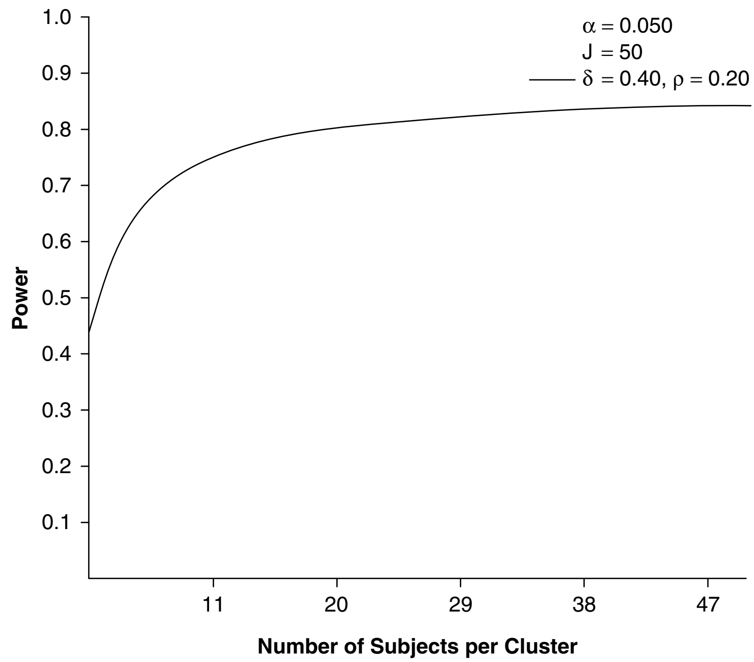

Figure 2.3 The Relationship Between Cluster Size and Power to Detect an Effect Size of .40, With 50 Clusters Total, an ICC of .20, and α = .05 (graph generated using optimal design software)

Figure 2.4 depicts the relationship between the number of clusters and the power to detect an effect size of .40 at α = .05 for a study with 15 participants per cluster and an ICC of .20. As that graph shows, a power of .80 to detect this effect size is only achieved when the total number of clusters is above 50, and it requires 82 clusters for .95 power. In many research contexts, collecting data from so many clusters may be impractical and other techniques for attaining adequate power must be employed.

Optimizing Power in a Multilevel Design

The techniques for maximizing statistical power in single-level analyses also apply, with appropriate adaptations, to multilevel analyses. Power can be increased by relaxing the alpha level or increasing the sample size (in this case, mainly the number of clusters). Also, adding covariates to the analysis is an effective way to increase power. In multilevel analysis, covariates measured at either the participant level or the cluster level (or both) can be used. Cluster-level covariates are often easier to obtain because each individual participant need not be measured and may be as helpful for increasing power as participant-level covariates (Bloom, 2005; Murray & Blitstein, 2003). As in single-level analysis, one of the best covariates, when available, is the pretest score on the same measure as the outcome variable or

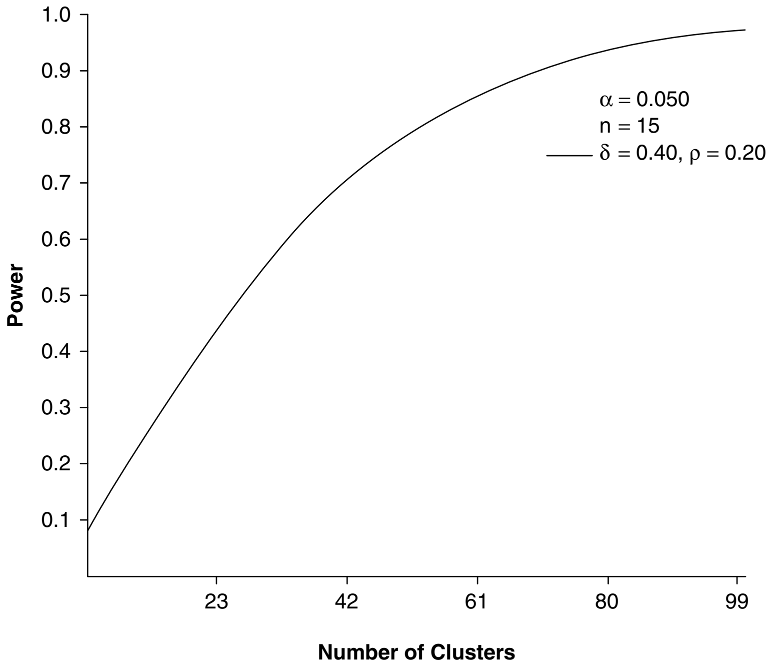

Figure 2.4 The Relationship Between Number of Clusters and Power to Detect an Effect Size of .40, With 15 Participants per Cluster, an ICC of .20, and α = .05 (graph generated using optimal design software)

a closely related one. Including a pretest covariate can reduce the number of clusters required to achieve adequate power anywhere from one half to one tenth and cluster-level pretest scores (the mean for each cluster) may be just as useful as participant-level pretest scores (Bloom, Richburg-Hayes, & Black, 2005).

Figure 2.5 illustrates the change in power associated with adding a cluster-level covariate that accounts for varying proportions of the between-cluster variance on the outcome variable. Without a covariate, 52 clusters (26 each in the treatment and control groups) with 15 participants per cluster and an ICC of .20 are required to detect an effect size of .40 at α = .05 with .80 power. With the addition of a cluster-level covariate that accounts for 66% of the between-cluster variance (i.e., correlates about .81), the same power is attained with half as many clusters (26 total). Accounting for that proportion of between-cluster variance would require a strong covariate (or set of covariates), but not so strong as to be unrealistic for many research situations.

Planning a Multilevel Study With Adequate Power

Estimating the power of a multilevel study requires taking into account the minimum meaningful effect size that the researcher would like to detect, the alpha level for the statistical test, the number of clusters, the number of participants within

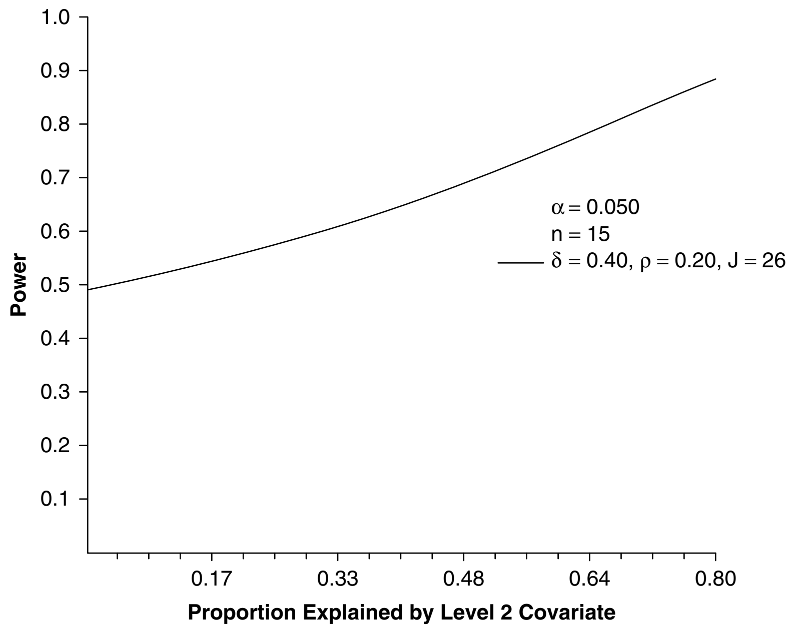

Figure 2.5 Power for Detecting an Effect Size of .40, With 26 Clusters, 15 Participants per Cluster, ICC of .20, and α = .05, as Influenced by the Addition of a Cluster-Level Covariate of Various Strengths (graph generated using optimal design software)

each cluster, the ICC associated with those clusters, and any covariates or blocking factors involved in the design. Given all these considerations, it is not surprising that computing power estimates is rather complicated (see Raudenbush, 1997; Snijders & Bosker, 1993, for examples of computational techniques). Fortunately, there is software available that facilitates these computations. One of the best documented and easiest to use is Optimal Design, based on the calculations outlined in Raudenbush and Liu (2000) (available without cost at the time this chapter was written at http://sitemaker.umich.edu/group-based/optimal_design_software). Optimal Design was used to generate the graphs in Figures 2.2, 2.3, 2.4, and 2.5. Power Analysis in Two-Level designs (PINT), developed by Snijders and his colleagues and using the formulas derived in Snijders and Bosker (1993), is another package that provides similar power calculations, but is currently more limited in the research designs that it can accommodate (PINT is available at the time this chapter was written at http://stat.gamma.rug.nl/snijders).

Design Strategy to Enhance Power

Perhaps the most important point to be gleaned from the above discussion about statistical power is that nearly all the factors that influence it can be manipulated to increase power. A research design that is sensitive to intervention effects, therefore, is achieved through the integration of decisions about all these factors in a way that is appropriate and practical for the particular research circumstances. This requires awareness of statistical power issues during the planning phase of a study, incorporation of procedures to enhance power in the design, and an analysis and interpretation of study results that reflects statistical power considerations.

The general strategy for optimizing power in intervention research necessarily begins with a decision about the minimum effect size that the research should be designed to detect reliably (Bloom, 1995). This minimum detectable effect should be set as a threshold value such that below that level, intervention effects are considered too small to be important, but above that level, they are potentially meaningful and thus should be detected by the research. It is at this point that the researcher must consider the various issues related to the effect sizes, such as what treatment versus control contrast will be represented in that effect size. This requires decisions about the "dosage" for the intervention, the nature of the control group (no treatment, placebo, service as usual, and so on), and the character of the dependent variable(s) (e.g., psychometric vs. edumetric).

Given decisions on these points, the researcher must then decide what numerical value of the effect size under the planned research circumstances represents a meaningful minimum to be detected. This usually involves a complex judgment regarding the practical meaning of effects within the particular intervention context. The next section provides some suggestions for framing this issue. For now, suppose that a threshold value has been set: Say that $ES = .20$ is judged the smallest effect size that the research should reliably detect. The next question is how reliably the researcher wishes to be able to detect that value—that is, what level of statistical power is desired. If the desired power is .80, for instance, statistically significant results would be found 80% of the time an effect of .20 was actually present in the populations sampled for the research, and null results would occur 20% of the time despite the population effect. If greater reliability is desired, a higher level of power must be set. Setting the desired power level, of course, is equivalent to setting the beta level for risk of Type II error. Alpha level for Type I error should also be set at this time, using some rational approach to weighing the risks of Type I versus Type II error, as discussed earlier.

With a threshold effect size value and a desired power level in hand, the researcher is ready to address the question of how to actually attain that power level in the research design. At this juncture it is wise to consider what variance control statistics might be used. These can generally be applied at low cost and with only a little extra effort to collect data on appropriate covariate variables or implement blocking. Using the formulas and discussion provided above in the subsection on the statistical test, the researcher can estimate the operative effect size with a variance control design and determine how much larger it will be than the original threshold value. With an ANCOVA design using the pretest as a covariate, for instance, the pretest-posttest correlation might be expected to be at least .80, increasing the operative effect size from the original .20 to a value of .33 (see Table 2.3). Analogous assessments of covariates can be made for multilevel designs by using appropriate statistical power software.

With an operative effect size and a desired power level now established, the researcher is ready to turn to the question of the size of the sample in each experimental group. This is simply a matter of looking up the appropriate value using a statistical power chart or computer program. If the result is a sample size the researcher can achieve, then all is well.

If the required sample size is larger than can be attained, however, it is back to the drawing board for the researcher. The options at this point are limited. First, of course, the researcher may revisit previous decisions and further tune the design—for example, enhancing the treatment versus control contrast, improving the sensitivity of the dependent measure, or applying a stronger variance control design. If this is not possible or not sufficient, all that remains is the possibility of relaxing one or more of the parameters of the study. Alpha or beta levels, or both, might be relaxed, for instance. Because this increases the risk of a false statistical conclusion, and because alpha levels particularly are governed by strong conventions, this must obviously be done with caution. Alternatively, the threshold effect size that the research can reliably detect may be increased. This amounts to reducing the likelihood that effects already assumed to be potentially meaningful will be detected.

Despite best efforts, the researcher may have to proceed with an underpowered design. Such a design may be useful for detecting relatively large effects but may have little chance of detecting smaller, but still meaningful, effects. Under these circumstances, the researcher should take responsibility for communicating the limitations of the research along with its results. To do otherwise encourages misinterpretation of statistically null results as findings of "no effect" when there may be a reasonable probability of an actual effect that the research was simply incapable of detecting.

As is apparent in the above discussion, designing research sensitive to intervention effects depends heavily on an advance specification of the magnitude of statistical effect that represents the threshold for what is important or meaningful in the intervention context. In the next section, we discuss some of the ways in which researchers can approach this judgment.

What Effect Size Is Worth Detecting?

Various frameworks can be constructed to support reasonable judgment about the minimal effect size that an intervention study should be designed to detect. That judgment, in turn, will permit the researcher to consider statistical power in a systematic manner during the design phase of the research. Also, given a framework for judgment about effect size, the researcher can more readily interpret the statistical results of intervention research after it is completed. Below, we review three frameworks for judging effect size: the actuarial approach, the statistical translation approach, and the criterion group contrast approach.

The Actuarial Approach

If enough research exists similar to that of interest, the researcher can use the results of those other studies to create an actuarial base for effect sizes. The distribution of such effect size estimates can then be used as a basis for judging the likelihood that the research being planned will produce effects of a specified size. For example, a study could reliably detect 80% of the likely effects if it is designed to have sufficient power for the effect size at the 20th percentile of the distribution of effect sizes found in similar studies.

Other than the problem of finding sufficient research literature to draw on, the major difficulty with the actuarial approach is the need to extract effect size estimates from studies that typically do not report their results in those terms. This, however, is exactly the problem faced in meta-analysis when a researcher attempts to obtain effect size estimates for each of a defined set of studies and do higher-order analysis on them. Books and articles on meta-analysis techniques contain detailed information about how to estimate effect sizes from the statistics provided in study reports (see, e.g., Lipsey & Wilson, 2000).

A researcher can obtain a very general picture of the range and magnitude of effect size estimates in intervention research by examining any meta-analyses that have been conducted on similar interventions. Lipsey and Wilson (1993) reported the distribution of effect sizes from more than 300 meta-analyses of research on psychological, behavioral, and educational research. That distribution had a median effect size of .44, with the 20th percentile at .24 and the 80th percentile at .68. These values might be compared with the rule of thumb for effect size suggested by Cohen (1977, 1988), who reported that across a wide range of social science research, *ES* = .20 could be judged as a "small" effect, .50 as "medium," and .80 as "large."

The Statistical Translation Approach

Expressing effect sizes in standard deviation units has the advantage of staying close to the terms used in statistical significance testing and, thus, facilitating statistical power analysis. However, that formulation has the disadvantage that in many intervention domains there is little basis for intuition about the practical meaning of a standard deviation's worth of difference between experimental groups. One approach to this situation is to translate the effect size index from standard deviation units to some alternate form that is easier to assess.

Perhaps the easiest translation is simply to express the effect size in the units of the dependent measure of interest. The *ES* index, recall, is the difference between the means of the treatment and control groups divided by the pooled standard deviation. Previous research, norms for standardized tests, or pilot research is often capable of providing a reasonable value for the relevant standard deviation. With that value in hand, the researcher can convert to the metric of the specific variable any level of *ES* he or she is considering. For example, if the dependent variable is a standardized reading achievement test for which the norms indicate a standard deviation of 15 points, the researcher can think of *ES* = .50 as 7.5 points on that test. In context, it may be easier to judge the practical magnitude of 7.5 points on a familiar test than .50 standard deviations.

Sometimes, what we want to know about the magnitude of an effect is best expressed in terms of the proportion of people who attained a given level of benefit as a result of intervention. One attractive way to depict effect size, therefore, is in terms of the proportion of the treatment group, in comparison to the control group, elevated over some "success" threshold by the intervention. This requires, of course, that the researcher be able to set some reasonable criterion for success on the dependent variable, but even a relatively arbitrary threshold can be used to illustrate the magnitude of the difference between treatment and control groups.

One general approach to expressing effect size in success rate terms is to set the mean of the control group distribution as the success threshold value. With symmetrical normal distributions, 50% of the control group will be below that point and 50% will be above. These proportions can be compared with those of the treatment group distribution below and above the same point for any given difference between the two distributions in standard deviation units. Figure 2.6 depicts the relationship for an effect size of $ES = .50$. In this case, 70% of the treatment group is above the mean of the control group, or, in failure rate terms, only 30% of the treated group is below the control group mean. There are various ways to construct indices of the overlap between distributions to represent effect size. This particular one corresponds to Cohen's (1977, p. 31) *U3* measure.

A variation on the percentage overlap index has been offered by Rosenthal and Rubin (1982), who used it to construct something that they call a "binominal effect size display" (BESD). They suggest that the success threshold be presumed to be at the grand median for the conjoint control and treatment distribution (line M in Figure 2.6). Though use of the grand median as a success threshold is somewhat arbitrary, it confers a particular advantage on the BESD. With normal distributions, the difference between the "success" proportions of the treatment and control groups has a simple relationship to the effect size expressed in correlational terms. In particular, when we express effect size as a correlation (r), the value of that correlation corresponds to the difference between the proportions of the respective distributions that are above the grand median success threshold. Effect size in standard deviation units can easily be converted into the equivalent correlation using the following formula:

$$r = \frac{ES}{\sqrt{ES^2 + 4}}.$$

For example, if the correlation between the independent variable and the dependent variable is .24, then the difference between the success proportions of the groups is .24, evenly divided around the .50 point, that is, $.50 \pm .12$, or 38% success in the control group, 62% in the treatment group. More generally, the distribution with the lower mean will have $.50 - (r/2)$ of its cases above the grand median success threshold, and the distribution with the greater mean will have $.50 + (r/2)$ of its cases above that threshold. For convenience, Table 2.4 presents the BESD terms for a range of *ES* and *r* values as well as Cohen's *U3* index described above.

The most striking thing about the BESD and the *U3* representations of the effect size is the different impression that they give of the potential practical significance of a given effect from that of the standard deviation expression. For

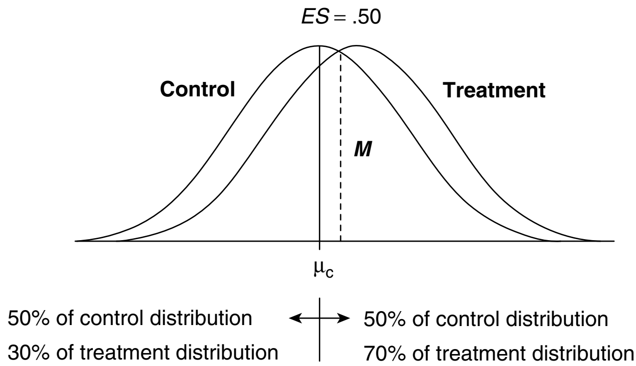

Figure 2.6 Depiction of the Percentage of the Treatment Distribution Above the Success Threshold Set at the Mean of the Control Distribution

Table 2.4 Effect Size Equivalents for *ES, r, U3,* and BESD

ES	*r*	*U3: % of T Above X_c*	*BESD C Versus T Success Rates*		*BESD C Versus T Differential*
.10	.05	54	.47	.52	.05
.20	.10	58	.45	.55	.10
.30	.15	62	.42	.57	.15
.40	.20	66	.40	.60	.20
.50	.24	69	.38	.62	.24
.60	.29	73	.35	.64	.29
.70	.33	76	.33	.66	.33
.80	.37	79	.31	.68	.37
.90	.41	82	.29	.70	.41
1.00	.45	84	.27	.72	.45

example, an effect size of one fifth of a standard deviation (*ES* = .20) corresponds to a BESD success rate differential of .10—that is, 10 percentage points between the treatment and control group success rates (55% vs. 45%). A success increase of 10 percentage points on a control group baseline of 45% represents a 22% improvement in the success rate (10/45). Viewed in these terms, the same intervention effect that may appear rather trivial in standard deviation units now looks potentially meaningful.

The Criterion Contrast Approach

Although actuarial and statistical translation approaches to assessing effect size may be useful for many purposes, they are somewhat removed from the specific context of any given intervention study. Often, the best answer to the question of what effect size has practical significance is one that is closely tied to the particular problems, populations, and measures relevant to the intervention under investigation. For example, if we could identify and measure a naturally occurring effect in the intervention context whose practical significance was easily recognized, it could be used as a criterion value or benchmark against which any expected or obtained intervention effect could be compared. What is required in the criterion group contrast approach is that some such comparison be identified and represented as a statistical effect size on the dependent measure relevant to the intervention research.

The criterion group contrast approach is best explained by an example. Consider a community mental health center in which prospective patients receive a routine diagnostic intake interview and are sorted into those judged to need, say, inpatient therapy versus outpatient therapy. This practice embodies a distinction between more serious and less serious cases and the "size" of the difference between the severity of the symptoms for these two groups that would be well understood at the practical level by those involved in community mental health settings. If we administer a functional status measure that is of interest as an outcome variable for both these groups, we could represent the difference between them as an effect size—that is, the difference between their means on that measure divided by the pooled standard deviations. Though this effect size does not represent the effect of intervention, we can nonetheless think of it in comparison with an intervention effect. That is, how successful would we judge a treatment to be that, when applied to clients as severe as the inpatient group, left them with scores similar to those of the outpatient group? Such an effect may well be judged to be of practical significance and would have recognized meaning in the treatment context. Real or anticipated intervention effects can thus be compared with this criterion contrast value as a way of judging their practical significance.

Reasonable criterion comparisons are often surprisingly easy to find in applied settings. All one needs to create a criterion contrast are, first, two groups whose difference on the variable of interest is easily recognized and, second, the result of measurement on that variable. It is also desirable to use groups that resemble, as much as possible, those samples likely to be used in any actual intervention research. Some of the possibilities for criterion contrasts that frequently occur in practical settings include the following:

- Eligible versus ineligible applicants for service where eligibility is determined primarily on the basis of judged need or severity. For example, a contrast on economic status might compare those who do not qualify for food stamps with those who do.

- Sorting of intervention recipients into different service or diagnostic categories based on the severity of the problems to be treated. For example, a contrast on literacy might compare those adult education students enrolled in remedial reading classes with those enrolled in other kinds of classes.

- Categories of termination status after intervention. For example, a contrast on functional status measures might compare those patients judged by physical therapists to have had successful outcomes with those judged to have had unsuccessful outcomes.

- Comparison of "normal" individuals with those who have the target problem. For example, a contrast on delinquent behavior could compare the frequency of self-reported delinquency for a sample of males arrested by the police with that of similar-age males from a general high school sample.

- Maturational differences and/or those occurring with usual service. For example, a contrast on mathematics achievement might compare the achievement test scores of third graders with those of fifth graders.

Conclusion

Attaining adequate statistical power in intervention research is not an easy matter. The basic dilemma is that high power requires a large effect size, a large sample size, or both. Despite their potential practical significance, however, the interventions of interest all too often produce modest statistical effects, and the samples on which they can be studied are often of limited size. Intervention researchers need to learn to live responsibly with this problem. The most important elements of a coping strategy are recognizing the predicament and attempting to overcome it in every possible way during the design phase of a study. The keys to designing sensitive intervention research are an understanding of the factors that influence statistical power and the adroit application of that understanding to the planning and implementation of each study undertaken. As an aid to recall and application, Table 2.5 lists the factors discussed in this chapter that play a role in the statistical power of experimental research along with some others of an analogous sort.

Table 2.5 Factors That Work to Increase Statistical Power in Treatment Effectiveness Research

Independent variable
Strong treatment, high dosage in the treatment condition
Untreated or low-dosage control condition for high contrast with treatment
Treatment integrity; uniform application of treatment to recipients
Control group integrity; uniform control conditions for recipients
Study participants
Large sample size (or number of clusters in the case of multilevel research) in each experimental condition

(Continued)

Table 2.5 (Continued)

Deploying limited participants into few rather than many experimental groups
Little initial heterogeneity on the dependent variable
Measurement or variance control of participant heterogeneity
Differential participant response accounted for statistically (interactions)
Dependent variables
Validity for measuring characteristic expected to change
Validity, sensitivity for change on characteristic measured
Fine-grained units of measurement rather than coarse or categorical
No floor or ceiling effects in the range of expected response
Mastery or criterion-oriented rather than individual differences measures
Inherent reliability in measure, unresponsiveness to irrelevant factors
Consistency in measurement procedures
Aggregation of unreliable measures
Timing of measurement to coincide with peak response to treatment
Statistical analysis
Larger alpha for significance testing
Significance tests for graduated scores, not ordinal or categorical
Statistical variance control; blocking, ANCOVA, interactions

Discussion Questions

1. In your area of research, which type of error (Type I or Type II) typically carries more serious consequences? Why?

2. In your field, would it ever be sensible to perform a one-tailed significance test? Why or why not?

3. In your field, what are some typical constructs that would be of interest as outcomes, and how are those constructs usually measured? What are the pros and cons of these measures in terms of validity for measuring change, reliability, and discrimination of individual differences?

4. In your research, what are some extraneous factors that are likely to be correlated with your dependent variables? Which of these are measurable so that they might be included as covariates in a statistical analysis?

5. What are some ways that you might measure implementation of an intervention in your field of research? Is it likely that interventions in your field are delivered uniformly to all participants?

6. Is the use of "no treatment" control groups (groups that receive no form of intervention) typically possible in your field? Why or why not?

7. In your field, are interventions typically delivered to individual participants, or to groups of participants such as classrooms, neighborhoods, etc.? If interventions are delivered to groups, do researchers normally use analytical techniques that take this into account?

8. If you were designing a study in which an intervention was to be delivered to groups (clusters) of participants, would you be better off, in terms of statistical power, collecting data on a large number of individuals within each cluster or on a smaller number of individuals in a larger number of clusters?

9. Imagine you conduct a study testing an intervention that is designed to increase the intelligence of children. You have access to a very large number of children and, thus, have adequate power to detect an effect size of .03. At the end of the intervention, the average IQ score of children in your control group is 100.0, and the average IQ score of children in your intervention group is 100.5. This difference in IQ scores is statistically significant. What do you conclude from your study?

Exercises

1. Look up four or five recent studies with treatment/control comparisons in your area of research and calculate the effect sizes they report. What is the average effect size, and what is the range of effect sizes? If you were designing a similar study, what is the minimum effect size that you would consider meaningful to detect?

2. Using the power chart in Figure 2.1, determine the power to detect an effect size of .70 with 20 participants per group, given a two-tailed α of .05. How many participants per group would you need to attain .90 power to detect the same effect size?

3. You are designing a study examining gains on a standardized test of academic achievement and your research leads you to believe that you can expect an effect size of .30 (assume the intervention group mean will be 105, the control group mean 100, and the shared standard deviation 15). Unfortunately, constraints on your resources require a design that is able to detect a minimum effect size of .60. If you were to add a covariate to your model to increase power, how strongly must that covariate be correlated with academic achievement to give you adequate power, given your design constraints?

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CHAPTER 3

Practical Sampling

Gary T. Henry

Conducting an applied research project that involves primary data collection requires that the study team develop and implement a sampling plan that includes deciding how individuals or other units will be selected, carrying out the selection process, encouraging participation of those selected, and assessing the extent to which departures from the expectations set when planning the sampling process may affect the study findings. When a subset of a study population is to be selected for data collection, the selection process is known as *sampling.* Not all studies involve sampling, for example, *census surveys* in which the entire *study population* is selected for the study do not require sampling. However, even when census surveys are to be used, many of the planning and implementation procedures related to sampling, which are described in this chapter, such as obtaining an accurate listing of the study population and evaluating the impact of nonresponse, are germane.

The *study population* for an applied research project can be individuals or other units, such as cities, hospitals, or defined geographic areas such as census tracts. When individuals are the focus of a study, they can be members of a *general population,* which are defined by age and place of residence at a specific time, for example, adults living in New York between October 1 and October 27, 2006, or members of a *special population.* Special populations are usually defined by participation or membership in a specific group during a prescribed time period, such as eighth graders enrolled in public schools in North Carolina during the 2005–2006 school year or adult mental health service consumers in Seattle who initiated service in 2006. In most cases, evaluations and other applied studies focus on special populations, often on populations who are eligible to participate in a certain program or those who have actually received services. However, there are examples of general population surveys that are used for applied research purposes such as statewide polls reported in the news media or surveys for assessing specific needs or measuring attitudes of the population concerning their support for a new program or policy.

Sampling or selecting a subset of the population is a part of most applied research projects. Sampling is required when not all members of the study population can be surveyed or included in the data collection. Constraints on time and budget often limit the number of members of the population who can be the subjects of the data collection and, thereby, require that only a subset of the population be selected for a study. However, it is important to note that on the few occasions when resources permit collecting data from the entire study population, it can be more accurate to collect data from a sample than to conduct a census survey. Greater accuracy can be achieved when choosing a subset of the population allows the researchers to use their resources to encourage more of the selected members of the population to participate in the data collection, thereby reducing the amount of missing data (Dillman, 1999; Fowler, 1993), or to improve measurement techniques in ways that could not have been done if they had attempted to collect data from all members of the study population. For example, evaluations of early childhood education programs often face the choice of using teacher ratings of the children's skills, which are collected for the entire population, or direct assessments of a sample of the children who attend these programs. Because of the bias that can result in having teachers rate the skills of their students (Mashburn & Henry, 2004), scores on direct assessments from a sample of children can be more accurate measures of the children's skills than teachers' rating on the entire population of children.

Probability and Nonprobability Sampling

Samples are generally categorized as either *probability samples* or *nonprobability samples*. The distinction between the two is that *probability samples* use random processes rather than human judgments to select the individuals or other units for a study. *Nonprobability samples* allow human judgments, either purposefully or unintentionally, to influence which individuals or units are selected for a study. Researchers using probability samples forego exercising their judgments about which individuals are selected for a sample by allowing a random process to decide which members of the study population are designated for participation in the study. Relying on random processes to choose the members of the study population who are selected for the study allows researchers to use well-grounded theories and methods to estimate the characteristics of the study population from the sample data or to test hypotheses about the study population. In addition, using random procedures to select the sample for a study eliminates a very important source of bias from the study. The use of probability sampling techniques can enhance the accuracy and credibility of the study findings.

The major benefit of eliminating human judgments in the selection process is that the *probability sample* that results is a statistical model of the study population. *Probability samples* make it possible to estimate averages or percentages for the study population (as well as other population parameters), estimate the range around the sample average (or other population parameter) within which the true average for the population is likely to occur, test hypotheses about the study population, and calculate indicators of sample bias when bias cannot be entirely eliminated. It is possible to calculate these estimates because probability sampling rests on probability theory. Probability theory requires that every member of the study population must have a known, nonzero chance of being included in the sample. This means that no known member of the target population is excluded from the possibility of being included in the sample and that all members have a known probability of selection. Major purposes of probability samples are to estimate characteristics of the population from the sample data or to use sample data to infer that a difference exists between two groups in the study population or between members of the study population at two time periods. Put another way, *probability samples* exist to provide information about the study population and to allow researchers familiar with the particular study population and measures to assess the adequacy of the sample from which the data were drawn for the purpose of the study.

In contrast, nonprobability samples are best used to provide information about specific cases or members of the study population that are intrinsically interesting or important for the study. Nonprobability samples are used to guide data collection about the specific experiences of some members of the study population, to explore a perceived social problem or issue, or to develop theories that are grounded in the actual experiences of some actual members of the study population. Often the cases selected through *purposeful nonprobability sampling* have particular theoretical or practical significance and can be used for developing theories or to generate explanations for the ways in which interesting or high-performing cases differ from other cases. When nonprobability samples are used, it is not reasonable to attribute the results to the entire study population. This limit on attributing the sample results to the study population is imposed since the judgments that led to selection of the sample, whether purposeful or merely convenient, can create bias. That is, the selected cases can be systematically different from the others in the study population, and there is no means to adjust or estimate how similar or different these cases selected through nonprobability sampling may be. This situation contrasts with probability samples, where the "random chance" of selection allows the sample to model the study population.

Perhaps, the most infamous case in which the characteristics of nonprobability samples were attributed to the study population occurred in the polling done to predict the 1948 presidential election in the United States. Three prominent polling firms, all of which used a form of nonprobability sampling known as quota sampling, were convinced that Thomas Dewey would defeat Harry Truman by a significant margin. Truman actually received 50% of the population vote compared with Dewey's 45%. The subjective bias of interviewers tilted toward the selection of more Republicans for interviews, even though the sample proportions matched the voting population proportions in terms of location, age, race, and economic status. The unintended bias affected the accuracy and credibility of the polls and caused polling firms to begin to use more costly probability samples. It will be interesting to follow the use of Internet surveys to predict elections to see if they suffer a similar fate. These types of Internet surveys use nonprobability sampling procedures, and it remains to be seen if the polling organizations are able to model the processes by which individuals are selected for the surveys, agree to participate in them, and the relationship between their responses and the actual vote can be used to predict the voting totals accurately.

Just as the researchers can exercise judgment in the selection processes, the individuals selected have a right to choose if they will participate in a study. Individuals, whether they have been selected by random processes or human judgments, have a right to exercise their own judgments about participation in the study. While probability samples eliminate researchers' judgments about which individuals will be selected to participate in a study, both probability and nonprobability samples have the potential for systematic error, also referred to as *bias,* in attributing sample characteristics to the entire study population when individuals decide not to participate in a study. An important difference between the use of probability samples and nonprobability samples is in the rigorous tracking and reporting of the potential for bias from probability samples. For example, it is often required or at least commonly expected that researchers using probability samples will use standard definitions for calculating *response rates,* such as those that have been promulgated by the American Association of Public Opinion Research (2006). *Response rates* are the selected sample members that participated in the study divided by the total sample and expressed in percentage terms. Reporting the response rates using the standard calculation methods makes the potential for bias transparent to the reader. It is very difficult, if not impossible, to specify what response rates are necessary to reduce bias to a negligible amount. For example, Keeter, Miller, Kohut, Groves, and Presser (2000) show that it is extremely rare for findings to differ in a statistically significant way between a survey with an exceptionally high responses rate (60.6%) and one with a more common response rate (36.0%). While similar monitoring and reporting procedures could be applied to nonprobability samples, presenting information about participation rates is highly variable and much less standardized.

As this discussion begins to show, probability and nonprobability samples differ in very fundamental and significant ways. Perhaps, the most significant difference is whether the sample data present a valid picture of the study population or rather is used to provide evidence about the individual or cases in the sample itself. Before beginning to develop a sampling plan, the research team must make a definitive statement about the purpose for which the study is undertaken. For studies that are undertaken to describe the study population or test hypotheses that are to be attributed to the membership of the study population, probability samples are required. Nonprobability sampling is appropriate when individuals or cases have intrinsic interest or when contrasting cases can help to develop explanations or theories about why differences occur. The evaluation literature is filled with exemplary or "successful case" studies and studies that seek to contrast successful cases and unsuccessful ones. Using nonprobability samples for these studies makes good sense and can add explanatory evidence to the discussion about how to improve social programs. However, once the decision is made to use nonprobability sampling methods, it is inappropriate to present the findings in ways that suggest that they apply to the study population. Conversely, probability samples will not always produce sharp contrasts that allow for the development of explanatory theories. Therefore, the next section of the chapter provides some guidance about the types of nonprobability samples that applied research could consider and the methods for implementing them. Then, we will turn to an in-depth coverage of probability sampling methods because these methods have been more extensively developed.

Nonprobability Sampling

Nonprobability samples are important tools for applied research that can be used to

- choose cases that can be used to construct socially or theoretically significant contrasts;
- obtain evidence about individuals whose experiences are particularly relevant to the study's research questions;
- obtain data at a low cost that motivates more extensive, systematic research;
- establishes the feasibility of using particular instruments or survey procedures for more costly research may motivate using probability samples; or
- collect data about a group for whom it would be too costly or too difficult to use probability sampling techniques for a specific study.

A very important but perhaps underutilized nonprobability sampling method is to select cases that allow the researchers to contrast high-performing cases (or individuals) with lower-performing cases (or individuals) in order to find differences between the two. Using this approach, which falls under the umbrella of contrasting cases designs, allows researchers to gather evidence on the characteristics or processes that differ between the higher- and lower-performing cases. These provide empirically grounded explanations of the differences that can be used as a basis for theory and further systematic assessment.

Contrasting cases along with five other nonprobability sampling designs that are used frequently in social research are listed in Table 3.1, along with descriptions of their selection strategies (each of these designs is described more fully in Henry, 1990). Nonprobability samples are often used very effectively in qualitative research designs (see Maxwell, Chapter 7, this volume), but their utility is certainly not limited to qualitative studies. Perhaps, the most frequently used type of nonprobability sample is the convenience sample. Convenience samples, although somewhat denigrated by their label, often capitalize on identifying individuals who are readily available to participate in a study or individuals for whom some of the needed study data have already been collected. Often, convenience samples are used for studies where high degrees of internal validity or unbiased estimates of a program's effects are needed, but it is impractical to conduct the research in a way that allows for extrapolating the results to the entire population served by the program. An example of this type of sample is the study of the impact of the prekindergarten in Oklahoma that used data collected about children enrolled in the pre-k program operated by Tulsa Public Schools (Gormley & Gayer, 2005), which will be discussed in more detail later. Gormley and Gayer made strategic use of available data and were able to calculate program impacts in ways that have enhanced knowledge about the impacts of state sponsored prekindergarten programs. However, the estimates of effects cannot be extrapolated beyond the Tulsa Public School population.

Table 3.1 Nonprobability Sample Designs

Type of Sampling	*Selection Strategy*
Convenience	Select cases based on their availability for the study and ease of data collection
Contrasting cases	Select cases that are judged to represent very different conditions; often well used when a theoretically or practically important variable can be used as the basis for the contrast
Typical cases	Select cases that are known beforehand to be useful and not to be extreme
Critical cases	Select cases that are key or essential for overall acceptance or assessment
Snowball	Group members identify additional members to be included in sample
Quota	Interviewers select sample that yields the same proportions as in the population on easily identified variables

To illustrate the use of convenience samples, let's consider a hypothetical example that is similar to actual studies in many fields. Psychologists interested in the relationship between violence in movies and aggressive behaviors by the American public may choose to recruit volunteers from an introductory psychology class in an experiment. The researchers may survey the students about their attitudes and behaviors relating to violence and then show them a movie containing graphic violence. After the movie, the researchers could administer the same survey a second time, which fits the schema of a simple pretest-posttest design (see Bickman & Rog, Chapter 1, this volume; Mark & Reichardt, Chapter 6, this volume).

To expose and then clarify a point of confusion that often arises when discussing random samples (which I label probability samples, in part, to avoid this confusion) and random assignment, I will add a *randomly assigned control group* to this design. Before the treatment is administered—in this case, before the movie is shown—each student is randomly assigned to either a treatment group, a movie with graphic violence, or a nontreated group that receives a placebo, a movie without violence. *Random assignment* means that the students are assigned by some method that makes it equally likely that each student will be assigned to either the treatment group or the placebo group (Boruch, Weisburd, Turner, Karpyn, & Littell, Chapter 5, this volume). In this case, the design employs random assignment from a convenience sample. The strength of this design is in its ability to detect differences in the two groups that are attributable to the treatment, which in this example is watching a violent movie.

Although this type of design can rate highly in isolating the effect of violent movies, the convenience sample restricts the researchers' ability to extrapolate or generalize the findings to the general population. The generalizability of findings refers to the external validity of the findings. If we are interested in the effect of violent movies on the U.S. population, the use of a convenience sample severely constrains the study's external validity. The differences in these two groups cannot be used to formally estimate the impact of violent movies on the U.S. population. Other conditions, such as age, may alter responses to seeing violent movies. The students in this sample are likely to be in their teens and early 20s if they were attending a traditional college or university, and their reactions to the violent movie may be different from the reactions of older adults. Applying the effects found in this study to the entire U.S. population could be misleading. The randomized assignment that was used increases the internal validity of a study, but it should not be confused with random sampling. Random sampling is a probability sampling technique that increases external validity. Although applied studies can be designed to provide high levels of both internal validity and generalizability, most prioritize one over the other due to practical concerns such as costs or study purposes or because there are gaps in the current knowledge about the topic that the research sets out to examine that lead to developing strategies to fill an important gap.

Convenience sampling and contrasting cases sampling are but two of the many types of nonprobability sampling that are frequently used in applied social research. Quota sampling, which was mentioned earlier, was frequently used by polling firms and other survey research organizations but has been largely discarded. Quota samples exactly match the study population on easily observed characteristics, but because the interviewers select the respondents, bias can produce significant differences between the sample and the study population. Snowball samples are very commonly used for studies where the study population members are not readily identified or located. Examples of these types of populations are individuals involved with gangs, drugs, or other activities that are not condoned by society or populations that may be stigmatized or potentially suffer other consequences if their membership in the group is known, such as individuals living with HIV/AIDS or undocumented workers. Snowball sampling involves recruiting a few members of the study population to participate in the study and asking them to identify or help recruit other members of the study population for the study. Snowball samples may be significantly biased if the individuals recruited for the study have limited knowledge of other members of the group. However, snowball samples may be used to obtain evidence about some members of the study population, when time and resources are limited or when developing a list of the members is considered unethical.

Probability Samples

As I stated earlier, probability samples have the distinguishing characteristic that each unit in the population has a known, nonzero probability of being selected for the sample. To have this characteristic, a sample must be selected through a random mechanism. Random selection mechanisms are independent means of selection that are free from human judgment and the other biases that can inadvertently undermine the independence of each selection.

Random selection mechanisms include a lottery-type procedure in which balls on which members of the population have been identified are selected from a well-mixed bowl of balls, a computer program that generates a random list of units from an automated listing of the population, and a random digit-dialing procedure that provides random lists of four digits matched with working telephone prefixes in the geographic area being sampled (see, e.g., Lavrakas, Chapter 16, this volume). Random selection requires ensuring that the selection of any unit is not affected by the selection of any other unit. The procedure must be carefully designed and carried out to eliminate any potential human or inadvertent biases. Random selection does not mean arbitrary or haphazard selection (McKean, 1987).

The random selection process underlies the validity, precision, power, and credibility of sample data and statistics. The validity of the data affects the accuracy of generalizing sample results to the study population and drawing correct conclusions about the population from the analytical procedures used to establish differences between two groups or covariation. Sampling theory provides the basis for calculating the precision of statistics for probability samples. Because sampling variability has an established relationship to several factors (including sample size and variance), the precision for a specific sample can be planned in advance of conducting a study. Power is closely related to precision. Precision applies to the size of the confidence interval around a parameter estimate such as the mean or a percentage. The *confidence interval* is the interval around the sample mean estimate in which the true mean is likely to fall given the degree of confidence specified by the analyst. For example, when a newspaper reports that a poll has a margin of error of ±3%, it is a way of expressing the precision of the sample. It means that the analyst is confident that 95 out of 100 times, the true percentage will fall within 3 percentage points of the percentage estimated for the sample. *Power* refers to the probability of detecting a difference of a specified size between two groups or a relationship of a specified size between two variables given a probability sample of a specific size. The principal means of increasing precision and power is increasing sample size, although sample design can have a considerable effect as will be discussed later in this chapter. Credibility, in large measure, rests on absence of perceived bias in the sample selection process that would result in the sample being systematically different from the study population. Probability sampling can increase credibility by eliminating the potential bias that can arise from using human judgment in the selection process. Credibility is a subjective criterion while validity, precision, and power are objective criteria and have widely agreed on technical definitions.

A distinct advantage of probability samples is that sampling theory provides the researcher with the means to decompose and in many cases calculate the probable error associated with any particular sample. One form of error is known as bias. *Bias,* in sampling, refers to systematic differences between the sample and the population that the sample represents. Bias can occur because the listing of the population from which the sample has been drawn (sampling frame) is flawed or because the sampling methods cause some populations to be overrepresented in the sample. Bias is a direct threat to the external validity of the results.

The other form of error in probability samples, *sampling variability,* is the amount of variability surrounding any sample statistic that results from the fact that a random subset of cases is used to estimate population parameters. Because a probability sample is chosen at random from the population, different samples will yield somewhat different estimates of the population parameter. Sampling variability is the expected amount of variation in the sample statistic based on the variance of the variable and the size of the sample. Taken together, bias and sampling variability represent *total error* for the sample. Error can arise from other sources, as other contributors to this volume point out, but here the focus is on total error that arises from the design and administration of the sampling process. In the next section, I describe the sources of total error in some detail.

Sources of Total Error in Sampling Design

The researcher can achieve the goal of practical sampling design by minimizing the amount of total error in the sample selection to an acceptable level given the purpose and resources available for the research. *Total error* is defined as the difference between the true population value for the target population and the estimate based on the sample data. Total error has three distinct components:

- *Nonsampling bias:* systematic error not related to sampling, such as differences in target and study populations or nonresponse.
- *Sampling bias:* systematic error in the actual sampling that produces an overrepresentation of a portion of the study population, such as a sampling frame that lists some population members more than once.
- *Sampling variability:* the fluctuation of sample estimates around the study population parameters that results from the random selection process.

Each component of error generates specific concerns for researchers and all three sources of error should be explicitly considered in the sampling plan and adaptation of the plan during the research process. Each of the three components of total error and some examples of the sources of each are illustrated in Figure 3.1. Because sample design takes place under resource constraints, decisions that allocate resources to reduce error from one component necessarily affect the resources available for reducing error from the other two components. Limited resources force the researcher to make trade-offs in reducing total error. The researcher must be fully aware of the three components of error to make the best decisions based on the trade-offs to be considered in reducing total error. I describe below each of the three sources of error and then return to the concept of total error for an example.

Nonsampling Bias

Nonsampling bias is the difference between the true target population value and the population value that would be obtained if the data collection procedures were administered with the entire population. Nonsampling bias results from decisions as well as implementation of the decisions during data collection efforts that are not directly related to the selection of the sample. For example, the definition of the study population may exclude some members of the target population that the researcher would like to include in the study findings. Even if data were collected on the entire study population, in this case, the findings would be biased because of

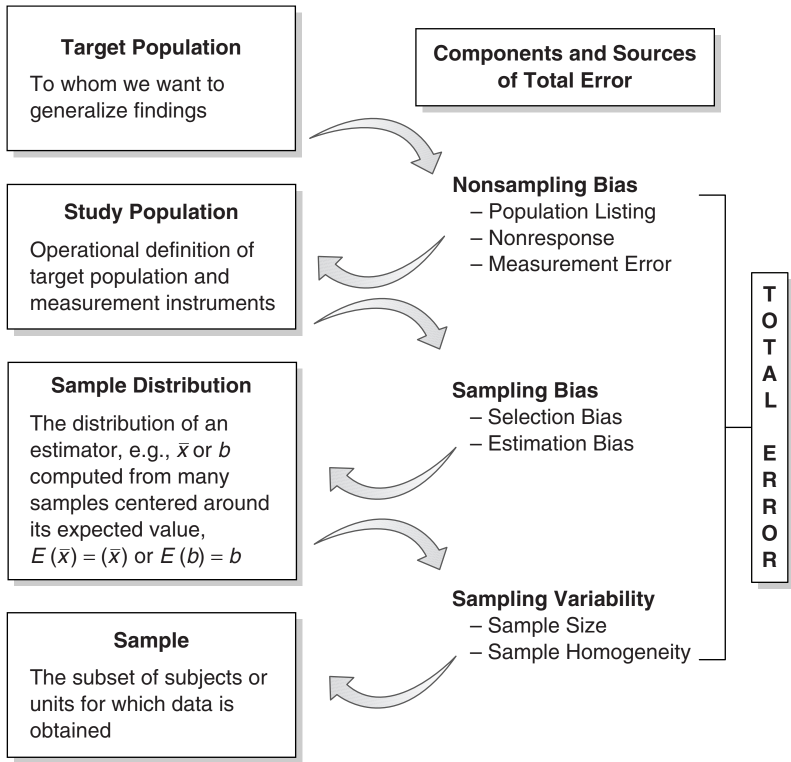

Figure 3.1 Decomposing Total Error

the exclusion of some target population members. For example, using the Atlanta telephone directory as the sampling frame for the current residents of the Atlanta metropolitan area would produce biased estimates of household characteristics due to unlisted numbers, households with phone service established after the phone book went to press, and residents without phones, including the homeless and those who rely exclusively on cellular phones.

Differences in the true mean of the population and the survey population mean arise from several sources. A principal difference relevant to sample design is the difference between the target population and the study population. The target population is the group about which the researcher would like to make statements. The target population can be defined based on conditions and concerns that arise from the theory being tested or factors specific in the policy or program being evaluated, such as eligibility criteria. For instance, in a comprehensive needs assessment for homeless individuals, the target population should include all homeless individuals, whether served by current programs or not. On the other hand, an evaluation of the effectiveness of community mental health services provided to the homeless should include only homeless recipients of community mental health care, which may exclude large numbers of the homeless. The target population for the needs assessment is more broadly defined and inclusive of all homeless.

Also, nonresponse creates nonsampling bias. Nonresponse results from the researcher's inability to contact certain members of the population or from some target population members' choice to exercise their right not to participate in a survey or provide other data for the research. If nonresponse is truly random, it does not represent a bias, but this is frequently not the case and nonresponse should never be assumed to be missing at random or even ignorable without careful examination. More frequently, nonrespondents come from a definable subgroup of the population that may regard the research project as less salient or more of an intrusion than others. The omission of subgroups such as these from the data that are actually collected creates a bias in the results.

Sampling Bias

Sampling bias is the difference between the study population value and the expected value for the sample. The expected value of the mean is the average of the means obtained by repeating the sampling procedures on the study population. The expected value of the mean is equal to the study population value if the sampling and calculation procedures are unbiased. Sampling bias can be subdivided into two components: selection bias and estimation bias. Selection bias occurs when not all members of the study population have an equal probability of selection. Estimation procedures can adjust for the unequal probabilities when the probabilities of selection are known. When the probability of selection is not equal, researchers adjust the estimates of the population parameters by using weights to compensate for the unequal probabilities of selection.

An illustrative example of selection bias is a case in which a sample is selected from a study population list that contains duplicate entries for some members of the population. In the citizen survey example presented in Henry (1990), two lists are combined to form the study population list: state income tax returns and Medicaid-eligible clients. An individual appearing on both lists would have twice the likelihood of being selected for the sample. It may take an inordinate amount of resources to purge such a combined list of all duplicate listings, but it could be feasible to identify sample members that appeared on both lists and adjust for the unequal probability of selection that arises.

To adjust for this unequal probability of selection, a weight (w) equal to the inverse of the ratio of the probability of selection of unit to the probability of selection of units only listed once (r) should be applied in the estimation process:

$$w = 1/p = 1/2 = .5$$

The probability of selection for this individual was twice the probability of selection for the members of the study population appearing on the list only once. Therefore, this type of individual would receive only one half the weight of the other population members to compensate for the increased likelihood of appearing in the sample. The logic here is that those with double listings have been overrepresented by a factor of two in the sample and, therefore, must be given less weight in the estimation procedures to compensate.

Estimation bias occurs when the average calculated using an estimation technique on all possible simple random samples from a population does not equal the study population value. For example, the median is a biased estimate of the central tendency for the population. This is due to the fact that the expected value of the median of the sample means is not equal to the true study population mean. Generally, biased estimators, such as the median, are used to overcome other issues with the data and, therefore, the estimation bias is outweighed by other factors. For example, the median income of a population is often estimated rather than the mean income because relatively few very high income individuals can cause the mean to be high relative to median and the income that most members of the population actually receive.

Sampling Variability

The final component of total error in a sample is directly attributable to the fact that statistics from randomly selected samples will vary from one sample to the next due to chance. In any particular sample, some members of the study population will be included and others will be excluded, which produces this variation. Because it is rare for sample estimates to be exactly equal to the study population value, it is useful to have an estimate of their likely proximity to the population value, or in the terms that I have used before, the precision of the sample estimate.

Sampling theory can be used to provide a formula to estimate the precision of any probability sample based on information available from the sample. Two factors have the greatest influence sampling on the standard error: the amount of variation around the mean of the variable (standard deviation or square root of the variance) and the size of the sample. Smaller standard deviations reduce the sampling error of the mean. The larger the sample, the smaller the standard deviation of the sampling distribution.

Because the standard deviation for the population can be estimated from the sample information and the sample size is known, a formula can be used to estimate the standard deviation of the sampling distribution, referred to hereafter as the standard error of the estimate, in this particular case, the standard error of the mean:

$$s_{\bar{x}} = \frac{s}{n^{1/2}},$$

where $s_{\bar{x}}$ is the estimate of the standard error of the mean, s is the estimate of the standard deviation, and n is the sample size. Using this formula allows the researcher to estimate the standard error of the mean, the statistic that measures the final component of total error, based solely on information from the sample.

The standard error is used to compute a confidence interval around the mean (or other estimate of a population parameter), or the range which is likely to include the true mean for the study population. The likelihood that the confidence interval contains the true mean is based on the product of the t statistic chosen for the following formula:

$$I = \overline{x} \pm t(s_{\overline{x}}).$$

The confidence interval is the most popular direct measure of the precision of the estimates, and it is common practice to use the value that represents 95% confidence, 1.96, for *t*. In most cases, the researcher should report the confidence interval along with the point estimate for the mean to give the audience an understanding of the precision of the estimates.

Two more technical points are important for discussion here. First, probability sampling design discussions thus far in this chapter have assumed that the sample would be selected without replacement; that is, once a unit has been randomly drawn from the population to appear in the sample, it is set aside and not eligible to be selected again. Sampling without replacement limits the cases available for selection as more are drawn from the population. If a sample is drawn from a finite population, sampling without replacement may cause a finite population correction (FPC) factor to be needed in the computation of the standard error of the estimate.

For the standard error of the mean, the formula using the FPC is

$$s_{\overline{x}} = (1 - n/N)\frac{s}{n^{1/2}}.$$

As a rule of thumb, the sample must contain more than 5% of the population to require the FPC. This is based on the fact that the FPC factor is so close to 1 when (n/N) the sampling fraction is less than .05 that it does not appreciably affect the standard error calculation.

Second, standard error calculations are specific to the particular population parameter being estimated. For example, the standard error for proportions is also commonly used:

$$s_p = [(pq)/n]^{1/2}.$$

where s_p is the standard error for the proportion, p is the estimate of the proportion, and $q = 1 - p$. Most statistic textbooks present formulas for the standard error of several estimators, including regression coefficients. Also, they are calculated for the statistic being used by almost any statistical software package. These formulas, like the formulas presented above, assume that a simple random sample design has been used to select the sample. Formulas must be adjusted for more complex sampling techniques (Henry, 1990; Kish, 1965).

One further note on terminology: The terms *sampling error* and *standard error* are used interchangeably in the literature. They are specific statistics that measure the more general concept of sampling variability. *Standard error,* however, is the preferred term. The common use of *sampling error* is unfortunate for two reasons. First, it implies an error in procedure rather than an unavoidable consequence of sampling. Second, the audience for a study could easily assume that sampling error is synonymous with total error concept, which could lead to the audience's ignoring other sources of error. For example, when newspapers report the margin of error for polling results that they publish (usually $s_p \times 1.96$), they typically ignore other sources of error, such as nonresponse that could be indicated by calculating and publishing the response rate using the appropriate formulas published by the American Association of Public Opinion Research (2006).

Total Error

Total error combines the three sources of error described above. Sample design is a conscious process of making trade-offs to minimize these three components of total error. Too frequently, reducing the standard error becomes the exclusive focus of sample design because it can be readily estimated. Because the two bias components cannot be calculated as readily, they are often given short shrift during the design process. When this occurs, sampling planning is reduced to the calculation of sample size and selection of the type of probability sample to be used. However, failing to consider and to attempt to reduce all three components of total error sufficiently can reduce the validity and credibility of the study findings. In the next section of this chapter, the practical sampling design framework will be described. By answering the questions presented in the framework, applied researchers can assess the options available to reduce total error while developing a sample plan and adapting the plan to the unexpected events that occur when the plan is being implemented.

Practical Sampling Design Framework

The framework for practical sampling design is a heuristic tool for researchers and members of the audience for research findings to use in sample design as well as an aid in interpretation of the findings. The framework is, in essence, a series of choices that must be made, with each choice having implications for the validity and integrity of the study. While much of the framework applies to nonprobability samples, especially the presampling questions, the framework was originally developed for probability samples. My purpose in providing the framework here is to help researchers and consumers of research structure their thinking about design choices and the effects of their choices on total error. No single sample design will accomplish all the goals for studying a particular population and choices may be made differently by different research teams. The process involves both calculations and judgment. As researchers work through the choices presented in the framework, issues may be raised, which may cause them to reassess earlier decisions. In some situations or with certain populations, some types of error raise greater concerns than others, so knowledge of prior research, including the sampling designs used in previous studies of the target population, may add important information to the sample planning process to fill in important gaps in knowledge about the population or program, to avoid problems experienced with the earlier studies or to adhere to commonly accepted practices.

The framework includes three phases of the overall design of the research project, which have been further subdivided into 14 questions (see Table 3.2).

Table 3.2 Questions for Sample Design

Presampling choices
What is the nature of the study—exploratory, developmental, descriptive, or explanatory?
What are the variables of greatest interest?
What is the target population for the study?
Are subpopulations important for the study?
How will the data be collected?
Is sampling appropriate?
Sampling choices
What listing of the target population can be used for the sampling frame?
What is the precision or power needed for the study?
What sampling design will be used?
Will the probability of selection be equal or unequal?
How many units will be selected for the sample?
Postsampling choices
How can the impact of nonresponse be evaluated?
Is it necessary to weight the sample data?
What are the standard errors and related confidence intervals for the study estimates?

The answers to these questions will result in a plan to guide the sampling process, assist the researchers in analyzing the data correctly, and provide ways to assess the amount of error that is likely to be present in the sample data. In the next three sections, we will focus on making choices that impact sample planning and implementation as well as understanding some of the implications of those choices. More detail on the implications of the various choices, as well as four detailed examples that illustrate how choices were actually made in four sample designs, is provided in Henry (1990). In addition, other chapters in this *Handbook* provide discussion of the other issues.

Presampling Choices

What Is the Nature of the Study: Exploratory, Developmental, Descriptive, or Explanatory?

Establishing the primary purpose of the study is one of the most important steps in the entire research process (see Bickman & Rog, Chapter 1, this volume).

Exploratory research is generally conducted to provide an orientation or familiarization with the topic under study. It serves to orient the researcher to salient issues, helps focus future research on important variables, and generates hypotheses to be tested. Exploratory research is often conducted on newly emerging social issues or recently developed social program. In these cases, the research base is often slim or not much is known about the issue or program in the specific area or region in which the study has been commissioned. In some cases, exploratory studies are undertaken in the early phases of an evaluation and the findings are used to develop a plan for more thorough-going evaluation studies. Sampling approaches for exploratory studies are quite reasonably limited by resource and time constraints placed on them. Preferred sampling methods include those that ensure a wide range of groups are covered in the study rather than those that reduce error, because estimates, such as averages and percentages, are not reasonable study products. Sample designs that ensure coverage of a wide range of groups or, said another way, intentionally heterogeneous samples are purposeful samples or small stratified samples. These approaches can yield a diverse sample at relatively low cost.

Developmental studies are a recent addition to the list of study purposes to emphasize the importance of studies that are commissioned for theory development or methodological development. For example, in the field of early childhood education, there is a growing need to assess the language, cognitive, and social skills of children who do not speak English at home, but we have few assessment instruments and little evidence about how to assess these children. Should the children who do not speak language at home be assessed in both their home language and English or only one? What are the implications for the length of the assessments and test fatigue if children are tested in both languages? To gather evidence to address questions of this sort, the organization that oversees the prekindergarten program for Los Angeles, California, recently commissioned a developmental study of measurement issues as the first phase of an evaluation of two of the prekindergarten programs operating in LA County. The sampling plan for the developmental phase calls for oversampling children who do not speak English at home to compare the strengths and weaknesses of alternative measures and measurement protocols.

Developing theories or explanations for socially or theoretically important phenomena can require studies with special sampling strategies. One option for studies designed to develop theory, which was mentioned earlier, is the contrasting cases of nonprobability design. This design can be extremely useful for evaluations that attempt to explain why some programs or program administrative units (e.g., schools or clinics) perform better than others. A nonprobability design might select only high-performing and low-performing units for the purpose of collecting qualitative data to contrast these two groups. Alternatively, a probability sampling approach could be adopted that first divides the units into high-, "average-," and low-performing units and then samples a higher proportion of high- and low-performing units but fewer "average" performers as well. Data collection could be either qualitative or quantitative depending on the existing state of theoretical development in the field. One advantage of the probability sampling approach is that once the organizational level or other factors correlated with performance are identified, an estimate of the frequency with which the factors occur in the study population could be calculated from the available sample data.

Descriptive research is the core of many survey research projects in which estimates of population characteristics, attributes, or attitudes are study objectives (see Fowler & Cosenza, Chapter 12; Lavrakas, Chapter 16, this volume). In fact, probability sampling designs were originally developed for this type of research. Therefore, most sampling texts, especially older ones, emphasize the use of sample data to develop estimates of the characteristics of the study population, such as averages and percentages. But it has become common for probability studies to be used for explanatory research purposes as well. Explanatory research examines expected differences between groups and/or relationships between variables, and the focus of these studies is explaining variation in one or more variables or estimating the difference between two groups. Typically, the emphasis for descriptive studies will be the precision of the estimates, while analytical studies will need to pay attention to the power to detect effects if the effects actually occur. In practice, many studies attempt both descriptive and explanatory tasks, which mean that the researchers may need to assess both precision and power as decisions about sample design and power are being considered.

In addition, it is common that practical considerations lead researchers to conduct their explanatory studies in more limited geographic areas than the entire area in which certain services are provided or programs operate. For example, Gormley and Gayer (2005) focused their evaluation of the impact of the prekindergarten program in Oklahoma on the children who participated in the program in Tulsa Public Schools. Even if a complete census survey of prekindergarteners attending Tulsa Public Schools had been possible, the effects that were estimated would only formally generalize the children who attended the Tulsa Public Schools program, not the other children attending the state-sponsored prekindergarten in Tulsa or the children served in the prekindergarten programs operated by the other 493 school districts in the state of Oklahoma. In cases such as these, it requires substantive expertise and knowledge of the populations being served in the locality chosen for the study to assess the reasonableness of suggesting that the effects would be similar for other children in the target population who were not eligible for participation in the study. This is an example of researchers placing greater emphasis on their ability to accurately estimate the size of the effect attributable to a program for a subset of the participants of the entire program than on the external validity or generalizability of the effect to the entire population served by the program. Often, such choices are fruitful and well justified, as was the case with Gormley and Gayer's study, so that gaps in existing knowledge can be reduced, and the state of knowledge in a field move forward. It is the slow and steady increments to knowledge rather than the "ideal" that will often shape the decision for the type of study to be conducted at a particular time and in specific circumstances.

Both descriptive and explanatory studies are concerned with reducing total error. Although they have similar objectives for reducing both types of bias, the sampling variability component of total error is quite different. For descriptive studies, the focus is on the precision needed for estimates. For explanatory studies, the most significant concern is whether the sample will be powerful enough to allow the researcher to detect an effect, given the expected effect size. This is done through a power analysis (see Lipsey & Hurley, Chapter 2, this volume). Explanatory and descriptive studies will be the primary focus in the responses to the remaining questions.

What Are the Variables of Greatest Interest?

Selecting the most important variables for a study is an important precursor to the sampling design. Studies often have multiple purposes. For instance, a study of student performance may seek to assess the impacts of a program on both achievement and retention in grade. Measuring the dependent variables as well as program participation and any control variables will need to be considered. Moreover, the researcher may envision including many descriptive tables in the write-up or using several statistical tools to examine expected relationships. Choosing the variable of greatest interest is a matter of setting priorities. Usually, the most important dependent variable in an applied study will be the one of greatest interest. At times, applied researchers must default to practical considerations such as choosing a dependent variable that can be measured within the study's time frame, even though other important variables must be reduced to secondary priorities as a result of the practical priorities. The variables of greatest interest are then used to develop responses to the questions that come later in the design process.

What Is the Target Population for the Study?

The target population for a study is the group about which the researcher would like to be able to speak in the reports and presentations that they develop from the findings. The population can be individuals (residents of North Carolina or homeless in Los Angeles), groups of individuals (households in Richmond or schools in Wisconsin), or other units (invoices, state-owned cars, schools, or dwelling units). In many cases, the study sponsor may be interested in a particular target population. For example, a state agency responsible for the administration of a statewide pre-k program may want the study findings to generalize to the entire state but a local program operator may be more focused on the program in her particular locality. Decisions about target population definitions should be made with both researchers and study sponsors fully aware of the limitations on extrapolating the findings beyond the target population once the study is completed.

Are Subpopulations Important for the Study?

Often, a researcher will choose to focus on a part of the target population for additional analysis. For example, households headed by single, working females were of particular interest to some scientists examining the impact of income maintenance experiments (Skidmore, 1983). It is most important to identify the subgroups for which separate analyses are to be conducted, including both estimating of characteristics of the subpopulation using the sample data and explanatory analyses. When subgroups are important focal points for separate analyses, later sampling design choices, such as sample size and sampling technique, must consider this. A sample designed without taking the subpopulation into account can yield too few of the subpopulation members in the sample for reliable analysis. Increasing the overall sample size or disproportionately increasing the sample size for the subpopulation of interest, if the members of subpopulation can be identified before sampling, are potential remedies, as will be discussed later.

How Will the Data Be Collected?

Certain sampling choices can be used only in conjunction with specific data collection choices. For example, random-digit dialing, a technique that generates a probability sample of households with working phones, is an option when interviews are to be conducted over the phone (see Lavrakas, Chapter 16, this volume). A probability sample of dwelling units is useful mainly for studies in which on-site fieldwork, usually in the form of personal interviews, is to be used. The collection of data from administrative records or mailed questionnaires also poses specific sampling concerns. For example, mailed questionnaires can have a high proportion of nonrespondents for some populations (see Mangione & Van Ness, Chapter 15, this volume). Nonresponse affects sampling variability and will cause nonsampling bias to the extent that the members of the sample who choose not to respond are different from those who do. In making a decision about sample size, which comes a bit later in these questions, the researcher should factor nonresponse into the final calculation. Because the sampling error depends on the number who actually respond, not the number surveyed, it is common to divide the desired sample size by the proportion expected to respond. For example, a desired sample size of 500 with an expected response rate of .80 will require an initial sample size of 625. If an alternative method of administering the instrument is expected to reduce response rates, it will increase the sample size required for the same number of completes.

Is Sampling Appropriate?

The decision to sample rather than conduct a census survey should be made deliberatively. In most cases, resources available for the study mandate sampling. Once again, it is important to note that when resources are limited, sampling can produce more accurate results than a population or census-type study. Often, resources for studies of entire populations are consumed by attempts to contact all population members. Response to the first contact is often far less than 50%, raising the issue of substantial nonsampling bias. Sampling would require fewer resources for the initial survey administration and could allow the investment of more resources in follow-up activities designed to increase responses, paying dividends in lowering nonsampling bias. In addition, when access to the target population is through organizations which serve the population, gaining access can require substantial resources. For instance, many organizations such as school districts have research review committees that require proposals to be submitted, reviewed, and approved, which can require substantial revisions, before access can be gained. Obviously, these increase the time and resources required for data collection. Even when automated databases that contain all members of the population are being used, sampling can improve the accuracy of results. Missing data are a frequent problem with automated databases. Missing data are another form of nonresponse bias, because the missing data cannot be assumed to be missing at random. The cost of collecting the data missing from the data base or supplementing information for variables that have not been collected will be less for the sample than for the entire population, in nearly every case.

On the other hand, small populations and use of the information in the political environment may weigh against sampling. For studies that may affect funding allocations or when there is expert knowledge of specific cases that may appear to be "unusual" or "atypical," the use of a sample can affect the credibility of a study. Credibility is vital when study results are used to inform policy or program decisions. Because program decisions often determine winners and/or losers, credibility rather than validity may be the criterion on which the use of the findings turns.

Sampling Choices

What Listing of the Target Population Can Be Used for the Sampling Frame?

The sampling frame, or the list from which the sample is selected, provides the definition of the study population. Differences between the target population and the study population as listed in the sampling frame constitute a significant component of nonsampling bias. The sampling frame is the operational definition of the population, the group about which the researchers *can* reasonably speak.

For general population surveys, it is nearly impossible to obtain an accurate listing of the target population. A telephone directory would seem to be a likely explicit sampling frame for a study of the population in a community. However, it suffers from all four flaws that are commonplace in sampling frames:

- *Omissions:* target population units missing from the frame (e.g., new listings and unlisted numbers)
- *Duplications:* units listed more than once in the frame (e.g., households listed under multiple names)
- *Ineligibles:* units not in the target population (e.g., households recently moved out of the area)
- *Cluster lists:* groupings of units listed in the frame (e.g., households, not individuals, listed)

The most difficult flaw to overcome is the omission of part of the target population from the sampling frame. This can lead to a bias that cannot be estimated for the sample data. An alternative would be to use additional listings that include omitted population members to formulate a combination frame or to choose a technique that does not require a frame, such as random-digit dialing instead of the phone book.

Duplications, or multiple listings of the same unit, increase the probability of selection for these units. Unchecked duplications result in sampling bias. For random-digit dialing, households with two or more phones are considered duplications, since the same household is listed two or more times. In some evaluations of program services, duplications can occur because lists of program participants are actually lists of enrollees, and individuals may be enrolled at some time during the study period in more than one program. In some cases, researchers can address duplications by removing them from the list before sampling. In other cases, weights can be calculated based on the number of duplications for each case in the sample (Henry, 1990) and used to adjust estimates.

Ineligibility occurs when cases that are not members of the target population appear on the sampling list. When ineligibles can be screened from the list or from the sample, the only concerns are the cost of screening and the reduction of the expected sample size. The cost of screening for a telephone survey includes making contact with someone in the household to determine eligibility. This can require several phone calls and can become quite costly, especially when the proportion of ineligibles is large. In addition to screening, it is likely that the sample size will need to be increased so that sampling errors will not increase due to the screening.

Cluster listings are caused by sampling frames that include groups of units that are to be analyzed, rather than the units themselves. Many general population surveys, such as random-digit dialing telephone surveys, actually sample households. Listings for special population surveys may also contain multiple units. For example, welfare rolls may actually be listings of cases that include all members of affected families. The primary issues with cluster listings are the selection of the unit of the study population from each listing and adjusting the probability of selection based on the number of units in the listing. In most cases, information is sought only from one individual per cluster listing. If the selection of the individual is done randomly, a correction may be needed to compensate for the probability of selection if the clusters are unequal in size. To return to the telephone survey example, a member of a household with four adults is half as likely to be selected out of that household as is a member of a household with two adults. If the selection is not done randomly, a systematic bias may be introduced.

What Is the Sampling Variability That Can Be Tolerated for the Study?

The sampling variability affects the precision of the estimates for descriptive studies and the power to detect effects for explanatory studies. *Precision* refers to the size of the confidence interval that is drawn around the sample mean or proportion estimates. The level of precision required relates directly to the purpose for which the study results will be used. A confidence interval of ±5% may be completely satisfactory for a study to assess the need for a particular type of service within a community but entirely too large for setting a mayoral candidate to decide on spending funds on more advertising in the midst of a campaign in the same locality.

Precision requirements are used in the calculations of efficient sample sizes. The objective of the researcher is to produce a specified interval within which the true value for the study population is likely to fall. Sample size is a principal means by which the researcher can achieve this objective. But the efficiency of the sampling design can have considerable impact on the amount of sampling error and the estimate of desired sample size.

For explanatory studies, the sample variability that can be tolerated is based on the desire to be able to detect effects or relationship if they occur. A power analysis is conducted to assess the needs for a particular study (see Lipsey & Hurley, Chapter 2, this volume, for more detail). The power analysis requires that the researchers have an estimate of the size of the effect that they expect the program or intervention to produce and the degree of confidence that they would like to be able to have to detect the effects. Effect sizes are stated in standard deviation units, for example an effect size of .25 means that the effect is expected to be one quarter of a standard deviation unit. In practice, it has become common to specify an 80% chance of detecting the effect. Power analysis software is available from several sources to determine what sample size would be required to detect an effect of a specified size.

What Types of Sampling Designs Will Be Used?

The five probability sampling designs are simple random sampling, systematic sampling, stratified sampling, cluster sampling, and multistage sampling. However, the multistage sampling design, which is also referred to as complex sample design, has many variations and is best considered a category of designs rather than a particular design. The choice of a design will depend on several factors, including availability of an adequate sampling frame, the cost of travel for data collection, and the availability of prior information about target population. However, the choices do not end with the selection of a design.

Choices branch off independently for each design. If stratified sampling is chosen, how many strata should be used? If cluster sampling is chosen, how should the clusters be defined? For multistage samples, how many sampling stages should be used? Table 3.3 presents the definitions of all five types of sampling techniques, as well as their requirements and benefits. For illustrative purposes, a two-stage sample is described in the table.

Will the Probability of Selection Be Equal or Unequal?

Choices about the probability of selection will also affect sampling bias. For simple random sampling, the probability of selecting any individual unit is equal to the sampling fraction or the proportion of the population selected for the sample (n/N). The probability of selecting any unit is equal to the probability of selecting any other unit. For stratified sample designs, the probability of selection for any unit is the sampling fraction for the stratum in which the unit is placed. Probabilities using a stratified design can be either equal or unequal as can multistage sample designs. It is also common to use stratified cluster sampling, in which the

Table 3.3 Probability Sampling Techniques

	Simple Random	*Systematic*	*Stratified*	*Cluster*	*Multistage (two stage)*
Definition	Equal probability of selection sample where n units are drawn from population list	Equal probability of selection sample where a random start that is less than or equal to the sampling interval is chosen, and every unit that falls at the start and at the interval from the start is selected	Either equal or unequal probability of selection sample where population is divided into strata (or groups) and a simple random sample of each stratum is selected	Clusters that contain members of the study population are selected by a simple random sample, and all members of the selected clusters are included in the study	First, clusters of study population members are sampled, then study population members are selected from each of the sampled clusters, both by random sampling
Requirements	List of study population	List of physical representation of study population	List of study population divided into strata	List of clusters in which all members of study population are contained in one and only one cluster	List of primary sampling units
					List of members for selected primary sampling units
	Count of study population (N)	Approximate count of study population (N)	Count of study population for each stratum	Count of clusters (C)	Count of primary sampling units
	Sample size (n)	Sample size (n)	Sample size for each stratum	Approximate size of clusters (N_c)	Number of primary sampling units to be selected

(Continued)

Table 3.3 (Continued)

	Simple Random	*Systematic*	*Stratified*	*Cluster*	*Multistage (two stage)*
Requirements	Random selection of individuals or units	Sampling interval ($I = N/n$ rounded down to integer)		Number of clusters to be sampled (c)	Number of members to be selected from primary sampling units
		Random start R such that $R \leq I$		Random selection mechanism	Random selection mechanism for primary sampling units and members
Benefits	Easy to administer	Easy to administer in field or with physical objects, such as files or invoices, when list unavailable	Reduces standard error	List of study population unnecessary	Same benefits as for cluster, plus may reduce standard error
	No weighting required		Disproportionate stratifications can be used to increase sample size of subpopulations		
	Standard error calculation is automatic in most software			Limits costs associated with travel or approvals from all clusters	Most complex but most efficient and flexible
				Clusters can be stratified for efficiency	

clusters, such as schools or clinics, are placed into strata and then sampled, either proportionately or disproportionately. If separate estimates or explanatory analyses are needed for certain subpopulations or some strata are known to have much higher variability for important variables, a disproportionate sampling strategy should be considered, which would result in unequal probability of selection.

How Many Units Will Be Selected for the Sample?

Determining the sample size is where many discussions of sampling begin, but as this framework points out, the research team needs a great deal of information before the sample size is determined for the study. In descriptive studies, researchers must answer this question: What sample size will produce estimates that are precise enough to meet the studies purpose, given the sampling sample design? Precision, from the sampling perspective, is a function size of the confidence interval, which is influenced primarily by three variables: the standard deviation of the variable of interest, the sample size, and level of confidence required (represented by the t statistic). In cases when the population is relatively small, it is influenced by sampling fraction as a result of the FPC, also. The researcher directly controls only the same sample size; to produce an estimate from the sample that is precise enough for the study objectives, the researcher can adjust the sample size. But increasing the sample size means increasing the cost of data collection. Trade-offs between precision and cost are inherent at this juncture.

For a descriptive study, assuming a simple random sample, the sample size calculation is done using the following formulas:

$$n' = \frac{s^2}{(\mathrm{te}/t)^2},$$

$$n = \frac{n'}{1+f},$$

where n' is the sample size computed in the first step, s is the estimate of the standard deviation, te is the tolerable error, t is the t value for the desired probability level, n is the sample size using the FPC error factor, and f is the sampling fraction.

The most difficult piece of information to obtain for these formulas, considering it is used prior to conducting the actual data collection, is the estimate of the standard deviation. A number of options are available, including prior studies, small pilot studies, and estimates using the range.

Although the sample size is the principal means for influencing the precision of the estimate once the design has been chosen, an iterative process can be used to examine the impact on efficient sample size if an alternative design were used. Stratification or the selection of more primary sampling units in multistage sampling can improve the precision of a sample without increasing the number of units in the sample. Of course, these adjustments may increase costs also, but perhaps less than increasing the sample size would.

In addition, other sample size considerations should be brought to bear at this point. For example, will the number of members of subpopulations that the sample can be expected to yield be sufficient precision for the subpopulation estimates? Determining the sample size is generally an iterative process. The researcher must consider and analyze numerous factors that may alter earlier choices, for example, the expected response rate or the percentage of ineligibles that may be included in the sampling frame. It is important for the researcher to review the proposed alternatives carefully in terms of total error, changes in the study population definition from using different sampling frames, and feasibility.

Postsampling Choices

How Can the Impact of Nonresponse Be Evaluated?

Nonresponse for sampling purposes means the number of sampled individuals who did not provide useable responses, calculated by subtracting the response rate from 1. Nonresponse can occur when a respondent refuses to participate in the survey or when a respondent cannot be contacted. If the nonresponding portion of the population is reduced, the nonsampling bias is reduced (Kalton, 1983). Also, nonresponse can occur when an individual who is participating in a survey cannot or will not provide an answer to a specific question. Fowler (1993; see also Chapter 12, this volume) and Dillman (1999) discuss several ways of reducing nonresponse. It is often necessary for the researcher to evaluate the impact of nonresponse by conducting special studies of the nonrespondents, comparing the sample characteristics with known population parameters, or examining the sensitivity of the sample estimates to weighting schemes that may provide greater weight to responses from individuals who are considered to have characteristics more like the nonrespondents (Henry, 1990; see also Braverman, 1996; Couper & Groves, 1996; Krosnick, Narayan, & Smith, 1996).

Is It Necessary to Weight the Sample Data?

Weighting is usually required to compensate for sampling bias when unequal probabilities result from the researcher's sampling choices. Unequal probabilities of selection can occur inadvertently in the sampling process, as with duplicates on the sampling frame or cluster listings. They can also arise from deliberate choices, such as disproportionate stratification. Generally, weights should be applied in all these cases. In addition, when the response rates are higher for some subgroups within the sample than others, many survey organizations increase the weights for the groups with lower response rates such that the proportions of each subgroup in the sample estimates equals the proportional representation of that subgroup in the study population. (For a discussion of the calculation of appropriate weights, see Henry, 1990.)

What Are the Standard Errors and Related Confidence Intervals for the Study Estimates?

The precision of the estimates and the power of hypothesis tests are determined by the standard errors. It is important to recognize that the sampling error formulas are different for the different sampling techniques. Formulas for calculating the standard error of the mean for simple random samples were presented earlier in the chapter. Other sampling techniques require modifications to the formula and can be found in Henry (1990), Kalton (1983), Sudman (1976), and Kish (1965).

However, some general guidance can be provided. Stratification lowers the sampling error, all other things held constant, when compared with simple random samples. Sampling error can be further lowered when larger sampling fractions are allocated to strata that have the highest standard deviations. Cluster sampling inflates the standard error of the estimates relative to simple random sampling. This occurs because the number of independent choices is the number of clusters in cluster sampling, not the number of units finally selected. The effect is reduced when clusters are internally heterogeneous on the important study variables (large standard deviations within the clusters) or cluster means do not vary. The standard error for a cluster sample can often be reduced by stratification of the clusters before selection. This means that the clusters must be placed into strata before selection, and the variables used to define the strata must be available for all clusters. This type of sampling strategy can result in standard errors very close to those associated with simple random samples when the sample is properly designed.

Summary

The challenge of sampling lies in making trade-offs to reduce total error while keeping study goals and resources in mind. The researcher must act to make choices throughout the sampling process to reduce error, but reducing the error associated with one choice can increase errors from other sources. Faced with this complex, multidimensional challenge, the researcher must concentrate on reducing total error. Error can arise systematically from bias or can occur due to random fluctuation inherent in sampling. Error cannot be eliminated entirely. Reducing error is the practical objective, and this can be achieved through careful design.

Discussion Questions

1. What are the main differences in probability and nonprobability samples?

2. For probability samples, what are the main alternatives to simple random samples? Name one circumstance in which each one might become a preferred option for the sampling design.

3. What is a confidence interval? What does it measure?

4. How would you go about determining the variable of greatest interest for an evaluation of adolescent mental health programs?

5. What sample plan would you develop for describing the uninsured population of your state?

6. In what circumstances might you choose a convenience sample over a probability sample?

7. What are the major factors that contribute to standard error of the mean? Which of the factors can be most easily controlled by researchers?

Exercises

1. Find an evaluation report for which survey data have been collected from a sample of the population. Answer the following questions:
   a. What is the target population?
   b. What is the study population?
   c. What target population members are omitted from the study population?
   d. Was a listing used as the sampling frame? Other than the omissions, are their issues with the sampling frame that might bias the findings?
   e. What sampling design was used for the evaluation?

2. Find a survey conducted by a federal agency and made available on the Internet. Look at the technical description of the sample. What was the sampling design that was used? What was the sample size? What factors affected the sample size? Did the survey researchers oversample to compensate for nonresponse? Did the researchers oversample a subpopulation or a strata of the population for other reasons? If so, what were the reasons?

3. Draw up two approaches for sampling teachers in your home state. The target population is full-time classroom teachers in public schools in the state. Assume that you are going to survey the teachers using a mailed survey. One approach should use a sampling frame. The other approach should use a sample design that does not require a sampling frame. Compare the nonsampling bias, sampling bias, and sampling variability of the two approaches. To compare the sampling variability, assume that the variable of interest is the percentage of teachers planning to leave teaching within the next 5 years. Are there differences in costs or in feasibility that might lead to choosing one of the approaches over the other?

4. Look carefully at the results and description of a national, statewide, or citywide poll based on a probability sample (surveys of readers should be excluded) that you see reported in the media. If reported in print media, you may find more detail about the survey online. What is the "margin of error" or confidence interval around the percentages reported? What other sources of error seem to have occurred, if any? What was the response rate? What would you like to know about the poll that is not mentioned in the descriptions?

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CHAPTER 4

Planning Ethically Responsible Research

Joan E. Sieber

Applied researchers examine and experiment with issues that directly affect people's lives—issues such as education, health, family life, work, finances, and access to government benefits, and must respect the interests of subjects and their communities. There is a practical, as well as a moral, point to this. Unless all parties concerned are recognized and respected, it is likely that research questions may be inappropriately framed, participants may be uncooperative, and findings may have limited usefulness. Consequently, investigators who are thoughtless regarding ethics are likely to harm themselves and their research as well as those that they study.

This chapter focuses on research planning and ethical problem solving, not on details of federal or state law governing human research or on preparing research protocols for institutional review boards (IRBs). Readers may wish to refer to www.hhs.gov/ohrp for the current federal regulations governing human research. Details on approaches to compliance with various aspects of federal law, and how to write a research protocol in compliance with IRB and federal requirements, are presented on the Web sites of many IRBs and in *Planning Ethically Responsible Research* (Sieber, 1992) in the Applied Social Research Methods Series published by Sage Publications. The reader's own IRB can provide information on its specific requirements.

An Introduction to Planning

The ethics of social and behavioral research is about creating a mutually respectful, win-win relationship in which important and useful knowledge is sought, participants are pleased to respond candidly, valid results are obtained, and the community considers the conclusions constructive. This requires more than goodwill or adherence to laws governing research. It requires investigation into the perspectives and cultures of the participants and their community early in the process of research design, so that their needs and interests are understood and served.

In contrast, a researcher who does not investigate the perspectives of the participants and plan accordingly may leave the research setting in pandemonium. The ensuing turmoil may harm all the individuals and institutions involved, as illustrated by the following example, adapted from an actual study.

A researcher sought to gather information that would help local schools meet the needs of children of migrant farm workers. He called on families at their homes to ask them, in his halting Spanish, to sign a consent form and to respond to his interview questions. Most of the families seemed not to be at home, and none acknowledged having children. Many farm workers are undocumented, and they assumed that the researcher was connected with the U.S. Immigration and Naturalization Service (INS). News of his arrival spread quickly, and families responded accordingly—by fleeing the scene.

> A more skilled scientist would have understood that community-based research cannot be planned or conducted unilaterally. He or she would have enlisted the help of community leaders in formulating the research procedures. Steps would have been taken to understand and allay respondents' fears. Perhaps, the researcher would have obtained a Certificate of Confidentiality[1] to prevent subpoena of the data by the INS or other authorities. Members of the community would have been employed and trained to conduct the interviews. Effective communication and informed consent would have occurred informally, through a correctly informed community "grapevine." The researcher would have developed the formal consent statement to language appropriate to this community, which is not fluent in English, with the help of community leaders, and would have communicated its contents to the community at an enjoyable meeting, perhaps a picnic provided by the researcher. The researcher would have learned what respondents would like to receive in return for their participation and likely would have arranged a mutually rewarding relationship so that he or she would have been welcome to create an ongoing research and development program for the community.

Such enlightened, ethical research practices make for successful science, yet many researchers have been trained to focus narrowly on their research agendas and to ignore the perceptions and expectations of their participants and of society at large. When one is narrowly focused on completing a research project, it is easy to overlook some of the interests and perspectives of the subjects and of society at large. The result would likely be a failed research program as well as a community that learned to disrespect researchers.

Ethical research practice entails skillful planning and effective communication, reduction of risk, and creation of benefits, as these issues pertain to the stakeholders in the research. *Stakeholders* include any persons who have interests in the research. Especially in field research, it is important that researchers try to identify all stakeholders early in the planning process. These might include the potential participants and their families, guardians, employers, institutions, and community leaders; the researchers and their institutions and funders; and, depending on the nature and scope of the research, social advocates and the mass media.

Stakeholders are not just those whom the researcher wishes to consult. They are also those who expect the researcher to consult them. For example, a researcher investigating the effects on learning of extensive parental involvement in the classroom readily perceives that parents, teachers, and school administrators are stakeholders who should be involved. But what of the teachers' union? What of the parents who are known to be skeptical of any new approaches to education? If the interest of potential stakeholders are not identified and considered at the outset, the chances for successful completion of the research may be diminished. Identifying all significant stakeholders and their interests in the research may require the researcher to conduct considerable ethnographic inquiry, networking, focus groups and consultation, and to do so with cultural sensitivity.

The researcher also should consult the federal regulations that govern human research (http://ohsr.od.nih.gov/guidelines/45cfr46.html).[2] These regulations govern each institution's Human Research Protection Program (HRPP), which then is empowered to develop the policies and procedures by which its human research ethics committee or IRB and other elements operate. The mandate of HRPPs and IRBs is to oversee *human research,* which is defined as the systematic gathering of scientific data with publication in mind. Hence, it typically does not include classroom demonstrations, research activities of students as a course assignment, administrative data gathering, or program evaluation, although the boundaries between research and these other activities are unclear. It is also unclear when *going out and talking to people* is research; see Howard (2006) on the debate about oral history and IRB review. Each institution's HRPP decides what requires review there.

To save time and prevent frustration, the early planning and integration of ethical concerns with methodological and procedural planning should be conducted in consultation with an appropriate representative of the HRPP. Then, the development of the IRB application (protocol) is largely a formality. In some cases, an IRB may propose inappropriate procedures (such as the requirement of signed consent when this would be impracticable or would jeopardize participants, or the requirement of consent language that is inappropriate), and the researcher would need to defend the appropriate procedures by presenting the literature that documents what is appropriate under the specific conditions or by conducting empirical research that illustrates the problem and a solution. For example, your IRB may want you to use a consent statement that contains legal jargon that they think protects the institution, but you may realize that this will be meaningless to your subjects and hence foolish, counterproductive, unethical, and in violation of federal regulations. You might use the cognitive interviewing methods described by Willis (2006, available at http://caliber.ucpress.net/loi/jer, March issue) to show what subjects do and do not understand. Most IRBs provide guidelines or templates for developing one's protocol; however, these should be adapted to your particular research requirements. Before leaving the topic of protocols, however, it is important to note the protections that they offer to the researcher.

The protocol has legal status as a "control document." It is the paper trail showing that the research is acceptable to a legally constituted board of reviewers. Should anyone raise questions about the project, the approved protocol shows that the project is deemed to be of sufficient value to justify any risks involved. Hence, the protocol must reflect what is actually done in the research. Once the IRB has approved a protocol for a particular project, the investigator must follow that procedure, have any desired changes approved by the IRB, or risk a disaster such as the following:

> Dr. Knowall interviewed schoolchildren about their understanding of right and wrong. A parent who gave permission for his child to participate in the research later felt that the project sought to change his child's religious beliefs. He called the newspaper, the ACLU, the mayor, the school board, and the governor to complain that Dr. Knowall's research violated the separation of church and state. The university, required to respond, proffered the approved protocol, which should have been powerful evidence in any legal proceeding that the project was socially and legally acceptable—except for one thing: The researcher had slipped in a few questions about religion after receiving IRB approval. The researcher found himself in serious trouble, and without enthusiastic backing from his institution.

HRPPs and IRBs: Origin and Evolution

The history of U.S. policies and regulations of human research has been discussed extensively elsewhere (e.g., Citro, Ilgen, & Marrett, 2003; Katz, 1972; National Bioethics Advisory Commission, 2001). Very briefly, after the Western world witnessed crimes against humanity committed in the name of science by Nazi scientists, the principle of voluntary informed consent was set forth. However, the United States was insensitive to its own lack of adherence to this principle until it was learned that a study begun in 1932 to discover the course of syphilis from inception to death continued the study of poor black men in Tuskegee, Alabama, long after penicillin was identified as a cure for syphilis in 1943 (Jones, 1981). When the study was discontinued in 1973, the National Commission for the Protection of Human Subjects in Biomedical and Behavioral Research was established to examine human research practices. One product of the National Commission is the Belmont Report, which enunciates the principles that should govern human research. Summarizing very briefly, three principles were set forth to govern human research:

- *Beneficence:* maximizing good outcomes for science, humanity and the individual research participant, while avoiding or minimizing unnecessary risk, harm or wrong.
- *Respect for subjects:* protecting the autonomy of (autonomous) persons, and treating the nonautonomous with respect and special protections.
- *Justice:* ensuring reasonable, nonexploitative, and carefully considered procedures and their fair administration.

Operationalizing these principles means employing valid research designs and procedures, having researchers capable of carrying out those procedures validly, assessing risks and benefits and adjusting procedures to minimize risk and maximize benefit, selecting the appropriate kind and number of subjects, obtaining voluntary informed consent, and compensating subjects for injury or at least informing them whether compensation will be available.

The interpretation of regulations needs to evolve as necessitated by new research challenges that need to be met. The IRB (a committee) is governed by the HRPP (the administrative policies and program that specify the role of the IRB and other elements of the system such as education of investigators, students, and IRB members). The HRPP should take advantage of the flexibility permitted by the federal regulations to modify the role of the IRB as circumstances require (Rubin & Sieber, 2006). For example, the HRPP may mandate that the IRB *not* review minimal risk research, but that these be reviewed outside the IRB, perhaps by IRB members who expedite the review of minimal risk or exempt protocols within their department or area of expertise. Researchers who observe the need for more ethical interpretations of regulations might work with their IRB to empirically test the efficacy of alternative procedures, as suggested by Levine (2006), for example. Thus, empirical research to determine what works to satisfy ethical principles can play an important role in ensuring that regulations are interpreted in ways that are sensible and ethical.

We turn now to three major aspects of ethical problem solving: consent (including debriefing and deception), privacy/confidentiality, and risk/benefit, and finally to the special needs of vulnerable populations, including children.

Voluntary Informed Consent

The informed consent statement should explain the research that is to be undertaken and should fulfill legal requirements (see www.research.umn.edu/consent, www.socialpsychology.org/consent.htm, or some of the other outstanding online tutorials for tips on developing an adequate informed consent). The consent statement should be simple and friendly in tone and should translate a scientific proposal into a language that potential participants understand and at a reading level that matches their ability, omitting details that are unimportant to the subjects, but including details that a reasonable person would want to know. The consent statement should be free of jargon and legalese. The researcher needs to learn what information would be important to the potential subjects and how to express that in ways that they understand. To do so, one needs to identify *surrogate subjects,* that is, persons who are representative of the subject population, who are willing to examine the research procedure and comment on what they would want to know if they were considering participation and to evaluate other aspects of the research procedure (see Fost, 1975).

Voluntary informed consent is *not* a consent form. It goes beyond the statement that is prepared and administered in the so-called consent procedure. It should begin as a conversation—an ongoing, two-way communication process between research participants and the investigator. After this discussion, the investigator may introduce the written consent form and explain how it covers the topics described. The consent form is a formal agreement about the conditions of the research participation, but it is not necessarily the final communication about the conditions of the research. Often, questions and concerns occur to the participants only after the research is well under way. Sometimes, it is only then that meaningful communication and informed consent can occur. The researcher must be open to continuing two-way communication throughout the study and afterward as questions occur to the participants.

Voluntary means without threat or undue inducement. When consent statements are presented as a plea for help or when people are rushed into decisions, they may agree to participate even though they would rather not. They are then likely to show up late, fail to appear, or fail to give the research their full attention. To avoid this, the researcher should urge each subject to make the decision that best serves his or her own interests. Also, the researcher should not tie participation to benefits that the subjects could not otherwise afford such as health services, especially if participants are indigent or otherwise vulnerable to coercion. And, participants need to know that they can quit at any time without repercussion.

Informed means knowing what a reasonable person in the same situation would want to know before giving consent, including who the researcher is and why the study is being done. Mostly, people want to know what they are likely to experience, including the length of time required, and how many sessions are involved. If the procedure is unusual or complicated, a videotape of the procedure may be more informative than a verbal description. People need to be informed in language that they understand. Two methods of learning the terminology that subjects would use and understand are described by Willis (2006), the think aloud method and the verbal probing method. In the *think aloud* method (surrogate) subjects are asked to externalize their thought processes ("Tell me what you are thinking.") as they respond to materials. For example, as the surrogate subject reads each element of the informed consent, he is to say out loud what it makes him think. In the *verbal probing* method, the subject is asked to explain each part and probes such as the following are used: "Tell me more about that . . ." "What does . . . (particular term) mean to you? When someone tells you that, what would you want to know?"

Although the competence to understand and make decisions about research participation is conceptually distinct from voluntariness, these qualities become blurred in the case of some populations. Children, adults with intellectual disabilities, the poorly educated, and prisoners, for instance, may not understand their right to refuse to participate in research when asked by someone of apparent authority. They may also fail to grasp details relevant to their decision. The researcher may resolve this problem by injecting probes (as in cognitive interviewing) into the informed consent process for each subject, or by appointing an advocate for the research subject, in addition to obtaining the subject's assent. For example, children cannot legally consent to participate in research, but they can "assent" to participate, and must be given veto power over parents or other adults who give permission for them to participate.

Consent means explicit agreement to participate. Competence to consent or assent and voluntariness are affected by the way the decision is presented (Melton & Stanley, 1991). An individual's understanding of the consent statement and acceptance of his of her status as an autonomous decision maker will be most powerfully influenced not by what the individual is told, but by how he or she is engaged in the communication. There are many aspects of the investigator's speech and behavior that communicate information to subjects. Body language, friendliness, a respectful attitude, and genuine empathy for the role of the subject are among the factors that may speak louder than words. To illustrate, imagine a potential subject who is waiting to participate in a study:

> *Scenario 1:* The scientist arrives late, wearing a rumpled lab coat, and props himself in the doorway. He ascertains that the subject is indeed the person whose name is on his list. He reads the consent information without looking at the subject. The subject tries to discuss the information with the researcher, who seems not to hear. He reads off the possible risk. The nonverbal communication that has occurred is powerful. The subject feels resentful and suppresses an urge to storm out. What has been communicated most clearly is that the investigator does not care about the subject. The subject is sophisticated and recognizes that the researcher is immature, preoccupied, and lacking in social skills, yet he feels devalued. He silently succumbs to the pressures of this unequal status relationship to do "the right thing"; he signs the consent form amid a rush of unpleasant emotions.

> *Scenario 2:* The subject enters the anteroom and meets a researcher who is well-groomed, stands straight and relaxed, and invites the subject to sit down with him. The researcher's eye contact,[3] easy and relaxed approach, warm but professional manner, voice, breathing, and a host of other cues convey that he is comfortable communicating with the subject. He is friendly and direct as he describes the study. Through eye contact, he ascertains that the subject understands what he has said. He invites questions and responds thoughtfully to comments, questions, and concerns. When the subject raises scientific questions about the study (no matter how naive), the scientist welcomes the subject's interest in the project and enters into a brief discussion, treating the subject as a respected peer. Finally, the researcher indicates that there is a formal consent form to be signed and shows the subject that the consent form covers the issues they have discussed. He mentions that it is important that people not feel pressured to participate, but rather should participate only if they really want to. The subject signs the form and receives a copy of the form to keep.

Though the consent forms in these two cases may have been identical, only the second scenario exemplifies adequate, respectful informed consent. The second researcher was respectful and responsive; he facilitated adequate decision making. Congruence, rapport, and trust were essential ingredients of his success.

Congruence of Verbal and Body Language. The researcher in Scenario 1 was incongruent; his words said one thing, but his actions said the opposite. The congruent researcher in Scenario 2 used vocabulary that the research participant easily understood, spoke in gentle, direct tones, breathed deeply and calmly, and stood or sat straight and relaxed. To communicate congruently, one's mind must be relatively clear of distracting thoughts.

Rapport. The researcher's friendly greeting, openness, positive body language, and willingness to hear what each subject has to say or to ask about the study are crucial to establishing rapport. When consent must be administered to many participants, the process can turn into a routine delivered without a feeling of commitment; this should be avoided.

Trust. If participants believe that the investigator may not understand or care about them, there will not be the sense of partnership needed to carry out the study satisfactorily. The issue of trust is particularly important when the investigator has higher status than the subject or is from a different ethnic group. It is useful for the researcher to ask members of the subject population, perhaps in a focus group, to examine the research procedures to make sure that they are respectful, acceptable, and understandable to the target population.

There are many ways to build respect, rapport, and trust, as the following examples illustrate:

> *Example 1:* A Caucasian anthropologist sought to interview families in San Francisco's Chinatown to determine what kinds of foods they eat, how their eating habits have changed since they immigrated here, and what incidence of cancer has been experienced in their families. She employed several Chinese American women to learn whether her interview questions were appropriate and to translate them into Mandarin and Cantonese. The research assistants worked on the basis of their personal knowledge of the language and culture of Chinatown, then tested their procedures on pilot subjects. There was confusion among pilot subjects about the names of some Chinese vegetables; consequently, the researchers devised pictures of those vegetables so that subjects could confirm which ones they meant. The Chinese American research assistants rewrote the questions and the consent statement until they were appropriate for the population that was to be interviewed, and then conducted the interviews. Their appearance, language, and cultural background engendered a level of trust, mutual respect, and communication that the researcher herself could not have created.

> *Example 2:* A researcher studying safe-sex knowledge and behavior of gay men identified legitimate leaders in the local gay community—gay physicians and other leaders concerned about the health and welfare of their community. He worked with them to develop a useful survey, an acceptable sampling and recruitment procedure, and ways to let the community know what safeguards to confidentiality were built into the study and what benefits from the study would flow back to the participating community.

> *Example 3:* A researcher studying infant nutrition offered to share data with the host community for its own policy-making purposes (e.g., Pelto, 1988). The community leaders were invited to request that any items of interest to them be added to survey, and they were then assisted with analyses and

> interpretations of the data. The result was a collaborative effort to achieve a shared goal—improved health and nutrition in that community.

There are many ways to enhance communication, rapport, respect, and trust, and to increase the benefits to subjects of a research project, depending on the setting and circumstances. When planning research, especially in a field setting, it is useful for researchers to conduct focus groups drawn from the target population, to consult with community gatekeepers, or to consult with pilot subjects to learn their reactions to the research procedures and how to make the research most beneficial and acceptable to them (see Stewart, Shamdasani, and Rook, Chapter 18, this volume, for discussion of uses of focus groups). For example, learn what terms to use when obtaining demographic information such as ethnicity and gender orientation. In some cases, this consultation should extend to other stakeholders and community representatives. The rewards to the researcher for this effort include greater ease of recruiting cooperative participants, a research design that will work, and a community that evinces goodwill.

In summary, it is important for the researcher to determine what the concerns of the subject population actually are. Pilot subjects from the research population, as well as other stakeholders, should have the procedure explained to them and should be asked to try to imagine what concerns people would have about participating in the study. Often some of these concerns turn out to be very different from those that the researcher would imagine, and they are likely to affect the outcome of the research if they are not resolved, as illustrated by the following case of misinformed consent:

> A PhD student interviewed elderly persons living in a publicly supported geriatric center on their perceptions of the center. At the time of the research, city budget cuts were occurring; rumors were rampant that eligibility criteria would change and many current residents would be evicted. Mrs. B, an amputee, was fearful that she would be moved if she were perceived as incompetent. After she signed the informed consent form, the researcher asked her several questions:
>
> *Researcher:* "Can you recite the alphabet?"
>
> *Mrs. B:* "Backwards or forwards?" (Seeking to demonstrate her intellectual competence.)
>
> *Researcher:* "How do you like the service here?"
>
> *Mrs. B:* "Oh it's great!" (She constantly complained to her family about the poor service.)
>
> *Researcher:* "How do you like the food here?"
>
> *Mrs. B:* "It's delicious."
>
> Mrs. B's anxiety was rising; midway through the questioning she asked, "Did I pass the test?"

Researcher: "What test?"

Mrs. B: "The one for whether I can stay in the hospital."

Researcher: "I'm not working for the hospital."

Mrs. B spun her chair around and wheeled herself away. (Fisher & Rosendahl, 1990, pp. 47–48)

Should consent be obtained in writing and signed? Signed (or documented) consent proves that consent was obtained, and probably does more to protect the institution than to protect the subjects. Most IRBs require *signed* consent for most kinds of research, except in the following situations (as specified in the federal regulations): (a) when signed consent is offensive to subjects or inconvenient, and subjects can easily refuse (e.g., by hanging up on a phone interviewer or by throwing out a mailed survey), (b) when signed consent would jeopardize the well-being of subjects, as in research on illegal behavior, for example, in which it would not be in subjects' best interest for the researcher to have a record of their identities, and (c) for minimal risk anonymous surveys.

However, just because *signed* consent is not required does not mean that consent is not necessary. Consent is necessary, and a copy of the consent statement may be given to the subject; only the *signed* agreement to participate is waived in such a situation. Alternatively, if the presence of the written consent statement might jeopardize the safety of the subject, as in interviews of victims of domestic violence, a written document should not be used.

Debriefing

The benefits of research include its educational or therapeutic value for participants. Debriefing provides an opportunity for the researcher to consolidate the value of the research to subjects through conversation and handouts. The researcher can provide rich educational material immediately, based on the literature that forms the foundation of the research. Debriefing also offers an opportunity for the researcher to learn about subjects' perceptions of the research: Why did they respond as they did—especially those whose responses were unusual? How do their opinions about the usefulness of the findings comport with those of the researcher? Typically, the interpretation and application of findings are strengthened by researchers' thoughtful discussions with participants. Many a perceptive researcher has learned more from the debriefing process than the data alone could ever reveal.

If the researcher or IRB have any concerns about whether subjects experience misgivings about the research, it is useful to know if, in fact, misgivings or upset do occur, and whether it is an idiosyncratic concern of just one or a few or a concern of a substantial proportion of the subjects. It is a mistake to confuse the misgivings of one or a few with the notion that the research is risky. Newman, Risch, and Kassam-Adams (2006) summarize research on trauma survivors to show that while most find it quite beneficial to be interviewed by an experienced professional about their trauma, a small percentage of subjects may find it a negative experience. A small percentage of persons find almost any new experience negative and may represent the small percentage of any population who are in poor mental health for reasons often unknown to the investigator or to anyone else. The Reactions of Research Participants Questionnaire (RRPQ, available at www.personal.utulsa .edu/~elana-newman) is a useful measure of the baseline experience of research participants and helps investigators and IRBs understand whether, on balance, most persons find the research experience positive. It can also indicate whether the research procedure is likely to adversely affect some portion of participants and, if so, what kinds of warning should appear in the informed consent or what kind of screening of potential subjects should occur (Newman, Willard, Sinclair, & Kaloupek, 2001). Persons who would be rendered highly fearful, anxious, paranoid, or angry by the research procedure not only may be harmed by it but also most likely would not yield interpretable data.

Deception

In deception research, the researcher studies reactions of subjects who are purposely led to have false beliefs or assumptions. This is generally unacceptable in applied research, but consent to concealment may be defensible when it is the only viable way (a) to achieve stimulus control or random assignment, (b) to study responses to low-frequency events (e.g., fights, fainting), (c) to obtain valid data without serious risk to subjects, or (d) to obtain information that would otherwise be unobtainable because of subjects' defensiveness, embarrassment, or fear of reprisal. An indefensible rationale for deception is to trick people into research participation that they would find unacceptable if they correctly understood it. If it is to be acceptable at all, deception research should not involve people in ways that members of the subject population would find unacceptable.

Deception studies that involve people in doing socially acceptable things, and pose no threat to persons' self-esteem are little different from many other everyday activities. The few deception studies that have been regarded as questionable or harmful, such as Milgram's (1974) study of obedience in which persons thought that they were actually delivering high voltage electric shock to others, are ones in which persons were strongly induced to commit acts that are harmful or wrong, or were surreptitiously observed engaging in extremely private acts (e.g., Humphreys, 1970).

There are three kinds of deception that involve consent and respect subjects' right of self-determination:

1. *Informed consent to participate in one of various specified conditions:* The various conditions to which subjects might be assigned are clearly described to subjects ahead of time. For example, most studies employing placebos use this consent approach. Subjects know that they cannot be told the *particular* conditions to which they will be assigned, as this knowledge would affect their responses. Complete debriefing is given afterward. Subjects who do not wish to participate under these conditions may decline to participate.

2. *Consent to deception:* Subjects are told that there may be misleading aspects of the study that will not be explained to them until after they have participated. A full debriefing is given as promised.

3. *Consent to waive the right to be informed:* Subjects waive the right to be informed and are not explicitly forewarned of the possibility of deception. They receive a full debriefing afterward.

Privacy, Confidentiality, and Anonymity

Privacy is about people. Confidentiality is about data. Anonymity means no identifiers.

Privacy refers to persons' interest in controlling the access of others to themselves. It is not necessarily about their wanting to be left alone. Privacy concerns tend to be highly idiosyncratic; experiences that some persons would welcome, others would want to avoid. Thus, informed consent should give subjects an adequate understanding of what they will experience, so that they can judge for themselves whether they want to "go there" or would rather be left alone. *Confidentiality* is an extension of the concept of privacy; it concerns data about the person and an agreement as to how the data are to be handled in keeping with the subjects' interest in controlling the access of others to information about themselves. The confidentiality agreement is typically handled in the informed consent, and it states what may be done with the information that the subject conveys to the researcher. The terms of the confidentiality agreement need to be tailored to the particular situation. *Anonymity* means that the names and other unique identifiers of the subjects (such as their social security number or address) are never attached to the data or known to the researcher, hence technically, the data would not meet the definition of *human* subjects' data.

This section introduces the reader to some basic concepts of privacy and confidentiality in human research.

The most comprehensive, sophisticated, and up-to-date source on privacy and confidentiality is the American Statistical Association's (2004) Web site www.amstat.org/comm/cmtepc.

Privacy

What one person considers private, another may not. We certainly know when our own privacy has been invaded, but the privacy interests of another may differ from ours. Thus, while researchers should be sensitive to the topics that might be regarded as private by those they plan to study, to judge what another considers private based on one's own sense of privacy is to set a *capricious* and *egocentric/ethnocentric* standard for judging privacy. One must let subjects and members of their community judge for themselves what is appropriate to ask or do in research and how subjects are to be given an opportunity to control the access of the researcher to themselves.

What is private depends greatly on context and on what we consider to be the other person's business. The kinds of things we consider appropriate to disclose to our physician differ from what we disclose to our banker, accountant, neighbor, and so on. If a highly professional interviewer establishes that a socially important piece of research hinges on the candid participation of a random sample of the population, many would disclose details that they might never tell others. However, a researcher who took a less professional approach, or sought to do trivial research, would receive a different reception.

Respecting Privacy. How can investigators protect subjects from the pain of having their privacy violated? How can investigators guard the integrity of their research against the lies and subterfuges that subjects will employ to hide some private truths or to guard against intrusions? Promises of confidentiality and the gathering of anonymous data may solve some of these problems, but respecting privacy is more complex than that. An understanding of the privacy concerns of potential subjects enables the researcher to communicate an awareness of, and respect for, those concerns, and to protect subjects from invasion of their privacy. Because privacy issues are often subtle, and researchers may not understand them, appropriate awareness may be lacking with unfortunate results, such as the following:

> *Scenario 1:* To study the experiences of adults who are survivors of childhood sexual abuse, an investigator joins an online chat room of survivors.[4] He "lurks" and gathers extensive data, confident that his subsequent use of pseudonyms and the fact that this is ostensibly a public venue means that he is not violating privacy interests. He decides to send members of the chat room some feedback based on his observations. So horrified are the members of the chat room at his invasion of space they regarded as private that most quit and never again will seek the comfort and validation that they thought the chat room would offer.
>
> *Scenario 2:* A researcher gains access to medical records, discovers which persons have asthma, and contacts them directly to ask them to participate in research on coping strategies of asthmatics. "How did you get my name?" "What are you doing with my medical records?" were possibly the thoughts, if not the actual questions, of most of those called. Most refused to participate. The researcher should have asked physicians to send their asthmatic patients a letter (drafted and paid for by the researcher) asking if they would be interested in participating in the research, and saying that, if so, the physician would release their names to the researcher.
>
> *Scenario 3:* A researcher interviews children about their moral beliefs. Believing that the children would want privacy, he interviews 5-year-olds alone. However, the children are sufficiently shy or afraid to be alone with the researcher that they do not respond as well as they would, had their mothers been present. Recognizing his error, the researcher then makes sure that subjects from the next group, 12-year-olds, are accompanied by their mothers. However, the 12-year-olds have entered that stage of development in which some privacy from parents is important. Consequently, they do not answer all the questions

honestly. This researcher should have invested time in better scholarship into the development of privacy needs in children (see Thompson, 1991).

Scenario 4: A researcher decides to use telephone interviews to learn about the health histories of older people of lower socioeconomic status, as the phone typically offers greater privacy than face-to-face interviews. She fails to recognize, however, that poor elderly people rarely live alone or have privacy from their families when they use the phone, and many keep health secrets from their families.

In each of the above cases, the researcher has been insensitive to privacy issues idiosyncratic to the research population and has not addressed the problems that these issues pose for the research. Had the researcher consulted the psychological literature, community gatekeepers, consumers of the research, or others familiar with the research population, he or she might have identified these problems and solved them in the design stage. Most of the topics that interest social scientists concern somewhat private or personal matters. Yet most topics, however private, can be effectively and responsibly researched if investigators employ appropriate sensitivity and safeguards.

Is There a Right to Privacy? The right to privacy from research inquiry is protected by the right to refuse to participate in research. An investigator is free to do research on consenting subjects or on publicly available information, including unobtrusive observation of people in public places, although the chat room case above illustrates that in some contexts a public venue should be treated as private. Researchers may videotape or photograph the behavior of people in public without consent. But if they do so, they should heed rules of common courtesy and should be sensitive to local norms. Intimate acts in public places, such as goodbyes at airports and intimate discussions in chat rooms, should be regarded as private, though done in a public venue.

Constitutional and federal laws have little to say directly about privacy and social/behavioral research. Except for HIPAA (see p. 128) which governs health data, the only definitive federal privacy laws governing social/behavioral research pertain to school research.

- *The Protection of Student Rights Amendment* (PPRA) is intended to protect the rights of parents and students in two ways pertinent to research: (1) Schools must make instructional materials available for inspection by parents if those materials will be used in connection with any U.S. Department of Education–funded survey, analysis, or evaluation in which their children participate; and (2) researchers must obtain written parental consent before minor students are required to participate in any U.S. Department of Education-funded survey, analysis, or evaluation that reveals information concerning political affiliations, mental and psychological problems potentially embarrassing to the student and his or her family, sexual behavior and attitude, illegal, antisocial, self-incriminating, and demeaning behavior, critical appraisals of other individuals with whom respondents have close family relationships, legally recognized privileged or analogous relationships, such as those of lawyers, physicians, and ministers; or income (other than that required by law to determine eligibility for participation in a program or for receiving financial assistance under such program). Parents or students who believe that their rights under PPRA may have been violated may file a complaint with the Department of Education by writing to the Family Policy Compliance Office. Complaints must contain specific allegations of fact, giving reasonable cause to believe that a violation of PPRA occurred.

- *The Family Educational Rights and Privacy Act* (FERPA, 1974) protects the privacy of student education records (hence, arguably, is about confidentiality rather than privacy). FERPA applies to all schools that receive funds under an applicable program of the U.S. Department of Education and is relevant to research for which schools must have written permission from the parent or student above 18 years to release any information to a researcher from a student's education record.

Researchers would be well-advised to consult their IRBs and relevant school administrators at the outset when planning research on schoolchildren. Local norms as well as federal and state laws must be considered.

Tort law provides a mechanism through which persons might take action against an investigator alleged to have invaded their privacy. In such an action, the law defines privacy in relation to other interests. It expects behavioral scientists to be sensitive to persons' claims to privacy but recognizes that claims to privacy must sometimes yield to competing claims. Any subject may file a suit against a researcher for "invasion of privacy," but courts of law are sensitive to the value of research as well as the value of privacy.

Important protections against such a suit are adequate informed consent statements signed by all participants, as well as parental permission for research participation by children. Persons other than research participants, however, may claim that their privacy has been invaded by the research. For example, family members of research participants may feel that the investigation probes into their affairs. If the research is socially important and validly designed, if the researcher has taken reasonable precautions to respect the privacy needs of typical subjects and others associated with the research, and if the project has been approved by an IRB, such a suit is likely to be dismissed.

A Behavioral Definition of Privacy

As a behavioral phenomenon, privacy concerns certain needs to establish personal boundaries; these needs seem to be basic and universal, but they are manifested differently depending on learning, cultural, and developmental factors (see Laufer & Wolfe, 1977, for a complete discussion of these factors as they relate to privacy). Privacy does not simply mean being left alone. Some people have too little opportunity to share their lives with others or to bask in public attention. When treated respectfully, many are pleased when an investigator is interested in hearing about their personal lives. Because of this desire on the part of lonely people for understanding and attention, competent survey investigators often have more difficulty exiting people's homes than entering.

Many claims to privacy are also claims to autonomy. For example, subjects' privacy and autonomy are violated when their self-report data on marijuana use become the basis for their arrest, when IQ data are disclosed to schoolteachers who would use it to track students, or when organizational research data disclosed to managers become the basis for firing or transferring employees. The most dramatic cases in which invasion of privacy results in lowered autonomy are those in which something is done to an individual's thought processes—the most private part of a person—through behavior control techniques such as psychopharmacology.

Privacy may be invaded when people are given unwanted information. For example, a researcher may breach a subject's privacy by showing him pornography or by requiring him to listen to more about some other person's sex life than he cares to hear. Privacy is also invaded when people are deprived of their normal flow of information, as when nonconsenting subjects (who do not realize that they are participating in a study) are deprived of information that they ordinarily would use to make important decisions.

Unusual personal boundaries were encountered by Klockars (1974), a criminologist, when he undertook to write a book about a well-known "fence." The fence was an elderly pawnshop owner who had stolen vast amounts earlier in his life. Klockars told the fence that he would like to document the details of his career, as the world has little biographical information about the lives of famous thieves. Klockars offered to change names and other identifying features of the account to ensure anonymity. The fence, however, wanted to go down in history and make his grandchildren *proud* of him. He offered to tell all, but only if Klockars agreed to publish the fence's real name and address in the book. This was done, and the aging fence proudly decorated his pawnshop with clippings from the book. (Thus confidentiality does not always involve a promise *not* to reveal the identity of research participants; rather, it entails *whatever* promise is mutually acceptable to researcher and participant.)

Privacy and Informed Consent. A research experience regarded by some as a constructive opportunity for self-disclosure may constitute an unbearable invasion of privacy for others. Informed consent provides the researcher with an important way to respect these individual differences. The investigator should specify the kinds of things that will occur in the study, the kinds of information that will be sought and given, and the procedures that will be used to assure anonymity or confidentiality. The subject can then decide whether to participate under those conditions. A person who considers a given research procedure an invasion of privacy can decline to participate and should know that it is acceptable to withdraw from the study at any time. However, informed consent is not the entire solution. A researcher who is insensitive to the privacy needs of members of the research population may be unprepared to offer the forms of respect and protection they want.

Gaining Sensitivity to Privacy Interests of Subjects. Although there is no way for researchers to be sure of the privacy interests of all members of a research population, they can learn how typical members would feel. If the typical member considers the research activity an invasion of privacy, the data are likely to be badly flawed; evasion, lying, and dropping out of the study are likely to occur, and those who answer honestly may worry about the consequences.

To learn about the privacy interests of a particular population, the researcher can (a) ask someone who knows that population (e.g., ask teachers and parents about the privacy interests of their children; ask a psychotherapist about the privacy interests of abused children; ask a social worker about the privacy interests of low-socioeconomic-status parents), (b) ask a researcher who works with that population, and (c) ask members of the population what they think *other* people in their group might consider private in relation to the intended study. (Asking what *other people* are likely to think is a graceful way to allow people to disclose their own thoughts.)

"Brokered" Data

If it would be too intrusive for an investigator to have direct access to subjects, a broker may be used. The term *broker* refers to any person who works in some trusted capacity with a population to which the researcher does not have access and who obtains data from that population for the researcher. For example, a broker may be a psychotherapist or a physician who asks patients if they will provide data for important research being conducted elsewhere. A broker may serve other functions in addition to gathering data for the researcher, as discussed below.

"Broker-Sanitized" Responses. Potential subjects may be concerned that some aspects of their responses will enable the investigator to deduce their identities. For example, if a survey is sent to organization leaders in various parts of the country, a postmark on an envelope might enable someone to deduce the identity of some respondents. To prevent this, a mutually agreed on third party may receive all the responses, remove and destroy the envelopes, and then send the responses to the investigator.

Brokers and Aliases. Sometimes, lists of potential respondents are unavailable directly to the researcher. For example, the researcher wishing to study the attitudes of psychiatric patients at various stages of their therapy may not be privy to their names. Rather, the individuals' treating psychiatrists may agree to serve as brokers. The psychiatrists would then obtain the informed consent of their patients and periodically gather data from those who consent. Each patient is given an alias. Each time data are gathered, the psychiatrist refers to a list for the alias, substitutes it for the patient's real name, and transmits the completed questionnaire back to the researcher.

Additional Roles for Brokers. A broker may (a) examine responses for information that might permit the researcher to deduce the identity of the respondent and, therefore, remove that information, (b) add information (e.g., a professional evaluation of the respondent), or (c) check responses for accuracy or completeness.

There should be some *quid pro quo* between researcher and broker. Perhaps the broker may be paid for his or her time, or the researcher may make a contribution to the broker's organization.

Confidentiality

Confidentiality refers to access to data, not access to people directly. The researcher should employ adequate safeguards of confidentiality, and these should be described in specific terms in the consent statement. For example, confidentiality agreements such as the following might be included in a consent letter from a researcher seeking to interview families in counseling.

- To protect your privacy, the following measures will ensure that others do not learn your identity or what you tell me: No names will be used in transcribing from the audiotape, or in writing up the case study. Each person will be assigned a letter name as follows: M for mother, F for father, MS1 for male first sibling, and so on.
- All identifying characteristics, such as occupation, city, and ethnic background, will be changed.
- The audiotapes will be reviewed only in my home or the office of my thesis adviser. The tapes and notes will be destroyed after my report of this research has been accepted for publication.
- What is discussed during our session will be kept confidential, with two exceptions: I am compelled by law to inform an appropriate other person if I hear and believe that you are in danger of hurting yourself or someone else or if there is reasonable suspicion that a child, elder, or dependent adult has been abused.[5]

Noteworthy characteristics of this agreement are that it (a) recognizes the sensitivity of some of the information likely to be conveyed, (b) states what steps will be taken to ensure that others are not privy to the identity of subjects or to identifiable details about individuals, and (c) states any legal limitations to the assurance of confidentiality.

Why Is Confidentiality an Issue in Research?

Confidentiality, like privacy, respects personal boundaries. Participants tend not to share highly personal information with a researcher unless they believe that their data will be kept from falling into the wrong hands, such as those who would gossip, blackmail, take adverse personnel action against the subjects, or subpoena the data. However, people tend to overestimate the risk of confidentiality breaches (Singer, 2003). Assurances of confidentiality by the researcher have little direct effect on willingness to participate in research and may also sensitize subjects so much to possible risks that they have an effect opposite to that intended by the researcher (Singer, Hippler, & Schwarz, 1992). To allay such fears, the researcher could gather the data anonymously—that is, without gathering any unique identifiers whenever feasible. When designing the research, the researcher should decide whether the data can be gathered anonymously. Four major reasons for gathering unique identifiers, such as names and addresses, are as follows:

1. They make it possible for the researcher to recontact subjects if their data indicate that they need help or information.
2. They make it possible for the researcher to link data sets from the same individuals. (This might also be achieved with code names.)
3. They allow the researcher to mail results to the subjects. (This might also be achieved by having subjects address envelopes to themselves, which are then stored apart from the data. After the results are mailed out, no record of the names of subjects would remain with the researcher.)
4. They make it possible for the researcher to screen a large sample on some measures in order to identify a low-base-rate sample (e.g., families in which there are twins).

Note that for the first two reasons, the issue is whether to have names associated with subjects' data; for the third reason, the issue is whether to have names on file at all. In the fourth case, identifiers may be expunged from the succeeding study as soon as those data are gathered. If the data can be gathered anonymously, subjects will be more forthcoming, and the researcher will be relieved of some responsibilities connected with assuring confidentiality. If the research cannot be done anonymously, the researcher must consider procedural, statistical, and legal methods for assuring confidentiality.

Some Procedural Approaches to Assuring Confidentiality or Anonymity

Certain procedural approaches eliminate or minimize the link between the identifiers and the data, and may be appropriate, depending on whether the research is cross-sectional or longitudinal. If unique identifiers are needed, they might be constructed identifiers, such as initials, date of birth, or the last four digits of a phone number. If there is no need to link individual data gathered at one time to data gathered at another, some simple methods of preventing disclosure in cross-sectional research are as follows:

- *Anonymity:* The researcher has no record of the identity of the respondents. For example, respondents mail back their questionnaires or hand them back in a group, without names or other unique identifiers.

- *Temporarily identified responses:* It is sometimes important to ensure that only the appropriate persons have responded and that their responses are complete. After the researcher checks the names against a list or makes sure that responses are complete, the names are destroyed.

- *Separately identified responses:* In mail surveys, it is sometimes necessary to know who has responded and who has not. To accomplish this with an anonymous survey, the researcher may ask each respondent to mail back the completed survey anonymously and to mail separately a postcard with his or her name on it (Dillman, 1978). This method enables the researcher to check off those who have responded and to send another wave of questionnaires to those who have not.

Any of these three methods can be put to corrupt use if the researcher is so inclined. Because people are sensitive to corrupt practices, the honest researcher must demonstrate integrity. The researcher's good name and that of the research institution may reduce the suspicion of potential respondents.

Different procedures are needed if individuals' data files are to be linked permanently, as in longitudinal research, or linking of other independently stored files:

Longitudinal Research. Here, the researcher must somehow link together the various responses of particular persons over time. A common way to accomplish this is to have each subject use an easily remembered code, such as mother's maiden name as an alias. The researcher must make sure that there are no duplicate aliases. The adequacy of this method depends on subjects' ability to remember their aliases. In cases where a subject is mistakenly using the wrong alias might seriously affect the research or the subject (e.g., the subject gets back the wrong HIV test result), this method of linking data would be inappropriate.

Other File Linking. Sometimes, a researcher needs to link each person's records with some other independently stored records on those same persons (exact matching) or on persons who are similar on some attributes (statistical matching). A researcher can link files without disclosing the identity of the individuals by constructing identifications based on the files, such as a combination of letters from the individual's name, his or her date of birth and gender, and the last four digits of the person's social security number.

Another approach to interfile linkage would be through use of a broker, who would perform the linkage without disclosing the identity of the individuals. An example would be court-mandated research on the relationship between academic accomplishment and subsequent arrest records of juveniles who have been sentenced to one of three experimental rehabilitation programs. The court may be unwilling to grant a researcher access to the records involved but may be willing to arrange for a clerk at the court to gather all the relevant data on each subject, remove identifiers, and give the anonymous files to the researcher. The obvious advantages of exact matching are the ability to obtain data that would be difficult or impossible to obtain otherwise and the ability to construct a longitudinal file.

Certificates of Confidentiality

Under certain circumstances, priests, physicians, and lawyers may not be required to reveal to a court of law the identities of their clients or sources of information. This privilege does *not* extend to researchers. Prosecutors, grand juries, legislative bodies, civil litigants, and administrative agencies can use their subpoena powers to compel disclosure of confidential research information. What is to protect research from this intrusion? Anonymous data, aliases, colleagues in foreign countries to whom sensitive data can be mailed as soon as it is gathered, and statistical strategies are not always satisfactory solutions. The most effective protection against subpoena is the Certificate of Confidentiality.

In 1988, the U.S. Congress enacted the Public Health Service Act, providing for an apparently absolute researcher-participant privilege when it is covered by a Certificate of Confidentiality issued by units of the Department of Health and Human Services. The Certificate of Confidentiality is designed to protect identifiable sensitive data against compelled disclosure in any federal, state, or local civil, criminal, administrative, legislative, or other proceeding (see http://grants1.nih.gov/grants/policy/coc/background.htm). Wolf and Zandecki (2006) recently surveyed National Institutes of Health (NIH)–funded investigators to learn about their experience of using Certificates of Confidentiality and found that while most investigators prefer using them, they cannot gauge how research participants regard them, and some investigators found them too complex to explain to participants. Singer (2004) found that mention of a Certificate of Confidentiality increases the perception of harm, especially among younger respondents.

Confidentiality and Consent

An adequate consent statement shows the subject that the researcher has conducted a thorough analysis of the risks to confidentiality and has acted with the well-being of the subject foremost in mind. The consent statement must specify any promises of confidentiality that the researcher cannot make. Typically, these have to do with reporting laws pertaining to child abuse, child molestation, and threats of harm to self and others. Reporting laws vary from state to state, so the researcher should be familiar with the laws in the state(s) where the research is to be conducted. Thus, the consent statement warns the subject not to reveal certain kinds of information to the researcher. This protects the researcher as well, since a skilled researcher can establish rapport and convince subjects to reveal almost anything, including things that the researcher may not want to be responsible for knowing.

There are many ways in which confidentiality or anonymity might be discussed in a consent statement. A few examples follow:

> *Example 1:* To protect your privacy, this research is conducted anonymously. No record of your participation will be kept. Do not sign this consent or put your name on the survey.

> *Example 2:* This is an anonymous study of teacher attitudes. No names of people, schools, or districts will be gathered. The results will be reported in the form of statistical summaries of group results.

> *Example 3:* The data will be anonymous. You are asked to write your name on the cover sheet so that I can make sure your responses are complete. As soon as you hand in your questionnaire, I will check your responses for completeness and ask you to complete any incomplete items. I will then tear off and destroy the cover sheet. There will then be no way anyone else can associate your name with your data.

Example 4: This survey is anonymous. Please complete it, and return it unsigned in the enclosed, postage-paid envelope. At the same time, please return the postcard bearing your name. That way we will know you responded, but we will not know which survey is yours.

Example 5: This anonymous study of persons who have decided to be tested for HIV infection is being conducted by Dr. John Smith at Newton University. Because we do not want to intrude on your privacy in any way, a physician at the AIDS Testing Center has agreed to ask you if you would be willing to respond to this survey. Please look it over. If you think you would be willing to respond, take it home, answer the questions, and mail it back to me in the attached, stamped, self-addressed envelope. If you are interested in knowing the results of the study, please write to me at the above address, or stop by the AIDS Testing Center and ask for a copy of the results of the survey which will be available after May 1.

Example 6: Because this is a study in which we hope to track your progress in coping with an incurable disease and your responses to psychotherapy designed to help you in that effort, we will need to interview you every 2 months and match your new interview data with your prior data. To keep your file strictly anonymous, we need to give you an alias. Think of one or more code names you might like to use. Make sure it is a name you will remember, such as the name of a close high school friend, a pet, or a favorite movie star. You will need to check with the researcher to make sure that no other participant has chosen the same name. The name you choose will be the only name that is ever associated with your file. We will be unable to contact you, so we hope you will be sure to keep in touch with us. If you decide to drop out of the study, we would be grateful if you would let us know.

Example 7: In this study, I will examine the relationship between your child's SAT scores and his attitude toward specific areas of study. We respect the privacy of your child. If you give me permission to do so, I will ask your child to fill out an attitude survey. I will then give that survey to the school secretary, who will write your child's SAT subscores on it, and erase your child's name from it. That way, I will have attitude and SAT data for each child, but will not know the name of any child. The data will then be statistically analyzed and reported as group data.

These are merely examples. The researcher needs to give careful consideration to the content and wording of each consent statement.

Data Sharing

If research is published, the investigator is accountable for the results, and is normally required to keep the data for 5 to 10 years. The editor of the publication in which the research is reported may ask to see the raw data to check its veracity.

Some funders (e.g., NIH, 2003) require that the documented data be archived in user-friendly form and made available to other scientists. When data are shared via a public archive, the researcher must ensure that all identifiers are removed and that there is no way for anyone to deduce subjects' identities.

A variety of techniques have been developed by the Federal government (which has an obligation to provide to other users the data collected at taxpayer expense) to transform raw data into a form that prevents deductive disclosure (Zarate & Zayatz, 2006). The objective is always to preserve the analytical value while removing the characteristics of that data that would enable one to reidentify the ostensibly deidentified data. Variables or cases with easily identifiable characteristics are removed. Microaggregation can be employed by ordering microdata along a single variable then aggregating adjacent records in groups of three or more. Within each grouping, the reported (actual) value on all variables is replaced by the average value of the group for each variable. For details of microaggregation see O'Rourke et al. (2006) who provide detailed descriptions of other techniques as well. If the analytical value of data would be destroyed by using techniques such as those described by O'Rourke et al., one may provide limited access to the raw data to persons who meet stringent requirements such as administration of the sharing arrangement by their institution, signing of contractual or licensing agreements, and so on (see Rodgers & Nolte, 2006, for details of these procedures).

When health data are to be shared, the Privacy Rule of the Health Insurance Portability and Accountability Act of 1996 (HIPAA)—which is really about confidentiality—permits a holder of identified health data to release those data without the individual's authorization if it meets certain conditions. Either it must delete any of the 18 identifiers specified in HIPAA or one can have a disclosure expert determine whether data elements, alone or combined with others, might lead to identification of a specific person (for details of HIPAA, see www.hhs.gov/ocr/combinedregtext.pdf; for details on compliance with HIPAA, see DeWolf, Sieber, Steel, & Zarate, 2006).

Recognizing Elements of Research Risk

Risk assessment is not intuitively easy. Most investigators are sensitive only to the risks that they have already encountered and may fail to assess major risks in new settings. The goal of this brief section is to help researchers recognize kinds and sources of risk.

Kinds of Risk. Risk, or the possibility of some harm, loss, or damage, may involve mere inconvenience (e.g., boredom, frustration, time wasting), physical risk (e.g., injury), psychological risk (e.g., insult, depression, upset), social risk (e.g., embarrassment, rejection), economic risk (e.g., loss of job, money, credit), or legal risk (e.g., arrest, fine, subpoena).

What Aspect of Research Creates Risk? Risk may arise from (a) the theory, which may become publicized and may blame the victim or create wrong ideas; (b) the research process; (c) the institutional setting in which the research occurs, which may be coercive in connection with the research; and (d) the uses of the research findings.

Who Is Vulnerable? Documents regarded as basic to understanding the ethics and regulations of human research such as the Belmont Report (www.med.umich .edu/irbmed/ethics/Belmont/BELMONTR.HTM) and the Federal Regulations of Human Research (www.hhs.gov/ohrp/requests/com101105. html) list categories of persons who are vulnerable, such as children, prisoners, pregnant women, military enlistees, and so on, and one could add many other kinds of people to this list, such as psychology undergraduates and illegal aliens. But are such people vulnerable to any kind of research? Is a pregnant woman vulnerable when interviewed about the kind of baby food she plans to buy? In short, what is it about these so-called vulnerable people that makes them vulnerable, and what should a researcher do when faced with such potential research participants?

Kipnis (2001, 2004) has advanced our understanding of vulnerability considerably by recognizing the folly of listing such categories of persons and instead answering the question: What makes a person vulnerable? He has developed the following useful taxonomy of six kinds of vulnerability:

- *Cognitive vulnerability:* Does the person have the capacity to decide whether to participate?
- *Juridic vulnerability:* Is the person liable to the authority of others who may have an independent interest in their research participation?
- *Deferential vulnerability:* Does the person have patterns of deferential behavior that may mask an unwillingness to participate?
- *Medical vulnerabililty:* Has the person been selected for having a serious health-related condition for which there are no satisfactory remedies?
- *Allocational vulnerability:* Does the person lack important social goods that will be provided in return for research participation?
- *Research infrastructure:* Does the political, organizational, economic, social context of the research have the integrity and resources needed to manage the study responsibly?

When such vulnerabilities appear to exist, Kipnis (2001, 2004) recommends conducting further inquiry and implementing compensating measures. Researchers should brainstorm with colleagues, gatekeepers, community members, and others who understand the risks inherent in the particular research setting. They should also be aware of their own biases and of alternative points of view. Researchers should also consider the assumptions underlying their theories and methods, limitations of the findings, and how the media and opinion leaders may translate the researcher's statements into flashy and dangerous generalizations.

Judging who can help to identify vulnerability is not always easy. Even one who has been a member of the vulnerable population to be studied (e.g., the researcher of homeless people who has, herself, been homeless; the researcher of victims of domestic violence who was once a victim of domestic violence) may not be able to empathize with the current concerns of members of that population, for the researcher is no longer an insider to that population.

Most of the social research that is regarded as high priority by funders and society today is concerned with vulnerable populations—drug abusers, runaways, prostitutes, persons with AIDS, victims of violence, and so on. The preceding discussion about communication, risk/benefit assessment, and privacy/confidentiality is doubly important for such populations. Furthermore, members of many stigmatized and fearful populations are especially unwilling to be candid with researchers who are interested primarily in discovering scientific truth, rather than helping the individuals being studied. Contrary to the usual scientific directive to be objective, the researcher who investigates the lives of runaways, prostitutes, or victims of domestic violence or spousal rape often must be an advocate for those subjects to gain their trust and cooperation and must relate in a personal and caring manner if candor and participation are to be forthcoming from members of the research population. However, the devil is in the details. General prescriptions pale alongside accounts of ethical issues in specific contexts. Each vulnerable research population has its own special set of fears, its own reasons for mistrusting scientists, and its own culture, which outsiders can scarcely imagine. Interested readers are referred to Renzetti and Lee (1993) for further discussion.

Maximizing the Benefits of Research

When researchers vaguely promise benefit to science and society, they approach being silly; a single research project, even if published, rarely benefits science and society. It is only after considerably lengthy research and development of a successful line of work that a project can reach such ultimate goals. Researchers typically overlook the more likely and more immediate benefits that are the precursors of societal and scientific benefit. Some of the most immediate benefits are those to subjects and—in the case of some community-based research—to their communities. These are not only easy to bring about but are also owed and may facilitate future research access to that population. The intermediate benefits—to the researcher, the research institution, and the funder, if any—are ones that any talented investigator with an ongoing research program can produce in some measure. It is on these immediate and intermediate goals or benefits that any ultimate scientific and societal benefits are likely to be based, as the following example illustrates:

> A researcher started an externally funded school-based experiment with instructional methods designed to improve the performance of students identified as learning disabled. Each method was designed to develop diagnostic and teaching procedures that could ultimately be used by school personnel. The researcher began by discussing her intervention and research plans with school administrators, teachers, parents, and students, and asking them to describe problems with which they would like to have assistance. Where feasible, she made slight alterations in her program to accommodate their wishes. She integrated the research program with a graduate course so that her students received extensive training in the conduct of research in the school setting, under her rigorous supervision.

She provided the school faculty with materials on learning disabilities, and gave bag-lunch workshops and presentations on her project. She worked with teachers who were interested in trying her approaches in their classrooms, urging them to adapt and modify her approaches as they deemed appropriate, and asked that they let her know the outcomes. Together, the researcher and the teachers pilot tested adaptations of the methods concurrently with the formal experiments. All learning disabled children who participated received special recognition and learned how to assist other students with similar problems. Two newspaper articles about the program brought favorable publicity to the researcher, the school, and the researcher's university. This recognition further increased the already high morale of students, teachers, and the researcher.

Of the six procedures examined, only two showed significant long-term gains on standardized tests of learning. However, the teachers who had gotten involved with pilot testing of variations on the treatments were highly enthusiastic about the success of these variations. When renewal of funding was sought, the funder was dissatisfied with the formal findings, but impressed that the school district and the university, together, had offered to provide in-kind matching funds. The school administrators wrote a glowing testimony to the promise of the new pilot procedures and of the overall approach, and the funder supported the project for a second year. The results of the second year, based on modified procedures, were much stronger. Given the structure that had been created, it was easy for the researcher to document the entire procedure on videotape and to disseminate it widely. The funder provided seed money to permit the researcher, her graduate students, and the teachers who had collaborated on pilot testing to start a national-level traveling workshop, which quickly became self-supporting. This additional support provided summer salary to the researcher, teachers, and graduate students for several years.

This tale of providing benefits to the many stakeholders in the research process is not strictly relevant to all research. Not every researcher does field research designed to benefit a community. In some settings, too much missionary zeal to include others in "helping" may expose some subjects to serious risk such as breach of confidentiality. Not all research is funded or involves student assistants. Many researchers engage in simple, unfunded, unassisted, one-time laboratory studies to test theory. Even in such uncomplicated research, however, any benefit to the institution (e.g., a Science Day research demonstration) may favorably influence the institution to provide resources for future research, and efforts to benefit subjects may be repaid with their cooperation and respect.

Significant contributions to science and society are not the results of one-shot activities. Rather, such contributions typically arise from a series of competently designed research or intervention efforts, which themselves are possible only because the researcher has developed appropriate institutional or community rapport and infrastructures and has disseminated the findings in a timely and effective way. Benefit to society also depends on widespread implementation, which, in turn, depends on the goodwill, support, and collective wisdom of many specific individuals, including politicians, funders, other professionals, and community leaders. Thus, the successful contributor to science and society is a builder of many benefits and a provider of those benefits to various constituencies, even if the conduct of the research, per se, is a solo operation.

As shown in Table 4.1, research benefits may be divided into seven (nonexclusive) categories, ranging from those that are relatively easy to provide through those that are extremely difficult. These seven kinds of benefits, in turn, might accrue to any of seven kinds of recipients—subjects, communities, investigators, research institutions, funders, science, and society in general. The seven categories of benefit are described below as they might pertain to a community that is the site of field research:

- *Valuable relationships*: The community establishes ties with helping institutions and funders.
- *Knowledge or education:* The community develops a better understanding of its own problems.
- *Material resources:* The community makes use of research materials, equipment, and funding.
- *Training, employment, opportunity for advancement:* Community members receive training and continue to serve as professionals or paraprofessionals within the ongoing project.
- *Opportunity to do good and to receive the esteem of others:* The community learns how to serve its members better.
- *Empowerment (personal, political, etc.):* The community learns to use findings for policy purposes and gains favorable attention from the press, politicians, and others.
- *Scientific/clinical outcomes:* The community provides effective treatment to its members (assuming that the research or intervention is successful).

Note that even if the experiment or intervention yields disappointing results, all but the last benefit might be available to the community, as well as to individual subjects. Let us now consider the seven kinds of beneficiaries.

The *subjects* may enjoy such benefits as the respect of the researcher, an interesting debriefing, money, treatment, or future opportunities for advancement.

The community or institution that provides the setting for the field research may include the subjects' homes, neighborhood, clinic, workplace, or recreation center. A community includes its members, gatekeepers, leaders, staff, professionals, clientele, and peers or family of the subjects. Benefits to the community are similar to those for the subjects. Sometimes, community members also serve as research assistants and so would receive benefits associated with those of the next category of recipients as well.

The *researcher,* as well as research assistants and others who are associated with the project, may gain valuable relationships, knowledge, expertise, access to funding, scientific recognition, and so on, if the research is competently conducted, and

Table 4.1 Benefit Table of a Hypothetical Learning Research Project

Benefit	*Subjects*	*Community*	*Researcher*	*Institution*	*Funder*	*Science*	*Society*
Relationships	Respect of researcher	Ties to university	Future access to community	Improved town-gown relationships	Ties with a successful project	Ideas shared with other scientists	Access to a new specialist
Knowledge/ education	Informative debriefing	Understanding of relevant learning problems	Knowledge	Improved graduate research	Outstanding final report	National symposium	Media presentation
Material resources	Workbook	Books	Grant support	Videotapes of research	Instructional materials	Refereed publications	Useful popular literature
Training opportunity	Tutoring skills	Trained practitioners	Greater research expertise	Student training program	Model project for future grant applicants	Workshop at national meetings	Training for practitioners nationally
Do good/ earn esteem	Esteem of peers	Local enthusiasm for project	Professional respect	Esteem of community	Satisfaction of funder overseers	Recognition of scientific contribution	Greater respect for science
Empowerment	Earn leadership status	Prestige from the program	National reputation with funder	Good reputation	Congressional increase in funding	Increased prestige of discipline	Increased power to help people
Scientific/ clinical success	Improved learning ability	Effective program	Leadership opportunities in national program	Headquarters for national teacher program	Proven success of funded treatment	Improved training via workshops	Nationally successful programs

especially if it produces the desired result or some other dramatic outcome. By creating these benefits for themselves, the investigators gain the credibility needed to go forward with a research program and to exert a significant influence on science and society.

The *research institution* may benefit along with the researcher. Institutional benefits are likely to be described as good university-community relations, educational leadership, funding of overhead costs and equipment, and a good scientific reputation for scientists, funders, government, and the scientific establishment. Such benefits increase a university's willingness to provide the kinds of support (e.g., space, clerical assistance, small grants, equipment, matching funds) that enable the researcher to move the research program forward.

The *funder* is vital to the success of a major research program and hopes to receive benefits such as the following: ties to a good project and its constituents, well-written intellectual products promptly and effectively disseminated, good publicity, evidence of useful outcomes, good ideas to share with other scientists, and good impressions made on politicians and others who have power to fund the funder. Such benefits will result in a funder favorably disposed to funding future research of that investigator.

Science refers to the discipline(s) involved, to the scientists within them, their scientific societies, and their publications. Benefits to science parallel benefits to funders and depend on the rigor and usefulness of the research. Development of useful insights and methods may serve science, even in the absence of findings that might benefit society. Initial papers and symposia give way to publications and invited addresses. Others evaluate, replicate, promote, and build on the work, thus earning it a place in the realm of scientific ideas. A single publication on which no one builds is hardly a contribution to science.

Society, including the target population from which subjects were sampled and to which the results are to be generalized, benefits only when the hoped-for scientific outcome occurs and is generalizable to other settings. This represents the most advanced developmental stage of any given research project. By the time benefits of this magnitude have accrued, the researcher or others have already implemented the idea broadly in society. The idea has begun to take on a life of its own, to be modified to a variety of uses, and to be adapted, used, and even claimed by many others.

The conjunction of the seven kinds of benefits and seven kinds of beneficiaries described above yields a 49-cell table that is useful in research planning. This table suggests that turning a research idea into a scientific and social contribution requires that benefits be developed at each stage of the process. It is useful for the researcher to design a tentative table of benefits as the basic research idea and design are being formulated and to continue planning the benefits as the project proceeds. Many valuable benefits may be easily incorporated, provided the researcher is attuned to opportunities for doing so.

These benefits are augmented if the project's progress is made available via the Internet and if all products are published in accessible media and in language that is understandable to the audiences who might be interested in it.

Research on Children and Adolescents

As a research population, minors are special in several respects: (a) They have limited psychological, as well as legal, capacity to give informed consent; (b) they may be cognitively, socially, and emotionally immature, and consequently, the law expects scientists to use knowledge of human development to reduce risk and vulnerability;[6] (c) there are external constraints on their self-determination and independent decision making; (d) they have unequal power in relation to authorities, such as parents, teachers, and researchers; (e) their parents and certain institutions, as well as the youngsters themselves, have an interest in their research participation; and (f) national priorities for research on children and adolescents include research on drug use, the problems of runaways, pregnancy among teenagers, and other sensitive topics, compounding the ethical and legal problems surrounding research on minors. Federal, state, and local laws governing research respond to these characteristics of youngsters by requiring that they have special protections and that parental rights be respected.

Even quite young children should be given an opportunity to assent (to decide for themselves) to participate in research. The assent procedure should be tailored to the cognitive and social/emotional level of the child. Both child assent and parent/guardian permission are required, and either child or parent/guardian may veto the child's participation in research. Parental or guardian permission may be waived only in some low-risk research that could not be conducted otherwise or when a parent or guardian is not situated to act in the minor's best interests.

In consequence of the tendency of troubled youngsters to defy their parents or to run away, the law recognizes that parental consent may be waived by an IRB under certain circumstances. In most cases, such research is conducted within an institution such as an HIV testing site, an abortion clinic, a youth detention center, a shelter for runaway children, or a drug treatment center. The problems of obtaining meaningful consent are manifold. These problems have been discussed extensively by Grisso (1991), who focuses on issues surrounding waiver of parental permission, and by Rotheram-Borus and Koopman (1991), who are concerned primarily with consent issues in research and treatment of runaway gay and sexually active youth, whose relationships with their parents are often marked by secrecy, conflict, and long absences. The following list summarizes some of their main points:

1. The youngster is unlikely to believe that the research is independent of the institution or that he or she may decline to participate with impunity.
2. The youngster is unlikely to believe promises of confidentiality, especially when he or she is in trouble with his or her parents and other authorities.
3. Issues of privacy, which are normally salient for adolescents, are likely to be even more heightened for this population.
4. Maltreated youngsters are likely to experience the research as more stressful than are normal children. If the researcher effectively establishes rapport, the youngster may reach out for help; the researcher must be prepared to respond helpfully.

The complexities of research on children are significant, and space constraints preclude their treatment here. The reader is referred to the Code of Ethics of the Society for Research in Child Development, at www.srcd.org/ethicalstandards.html, research guidelines by the Institute on Chronic Poverty at www.chronicpoverty.org/CPToolbox/Children.htm, and to literature on sensitive child populations, including ethnic minorities in the child welfare system (Elliott & Urquiza, 2006) and vulnerable adolescents (Cauce & Nobles, 2006).

Vulnerable Populations

Most high-priority social research is concerned with vulnerable populations—drug abusers, runaways, prostitutes, persons with AIDS, victims of violence, the mentally ill, and so on. The foregoing discussions about communication, risk/benefit assessment, and privacy/confidentiality are doubly applicable to these populations. Additionally, members of many stigmatized and fearful populations are unwilling to be candid with researchers who are interested primarily in discovering scientific truth, rather than helping the individuals being studied. Contrary to the usual scientific directive to be objective, the researcher who investigates the lives of such people as runaways, prostitutes, or victims of domestic violence or spousal rape must be an advocate for those studied to gain their trust and cooperation (Renzetti & Lee, 1993). The investigators must relate in a personal and caring manner if candor and participation are to be forthcoming from members of such research populations. Critical to success is understanding the ways in which members of such populations may be vulnerable. Application of Kipnis's categories of vulnerability discussed above (p. 129) is critically important when analyzing the ways in which such populations are vulnerable in the research setting, and seeking to minimize those vulnerabilities.

Discussion Questions

1. Ethics is a win-win matter. Discuss the ways that researchers who are thoughtful can benefit the many stakeholders in human research (including the seven categories of stakeholders listed in Table 4.1). Discuss ways that researchers who are thoughtless of ethics might destroy opportunities to do useful research and negate possible benefits of research.

2. Discuss ways empirical research can enable investigators and IRBs to establish truly ethical interpretations of the Belmont principles. (*Hint:* How can they create informed consent statements and procedures that are correctly understood by the target research population; how can they learn what fears subjects have about breach of confidentiality (whether warranted or not); how can they understand the privacy interests of some subjects? How can they learn what kinds of benefits subjects would really like to have? How can they learn how subjects respond to the experience of participating in their research?)

3. What are some of the things one should consider when preparing the informed consent procedure? Why might this matter? Arguably, the manner of delivery of the consent procedure is more important than the verbal content of the statement; explain.

4. Debriefing should be a two-way communication. What do you think are some of the things that the researcher should seek to learn about the research and the subjects in the debriefing process?

5. When is deception justified? When not? What are some approaches that respect subjects' rights of self-determination? Describe a way in which a deception study can have a "learning not to be fooled" element added to it.

6. Distinguish between privacy, confidentiality, and anonymity. Why are privacy interests of others difficult to judge? What is the role of informed consent in respecting privacy? Describe several ways to explore the likely privacy interests of your research population.

7. Assume that you have plans to gather survey data. What are some of the confidentiality issues you might explore? What might be the advantages of anonymity? The disadvantages?

8. What are the provisions of PPRA and FERPA? What are the implications for planning educational research?

9. Describe several kinds of research in which you may need to use a broker. How might you organize the brokering procedure in each situation?

10. What are the kinds of risk possibly inherent in research? What are ways, according to Kipnis, in which one might be vulnerable?

11. Describe some of the kinds of benefits that might be received directly by subjects when they participate in research? Why would it matter whether your institution or funder benefited?

12. Minors, as research subjects, are different from adults. What are some of the ways they are different? Why are troubled youth a particular challenge to study?

Exercises

For purposes of convenience, the exercises presented here are based on material available on the Internet. Three of the articles you will draw on appear in the March issue of the *Journal of Empirical Research on Human Research Ethics (JERHRE,* pronounced Jerry). Articles in the March issue of *JERHRE* can be downloaded free of charge from http://caliber.ucpress.net/loi/jer.

1. Formulate a focus-group study of scientific misbehavior in which you will ask persons involved in social/behavioral research what behaviors they believe to be most threatening to the integrity of the research enterprise. This exercise is

patterned after the focus group research conducted by Raymond DeVries, Melissa Anderson, and Brian Martinson (2006), "Normal Misbehavior: Scientists Talk About the Ethics of Research" (available at http://caliber.ucpress.net/loi/jer). Peruse this brief article to understand the purpose of the study on which your first practice exercises will be based.

2. Identify some people who are involved in research, who could serve as surrogate subjects in your exercise.

3. Review "Tips on Informed Consent" at www.socialpsychology.org/consent.htm/. Notice that the U.S. government regulations offered in the first set of tips appear to be designed primarily for biomedical research and are less focused on social and behavioral research than the second set of tips by the American Psychological Association. Note that this site also offers tips on developing a consent form for a Web-based study. At the bottom of this Web page, click on *Sample Consent Form,* which is a good example of a consent form that would be clear and understandable to members of an academic community. Using the ideas presented at this Web site, draft your consent statement.

4. Describe how you will use cognitive interviewing, both the *think aloud* and the *verbal probing* procedures, to examine whether your surrogate subjects understand the consent statement you have drafted. A detailed discussion of the use of these procedures may be found in an article by Gordon Willis (2006) titled "Cognitive Interviewing as a Tool for Improving the Informed Consent Process," in *JERHRE* (available at http://caliber.ucpress.net/loi/jer). Recognizing that your research topic is a rather unusual one, consider what aspects of it your subjects are likely to misunderstand based on your consent statement. Think especially about how you will focus on these areas of likely misunderstanding in your cognitive interview.

5. Conduct sequential cognitive interviews with your surrogate subjects until you feel you have addressed the areas of misunderstanding or ambiguity in your consent statement, and have arrived at a statement that your subjects correctly understand.

6. Conduct the focus group. After your focus group of surrogate subjects has generated a list of behaviors that they believe to be most threatening to the integrity of the research enterprise, use their experience to generate your debriefing material. (a) Ask the surrogate subjects to discuss what they thought of their research experience, and what kind of debriefing discussion they think people would want. (b) Take careful notes on what they say. (c) Probe and ask what privacy interests subjects participating in the focus groups might have. (d) Ask what other kinds of risks participants might be concerned about or be exposed to. (e) Ask what benefits they think participants might enjoy from the experience. (f) Administer the Reactions to Research Participation Questionnaire—RRPQ (which can be downloaded from www.personal.utulsa.edu/~elana-newman) asking that respondents not identify themselves on the questionnaire. (g) Ask if they have any further reactions that they would like to share with the group. (h) After thanking and dismissing the participants, examine the RRPQ for further ideas about what to add to the debriefing procedure. (i) Write out the debriefing procedure.

7. Revisit your informed consent statement, taking into account what you have learned. Can you better describe what people will experience and what risks or benefits they might perceive from the experience? Do you think that there will be people who are likely to want to opt out of participating if they fully understand what they will experience? Have you written the statement to give them that opportunity? There are good scientific and practical reasons not to include such people in your focus groups; if so, state some of these reasons.

8. Suppose that you are now going to conduct a survey of scientists to discover what percentage of them have committed any of the 10 scientific misbehaviors described in Brian Martinson, Melissa Anderson, Lauren Crain, and Raymond DeVries (2006, table 2, p. 58), "Scientists' Perceptions of Organizational Justice and Self-Reported Misbehavior" (available at http://caliber.ucpress.net/loi/jer). Since you would be asking people to disclose such egregious wrongdoing as falsifying data, and ignoring human subjects' requirements, what confidentiality concerns would you have? What confidentiality concerns do you think your subjects would have? What procedure did Martinson et al. employ to resolve confidentiality concerns? Can you think of a different procedure that would work as well or better?

9. Furthermore, suppose that you conducted this survey over the Internet and that to better understand the reasons why anyone would commit any of these 10 misbehaviors, you further asked your subjects whether you might interview them by phone and if so they should contact you. While there is much you could do to ensure that the data were kept in an anonymous form, you worry that there could be risk of subpoena of data. Go to http://grants1.nih.gov/grants/policy/coc/background.htm and learn what would be involved in obtaining a Certificate of Confidentiality that would protect the data from subpoena. Identify two ways in which your interview subjects might be vulnerable, from Kipnis's vulnerability factors; see http://www.onlineethics.diamax.com/cms/8087.aspx.

10. Using Table 4.1, identify kinds of benefits you could offer to each of the seven categories of potential benefit recipients in connection with the hypothetical study based on Martinson et al. (2006).

11. Do you think your focus group project is a minimal risk project? How might you be sure whether it is? How would you demonstrate your conclusion to your IRB? Do you think that the hypothetical second project is a minimal risk project? Why or why not?

Notes

1. For discussion of Certificates of Confidentiality and how they may be obtained from a federal agency, see http://grants1.nih.gov/grants/policy/coc/background.htm.

2. Federal regulations governing human research are written largely for biomedical research and may be difficult to interpret. For an interpretation of the regulations that provides user-friendly instruction, see excellent online materials created by institutional HRPPs, such as the Web site from the University of Minnesota, www.research.umn.edu/consent, which

presents separate guidance for biomedical and social/behavioral research focusing primarily on informed consent and understanding the IRB, and an orientation to the rest of the HRPP Web site www.research.umn.edu/irb/guidance, which discusses many other issues in depth.

3. The researcher should be aware that the significance of eye contact varies with culture. Direct eye contact conveys honesty in some cultures, whereas in others it is construed as a sign of disrespect.

4. The Internet provides many kinds of opportunities for recruiting subjects, doing online experiments, and observing behavior online. A full discussion of the ways in which the Internet has changed human research and the distinctive ethical questions raised by these innovations are beyond the scope of this chapter. An excellent article summarizing these new opportunities and challenges may be found in a key article by Kraut, Olson, Banaji, Bruckman, Cohen, and Couper (2004).

5. This example, adapted from a statement developed by David H. Ruja is discussed in Gil (1986).

6. See Thompson (1991) for discussion of developmental aspects of vulnerability to research risk.

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PART II

Applied Research Designs

In this section of the handbook we move from the broader design and planning issues raised in Part I to more specific research designs and approaches. In Part I, the contributors noted the unique characteristics of applied research and discussed issues such as sampling, statistical power, and ethics. In Part II, the focus narrows to particular types of designs, including experimental and quasi-experimental designs, case studies, needs analysis, cost-effectiveness evaluations, and research synthesis.

In Chapter 5, Boruch and his co-authors focus on one type of design, the randomized experiment. The randomized study is considered the gold standard for studying interventions, both in applied settings and more basic research settings. Boruch et al. provide justifications for this widespread belief, noting the investigations that have demonstrated the relative strengths of randomized studies over quasi-experiments. However, implementing a randomized design in field settings is difficult. Through the use of multiple examples, the chapter describes some of the best ways to implement this design. The authors note the need to conduct pipeline studies, as well as the need for careful attention to the ethical concerns raised by randomized experiments. They also discuss the management requirements of a randomized design and issues concerning the reporting of results. Through the use of examples they illustrate how to plan and implement a randomized experiment.

Although randomized experiments represent the gold standard, it is not always possible to conduct such research. In Chapter 6, Mark and Reichardt move us from the simpler, but elegant, randomized design to a discussion of quasi-experiments. They reconceptualize the traditional ways of thinking about the several forms of validity. Their approach clarifies many of the problems of previous schemes for describing the variety of quasi-experiments. Chapter 6 can serve as a guide for researchers who want to avoid some of the difficulties in planning quasi-experiments and interpreting their results.

When randomized experiments are not feasible to establish causality then it may be possible to implement a quasi-experiment. Mark and Reichardt provide a grand tour of the variety of quasi-experiments that can be used. However, the authors are quick to point out the limitations that all quasi-experiments have. The weakest ones, that they call queasy-experiments, have many alternative explanations for an effect other than the intervention. However, there are quasi-experiments such as the regression-discontinuity design and the interrupted time series design with a control group that can be used with more confidence that the results obtained are caused by the intervention. In addition to alternative designs the authors review several statistical techniques that can improve the strength of both randomized and quasi-experiments. In the end, however, it is the logic of the study and the insight and creativity of the researcher that provides the basis for causal conclusions.

In Chapter 7, Maxwell presents a new model of research design, representing the logic and process of qualitative research. Calling it an "interactive" model of research design, Maxwell outlines five key components in the model: goals, conceptual framework, research questions, methods, and validity. Although these components are common to other models of applied research design, Maxwell contends that what is unique is conceptualizing the relationships among the components as forming an integrated and interacting whole. For example, research questions should be related to the study purposes and informed by the conceptual framework. Similarly, the purposes should be informed by the conceptual knowledge, and the relevant theory and knowledge needed to develop the conceptual framework should be driven by the purposes of the research and the questions that are posed. Qualitative design is consequently flexible, due to the simultaneous nature of many of the research activities. Despite this flexibility, Maxwell demonstrates, it is important for the researcher to have an underlying scheme that provides some guidance for a coherent research study. For example, the researcher may have a tentative plan that has considerable detail for sampling (times, settings, people), data collection, and analysis, but should remain open to revising these based on emergent insights as the study unfolds. Maxwell provides considerable attention to these design decisions, especially those about data analysis, as they are key to research planning and also need to be reconsidered throughout the study.

Yin's contribution in Chapter 8 concentrates on helping researchers improve their practice of case study research. In contrast to the chapter in the first edition of the *Handbook*, this chapter does not provide a full overview of case study topics, but rather is focused on four steps that are the most challenging. First, Yin reviews practical and substantive considerations for defining and selecting the case for a case study. He then discusses how to strengthen the case study by incorporating two or more cases in the same study, and using replication logic to expand the generalizability of the findings. A third step in the case study approach that has proved challenging is collecting the evidence needed. The goal is to collect and integrate multiple sources of credible data that will ideally triangulate, and thus provide a stronger evidentiary base for the findings. Possible data sources include direct observations, archival records, and interviews. Finally, a methodological analysis of these data, using qualitative and/or quantitative methods, will then lead to more defensible findings and conclusions. Yin provides four examples of analytic strategies, including pattern-matching, explanation building, chronological analysis, and constructing and testing logic models. The chapter draws upon numerous examples from several fields to cover these topics and provide concrete and operational advice for readers.

In Chapter 9, Tashakkori and Teddlie note the increasing frequency of mixed methods designs in applied social research. The widespread population of mixed methods is seen in the number of texts written, the growing number of references on the internet, and even a journal devoted to the field, *Journal of Mixed Methods Research.* The authors broadly define mixed methods as research in which the researcher collects and analyzes data from both qualitative and quantitative approaches, integrates the findings and draws inferences from the analysis. In this chapter, the authors begin by offering the assumptions that guide their approach to mixed methods, with an emphasis on believing that qualitative and quantitative methods are not dichotomous or discrete, but are on a continuum of approaches. They then provide an overview of various integrative approaches to sampling, data collection, data analysis, and inferences, and end with a discussion of the issues involved in evaluating the inferences made based on the results.

Michael Harrison in Chapter 10 offers an introduction to organizational diagnosis, the use of conceptual models and applied research methods to conduct an assessment of an organization that can inform decision-making. Similar to evaluation research, organizational diagnosis is practically oriented and can involve a focus on both implementation and effectiveness. What distinguishes organizational diagnosis is that its focus is typically broader than a program evaluation, with an examination of organizational features and a wide range of indicators of effectiveness. To provide both useful and valid information for a client, Harrison highlights three key aspects of diagnosis—process, modeling, and methods. Process involves interacting with the clients and other stakeholders over the course of a study. Modeling refers to using research-based models to guide the study, including models and frames for identifying what to study, framing the problem, choosing effectiveness criteria, determining which organizational conditions to examine for their influence on effectiveness, and organizing and providing feedback to the clients. Methods refers to techniques for gathering, summarizing, and analyzing data that can provide both rigorous and valid results. Harrison stresses that there is no step-by-step guide to conducting a diagnosis, but rather a set of choices that the diagnosis practitioner must make. The ultimate task is to use methods and models from the behavioral and organization sciences to help identify what is going on in an organization and to help guide clients in making decisions based on this information.

As we noted in our introduction, a major theme of this handbook is the importance of accumulating knowledge in substantive areas so as to make possible more definitive answers to key questions. Do we have the tools and methods in applied research to pull together the vast number of studies that have been completed? In Chapter 11, Cooper, Patall, and Lindsay summarize a number of useful meta-analytic techniques to produce quantitative summaries of often hundreds of studies. Although most of these techniques have been developed in the past 20 to 25 years, the authors, in a brief history of research synthesis and meta-analysis, note that the first meta-analysis was actually published in 1904 by Karl Pearson and was followed by more than a dozen papers on techniques for statistical combination of findings prior to 1960. In recent years, there has been an explosion of meta-analyse published, and two networks—the Cochrane Collaboration and the Campbell Collaboration—are the leading producers in research syntheses in health care and social policy, respectively, and are considered the gold standard for determining the effectiveness of different interventions in these areas.

In addition to presenting a brief history of the method and an overview of a number of statistical strategies for combining studies, Cooper et al. review the stages of research synthesis, including problem formulation, literature search, data evaluation, analysis and interpretation, and public presentation. With an overriding purpose of the chapter to help researchers distinguish good from bad syntheses, the authors discuss the difficult decisions that researchers face in conducting a meta-analysis (e.g., handling missing data), and address the criteria that need to be considered in evaluating the quality of both knowledge syntheses more generally and meta-analysis in particular.

CHAPTER 5

Randomized Controlled Trials for Evaluation and Planning

Robert F. Boruch

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Julia Littell

Suppose you were asked to determine the effectiveness of a new police strategy to reduce crime and disorder at crime hot spots. The police had determined that a limited number of blocks in the city were responsible for a large proportion of crime and disorder and had decided to crack down on those high crime areas. The strategy involved concentrating police patrol at the hot spots, rather than simply having the police spread their resources thinly across the city. A study of the topic would require comparing the crime rates and disorder at the hot spots after police intervention, with rates of crime at places that did not receive the intervention. The study's objective is to establish whether concentrating patrol at hot spots will reduce crime and disorder at those places.

In an uncontrolled or observational study, particular hot spots would be targeted based on the preferences of police commanders who are often pressured by citizens to do something about crime on their block. This *selection factor,* born of commanders' preferences, leads to two groups of hot spots that are likely to differ systematically. Those hot spots that receive the innovative policing program may, for example, have higher rates of crime or disorder. The targeted hot spots, for instance, may be places with wealthier citizens who are perhaps more able to apply pressure to the police, or places in which citizens are simply better organized and, thus, with more contacts with the department. They may be in certain areas of the city where police patrol is ordinarily concentrated, or areas close to businesses, schools, or community centers that are seen as deserving special police attention. Each of these factors, of course, may influence the primary outcomes of interest—crime and disorder—and may affect how effective or ineffective the police are in doing something about these problems.

The differences between the groups that evolve from natural processes, rather than a controlled study, will then be inextricably tangled with the actual effect of police patrol on crime, if indeed there is an effect. A simple difference in crime between the two naturally occurring (nonrandomized) groups, one that received the intervention and one that did not, will not then register the effect of the intervention alone. It will reflect the effect of police patrol at hot spots and the combined effect of all selection factors: commanders' preferences, political clout, socioeconomic factors, the location of institutions thought important to the police, and so on. As a consequence, the estimate of the effect of police patrol at hot spots based on a simple difference between the groups is equivocal at best. It may be misleading at worst.

Crime in the self-selected hot spots policing area, for instance, may be higher following the intervention, making it appear that hot spots patrol increases crime, when in fact it had no effect. For example, burglaries may be higher in the hot spots patrol area because the places targeted included people with higher incomes. Their relative wealth might have given them preference when the program was initiated, but it also might suggest higher burglary rates since such places will naturally be more attractive targets—they have more goods that can be stolen. The point is that a simple observational study comparing crime hot spots that received extra patrol and those that did not will yield a result that cannot be interpreted easily.

Eliminating the selection factors in evaluations that are designed to estimate the relative effectiveness of alternative approaches to reducing the incidence of violence is difficult. Hot Spots Policing experiments described by Weisburd (2005) met this challenge through randomized trials. Related kinds of selection issues affect nonrandomized studies that are used to assess the impact of initiatives in human resources training programs, health care, education, and welfare, among others. It also affects studies that purport to match places or individuals in each group to the extent that matching is imperfect or incomplete in ways that are unknown or unknowable.

That many applied research and evaluation projects cannot take selection factors into account does not mean such studies are useless. Some of them are, of course. It does imply that, where appropriate and feasible, researchers ought to exploit valid methods for estimating the relative effects of initiatives, methods that are not vulnerable to selection problems and do not lead to estimates that are equivocal or biased in unknown ways. Randomized field trials, the focus of this chapter, are less vulnerable to such problems.

This chapter covers basic definitions and aims of randomized trials and the distinction between this approach and others that purport to estimate effects of interventions. Illustrations are considered next, partly to show how trials are mounted in different arenas, partly to provide evidence against naive academic, institutional, and political claims that such trials are not feasible. We next consider basic ingredients of a randomized trial; each ingredient is handled briefly. The final section summarizes efforts to develop capacity. This chapter updates one that appeared in the earlier edition of Bickman and Rog (1998); the update is no easy task given the remarkable expansion in trials over the past decade in education, crime and justice, social services, and other areas.

Randomized Controlled Trials: Definitions and Distinctions

In the simplest randomized trial, individuals or entities are randomly assigned to one of two or more intervention groups. The groups so composed are, roughly speaking, equivalent. In statistical parlance, they do not differ systematically although they may differ by chance.

The various interventions that are applied to each group may eventually produce an important difference in the state of each group, the trial being designed so as to discern such a difference. In the Minneapolis Hot Spots Patrol Experiment (Sherman & Weisburd, 1995), for instance, crime hot spots were randomly assigned either to an experimental group that received greater police patrol than normal, or to a control group that received a standard emergency police service. In this latter "control" condition, police did not visit the hot spots unless citizens called the police for assistance. The object was to learn whether increased and targeted police patrol would reduce crime and disorder at crime hot spots.

In some trials, a sample of individuals, institutions, or entities may be matched into pairs or put into similar categories and then randomly assigned to intervention conditions. Such strategies can maximize the statistical power of an experiment. That is, intervention effects are rendered more detectable in a trial that employs matching, blocking, or other precision-enhancement tactics. In the Minneapolis Hot Spots Patrol Experiment, the researchers were particularly concerned that the two groups studied would be as alike as possible and that the trial would yield evidence about a dependable effect. Accordingly, they first divided the hot spots into groups based on how much crime and disorder had been found in prior years. Regardless of these tactics, the randomization assures that unknown influences on behavior do not differ on average across the intervention groups, including the control group, regardless of the effectiveness of matching.

The first of two principal benefits of randomized trials is that they permit fair comparison. That is, the statistical estimates of the intervention's effect will not be tangled with competing explanations of what caused the difference in observed outcome. The virtue of a comparison that leads to clearly interpretable results was recognized more than a century ago by Jastrow and Pierce in psychophysical laboratory experiments. It is a virtue in planning and evaluating programs in the social arena to judge from increased use of the randomized trials in policy research.

The second benefit of a randomized trial is a statistical statement of the researcher's confidence in the results. This depends on recognizing that the trial's results are subject to ordinary variability in human, organizational, and regional behavior and, moreover, that this variability needs to be taken into account. The ability to make such a statement is important on scientific grounds. We know that we will err, at times, in judging a treatment's effectiveness simply because ordinary chance variability can be substantial. Understanding the character of the random error and delimiting its magnitude are then important.

Texts on designing trials that involve individuals as the units of random allocation are readily accessible. Newer technical books handle scenarios in which entities, such as schools or hospitals, are the randomization units, and outcome data on individuals or other entities at the entity level and within the entities are also analyzed (see, e.g., Bloom, 2005; Donner & Klar, 2000; Murray, 1998; Raudenbush & Bryk, 2002). A special issue of the *Annals of the American Academy of Political and Social Science* dedicated cross-discipline and cross-national attention to the topic (Boruch, 2005).

Distinctions

Randomized trials are different from *observational studies* in which there is an interest in establishing cause-effect relations, but there is no opportunity to assign individuals to alternative interventions using a randomization plan (Cochran, 1983; Rosenbaum, 2002). Such studies are often based on survey samples and depend on specialized methods for constructing comparison groups and estimates the effects of interventions.

Observational studies can and often do produce high-quality descriptive data on the state of individuals or groups. They can provide promissory notes on what works or what does not, conditional on assumptions that one might be willing to make. They cannot always sustain defensible analyses of the relative effects of different treatments, although they are often employed to this end. Statistical advances in the theory and practice of designing better observational studies, and in analyzing resultant data and potential biases in estimates of an intervention's effects, are covered by Rosenbaum (2002).

Randomized field tests also differ from quasi-experiments. Quasi-experiments have the object of estimating the relative effectiveness of different interventions that have a common aim, just as randomized experiments do. But the quasi-experiments depend on methods other than randomization to rule out competing explanations for differences in the outcomes of competing interventions or to recognize bias in the estimates of a difference. In some respects, quasi-experiments aim to approximate the results of randomized field tests (Campbell & Stanley, 1966; Cochran, 1983; Shadish, Cook, & Campbell, 2002).

Important statistical approaches have been invented to try to isolate the relative effects of different interventions based on analyses of data from observational surveys and quasi-experiments of the interventions. These approaches attempt to recognize all the variables that may influence outcomes, including selection factors to measure them and to separate the intervention effects from other factors. Advances in this arena fall under the rubrics of *structural models, selection models,* and *propensity scores.* Antecedents and augmentations to these approaches include ordinary least square regression/covariance analysis and matching methods.

The scientific credibility of some of these techniques is reviewed on empirical grounds by Glazerman, Levy, and Myers (2003) in the context of employment, training, and education. See Weisburd, Lum, and Petrosino (2001) for criminological research comparing results of randomized trials, including quasi-experiments with the results of nonrandomized trials; and Chalmers (2003) and Deeks et al. (2003) for analogous comparisons of studies of effects of health interventions. Victor's (2007) dissertation gives a review of statistical matching methods in quasi-experiments and reports on simulation studies on how propensity scores and ordinary least squares regression can produce better estimates of effect than competing models/analyses in such quasi-experimental designs. The general conclusion one reaches based on such empirical work is that estimates of an intervention's effect based on randomized trials often differ in both magnitude and variability from those based on nonrandomized studies. The reasons for such differences are an important target for new methodological research.

In this chapter, the phrases *randomized experiment* and *randomized trial* will be used interchangeably with other terms that have roughly the same meaning and are common in different research literatures. These terms include *randomized test* and *randomized social experiments,* used frequently during the 1970s and 1980s. They also include *randomized clinical trials,* a phrase often used to describe the same design for evaluating the relative effectiveness of medical or pharmaceutical treatments, for example, Piantadosi (1997) and Donner and Klar (2000). Similarly, the phrases "cluster randomized," "place randomized," and "group randomized" are used interchangeably when independent entities or independent assemblies of related individuals or entities are randomly assigned to different regimens.

Experiments in Context

The main benefit of a randomized trial is unbiased estimates of the relative effect of interventions coupled with a statistical statement of one's confidence in results. The benefit must be put into the broader context of applied social research, of course. Addressing questions about the nature of the phenomenon or problem at hand, and producing evidentiary answers, precedes any good trial. Determining how interventions may be constructed and deployed, and generating evidence on such determinations, must also precede such trials. It is only after such questions are addressed that it makes sense to undertake controlled trials so as to answer questions about "effect."

Understanding which questions to address, in what conditions, and when, is an ingredient of research policy. The need to arrange one's thinking about this understanding has been reiterated and elaborated in recent tracts on applied research on crime prevention (Lipsey et al., 2005), in education (Shavelson & Towne, 2002), and in the context of federal policies more generally (Julnes & Rog, 2007). The message in these and others is that the question to be addressed drives the methods to be used to generate dependable evidence.

Further questions depend on having answered questions about problem scope, program deployment, and program effect. "What is the cost effectiveness ratio for programs that have been tested? How can the evidence on any question be employed well in systematic reviews, legislation, and generation of practice guidelines? How can the trialists keep abreast of the state of the art in each question category?"

This chapter focuses mainly on randomized trials. Other chapters in this *Handbook* carry the weight in addressing other related topics. See also Rossi, Lipsey, and Freeman (2004) and Stufflebeam and Shrinkfield (2007) for randomized trials in a broader evaluation context.

Illustrative Experiments From Different Fields

Randomized trials in the health arena are far more common than in the social sector. The Cochrane Collaboration's register on reports of such trials, for instance, includes about 500,000 entries (http://cochrane.org). The Campbell Collaboration's newer and, therefore, more incomplete register in the social sector includes about 14,000 entries on reports on randomized and possibly randomized trials (http://campbellcollaboration.org).

Regardless of this disparity, the number of randomized trials in the social, educational, criminal and civil justice, and other arenas is increasing (see the charts in Mosteller & Boruch, 2005). The following section provides illustrations from different fields partly to emphasize the idea that the method transcends academic discipline and geopolitical boundaries.

Education

In education as in other arenas, researchers may randomly allocate eligible and willing teachers, individuals, classrooms, schools, and other entities to different interventions in order to produce good estimates of their relative effectiveness. The choice of the experiment's *unit* of assignment in education, as in other social sectors depends on the nature of the intervention and on whether the units can be regarded as statistically independent. For instance, entire schools have been randomly assigned to alternative regimens in dozens of studies designed to determine whether schoolwide campaigns could delay or prevent youngsters' use of tobacco, alcohol, and drugs (e.g., Flay & Collins, 2005). In a milestone experiment on class size, students and teachers were randomly assigned to small classes or to regular classes in Tennessee to learn whether smaller classes would yield higher achievement levels and for whom (Finn & Achilles, 1990; Mosteller, Light, & Sachs, 1995). See Stufflebeam and Shinkfield (2007) for a description of this and other remarkable precedents.

Over the past decade, the role of randomized trials in education has changed remarkably. Between 1999 and 2006, for instance, the Interagency Education Research Initiative funded about 20 small-, moderate-, and large-scale trials. This joint effort to develop and evaluate programs in science, mathematics, and reading involved thousands of students in at least a dozen states over 5 years (Brown, McDonald, & Schneider, 2006). In the United States, the Institute of Education Sciences (IES) began in 2000 to lead the way toward more dependable evidence on effects of interventions on randomized trials in the face of notable criticism. The IES Director's Report to the Congress notes that only *one* substantial trial was underway in 2000 (U.S. Department of Education, 2007). Spybrook's (2007) fine dissertation on statistical power in certain kinds of trials identified nearly 60 trials supported by IES between 2001 and 2006. This is a lower bound on the number of recently sponsored trials in that Spybrook focused only on group randomized trials in her research and could not handle trials undertaken by Regional Education Laboratories during 2006–2007.

William T. Grant Foundation (2007) played a leadership role in the private foundation sector through its support of randomized trials and its building the research community's capacity to implement such trials. After school programs and summer programs in math and reading, for instance, have been a special focus. Large-scale cluster trials have been supported on schoolwide mentoring, socioemotional learning, literacy, positive youth development, school-based prevention, and reading.

Welfare, Employment, and Training

Moffitt's (2004) tidy but selective review of randomized trials in the welfare sector covers the 1960s, 1970s, and 1980s. For Moffitt, and other welfare and labor economists, the interventions subject to evaluation include tax plans and guarantees, structured and conditional job search, job training, education, case management, among others. Each intervention of course can be packaged in different ways and with different incentives.

The geographic scope of these economic experiments has been substantial. Moffitt's (2004) listing identifies more than 20 massive trials. They include one "national" randomized trial and several regional ones, and state- (or municipality-) based trials run in California and Maine, Washington, and the Virginias, and New Jersey/New York to Florida.

Moffitt (2004) reminds us that "few" randomized trials have been conducted in this welfare arena since the 1990s. Bloom's (2005) book covering newer trials, supported by both federal and foundation (Rockefeller) resources, is an important resource. Moffitt (2004) does explain that a plausible reason for the decline is the shift, since 1996, from federal to state responsibility for evaluating the effects of interventions. An implication is that state agencies in the United States need to educate themselves about evidence. Here, we acknowledge the state of Washington's remarkable leadership role (Aos, 2007).

The randomized experiments in this sector eliminated the problem of creaming—that is, selecting superior applicants in the interest of making a program look good. This selection factor was chronic in evaluations of welfare and employment programs prior to the 1980s. Furthermore, the randomization helped to avoid a major problem encountered in earlier attempts to evaluate such programs. In particular, it is difficult or impossible to disentangle the average effect of a new program from the characteristics of eligible individuals who elect (or do not elect) to enter a new program, unless a controlled randomized trial is done.

Tax Administration

The interests of the U.S. Internal Revenue Service (IRS) and of tax agencies in other countries lie partly in understanding how citizens can be encouraged to pay the proper amount of taxes. Randomized trials in this arena have also been undertaken. For example, delinquent taxpayers identified by the IRS have been randomly assigned to different strategies to encourage payment, and they are then tracked to determine which strategies yielded the best returns on investment (Perng, 1985). Other experiments have been undertaken to determine how tax forms may be simplified and how taxpayer errors might be reduced through various alterations in tax forms (e.g., Roth, Scholz, & Witte, 1989). Such research extends a remarkable early experiment by Schwartz and Orleans (1967) to learn how people might be persuaded to report certain taxable income more thoroughly. In an ambitious update of this work, Koper, Poole, and Sherman (2006) focused on 7,000 businesses in Pennsylvania that had not complied with the state's sales tax code. Moral appeals, personal letters, as well as threats were tested in a randomized trial to understand whether they have appreciable effects on payment.

Civil and Criminal Justice

The Minneapolis Hot Spots Patrol Experiment was designed to determine how police patrol resources could be best allocated to do something about crime problems (Sherman & Weisburd, 1995). The study used computerized mapping of crime calls to identify 110 "hot spots," roughly of street block length. Police patrol was doubled on average for the experimental sites over a 10-month period. The object was to determine whether the increase in police patrol would lead to a significant relative decline in crime and disorder in the places where police were concentrated. While this theory is now well supported through fair randomized trials, when the study was conducted, there was widespread belief among scholars and the police, and evidence from a series of observational studies, that police patrol, however applied, would not have an impact on crime (Bayley, 1994; Gottfredson & Hirschi, 1990).

The credibility of the Minneapolis results depended heavily on the random allocation of cases assigned to the new intervention versus the control condition (ordinary patrol practice). That is, the cases in the intervention and control groups did not differ systematically on account of the random assignment. Competing explanations that were common in earlier nonrandomized studies could then be ruled out. The study found that the experimental intervention as compared with the control condition experienced statistically significant reductions in crime calls and observed disorder. The Minneapolis Experiment served to challenge the strongly held assumption that the police could not have substantive impact on crime problems and led to a series of experimental studies of crime hot spots (see Braga, 2005; Weisburd & Eck, 2004). In good part because of the experimental evidence for hot spots policing, a recent review of police practices and policies by a National

Academy of Sciences panel concluded that "studies that focused police resources on crime hot spots provide the strongest collective evidence of police effectiveness that is now available" (Skogan & Frydl, 2004; see also Weisburd & Eck, 2004).

Trialists have undertaken several substantial reviews of randomized field experiments in civil and criminal justice. Dennis (1988), for instance, analyzed the factors that influenced the quality of 40 such trials undertaken in the United States. His dissertation updated Farrington's (1983) examination of the rationale, conduct, and results of randomized experiments in Europe and North America. Farrington and Welsh's (2005) review covers more than 80 trials.

The range of interventions whose effectiveness has been evaluated in these randomized controlled trials is remarkable. They have included efforts to appraise relative effects of different appeals processes in civil court, telephone-based appeals hearings, restorative justice programs, victim restitution plans, jail time for offenders, diversion from arrest, arrest versus mediation, juvenile diversion and family systems intervention, probation rules, bail procedures, work-release programs for prisoners, and sanctions that involve community service rather than incarceration.

Abused and Neglected Children

A stream of randomized controlled experiments has been undertaken to understand how to prevent out-of-home placement of neglected and abused children. In Illinois, for instance, the trial involved randomly assigning children at risk of foster care to either conventional welfare services, which include foster care, or a special Family First program, which leaves the child with the parents but provides intensive services from counselors and family caseworkers. Related research has been undertaken in many states (Littell & Schuerman, 1995). Schuerman, Rzepnicki, and Littell (1994), who investigated the Illinois experiment, found that the program was actually targeted at families with children at low risk of out-of-home placement, rather than high-risk families, virtually guaranteeing that no treatment differences would appear in placement outcomes. The need to produce good evidence in this arena is driven partly by political and professional interest in learning whether foster care can be avoided. Following the Illinois Family First trial, the U.S. Department of Health and Human Services funded similar experiments in multiple sites in New Jersey, Kentucky, Tennessee, and Pennsylvania (Westat, Inc., 2002).

Nutrition

With rates of obesity approaching 20% for children and 60% for adults in the United States, there is increasing interest in understanding effective prevention and intervention strategies (University of Virginia Health Systems, 2008). Programs that demonstrably prevent overweight and obesity are of interest in school and community settings. As a result, randomized trials have been undertaken to assess school-based nutrition education and environmental change efforts, programs to maximize nutrition and health prevention efforts among those receiving federal assistance program benefits and work site interventions.

For example, a randomized longitudinal study of schools participating in an innovative approach to health promotion and obesity prevention was undertaken in public schools in Philadelphia to understand the extent to which the program was able to effectively prevent youth in grades 4 to 8 from becoming overweight or obese (Foster et al., 2006). This School Nutrition Policy Program involved teacher training and integration of nutrition education into the curriculum and the reduction of high-sugar beverages and high-fat snack foods in the school cafeterias. The study first identified criteria for a school's inclusion, such as serving a primarily low-income community. Schools were then matched based on characteristics of cafeteria and randomly assigned to a control or treatment condition. This work builds on groundbreaking experimental studies of school-based nutrition work conducted by Gortmaker and colleagues (1999) in the study of Planet Health, as well as that of Nicklas, Johnson, Myers, Farris, and Cunningham (1998) assessing outcomes of Gimme 5.

A place randomized trial of the Teens Eating for Energy and Nutrition in Schools (TEENS) investigated the differential impact of varying components of a school-based nutrition program. The researchers assigned schools to either a control or intervention group and students within intervention schools to one of three intervention types: (1) school environment interventions only, (2) classroom curriculum plus school environment interventions, or (3) peer leaders plus classroom curriculum plus school environment interventions. Findings showed stable consumption patterns among control school students and incremental differences in consumption patterns among the group assignments (Birnbaum, Lytle, Story, Perry, & Murray, 2002).

Efforts to understand the usefulness of programs aimed at increasing positive dietary behaviors, such as fruit and vegetable consumption, have been undertaken among those with young children receiving benefits from the Woman and Infant Care (WIC) program. For example, a study of the Maryland WIC Food for Life program (Havas et al., 2003) used a randomized crossover design in which each WIC site was the unit of analysis and served as its own control. Half of the sites were randomly assigned to receive the intervention, while the other half remained as controls. After the first 12 months of the program, the assignments switched. Results identified changes in consumption of fruits and vegetables, fiber intake, and the percentage of calories derived from fat.

The interest in studying the impacts of interventions that aimed at improving diets also includes several studies of workplace interventions for middle-income populations. Programs such as Working Well (Sorensen et al., 1996) and The Next Step Trial (Tilley et al., 1999) randomly assign workplace sites to receive a program consisting of educational outreach, food sampling, changes in food availability, and self-help materials.

Marketing and Campaign Research

The for-profit business sector is vulnerable to many of the same evidential issues as the public sector is. As Pfeffer and Sutton (2006) suggest, managers are often confronted with heaps of data whose quality is variable, whose relevance is often unclear, and in which biases may be chronic. Their paper reviews some efforts to mount controlled trials in the interest of better evidence for evidence-based management. Such trials, they report, have been undertaken at times to understand the effects of different marketing strategies in the hotel and legal gambling business, global Web-based services industry, convenience store chains, and elsewhere. Individual customers, or corporate units, or stores, etc. may be the units of random allocation. The authors' examples are brief but provocative. However, it is difficult to gauge scope and quality in this arena of applications on account of proprietary aspects of the research.

In another kind of marketing arena, Gerber (2004) points out that "virtually all of the work on candidate spending effects have been based on non experimental evidence" (p. 544). His article reviews the few and very recent efforts to assess effects of political campaign spending (and different campaign programs) on vote share, voter preferences, and other election outcomes. Randomization in some experiments is at the household level; in others it is at the ward level. Gerber's handling of the topic is distinctive in trying to synthesize and reconcile results of both the trials and related nonexperimental studies and building more nuanced theory (models) of when and how incumbent spending has positive, negative, or no effects.

Elements of a Randomized Controlled Trial

The basic elements of a randomized test for learning "what works" or "what works better" are discussed briefly in this section. The description is based mainly on controlled field tests of hot spots policing. Other substantive examples, such as tests of employment and training, and education projects, are used to reiterate the fundamental character of the elements. In general, the elements of a randomized field experiment are as follows:

- The basic questions and the role of theory
- Theory: statistical and substantive
- Assuring ethical propriety
- The experiment's design
- Management
- The analysis and reporting of the results

The first three topics are considered next. The subsequent topics are considered in the following section under the rubric of the Experiments Design.

The Basic Question and the Role of Theory

Put bluntly, the questions best addressed by a randomized controlled experiment are as follows: What works better? For whom? And for how long? The primary question must, of course, be framed more specifically for the particular study. Secondary questions are often important for science or policy and their lower priority needs to be made plain.

In the Hot Spots Patrol experiments, for example, the *primary* question was, "Does the focus of police resources such as preventative patrol in specific areas where crime is high, as opposed to a more even spread of policing activities in a city, lead to crime prevention benefits?" The question was developed from theoretical debate and empirical evidence that crime is tightly clustered in urban areas and that such clustering is due to the presence of specific opportunities for crime and the presence of motivated offenders at crime hot spots.

Cohen and Felson's (1979) theory of routine activities was an important catalyst for the hot spots policing studies (Weisburd, 2005). Prior theorizing in criminology had focused on individual offenders and the possibilities for decreasing crime by focusing criminal justice resources either on their incapacitation, rehabilitation, or in deterring them from future offending. This offender-based criminology dominated crime and justice interventions for most of the past century, but it was criticized extensively beginning in the 1970s for failing to provide the crime prevention benefits that were often promised (Brantingham & Brantingham, 1975; Martinson, 1974). Cohen and Felson (1979) observed that for criminal events to occur, there is need not only of a criminal but also of a suitable target and the absence of a capable guardian. Their theory suggested that crime rates could be affected by changing the nature of targets or of guardianship, without a specific focus on offenders themselves. Drawing on similar themes, British scholars led by Ronald Clarke began to explore the theoretical and practical possibilities of situational crime prevention (Clarke, 1983, 1992, 1995; Cornish & Clarke, 1986). Their focus was on criminal contexts and the possibilities for reducing the opportunities for crime in very specific situations. Their approach, like that of Cohen and Felson, placed opportunities for crime at the center of the crime equation. One natural outgrowth of these perspectives was that the specific places where crime occurs would become an important focus for crime prevention researchers (Eck & Weisburd, 1995; Taylor, 1997).

In the mid- to late 1980s, a group of criminologists began to examine the distribution of crime at places such as addresses, street segments and small clusters of addresses or street segments. Perhaps the most influential of these studies was conducted by Sherman, Gartin, and Buerger (1989). Looking at crime addresses in the city of Minneapolis, they found a concentration of crime there that was startling. Only 3% of the addresses in Minneapolis accounted for 50% of the crime calls to the police. Similar results were reported in a series of other studies in different locations and using different methodologies, each suggesting a very high concentration of crime in microplaces (e.g., see Pierce, Spaar, & Briggs, 1986; Weisburd, Bushway, Lum, & Yang, 2004; Weisburd & Green, 1994; Weisburd, Maher, & Sherman, 1992). This empirical research reinforced theoretical perspectives that emphasized the importance of crime places and suggested a focus on small areas, often encompassing only one or a few city blocks that could be defined as crime hot spots.

While the Minneapolis Hot Spots Patrol Experiment (Sherman & Weisburd, 1995) examined whether extra police presence would have crime prevention impact at hot spots, other studies began to study whether different types of police strategies such as problem-oriented policing would enhance crime prevention benefits at hot spots (see, e.g., Braga, Weisburd, Waring, & Mazerolle, 1999; Weisburd

& Green, 1995). Importantly, later studies also examined the theory that crime would simply be displaced to other areas near the targeted hot spots. If crime simply "moved around the corner," then such hot spots approaches would not be very useful for decreasing crime and disorder more generally in a city (Weisburd et al., 2006). In the Jersey City Drug Market Analysis Experiment (Weisburd & Green, 1995), for example, displacement within two block areas around each hot spot was measured. No significant displacement of crime or disorder calls was found. Importantly, however, the investigators found that drug-related and public morals calls actually declined in the displacement areas. This "diffusion of crime control benefits" (Clarke & Weisburd, 1994) was also reported in the New Jersey Violent Crime Places experiment (Braga et al., 1999) and the Oakland Beat Health experiment (Mazerolle & Roehl, 1998).

Rossi et al. (2004) and Stufflebeam and Shinkfield (2007) elaborated on the role of theory in the context of randomized trials and *other* types of evaluation that address questions that precede or succeed an impact evaluation. Wittman and Klumb (2006) provided counsel about how researchers might deceive themselves about testing theory in the context of randomized experiments considering the topics of history since the 1950s.

Theory: Statistical and Substantive

Contemporary statistical textbooks on the design of randomized experiments do not often handle the substantive theory or logic model underlying the relation between the intervention being tested in an experiment and the intervention's expected outcomes. Statistical texts depend on basic statistical theory. Nonetheless, the substantive theory must be addressed.

A substantive theory (or several theories) should drive the selection of interventions that are tested in randomized trials. For example, a rudimentary routine activities theory helped the researchers in the Minneapolis Hot Spots Patrol Experiment to identify increased police patrol as a potentially effective approach for reducing crime at hot spots. The theory predicted that increased guardianship at hot spots, as evidenced by increased police patrol presence, would lead to less crime and disorder. The Jersey City Drug Market Experiment (Weisburd & Green, 1995) drew on elements of situational crime prevention to develop a series of police interventions at drug hot spots, including not just crackdowns and increased guardianship but also cooperation between store owners and the police, and environmental interventions to reduce opportunities for crime.

Both statistical theory and substantive theory must also drive the identification of the units of allocation in a randomized field experiment. Good substantive theory or a logic model, for instance, posits plainly who or what should be the target of the program and, by implication, the unit of random allocation in a trial. Statistical theory is pertinent here inasmuch as statistical analyses depend on the assumption that the units of allocation in the experiment are independent of one another (Mosteller, 1986). When they are not independent, specialized analyses are necessary to take the dependence among units into account (e.g., Hedges & Hedberg, 2007; Raudenbush & Bryk, 2002).

Substantive theory, implicit or explicit, also drives the choice of outcome variables to be measured in a randomized trial. In the Crime Hot Spots experiments, researchers relied on emergency calls for police service to measure program outcomes, because such calls were seen as a direct measure of criminal activity in the hot spots. The question was not whether individual offenders reduced their motivations to commit crime which would have been best noted in surveys or interviews with offenders, but whether crime and disorder was reduced. In Tennessee's experiments on class size, Finn and Achilles (1990) measured student achievement as an outcome variable based on theory and earlier research about how class size might enhance children's academic performance.

Well-articulated theory can also help to determine whether and which context (setting) variables need to be measured. For instance, most trials on new employment and training programs have measured the local job market in which the program is deployed. This is based on rudimentary theory of demand for and supply of workers. Knowing that there are no jobs available in an area, for example, is important for understanding the results of a trial that compares wage rates of participants in new training programs against wages of those involved in ordinarily available community employment and training programs.

Finally, theory may also drive how one interprets a simple comparison of the outcomes of two programs, deeper analyses based on data from the experiment at hand and broader analyses of the experiment in view of research in the topical area generally. Rossi et al.'s (2004) discussed different kinds of hypotheses. The implication is that we ought to have a theory (an enlarged hypothesis or hypothesis system) that addresses people and programs in the field, a theory about the interventions in the trial given the field theory, and a theory about what would happen if the results of the trial were exploited to change things in the field.

A bottom line for trialists is that the theory or logic about how the intervention is supposed to work ought to be explicit. It is up to the design team for the randomized trial to draw that theory into the open, so as to assure that the trial exploits all the information that must be exploited in designing the trial.

Assuring Ethical Propriety

Whether a randomized trial is ethical depends on a variety of criteria. The medical, social, and behavioral sciences and education have been energetic in producing ethical guidelines for research and monitoring adherence to them. Only two kinds of standards are considered here.

One first set of standards, developed by the Federal Judicial Center (FJC; 1983), involves general appraisal of the social ethics of randomized trials. The FJC's threshold conditions for deciding whether an experiment ought to be considered involve addressing the following questions:

- Is the problem severe and is there need for improvement?
- Is the effectiveness of proposed improvements uncertain?
- Will a randomized experiment yield more defensible evidence than alternatives?
- Will the results be used?
- Will the rights of participants be protected?

Affirmative responses to all these questions invite serious consideration of a randomized trial. Negative answers to all the questions, or most, invite terminating consideration of a randomized trial.

The second set of ethics standards come under the rubric of the institutional review boards (IRBs). In any institution receiving federal research funds in the United States, an IRB is responsible for reviewing the ethical propriety of research, including field experiments. Countries other than the United States, including the Nordic countries and some European Union countries have similar entities. IRB standards and processes are explicit, demanding, and important.

In a series of criminal justice experiments, termed the Spouse Assault Replication Program (SARP; Garner, Fagen, & Maxwell, 1995), researchers tested the impact of arresting offenders for misdemeanor spouse assault, rather than using more traditional approaches (at that time) of simply separating spouses or providing some type of counseling. In these experiments, discussions of each of the FJC's threshold questions were undertaken by the National Institute of Justice (the funder) and its advisers and at the local level, for example, by the Milwaukee City Council, the Milwaukee Police Department, and the city's various advocacy groups. An independent IRB also reviewed the experiment's design in accordance with the federal legal requirement to do so. The principal investigator has the responsibility to explain matters to each group and to develop a design that meets local concerns about the ethical appropriateness of the experiment.

In the Minneapolis Hot Spots Experiment, as in many place randomized trials, fewer ethical questions were raised since the subject of intervention was not individuals but rather places (Weisburd, 2000, 2005). Nonetheless, in Minneapolis, one city council member was concerned that the concentration of police patrol in specific areas of the city might leave other areas unprotected. The researchers in this case agreed to monitor burglary rates, the main crime noted, outside the experimental and control areas so that any spikes in crime could be observed and then dealt with.

Sieber (1992) and Stanley and Sieber (1992; see also Chapter 4, this volume) provide general guidance for meeting ethical standards in social research. In 2006, Joan Sieber created the *Journal of Empirical Research on Human Research Ethics*. This initiative was undertaken to understand whether and how *empirical* research on ethical issues can inform ethical decisions. When, for instance, is informed consent "informed"? When does belief run contrary to dependable evidence? And how do we know? The aim is to help inform dialogue between ethicists and researchers.

The Experiment's Design

The design of a randomized field trial involves specifying the following elements:

- Population, statistical power, and pipeline
- Interventions and methods for their observation
- The method of random assignment and checks on its integrity

- The response or outcome variables and their measurement
- Analysis and reporting

Each of these topics is considered below.

Population, Power, and the Pipeline

Many randomized trials undertaken in the United States focus on individuals as the unit of random allocation to interventions. Many human services programs, for instance, target eligible service recipients in tests of alternative services (e.g., Alexander & Solomon, 2006). Institutions or other entities, at times, are allocated randomly to different regimens in larger-scale trials. Eligible and willing schools, for instance, have been randomly assigned to substance use prevention programs and to control conditions. The policy justification for doing so is that interventions are delivered at the entity level. The statistical justification for randomization at any given level lies in the assumption that the units are independent.

In randomized trials in criminal justice, medicine, employment and training, and other areas, the *target population* depends heavily on theory about what kinds of individuals (or entities) are expected to benefit from the interventions being compared. The SARP studies, for instance, included only adult offenders partly because handling juvenile offenders entails different legal procedures and social values. Similarly, police had to establish the existence of probable cause evidence to believe that a misdemeanor crime had been committed for a case to be eligible, the arrest treatment being irrelevant to noncriminal events (Garner et al., 1995). In contrast, the hot spots policing studies engendered fewer barriers to identifying the population of units for study. They were defined simply as geographic areas with empirically high rates of crime.

Eligibility criteria that are used to define the target population in contemporary trials are usually specified on the basis of relevant law, theory, or regulation. In education, for instance, a decision about whether to randomly allocate schools or classrooms within schools hinges on whether the interest is in the impact of the intervention in schoolwide or in classroomwide effect. At times, the implicit theory is found to be weak once the experiment is done. For instance, Schuerman et al. (1994) discovered that the main eligibility standard for Family First programs in Illinois, a child's "imminent risk of placement" into a foster home as judged by case workers, was of dubious value in identifying such children. That is, children so identified were no more at risk than others in the system not identified as such.

Eligibility and exclusionary criteria substantially define the target population and the sample drawn from it. This in turn helps characterize the generalizability of the trial's results. The criteria also influence the statistical power of the trial through their effect on producing a heterogeneous or a homogeneous sample and their influence on restriction of sample size. It is to this topic that we turn next, emphasizing sample size issues.

Statistical power analysis refers to the experiment's capacity to detect important differences between groups on outcomes of interest. Power depends, of course, on the specific null hypothesis and alternative, and on the particular test statistic and its assumptions, and should be calculated as part of the experiment's design. Indeed, few, if any, trials nowadays are funded under contracts or grants from U.S. federal agencies or by grants from well-informed private foundations such as W. T. Grant unless a competent power analysis is provided in the proposal. Spybrook (2007) reviews such proposals in the context of awards made by the IES. In criminal justice research, the sample size of the Minneapolis Hot Spots Patrol Experiment was chosen and the study funded, following a power analysis requiring that there would be at least an 80% probability of detecting a moderate effect of police patrol on hot spots at an alpha level of 0.10. It is not yet common to incorporate specific information about the reliability of outcome measurement or about the level of deployment of programs into power analyses; doing so is likely to be important in the future.

Commercial software packages, such as *Power and Precision* (www.biostat.com), and high-quality free software, such as *Optimal Design* (www.wtgrantfdn.org), among others, are readily available at the time of writing. The former can be used to calculate statistical power for a large array of experimental and nonexperimental designs in which individuals are the unit of random assignment, and the test is a conventional one on the null hypothesis or one for equivalence. Optimal Design is particularly useful for calculating statistical power for multilevel experiments, for example, in which schools or classrooms are the units of random assignment and students are nested within them. Simple power tables are of course also given in biomedical texts such as Piantadosi's (1997). Schochet (2008) provides tables for complex hierarchical setups. Rules of thumb in simpler designs are important. St. Pierre (2004) reminds us that using a covariate whose correlation with the outcome variable is about .7 can reduce the required sample size by half; this is not a trivial matter. Statistical power issues for hypotheses *other* than the traditional null hypothesis and some related software are considered briefly in Boruch (2007).

Over the past decade, progress in understanding how to enhance power in cluster, group, or place randomized trials has been remarkable. See Raudenbush and Bryk (2002), Bloom, Richburg-Hayes, and Black (2007), and Hedges and Hedberg (2007), and references therein, on the mathematical and empirical underpinnings for planning the use of matching, blocking, and covariance and the role of intraclass correlation. Empirical studies of the statistical power of randomized trials are important, but they are uncommon. Building on earlier work, Spybrook (2007), for example, focused on more than 50 trials in education and found (a) remarkable improvements over a 5-year period and (b) remarkably complex trial designs that depend on more complex power analyses. She also reviewed contemporary empirical bases for calculating statistical power.

A *pipeline study* directs attention to how many individuals or entities or other units of randomization may be entrained in the experiment. Moreover, a pipeline description characterizes the eligible and ineligible target population over time. It helps anticipate the sample size and statistical power that can be achieved.

In the hot spots policing studies and similar place-based randomized trials, for instance, it is often possible to define the number of units in a study with accuracy at the outset. However, in many studies that involve complex chains of events leading to eligibility and eventual engagement, it is important to conduct careful studies of the pipeline of cases. For instance, each of the investigators in the SARP studies (Boruch, 1997; Garner et al., 1995) developed such a study prior to each of six experiments. In most, the following events and relevant numbers constituted the evidential base: total police calls received, cases dispatched on call, cases dispatched as domestic violence cases, domestic cases that were found on site actually to be domestic violence cases, and domestic cases in which eligibility requirements were met. In one site over a 2-year period, for example, nearly 550,000 calls were dispatched; 48,000 of these were initially dispatched as domestic cases. Of these, only about 2,400 were actually domestic disputes and met eligibility requirements. That is, the cases that involved persons in spouselike relationships, in which there were grounds for believing that misdemeanor assaults had occurred, and so on, were far fewer than those initially designated as "domestic" by police dispatchers.

Pipeline studies have been undertaken in other social experiments. See Bickman and Rog (1998; the earlier edition of this *Handbook*) for examples from the 1980s and 1990s. Generally, a pipeline study would describe in quantitative and qualitative terms eligible target populations, obtained samples, and rates of nonparticipation, crossovers, and attrition. St. Pierre (2004) gives informative examples from education and economic trials that would be incorporated into a pipeline study. The pipeline is sufficiently important that CONSORT statement recommends routine reporting on this matter in health care trials (Mohler, Schultz, & Altman, 2001). Flay et al. (2005) make a similar recommendation for the behavioral and education sciences.

Population, power, and pipeline are intimately related to one another in randomized field trials. Considering them together in the study's design is essential. Where this consideration is inadequate or based on wrong assumptions, and especially when early stages of the trial show that the flow of cases into the trial is sparse, drastic change in the trial's design may be warranted. Such changes might include terminating the study, of course. Change might include extending the time frame for the trial so as to accumulate adequate sample sizes in each arm of the trial. Intensifying outreach efforts so as to identify and better engage target cases is another common tactic for assuring adequate sample size.

Interventions

Interventions here mean the programs or projects, program components, or program variations whose relative effectiveness is of primary interest in a randomized trial. In the simplest case, this implies verifying and documenting activity undertaken in *both* the program being evaluated and the control condition in which that program is absent.

Interventions are, of course, not always delivered as they are supposed to be. Math curricula have been deployed in schools but teachers have not always delivered the curriculum as intended. Fertility control devices designed to reduce birthrates have not been distributed to potentially willing users. Human resources training projects have not been put into place, in the sense that appropriate staffs have not been hired. Drug regimens have been prescribed for tests, but individuals assigned to a drug do not always comply with the regimen.

To assure recognition and handling of such a problem, the Minneapolis Hot Spots Patrol Experiment conducted almost 6,500 twenty-minute observations of the hot spots to identify whether the treatment hot spots actually received more police patrol than the control areas. The importance of this effort was illustrated in analysis of study data. The experiment had a fairly consistent impact on crime and disorder for the first 9 months of the study. However, in the summer months, the observed effect disappeared. This makes sense as many police take vacations during those months and the school vacation, hot weather, and other factors contribute to higher demand for police service. Analysis of the observational data showed that the difference in the dosage of police patrol between experimental and control hot spots became negligible during that period, which provided a strong explanation for the variability of the intervention's across time.

Understanding whether and how to assure that interventions are delivered properly falls under the rubric of "compliance" research in drug trials and some medical experiments. In small experiments, the research team usually develops "manipulation checks." In effectiveness trials and scale-up studies, program staff rather than experimenters are responsible for the intervention's delivery. Assuring fidelity of implementation is then usually handled through program guidelines, and manualization, training sessions, briefings, and the like.

In most applied social research, the "control" condition is *not* one in which any intervention is absent. Rather, the label usually denotes a condition in which a conventional or customary intervention is delivered. This being the case, the composition and activity of the control group must be measured and understood, as that of the new intervention group must be. For instance, in the Minneapolis Hot Spots Patrol Experiment, there was no consideration of withdrawing all police service from the control hot spots. These sites received normal emergency service from the police. In the Jersey City Drug Market Analysis Experiment, the same number of detectives was assigned to the control and treatment hot spots. What differentiated the groups was the introduction of a problem-oriented policing approach in the experimental sites.

Similarly, experiments on classroom size have included a control condition in which classrooms are of customary large size, with observations being made on what happens in these as on what happens in smaller classrooms. Well-done employment and training experiments verify that the same new program is not delivered to control group members and, moreover, document processes and events in the latter as in the intervention conditions.

Activity in the intervention and control conditions must be sufficiently different to justify expecting differences in outcome. Datta (2007), for instance, argues persuasively that a national trial on Head Start, mandated in 2000, was inappropriate because many control group children did have access to non–Head Start preschool programs with similar ingredients.

The main point is that interventions including control conditions need to be understood. In the absence of such understanding, a randomized field experiment is useless. With such understanding, clear statements of what works, or what works better, are far more likely.

Random Assignment

Technical advice on how to assign individuals or entities randomly to interventions is readily available in statistical textbooks on design of experiments. Technical advice is necessary but insufficient. Researchers must also recognize the realities of field conditions. Inept or subverted assignments are, for example, distinct possibilities. See Boruch (1997) for early examples that are becoming admirably less frequent.

Contemporary good practice focuses on who controls the random assignment procedure, when the procedure is employed, and how it is structured. Practice is driven by scientific standards that demand that the random assignment cannot be anticipated by service providers, for instance, and therefore subverted easily. Contemporary standards require that the assignments cannot be subverted post facto and cannot be manipulated apart from the control exercised by a blind assignment process. As a practical matter, these standards usually preclude processes that are easily subverted, such as coin flips and card deck selections.

In studies such as the Hot Spots Policing experiments, cases that are eligible are often known in advance to trialists and so the trialist can randomize cases before the experiment even begins. In this scenario and others, contemporary experiments employ a centralized randomization procedure that assures quality control and independence of the intervention's delivery. Trials undertaken to test mathematics curriculum packages by the Mid Atlantic Regional Laboratory, for instance, include centralized assignment of schools based on well-defined eligibility criteria (Turner, 2007). The Mid Atlantic Regional Education Laboratory's various trials on Odyssey Math involved 32 schools, 24 classrooms, and 2,800 students. In one such trial, eligible classrooms were randomly assigned to interventions within schools using a random assignment algorithm that was commercially available (Excel's "random function"), which was tested by the Laboratory's Technical Group and then applied by an independent organization, Analytica Inc. (Turner, 2007).

The random allocation's timing is important in several respects. A long interval between the assignment and the intervention's delivery can engender the problem that assigned individuals disappear, engage in alternative interventions, and so on. For example, individuals assigned to one of two different employment programs may, if engagement in the programs is delayed, seek other options. The experiment then is undermined. A similar problem can occur in tests of programs in rehabilitation, medical services, and civil justice. The implication is that assignment should take place as close as possible to the point of entry to the intervention.

The random assignment process must be structured so as to meet the demands of both the experiment's design and the field conditions. The individual's or entity's eligibility for intervention, for instance, must usually be determined prior to assignment. Otherwise, there may be considerable wastage of effort and opportunity for subversion of the trial. Moreover, individuals or entities such as schools or hospitals may have to be blocked or stratified on the basis of demographic characteristics prior to their assignments. This is partly to increase precision in (say) a randomized block design. It may also be done to reduce volatility of issues that the trial might otherwise engender. For example, in the Odyssey math trial, each of the 32 schools were used as a blocking factor, and classrooms within schools were then assigned randomly to Odyssey Math and to the control condition. This was done partly to increase power; half as many schools were needed as compared with school randomization design. The design also alleviated school principals' concerns that their schools might be denied the opportunity to obtain the Odyssey curriculum.

Blocking prior to randomization is also done at lower levels to address volatile field issues. For example, the trialist involved in an employment experiment may group four individuals into two blocks consisting of two individuals each, one block containing two African Americans and the second containing two Hispanics. The randomization process then involves assigning one African American to one of the interventions and the second individual to the remaining one. The randomization of Hispanics is done separately, within the Hispanic block. This approach assures that chance-based imbalances will not occur. That is, one will not encounter a string of Hispanics being assigned to one intervention rather than another. This, in turn, avoids local quarrels about favoritism. It also enhances the statistical power of the experiment to the extent that ethnic or racial characteristics influence individuals' responses to the intervention.

Simple random allocation of half the eligible units to intervention A and half to intervention (control) B is common. This tactic maximizes statistical power also, but good reasons for departing from this simple 1:1 allocation scheme often appear in the field. The demand for one intervention may be especially strong, and the supply of eligible candidates for intervention may be ample. This scenario justifies consideration of allocating in a (say) 2:1 ratio in a two-arm experiment. Allocation ratios different from 1:1 are of course legitimate and, more important, may resolve local constraints. They can do so without appreciably affecting the statistical power of the experiment, if the basic sample sizes are adequate and the allocation ratio does not depart much from 60:40. Larger differences in ratio require increased sample size.

A final aspect of the structuring of the random assignment, and the experiment's design more generally, involves a small sample size. For example, experiments that involve organizations, communities, or crime hot spots (e.g., see Weisburd & Green, 1995) as the primary unit of random assignment and analysis can often engage far fewer than 100 entities. Some experiments that focus on individuals as the unit of random assignment must also contend with small sample size, for example, local tests of interventions for those who attempt suicide, people who sexually abuse children, abusers of some controlled substances.

Regardless of what the unit of allocation is, a small sample presents special problems. A simple randomization scheme may, by chance, result in imbalanced assignment; for example, eight impoverished schools may be assigned to one health program and eight affluent schools assigned to a second. The approaches recommended by Cox (1958) are sensible. First, if it is possible to match or block prior to randomization, this ought to be done. This approach was used both in the Minneapolis Hot Spots Patrol Experiment and the Jersey City Drug Market Analysis Experiment. Second, one can catalog all random allocations that are possible, eliminate beforehand those that arguably would produce peculiarly uninterpretable results, and then choose randomly from the remaining set of arrangements. This approach is more complex and, on this account, seems not in favor.

Third, one can incorporate into the experiment's design strategies that can enhance analytic precision despite small sample size. See, for instance, Raudenbush and Bryk (2002) on matching prior to randomization and on the value of covariates. And see Bloom et al. (2007) and Schochet (2008) on using covariates when schools are the units of random allocation. The bottom line is that covariates can be valuable and often inexpensive in place randomized trials.

Observation and Measurement

The targets for observation and measurement in randomized trials include response (outcome) variables, intervention variables, baseline information (pretest, eligibility), context (settings), cost, and "missingness." Theory about how interventions are supposed to work, and for whom, is essential to specifying what variables in what category are to be observed. In rehabilitation programs, for instance, rudimentary theory suggests that certain outcomes, such as functional level, are influenced by certain kinds and duration of treatments (e.g., sheltered long-term workshops vs. conventional approaches). These are also affected by contextual factors, such as living arrangements and family, and may depend on pretreatment condition (baseline) of the individuals who are engaged in the treatments. The array of potential variables that could inform analyses beyond simple "intent to treat" (ITT) is large. Cordray (2000) provides a perspective related to one summarized here and gives more detail.

The basis for choosing a measure of the response variables and other variables lies partly in the variables' theoretical relevance to the intervention being tested. It lies also in conventional criteria for measurement quality, such as the reliability and validity of the observational method and how quality might vary over time and across intervention groups. In the Hot Spots Policing experiments, emergency calls for police were used as a primary measure because they were assumed to have less bias than police incident reports or police arrests, which are filtered through police activities. Systematic social observations have been seen as a reliable method for gaining information on street-level disorder, but were not used in the majority of these studies because of their very high expense.

Learning about how well response variables are measured in experiments, at times, entails qualitative observation. "Ride alongs" were carried out with police officers in the Hot Spots Policing experiments, for instance. This is not easy inasmuch as it requires body armor, possessed by at least two authors of this chapter. Ride alongs illuminated what the variable called "police patrol" or "problem oriented" policing meant, how they varied across the sites, and how arrests were made.

In principle, nothing prevents researchers from obtaining different kinds of information on outcomes and the processes that underlie experiments. Contemporary experiments often include both quantitative and qualitative approaches. However, good reports that combine both have, until recently, been difficult to find. Weisner's (2005) book is exceptional. It provides informative examples under the rubric of "mixed methods" in field research that embeds ethnographic work in statistical surveys and quasi-experiments. From one of the chapters on the randomized field test of the New Hope program for low-income working families, for instance, we learn about a puzzling statistical result and ethnographic approaches to this answer (Gibson-Davis & Duncan, 2005). Boys seem to benefit more than girls in the sense of statistically reduced problem behavior, apparently on account of mothers' investing more resources (day care) in them so as to avert all the higher risks that mothers perceive.

The frequency and periodicity of observing outcomes on intervention and control groups is important. For instance, theory and prior research may suggest that an intervention's effects decay or appear late, or that particular responses to one intervention appear at different rates than responses to another. We already noted the importance of social observations of hot spots in the Minneapolis Hot Spots Experiment in understanding the decline of the program's effects during the summer months. No consolidated handling of this matter is available yet in the context of social experiments. Nonetheless, if the trialist thinks about the arms of a randomized trial as two or more parallel surveys, then one can exploit contemporary advances in survival analysis, event history analysis, and in longitudinal data analysis. See Singer and Willett (2003) and references therein, generally, and Raudenbush and Bryk (2002) on multilevel models in which one level involves measures on the same entities over time.

It is obvious that the interventions that were assigned randomly to people or entities ought to be recorded, and the interventions that were actually delivered also ought to be recorded. The simplest recording is a count. In the Minneapolis Hot Spots trial, for instance, researchers measured the level of police presence each month through observations and used these data as a method of monitoring the dosage of police patrol. But measures on at least two deeper levels are commonly made to inform policy and science on the character of the interventions that are under scrutiny in the trial. At the study level, the counts on departures from randomization are, as a matter of good practice, augmented by qualitative information. In the SARP, for instance, departures were monitored and counted at each site to assure proper execution of the basic experiment's design and to learn about how departures occurred through qualitative interviews with police officers. At the intervention provider level, measures may be simple—for example, establishing how many police officers in the SARP contributed how many eligible cases and with what rate of compliance with assigned treatments. In large-scale education and employment experiments, measures are often more elaborate. They attend to duration, character, and intensity of training and support services, and to staff responsible for them (see, e.g., Gueron & Pauly, 1991; St. Pierre, 2004; and references therein).

Baseline or pretest measures in a randomized field experiment function to provide evidence that interventions are delivered to the right target individuals or entities, to reassure the trialist about the integrity of the random assignment process, to enhance the interpretability of the experiments, and to increase precision in analysis. Each function is critical and requires a different use of the baseline data.

In the Hot Spots Patrol experiment, for instance, data were generally collected for more than a year before eligibility was defined to make sure that police efforts were focused on places that had consistently high levels of crime and disorder. In the Minneapolis Experiment, researchers required a high level of stability in crime rates across time, since variability in prior measurement of crime is likely to be reflected in future measurement.

Consider next what trialists must observe on the trial's context. In experiments on training and employment programs that attempt to enhance participants' wage rates, it is sensible to obtain data on the local job market. This is done to understand whether programs being evaluated have an opportunity to exercise any effect. The measurement of job markets, of course, may also be integrated with employment program operations. Studies of programs designed to prevent school dropout or to reduce recidivism of former offenders might also, on theoretical grounds, attend to job markets, though it is not yet common practice to do so.

In some social experiments, measurement of costs is customary. Historically, trials on employment and training programs, for example, have addressed cost seriously, as in the Rockefeller Foundation's experiments on programs for single parents (Gordon & Burghardt, 1990) and work-welfare projects (e.g., Gueron & Pauly, 1991; Hollister, Kemper, & Maynard, 1984). Producing good estimates of costs requires resources, including expertise, that are not always available in other sectors. None of the Hot Spots Policing experiments, for example, focused measurement attention on cost; the focus was on the treatments' effectiveness. This is despite the fact that the interventions being tested involved substantial and expensive investments of police resources and might have negative as well as positive impacts on the communities living in the hot spots (Rosenbaum, 2006; Weisburd & Braga, 2006). Trials sponsored by the IES in education since 2002 seem also not to include much attention to costs.

Guidelines on measuring different kinds of costs are available in textbooks on evaluation (see, e.g., Rossi et al., 2004). Illustrations and good advice are contained in such texts, in reports of the kind cited earlier, and in monographs on cost-effectiveness analysis (e.g., Gramlich, 1990). Part of the future lies in trialists doing better at reporting on costs and in journal editors assuring that costs get reported uniformly.

Missingness here refers to failures to obtain data on who was assigned to and received what interventions, on what the outcome measurement was for each individual or unit, and on baseline characteristics of each participant. A missing data registry, a compilation of what data are missing from whom at what level of measurement, is not yet a formal part of a measurement system in many randomized controlled trials. The need for such registries is evident. The rate of follow-up on victims in ambitious police experiments such as SARP, for example, does not exceed 80%. On the other hand, follow-up in studies such as the Hot Spots Policing experiments based on police records is nearly perfect; missingness is negligible.

Understanding the missingness rate and especially how the rate may differ among interventions (and can be affected by interventions) is valuable for the study at hand and for designing better trials. The potential biases in estimates of effect are a fundamental reason why the What Works Clearinghouse [WWC] (2007) takes differential attrition into account in its standards of evidence. Understanding why data are missed is no less important. But the state of the art in reporting on missingness in experiments is not well developed. This presents an opportunity for young colleagues to get beyond precedent.

Management

Three features of the management of experiments are important. The first involves identifying and recruiting competent partners. In the Jersey City hot spots trial, the strong involvement of a senior police commander as a principal investigator in the study played a crucial role in preventing a complete break down of the experiment after 9 months (Weisburd, 2005). This suggests the importance of the integration of clinical work and research work in criminal justice, much as they are integrated in medical experiments (see Shepherd, 2003).

A second important feature in medium- and larger-scale efforts is the formation of advisory groups. Contemporary trialists depend on a committee to help assure that the experiment is run well. The counsel, at best, advises on technical, local, managerial, and political issues. The counsel, at best, helps meet naive as well as informed attempts to attack a fragile but important effort to get evidence. In some of the SARP sites, for example, representatives of community groups such as victims' advocates for the local police department and social services agencies advised and facilitated the experiment's emplacement. In multisite, large-scale evaluations, an oversight group may be formed by the experiments' sponsor (Reiss & Boruch, 1991).

Third, consider the actual task of management. Texts on management of randomized trials do not yet exist in the social, educational, and criminological areas. However, fine descriptions have appeared, at times, in reports issued by experiment teams. See, for instance, Weisburd et al. (2006) on managing Hot Spots Policing experiments, Sherman, Schmidt, and Rogan (1992) on managing the Milwaukee SARP, and Doolittle and Traeger (1990) on the Job Training Partnership Act study.

For large-scale trials, working with organizations that have developed the managerial and institutional skills to undertake such trials is essential. Learning about this has become a bit easier in recent years. For instance, over a third of the entries in the WWC's Register of Evaluators include research firms with documentable track records in managing large trials in education and, often, in other social sectors (www.whatworks.ed.gov). IES's reformation of Regional Educational Laboratories (RELs) in the United States has led to a buildup in the RELs' sophisticated studies typically in partnership with large and small firms, and with universities.

Managerial resources exist in some universities. Faculty at University of Pennsylvania, Vanderbilt, Northwestern, and many others have laid the groundwork for trials in their jurisdictions, and have collaborated with nonprofit firms and some for-profit firms to mount high quality trials. But there is considerable variability within and across academic institutions, and no central listing appears to exist. Typically, the principal authors of reports of the kind cited in this chapter have some of the requisite skills. Documentation on management is sparse. See the articles in Boruch (2005) for some exceptions.

Understanding what tasks need to be done, by whom, when, and how is basic to management in this arena. The tasks fall to the study's sponsor and the experiment's team, including the service providers. The challenges lie in clarifying the role of each and in developing partnerships and advisory groups. Partly because experience in this arena is so difficult to document, documentation is sparse.

Analysis

Contemporary randomized trials in the social sector usually involve at least four classes of analyses. The first class focuses on quality assurance. It entails developing information on which interventions were randomly assigned to which individuals or entities, which interventions were actually received by each, and analyses of departures from the random assignment. Each experiment in the SARP, for instance, engaged these tasks to assure that the experiments were executed as designed and to assess the frequency and severity of departures from design during the study and at its conclusion. Quality assurance also usually entails examination of baseline (pretreatment) data to establish that, indeed, the randomized groups do not differ appreciably from one another prior to the intervention. Presenting numerical tables on the matter in final reports is typical in peer-reviewed reports to government (good) in peer-reviewed journals (poor).

Core analysis here refers to the basic comparisons among interventions that were planned prior to the experiment. The fundamental theme underlying the core analysis is to "analyze them as you have randomized them." In statistical jargon, this is an "intent to treat" analysis. That is, the groups that are randomly assigned to each intervention are compared regardless of which intervention was actually received. At this level of analysis, departures from assignment are ignored.

ITT is justified by the statistical theory underlying a formal test of hypothesis and by the logic of comparing groups that are composed through randomization so as to undergird fair comparison. It also has a policy justification. Under real field conditions, one can often expect departures from an assigned treatment. In the SARP, for instance, some individuals who were assigned to a mediation treatment then became obstreperous and were then arrested; arrest was a second randomized treatment. Such departures occur normally in field settings. Comparing randomly assigned groups regardless of actual treatment delivered recognizes that a reality of core analysis is basic in medical and clinical trials (e.g., Friedman, Furberg, & DeMets, 1985) as in the social and behavioral sciences (Riecken et al., 1974).

The product of the ITT analysis is an estimate of the relative effect of intervention. This product addresses the question, "What works?" and a statistical statement of confidence in the result, based on randomized groups. Where departures from random assignment are substantial, the researcher has to decide whether any ITT analysis is warranted and indeed whether the experiment has been executed at all. The experiment or core analysis, or both may have to be aborted. If information on the origins or process of departures from random assignment has been generated, the researcher may design and execute a better experiment. This sequence of failure and trying again is a part of science. See, for instance, Silverman's (1980) descriptions of research on retrolental fibroplasia that covers blindness of premature infants as a function of enriched oxygen environments.

Deeper levels of analysis than ITT are often warranted on account of the complexity of the phenomenon under study or on account of unanticipated problems in the study's execution. For example, finding "no differences" among interventions may be a consequence of using interventions that were far less different from one another than the researcher anticipated or inadequate statistical power. A no difference finding may also be on account of unreliable or invalid measures of the outcomes on each randomized group. Interactions between intervention type and subgroup, of course, can lead to a naive declaration of "no difference." The topic is understudied, but good counsel has been developed by Yeaton and Sechrest (1986, 1987), and Julnes and Mohr (1989).

A final class of analysis directs attention to how the results of the trial at hand relates to the results of similar studies. Exploring how a given study fits into the larger scientific literature on related studies is demanding. One disciplined approach to the task lies in exploiting the practice underlying the idea of systematic reviews and meta-analyses. That is, the researcher does a conscientious accounting for each study of who or what was the target (eligibility for treatments, target samples, and population), what variables were measured and how, the character of the treatments and control conditions, how the specific experiment was designed, and so on. The U.S. General Accounting Office (1994), now called the Government Accountability Office, formalized such an approach to understand the relative effectiveness of mastectomy and lumpectomy on 5-year survival rates of breast cancer victims. See Pettigrew and Roberts (2006), and the U.S. General Accounting Office (1992, 1994) more generally on the topic of synthesizing the results of studies. Each contains implications for understanding how to view the experiments at hand against earlier work.

Reporting

The medical and health sciences led the way in developing standards for reporting on randomized trials (e.g., Chalmers et al., 1981). Later, Boruch (1997) provided a checklist that depended partly on one prepared for reports on medical clinical trials. The Consolidated Statement on Reporting of Trials (CONSORT) Statement is one of the best articulated statements of its kind (Mohler et al., 2001). One of CONSORT's innovations is the requirement that authors provide a flowchart that details case flow into and out of the trial. The flowchart is a numerical and graphical portrayal of the pipeline discussed earlier in this Chapter. The CONSORT guidelines have been updated and revised to foster standardized and thorough reporting on cluster randomized trials (Campbell, Elbourne, & Altman, 2004).

CONSORT's ingredients have informed the WWC's (2007) guidance on how to report and what to report on controlled trials in education (http://ies.ed.gov/ncee/wwc). The WWC, a unit of the IES in the United States, has also built on standards of evidence work by the Society for Prevention Research, the Campbell Collaboration, and others to develop its standards of evidence. The production and revision of nongovernmental standards of reporting have begun, in turn, to depend on the WWC. The sheer volume of research publications (20,000 year in education alone) has provoked a move toward standardized abstracts that contain brief statements about the experiment's design elements and results (Mosteller, Nave, & Miech, 2004).

Capacity Building

Developing better capacity to design randomized trials and to analyze results is not new in one sense. Excellent texts on statistical aspects of randomized trials, and new ones that cover remarkable advances in the field, such as Raudenbush and Bryk (2002), Piantadosi (1997), and Donner and Klar (2000) are readily available and are used in many graduate courses.

Capacity building in the sense of educating ourselves and others about managing and executing such trials, and handling the political and institutional problems that they engender, has only recently received serious attention. The World Bank's International Program for Development Evaluation Training (IPDET) included such matters in 2004 and 2005 after years of neglect. NIMH's summer institutes on trials and the workshops on trials at professional society meetings run by the American Institutes for Research, Manpower Demonstration Research Corporation, and others are illustrative. William T. Grant Foundation invested substantially in special seminars on the topic for senior and midlevel researchers and civil servants. Beginning in 2007, the IES invested substantially in training institutes and conferences, in predoctoral and postdoctoral fellowship programs that focused heavily (not entirely) on randomized trials (U.S. Department of Education, 2007). Participants have typically included researchers, people from local, state, and federal agencies, and service providers.

Of course, capacity building includes providing resources to different entities to run trials. The entities include schools, police departments, etc., whose cooperation is essential in generating better evidence. See the examples given earlier. The challenges for the future include learning how to institutionalize and cumulate the learning by professionals in these organizations and to assure that the learning leads to decisions that will inform. This particular challenge is also not new, but the refreshed interest over the last decade in randomized trials will help to drive more sophisticated uses of evidence and ways to think about use.

Conclusion

During the 1960s, when Donald T. Campbell developed his prescient essays on the experimenting society, fewer than 100 randomized field experiments in the social sector had been mounted to test the effects of domestic programs. The large number of randomized trials undertaken since then is countable, but not without substantial effort. Registers of such trials, generated with voluntary resources, such as the Campbell Collaboration (http://campbellcollaboration.org), yield more than 14,000 entries and the actual number is arguably far larger. Executing randomized controlled trials help us to transcend debates about the quality of evidence and, instead, inform social choices based on good evidence. In the absence of randomized controlled experiments on policy and programs, we will, in Walter Lippman's (1963) words, "Leave matters to the unwise . . . those who bring nothing constructive to the process and who greatly imperil the future. . . . by leaving great questions to be fought out by ignorant change on the one hand, and ignorant opposition to change on the other" (p. 497).

Exercises and Questions

1. For a specific arena of interest, develop a briefing to address the overreaching question: Which research question should be addressed and why?

2. For a specific arena of interest, where questions about intervention effects are important, develop a briefing to address the questions: Is an impact evaluation warranted and should a randomized trial be considered?

3. For a specific arena of interest and context, develop a briefing to address the question: How should the randomized trial be designed?

4. For a specific arena of interest and context, and for a scenario involving one or two experiment design options, develop a briefing to address the question: Who would be able, under what circumstances, to implement the designs for the randomized trials?

5. For a specific arena of interest, and for a scenario involving a randomized trial, develop a briefing to address the question: "What theory or logic model is being invoked in the decision to mount the trial, frame the relation between outcome variables, the interventions being tested, the baselines, and the context?"

6. Why is random assignment a prerequisite for obtaining unbiased estimates of an intervention policy, program, or practice?

7. Why is it important to distinguish between the random assignment of individuals and the random assignment of groups (or clusters) of individuals? What are some of the names used in the literature to describe the latter?

8. What is a power analysis? Why is it fundamental to the design of a randomized controlled trial?

9. Why is it important to differentiate between the parameter used in power analysis for the randomized controlled trial and those used for power analysis for a cluster randomized controlled trial? What statistical parameter is assumed to be greater than zero in a cluster randomized trial?

10. In submitting a funding proposal for a cluster randomized control trial for a large grant, you have been asked to include a power analysis for a within-school design, where classrooms are randomly assigned to intervention and control conditions within each school, with the following assumptions:
    - Statistical power is 80%.
    - Statistical significance level is at $\alpha = 0.05$ for a two-tailed test.
    - Each classroom includes 25 students.
    - Balanced allocation with four classrooms per school.
    - Minimum detectable effect size (MDE) of 0.20.

- Explanatory power (R2) classroom level covariates (math pretest of the math outcome measure) of .56.
- Intraclass correlation (*p*) values of .15.
- Use a random effects models.

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CHAPTER 6

Quasi-Experimentation

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Applied social science researchers often try to assess the effects of an intervention of interest, also known as a treatment. To take just a few examples, educational researchers have estimated the effects of preschool programs, economists have examined the consequences of an increase in the minimum wage, psychologists have assessed the psychological effects of living through a natural disaster, and legal scholars have studied the results of legal changes such as laws mandating helmets for motorcycle riders. When an applied social researcher is interested in estimating the effects of a treatment, a range of research options exists. One option is to employ a randomized experiment. In a randomized experiment, a random process, such as a flip of a fair coin, decides which participants receive one treatment condition (e.g., a new state-supported preschool program) and which receive no treatment or an alternative treatment condition (e.g., traditional child care). The randomized experiment is the preferred option for many applied researchers, and sometimes is held out as the "gold standard" for studies that estimate the effect of a treatment. In applied social research, however, practical or ethical constraints often preclude random assignment to conditions. For instance, it will usually not be feasible to randomly assign people or states to a law that mandates helmets for motorcyclists. When random assignment to conditions is not feasible—as will often, but hardly inevitably, be the case in applied research—a quasi-experiment may be the method of choice.

"Quasi" is a Latin term meaning "as if." Donald Campbell, the original architect of the logic of quasi-experimentation (e.g., Campbell & Stanley, 1966; Cook & Campbell, 1979; Shadish, Cook, & Campbell, 2002), coined the term *quasi-experiment.* It means an approximation of an experiment, a "near experiment." Like randomized experiments, quasi-experiments are used to estimate the effects of one or more treatments on one or more outcome variables. The difference is that quasi-experiments do not have random assignment to treatment conditions. Instead, the treatment effect is estimated by making comparisons across cases that are exposed to different treatment conditions in some nonrandom fashion, and/or by comparisons across time (before and after treatment implementation), and/or by other kinds of comparisons discussed later.

How successful these nonrandom comparisons are in providing an accurate treatment effect estimate is a matter of some controversy. Ironically, this controversy is indirectly reflected in the very name "quasi-experiment." Commenting on a political columnist's reference to "*The Wall Street Journal*'s quasi-*wingnut* editorial page," language maven William Safire (2006) stated that quasi, "when used as a prefix means 'seemingly.'" In this light, the term *quasi-experiment* could be taken as implying that, while quasi-experiments might *seemingly* resemble experiments, they fall short. Indeed, Campbell himself occasionally made punning reference to "queasy-experiments."

Are quasi-experiments so "queasy" that applied social researchers should forego their use? Or do they approximate experiments closely enough that researchers can draw confident conclusions from them? As the answer so often is, it depends. In this chapter, we review a set of classic quasi-experimental designs, showing why some are generally queasier than others. We also briefly review the logic of quasi-experimental design, showing that skilled quasi-experimentalists do not simply pull an existing design off the shelf. Rather, they show creativity in identifying comparisons that will provide the best estimate of the treatment effect possible under the circumstances. In addition, in this chapter, we review methodological and statistical developments that have occurred since Campbell and his colleagues outlined the basic quasi-experimental design options. Use of these procedures strengthens the inferences that a quasi-experiment can support. We also briefly review literature that compares the results of quasi-experiments with those of randomized experiments and from that draw suggestions for the conduct of quasi-experimentation.

A Review of Alternative Quasi-Experimental Designs

In this, the longest section of the chapter, we review four quasi-experimental designs: the one-group pretest-posttest design, the nonequivalent group design, the interrupted time-series design, and the regression-discontinuity design. In the context of these designs, we introduce several potential threats to the validity of inferences from quasi-experiments. We begin with relatively "queasy" designs that generally do not provide sufficiently confident causal inferences in applied social research. Even here, however, the adequacy of a design is not preordained, but depends. We then move to more compelling quasi-experimental designs and to additional comparisons that can facilitate causal inference.

The One-Group Pretest-Posttest Design

Until recently, hormone replacement therapy (HRT) was commonly prescribed for menopausal and postmenopausal women. In 2002, the Women's Health Initiative (WHI) study showed that women who were taking HRT had an increased risk of breast cancer, as well as heart disease and stroke. The use of HRT quickly plummeted. If they recommended it at all, most physicians suggested HRT only as a shorter-term treatment for these women experiencing severe menopausal symptoms. In 2003, the number of women diagnosed with breast cancer declined 7.2% relative to 2002, representing roughly 14,000 fewer cases than expected. In headline news articles, the decline in breast cancer cases was attributed to the reduction in HRT stimulated by the WHI study findings (e.g., MSNBC News Services, 2006).

Data on the HRT-cancer relationship can be viewed as a quasi-experiment, specifically an instance of the one-group pretest-posttest design. In this design, the effect of a treatment is estimated by comparing (a) what happened before the treatment was implemented with (b) what happened after the treatment was implemented. Using notation popularized by Campbell and his colleagues, where O represents an observation, X represents a treatment, and time runs from left to right, a one-group pretest-posttest design can be diagrammed as

O X O.

In other words, a comparison is made across time in an effort to estimate the effects of an intervention, such as the effect of the WHI study and the consequent drop in HRT on the number of breast cancer cases.

Although the one-group pretest-posttest design is easily implemented and therefore widely used, it is usually susceptible to a variety of alternative interpretations. In other words, typically the design is rather queasy in terms of providing a good estimate of the treatment's effects. Using terminology popularized by Campbell and his associates, we now describe the generic alternative interpretations that commonly plague the one-group pretest-posttest design.

History refers to the possibility that a specific event, *other than* the treatment of interest, occurred between the pretest and posttest observations and caused change in the outcome(s) of interest. For example, some other change in medical treatment might have happened in late 2002 or early 2003 that caused the decline. (While perhaps not plausible in the HRT-cancer case, history often does threaten the one-group pretest-posttest design.)

Maturation refers to processes that occur over time within study participants, such as growing older, becoming hungrier, growing more fatigued, and growing wiser. Maturation typically involves relatively continuous processes emanating naturally from within study participants; history, in contrast, involves more discrete, external events. In the HRT-cancer study, maturation could be a threat if there were steady shifts in cancer rates over time due to demography, gradual shifts in nutrition, or the like. Imagine that there was a long-term decline in breast cancer cases, averaging about 7% a year. Given such a long-term trend, the observed decline in cancer cases between 2002 and 2003 would not imply an effect of the reduced use of HRT.

Instrumentation can lead to inaccurate inferences about a treatment's effects when an apparent effect is instead the result of a change in a measuring instrument. One reason that instrumentation can occur is because of changes in the definition of an outcome variable. Paulos (1988) gave an example, noting that "Government employment figures jumped significantly in 1983, reflecting nothing more than a decision to count the military among the employed" (p. 124). Instrumentation would be a problem in the HRT-cancer study if, for example, the official definition of breast cancer changed, say, with some of cases that in 2002 would have been classified as breast cancer instead defined in 2003 as lymph node cancer. Instrumentation can also be a problem when there is not a formal change in definition, if the procedures or standards of those who record the observations shift over time.

The threat of *testing* arises when the very act of measuring the pretest alters the results of the posttest. For example, individuals unfamiliar with tests such as the SAT may score higher on a second taking of the test than they did the first time, simply because they have become more familiar with the test format. In the HRT-cancer investigation, testing appears to be an implausible threat, but it would be a problem if many women had mammograms in 2002 and by some biological process this screening itself offered protection against cancer.

Regression toward the mean is an inferential threat that occurs most strongly when the pretest observation is substantially different than usual, either higher or lower. When things are unusual at the pretest, the posttest observation often will return to a more average or "normal" level even in the absence of a treatment effect. This kind of pattern is called *spontaneous remission* in medical treatments or psychotherapy. That is, people often seek out treatment when their physical or emotional conditions are at their worst and, because many conditions get better on their own, patients often improve without any intervention. In theory, an unusual form of regression toward the mean could have occurred in the HRT-cancer study. Publicity about the WHI study results could have created a stampede of women to get mammograms, including women who otherwise would have not have had a mammogram until 2003 or after. The 2002 tally of breast cancer cases thus might have been unusually high, with a decline in 2003 to be expected even without any real effect of the reduction in HRT.

Attrition, alternatively labeled *experimental mortality,* refers to the loss of participants in a study. Such a loss can create a spurious difference in a pretest-posttest comparison. For example, the average test scores of college seniors tend to be higher than the average test scores of college freshmen, simply because poor-performing students are more likely than high-performing students to drop out of school. A form of attrition could have threatened internal validity in the HRT-cancer study if fewer women, especially those high at risk, were screened for cancer in 2003 than in 2002. Hypothetically, publicity about the WHI might have made some women too anxious to be screened or given a false sense of security to women not on HRT. The WHI study and the associated reduction in HRT therapy would not have caused a real drop in breast cancer, but would have only reduced detection via attrition from screening (and thus from the study data).

Lessons From the One-Group Pretest-Posttest Design

The one-group pretest-posttest design is relatively easy to implement. All that is required is that an outcome be measured both before and after a treatment is implemented. Because of its relative ease, use of this quasi-experimental design may be appealing. However, for many applied social research questions, the design will be queasy at best. In terminology developed by Campbell and his associates, the design is susceptible to a number of "threats to internal validity," specifically, history, maturation, and the other problems just noted. *Internal validity* refers to the extent to which accurate conclusions can be drawn about whether and to what degree the treatment-as-manipulated makes a difference in the outcome-as-measured (Shadish et al., 2002). Internal validity, it is important to recognize, refers to getting the causal inference correct in the time and place of the study. *External validity,* in contrast, refers to the accuracy of inferences about the generalizability of findings across persons, settings, and times. Campbell's original argument, still persuasive to many, was that internal validity should be the researcher's first concern, because it would be of dubious value to enhance external validity unless you are confident that you have the causal inference correct in the first place (see Cronbach, 1982, for an alternative position).

Threats to internal validity refer to general categories of factors, such as history and regression toward the mean, which can lead to inaccurate conclusions about the causal relationship (within the context of the study). Internal validity threats, however, are *not a given.* We emphasize this point here because some practitioners of quasi-experimentation seem to believe that, if a design is susceptible to an internal validity threat in general, this inevitably makes findings from the design ambiguous in every case. On the contrary, the plausibility of an internal validity threat depends not only on the research design but also on the particular content and context of the study. For example, history would not plausibly account for the results of a memory study in which participants were asked to learn a list of nonsense syllables. That is, in general, it is implausible that some historical event would have occurred that would explain why participants recall the right nonsense syllables at the posttest. Similarly, in chemistry courses, threats such as history and maturation are not taught as important concerns, simply because they are not likely to be plausible alternative explanations for the chemical reactions observed in the chem lab. Of course, history, maturation, and the other threats are far more likely to affect most of the kinds of outcomes that applied social researchers examine.

Eckert (2000), however, has argued that for at least one applied social research question, the effect of training programs on immediate outcomes such as knowledge gain, the threats of history, maturation, and so on often are not plausible. Akin to the nonsense syllable example, Eckert argues that it is implausible that history, maturation, and the other threats could create detectable increases in knowledge in the short time frame of the studies. Notably, Eckert does not argue that this would hold for other, less immediate outcomes such as improved organizational performance. Because an effect on organizational performance would take longer, and because it is influenced by many forces, history and other threats would be more plausible for organizational performance as an outcome than they are for immediate outcomes such as knowledge.

Eckert's argument highlights several "take home messages" about quasi-experimentation. First, to reiterate, threats do not automatically cripple a quasi-experiment based solely on its design. The specifics of a study, including its context and content, including what the outcome measure is, determine whether a threat applies in a particular case. Second, quasi-experimentation should not be seen—or practiced—as a mindless or automatic process of selecting from a preexisting menu of quasi-experimental designs. One consideration in thoughtfully selecting a quasi-experimental design is the plausibility of internal validity threats in the specific circumstances of the study. For instance, if Eckert is right that the one-group pretest-posttest design suffices for evaluating the immediate learning effects of certain training programs, then it could be a waste of resources to implement a more complex design. Of course, this argument rests on the assumption that the risk of the various internal validity threats can be assessed reasonably well in advance. Moreover, the higher costs of a more rigorous design may sometimes be practically inconsequential, in which the stronger design would of course be preferred. Or the more rigorous design may be more costly (in terms of dollars, time, or other resources), but this cost could be outweighed by the importance of having a strong evidentiary base to convince skeptics. Again, the selection of a particular quasi-experimental design, or the selection of a quasi-experiment versus a randomized experiment, involves judgment and consideration of trade-offs. A third implication is that the quasi-experimental researcher often has a larger burden than the researcher conducting a randomized experiment. Rather than simply reporting the results of a pretest-posttest evaluation of the effects of a training program on knowledge, for example, Eckert would need to offer evidence and argument to rule out the validity threats to which the design is generically susceptible.

Sometimes the evidence that a quasi-experimentalist might add to his or her argument is relatively direct evidence about the plausibility of a particular validity threat. For instance, Ross (1973), in a study of a British intervention directed at road safety, used a variety of sources to see if there were actual history threats such as other legislation or shifts in gasoline prices. In the HRT-cancer example, the threat of attrition could be directly assessed by examining whether there was a decline from 2002 to 2003 in the number of women screened for breast cancer by mammograms.

Alternatively, the quasi-experimentalist might seek to rule out threats less directly, by creatively identifying additional comparisons that help render relevant validity threats implausible. For instance, consider a one-group pretest-posttest evaluation of a training program. The researcher could create two knowledge scales, one closely reflecting the training program's content and the other measuring related knowledge that the program did not teach—but that would be expected to change if maturation occurred. If the posttest showed improvement on the first but not the second measure, this would further support the conclusion that the training worked (vs. the alternative explanation that maturation occurred). In the HRT-cancer study, a similar strategy was employed. Investigators found that the decline in cancer cases occurred primarily among women in the age group previously targeted for HRT therapy and in the types of cancer sensitive to estrogen, a component of HRT. The logic of adding such comparisons is addressed further in a later section.

Despite the preceding discussion, in most circumstances the one-group pretest-posttest design will not be adequate for applied social research. This is because one or more of the previously described threats to internal validity are likely to be sufficiently plausible and sufficiently large in size as to render results from the design ambiguous. Thus, we turn to other quasi-experimental designs.

Nonequivalent Group Quasi-Experiments

In the one-group pretest-posttest design, the researcher estimates the treatment effect by comparing the same individuals (or at least the same pool of individuals) at different points in time, before and after the treatment. The other primary means of estimating a treatment effect is by comparing different groups of individuals at the same time. In general, such designs are called between-group designs. In a quasi-experimental context, they are called nonequivalent group designs. This is because, in the absence of random assignment to groups, there is no a priori reason to believe that the two groups will initially be equivalent (in contrast to randomized experiments).

In the simplest nonequivalent group design, *the posttest-only nonequivalent group design,* individuals (or other units) fall into two groups. One, the treatment group, receives the treatment, while a control or comparison group does not. Or the two groups might receive alternative treatments. In nonequivalent group designs, the groups might have been created by self-selection (e.g., by individuals who decide whether to receive the treatment or not), by administrative decisions, or by some other nonrandom process. In the posttest-only nonequivalent group design, the two groups are observed only after the treatment has been administered. Such a design can be represented as

$$\frac{\text{X O}}{\text{O}},$$

where the broken line denotes that the groups are nonequivalent, which simply means that group assignment was not random.

The posttest difference between the groups on the outcome variable is used to estimate the size of the treatment effect. However, the internal validity threat of *selection* usually makes the results of the posttest-only nonequivalent group design uninterpretable in applied social research. Selection refers to the possibility that initial differences between groups, rather than an actual treatment effect, are responsible for any observed difference between groups on the outcome measure. When nonequivalent groups are compared, the selection threat is usually sufficiently plausible that the posttest-only nonequivalent group design is not recommended for applied social research. That is, differences on the outcome variable seem likely to result from self-selection or whatever the nonrandom process is that created the groups, which would of course obscure the effects of the intervention in the posttest-only design.

In a more prototypical nonequivalent group design, the groups are observed on both a pretest and a posttest. Diagrammatically, this *pretest-posttest nonequivalent group design* is represented as

$$\frac{\text{O X O}}{\text{O}\quad\text{O}},$$

where the dashed line again denotes nonequivalent groups. With this design, the researcher can use the pretest to try to take account of initial selection differences. The basic logic of the pretest-posttest nonequivalent group design can perhaps most easily be seen from the vantage of one potential data analysis technique, gain score analysis. Gain (or change) score analysis focuses on the average pretest-to-posttest gain in each group. The difference between the two groups in terms of change (i.e., the difference between groups in the average pretest-posttest gain) serves as the estimate of the treatment effect. That is, the treatment effect is estimated by how much more (or less) the treatment group gained on average than the control group. Unlike the posttest-only design, the pretest-posttest nonequivalent group design at least offers the possibility of controlling for the threat of selection—using the pretest to represent the initial difference that is due to selection.

Gain score analysis, however, controls only for a simple main effect of initial selection differences. For example, imagine that (a) the treatment group begins 15 points higher than the control group at the pretest and (b) it would remain 15 points ahead at the posttest unless there is an effect of the treatment. In this case, gain score analysis would perfectly adjust for the effect of the initial selection difference. However, the analysis does not control for interactions between selection and other threats. In particular, gain score analysis of data from the pretest-posttest nonequivalent group design does not control for a selection-by-maturation interaction, whereby one of the groups improves faster than the other group (i.e., matures at a different rate) even in the absence of a treatment effect.

Functionally, there are two ways to think about why a selection-by-maturation interaction would occur. One is captured in the old expression, "The rich get richer." Certain maturational processes are characterized by increasingly larger gaps over time between the best and the rest. For example, skill differences are usually less pronounced among younger children and more pronounced among older children. When such a pattern holds, a gain score analysis will not remove the differential maturation across groups. That is, the initially higher-scoring group would be further ahead of the other group at the posttest ("the rich get richer"), even in the absence of a treatment effect. A second (and conceptually related) reason for the selection-by-maturation pattern is that the pretest might not capture all the relevant initial differences between groups in the face of certain maturational processes. Consider the case of a quasi-experimental evaluation of a program intended to prevent drug use in early adolescents. If the two groups had similar levels of drug use at the pretest, while at the posttest the comparison group used drugs more than treatment group youths, a gain score analysis would suggest that the program was effective. However, the groups might have appeared similar at the pretest because that measurement took place at an age before many youths have begun to use drugs. But if the two groups differed on risk factors such as community levels of drug use, then divergence between the two groups over time may be expected even if no treatment effect occurred. More generally, a single pretest (measured in the same way as the posttest) may not represent all the factors that should be controlled for.

The task of controlling for initial selection differences can be approached in several different ways through alternative statistical analyses (Reichardt, 1979; Shadish et al., 2002). Another common analytic procedure is the *analysis of covariance* (ANCOVA). In controlling for initial selection differences, in essence ANCOVA statistically matches individuals in the two treatment groups on their pretest scores and uses the average difference between the matched groups on the posttest to estimate the treatment effect. Unlike gain score analysis, ANCOVA allows the use of covariates that are not operationally identical to the posttests, as well as the use of multiple covariates. However, measurement error in the pretest scores will introduce bias into the ANCOVA's estimate of the treatment effect, because the statistical adjustment would not control for the true initial differences. Bias will also arise if the statistical model does not include all the variables that both affect the outcome variable and account for initial selection differences. There is seldom any way to be confident that all such variables have been appropriately included in the analysis. So the possibility of bias due to initial selection differences usually remains.

Because measurement error in the pretest will introduce bias in ANCOVA (Reichardt, 1979), *latent variable structural equation models* are sometimes used instead (Magidson & Sorbom, 1982; Ullman & Bentler, 2003). These models use multiple measures of the construct thought to affect the outcome variable and account for initial selection differences, and these measures are essentially factor analyzed in an effort to obtain an estimate of the "latent variable" that effectively is without measurement error. (Latent variable structural equation models also nicely support the testing of mediational models, discussed below.) However, the validity of the estimates that result from these models depends on the accuracy and thoroughness of the model, and applied social researchers often cannot be confident that they have specified a model accurately.

An alternative approach, *propensity score analyses,* is gaining in popularity of late. In this approach, the predicted probability of being in the treatment (rather than the control) group is generated by a logistic regression (Little & Rubin, 2000; Rosenbaum, 1995; Rosenbaum & Rubin, 1983). An advantage, relative to the simpler ANCOVA, is that the influence of numerous covariates can be captured in a single propensity score. Cases are then usually stratified into subgroups (commonly five subgroups) based on their propensity scores, and the treatment effect computed as a weighted average based on the treatment and control group means within each subgroup. Alternatively, the propensity score can be treated as a covariate in ANCOVA. Winship and Morgan (1999) provide a useful review of several of these techniques (also see Little & Rubin, 2000; Shadish et al., 2002).

Much uncertainty remains about how to tailor an adequate statistical analysis for the pretest-posttest nonequivalent group design under different research conditions.

As a result, three recommendations seem especially sensible. First, where possible, it is desirable to conduct sensitivity analyses, that is, analyses that assess how robust a given finding is to different assumptions within a single form of analysis (Rosenbaum, 1995). In particular, recent forms of sensitivity analyses can assess how large the biasing effect of an unmeasured (or "hidden") covariate would have to be to change the conclusions from an analysis. Sensitivity analysis, common, for example, in the econometric tradition, constitutes a promising addition to the practice of quasi-experimental analysis. Leow, Marcus, Zanutto, and Boruch (2004) discuss and provide an example of sensitivity analysis in the context of propensity score analysis. Second, confidence will be enhanced if different forms of analysis are employed and the results converge reasonably well on an estimate of the treatment effect (e.g., Reynolds & Temple, 1995). The recommendation to conduct multiple analyses to bracket the real treatment effect in a quasi-experiment is hardly a new one (e.g., Wortman, Reichardt, & St. Pierre, 1978) but deserves to be put into practice more often. Third, rather than relying exclusively on statistical adjustments, it is preferable to develop a stronger research design (Shadish et al., 2002). This advice is not news to those well trained in the Campbellian tradition but appears to diverge from the focus of many researchers from other traditions that focus on statistical controls.

In addition to these three recommendations, researchers should keep in mind that the plausibility of selection as an explanation for a study's findings also may depend on the size and pattern of findings. If a treatment effect is large enough, it may be implausible that selection or another internal validity threat could be responsible for it. Of course, the applied social researcher does not have free reign in selecting the size of treatment effects (but may be able to improve the odds of detectable effects by advocating for things such as consistent implementation of an adequately sized treatment and the use of outcome measures sensitive to change; see Lipsey, 1990). The pattern of observed effects also has implications for the plausibility of selection as a threat (Cook & Campbell, 1979). For instance, a crossover (or X-shaped) interaction pattern can often be plausibly interpreted as a treatment effect. As an example, Braucht et al. (1995) examined the effects of a continuum of services on the use of alcohol by homeless substance abusers. As Figure 6.1 reveals, those who received more services used more alcohol at the time of the pretest than those who received fewer services, but this difference was reversed at the two posttest times. Such a crossover interaction usually will not result from common internal validity threats, such as selection-by-maturation effects and differential regression toward the mean. In particular, maturational processes rarely appear to result in "the poor becoming the rich." Of course, although a crossover pattern, if it occurs, can reduce the plausibility of selection and other threats, the applied social researcher who is planning a study cannot count on such a pattern to arise.

Interrupted Time-Series Designs

Recall the question addressed earlier: Did the reduction in the use of HRT therapy starting in 2002, following publication of the WHI study, result in a decrease in breast cancers? Future investigators should be able to address this question using an

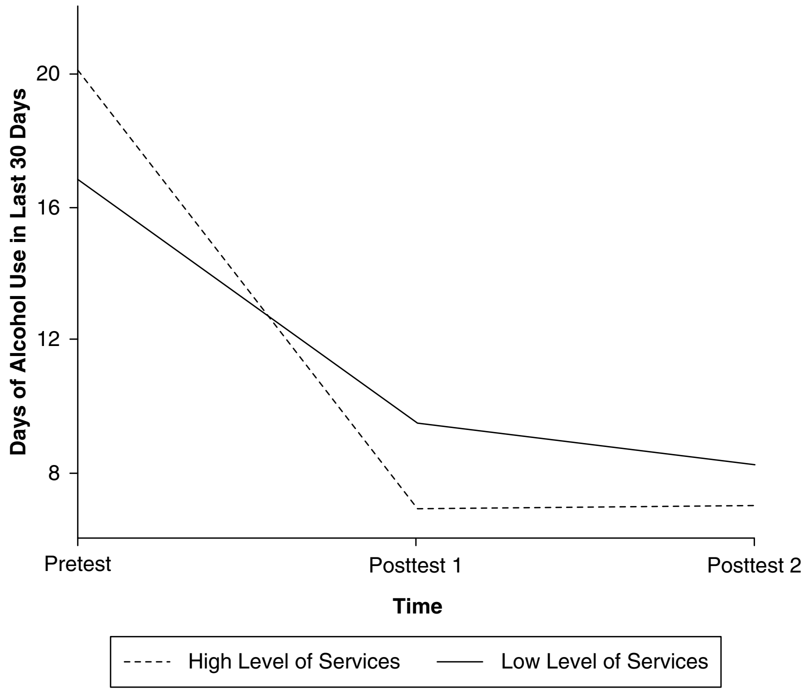

Figure 6.1 Number of Days of Alcohol Use Both Before and After Two Groups of Homeless Individuals Received Different Amounts of Substance Abuse Treatment

SOURCE: Adapted from Braucht et al. (1995, p. 103) by permission. Copyright by Haworth Press, Inc.

interrupted time-series (ITS) design. Using the X and O notation introduced previously, a simple, one-group ITS design can be represented as

O O O O O OXO O O O O O.

In the simple ITS design, a series of observations is collected over a period of time, a treatment is introduced, and the series of observations continues. In essence, the trend in the pretreatment observations is estimated and projected forward in time so as to provide an assessment of what the outcome data would have been if there had been no treatment. The actual trend in the posttest observations is then compared with the projected trend, and the difference between them provides an estimate of the treatment effect. When the actual observations differ from the projection, as in Figure 6.2, the inference is that the treatment had an effect. Figure 6.2 shows a hypothetical result for a future ITS study, showing a permanent decline in the number of breast cancer cases following the reduced use of HRT starting in 2002.

Unlike other kinds of designs, ITS designs make it possible to detect the temporal pattern of the treatment effect. A treatment could change the *level* of the

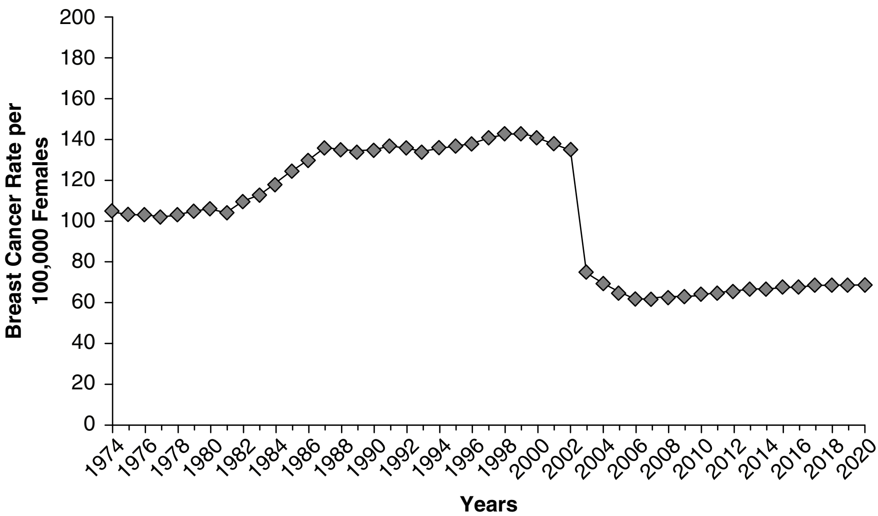

Figure 6.2 Time Series From Hypothetical Study of Reduced HRT and Breast Cancer

outcome variable, as in the hypothetical findings in Figure 6.2, where the intervention appears to have reduced breast cancer cases by a relatively constant amount over the posttreatment period. Change can also occur in *slope,* either alone or in association with a change in level. For instance, a future HRT-cancer time-series study might show both a reduced level and a declining slope (imagine Figure 6.2 with a downward slope after the intervention). Moreover, a treatment effect could be either immediate or delayed, and could also be either permanent or temporary. However, validity threats (history and maturation, respectively, as will be discussed later) are often more plausible for both delayed and gradual effects than for an immediate, abrupt effect. The temporal pattern of the effect also can have serious implications for judgments about the importance of the effect. For example, if the effects of reduced HRT lasted only 1 year, most observers would judge this as less important than if the effects were permanent.

How does the simple ITS design fare with respect to internal validity threats? Like the one-group pretest-posttest design, the simple ITS design estimates the treatment effect by comparing the same individuals (or the same aggregate group) at different points in time, before and after the treatment. However, the ITS design does far better in terms of ruling out several validity threats. Consider the six validity threats introduced in the earlier discussion of the one-group pretest-posttest design. While *maturation* is a plausible threat in the one-group pretest-posttest design, the pretreatment observations in a time series can allow the researcher to estimate the pattern of maturation. For example, if maturation follows a simple linear trend, the researcher can see (often literally) the pattern of maturation and model it in the statistical analysis. The pretreatment observations in a simple ITS also can reveal the likely degree of *regression toward the mean.* That is, with a series of pretreatment observations it is possible to see whether the observation(s) immediately prior to the treatment are unusually high or low and, if so, to remove the validity threat by assessing the "normal" or average level to which the posttreatment observations should regress—something the simple pretest-posttest design does not allow. *Testing* is also unlikely to be a threat to the ITS design. With repeated observations before the intervention, testing effects are likely to diminish over time and are unlikely to be powerful at the time the treatment is introduced.

Although time-series data can help rule out maturation, regression to the mean, and testing, other threats that apply to the simple one-group pretest-posttest design also may threaten the simple ITS design. *If* the length of the time interval between observations is the same in the two kinds of designs, then *history* effects are as likely in the simple ITS designs as in the one-group pretest-posttest designs. However, history will generally be less plausible if the time interval between observations is shorter, and sometimes time-series designs have shorter intervals than the one-group pretest-posttest design. *Instrumentation* can also be a plausible threat to validity in an ITS design if the intervention is associated with changes in the way observations are defined or recorded. For example, estimating the effects of changes in sexual assault laws with an ITS design can be biased if there are corresponding changes in the way sexual assaults are defined and measured (Marsh, 1985). Careful analysis of definitions and record-keeping procedures may be necessary to determine the plausibility of threats due to changes in instrumentation. Finally, *attrition* can sometimes be a threat to validity in the ITS design, just as in the one-group pretest-posttest design. If the amount of attrition follows a relatively smooth and continuous pattern over time, the researcher can take the effect of attrition into account in much the same way as maturation is taken into account, by modeling the trend in the pretreatment observations. However, in certain studies, attrition may be induced by the treatment itself, as would happen if publicity about the WHI findings led to a decline in the number of women being screened for breast cancer. In such circumstances, taking account of attrition may require examining another time series which represents the number of individuals who contribute to each time period's observation (so, e.g., one could calculate a breast cancer rate per 1,000 women screened).

In short, elaborating a one-group pretest-posttest design into a simple ITS design can help make several internal validity threats less plausible. Nevertheless, some threats, particularly history, will often remain plausible. Instrumentation and attrition will sometimes be plausible as well, depending on the specifics of the study. As noted previously, direct assessment of potential validity threats can be undertaken. For example, attrition could be assessed by determining whether fewer women were screened for breast cancer after the WHI study was publicized. In addition to assessing potential threats directly, the plausibility of threats will generally be reduced by moving to a more elaborate ITS design, such as by adding a control time series or by removing and repeating the treatment, as described next.

An *ITS design with a control time series* can be represented diagrammatically as follows:

O O O OXO O O O

O O O O O O O O

The top line of Os represents data from the experimental subjects who receive the treatment, whereas the bottom line of Os represents data from the control subjects who do not receive the treatment. The broken line indicates that the two time series of observations did not come from randomly assigned groups. Ideally, the control time-series of observations would be affected by everything that affects the experimental time series, except for the treatment. To the extent this is the case, the control series increases one's knowledge of how the experimental series would have behaved in the absence of a treatment, and thereby increases one's confidence in the estimate of the treatment effect. For example, if the two groups have similar maturational patterns, then the control time series can be used in modeling the pretreatment trend and projecting it into the future. Furthermore, a control time series can take account of the validity threat of history, to the extent the control time series is affected by the same history effects. In this case, the treatment effect is estimated as the size of the change in the experimental series after the treatment is introduced, minus the size of the change in the control series at the same point in time.

For example, Wagenaar (1981, 1986) was interested in the effect that an increase in the drinking age had on traffic accidents. In 1979, the drinking age in Michigan was raised from 18 to 21 years. To assess the effect of this change, Wagenaar (1981) plotted one experimental time series (for the number of drivers aged between 18 and 20 years who were involved in a crash) and two control series (for the number of drivers aged between 21 and 24 years or between 25 and 45 years who were involved in crashes). These time series are reproduced in Figure 6.3. A drop in fatalities

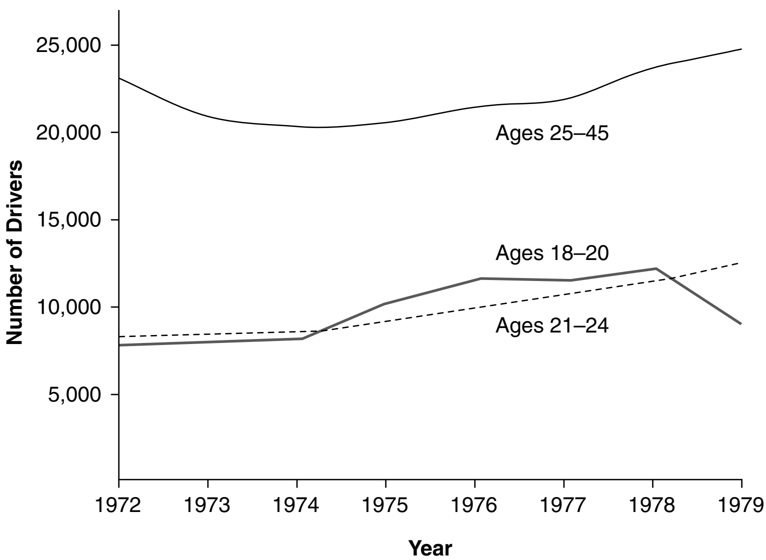

Figure 6.3 The Number of Drivers Involved in Crashes While Drinking, Plotted Yearly Both Before and After the Legal Drinking Age Was Raised in 1979 from 18 to 21

SOURCE: Adapted from Wagenaar (1981) by permission. Copyright by The University of Michigan Transportation Research Institute.

occurred in 1979 only for the experimental time series—that is, only for the data from the 18- to 20-year-old drivers, which is the only time series of observations that should have been affected by the treatment intervention. The two control series add to our confidence that the dip in the experimental series is an effect of the treatment and not due to other factors that would also affect the control series, such as changes in the severity of weather patterns or changes in the price of gasoline. As noted earlier, in the case of the HRT-breast cancer relationship, it will be useful to compare the time series of breast cancer cases for women of the age typical for HRT with the time series for women of other ages. It would also be useful to compare time series for estrogen-sensitive cancers (which should be affected by HRT) and nonestrogen-sensitive cancers (which should not be affected by HRT). This can be labeled a *non-equivalent dependent variables ITS design* (Cook & Campbell, 1979), because a comparison times series of observations exists that consists of a different dependent variable than the primary dependent, time-series variable.

Other design elaborations can also be useful. When the treatment's effects are transitory (i.e., they disappear when the treatment is removed), one potentially useful option is the *ITS with removed and repeated treatment.* Such a design is diagrammatically depicted as

O O O X O O O –X O O O X O O O –X O O O,

where X indicates that the treatment was introduced and –X indicates that the treatment was removed. For example, Schnelle et al. (1978) estimated the effects of police helicopter surveillance, as an adjunct to patrol car surveillance, on the frequency of home burglaries. After a baseline of observations was collected with patrol car surveillance alone, helicopter surveillance was added for a while, then removed, and so on. In general, the frequency of burglaries decreased whenever helicopter surveillance was introduced, while burglaries increased when helicopter surveillance was removed. The repeated introduction and removal of the treatment can greatly lessen the plausibility of the threat of history. In the Schnelle et al. study of helicopter surveillance, for example, it is unlikely that historical events that decrease burglaries would happen to coincide repeatedly with the multiple introductions of the treatment, while the multiple removals of the treatment would happen repeatedly to coincide with historical events that increased burglaries.

The statistical analysis of time-series data generally raises complexities. In a time series, data points that are adjacent in time are likely to be more similar than data points that are far apart in time. This pattern of similarity, called *autocorrelation,* violates the assumptions of typical parametric analyses such as multiple regression analysis. In short, autocorrelation can bias significance tests and confidence intervals. In ITS studies that examine aggregate data, such as annual number of breast cancer cases in the United States, autoregressive integrated moving average (ARIMA) models are frequently suggested (e.g., Box, Jenkins, & Reinsel, 1994; Box & Tiao, 1975). However, the number of time points must be relatively large, perhaps as large as 50 to 100 observations. When there is a control ITS, ARIMA models could be fit separately to each of the different time series of observations.

Alternatively, when data are collected over time from numerous cases (e.g., annual test scores collected from many students), a variety of techniques can be used to analyze the data. Importantly, the analysis of such "N much greater than 1" ($N >> 1$) designs can require far fewer than the 50 to 100 time points of observations that are necessary for ITS designs that have only a single case (i.e., $N = 1$ designs), the latter having to meet the demands of the ARIMA analysis strategy. In other words, having a large number of observations (i.e., cases) at any one point in time can reduce the number of different time points of observation that are required. For numerous cases ($N >> 1$) designs, the most frequently recommended analysis strategy in the past was derived from multivariate analysis of variance (MANOVA; Algina & Olejnik, 1982; Algina & Swaminathan, 1979; Simonton, 1977, 1979). The MANOVA approach allowed the autocorrelation structure among observations to have any form over time but fit the same model to the data for each individual.

More recently, two additional statistical approaches have been developed. These newer approaches model the trajectory of growth for each case (e.g., student) individually, which means these two statistical approaches allow trajectories to differ across the individual cases and allow these differences in trajectories to be explained using other variables in the model. In addition, different models of the treatment effect can be fit to each case and differences across cases in the effects of the treatment can be assessed. The first of the two newer approaches has been given a variety of names, including multilevel modeling and hierarchical linear modeling (HLM; Raudenbush & Bryk, 2001). An example using HLM with a short time series is provided by Roderick, Engel, Nagaoka, and Jacob (2003), who evaluated the effects of a summer school program in the Chicago school district. They provide an accessible explanation of the benefits of the HLM approach for accounting for statistical regression in the context of a short time series. The second approach is called latent growth curve modeling (LGCM; Duncan & Duncan, 2004; Muthén & Curran, 1997) and is implemented using software for structural equation modeling. Under a range of conditions, the HLM and LGCM analyses are equivalent and produce the same estimates of effects (Raudenbush & Bryk, 2001).

To sum up regarding ITS designs, in these quasi-experiments a series of observations is collected over time both before and after a treatment is implemented. Essentially, the trend in the pretreatment observations is projected forward in time and compared with the trend in the posttreatment observations, and differences between these two trends are used to estimate the treatment effect. The ITS design often has the greatest credibility when the effect of the treatment is relatively immediate and abrupt. Some of the advantages of the ITS design are that it (a) can be used to assess the effects of the treatment on a single individual (or a single aggregated unit, such as a city), (b) can estimate the pattern of the treatment effect over time, and (c) can be implemented without the treatment's being withheld from anyone. The researcher can often strengthen the design by removing and then repeating the treatment at different points in time, adding a control time series, or both. The ITS design, especially with a control group or other elaborations, is generally recognized as among the strongest quasi-experimental designs. With more recent advances in analysis (e.g., the use of HLM for growth curve modeling), the use of shorter time series with multiple cases appears to have become more commonplace.

The Regression-Discontinuity Design

The regression-discontinuity (R-D) design is another quasi-experimental design recognized as relatively strong in terms of internal validity (Shadish et al., 2002). In the R-D design, participants are assigned to treatment groups based on their scores on a measure that can be called the quantitative assignment variable or QAV. The participants who score above a specified cutoff value on the QAV are assigned to one treatment group, while the participants who score below the cutoff value are assigned to the other group. With schools as the participating units, for instance, the QAV could be average absenteeism levels, with schools above the cutoff assigned to a new antiabsenteeism program and schools below the cutoff serving as a comparison group. (With more than two treatment groups, more than one cutoff value would be used.) Subsequently, all participants are assessed on the outcome measure, such as postprogram absenteeism rates. Interestingly, the R-D design was independently invented by Thistlethwaite and Campbell (1960; Campbell & Stanley, 1966), Goldberger (1972), and Cain (1975), with the latter two inventors apparently unaware of the design's prior genesis.

As an example, Mark and Mellor (1991) used the R-D design to examine the effect of a job layoff on plant workers, focusing on a set of plants where being laid off was determined by the workers' seniority. The number of years worked in the plant served as the QAV, with those having 19 or fewer years of seniority being laid off and those with 20 or more years not being laid off. Mark and Mellor found that those who were laid off were relatively less likely to report that the layoff was foreseeable. Although that and other R-D studies are relatively strong in internal validity, the design has been used relatively infrequently (Shadish et al., 2002). However, recent attention to the design, largely in the field of education, may lead to increased use (see, e.g., Gormley, Gayer, Phillips, & Dawson, 2005).

To estimate the treatment effect in an R-D design, a regression line is fit separately to the data in each treatment group. The treatment effect is estimated as the difference, or discontinuity, between the regression lines in the two groups. Figures 6.4 and 6.5 present hypothetical data from an R-D design. In both figures, scores on the QAV are plotted along the horizontal axis while scores on the outcome measure are plotted on the vertical. The ellipse represents the bivariate distribution of the scores from the two variables, although the individual scores are not shown. The vertical line at 10 on the QAV marks the cutoff value, with individuals above the cutoff being in the experimental group and individuals below the cutoff being in the control group (as might occur for a meritocratic rather than a compensatory treatment, such as awarding of fellowships or grants). Separate regression lines for the regression of the outcome scores on the QAVs are shown for each group. The R-D design is particularly well suited to circumstances in which a treatment is to be assigned on the basis of measured merit or, conversely, measured need.

Figure 6.4 depicts no treatment effect. The lack of a treatment effect is revealed by the fact that the regression lines are not displaced vertically relative to each other—they intersect as though they fall on a continuous straight line. In contrast, Figure 6.5 presents hypothetical data depicting a treatment effect. In Figure 6.5, the

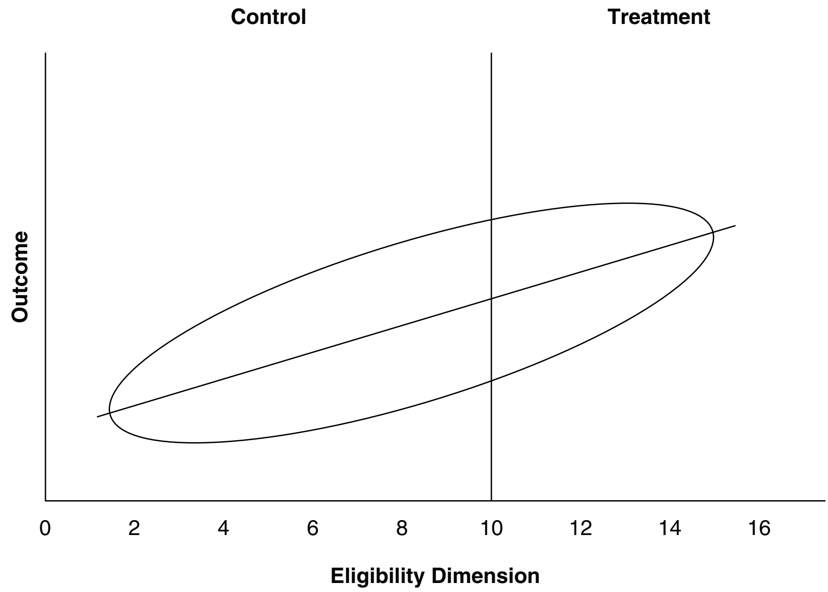

Figure 6.4 Hypothetical Data From an R-D Design (depicting no treatment effect)

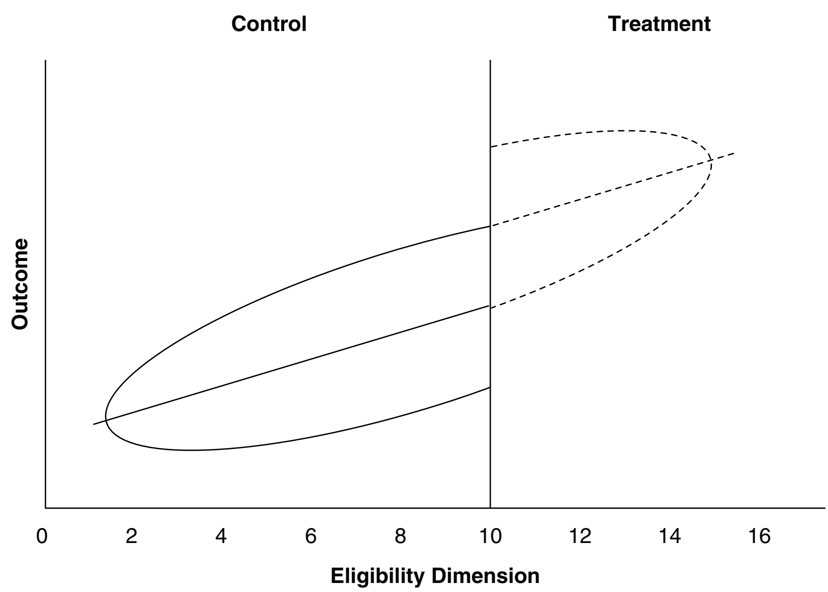

Figure 6.5 Hypothetical Data From an R-D Design (depicting positive treatment effect)

treatment effect is positive, with the regression line in the experimental group displaced above the regression line in the control group—the treatment group scores are higher than you would expect relative to the regression line in the control group. The estimate of size of the treatment effect is equal to the vertical displacement between the two regression lines.

The graphical representation of an R-D study's findings, as illustrated in Figure 6.5, highlights the source of the design's inferential strength. In general, it is implausible that any threat to validity, whether selection, statistical regression, or any other threat, would produce a discontinuity precisely at the cutoff between the treatment conditions. Put informally, the question is: How likely is it that there would be a jump in scores on the outcome variable that coincides precisely with the cutoff on the eligibility criterion, unless there really is a treatment effect? Unless the treatment really makes a difference, why would individuals who score just below the eligibility criterion look so different on the outcome than those who score just above it, and why would this difference between individuals just above and below the cutoff be so much greater than the difference, say, between those who score right below the cutoff as compared with those who score just below that? Because there are usually few plausible answers to these questions, the R-D design has relatively strong internal validity, approaching that of a randomized experiment (Shadish et al., 2002).

The conventional statistical analysis of the R-D design involves predicting the outcome variable using regression analyses, where the predictors are (a) the QAV (transformed by subtracting the cutoff value, so that the treatment effect is estimated at the cutoff point), (b) a dummy variable representing condition (e.g., 1 = treatment vs. 0 = comparison), and (c) a term representing the interaction of condition and the QAV. The regression coefficient for the dummy variable estimates the treatment effects (seen visually as the vertical displacement of the regression lines in Figure 6.5). The interaction term assesses whether the size of the treatment effect varies across the QAV. For example, imagine that the treatment in Figure 6.5 is more effective for those who initially scored the highest. If so, the two regression lines would no longer be parallel, and the experimental group's regression line would be higher on the right side than it is in Figure 6.5.

Curvilinearity in the relationship between the QAV and the outcome variable is one potential source of bias in an R-D design's estimate of the treatment effect. If the underlying relationship is curvilinear, but a linear relationship is fit to the data, a spurious effect may be observed (Exercise 2). To address this problem, curvilinearity in the data should be modeled in the analysis. Typically, in practice, this would be done after visual inspection for curvilinearity in the original and smoothed data. In the regression analysis, polynomials terms of the (transformed) QAV and interaction are added. Inclusion of the polynomials serves to test for the possibility that a nonlinear relationship exists that could otherwise masquerade as a treatment effect. Trochim (1984) and Reichardt, Trochim, and Cappelleri (1995) discuss procedures for modeling interactions and curvilinearity, and for performing the regression analysis. The R-D design has substantially less power than a randomized experiment (Cappelleri, Darlington, & Trochim, 1994). For example, to have the same precision and power as randomized experiment (assuming that a measure analogous to the QAV is used as a covariate), the R-D design must have at least 2.7 times as many participants (Goldberger, 1972). Alternative analysis approaches have been explored in recent years (e.g., Hann, Todd, & Van der Klaauw, 2001) and deserve attention.

The Logic of Ruling Out Threats to Validity

As we have seen, quasi-experimental designs fail to rule out all internal validity threats a priori, a statement especially true of the "queasier" designs. As we have also seen, validity threats can sometimes be assessed directly (as when the researchers checks on attrition by seeing if fewer women had mammographies at the posttest than at the pretest) and sometimes by using a more complex design. A general logic applies to both of these options. A researcher can rule out threats to validity by (a) thinking through the implications of the treatment as to discover ones that conflict with implications of the threat to validity and (b) obtaining data to see whether it is the implications of the treatment or of the validity threat that hold true. In other words, when a comparison is susceptible to a threat to validity, the researcher can assess the plausibility of the threat by adding a comparison that puts the treatment and the alternative explanation into competition. We call this process *competitive elaboration*. The essence of this logic has been advocated by methodological experts including Campbell (1975), Rosenbaum (1984), and R. A. Fisher, and has been explicated in-depth by Reichardt (2000). For example, Cochran (1965, p. 252) reported that, when asked what can be done in nonexperimental studies "to clarify the step from association to causation, Sir Ronald Fisher replied, 'Make your theories elaborate.'"

Many of the design features previously discussed in this chapter operate by competitive elaboration. For example, competitive elaboration explains how a control time series rules out history effects in an ITS design. Consider the data from Wagenaar (1981, 1986), shown previously in Figure 6.3. The experimental time series (between 18 and 20 years of age) and the two control time series (between 21 and 24 and between 25 and 45 years of age) should share many of the same history effects. So to the extent history effects are a threat to validity, one would predict similar patterns of change in the control and experimental series at the point the treatment is introduced. The same holds for biases introduced by instrumentation and testing. In contrast, if the treatment is effective, one would predict a different posttreatment pattern of change for the experimental and control series, because the treatment should affect only the experimental series. Because the pattern across time in the experimental and control series in Figure 6.3 is similar before the treatment is introduced but quite different afterward, the difference is more plausibly attributed to the effect of the treatment than to history.

Reichardt (2006), in describing the "principle of parallelism," has recently pointed out that researchers can make comparisons across different kinds of factors to better assess the effects of a treatment. As the Wagenaar (1981, 1986) example illustrates, one way to put the treatment effect and validity threats into competition is by drawing comparisons across subgroups (or types of "recipients," in the language of Reichardt, 2006). In the HRT-breast cancer study discussed earlier, it is helpful to compare findings for women of the age typically treated with HRT versus findings for women of other ages. Alternatively, comparisons can be drawn across measures, as Cook and Campbell (1979) demonstrated with the so-called nonequivalent dependent variable (a comparison across outcome variables, in Reichardt's language). In the HRT-breast cancer example, a treatment effect would predict a decline in estrogen-sensitive cancers only, while most alternative explanations would predict a decline in both estrogen-sensitive and nonestrogen-sensitive cancers. As Reichardt (2006) has noted, competitive elaboration can also take place with respect to comparisons across variations in settings and times. See Reichardt (2006) for examples and further discussion.

Ancillaries to Quasi-Experimental Designs

Since Campbell and his colleague's well-known presentations of quasi-experimental design, a number of procedures have become relatively common as adjuncts both to experiments and quasi-experiments. This section describes three such ancillary procedures: implementation assessment, tests of mediation, and the study of moderation. These three ancillary methods have notable benefits in their own right. In addition, they often can strengthen causal inference in quasi-experiments. (Other ancillary procedures exist, including power analyses, newer techniques for dealing with missing data, and methods for minimizing and estimating the effects of attrition. Many of these are described elsewhere in this *Handbook*.)

Implementation Assessment

In early applied social research, researchers often failed to assess systematically what the "treatment" and the "comparison" (or control) actually consisted of in practice. For example, an early evaluator of the effects of bilingual education probably would not have observed the education of second-language learners in the so-called bilingual education schools, nor what transpired in the so-called comparison group schools. But without attention to the specifics of treatment implementation, sensible conclusions are hard to reach. For example, if no treatment effect is observed, the implications would be quite different (a) if bilingual education was not implemented than if (b) bilingual education was well implemented but nonetheless ineffective.

Systematic assessment of a treatment's implementation is more commonplace nowadays than in early applied social research. Several approaches to implementation assessment have been employed (Mark & Mills, 2007). For example, interventions sometimes have a relatively detailed implementation plan, as is the case for many school-based prevention programs and psychological therapies. In such cases, implementation assessment may consist of checks, preferably by observation but perhaps by self-report from program implementers or recipients, on the extent to which the intervention was implemented with fidelity to the plan. Checks should also be made about whether the same or similar activities are carried out in the comparison or control group. For example, a study of bilingual education should assess not only the fidelity to the program plan in treatment group schools but also the extent to which similar activities did not occur in the comparison group. (See Mark & Mills, 2007, for discussion of alternative models of implementation assessment.)

Information from an implementation assessment is valuable, as already noted, in terms of facilitating more sensible interpretation of no-effect findings. Implementation analyses, by allowing better description of the actual intervention, are also valuable in facilitating dissemination of effective treatments. In some instances, implementation assessment results can also strengthen causal inference in a quasi-experiment. For example, there is often variation within the treatment group in terms of the degree or nature of the exposure to the treatment. Based on a simple dose-response logic, researchers may seek to test the hypothesis that there are larger effects for clients who received higher doses of the treatment. However, potential selection effects may bias this comparison. That is, clients may have self-selected into different amounts of treatment exposure, and these self-selected subgroups may differ initially in important ways. Propensity scores or other forms of statistical adjustment can be used to try to alleviate this bias. See Yoshikawa, Rosman, and Hsueh (2001) for a related example.

Mediational Tests

A *mediator* is a variable that "falls" between two other variables in a causal chain, such as between a program and its outcome. Substantively and statistically, the mediator accounts for or is responsible for the relationship between an intervention and its outcome. To take an example, for many years the drug abuse prevention program DARE (Drug Abuse Resistance Education) was based on a mediational model assuming that the program activities, its lessons and exercises, would cause an increase in students' refusal skills, the mediator, and these enhanced refusal skills would in turn translate into reduced drug use by the students, the intended outcome. In many areas of social research, whether basic or applied, it has become commonplace to test mediational models. For example, theory-driven evaluation, a popular approach to program and policy evaluation, includes mediational analyses as a routine practice (Donaldson, 2003). Mediational tests are often conducted via structural equation modeling (SEM; e.g., Ullman & Bentler, 2003) or simpler statistical procedures (e.g., Baron & Kenny, 1986), although more qualitative methods are sometimes used (e.g., Weiss, 1995). Although these techniques have limits, they can be useful at least in probing mediation.

A mediational model may contain only one mediator, as in the model held by the original advocates of DARE. Or there may be multiple mediators. Indeed, research on programs such as DARE have demonstrated that their program activities influence more than one mediator. In particular, although DARE and similar programs increase refusal skills, they also make drug use seem more common, and unfortunately, making drug use seem more common or "normative" is associated with a higher level of drug use (e.g., Donaldson, Graham, Piccinin, & Hansen, 1995). This example illustrates some of the benefits of mediational analyses. Like implementation assessment, mediational tests can facilitate interpretation of the treatment effect results. For instance, if a study found DARE to be ineffective, the implications would differ if (a) the program failed to increase refusal skills versus (b) refusal skills were increased but the program nevertheless failed to achieve reduced drug use. In addition, the finding that DARE and similar programs affected perceived norms provides guidance about how to revise DARE.

Mediational analyses can also strengthen confidence that the treatment, rather than a validity threat, accounts for the observed differences between groups in a quasi-experiment. This follows from the idea of competitive elaboration discussed in the previous section. When a theory of the treatment predicts a particular mediational pattern *and* findings are consistent with that pattern, causal inference is strengthened to the extent plausible validity threats would not account for the same pattern. Mediational evidence can also make quasi-experimental (or experimental) findings easier to communicate and more persuasive. For instance, being able to explain *why* DARE is ineffective is likely to be more compelling than simply stating it *is* ineffective. Testing mediation also can erase the distinction between applied and basic research, as when the evaluation of a real-world program includes a test of a theoretical hypothesis about social norms.

Moderators: Testing for Differences in Effects Across Groups

A *moderator,* in contrast to a mediator, is a variable that modifies the strength or direction of the relationship between two other variables. For example, we might hypothesize that a refusal skills drug prevention program would be more effective with students who have higher self-esteem than with students who have lower self-esteem. The argument would be that those with low self-esteem would be less likely to employ their refusal skills when offered drugs. Using terminology from an analysis of variance tradition, the hypothesis is that students' self-esteem level will interact with the treatment.

Tests of moderation can be useful in several ways. When it is possible to use alternative interventions with different individuals or communities, findings about moderators can assist in matching the right intervention to the right cases. In certain areas of applied social research, tests of moderation are important in terms of equity considerations. For example, such tests clarify whether an educational program reduces or exacerbates achievement gaps across racial, ethnic, gender, social class, or other groupings. Of interest to quasi-experimental researchers, tests of moderation may strengthen causal inference from a quasi-experimental design. Theories of the intervention may provide hypotheses about moderation (as in our hypothetical example about self-esteem and refusal skills programs). If one tests the theory-based moderation hypothesis, and if moderation is in fact observed as predicted, then this more elaborated pattern of findings offers stronger evidence than a simpler treatment-comparison group contrast. Of course, the researcher also should consider whether any plausible validity threat would account for the same pattern of moderation.

In general, testing for moderation requires a pretreatment measurement of the potential moderator. For example, one could not test the hypothesis about self-esteem moderating the effectiveness of training in refusal skills, unless a prior measure of self-esteem is available (posttreatment measures are less desirable, because the treatment itself could have affected self-esteem). However, researchers have recently been exploring techniques for estimating the consequences of moderators that are not measured in advance. See Hill, Waldfogel, and Brooks-Gunn (2002) for an example, in which propensity score methods were used to construct subgroups on a factor that had not been directly measured before the treatment.

Summary

Implementation assessment, mediational tests, and the study of moderation have each become more commonplace in applied social research. These procedures have specific benefits as ancillaries to both experiments and quasi-experiments. For the quasi-experimentalist, it is important to note that these procedures, in at least some cases, can also strengthen causal inference. This will especially occur if the researcher implements these procedures thoughtfully from the perspective of competitive elaboration.

Comparisons of Quasi- and Randomized Experiments, and Their Implications

A small but growing literature exists in which researchers compare the results of quasi-experiments with the results of randomized experiments. One version arises from meta-analysis, that is, the quantitative synthesis of a research area. Many meta-analysts have compared the average effect size from randomized experiments with that from quasi-experiments. In other words, in part of the literature comparing study types, researchers synthesize the findings from multiple quasi-experimental investigations of a particular treatment and compare them with the findings from a set of experimental studies of the same treatment.

Lipsey and Wilson (1993), in a classic paper, did this one better. They synthesized findings from more than 300 meta-analyses of psychological, educational, and behavioral interventions. Lipsey and Wilson found that, averaging across a large number of types of treatment, experiments and quasi-experiments gave similar results. This finding, while interesting, does not answer the practical question about the likelihood that a quasi-experiment examining a particular treatment will give similar results to an experiment examining the same treatment. On this question, the findings are not so optimistic. For many of the specific interventions, Lipsey and Wilson report, quasi-experiments on average gave different answers than experiments. Sometimes quasi-experiments provided a larger treatment effect than randomized experiments, and other times a smaller treatment effect. This inconsistency, whereby quasi-experiments gave more positive answers in some treatment domains and more negative answers in others, suggests that the dominant validity threats and their effects vary across research areas. That is, it appears that, in certain research areas, there is an "upward" bias from the dominant validity threats that apply to the quasi-experiments that were conducted; in other research areas, the typical bias is "downward." For example, in one research area nonequivalent group designs might be beset by a selection bias that cause the quasi-experiments on average to overestimate the real treatment effects, while in another research area the typical selection bias might lead to an underestimate. And in yet other areas there may not be a consistent direction of bias. For instance, a particular research area might not be plagued by consistent selection effects, but history effects might apply. Given the vagaries of history, this threat would sometime lead to an overstatement of the true treatment effect and at other times to an underestimate. A related finding from Lipsey and Wilson is that quasi-experiments were associated with more variability in effect size estimates. That is, in a given research area, there was less consistency across studies in the treatment effect estimates from quasi-experiments than from randomized experiments. This does not seem surprising, in that the validity threats that generally apply to the quasi-experiments in a given research area are not likely to operate to the same degree in every study. For example, if history is an applicable threat, the vagaries of history are in essence adding random error to the treatment effect estimates across quasi-experimental studies.

Altogether, then, the findings of Lipsey and Wilson (1993) do not inspire confidence that the results of a quasi-experiment will match the results that would have arisen if a randomized experiment were done instead—although they may do well in some research areas. Aiken, West, Schwalm, Carroll, and Hsiung (1998) and Cook and Wong (2008) have summarized other research that compares results from a set of quasi-experiments and a set of randomized experiments investigating a particular treatment. In short, their conclusions seem compatible with the findings of Lipsey and Wilson. As both Aiken et al. and Cook and Wong (2008) point out, however, comparisons of this kind are themselves subject to bias. That is, many differences on average may exist between the quasi-experimental and the experimental studies in a given research area, including differences in the way the treatments are implemented, differences in the type of individuals receiving the treatment, differences in the way outcomes are measured, differences in the settings in which the two types of studies are implemented, and so on.

Other comparisons of study types have taken a more local or "within-study" approach (Cook & Wong, 2008). In some cases, the researchers have constructed both a randomized experimental test and one or more quasi-experimental tests in the same context (e.g., Aiken et al., 1998; Lipsey, Cordray, & Berger, 1981). In other studies, the researcher has conducted a randomized experiment; for the quasi-experiment, data from the randomized experiment's treatment group are compared with data from another source, typically a large national data set. One problem with this approach is that, as has been emphasized throughout this chapter, quasi-experiments are not all alike. Some are queasier than others. And, as Cook and Wong suggest, an argument can be made that in many of the local comparisons across study types, a well-designed randomized experiment has been compared with a mediocre quasi-experiment.

Cook and Wong (2008) indicate that, in those few instances in which randomized experiments are compared with the *strongest* of the quasi-experiments, the results are similar. In the case of R-D designs, for instance, Aiken et al. (1998) found similar results for an R-D quasi-experiment and a randomized experiment studying the effects of a remedial writing course. Lipsey et al. (1981) similarly found convergence between the results of an R-D and an experimental investigation, specifically of the effects of a juvenile justice diversion program. Likewise, the R-D design gave similar results to those of a randomized experiment in two other unpublished studies that Cook and Wong (2008) described. Cook and Wong also reviewed the one study they found comparing findings from an ITS design with those from a comparable randomized experiment. Bloom, Michalopoulos, and Hill (2005), in the context of a job training program, conducted a randomized experiment in five locations, with a short time series combined with the experiment. They also constructed a nonequivalent time-series comparison group drawing on untreated individuals from a nearby location. Cook and Wong conclude that, combining across the five ITS designs with nonequivalent comparison groups, these quasi-experiments give the same answer as the randomized experimental design. Bloom et al. (2005) offer a less optimistic interpretation, but their conclusions can be reinterpreted as consistent with Lipsey and Wilson's finding that quasi-experiments have more variable findings than do randomized experiments.

When randomized experiments are compared with weaker quasi-experiments, the picture is somewhat more mixed. For example, widely cited comparisons between randomized experiments and nonequivalent group designs in the area of job training found differences in results from the two kinds of studies (e.g., LaLonde, 1986). In these studies, statistical adjustments were the only way of attempting to account for selection bias. In addition, the comparison groups in these studies were typically drawn from existing data sets, and so selection differences may both have been nontrivial and difficult to control for adequately. In contrast, other researchers, such as Shadish, Luellen, and Clark (2006) and Shadish and Ragsdale (1996) have attempted to assess the implications of how the nonequivalent group is constructed. These researchers have found that nonequivalent group designs with "internal" control groups, which are drawn from the same general pool of individuals as the treatment group (e.g., from the same neighborhood), at least sometimes, better match the findings of randomized experiments, relative to nonequivalent group designs with "external" control groups (e.g., individuals from another community across the state). An alternative approach that can sometimes increase the comparability of a comparison group is by using a cohort control, such as younger or older siblings or the previous sixth-grade class in an educational context (see Cook & Campbell, 1979). Another lesson from the same studies is that having a good model of the selection process (i.e., the process whereby individuals end up in the treatment or in the comparison group) facilitates statistical analyses that increase the correspondence between the nonequivalent group design's results and the findings of randomized experiments (Shadish et al., 2005). Analogous to the classic "play within a play," Shadish and Clark (2007) randomly assigned participants to either of two study designs, a randomized experiment or an otherwise comparable nonequivalent group design. With numerous measures of potential self-selection processes in the nonequivalent group design, adjustment via propensity scores led to comparable results as in the randomized experiment.

In short, although not yet conclusive, the literature that compares findings from randomized experiments and quasi-experiments suggests several lessons. Many of these echo points from earlier in this chapter. First, use of a stronger quasi-experimental design, rather than a queasier one, appears to be highly desirable. Second, not all comparison groups are alike, and procedures such as using an internal control group or a cohort control—by creating a comparison group more initially similar to the treatment group—may result in more accurate findings. Third, statistical controls for selection bias will be enhanced to the extent the researcher has a good understanding of the selection process and measures the variables that are involved. Fourth, rather than relying only on statistical adjustments, the quasi-experimentalist should rely on the logic of competitive elaboration, considering the full range of comparisons that can be used to try to deal with selection and other validity threats (e.g., nonequivalent dependent variables and theory-driven subgroup analyses). Fifth, although the argument for replication is important in research generally, it may be stronger for research using quasi-experiments given the possibility not only of bias despite the researcher's best efforts, but also of more variability in treatment effect estimates.

Conclusion

A variety of designs are available for estimating the effects of a treatment. No single design type is always best. The choice among designs depends on the circumstances of a study, particularly on how well potential threats to validity and other criticisms can be avoided under the given circumstance. For this reason, researchers would be well-advised to consider a variety of designs before making their final choices. Researchers should evaluate each design relative to the potential validity threats that are likely to be most plausible in their specific research contexts.

Researchers should also be mindful that they can rule out threats to validity by adding comparisons that put the treatment and potential threats into direct competition. Sometimes, researchers can add such a comparison simply by disaggregating data that have already been collected. For example, in studying the HRT-breast cancer relationship, researchers could render threats implausible by disaggregating the available data into a subgroup of women of the age typically treated by HRT and of women of different ages. In other cases, researchers must plan ahead of time to collect data that allows the additional comparisons needed to evaluate threats to validity.

At its best, quasi-experimentation is not simply a matter of picking a prototypical design out of a book. Rather, considerable intellectual challenge is encountered in recognizing potential threats to validity and in elaborating design comparisons so as to minimize uncertainty about the size of the treatment effect. Indeed, the fact that it can be challenging to get the right answer with quasi-experiments, especially the queasier ones, is an argument for the use of randomized experiments. In this regard, researchers have recently attempted to integrate quasi-experiments with randomized experiments, such as using ITS designs in conjunction with small "*N*" experiments (Bloom et al., 2005; Riccio & Bloom, 2002). However, when random assignment is not feasible, implementing a strong quasi-experimental design and creatively employing the strategy of competitive elaboration is highly recommended. Indeed, an argument can be made that it is unethical to implement a weak design and thereby obtain biased results that could prolong the use of treatments that appear effective but are not (Rosenthal, 1994). In this regard, it is noteworthy that many treatments thought to be helpful have later been proven to be harmful, and many treatments thought to be harmful have later been proven to be helpful (Goodwin & Goodwin, 1984). Consider the widespread use of HRT as a treatment for menopause as a recent case in point.

Regardless of the chosen design and the elaborateness of comparisons, however, some uncertainty about the size of treatment effects will always remain. It is impossible to rule out completely all threats to validity. Ultimately, researchers must rely on accumulating evidence across multiple designs and the corresponding multiple estimates of effects. Usually, this accumulation is accomplished across research projects, but sometimes wise and adequately funded researchers are able to implement multiple designs and produce multiple estimates in a single research project. For example, the project reported by Lipsey, Cordray, and Berger (1981) remains exemplary in a number of ways, not the least of which is that their evaluation of the effects of juvenile diversion programs on criminal recidivism incorporated multiple comparisons, including an ITS design, nonequivalent group design, randomized experiment, and R-D design. The convergence of estimates across these designs enabled a more confident conclusion than would have been warranted based on any one of the designs alone.

Discussion Questions

1. Quasi-experiments are appropriate for certain research questions but not others. Generate four or five examples of research questions for which a quasi-experiment would make sense and also four or five research questions for which a quasi-experiment would not make sense.

2. Look at the two sets of research questions you generated in response to the previous question. What differentiates the two sets?

3. Discuss the assertion that, in certain circumstances, even a relatively queasy quasi-experiment should suffice. Try to describe an example (not one from the chapter) where a weaker quasi-experiment would be good enough.

4. Conversely, are there circumstances where you think only a randomized experiment would be adequate?

5. Think about what makes one quasi-experiment queasy and another one relatively rigorous. Explain.

6. The chapter discussed a possible future study of the effects of the recent rapid decline in hormone replacement therapy for menopausal women. Discuss the way that a more elaborate set of evidence could enhance causal inference in that study.

Exercises

1. Identify a real or hypothetical applied social research question that can be examined quasi-experimentally. Then, in Step 1, describe a relatively weak quasi-experiment (e.g., a one-group pretest-posttest design or a posttest-only nonequivalent group design) to examine the research question. In Step 2, apply a pretest-posttest nonequivalent group design to the same research question. In Step 3, try to apply a relatively rigorous quasi-experiment (some form of ITS design or a regression-discontinuity design). At each step, explain what key internal validity threats are plausible. For the second step (the pretest-posttest nonequivalent group design) and the third step (the ITS or R-D design), indicate how that design rules out threats that the weaker design did not.

2. Curvilinearity is a threat to the regression-discontinuity design. Draw a figure to show why this is the case (remember that a simple regression analysis fits straight lines).

3. Pretend you were one of the first researchers to try to study the hypothesis that smoking tobacco causes lung cancer. Using the logic of ruling out threats to validity, identify an elaborate set of comparisons you could make to assess the causal hypothesis.

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CHAPTER 7

Designing a Qualitative Study

Joseph A. Maxwell

Traditionally, works on research design (most of which focus on quantitative research) have understood "design" in one of two ways. Some take designs to be fixed, standard arrangements of research conditions and methods that have their own coherence and logic, as possible answers to the question, "What research design are you using?" (e.g., Campbell & Stanley, 1967). For example, a randomized, double-blind experiment is one research design; an interrupted time-series design is another. Beyond such broad categories as ethnographies, qualitative interview studies, and case studies (which often overlap), qualitative research lacks any such elaborate typology into which studies can be pigeonholed. In addition, typologies are usually based on a limited number of features of the study, and by themselves do little to clarify the actual functioning and interrelationship of the component parts of a design.

Other models present design as a logical progression of stages or tasks, from problem formulation to the generation of conclusions or theory, that are necessary in planning or carrying out a study (e.g., Creswell, 1997; Marshall & Rossman, 1999). Such models usually resemble a flowchart with a clear starting point and goal and a specified order for doing the intermediate tasks. Although some versions of this approach are circular or iterative (see, e.g., Bickman & Rog, Chapter 1, this volume), so that later steps connect back to earlier ones, all such models are linear in the sense that they are made up of one-directional sequences of steps that represent what is seen as the optimal order for conceptualizing or conducting the different components or activities of a study.

Neither of these models adequately represents the logic and process of qualitative research. In a qualitative study, "research design should be a reflexive process operating through every stage of a project" (Hammersley & Atkinson, 1995, p. 24);

the activities of collecting and analyzing data, developing and modifying theory, elaborating or refocusing the research questions, and identifying and dealing with validity threats are usually going on more or less simultaneously, each influencing all of the others. In addition, the researcher may need to reconsider or modify any design decision during the study in response to new developments or to changes in some other aspect of the design. Grady and Wallston (1988) argue that applied research in general requires a flexible, nonsequential approach and "an entirely different model of the research process than the traditional one offered in most textbooks" (p. 10).

This does not mean that qualitative research lacks design; as Yin (1994) says, "Every type of empirical research has an implicit, if not explicit, research design" (p. 19). Qualitative research simply requires a broader and less restrictive concept of "design" than the traditional ones described above. Thus, Becker, Geer, Hughes, and Strauss (1961), authors of a classic qualitative study of medical students, begin their chapter titled "Design of the Study" by stating,

> In one sense, our study had no design. That is, we had no well-worked-out set of hypotheses to be tested, no data-gathering instruments purposely designed to secure information relevant to these hypotheses, no set of analytic procedures specified in advance. Insofar as the term "design" implies these features of elaborate prior planning, our study had none.
>
> If we take the idea of design in a larger and looser sense, using it to identify those elements of order, system, and consistency our procedures did exhibit, our study had a design. We can say what this was by describing our original view of the problem, our theoretical and methodological commitments, and the way these affected our research and were affected by it as we proceeded. (p. 17)

For these reasons, the model of design that I present here, which I call an *interactive* model, consists of the components of a research study and the ways in which these components may affect and be affected by one another. It does not presuppose any particular order for these components, or any necessary directionality of influence.

The model thus resembles the more general definition of *design* employed outside research: "An underlying scheme that governs functioning, developing, or unfolding" and "the arrangement of elements or details in a product or work of art" (Frederick et al., 1993). A good design, one in which the components work harmoniously together, promotes efficient and successful functioning; a flawed design leads to poor operation or failure.

Traditional (typological or linear) approaches to design provide a model *for* conducting the research—a prescriptive guide that arranges the components or tasks involved in planning or conducting a study in what is seen as an optimal order. In contrast, the model presented in this chapter is a model *of* as well as *for* research. It is intended to help you understand the *actual* structure of your study as well as to plan this study and carry it out. An essential feature of this model is that it treats research design as a real entity, not simply an abstraction or plan. Borrowing

Kaplan's (1964, p. 8) distinction between the "logic-in-use" and "reconstructed logic" of research, this model can be used to represent the "design-in-use" of a study, the *actual* relationships among the components of the research, as well as the intended (or reconstructed) design (Maxwell & Loomis, 2002).

This model of research design has five components, each of which addresses a different set of issues that are essential to the coherence of a study:

1. *Goals:* Why is your study worth doing? What issues do you want it to clarify, and what practices and policies do you want it to influence? Why do you want to conduct this study, and why should we care about the results?

2. *Conceptual framework*: What do you think is going on with the issues, settings, or people you plan to study? What theories, beliefs, and prior research findings will guide or inform your research, and what literature, preliminary studies, and personal experiences will you draw on for understanding the people or issues you are studying?

3. *Research questions*: What, specifically, do you want to learn or understand by doing this study? What do you not know about the things you are studying that you want to learn? What questions will your research attempt to answer, and how are these questions related to one another?

4. *Methods*: What will you actually do in conducting this study? What approaches and techniques will you use to collect and analyze your data, and how do these constitute an integrated strategy?

5. *Validity*: How might your results and conclusions be wrong? What are the plausible alternative interpretations and validity threats to these, and how will you deal with these? How can the data that you have, or that you could potentially collect, support or challenge your ideas about what's going on? Why should we believe your results?

I have not identified ethics as a separate component of research design. This isn't because I don't think ethics is important for qualitative design; on the contrary, attention to ethical issues in qualitative research is being increasingly recognized as essential (Christians, 2000; Denzin & Lincoln, 2000; Fine, Weis, Weseen, & Wong, 2000). Instead, it is because I believe that ethical concerns should be involved in *every* aspect of design. I have particularly tried to address these concerns in relation to methods, but they are also relevant to your goals, the selection of your research questions, validity concerns, and the critical assessment of your conceptual framework.

These components are not substantially different from the ones presented in many other discussions of qualitative or applied research design (e.g., LeCompte & Preissle, 1993; Lincoln & Guba, 1985; Miles & Huberman, 1994; Robson, 2002). What is innovative is the way the relationships among the components are conceptualized. In this model, the different parts of a design form an integrated and interacting whole, with each component closely tied to several others, rather than being linked in a linear or cyclic sequence. The most important relationships among these five components are displayed in Figure 7.1.

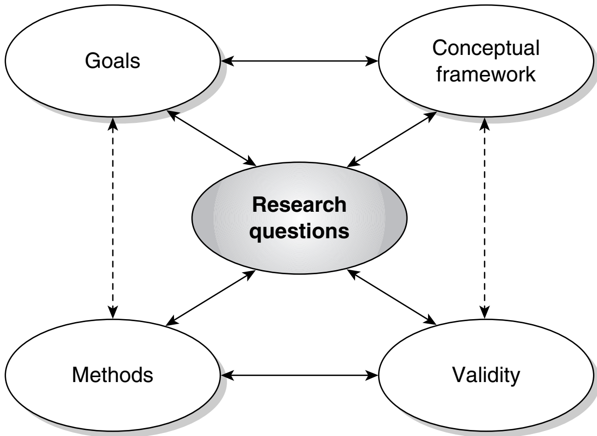

Figure 7.1 An Interactive Model of Research Design

SOURCE: From *Qualitative Research Design: An Interactive Approach,* by J. A. Maxwell, 2005. Copyright by SAGE.

There are also connections other than those emphasized here, some of which I have indicated by dashed lines. For example, if a goal of your study is to empower participants to conduct their own research on issues that matter to them, this will shape the methods you use, and conversely the methods that are feasible in your study will constrain your goals. Similarly, the theories and intellectual traditions you are drawing on in your research will have implications for what validity threats you see as most important and vice versa.

The upper triangle of this model should be a closely integrated unit. Your research questions should have a clear relationship to the goals of your study and should be informed by what is already known about the phenomena you are studying and the theoretical concepts and models that can be applied to these phenomena. In addition, the goals of your study should be informed by current theory and knowledge, while your decisions about what theory and knowledge are relevant depend on your goals and questions.

Similarly, the bottom triangle of the model should also be closely integrated. The methods you use must enable you to answer your research questions, and also to deal with plausible validity threats to these answers. The questions, in turn, need to be framed so as to take the feasibility of the methods and the seriousness of particular validity threats into account, while the plausibility and relevance of particular validity threats, and the ways these can be dealt with, depend on the questions and methods chosen. The research questions are the heart, or hub, of the model; they connect all the other components of the design, and should inform, and be sensitive to, these components.

There are many other factors besides these five components that should influence the design of your study; these include your research skills, the available resources, perceived problems, ethical standards, the research setting, and the data and

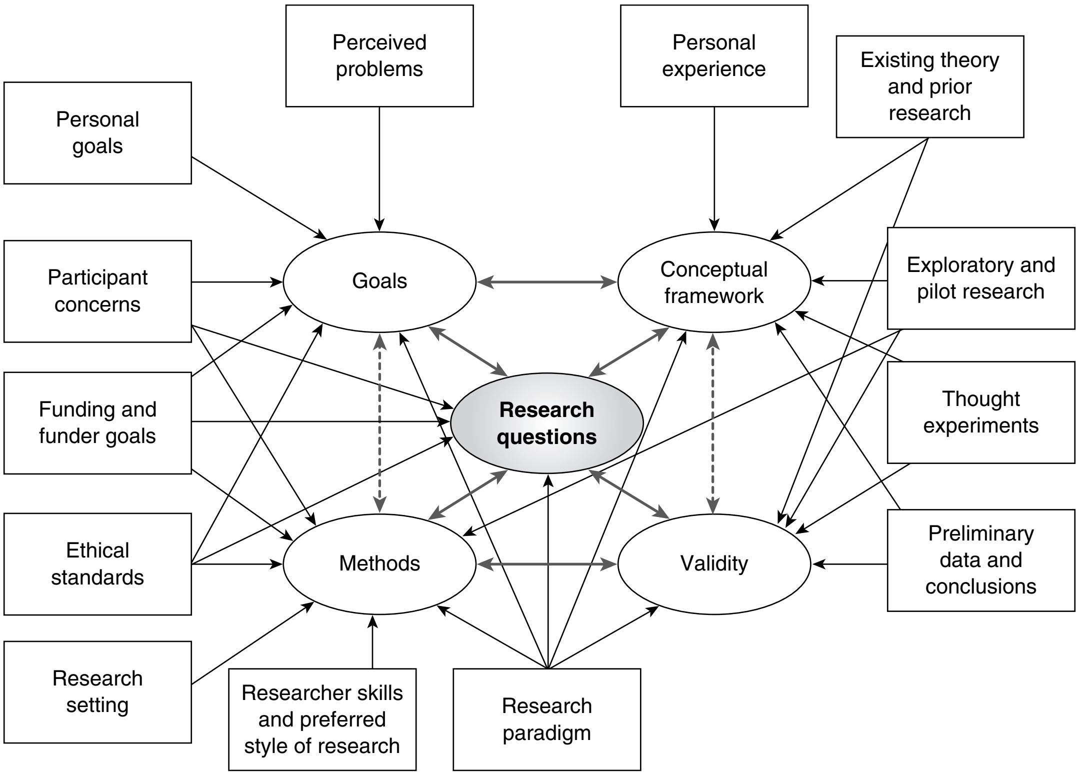

Figure 7.2 Contextual Factors Influencing a Research Design

preliminary conclusions of the study. In my view, these are not part of the *design* of a study; rather, they either belong to the *environment* within which the research and its design exist or are *products* of the research. Figure 7.2 presents some of the environmental factors that can influence the design and conduct of a study.

I do not believe that there is one right model for qualitative or applied research design. However, I think that the model I present here is a useful one, for three main reasons:

1. It explicitly identifies as components of design the key issues about which decisions need to be made. These issues are therefore less likely to be ignored, and can be dealt with in a systematic manner.

2. It emphasizes the interactive nature of design decisions in qualitative and applied research, and the multiple connections among the design components.

3. It provides a model for the structure of a proposal for a qualitative study, one that clearly communicates and justifies the major design decisions and the connections among these (see Maxwell, 2005).

SOURCE: From *Qualitative Research Design: An Interactive Approach,* by J. A. Maxwell, 2005. Copyright by SAGE.

Because a design for your study always exists, explicitly or implicitly, it is important to *make* this design explicit, to get it out in the open, where its strengths, limitations, and implications can be clearly understood. In the remainder of this chapter, I present the main design issues involved in each of the five components of my model, and the implications of each component for the others. I do not discuss in detail how to actually *do* qualitative research, or deal in depth with the theoretical and philosophical views that have informed this approach. For additional guidance on these topics, see the contributions of Fetterman (Chapter 17, this volume) and Stewart, Shamdasani, and Rook (Chapter 18, this volume) to this *Handbook*; the more extensive treatments by Patton (2000), Eisner and Peshkin (1990), LeCompte and Preissle (1993), Glesne (2005), Weiss (1994), Miles and Huberman (1994), and Wolcott (1995); and the encyclopedic handbooks edited by Denzin and Lincoln (2005) and Given (in press). My focus here is on how to design a qualitative study that arrives at valid conclusions and successfully and efficiently achieve its goals.

Goals: Why Are You Doing This Study?

Anyone can find an unanswered, empirically answerable question to which the answer isn't *worth* knowing; as Thoreau said, it is not worthwhile to go around the world to count the cats in Zanzibar. Without a clear sense of the goals of your research, you are apt to lose your focus and spend your time and effort doing things that won't contribute to these goals. (I use *goals* here in a broad sense, to include motives, desires, and purposes—anything that leads you to do the study or that you hope to accomplish by doing it.) These goals serve two main functions for your research. First, they help guide your other design decisions to ensure that your study is *worth* doing, that you get out of it what you want. Second, they are essential to *justifying* your study, a key task of a funding or dissertation proposal. In addition, your goals inevitably shape the descriptions, interpretations, and theories you create in your research. They therefore constitute not only important *resources* that you can draw on in planning, conducting, and justifying the research, but also potential *validity threats,* or sources of bias, that you will need to deal with.

It is useful to distinguish among three kinds of goals for doing a study: personal goals, practical goals, and intellectual goals. Personal goals are those that motivate *you* to do this study; they can include a desire to change some existing situation, a curiosity about a specific phenomenon or event, or simply the need to advance your career. These personal goals often overlap with your practical or research goals, but they may also include deeply rooted individual desires and needs that bear little relationship to your "official" reasons for doing the study.

It is important that you recognize and take account of the personal goals that drive and inform your research. Eradicating or submerging your personal goals and concerns is impossible, and attempting to do so is unnecessary. What *is* necessary, in qualitative design, is that you be *aware* of these concerns and how they may be shaping your research, and that you think about how best to deal with their consequences.

To the extent that you have *not* made a careful assessment of ways in which your design decisions and data analyses are based on personal desires, you are in danger of arriving at invalid conclusions.

However, your personal reasons for wanting to conduct a study, and the experiences and perspectives in which these are grounded, are not simply a source of "bias" (see the later discussion of this issue in the section on validity); they can also provide you with a valuable source of insight, theory, and data about the phenomena you are studying (Marshall & Rossman, 1999, pp. 25–30; Strauss & Corbin, 1990, pp. 42–43). This source is discussed in the next section, in the subsection on experiential knowledge.

Two major decisions are often profoundly influenced by the researcher's personal goals. One is the topic, issue, or question selected for study. Traditionally, students have been told to base this decision on either faculty advice or the literature on their topic. However, personal goals and experiences play an important role in many research studies. Strauss and Corbin (1990) argue that

> choosing a research problem through the professional or personal experience route may seem more hazardous than through the suggested [by faculty] or literature routes. This is not necessarily true. The touchstone of your own experience may be more valuable an indicator for you of a potentially successful research endeavor. (pp. 35–36)

A second decision that is often influenced by personal goals and experiences is the choice of a qualitative approach. Locke, Spirduso, and Silverman (1993) argue that "every graduate student who is tempted to employ a qualitative design should confront one question, 'Why do I want to do a qualitative study?' and then answer it honestly" (p. 107). They emphasize that qualitative research is *not* easier than quantitative and that seeking to avoid statistics bears little relationship to having the personal interests and skills that qualitative inquiry requires (pp. 107–110). The key issue is the compatibility of your reasons for "going qualitative" with your other goals, your research questions, and the actual activities involved in doing a qualitative study.

Besides your personal goals, there are two other kinds of goals that I want to distinguish and discuss, ones that are important for other people, not just yourself: practical goals (including administrative or policy goals) and intellectual goals. Practical goals are focused on *accomplishing* something—meeting some need, changing some situation, or achieving some goal. Intellectual goals, on the other hand, are focused on *understanding* something, gaining some insight into what is going on and why this is happening. Although applied research design places much more emphasis on practical goals than does basic research, you still need to address the issues of what you want to *understand* by doing the study and how this understanding will contribute to your accomplishing your practical goals. (The issue of what you want to understand is discussed in more detail below, in the section on research questions.)

There are five particular intellectual goals for which qualitative studies are especially useful:

1. Understanding the *meaning,* for participants in the study, of the events, situations, and actions they are involved with, and of the accounts that they give of their lives and experiences. In a qualitative study, you are interested not only in the physical events and behavior taking place, but also in how the participants in your study make sense of these and how their understandings influence their behavior. The perspectives on events and actions held by the people involved in them are not simply their accounts of these events and actions, to be assessed in terms of truth or falsity; they are part of the reality that you are trying to understand, and a major influence on their behavior (Maxwell, 1992, 2004a). This focus on meaning is central to what is known as the "interpretive" approach to social science (Bredo & Feinberg, 1982; Geertz, 1973; Rabinow & Sullivan, 1979).

2. Understanding the particular context within which the participants act and the influence this context has on their actions. Qualitative researchers typically study a relatively small number of individuals or situations and preserve the individuality of each of these in their analyses, rather than collecting data from large samples and aggregating the data across individuals or situations. Thus, they are able to understand how events, actions, and meanings are shaped by the unique circumstances in which these occur.

3. Identifying unanticipated phenomena and influences and generating new, "grounded" theories about the latter. Qualitative research has long been used for this goal by survey and experimental researchers, who often conduct "exploratory" qualitative studies to help them design their questionnaires and identify variables for experimental investigation. Although qualitative research is not restricted to this exploratory role, it is still an important strength of qualitative methods.

4. Understanding the processes by which events and actions take place. Although qualitative research is not unconcerned with outcomes, a major strength of qualitative studies is their ability to get at the processes that lead to these outcomes, processes that experimental and survey research are often poor at identifying (Maxwell, 2004a).

5. Developing causal explanations. The traditional view that qualitative research cannot identify causal relationships is based on a restrictive and philosophically outdated concept of causality (Maxwell, 2004b), and both qualitative and quantitative researchers are increasingly accepting the legitimacy of using qualitative methods for causal inference (e.g., Shadish, Cook, & Campbell, 2002). Such an approach requires thinking of causality in terms of processes and mechanisms, rather than simply demonstrating regularities in the relationships between variables (Maxwell, 2004a); I discuss this in more detail in the section on research questions. Deriving causal explanations from a qualitative study is not an easy or straightforward task, but qualitative research is not different from quantitative research in this respect. Both approaches need to identify and deal with the plausible validity threats to any proposed causal explanation, as discussed below.

These intellectual goals, and the inductive, open-ended strategy that they require, give qualitative research an advantage in addressing numerous practical goals, including the following.

Generating results and theories that are understandable and experientially credible, both to the people being studied and to others (Bolster, 1983). Although quantitative data may have greater credibility for some goals and audiences, the specific detail and personal immediacy of qualitative data can lead to the greater influence of the latter in other situations. For example, I was involved in one evaluation, of how teaching rounds in one hospital department could be improved, that relied primarily on participant observation of rounds and open-ended interviews with staff physicians and residents (Maxwell, Cohen, & Reinhard, 1983). The evaluation led to decisive department action, in part because department members felt that the report, which contained detailed descriptions of activities during rounds and numerous quotes from interviews to support the analysis of the problems with rounds, "told it like it really was" rather than simply presenting numbers and generalizations to back up its recommendations.

Conducting formative studies, ones that are intended to help improve existing practice rather than simply to determine the outcomes of the program or practice being studied (Scriven, 1991). In such studies, which are particularly useful for applied research, it is more important to understand the process by which things happen in a particular situation than to measure outcomes rigorously or to compare a given situation with others.

Engaging in collaborative, action, or "empowerment" research with practitioners or research participants (e.g., Cousins & Earl, 1995; Fetterman, Kaftarian, & Wandersman, 1996; Tolman & Brydon-Miller, 2001; Whyte, 1991). The focus of qualitative research on particular contexts and their meaning for the participants in these contexts, and on the processes occurring in these contexts, makes it especially suitable for collaborations with practitioners or with members of the community being studied (Patton, 1990, pp. 129–130; Reason, 1994).

A useful way of sorting out and formulating the goals of your study is to write memos in which you reflect on your goals and motives, as well as the implications of these for your design decisions (for more information on such memos, see Maxwell, 2005, pp. 11–13; Mills, 1959, pp. 197–198; Strauss & Corbin, 1990, chap. 12). See Exercise 1.

Conceptual Framework: What Do You Think Is Going On?

The conceptual framework of your study is the system of concepts, assumptions, expectations, beliefs, and theories that supports and informs your research. Miles and Huberman (1994) state that a conceptual framework "explains, either graphically or in narrative form, the main things to be studied—the key factors, concepts, or variables—and the presumed relationships among them" (p. 18). Here, I use the term in a broader sense that also includes the actual ideas and beliefs that you hold about the phenomena studied, whether these are written down or not.

Thus, your conceptual framework is a formulation of what you think is *going on* with the phenomena you are studying—a tentative *theory* of what is happening and why. Theory provides a model or map of *why* the world is the way it is (Strauss, 1995). It is a simplification of the world, but a simplification aimed at clarifying and explaining some aspect of how it works. It is not simply a "framework," although it can provide that, but a *story* about what you think is happening and why. A useful theory is one that tells an enlightening story about some phenomenon, one that gives you new insights and broadens your understanding of that phenomenon. The function of theory in your design is to inform the rest of the design—to help you assess your goals, develop and select realistic and relevant research questions and methods, and identify potential validity threats to your conclusions.

What is often called the "research problem" is a part of your conceptual framework, and formulating the research problem is often seen as a key task in designing your study. It is part of your conceptual framework (although it is often treated as a separate component of a research design) because it identifies something that is *going on* in the world, something that is itself problematic or that has consequences that are problematic.

The conceptual framework of a study is often labeled the "literature review." This can be a dangerously misleading term, for three reasons. First, it can lead you to focus narrowly on "literature," ignoring other conceptual resources that may be of equal or greater importance for your study, including unpublished work, communication with other researchers, and your own experience and pilot studies. Second, it tends to generate a strategy of "covering the field" rather than focusing specifically on those studies and theories that are particularly *relevant* to your research (Maxwell, 2006). Third, it can make you think that your task is simply descriptive—to tell what previous researchers have found or what theories have been proposed. In developing a conceptual framework, your purpose is not only descriptive, but also critical; you need to treat "the literature" not as an *authority* to be deferred to, but as a useful but fallible source of *ideas* about what's going on, and to attempt to see alternative ways of framing the issues (Locke, Silverman, & Spirduso, 2004).

Another way of putting this is that the conceptual framework for your research study is something that is *constructed,* not found. It incorporates pieces that are borrowed from elsewhere, but the structure, the overall coherence, is something that *you* build, not something that exists ready-made. Becker (1986, 141ff.) systematically develops the idea that prior work provides *modules* that you can use in building your conceptual framework, modules that you need to examine critically to make sure they work effectively with the rest of your design. There are four main sources for these modules: your own experiential knowledge, existing theory and research, pilot and exploratory studies, and thought experiments. Before addressing the sources of these modules, however, I want to discuss a particularly important part of your conceptual framework—the research paradigm(s) within which you situate your work.

Connecting With a Research Paradigm

One of the critical decisions that you will need to make in designing your study is the paradigm (or paradigms) within which you will situate your work. This use of the term *paradigm*, which derives from the work of the historian of science Thomas Kuhn, refers to a set of very general philosophical assumptions about the nature of the world (ontology) and how we can understand it (epistemology), assumptions that tend to be shared by researchers working in a specific field or tradition. Paradigms also typically include specific methodological strategies linked to these assumptions, and identify particular studies that are seen as exemplifying these assumptions and methods. At the most abstract and general level, examples of such paradigms are philosophical positions such as positivism, constructivism, realism, and pragmatism, each embodying very different ideas about reality and how we can gain knowledge of it. At a somewhat more specific level, paradigms that are relevant to qualitative research include interpretivism, critical theory, feminism, postmodernism, and phenomenology, and there are even more specific traditions within these (for more detailed guidance, see Creswell, 1997; Schram, 2005). I want to make several points about using paradigms in your research design:

1. Although some people refer to "the qualitative paradigm," there are many different paradigms within qualitative research, some of which differ radically in their assumptions and implications (see also Denzin & Lincoln, 2000; Pitman & Maxwell, 1992). You need to make explicit which paradigm(s) your work will draw on, since a clear paradigmatic stance helps guide your design decisions and to justify these decisions. Using an established paradigm (such as grounded theory, critical realism, phenomenology, or narrative research) allows you to build on a coherent and well-developed approach to research, rather than having to construct all of this yourself.

2. You don't have to adopt in total a single paradigm or tradition. It is possible to combine aspects of different paradigms and traditions, although if you do this you will need to carefully assess the compatibility of the modules that you borrow from each. Schram (2005) gives a valuable account of how he combined the ethnographic and life history traditions in his dissertation research on an experienced teacher's adjustment to a new school and community.

3. Your selection of a paradigm (or paradigms) is not a matter of free choice. You have already made many assumptions about the world, your topic, and how we can understand these, even if you have never consciously examined these. Choosing a paradigm or tradition primarily involves assessing which paradigms best fit with your own assumptions and methodological preferences; Becker (1986, pp. 16–17) makes the same point about using theory in general. Trying to work within a paradigm (or theory) that doesn't fit your assumptions is like trying to do a physically demanding job in clothes that don't fit—at best you'll be uncomfortable, at worst it will keep you from doing the job well. Such a lack of fit may not be obvious at the outset; it may only emerge as you develop your conceptual framework, research questions, and methods, since these should also be compatible with your paradigmatic stance.

Experiential Knowledge

Traditionally, what you bring to the research from your background and identity has been treated as "bias," something whose influence needs to be *eliminated* from the design, rather than a valuable component of it. However, the explicit incorporation of your identity and experience (what Strauss, 1987, calls "experiential data") in your research has recently gained much wider theoretical and philosophical support (e.g., Berg & Smith, 1988; Denzin & Lincoln, 2000; Jansen & Peshkin, 1992; Strauss, 1987). Using this experience in your research can provide you with a major source of insights, hypotheses, and validity checks. For example, Grady and Wallston (1988, p. 41) describe how one health care researcher used insights from her own experience to design a study of why many women don't do breast self-examination.

This is not a license to impose your assumptions and values uncritically on the research. Reason (1988) uses the term *critical subjectivity* to refer to

> a quality of awareness in which we do not suppress our primary experience; nor do we allow ourselves to be swept away and overwhelmed by it; rather we raise it to consciousness and use it as part of the inquiry process. (p. 12)

However, there are few well-developed and explicit strategies for doing this. The "researcher identity memo" is one technique; this involves reflecting on, and writing down, the different aspects of your experience that are potentially relevant to your study. Example 7.1 is part of one of my own researcher identity memos, written when I was working on a paper of diversity and community; Exercise 1 involves writing your own researcher identity memo. (For more on this technique, see Maxwell, 2005.) Doing this can generate unexpected insights and connections, as well as create a valuable record of these.

Example 7.1 Identity Memo on Diversity

I can't recall when I first became interested in diversity; it's been a major concern for at least the past 20 years . . . I do remember the moment that I consciously realized that my mission in life was "to make the world safe for diversity"; I was in Regenstein Library at the University of Chicago one night in the mid-1970s talking to another student about why we had gone into anthropology, and the phrase suddenly popped into my head.

However, I never gave much thought to tracing this position any further back. I remember, as an undergraduate, attending a talk on some political topic, and being struck by two students' bringing up issues of the rights of particular groups to retain their cultural heritages; it was an issue that had never consciously occurred to me. And I'm sure that my misspent youth reading science fiction rather than studying had a powerful influence on my sense of the importance of tolerance and understanding of diversity; I wrote my essay for my application to college on tolerance in high school society. But I didn't think much about where all this came from.

(Continued)

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It was talking to the philosopher Amelie Rorty in the summer of 1991 that really triggered my awareness of these roots. She had given a talk on the concept of moral diversity in Plato, and I gave her a copy of my draft paper on diversity and solidarity. We met for lunch several weeks later to discuss these issues, and at one point she asked me how my concern with diversity connected with my background and experiences. I was surprised by the question, and found I really couldn't answer it. She, on the other hand, had thought about this a lot, and talked about her parents emigrating from Belgium to the United States, deciding they were going to be farmers like "real Americans," and with no background in farming, buying land in rural West Virginia and learning how to survive and fit into a community composed of people very different from themselves.

This made me start thinking, and I realized that as far back as I can remember I've felt different from other people, and had a lot of difficulties as a result of this difference and my inability to "fit in" with peers, relatives, or other people generally. This was all compounded by my own shyness and tendency to isolate myself, and by the frequent moves that my family made while I was growing up.

The way in which this connects with my work on diversity is that my main strategy for dealing with my difference from others, as far back as I can remember, was *not* to try to be more *like* them (similarity-based), but to try to be *helpful* to them (contiguity-based). This is a bit oversimplified, because I also saw myself as somewhat of a "social chameleon," adapting to whatever situation I was in, but this adaptation was much more an *interactional* adaptation than one of becoming fundamentally similar to other people.

It now seems incomprehensible to me that I never saw the connections between this background and my academic work.

[The remainder of the memo discusses the specific connections between my experience and the theory of diversity and community that I had been developing, which sees both similarity (shared characteristics) and contiguity (interaction) as possible sources of solidarity and community.]

SOURCE: From *Qualitative Research Design: An Interactive Approach,* by J. A. Maxwell, 2005. Copyright by SAGE.

Existing Theory and Research

The second major source of modules for your conceptual framework is existing theory and research—not simply published work, but also unpublished papers and dissertations, conference presentations, and what is in the heads of active researchers in your field (Locke, Spirduso, & Silverman, 2000). I will begin with theory, because it is for most people the more problematic and confusing of the two, and then deal with using prior research for other purposes than as a source of theory.

Using existing theory in qualitative research has both advantages and dangers. A useful theory helps you *organize* your data. Particular pieces of information that otherwise might seem unconnected or irrelevant to one another or to your research questions can be related if you can fit them into the theory. A useful theory also *illuminates* what you are seeing in your research. It draws your attention to particular events or phenomena and sheds light on relationships that might otherwise go unnoticed or misunderstood.

However, Becker (1986) warns that the existing literature, and the assumptions embedded in it, can deform the way you frame your research, causing you to overlook important ways of conceptualizing your study or key implications of your results. The literature has the advantage of what he calls "ideological hegemony," making it difficult for you to see any phenomenon in ways that are different from those that are prevalent in the literature. Trying to fit your insights into this established framework can deform your argument, weakening its logic and making it harder for you to see what this new way of framing the phenomenon might contribute. Becker describes how existing theory and perspectives deformed his early research on marijuana use, leading him to focus on the dominant question in the literature and to ignore the most interesting implications and possibilities of his study.

Becker (1986) argues that there is no way to be sure when the established approach is wrong or misleading or when your alternative is superior. All you can do is try to identify the ideological component of the established approach, and see what happens when you abandon these assumptions. He asserts that "a serious scholar ought routinely to inspect competing ways of taking about the same subject matter," and warns, "Use the literature, don't let it use you" (p. 149; see also Mills, 1959).

A review of relevant prior research can serve several other purposes in your design besides providing you with existing theory (see Locke et al., 2004; Strauss, 1987, pp. 48–56). First, you can use it to develop a *justification* for your study—to show how your work will address an important need or unanswered question. Second, it can inform your *decisions about methods,* suggesting alternative approaches or revealing potential problems with your plans. Third, it can be a source of *data* that you can use to test or modify your theories. You can see if existing theory, the results of your pilot research, or your experiential understanding is supported or challenged by previous studies. Finally, you can use ideas in the literature to help you *generate* theory, rather than simply borrowing such theory from the literature.

Pilot and Exploratory Studies

Pilot studies serve some of the same functions as prior research, but they can be focused more precisely on your own concerns and theories. You can design pilot studies specifically to test your ideas or methods and explore their implications, or to inductively develop *grounded* theory. One particular use that pilot studies have in qualitative research is to generate an understanding of the concepts and theories held by the people you are studying—what I have called "interpretation" (Maxwell, 1992).

This is not simply a source of additional concepts for your theory; instead, it provides you with an understanding of the *meaning* that these phenomena and events have for the actors who are involved in them, and the perspectives that inform their actions. In a qualitative study, these meanings and perspectives should constitute an important focus of your theory; as discussed earlier, they are one of the things your theory is *about,* not simply a source of theoretical insights and building blocks for the latter.

Thought Experiments

Thought experiments have a long and respected tradition in the physical sciences (much of Einstein's work was based on thought experiments) but have received little attention in discussions of research design, particularly qualitative research design. Thought experiments draw on both theory and experience to answer "what if" questions, to seek out the logical implications of various properties of the phenomena you want to study. They can be used both to test your current theory for logical problems and to generate new theoretical insights. They encourage creativity and a sense of exploration and can help you make explicit the experiential knowledge that you already possess. Finally, they are easy to do, once you develop the skill. Valuable discussions of thought experiments in the social sciences are presented by Mills (1959) and Lave and March (1975).

Experience, prior theory and research, pilot studies, and thought experiments are the four major sources of the conceptual framework for your study. The ways in which you can put together a useful and valid conceptual framework from these sources are particular to each study, and not something for which any cookbook exists. The main thing to keep in mind is the need for integration of these components with one another and with your goals and research questions.

Concept Mapping

A particularly valuable tool for generating and understanding these connections in your research is a technique known as concept mapping (Miles & Huberman, 1994; Novak & Gowin, 1984). Kane and Trochim (Chapter 14, this volume) provide an overview of concept mapping but focus on using concept mapping with groups of stakeholders for organizational improvement or evaluation, employing mainly quantitative techniques. However, concept mapping has many other uses, including clarification and development of your own ideas about what's going on with the phenomena you want to study. Exercise 2 is designed to help you develop an initial concept map for your study (for additional guidance, see the sources above and Maxwell, 2005).

Research Questions: What Do You Want to Understand?

Your research questions—what you specifically want to learn or understand by doing your study—are at the heart of your research design. They are the one component that directly connects to all the other components of the design. More than any other aspect of your design, your research questions will have an influence on, and should be responsive to, every other part of your study.

This is different from seeing research questions as the *starting point* or primary determinant of the design. Models of design that place the formulation of research questions at the beginning of the design process, and that see these questions as determining the other aspects of the design, don't do justice to the interactive and inductive nature of qualitative research. The research questions in a qualitative study should not be formulated in detail until the goals and conceptual framework (and sometimes general aspects of the sampling and data collection) of the design are clarified, and should remain sensitive and adaptable to the implications of other parts of the design. Often, you will need to do a significant part of the research before it is clear to you what specific research questions it makes sense to try to answer.

This does not mean that qualitative researchers should, or usually do, begin studies with *no* questions, simply going into the field with "open minds" and seeing what is there to be investigated. Every researcher begins with a substantial base of experience and theoretical knowledge, and these inevitably generate certain questions about the phenomena studied. These initial questions frame the study in important ways, influence decisions about methods, and are one basis for further focusing and development of more specific questions. However, these specific questions are generally the *result* of an interactive design process, rather than the starting point for that process. For example, Suman Bhattacharjea (1994; see also Maxwell, 2005, p. 66) spent a year doing field research on women's roles in a Pakistani educational district office before she was able to focus on two specific research questions and submit her dissertation proposal; at that point, she had also developed several hypotheses as tentative answers to these questions.

The Functions of Research Questions

In your research design, the research questions serve two main functions: to help you focus the study (the questions' relationship to your goals and conceptual framework) and to give you guidance for how to conduct it (their relationship to methods and validity). A design in which the research questions are too general or too diffuse creates difficulties both for conducting the study—in knowing what site or informants to choose, what data to collect, and how to analyze these data—and for clearly connecting what you learn to your goals and existing knowledge (Miles & Huberman, 1994, pp. 22–25). Research questions that are precisely framed too early in the study, on the other hand, may lead you to overlook areas of theory or prior experience that are relevant to your understanding of what is going on, or cause you to pay too little attention to a wide range of data early in the study, data that can reveal important and unanticipated phenomena and relationships.

A third problem is that you may be smuggling unexamined assumptions into the research questions themselves, imposing a conceptual framework that doesn't fit the reality you are studying. A research question such as "How do elementary school teachers deal with the experience of isolation from their colleagues in their classrooms?" assumes that teachers *do* experience such isolation. Such an assumption needs to be carefully examined and justified, and without this justification it might be better to frame such a question as a tentative subquestion to broader questions about the nature of classroom teachers' experience of their work and their relations with colleagues.

For all these reasons, there is real danger to your study if you do not carefully formulate your research questions in connection with the other components of your design. Your research questions need to take account of what you want to accomplish by doing the study (your goals), and of what is already known about the things you want to study and your tentative theories about these phenomena (your conceptual framework). There is no reason to pose research questions for which the answers are already available, that don't clearly connect to what you think is actually going on, or that would have no direct relevance to your goals in doing the research.

Likewise, your research questions need to be ones that are answerable by the kind of study you can actually conduct. There is no value to posing questions that no feasible study could answer, either because the data that could answer them could not be obtained, or because any conclusions you might draw from these data would be subject to serious validity threats.

A common problem in the development of research questions is confusion between research issues (what you want to *understand* by doing the study) and practical issues (what you want to *accomplish*). Your research questions need to *connect* clearly to your practical concerns, but in general an empirical study cannot directly answer practical questions such as, "How can I improve this program?" or "What is the best way to increase students' knowledge of science?" To address such practical questions, you need to focus on what you don't *understand* about the phenomena you are studying, and investigate what is really going on with these phenomena. For example, the practical goal of Martha Regan-Smith's (1992) dissertation research was to improve the teaching of the basic sciences in medical school (see Maxwell, 2005, 117ff.). However, her research questions focused not on this goal but on what exceptional teachers in her school did that helped students learn science—something she had realized that she didn't know and that she believed would have important implications for how to improve such teaching overall.

A second confusion, one that can create problems for interview studies, is that between *research* questions and *interview* questions. Your research questions identify the things that you want to understand; your interview questions generate the data that you need to understand these things. This distinction is discussed in more detail below, in the section on methods.

There are three issues that you should keep in mind in formulating research questions for applied social research. First, research questions may legitimately be framed in particular as well as general terms. There is a strong tendency in basic research to state research questions in general terms, such as, "How do students deal with racial and ethnic difference in multiracial schools?" and then to "operationalize" these questions by selecting a particular sample or site. This tendency can be counterproductive when the goal of your study is to understand and improve some particular program, situation, or practice. In applied research, it is often more appropriate to formulate research questions in particular terms, such as, "How do students at North High School deal with racial and ethnic difference?"

Second, some researchers believe that questions should be stated in terms of what the respondents report or what can be directly observed, rather than in terms of inferred behavior, beliefs, or causal influences. This is what I call an instrumentalist or positivist, rather than a realist, approach to research questions (Maxwell, 1992; Norris, 1983). Instrumentalists formulate their questions in terms of observable or measurable data and are suspicious of inferences to things that cannot be defined in terms of such data. For example, instrumentalists would reject a question such as, "How do exemplary teachers help medical students learn science?" and replace it with questions such as, "How do medical students *report that* exemplary teachers help them learn science?" or "How are exemplary teachers *observed to teach* basic science?"

Realists, in contrast, don't assume that research questions about feelings, beliefs, intentions, prior behavior, effects, and so on need to be reduced to, or reframed as, questions about the actual data that one uses. Instead, they treat their data as fallible *evidence* about these phenomena, to be used critically to develop and test ideas about what is going on (Campbell, 1988; Maxwell, 1992).

The main risk of using instrumentalist questions is that you will lose sight of what you are really interested in, and define your study in ways that obscure the actual phenomena you want to investigate, ending up with a rigorous but uninteresting conclusion. As in the joke about the man who was looking for his keys under the streetlight (rather than where he dropped them) because the light was better there, you may never find what you started out to look for. An instrumentalist approach to your research questions may also make it more difficult for your study to address important goals of your study directly, and it can inhibit your theorizing about phenomena that are not directly observable.

My own preference is to use realist questions and to address, as systematically and rigorously as possible, the validity threats that this approach involves. The seriousness of these validity threats (such as self-report bias) needs to be assessed in the context of a particular study; these threats are often not as serious as instrumentalists imply. There are also effective ways to address these threats in a qualitative design, which I discuss below in the section on validity. The risk of trivializing your study by restricting your questions to what can be directly observed is usually more serious than the risk of drawing invalid conclusions. As the statistician John Tukey (1962) put it, "Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise" (p. 13).

One issue that is not entirely a matter of realism versus instrumentalism is whether research questions in interview studies should be framed in terms of the respondents' perceptions or beliefs rather than the actual state of affairs. You should base this decision not simply on the seriousness of the validity threats, but also on what you actually want to understand. In many qualitative studies, the real interest is in how participants make sense of what has happened, and how this perspective informs their actions, rather than determining precisely what took place.

Finally, many researchers (consciously or unconsciously) focus their questions on variance rather than process (Maxwell, 2004a; Mohr, 1982, 1995, 1996). Variance questions deal with difference and correlation; they often begin with "Is there," "Does," "How much," or "To what extent." For example, a variance approach to Martha Regan-Smith's (1992) study would ask questions such as, "Do exemplary medical school teachers differ from others in their teaching of basic science?" or "Is there a relationship between teachers' behavior and students' learning?" and attempt to measure these differences and relationships. Process questions, in contrast, focus on *how* and *why* things happen, rather than *whether* there is a particular difference or relationship or how much it is explained by other variables. Regan-Smith's actual questions focused on *how* these teachers helped students learn—that is, the process by which their teaching helped the students learn.

In a qualitative study, it can be dangerous for you to frame your research questions in a way that focuses on differences and their explanation. This may lead you to begin thinking in variance terms, to try to identify the variables that will account for observed or hypothesized differences, and to overlook the real strength of a qualitative approach, which is in understanding the process by which phenomena take place. Variance questions are often best answered by quantitative approaches, which are powerful ways of determining *whether* a particular result is causally related to one or another variable, and to *what extent* these are related. However, qualitative research is often better at showing *how* this occurred. Variance questions are legitimate in qualitative research, but they are often best grounded in the answers to prior process questions (Maxwell 2004a).

Qualitative researchers therefore tend to generate two kinds of questions that are much better suited to process theory than to variance theory: (1) questions about the *meaning* of events and activities to the people involved in them and (2) questions about the influence of the physical and social *context* on these events and activities. (See the earlier discussion of meaning and context as research goals.) Because both of these types of questions involve situation-specific phenomena, they do not lend themselves to the kinds of comparison and control that variance theory requires. Instead, they generally involve an open-ended, inductive approach to discover what these meanings and influences are and *how* they are involved in these events and activities—an inherently processual orientation.

Developing relevant, focused, answerable research questions takes time; such questions cannot be thrown together quickly, nor in most studies can they be definitively formulated before data collection and analysis begin. Generating good questions requires that you pay attention not just to the questions themselves but to their connections with all the other design components: the goals that answering the questions might serve, the implications for your questions of your conceptual framework, the methods you could use to answer the questions, and the validity threats you will need to address. As is true with the other components of your design, writing memos about these issues is an extremely useful tool for developing your questions (Maxwell, 2005, pp. 76–78).

Methods: What Will You Actually Do?

There is no "cookbook" for doing qualitative research. The appropriate answer to almost any question about the use of qualitative methods is, "It depends." The value and feasibility of your research methods cannot be guaranteed by your adhering to methodological rules; rather, they depend on the specific setting and phenomena you are studying and the actual consequences of your strategy for studying it.

Prestructuring a Qualitative Study

One of the most important issues in designing a qualitative study is how much you should attempt to prestructure your methods. Structured approaches can help ensure the comparability of data across sources and researchers and are therefore particularly useful in answering variance questions, questions that deal with *differences* between things and the explanation for these differences. Unstructured approaches, in contrast, allow the researcher to focus on the *particular* phenomena studied; they trade generalizability and comparability for internal validity and contextual understanding and are particularly useful for understanding the processes that led to specific outcomes, what Huberman and Miles (1988) call "local causality." Sayer (1992, 241ff.) refers to these two approaches as "extensive" and "intensive" research designs, respectively.

However, Miles and Huberman (1994) warn that

> highly inductive, loosely designed studies make good sense when experienced researchers have plenty of time and are exploring exotic cultures, understudied phenomena, or very complex social phenomena. But if you're new to qualitative studies and are looking at a better understood phenomenon within a familiar culture or subculture, a loose, inductive design is a waste of time. Months of fieldwork and voluminous case studies may yield only a few banalities. (p. 17)

They also point out that prestructuring reduces the amount of data that you have to deal with, functioning as a form of preanalysis that simplifies the analytic work required.

Unfortunately, most discussions of this issue treat prestructuring as a single dimension, and view it in terms of metaphors such as hard versus soft and tight versus loose. Such metaphors have powerful connotations (although they are different for different people) that can lead you to overlook or ignore the numerous ways in which studies can vary, not just in the *amount* of prestructuring, but in *how* prestructuring is used. For example, you could employ an extremely open approach to data collection, but use these data for a confirmatory test of explicit hypotheses based on a prior theory (e.g., Festinger, Riecker, & Schachter, 1956). In contrast, the approach often known as ethnoscience or cognitive anthropology (Werner & Schoepfle, 1987a, 1987b) employs highly structured data collection techniques, but interprets these data in a largely inductive manner with very few preestablished categories. Thus, the decision you face is not primarily *whether* or *to what extent* you prestructure your study, but *in what ways* you do this, and *why*.

Finally, it is worth keeping in mind that you can lay out a *tentative* plan for some aspects of your study in considerable detail, but leave open the possibility of substantially revising this if necessary. Emergent insights may require new sampling plans, different kinds of data, and different analytic strategies.

I distinguish four main components of qualitative methods:

1. The research relationship that you establish with those you study
2. Sampling: what times, settings, or individuals you select to observe or interview, and what other sources of information you decide to use
3. Data collection: how you gather the information you will use
4. Data analysis: what you do with this information to make sense of it

It is useful to think of all these components as involving *design* decisions—key issues that you should consider in planning your study and that you should rethink as you are engaged in it.

Negotiating a Research Relationship

Your relationships with the people in your study can be complex and changeable, and these relationships will necessarily affect you as the "research instrument," as well as have implications for other components of your research design. My changing relationships with the people in the Inuit community in which I conducted my dissertation research (Maxwell, 1986) had a profound effect not only on my own state of mind, but also on who I was able to interview, my opportunities for observation of social life, the quality of the data I collected, the research questions I was able to answer, and my ability to test my conclusions. The term *reflexivity* (Hammersley & Atkinson, 1995) is often used for this unavoidable mutual influence of the research participants and the researcher on each other.

There are also philosophical, ethical, and political issues that should inform the kind of relationship that you want to establish. In recent years, there has been a growing interest in alternatives to the traditional style of research, including participatory action research, collaborative research, feminist research, critical ethnography, and empowerment research (see Denzin & Lincoln, 2005; Fetterman et al., 1996; Oja & Smulyan, 1989; Whyte, 1991). Each of these modes of research involves different sorts of relationships between the researcher and the participants in the research and has different implications for the rest of the research design.

Thus, it is important that you think about the kinds of relationships you want to have with the people whom you study, and what you need to do to establish such relationships. I see these as *design decisions,* not simply as external factors that may affect your design. Although they are not completely under your control and cannot be defined precisely in advance, they are still matters that require systematic planning and reflection if your design is to be as coherent as possible.

Decisions About Sampling: Where, When, Who, and What

Whenever you have a choice about when and where to observe, whom to talk to, or what information sources to focus on, you are faced with a sampling decision. Even a single case study involves a choice of this case rather than others, as well as requiring sampling decisions *within* the case itself. Miles and Huberman (1994, pp. 27–34) and LeCompte and Preissle (1993, pp. 56–85) provide valuable discussions of particular sampling issues; here, I want to talk more generally about the nature and purposes of sampling in qualitative research.

Works on quantitative research generally treat anything other than probability sampling as "convenience sampling," and strongly discourage the latter. For qualitative research, this ignores the fact that most sampling in qualitative research is neither probability sampling nor convenience sampling, but falls into a third category: purposeful sampling (Patton, 1990, 169ff.). This is a strategy in which particular settings, persons, or events are deliberately selected for the important information they can provide that cannot be gotten as well from other choices.

There are several important uses for purposeful sampling. First, it can be used to achieve representativeness or typicality of the settings, individuals, or activities selected. A small sample that has been systematically selected for typicality and relative homogeneity provides far more confidence that the conclusions adequately represent the average members of the population than does a sample of the same size that incorporates substantial random or accidental variation. Second, purposeful sampling can be used to capture adequately the heterogeneity in the population. The goal here is to ensure that the conclusions adequately represent the entire *range* of variation rather than only the typical members or some subset of this range. Third, a sample can be purposefully selected to allow for the examination of cases that are critical for the theories that the study began with or that have subsequently been developed. Finally, purposeful sampling can be used to establish particular comparisons to illuminate the reasons for differences between settings or individuals, a common strategy in multicase qualitative studies.

You should not make sampling decisions in isolation from the rest of your design. They should take into account your research relationship with study participants, the feasibility of data collection and analysis, and validity concerns, as well as your goals and conceptual framework. In addition, feasible sampling decisions often require considerable knowledge of the setting studied, and you will need to alter them as you learn more about what decisions will work best to give you the data you need.

Decisions About Data Collection

Most qualitative methods texts devote considerable space to the strengths and limitations of particular data collection methods (see particularly, Bogdan & Biklen, 2006; Emerson, Fretz, & Shaw, 1995; Patton, 2000; Weiss, 1994), so I won't deal with these issues here. Instead, I want to address two key design issues in selecting and using data collection methods: the relationship between research questions and data collection methods, and the triangulation of different methods.

Although researchers often talk about "operationalizing" their research questions, or of "translating" the research questions into interview questions, this language is a vestigial remnant of logical positivism that bears little relationship to qualitative research practice. There is no way to convert research questions into useful methods decisions; your methods are the *means* to answering your research questions, not a logical transformation of the latter. Their selection depends not only on your research questions, but on the actual research situation and what will work most effectively in that situation to give you the data you need. For example, your interview questions should be judged not by whether they can be logically derived from your research questions, but by whether they provide the *data* that will contribute to answering these questions, an issue that may require pilot testing a variety of questions or actually conducting a significant number of the interviews. You need to anticipate, as best you can, how particular interview questions or other data collection strategies will actually work in practice. In addition, your interview questions and observational strategies will generally be far more focused, context-specific, and diverse than the broad, general research questions that define what you seek to understand in conducting the study. The development of a good data collection plan requires creativity and insight, not a mechanical translation of your research questions into methods.

In addition, qualitative studies generally rely on the integration of data from a variety of methods and sources of information, a general principle known as triangulation (Denzin, 1970). This strategy reduces the risk that your conclusions will reflect only the systematic biases or limitations of a specific method, and allows you to gain a better assessment of the validity and generality of the explanations that you develop. Triangulation is also discussed below in the section on validity.

Decisions About Data Analysis

Analysis is often conceptually separated from design, especially by writers who see design as what happens *before* the data are actually collected. Here, I treat analysis as a part of design (Coffey & Atkinson, 1996, p. 6), and as something that must itself be designed. Every qualitative study requires decisions about how the analysis will be done, and these decisions should influence, and be influenced by, the rest of the design.

A basic principle of qualitative research is that data analysis should be conducted simultaneously with data collection (Coffey & Atkinson, 1996, p. 2). This allows you to progressively focus your interviews and observations, and to decide how to test your emerging conclusions.

Strategies for qualitative analysis fall into three main groups: categorizing strategies (such as coding and thematic analysis), connecting strategies (such as narrative analysis and individual case studies), and memos and displays (for a more detailed discussion, see Coffey & Atkinson, 1996; Dey, 1993; Maxwell, 2005). These methods can, and generally should, be combined, but I will begin by discussing them separately.

The main categorizing strategy in qualitative research is coding. This is rather different from coding in quantitative research, which consists of applying a pre-established set of categories to the data according to explicit, unambiguous rules, with the primary goal being to generate frequency counts of the items in each category. In qualitative research, in contrast, the goal of coding is not to produce counts of things but to "fracture" (Strauss, 1987, p. 29) the data and rearrange it into categories that facilitate comparison between things in the same category and between categories. These categories may be derived from existing theory, inductively generated during the research (the basis for what Glaser & Strauss, 1967, term *grounded theory*), or drawn from the categories of the people studied (what anthropologists call "emic" categories). Such categorizing makes it much easier for you to develop a general understanding of what is going on, to generate themes and theoretical concepts, and to organize and retrieve your data to test and support these general ideas. (An excellent practical source on coding is Bogdan & Biklen, 2006.)

However, fracturing and categorizing your data can lead to the neglect of contextual relationships among these data, relationships based on contiguity rather than similarity (Maxwell & Miller, 2008), and can create analytic blinders, preventing you from seeing alternative ways of understanding your data. Atkinson (1992) describes how his initial categorizing analysis of data on the teaching of general medicine affected his subsequent analysis of the teaching of surgery:

> On rereading the surgery notes, I initially found it difficult to *escape* those categories I had initially established [for medicine]. Understandably, they furnished a powerful conceptual grid . . . The notes as I confronted them had been fragmented into the constituent themes. (pp. 458–459)

An important set of distinctions in planning your categorizing analysis is between what I call organizational, substantive, and theoretical categories (Maxwell, 2005). Organizational categories are generally broad subjects or issues that you establish prior to your interviews or observations, or that could usually have been anticipated. McMillan and Schumacher (2001) refer to these as *topics* rather than categories, stating that "a topic is the descriptive name for the subject matter of the segment. You are not, at this time, asking 'What is said?' which identifies the meaning of the segment" (p. 469). In a study of elementary school principals' practices of retaining children in a grade, examples of such categories are "retention," "policy," "goals," "alternatives," and "consequences" (p. 470). Organizational categories function primarily as "bins" for sorting the data for further analysis. They may be useful as chapter or section headings in presenting your results, but they don't help much with the actual work of making sense of what's going on.

This latter task requires substantive and/or theoretical categories, ones that provide some insight into what's going on. These latter categories can often be seen as subcategories of the organizational ones, but they are generally *not* subcategories that, in advance, you could have known would be significant, unless you are already fairly familiar with the kind of participants or setting you're studying or are using a well-developed theory. They implicitly make some sort of claim about the topic being studied—that is, they could be *wrong,* rather than simply being conceptual boxes for holding data.

Substantive categories are primarily *descriptive,* in a broad sense that include description of participants' concepts and beliefs; they stay close to the data categorized and don't inherently imply a more abstract theory. In the study of grade retention mentioned above, examples of substantive categories would be "retention as failure," "retention as a last resort," "self-confidence as a goal," "parent's willingness to try alternatives," and "not being in control (of the decision)" (drawn from McMillan & Schumacher, 2001, p. 472). Substantive categories are often inductively developed through a close "open coding" of the data (Corbin & Strauss, 2007). They can be used in *developing* a more general theory of what's going on, but they don't *depend on* this theory.

Theoretical categories, in contrast, place the coded data into a more general or abstract framework. These categories may be derived either from prior theory or from an inductively developed theory (in which case the concepts and the theory are usually developed concurrently). They usually represent the *researcher's* concepts (what are called "etic" categories), rather than denoting participants' own concepts ("emic" concepts). For example, the categories "nativist," "remediationist," or "interactionist," used to classify teachers' beliefs about grade retention in terms of prior analytic distinctions (Smith & Shepard, 1988), would be theoretical.

The distinction between organizational categories and substantive or theoretical categories is important because some qualitative researchers use mostly organizational categories to formally analyze their data, and don't systematically develop and apply substantive or theoretical categories in developing their conclusions. The more data you have, the more important it is to create the latter types of categories; with any significant amount of data, you can't hold all the data relevant to particular substantive or theoretical points in your mind, and need a formal organization and retrieval system. In addition, creating substantive categories is particularly important for ideas (including participants' ideas) that don't fit into existing organizational or theoretical categories; such substantive ideas may get lost, or never developed, unless they can be captured in explicit categories. Consequently, you need to include strategies for developing substantive and theoretical categories in your design.

Connecting strategies, instead of fracturing the initial text into discrete elements and re-sorting it into categories, attempt to understand the data (usually, but not necessarily, an interview transcript or other textual material) in context, using various methods to identify the relationships among the different elements of the text. Such strategies include some forms of case studies (Patton, 1990), profiles (Seidman, 1991), some types of narrative analysis (Coffey & Atkinson, 1996), and ethnographic microanalysis (Erickson, 1992). What all these strategies have in common is that they look for relationships that connect statements and events within a particular context into a coherent whole. Atkinson (1992) states,

> I am now much less inclined to fragment the notes into relatively small segments. Instead, I am just as interested in reading episodes and passages at greater length, with a correspondingly different attitude toward the act of reading and hence of analysis. Rather than constructing my account like a patchwork quilt, I feel more like working with the whole cloth . . . To be more precise, what now concerns me is the nature of these products as *texts.* (p. 460)

The distinction between categorizing and connecting strategies has important implications for your research questions. A research question that asks about the way events in a specific context are connected cannot be answered by an exclusively categorizing analysis (Agar, 1991). Conversely, a question about similarities and differences across settings or individuals, or about general themes in your data, cannot be answered by an exclusively connecting analysis. Your analysis strategies have to be compatible with the questions you are asking. Both categorizing and connecting strategies are legitimate and valuable tools in qualitative analysis, and a study that relies on only one of these runs the risk of missing important insights.

The third category of analytic tools, memos and displays, is also a key part of qualitative analysis (Miles & Huberman, 1994, pp. 72–75; Strauss & Corbin, 1990, pp. 197–223). As discussed above, memos can perform functions not related to data analysis, such as reflection on methods, theory, or goals. However, displays and memos are valuable *analytic* techniques for the same reasons that they are useful for other purposes: They facilitate your thinking about relationships in your data and make your ideas and analyses visible and retrievable. You should write memos frequently while you are doing data analysis, in order to stimulate and capture your ideas about your data. Displays (Miles & Huberman, 1994), which include matrices or tables, networks or concept maps, and various other forms, also serve two other purposes: data reduction and the presentation of data or analysis in a form that allows you to see it as a whole.

There are now a substantial number of computer programs available for analyzing qualitative data (Weitzman, 2000). Although none of these programs eliminate the need to read your data and create your own concepts and relationships, they can enormously simplify the task of coding and retrieving data in a large project. However, most of these programs are designed primarily for categorizing analysis, and may distort your analytic strategy to favor such approaches (see Example 7.2). So-called hypertext programs (Coffey & Atkinson, 1996, pp. 181–186) allow you to create electronic links, representing any sort of connection you want, among data within a particular context, but the openness of such programs can make them difficult for less experienced researchers to use effectively. A few of the more structured programs, such as ATLAS/ti and HyperRESEARCH, enable you not only to create links among data chunks, codes, and memos, but also to display the resulting networks.

Example 7.2 A Mismatch Between Questions and Analysis

Mike Agar (1991) was once asked by a foundation to review a report on an interview study that they had commissioned, investigating how historians worked. The researchers had used the computer program The Ethnograph to segment and code the interviews by topic and collect together all the segments on the same topic; the report discussed each of these topics and provided examples of how the historians talked about these. However, the foundation felt that the report hadn't really answered their questions, which

(Continued)

(Continued)

had to do with how individual historians thought about their work—their theories about how the different topics were connected, and the relationships that they saw between their thinking, actions, and results.

Answering the latter question would have required an analysis that elucidated these connections in each historian's interview. However, the categorizing analysis on which the report was based fragmented these connections, destroying the contextual unity of each historian's views and allowing only a collective presentation of shared concerns. Agar argues that the fault was not with The Ethnograph, which is extremely useful for answering questions that require categorization, but with its misapplication. He comments that "The Ethnograph represents a *part of* an ethnographic research process. When the part is taken for the whole, you get a pathological metonym that can lead you straight to the right answer to the wrong question" (p. 181).

SOURCE: From "The Right Brain Strikes Back, by M. Agar in *Using Computers in Qualitative Research* edited by N. G. Fielding and R. M. Lee, 1991. Copyright by SAGE.

Linking Methods and Questions

A useful technique for linking your research questions and methods (and also other aspects of your design) is a matrix in which you list your questions and identify how each of the components of your methods will help you get the data to answer these questions. Such a matrix displays the *logic* of your methods decisions. Figure 7.3 is an example of how such a matrix can be used; Exercise 3 helps you develop such a matrix for your own study.

Validity: How Might You Be Wrong?

Quantitative and experimental researchers generally attempt to design, in advance, controls that will deal with both anticipated and unanticipated threats to validity. Qualitative researchers, on the other hand, rarely have the benefit of formal comparisons, sampling strategies, or statistical manipulations that "control for" the effect of particular variables, and they must try to rule out most validity threats after the research has begun, by using evidence collected during the research itself to make these "alternative hypotheses" implausible. This approach requires you to identify the *specific* threat in question and to develop ways to attempt to rule out that particular threat. It is clearly impossible to list here all, or even the most important, validity threats to the conclusions of a qualitative study, but I want to discuss two broad types of threats to validity that are often raised in relation to qualitative

What do I need to know?	*Why do I need to know this?*	*What kind of data will answer the questions?*	*Where can I find the data?*	*Whom do I contact for access?*	*Timelines for acquisition*
What are the truancy rates for American Indian students?	To assess the impact of attendance on American Indian students' persistence in school	Computerized student attendance records	Attendance offices; assistant principal's offices for all schools	Mr. Joe Smith, high school assistant principal; Dr. Amanda Jones, middle school principal	August: Establish student database October: Update June: Final tally
What is the academic achievement of the students in the study?	To assess the impact of academic performance on American Indian students' persistence in school	Norm- and criterion-referenced test scores; grades on teacher-made tests; grades on report cards; student portfolios	Counseling offices	High school and middle school counselors; classroom teachers	Compilation #1: End of semester Compilation #2: End of school year
What is the English-language proficiency of the students?	To assess the relationship between language proficiency, academic performance, and persistence in school	Language-assessment test scores; classroom teacher attitude surveys; ESL class grades	Counseling offices; ESL teachers' offices	Counselors' test records; classroom teachers	Collect test scores Sept. 15 Teacher survey, Oct. 10–15 ESL class grades, end of fall semester and end of school year
What do American Indian students dislike about school?	To discover what factors lead to antischool attitudes among American Indian students	Formal and informal student interviews; student survey	Homeroom classes; meetings with individual students	Principals of high school and middle schools; parents of students; homeroom teachers	Obtain student and parent consent forms, Aug.–Sept. Student interviews, Oct.–May 30 Student survey, first week in May

Figure 7.3 Adaptation of the Data Planning Matrix for a Study of American Indian At-Risk High School Students *(Continued)*

What do I need to know?	*Why do I need to know this?*	*What kind of data will answer the questions?*	*Where can I find the data?*	*Whom do I contact for access?*	*Timelines for acquisition*
What do students plan to do after high school?	To assess the degree to which coherent post–high school career planning affects high school completion	Student survey; follow-up survey of students attending college and getting jobs	Counseling offices; Tribal Social Services office; Dept. of Probation; Alumni Association	Homeroom teachers; school personnel; parents; former students; community social service workers	Student survey, first week in May Follow-up survey, summer and fall
What do teachers think about their students' capabilities?	To assess teacher expectations of student success	Teacher survey; teacher interviews	—	Building principals; individual classroom teachers	Teacher interviews, November (subgroup) Teacher survey, April (all teachers)
What do teachers know about the home culture of their students?	To assess teachers' cultural awareness	Teacher interviews; teacher survey; logs of participation in staff development activities	Individual teachers' classrooms and records	Building principals; individual classroom teachers; assistant superintendent for staff development	Teacher interviews, November (subgroup) Teacher survey, April (all teachers)
What do teachers do to integrate knowledge of the student's home culture community into their teaching?	To assess the degree of discontinuity between school culture and home culture	Teachers' lesson plans; classroom observations; logs of participation in staff development activities	Individual teachers' classrooms and records	Building principals; individual classroom teachers; assistant superintendent for staff development	Lesson plans, Dec.–June Observations, Sept. 1–May 30 Staff development, June logs

Figure 7.3 Adaptation of the Data Planning Matrix for a Study of American Indian At-Risk High School Students

SOURCE: This figure was published in *Ethnography and Qualitative Design in Educational Research,* 2nd ed. by M. D. LeCompte & J. Preissle, with R. Tesch. Copyright 1993 by Academic Press.

studies: researcher bias, and the effect of the researcher on the setting or individuals studied, generally known as reactivity.

Bias refers to ways in which data collection or analysis are distorted by the researcher's theory, values, or preconceptions. It is clearly impossible to deal with these problems by eliminating these theories, preconceptions, or values, as discussed earlier. Nor is it usually appropriate to try to "standardize" the researcher to achieve reliability; in qualitative research, the main concern is not with eliminating *variance* between researchers in the values and expectations that they bring to the study but with understanding how a particular researcher's values influence the conduct and conclusions of the study. As one qualitative researcher, Fred Hess, has phrased it, validity in qualitative research is the result not of indifference, but of integrity (personal communication).

Reactivity is another problem that is often raised about qualitative studies. The approach to reactivity of most quantitative research, of trying to "control for" the effect of the researcher, is appropriate to a "variance theory" perspective, in which the goal is to prevent researcher *variability* from being an unwanted cause of variability in the outcome variables. However, eliminating the *actual* influence of the researcher is impossible (Hammersley & Atkinson, 1995), and the goal in a qualitative study is not to eliminate this influence but to understand it and to use it productively.

For participant observation studies, reactivity is generally *not* as serious a validity threat as many people believe. Becker (1970, 45ff.) points out that in natural settings, an observer is generally much less of an influence on participants' behavior than is the setting itself (though there are clearly exceptions to this, such as settings in which illegal behavior occurs). For all types of interviews, in contrast, the interviewer has a powerful and inescapable influence on the data collected; what the interviewee says is *always* a function of the interviewer and the interview situation (Briggs, 1986; Mishler, 1986). Although there are some things that you can do to prevent the more undesirable consequences of this (such as avoiding leading questions), trying to "minimize" your effect on the interviewee is an impossible goal. As discussed above for "bias," what is important is to understand *how* you are influencing what the interviewee says, and how to most productively (and ethically) use this influence to answer your research questions.

Validity Tests: A Checklist

I discuss below some of the most important strategies you can use in a qualitative study to deal with particular validity threats and thereby increase the credibility of your conclusions. Miles and Huberman (1994, 262ff.) include a more extensive list, having some overlap with mine, and other lists are given by Becker (1970), Kidder (1981), Guba and Lincoln (1989), and Patton (2000). Not every strategy will work in a given study, and even trying to apply all the ones that are feasible might not be an efficient use of your time. As noted above, you need to think in terms of *specific* validity threats and what strategies are best able to deal with these.

1. *Intensive, long-term involvement:* Becker and Geer (1957) claim that long-term participant observation provides more complete data about specific situations and events than any other method. Not only does it provide more, and more different kinds, of data, but the data are more direct and less dependent on inference. Repeated observations and interviews, as well as the sustained presence of the researcher in the setting studied, can help rule out spurious associations and premature theories. They also allow a much greater opportunity to develop and test alternative hypotheses during the course of the research. For example, Becker (1970, pp. 49–51) argues that his lengthy participant observation research with medical students not only allowed him to get beyond their public expressions of cynicism about a medical career and uncover an idealistic perspective, but also enabled him to understand the processes by which these different views were expressed in different social situations and how students dealt with the conflicts between these perspectives.

2. *"Rich" data:* Both long-term involvement and intensive interviews enable you to collect "rich" data, data that are detailed and varied enough that they provide a full and revealing picture of what is going on (Becker, 1970, 51ff.). In interview studies, such data generally require verbatim transcripts of the interviews, not just notes on what you felt was significant. For observation, rich data are the product of detailed, descriptive note-taking (or videotaping and transcribing) of the specific, concrete events that you observe. Becker (1970) argued that such data

> counter the twin dangers of respondent duplicity and observer bias by making it difficult for respondents to produce data that uniformly support a mistaken conclusion, just as they make it difficult for the observer to restrict his observations so that he sees only what supports his prejudices and expectations. (p. 53)

3. *Respondent validation:* Respondent validation (Bryman, 1988, pp. 78–80; Lincoln & Guba, 1985, refer to this as "member checks") is systematically soliciting feedback about one's data and conclusions from the people you are studying. This is the single most important way of ruling out the possibility of misinterpreting the meaning of what participants say and do and the perspective they have on what is going on, as well as being an important way of identifying your own biases and misunderstandings of what you observed. However, participants' feedback is no more inherently valid than their interview responses; both should be taken simply as *evidence* regarding the validity of your account (see also Hammersley & Atkinson, 1995).

4. *Searching for discrepant evidence and negative cases:* Identifying and analyzing discrepant data and negative cases is a key part of the logic of validity testing in qualitative research. Instances that cannot be accounted for by a particular interpretation or explanation can point up important defects in that account. However, there are times when an apparently discrepant instance is not persuasive, as when the interpretation of the discrepant data is itself in doubt. The basic principle here is that you need to rigorously examine both the supporting and discrepant data to assess whether it is more plausible to retain or modify the conclusion, being aware of all of the pressures to ignore data that do not fit your conclusions. In particularly

difficult cases, the best you may be able to do is to report the discrepant evidence and allow readers to evaluate this and draw their own conclusions (Wolcott, 1990).

5. *Triangulation:* Triangulation—collecting information from a diverse range of individuals and settings, using a variety of methods—was discussed earlier. This strategy reduces the risk of chance associations and of systematic biases due to a specific method and allows a better assessment of the generality of the explanations that one develops. The most extensive discussion of triangulation as a validity-testing strategy in qualitative research is by Fielding and Fielding (1986).

6. *Quasi-Statistics:* Many of the conclusions of qualitative studies have an implicit quantitative component. Any claim that a particular phenomenon is typical, rare, or prevalent in the setting or population studied is an inherently quantitative claim and requires some quantitative support. Becker (1970) coined the term *quasi-statistics* to refer to the use of simple numerical results that can be readily derived from the data. He argues that "one of the greatest faults in most observational case studies has been their failure to make explicit the quasi-statistical basis of their conclusions" (pp. 81–82).

Quasi-statistics not only allows you to test and support claims that are inherently quantitative, but also enable you to assess the *amount* of evidence in your data that bears on a particular conclusion or threat, such as how many discrepant instances exist and from how many different sources they were obtained.

7. *Comparison:* Although explicit comparisons (such as control groups) for the purpose of assessing validity threats are mainly associated with quantitative research, there are valid uses for comparison in qualitative studies, particularly multisite studies (e.g., Miles & Huberman, 1994, p. 237). In addition, single case studies often incorporate implicit comparisons that contribute to the interpretability of the case. For examples, Martha Regan-Smith (1992), in her "uncontrolled" study of how exemplary medical school teachers helped students learn, used both the existing literature on "typical" medical school teaching and her own extensive knowledge of this topic to determine what was distinctive about the teachers she studied. Furthermore, the students that she interviewed explicitly contrasted these teachers with others whom they felt were not as helpful to them, explaining not only what the exemplary teachers did that increased their learning, but *why* this was helpful.

Exercise 4 is designed to help you identify, and develop strategies to deal with, the most important validity threats to your conclusions.

Generalization in Qualitative Research

Qualitative researchers often study only a single setting or a small number of individuals or sites, using theoretical or purposeful rather than probability sampling, and rarely make explicit claims about the generalizability of their accounts. Indeed, the value of a qualitative study may depend on its *lack* of generalizability in the sense of being representative of a larger population; it may provide an account of a setting or population that is illuminating as an extreme case or "ideal type." Freidson (1975), for his study of social controls on work in a medical group practice, deliberately selected an atypical practice, one in which the physicians were better trained and more "progressive" than usual and that was structured precisely to deal with the problems that he was studying. He argues that the documented failure of social controls in this case provides a far stronger argument for the generalizability of his conclusions than would the study of a "typical" practice.

The generalizability of qualitative studies is usually based not on explicit sampling of some defined population to which the results can be extended, but on the development of a theory that can be extended to other cases (Becker, 1991; Ragin, 1987); Yin (1994) refers to this as "analytic," as opposed to statistical, generalization. For this reason, Guba and Lincoln (1989) prefer to talk of "transferability" rather than "generalizability" in qualitative research. Hammersley (1992, pp. 189–191) and Weiss (1994, pp. 26–29) list a number of features that lend credibility to generalizations made from case studies or nonrandom samples, including respondents' own assessments of generalizability, the similarity of dynamics and constraints to other situations, the presumed depth or universality of the phenomenon studied, and corroboration from other studies. However, none of these permits the kind of precise extrapolation of results to defined populations that probability sampling allows.

Conclusion

Harry Wolcott (1990) provided a useful metaphor for research design: "Some of the best advice I've ever seen for writers happened to be included with the directions I found for assembling a new wheelbarrow: *Make sure all parts are properly in place before tightening*" (p. 47). Like a wheelbarrow, your research design not only needs to have all the required parts, it has to work—to function smoothly and accomplish its tasks. This requires attention to the connections among the different parts of the design—what I call *coherence.* There isn't one right way to create a coherent qualitative design; in this chapter I have tried to give you the tools that will enable you to put together a way that works for you and your research.

Discussion Questions

The following questions are ones that are valuable to review before beginning (or continuing) with the design of a qualitative study.

1. Why are you thinking of doing a *qualitative* study of the topic you've chosen? How would your study use the strengths of qualitative research? How would it deal with the limitations of qualitative research?

2. What do you already know or believe about your topic or problem? Where do these beliefs come from? How do the different beliefs fit together into a coherent picture of this topic or problem?

3. What do you *not* know about your topic or problem that a qualitative study could help you understand?

4. What types of settings or individuals would be most productive to select for your study, in terms of answering your research questions? Why? What practical issues would you need to deal with to do this? What compromises might be required to make your study feasible and how would these affect your ability to answer your questions?

5. What relationships do you already have, or could you create, with potential settings or individuals you could select for your study? How could these relationships help or hinder your study? What relationships do you *want* to create with the individuals and settings you select?

6. What data collection methods would best provide the information you need to answer your research questions? Why? Could you combine different methods to better answer your questions?

7. How would you need to analyze your data to answer your questions? Why? If you use a categorizing approach, how would you develop and apply your coding categories? What could connecting strategies contribute to your analysis?

8. What are the most serious potential validity threats to the conclusions you might draw from your study? What could you do (in your design as a whole, not just data collection and analysis) to address these threats?

Exercises

These exercises give you an opportunity to work through several of the most important issues in designing a qualitative study. Other important issues are addressed in the discussion questions.

Exercise 1: Researcher Identity Memo

The purpose of this exercise is to help you identify the goals, experiences, assumptions, feelings, and values that are most relevant to your planned research and to reflect on how these could inform and influence your research (see Example 7.1).

I would begin working on this memo by "brainstorming" whatever comes to mind when you think about prior experiences that relate to your topic, and jotting these down without immediately trying to organize or analyze them. Then, try to identify the issues most likely to be important in your research, think about the implications of these, and organize your reflections. There are two broad types of questions that it is productive to reflect on in this memo.

1. What prior experiences have you had that are relevant to your topic or setting? What assumptions about your topic or setting have resulted from these experiences? What goals have emerged from these? How have these experiences, assumptions, and goals shaped your decision to choose this topic, and the way you are approaching this project?

2. What potential advantages do you think these goals, beliefs, and experiences have for your study? What potential disadvantages do you think these may create for you, and how might you deal with these?

Exercise 2: Developing Research Questions

This exercise involves both developing an initial set of research questions and trying to connect these questions to the other four components of your design. At this point, your ideas may still be very tentative; you can repeat this exercise as you get a better idea of what your study will look like.

1. Begin by thinking about your goals for this study. What could you learn in a research study that would help accomplish these goals? What research questions does this suggest? Conversely, how do any research questions you may already have formulated connect to your goals in conducting the study? How will answering these specific questions help you achieve your goals? Which questions are most interesting to you, personally, practically, or intellectually?

2. Next, connect these research questions to your conceptual framework. What would answering these questions tell you that you don't already know? Where are the places in this framework that you don't understand adequately or where you need to test your ideas? What could you learn in a research study that would help you better understand what's going on with these phenomena? What changes or additions to your questions does your framework suggest? Conversely, are there places where your questions imply things that should be in your framework, but aren't?

3. Now focus. What questions are most central for your study? How do these questions form a coherent set that will guide your study? You can't study everything interesting about your topic; start making choices. Three or four main questions are usually a reasonable maximum for a qualitative study, although you can have additional subquestions for each of the main questions.

4. In addition, you need to consider how you could actually answer the questions you pose. What methods would you need to use to collect data that would answer these questions? Conversely, what questions can a qualitative study of the kind you are planning productively address? At this point in your planning, this may primarily involve "thought experiments" about the way you will conduct the study, the kinds of data you will collect, and the analyses you will perform on these data. This part of the exercise is one you can usefully repeat when you have developed your methods and validity concerns in more detail.

5. Assess the potential answers to your questions in terms of validity. What are the plausible validity threats and alternative explanations that you would have to rule out? How might you be wrong, and what implications does this have for the way you frame your questions?

Don't get stuck on trying to precisely frame your research questions or in specifying in detail how to measure things or gain access to data that would answer your questions. Try to develop some meaningful and important questions that would be *worth* answering. Feasibility is obviously an important issue in doing research, but focusing on it at the beginning can abort a potentially valuable study.

A valuable additional step is to share your questions and your reflections on these with a small group of fellow students or colleagues. Ask them if they understand the questions and why these would be worth answering, what other questions or changes in the questions they would suggest, and what problems they see in trying to answer them. If possible, tape record the discussion; afterward, listen to the tape and take notes.

Exercise 3: Questions × Methods Matrix

This exercise (based on Figure 7.3) helps you display the logical connections between your research questions and your selection, data collection, and data analysis decisions. Doing this isn't a mechanical process; it requires thinking about *how* your methods can provide answers to your research questions. Start with your questions and ask what data you would need, how you could get these data, and how you could analyze them to answer these questions. You can also work in the other direction: Ask yourself *why* you want to collect and analyze the data in the way you propose—what will you learn from this?

Your matrix should include columns for research questions, selection decisions, data collection methods, and kinds of analyses, but you can add any other columns you think would be useful in explaining the logic of your design. You should also include a *justification* for the choices you make in the matrix, either as a separate discussion, by question, of the rationale for your choices in each row, or by including this as a column in the matrix itself (as in Figure 7.3). This exercise is intended to help you *make* your methods decisions, not as a final formulation of these, so it may require you to revise your questions, your planned methods, or both.

Exercise 4: Identifying and Dealing With Validity Threats

1. What are the most serious validity threats that you need to be concerned with in your study? In other words, what are the main ways in which you might be mistaken about what's going on, and what issues will your potential audiences be most concerned about? These threats can include alternative theories or interpretations of your data, as well as potential methodological flaws. Be as specific as you can, rather than just listing general categories. Also, think about *why* you believe these might be serious threats.

2. What could you do in your research design (including data collection and data analysis) to deal with these threats and increase the credibility of your conclusions? This includes ways of testing your interpretations and conclusions, and of investigating the existence and plausibility of alternative interpretations and conclusions (e.g., could your analysis of your data be biased by your preconceptions about your topic? How could you test this?). Start by brainstorming possible solutions, and then consider which of these strategies are *practical* for your study, as well as effective.

Remember that some validity threats are unavoidable; you will need to acknowledge these in your proposal or in the conclusions to your study, but no one expects you to have airtight answers to *every* possible threat. The key issue is how plausible and how serious these unavoidable threats are.

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CHAPTER 8

How to Do Better Case Studies

(With Illustrations From 20 Exemplary Case Studies)

Robert K. Yin

Whether you are starting as a novice or a seasoned investigator, this chapter will help you improve your case study research.[1] The chapter differs from other case study guides, and especially, the earlier case study chapter (Yin, 1998) in the first edition of this *Handbook,* in at least two ways.

First, this chapter does not attempt to cover the full range of case study topics. Such broader coverage was the scope of the earlier chapter and also of a full textbook written by the present author (Yin, 2003b).[2] Instead, this chapter's narrowed scope allows it to focus on the following four steps that seem to have been the most challenging in doing case study research:

Step 1: Defining and selecting the case(s) for a case study.

Step 2: Using multiple cases as part of the same case study.

Step 3: Strengthening the evidence used in a case study.

Step 4: Analyzing case study evidence.

Although other steps also are important in doing case study research, somehow these four have posed the most formidable demands. If you can meet them, you will be able to conduct high-quality case studies—ones that may be better and more distinctive than those of your peers. Because of the importance of the four steps, this advantage will prevail whether you are doing a dissertation, case study evaluations (e.g., U.S. Government Accountability Office, 1990), case studies of natural settings (e.g., Feagin, Orum, & Sjoberg, 1991), or more theory-based (e.g., George & Bennett, 2004; Sutton & Staw, 1995) or norm-based (e.g., Thacher, 2006) case study research.[3]

Second, the chapter goes beyond merely describing the relevant research procedures. It also refers to many exemplary examples from the existing case study literature.[4] The examples include some of the best case studies ever done, including a case study that is more than 75 years old but that is still in print. The richness of the examples permits the discussion of the four steps—and especially the fourth and most difficult step of doing case study analysis—to be deeper than commonly found in other texts. In this sense, this chapter should help you do more advanced case studies.

The exemplary examples come from different fields, such as community sociology, public health services, national and international politics, urban planning, business management, criminal justice, and education. The hope is that among these examples you will find case studies that cover not only methodologically important issues but also topics relevant to your interests.

Step 1: Defining and Selecting a Case Study

In a way, this first step of defining and selecting a case study entails the greatest risk in doing case study research. Significant cases will receive attention on their own right, somewhat independent of the quality of the research effort, and mundane cases are not likely to be cherished even if they reflect sound research procedures. When starting your own[5] case study, the goal is to set your sights high in the selection process. Seasoned case study investigators have learned to attend to both practical and substantive considerations.

Practical Considerations

From a practical standpoint, you will be devoting significant time to your case study. You therefore would like to reduce any likelihood of finding that, midstream, your case will not work out.

The most frequent surprise involves some disappointment regarding the actual availability, quality, or relevance of the case study data. For instance, you might have planned to interview several key persons as part of your case study but later found only limited or no access to these persons. Similarly, you might have planned to use what you had originally considered to be a rich source of documentary evidence, only later to find their contents to be unhelpful and irrelevant to your case study. Last, you might have counted on an organization or agency updating an annual data set, to provide a needed comparison to earlier years, only later to learn that the update will be significantly delayed. Any of these three situations could then cause you to search for another case to study, making you start all over again.

These and other practical situations need, as much as possible, to be investigated prior to starting your case study. A commonplace practice in other types of research, from laboratory experiments to surveys, is to carry out pilot work to refine research procedures. For case studies, doing a pilot study can likewise produce the same benefits and also can reduce the risks of defining and selecting the wrong case study. The pilot case can specifically tighten the link between your research questions and the likely availability of evidence. You can then decide better whether this is the type of case study you want to conduct. If you are unable to conduct a pilot study, assess the availability, relevance, and usefulness of your case-study-to-be as carefully as possible. Do your best to anticipate any problems that you will encounter in doing your case study.

Substantive Considerations

The selection process, however, should not dwell on practical considerations only. You should be ambitious enough to try to select a significant or "special" case for your case study, as a more mundane case may not produce an acceptable study (or even dissertation). Think of the possibility that your case study may be one of the few that you ever might complete and that you, therefore, would like to put your efforts into as important, interesting, or significant a case study as possible.

What makes a case special? One possibility arises if your case covers some distinctive event or condition, such as the revival or renewal of a major organization, the creation and confirmed efficacy of a new medical procedure, the discovery of a new way of reducing youth gang violence; a critical political election; some dramatic neighborhood change; or even the occurrence and aftermath of a natural disaster. By definition, these are likely to be remarkable circumstances. To do a good case study of any of them may produce an exemplary piece of research (see Case Studies 1 and 2).

CASE STUDIES 1 AND 2: TWO SPECIAL CASES

Two historically distinctive, if not unique, events were the Swine Flu Scare and the Cuban Missile Crisis. Both events became the subjects of now well-known case studies in the field of political science.

In the first case (Neustadt & Fineberg, 1983), the United States faced a threat of epidemic proportions from a new, and potentially lethal, influenza strain. As a result, the U.S. government planned and then tried to immunize the whole U.S. population. Over a 10-week period, the immunization effort reached 40 million people before the campaign was ended amidst controversy, delay, administrative troubles, and legal complications.

In the second case (Allison, 1971), a nuclear holocaust between the United States and the former Soviet Union threatened the survival of the entire world. The case study investigates how and why military and diplomatic maneuvers successfully eliminated the confrontation. With the later availability of new documentation after the fall of the Soviet Union, an entirely updated and revised version of the case study was written, corroborating but also refining the understanding of the key decisions (Allison & Zelikow, 1999).

But what if no such distinctive circumstances are available for you to study? Or what if you deliberately want to do a case study about a common and even "everyday" phenomenon? In these situations, you need to define some compelling theoretical framework for selecting your case. The more compelling the framework, the more your case study can contribute to the research literature, and in this sense, you will have conducted a special case study.

A compelling framework could be based on some historical context or some sociological insight. Around the context or insight, you would still need to amass the relevant existing literature, to show how your compelling framework would fit (or depart from) the literature, and how your case study would eventually extend that literature. These ingredients would lay the groundwork for your case study making a significant contribution to the literature (see Case Studies 3 and 4).

CASE STUDIES 3 AND 4: STRONG THEORETICAL FRAMEWORKS

Two "community" case studies have compelling theoretical frameworks and have achieved the status of classic case studies.

The first case study is about an average American city, but the framework highlights a significant development in American history—the transition from an agricultural to an industrial economy and how it occurred in the average American city (Lynd & Lynd, 1957).

The second case study is about the discovery of a social class structure within the average American city (Warner & Lunt, 1941). The terminology and concepts for describing this structure were new. However, they were later applied to virtually all American communities and the American social structure as a whole.

Alternatively, a compelling theoretical framework could call attention to organizational, community, group, or other types of social processes or outcomes. The purpose of your case study would be to develop new knowledge about these processes and outcomes, based on the facts of the case. Again, you need to review the existing literature carefully, to develop a refined conceptual niche so that your completed case study will contribute to that literature (see Case Study 5).

CASE STUDY 5: A "PROCESS" CASE STUDY

This case study is about a specific economic development program in a specific city, Oakland, CA (Pressman & Wildavsky, 1973). However, the case study's main contribution is not about urban economic development or about the city and its history.

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Rather, the case study's lasting value derives from its focus on the decisions made by officials trying to put a federal initiative (the economic development program) into place in a local community. The authors show how the decisions were numerous, complex, and interdependent. They use these decisions to define, operationally, a broader implementation process that, until that time, had not been fully appreciated in the field of public policy. Instead of being about the program or the city, the case study therefore is about a process. The lessons learned have been helpful for understanding other implementation experiences.

Exercise for Step 1

You have just selected the case for your case study. Describe the significance of the case to a colleague (or faculty adviser). Pretend you might even have completed the case study, and preview what you might have learned. Argue persuasively about the significant contribution(s) made by your case study. If your colleague appears unimpressed with the significance of the learnings from your case study, reconsider whether you have selected the best case possible or, as an alternative, defined the best theoretical framework.

Step 2: Using Multiple Cases as Part of the Same Case Study

This step favors doing "multiple-" rather than "single-" case studies (see Yin, 2003b, pp. 39–54). Even though the classic case study has been about single cases, your case study is likely to be stronger if you base it on two or more cases.

"Two-Case" Case Studies

No matter how well you do a single case, doing more than one can strengthen your case study. Even if you only do a "two-case" study, the second case offers the possibility of responding to a frequent complaint against single-case studies that the case was aberrant in some undesirable manner. Thus, you can use a second case to produce a direct replication of your first case (see Case Study 6).

CASE STUDY 6: REPLICATION CASES

Conventional wisdom attributed the remarkable growth of Japan's economy, in the latter part of the 20th century, to the role of Japan's national government in supporting Japanese industrial planning. The same conventional wisdom led to the belief that the United States's traditional free enterprise economy precluded a strong role by the U.S. government. Both beliefs led to the complaint that U.S. industries were disadvantaged in competing against Japanese industries.

Gregory Hooks's (1990) "two-case" study challenged the conventional wisdom. His first case pointed to the U.S. Department of Defense's relationship with the aeronautics industry. However, critics would argue that this industry long had a special relationship with defense.

Hooks's second case then showed how the department also played a similar role in the microelectronics industry, not usually considered as defense oriented. Together, the two cases provided a strong rationale for challenging the conventional wisdom.

The replication logic is analogous to that used in multiple experiments (see Yin, 2003b, p. 47–52). For example, on uncovering a significant finding from a single experiment, the immediate research goal would be to replicate this finding by conducting a second, third, and even more experiments. For "two-case" case studies, you may have selected both cases at the outset of your case study, anticipating that they will either produce similar findings (a *literal replication*) or produce contrasting results, but for predictable reasons (a *theoretical replication*). With more cases, the possibilities for more subtle and varied replications increase. Most important, the replication logic differs completely from the sampling logic used in survey research.

Case Studies Having More Than Two Cases

Multiple cases, compared to single-case studies, also can broaden the coverage of your case study. For instance, consider the benefits if you do a case study of school reform but include more than one school, varying the schools according to enrollment size. The variations permit you to examine whether reform occurs in similar fashion in large and small schools—or if reform strategies need to be tailored according to the size of the school. By leading to the opportunity (and need) to conduct a "cross-case" analysis, a multiple-case study can actually address a broad topic of contemporary interest (see Case Studies 7 and 8). Such breadth contrasts strongly with the limited scope of a single-case study.

CASE STUDIES 7 AND 8: TWO MULTIPLE-CASE STUDIES

Multiple-case studies provide more convincing data and also can permit the investigation of broader topics than single-case studies.

Case Study 7 (Magaziner & Patinkin, 1989) was one of nine cases amassed to describe various facets of a global but silent war, involving world economic competition at all levels. These include the United States's competition with low-wage countries, with developed countries, and in relation to future technologies.

Case Study 8 (Derthick, 1972) uses seven cases to illuminate the weakness of the federal government in addressing local affairs and attempting to respond to local needs. The federal objective was to implement new housing programs in seven different cities. The cross-case analysis, based on the experiences in all seven cities, readily pointed to common reasons for the problems that arose.

As the ability to expand the number of cases increases, you can start seeing the advantages of doing multiple-case studies. As part of the same case study, you might have two or three literal replications and two or three deliberately contrasting cases. Alternatively, multiple cases covering different contextual conditions might substantially expand the generalizability of your findings to a broader array of contexts than can a single-case study. Overall, the evidence from multiple-case studies should produce a more compelling and robust case study.

In principle, you will need more time and resources to conduct a multiple- rather than single-case study. However, you should note that the classic, single-case studies nevertheless consumed much time and effort. For instance, Case Study 3 involved a four-person research team living in the city under study for 18 months—just to carry out the data collection. Analysis and writing then took another couple of years. Other classic single-case studies have involved extensive time commitments made by single investigators. Doing a good single-case study should not automatically lead to reduced time commitments on your part.

Exercise for Step 2

From Section 1's discussion, you may have developed some preliminary ideas about defining and selecting a "case" for your case study. If not, recall some single-case study with which you are familiar—or even focus on one of the single cases presented earlier in this chapter.

Whether choosing your own case or the recalled case, now think of a companion case to match it. In what ways might the companion case's findings augment those of the first case? Could the data from the second case fill a gap left by the first case or respond better to some obvious shortcoming or criticism of the first case? Would the two cases together comprise a stronger case study? Could yet a third case make the findings even more compelling? The more you can address these and related questions, the more you will be on your way to thinking about the advantages and disadvantages of doing a multiple-case study.

Step 3: Strengthening the Evidence Used in Your Case Study

The case study method is not limited to any single type of evidence or data. Both qualitative (e.g., categorical or nominal) and quantitative (e.g., ratio, interval, and ordinal) data may be relevant and should be part of your case study. These different data will come as a result of using different data sources and techniques such as focus groups, ethnographies, participant observation, key interviews, documentary evidence, access to archival records, direct observations in the field, and surveys. Your case study may call on a combination of such techniques, thereby involving a combination of qualitative and quantitative data.

The goal is to use different types of evidence to triangulate or converge on the same research questions. The findings will then be less open to the criticism that they had resulted from and possibly been biased by a single data collection method. To take advantage of this principle, good case study investigators need to be adept at using different data collection methods.

Regardless of the type of evidence, the objective is to present it apart from any interpretation or assessment that you might then make of the evidence. This way, readers can judge the evidence for themselves. They then can agree or take issue with your interpretation and assessment, which are part of the analysis that comes later in the case study. Any mixing of the evidence with your interpretation is undesirable, and such mixing has been a continuing source of criticism of earlier case studies.

Direct Observations: Two Examples

Let's start with one of the most common methods: making direct observations in the field. If nothing else, the opportunity to make such observations is one of the most distinctive features in doing case studies.

The observational data can be qualitative or quantitative. The conventional manner of reporting qualitative data takes the form of a narrative text. The composing of this text must overcome the pitfall just discussed—by presenting the observational evidence as neutrally and factually as possible, and by minimizing your interpretation of, or judgment about, the evidence (see Case Study 9).

CASE STUDY 9: OBSERVATIONAL EVIDENCE AS PART OF A CASE STUDY

Part of a case study about the firms and working life in Silicon Valley called for the case study investigators to observe the "clean room" operations where silicon chips are made (Rogers & Larsen, 1984).The clean rooms are a

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key part of the manufacturing process for producing semiconductor chips. Among other features, employees wear "bunny suits" of lint-free cloth and handle extremely small components in these rooms. The case study observations showed how the employees adapted to the working conditions in these clean rooms, adding that, at the time, most of the employees were female while most of the supervisors were male.

Coroners' reports, with their dry and factually operational tone, may serve as a good model for the desired narrative. Note that such narrative—whose main function is to present observational evidence—is not the same as the interpretive narrative that will appear elsewhere in the case study. That narrative discusses evidence and interpretation together, and the case still may be told in a compelling manner. This latter narrative, in combination with the drier, operational narrative covering the observational evidence, parallels other types of research where numeric tables (the evidentiary portion) are accompanied by the investigator's interpretation of the findings (the interpretive portion). Again, the main point is that many case studies confuse the two presentations, and yours should not.

The separate presentation of narrative evidence can assume several forms. One, the use of *vignettes,* is illustrated in this very chapter by the material in the boxes about the individual case studies. Another, the use of *word tables,* is a table, arranged with rows and cells like any other table, but whose cells are filled with words (i.e., categorical or qualitative evidence) rather than the numbers found in numeric tables.

Going beyond this traditional, narrative form of reporting observational data, you can quantify observations by using a formal observational instrument and then report the evidence in numeric form (e.g., tables showing the frequency of certain observations). The instrument typically requires you to enumerate an observed activity or to provide one or more numeric ratings about the activity (see Case Study 10). Thus, observational evidence can be reported both as narrative and in the form of numeric tables.

CASE STUDY 10: QUANTIFYING OBSERVATIONAL EVIDENCE IN A CASE STUDY

An elementary school was the site for a case study of a new instructional practice, or "innovation" (Gross, Bernstein, & Giacquinta, 1971). To judge how well teachers were implementing the new practice, members of the research team made classroom observations and quantified their observations.

An observational instrument called for the use of a 5-point rating scale (from *high* to *low*) for 12 kinds of teachers' behaviors that reflected the new practice:

- making the materials in the classroom available to students;
- permitting students to move freely about the room, to choose their own activities and to decide whether they wanted to work individually, in pairs, or in groups; and
- acting as a guide, catalyst, or resource person between children.

The overall pattern of ratings, across all the desired behaviors, became the basis for assessing the degree of implementation of the new practice.

Archival Records

In contrast to direct observations in the field, case studies also can rely on archival data—information stored through existing channels, such as electronic records, libraries, and old-fashioned (paper) files. Newspapers, television, and the mass media are but one type of channel. Records maintained by public agencies, such as public health or police records, serve as another. The resulting archival data can be quantitative or qualitative (or both).

From a research perspective, the archival data can be subject to their own biases or shortcomings. For instance, researchers have long known that police records of *reported crime* do not reflect the actual amount of crime that might have occurred. Similarly, school systems' reports of their enrollment, attendance, and dropout rates may be subject to systematic under- or overcounting. Even the U.S. Census struggles with the completeness of its population counts and the potential problems posed because people residing in certain kinds of locales (rural and urban) may be undercounted.

Likewise, the editorial leanings of different mass media are suspected to affect their choice of stories to be covered (and not covered), questions to be asked (and not asked), and writing detail (and not detailed). All these editorial choices can collectively produce a systematic bias in what would otherwise appear to be a full and factual account of some important events.

Case studies relying heavily on archival data need to be sensitive to these possible biases and to take steps to counteract them. With mass media, a helpful procedure is to select two different media that are believed, if not known, to have opposing orientations. A more factually balanced picture may then emerge (see Case Study 11). Finding and using additional sources bearing on the same topic would help even more.

CASE STUDY 11: A CASE STUDY USING TWO ARCHIVAL SOURCES TO COVER THE SAME COMMUNITY EVENTS

One of the most inflammatory community events in the 1990s came to be known as the "Rodney King crisis." White police officers were serendipitously videotaped in the act of beating an African American male, but a year later they all were acquitted. The acquittal sparked a major civil disturbance in which 58 people were killed, 2,000 injured, and 11,000 arrested.

A case study of this crisis deliberately drew from two different newspapers—the major daily for the metropolitan area and the most significant newspaper for the area's African American community (Jacobs, 1996). For the pertinent period surrounding the crisis, the first newspaper produced 357 articles and the second (a weekly, not daily publication) 137 articles. The case study not only traces the course of events but also shows how the two papers constructed different but overlapping understandings of the crisis.

Open-Ended Interviews

A third common type of evidence for case studies comes from open-ended interviews. These interviews offer richer and more extensive material than data from surveys and especially the closed-ended portions of survey instruments. On the surface, the open-ended portions of surveys may resemble open-ended interviews, but the latter are generally less structured and even may assume a conversational manner.

The diminished structure permits open-ended interviews, if properly done, to reveal how case study interviewees construct reality and think about situations, not just giving answers to specific questions. For some case studies, the construction of reality provides important insights into the case. The insights gain even further value if the interviewees are key persons in the organizations, communities, or small groups being studied, not just the average member of such groups. For a case study of a public agency or private firm, for instance, a key person would be the head of the agency or firm. For schools, the principal or a department head would carry the same status. Because by definition such roles are not frequently found within an organization, the open-ended interviews also have been called "elite" interviews. A further requirement is that case study investigators need to be able to gain access to these elites. Such access is not always available and may hamper the conduct of the case study in the first place (see Case Study 12).

CASE STUDY 12: OPEN-ENDED INTERVIEWS AS A SOURCE OF CASE STUDY EVIDENCE

Professional life in entrepreneurial firms, such as electronic firms in Silicon Valley, can be highly demanding. Employees from the top to the bottom of the firms may dedicate long hours and hard thinking to their work. At the same time, because older firms may cease growing at a rapid pace and newer firms are continually getting started, employees' loyalties also are tested by their willingness to stay with their existing firms.

Describing these and other delicate conditions were an integral part of a case study of Silicon Valley (Rogers & Larsen, 1984). Some of the most relevant information could only be obtained through open-ended interviews, often with the key executives and supervisors in a firm. The case study's authors, who were local to the Silicon Valley area, used their professional and personal ties to gain access to these persons. In addition, the sensitivity of some of the information meant that the authors withheld the real names of some of the interviewees, referring to them with pseudonyms instead.

Integrating Evidence

The preceding paragraphs have covered three types of case study evidence. Other chapters in this *Handbook* actually cover some of the other types, such as the use of focus groups, surveys, and ethnographies. Together, you should now have a good idea of the different kinds of evidence that you can use in case studies.

More important than reviewing the remaining types at this juncture is the need to show how various sources of evidence might come together as part of the same case study. Recall that the preferred integration would position the evidence from each source in a way that converged with, or at least complemented, the evidence from other sources.

Such integration readily takes place in many existing case studies. The presentation of a case study can integrate (a) information from interviews (e.g., quotations or insights from the interviews appearing in the text, but citations pointing the reader to the larger interview database) with (b) documentary evidence (e.g., quotations or citations to specific written texts, accompanied by the necessary citations) and with (c) information drawn from direct observations. The resulting case study tries to see whether the evidence from these sources presents a consistent picture. The procedure involves juxtaposing the different pieces of evidence, to see whether they corroborate each other or provide complementary (or conflicting) details. If the case study is well documented, all the evidence contains appropriate footnotes and citations to data collection sources (e.g., the name and date of a document that was used), and the case study also includes a full description of the data collection methods, often appearing as an appendix to the case study.

Integrating and presenting the evidence in this manner can be a major challenge (see Case Studies 13 and 14). Although the final case study still may be criticized for having undesirable biases, the richness of the evidence should nevertheless shift any debate into a more empirical mode—that is, critics need to produce contrary evidence rather than simply make alternative arguments. The shift is highly desired, because case studies should promote sound social science inquiry rather than raw polemic argument.

CASE STUDIES 13 AND 14: TWO CASE STUDIES THAT BRING THE EVIDENCE TOGETHER

Two case studies exhibit similar methodological features by integrating data from direct observations, documentary sources, and extensive interviews of key informants. In both cases, the main author was a participant in the case being studied, and extensive additional evidence is cited and used, to offset the possible biases created by the participatory role.

Case Study 13 (Zigler & Muenchow, 1992) covers the *Head Start* program—a well-known federal initiative that boosts support for early childhood development. In its early years, the program was controversial, drawing sharp critics as well as supporters. In the long run, however, the program became a forerunner of many related initiatives, all aimed at improving the health and well-being of preschool children.

The lead author of the case study was one of the directors of the *Head Start* program. The director's role provided observational evidence for the case study, but the authors also buttressed this evidence with a wide array of other evidence, including data from hundreds of open-ended interviews, reviews of numerous program-related documents, and references to many other studies of the program conducted by eminent scholars. In their case study, the authors continually weave together the evidence from these various sources, trying to present an accurate picture of the program though not denying the director's role as a strong supporter of the program.

Case Study 14 (McAdams, 2000) has a similar flavor, as the author was a prominent member of the school board overseeing a large urban system during a critical period in the system's life. Again, citations to specific documentary sources, including newspaper accounts, as well as references to numerous interviews, demonstrate the author's concern with integrating the evidence and accurately depicting events as they transpired.

As an alternative strategy, you can bring the evidence together, from multiple sources, on an even grander scale than just described. Understanding this grander scale requires an appreciation of the concept of *embedded units of analysis* (see Yin, 2003b, pp. 42–45).

The concept applies when the data for a case study come from more than a single layer. For instance, a case study about an organization will certainly include data about an organizational layer (the organization's overall performance, policies, partnerships, etc.). However, depending on the research questions being studied, additional data may come from a second layer—the organization's employees. Data might come from an employee survey, which, if used alone, might have served to support a study of the employees. However, within the context of the case study of the organization, the employee layer would be an embedded unit of analysis, falling within the main unit of analysis for the case study, which is the organization as a whole.

You can imagine many situations where case studies will have embedded units of analysis: a neighborhood case study, where the services or the residents in the neighborhood might represent embedded units of analysis; a case study of a public or foundation program that consists of multiple, separately funded projects; a study of a new technology, with an assessment of the technology's multiple applications also being part of the case study; or a study of a health services marketplace, with different health service providers and clients being the embedded units.

In all these examples, the embedded units are embedded within the larger, main unit of the case study. The main unit is the single entity, covering a single-case. The embedded units are more numerous and can produce a large amount of quantitative data. Nevertheless, the data are still part of the same single case. The most complex case study design then arises when your case study may contain multiple cases (e.g., multiple organizations), each of which has an embedded unit of analysis.

In these situations, the multiple sources of evidence help cover the different units of analysis—the main and embedded units. In the example of an organization and its employees, the case study might be about the development of an organizational culture. At the main unit of analysis, only a single entity—the organization—exists, and the relevant data could include the kind of observations, key interviews, and documents review previously highlighted in Case Studies 13 and 14. At the embedded unit of analysis—a sample or universe of employees—the relevant data would include an employee survey or some analysis of employee records. In contrast to Case Studies 13 and 14, which did not have an embedded unit of analysis, Case Study 15 is an older but classic case study of a single organization (a labor union), with multiple layers and in fact, several levels of embedded units.

CASE STUDY 15: BRINGING THE EVIDENCE TOGETHER IN A MORE COMPLEX CASE STUDY

This case study is about a single trade union, the International Typographical Union, whose membership came from across the country (Lipset, Trow, & Coleman, 1956). Because of its national coverage, the union, like many other unions, was organized into a series of "locals," each local

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representing the members in a local area. Similarly, each local consisted of a number of "shops." Finally, each shop contained individual union members. From top to bottom, the organization therefore had four layers. As a case study, the case had one main unit (the union) and three embedded units. In this sense, the case study was complex.

The research questions called for information at every level. The three investigators, who ultimately became recognized as prominent scholars in their fields, designed a variety of data collection activities, ranging from key interviews with the top officials to observations of informal group behavior among the locals and shops to a survey of the individual members. For each of the three embedded levels, the investigators also had to define and defend their sample selection. The study took 4 years to complete, in addition to two earlier years when the senior author had begun preliminary queries.

Exercise for Step 3

Name five ways of collecting social science data. For each way, describe the method briefly and create an imagined application of the method as part of a case study.

Describe the strengths and weaknesses of each method, as it might have been used in this application. Where any weaknesses have been identified, indicate whether some other method's strengths can counteract all or most of the weaknesses. For instance, a major weakness of the survey method is that the survey data are limited to "self-reports" of respondents' own behavior. The accuracy of the self-reports could be checked by combining the survey data with investigators' direct observations of the respondents' actual behavior.

Step 4: Analyzing Case Study Evidence

Case study analysis takes many forms. Regardless of the form, the task is difficult because the analytic procedures are not usually formulaic, as they may be with other research methods. The absence of a strict routine leaves case study investigators with the need (some would say, "opportunity") to make critical procedural decisions when analyzing case study data. In doing so, investigators should document carefully the procedures used. As another alert, the course of the analysis may depend as much on the marshaling of arguments as on the tallying of data. Strong case study arguments will reflect a thoroughness in covering all relevant conditions combined with the explicit naming and entertaining of rival explanations (Yin, 2000).

The absence of any cookbook for analyzing case study evidence has only partially been offset by the development of prepackaged software to conduct computer-assisted tallies of large amounts of narrative text. The software helps code and categorize the words found in a text, as might have been collected from open-ended interviews or extracted from documents. However, the coding can only attend to the verbatim or surface language in the texts, potentially serving as a microlevel starting point for doing case study analysis. Yet the case study of interest is likely to be concerned with broader themes and events than represented by the surface language of texts. To this extent, you still need to have a broader analytic strategy, even if you have found the computer software to be a useful preliminary tool.

Discussed next are four examples of the broader analytic strategies (see also Yin, 2003b, pp. 116–133). The associated case study examples suggest that all the strategies can use either qualitative or quantitative data, or both. This duality reinforces the positioning of the case study method as a method not limited to either type of data. An important correlate is that case study investigators, including yourself, should not only be acquainted with collecting data from the variety of sources of evidence discussed in the preceding section but also with the analytic techniques now discussed in the present section.

Compare Expected and Actual Patterns

A pattern-matching procedure is the first type of case study analysis.

Many types of patterns can be relevant to a case study. Some patterns might cover a series of related *actions* or *events.* For instance, the conditions for transforming a business organization might include multiple changes, such as (National Institute of Standards and Technology, 1999, 2000) the implementation of new human resource and administrative practices; turnover in board or executive leadership; a retooling of product or service lines; and changed relationships in suppliers and the organization's supply chain. If you were doing a case study of such a transformation, you would start by hypothesizing the needed changes and their relationships. You would then collect data to see whether the changes and their relationships actually occurred, by matching the data against the predicted pattern.

Alternatively, the predicted pattern of events can be a pattern of *outcomes.* Cook and Campbell (1979, p. 118) defined such a pattern as the key ingredient in their quasi-experimental research design known as the *nonequivalent dependent variables* design. According to this design, an experiment or quasi-experiment may have multiple dependent variables—in other words, a variety of outcomes. The design pertains directly to case studies, as well. Whether as part of a quasi-experiment or a case study, the matching procedure would then pit an empirically observed or measured set of outcomes against those that had been predicted prior to the data collection.

For either of the preceding or other types of patterns, the specific pattern-matching technique depends on the nature of the data. If the pattern of outcomes includes some variables that enable you to compare the means and respective variances from two groups, you could perform statistical tests of significance. For instance, a study of math-science education reform might predict a pattern whereby students' test scores in math and science at different grade levels will improve compared to some baseline period, but that their reading scores at different grade levels will remain on the same trend lines compared to the same baseline period. In this example, you could conduct all the needed matching (comparisons) through statistical tests.

More commonly, the variables of interest are likely to be categorical or nominal variables. In this situation, you would have to judge the presence or absence of the predicted pattern by setting your own criteria (ahead of time) for what might constitute a "match" or a "mismatch." For instance, a case study investigating the presumed economic impact of a military base closing argues that the closing was *not* associated with the pattern of dire consequences that pundits commonly predicted would occur as a result of such closings (see Case Study 16).

CASE STUDY 16: PATTERN MATCHING TO SHOW WHY A MILITARY BASE CLOSURE WAS NOT CATASTROPHIC

Many military bases in the United States have been the presumed economic and residential driving forces of the local community. When such bases close, the strong belief is that the community will suffer in some catastrophic manner—leaving behind both economic and social disarray.

A case study of such a closure in California (Bradshaw, 1999), assembled a broad array of data to suggest that such an outcome did not, in fact, occur. The analytic strategy was to identify a series of sectors (e.g., retail markets, housing sales, hospital and health services, civilian employment, unemployment, and population turnover and stability) where catastrophic outcomes might have been feared, and then to collect data about each sector before and after the base closure. In every sector, and also in comparison to other communities and statewide trends, a pattern-matching procedure showed that the outcomes were much less severe than anticipated. The case study also presented potential explanations for these outcomes, thereby producing a compelling argument for its conclusions.

As but one example presented in Case Study 16, among the predicted consequences was a rise in unemployment. The case study tracked the seasonal pattern of unemployment for several years before and after the base closing and showed how, after observing seasonal variations, the overall rate did not appear to decline at all, much less in any precipitous manner. The case study especially called attention to the employment levels between January and April 1997, well after the base closing. The levels at these later times exceeded those of the January and April periods in the previous 5 years, when the base was still in operation (see Figure 8.1).

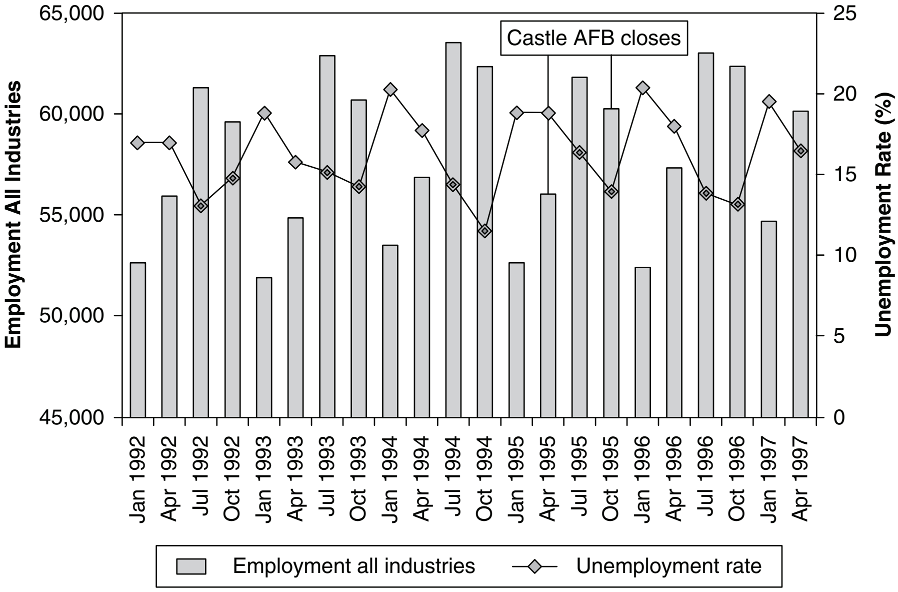

Figure 8.1 Employment and Unemployment Rate in Merced County

SOURCE: From "Communities not fazed: Why military base closures may not be catastrophic," by T. K. Bradshaw, 1999, *Journal of the American Planning Association, 65,* p. 201, fig. 1. Used with permission.

Important, too, was the breadth of possible consequences covered by the case study. Thus, the full case study did not rely on the unemployment outcome alone but showed that similar patterns existed in nearly every other important sector related to the community's economy. In this same manner, you would want to show that you had considered a broad array of relevant variables related to your research questions and also had defined and tested a variety of rival conditions—the more conditions, the better.

Use Evidence to Build an Explanation

This second analytic strategy comes directly from the explanatory role of case studies, based on their claimed advantage in addressing "how" and "why" questions (Shavelson & Townes, 2002). Following this strategy, you need to analyze your case study data by putting forth a convincing explanation for some set of events or conditions.

Unfortunately, building an explanation has no well-trodden template to emulate. You have to decide ahead of time what your case study is trying to demonstrate (if anything) and how you will meet the requirements for making such a demonstration convincing. Because all this may sound extremely vague, let's go into more detail with two illustrative case studies.

The explanation building in the first case study follows many situations in which an explanation is built "post-hoc," or after the fact. Such a label means that you try retrospectively to explain an event whose outcome already is known. In this first case study (see Case Study 17), the known outcome was that a Fortune 50 firm had gone out of business. The case study tried to explain why this outcome might have occurred. To do this, the case study posited the downside effects of several of the firm's "cultural tendencies." The case study then offered evidence in support of these tendencies and explained how they collectively left the firm without a critical "survival" motive.

CASE STUDY 17: EXPLANATION BUILDING: WHY A FORTUNE 50 FIRM WENT OUT OF BUSINESS

Business failure has been a common part of the American scene. Less common is when a failure occurs with a firm that, having successfully grown for 30 years, had risen to be the number two computer maker in the United States and, across all industries, among the top 50 corporations in size.

A professor at MIT served as a consultant to the senior management of the firm during nearly all its history. His case study (Schein, 2003) tries to explain how and why the company had a "missing gene," critical to the survival of the business.

As an important part of the explanation, the author argues that the gene needed to be strong enough to overcome the firm's other cultural tendencies, which included its inability to address layoffs that might have pruned deadwood in a more timely manner; set priorities among competing development projects (the firm developed three different PCs, not just one); and give more prestige to marketing and business as opposed to technological functions within the firm.

The case study cites much documentation and interviews but also includes supplementary chapters permitting key former officials of the firm to offer their own rival explanations.

The second case study took place in an entirely different setting. In New York City, a long-time rise in crime from 1970 finally peaked in the early 1990s, starting a new, declining trend from that time thereafter (see Figure 8.2). The case study (see Case Study 18) attempts to explain how actions taken by the New York City Police Department might have contributed to the turnaround. The case study builds a twofold explanation. First, it devotes several chapters to the nature of the police department's specific protective actions, showing how they could plausibly reduce crime. Second, it presents time-series data and suggests that the timing of the actions fit well the timing of the turnaround. In particular, the case study argues that, although a declining trend already had started in 1991, an even sharper decline in murder rates in 1994 coincided with the first full year of new police protection practices (see Figure 8.2).

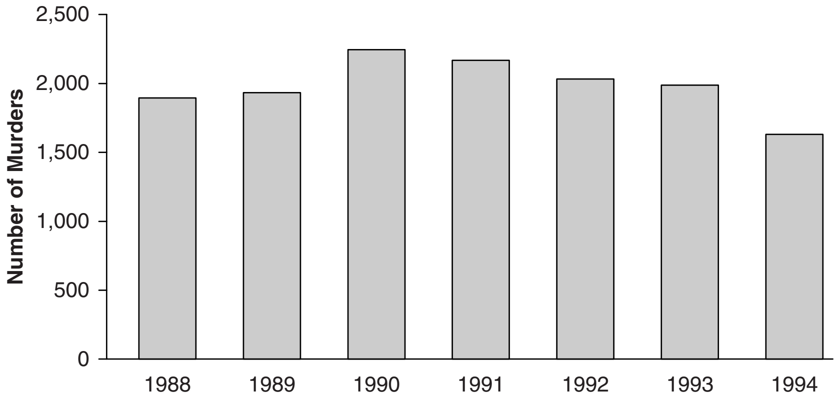

Figure 8.2 New York Murder Rate (1988–1994)

SOURCE: Reprinted with the permission of The Free Press, a Division of Simon & Schuster Adult Publishing Group, from FIXING BROKEN WINDOWS: Restoring Order and Reducing Crime in Our Communities by George M. Kelling and Catherine M. Coles. Copyright © 1996 by George L. Kelling and Catherine M. Coles. All rights reserved.

CASE STUDY 18: EXPLAINING THE DECLINE IN CRIME RATES IN NEW YORK CITY

In New York City, following a parallel campaign to make the city's subways safer, the city's police department took many actions to reduce crime in the city more broadly. The actions included enforcing minor violations ("order restoration and maintenance"), installing computer-based crime-control techniques, and reorganizing the department to hold police officers accountable for controlling crime.

Case Study 18 (Kelling & Coles, 1996) first describes all these actions in sufficient detail to make their potential effect on crime reduction understandable and plausible. The case study then presents time series of the annual rates of specific types of crime over a 7-year period. During this period, crime initially rose for a couple of years and then declined for the remainder of the period. The case study explains how the timing of the relevant actions by the police department matches the changes in the crime trends. The authors cite the plausibility of the actions' effects, combined with the timing of the actions in relation to the changes in crime trends, as part of their explanation for the reduction in crime rates in the New York City of that era.

Both of these examples show how to build explanations for a rather complex set of events. Each case study is book length. Neither follows any routine formula or procedure in the explanation-building process. However, the work in both case studies suggests the following characteristics that might mark the explanations in your own case study analyses:

- Thoroughness in identifying and incorporating data relevant to the testing of logical explanations
- Clarity, through the use of tables and exhibits where possible, in showing how the data collected were used to test at least the most important parts of the explanations
- Exploration of alternative or rival explanations
- A summary interpretation that directly compares the main and rival explanations

Ascertain and Array Key Events, Chronologically

A third strategy is more straightforward and applicable to many case studies. The strategy is based on the principle that, in explaining a series of events, an event claimed to be the cause of a second event needs to occur prior to the second event. For instance, a health service's new resources in Year 1 could only affect its performance after, and not prior to, Year 1. To this extent, organizing events chronologically can help develop a logical sequence for explaining how and why the case study's key events might have occurred (see Case Study 19).

CASE STUDY 19: A CHRONOLOGY SHOWING THE DELAYED START-UP OF A CONTROVERSIAL COMMUNITY PROGRAM

Controversies surround the opening and location of certain kinds of public services, such as those, like a methadone maintenance clinic, aimed at helping drug addicts. Communities fear that the services will bring undesirable "elements" into the community and also jeopardize the quality of related health services such as psychiatric services to nonaddicted clients.

Starting and running a methadone clinic in an urban community were the subjects of a case study (Nelkin, 1973). The case study refers extensively to specific chronological information in explaining the sources of significant delays in opening the clinic. The case study also refers to chronological information in explaining how resistance by some community members and by the medical staff of some related public health services led the methadone maintenance clinic to operate differently than its originally proposed configuration.

Chronologies offer the additional advantage that chronological data are usually easy to obtain. One value of using documentary evidence is that the documents frequently cite specific dates. But even in the absence of specific dates, having an estimated month or even season of occurrence may be sufficient to serve your case study's needs. If so, you need not depend solely on having relevant documentary evidence. You also can ask your interviewees to estimate when something might have happened. Such an inquiry does not require them to have been a chronicler. Rather, you can ask whether something happened before or after a well-known election, a holiday season, or some other benchmark such as the annual Super Bowl in professional football. Citing such a benchmark usually can help most people recall more readily the chronological occurrence of an event or even the chronology of a sequence of events.

Chronological data are sufficiently valuable that collecting such information should be a routine part of all the data collected for your case study. Tracking such chronologies requires you to take note of the dates that appear in documents and to ask interviewees *when* something might have occurred, not just whether it had transpired. Even if you had not identified the need for this information at the outset of your case study, in later analyzing your data you may find that the chronologies lead to surprising insights. Evidence about the timing of events also may help you reject some rival explanations, because they may not fit the chronological facts that you have amassed.

Construct and Test Logic Models

A logic model (Wholey, 1979) stipulates a complex sequence of events over time, covering presumed causal relationships among a host of independent, intervening, and dependent variables. This fourth analytic strategy has become extremely useful for doing case study evaluations but also can be used for case study research.

For evaluation, the logic model assists in assessing an intervention, which is supposed to produce a certain outcome or outcomes. However, most interventions are complex chains of events: Initial activities (e.g., employee training) have their own immediate outcomes (e.g., employees' new knowledge), which, in turn, produce some intermediate outcome (e.g., new practices by the employees), which, in turn, produce final or ultimate outcomes (e.g., improved business performance). The strength of the logic model is its requirement of an explicit conceptualization, or *theory of action,* of the chain of events.

After you develop operational definitions for the events in the logic model, you will then try to collect relevant data for your case study. Comparing the collected data with the previously stipulated sequence of events serves as the empirical test of the logic model and is the actual analytic step. The more the data support the original logic model, the more the original theory of action is to be favorably judged.

An illustrative logic model (see Figure 8.3) represents an increasingly common type used in case study evaluations. The model begins with the resources or support (see Box 1, Figure 8.3) needed to conduct the intervention (brokerage and technical assistance services—see Boxes 2 and 3). The actual case study data are needed to confirm this support as well as whether the intervention ultimately preceded a series of desired outcomes, culminating in changed business performance and related benefits (Boxes 8, 9, 10, and 11). Distinctive about this logic model is that it also has a place for two sets of rival explanations (Boxes 12 and 13), apart from the intervention of interest. The rivals hypothesize that the same outcomes might have occurred, but due to conditions other than the brokerage and technical assistance services. The collected data need to show whether these rival conditions existed and how they might have influenced the outcomes. The more the intervention of interest can be supported in the face of these rivals, the more positive will be the overall assessment.

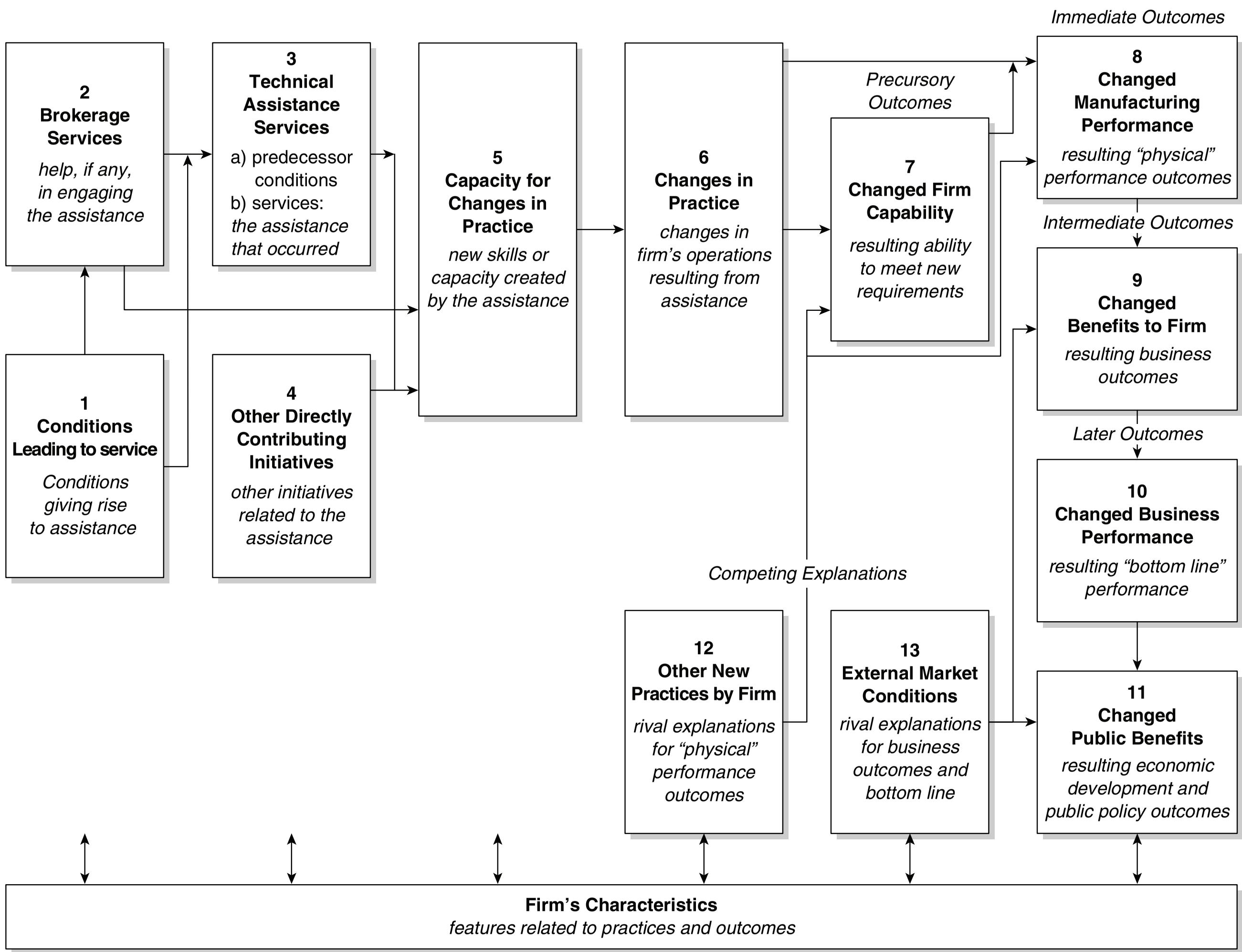

Figure 8.3 Changes in Performance in the Manufacturing Firm

SOURCE: Yin and Oldsman (1995).

The logic model framework has quantitative counterparts that take the form of structural equation models (SEMs) and path analyses. For example, schools' progress in implementing education reform was a major subject of a case study of a reforming school system. Although the single system was the subject of a single-case study (see Case Study 20), the size of the system meant that it contained hundreds of schools. The school-level data then became the subject of a path analysis. Figure 8.4 shows the results of the path analysis, enumerating all the original variables but then only showing arrows where the standardized regression coefficients were statistically significant.

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CASE STUDY 20: TESTING THE LOGIC OF A SCHOOL REFORM ACT

Case studies can include rather advanced quantitative analyses. The subject of the case study (Bryk, Bebring, Kerbow, Rollow, & Easton, 1998) is the attempted transformation of a major urban school system (a single case) that took place in the 1980s. A new law was passed to decentralize the system by installing powerful local school councils.

(Continued)

(Continued)

The case study includes qualitative data about the system as a whole and about the individual schools in the system. At the same time, the study also includes a major quantitative analysis that takes the form of structural equation modeling. The resulting path analysis tests a complex logic model whereby prereform restructuring is claimed to produce strong democracy, in turn producing systemic restructuring, and finally producing innovative instruction, all taking into account a context of basic school characteristics. The analysis is made possible because the single case (the school system) contains an embedded unit of analysis (individual schools), and the path model is based on data from 269 of the elementary schools in the system. The results of the path model do not pertain to any single school but represent a commentary about the collective reform experience across all the schools—in other words, the overall reform of the system (single case) as a whole.

In this example, the schools represented an embedded unit of analysis within the overall single-case study, and the collective experiences of the schools provided important commentary about the advances made by the system as a whole. Note the similarity between the variables used in the path analysis and those that might have been used in a logic model studying the same situation. Other investigators of school reform have used the same path analysis method to test the logic of reform in multiple school systems, not just single systems (see Borman & Associates, 2005).

Exercise for Step 4

Select one of your own empirical studies—but *not* a case study—in which you analyzed some data (if you cannot cite one of your own studies, choose one from the literature, related to a topic of interest to you). Examine and describe how the data were analyzed in this study. Was it a qualitative or quantitative analysis? Argue whether this same analysis, virtually in its same form, could be found as part of a case study. Do you think that quantitative analyses are less relevant to case studies than qualitative analyses?

Summary

This chapter has suggested ways of dealing with four steps that have been the most challenging in doing case study research. In the first step, investigators like yourself commonly struggle with how to choose a significant, not mundane, case or cases for their case studies.

In the second step, having multiple cases within your case study may require greater effort. However, the benefit will be a more strongly designed case study, where the cases may replicate or otherwise complement each other's experiences.

In the third and fourth steps, creating a strong evidentiary base will provide greater credibility for your case study, and methodically analyzing these data, using qualitative or quantitative methods, will then lead to more defensible findings and conclusions.

By covering these four steps, the chapter follows the spirit of handbooks that try to provide concrete and operational advice to readers. The chapter's descriptions of numerous, specific case studies add to the concreteness. If you can emulate some of these case studies, or if you can successfully implement the four steps more generally, you may markedly improve your own case studies.

In contrast, the chapter has not attempted another conventional use of handbooks—to provide a theoretical and historical perspective on the evolution of a topic such as case study research. Such a perspective already has been provided elsewhere by Jennifer Platt (1992), and readers interested in learning more about it would be well-advised to consult her work.[6]

Exercises

Different exercises may be relevant, depending on whether a class is at the preliminary or advanced end of the spectrum of doing social science research.

Exercise 1. Finding and Analyzing an Existing Case Study: Have each student retrieve an example of case study research from the literature.

- *Prelim. Class:* The case study can be on any topic, but it must have used some empirical method and presented some empirical data. Questions for discussion:
  1. Why is this a case study?
  2. What, if anything, is distinctive about the findings that could not be learned by using some other social science method focusing on the same topic?
- *Advanced Class:* The case study must have presented some numeric (quantitative) as well as narrative (qualitative) data. Questions for discussion:
  1. How were these data derived (e.g., from what kind of instrument, if any) and were they presented clearly and fairly?
  2. How were these data analyzed? What were the specific analytic procedures or methods?
  3. Are there any lessons regarding the potential usefulness of having both qualitative and quantitative evidence within the same case study?

Exercise 2. Designing Case Study Data Collection: Have each student design a case study on a topic with which he or she is familiar (my family, my school, my friends, my neighborhood, etc.).

- *Prelim. Class:* What are the case study's questions? Among the various sources of evidence for the case study, will interviews, documents, observations, and archival data all be relevant? If so, how?

- *Advanced Class:* Design a preliminary case study protocol (instrument), to collect data from the relevant sources of evidence relevant to the case study.

Exercise 3. Testing for Case Study Skills: Have each student present the following "claims," either in the form of a classroom presentation or written assignment.

- *Prelim. Class:* Why and with what distinctive skills, if any, does a student believe that he or she is adequately equipped (or *not* equipped) to do a case study? Where not well-equipped, what remedies does the student recommend for himself or herself?
- *Advanced Class:* Carry out the same exercise as that of the prelim class. In addition, however, ask two other students to prepare critiques of the first student's claims and permit the first student time for a brief response or rebuttal.

Notes

1. The chapter is based on and draws heavily from a case study anthology compiled by the author (see Yin, 2004). See also Yin (2005) for an anthology of case studies devoted solely to the field of education.

2. Aspiring case study investigators may, therefore, need to consult (and use) the earlier chapter and the full textbook, as well as several other directly related works by the present author: Yin (2003a) for in-depth applications of the case study method; Yin (2006a) for guidance in doing case studies in the field of education; and Yin (2006b) if case studies are to be part of a mixed methods research study. These other works can help investigators address such questions as "when and why to use the case study method" in the first place, compared to other methods.

3. These forms all fall within the domain of "case study research." In turn, many specialists consider case study research to fall within a yet broader domain of "qualitative research" (Creswell, 2007). However, the present approach to case study research resists any categorization under the broader domain, because case study research, as discussed throughout the present chapter, can include quantitative and not just qualitative methods.

4. The case study anthology (Yin, 2004) referenced in Footnote 1 contains lengthy excerpts of all the case studies described in the boxes throughout this chapter.

5. Case study evaluations are not necessarily the same as doing your own case studies. Clients and sponsoring organizations (e.g., private foundations) usually prespecify the research questions as well as the cases to be studied. In this sense, case study evaluators may not need to decide how to define and select their case studies as covered in the text.

6. Platt traces the evolution of case study research, starting with the work of the "Chicago School" (of sociology) in the 1920s. Despite this auspicious beginning, Platt explains why case study research became moribund during the post–World War II period—a period so barren that the term *case study* was literally absent from the methodological texts of the 1950s and 1960s. Platt then argues that the resurgence of case study research occurred in the early 1980s, crediting the resurgence to a fresh understanding of the benefits that may accrue when case study research is properly designed.

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CHAPTER 9

Integrating Qualitative and Quantitative Approaches to Research

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Integrating Qualitative and Quantitative Approaches to Research

Despite considerable differences in methodology, it is widely acknowledged by scholars across disciplines that research questions drive research methods. Contemporary research questions are more complex than ever, requiring complex methods for finding answers. For example, in international development studies, investigators need to expand their research questions beyond a thorough ethnographic understanding of villages and/or cultural groups, because they need to also generalize their understanding across broader social and political contexts. Or, in clinical trials, questions go beyond group differences in a specific variable (or a set of variables) and look for a wider possibility of differences in behaviors, cognitions, and social contexts (see Tashakkori & Creswell, 2007).

This chapter attempts to summarize the issues and procedures for integrating qualitative and quantitative approaches to research in order to answer such questions more effectively. Different scholars have used different terms (integrative, combined, blended, mixed methods, multimethod, multistrategy, etc.) to identify studies that attempt such integration. However, the term *mixed methods* seems to be accepted by most scholars across disciplines (Collins, Onwuegbuzie, & Jiao, 2007; Creswell & Plano Clark, 2007; Greene, 2007; Greene & Caracelli, 2003; Johnson &

Onwuegbuzie, 2004; Rao & Woolcock, 2004; Teddlie & Tashakkori, 2006). The term *mixed methodology* has been broadly used to denote the academic field or discipline of studying and presenting the philosophical, theoretical, technical, and practical issues and strategies for such integration (Teddlie & Tashakkori, in press). In the following sections, we provide an overview of mixed methodology.

The sections that follow will first examine our guiding assumptions for the chapter. We then introduce an overview of qualitative, quantitative, and integrated approaches to sampling, data collection, data analysis, and inference. The chapter will end with a discussion of issues in evaluating/auditing the inferences that are made on the basis of the results.

Our Guiding Assumptions

Before we start, we would like to iterate our previous assertions (e.g., Tashakkori & Teddlie, 1998, 2003b; Teddlie & Tashakkori, 2006) about mixed methods. We consider them our guiding assumptions, and we believe that they would also facilitate a holistic understanding of this chapter. They are as follows:

- We believe that qualitative and quantitative approaches to research are not dichotomous and discrete. Every component or aspect of a study (e.g., research questions, data, data collection and analysis techniques, inferences, recommendations) is on a continuum of qualitative-quantitative approaches. As a result, studies differ in their degree of inductive-deductive logic, subjectivity, cultural relativity, value-addedness, and emic-etic (from the perspective of the participants vs. the investigator's perspective).

- We believe that research questions must drive the methods of a study. As such, the utility of qualitative and quantitative approaches and methods must be evaluated in terms of the quality of potential answers. Therefore, we do not believe that mixed methods are suitable for answering all (or even most) research questions.

- We believe that data collection and analysis techniques should be distinguished from research methods and design. Consequently, we believe that, regardless of the approach or design of a study, most data collection techniques can potentially yield qualitative and quantitative data that may be analyzed qualitatively or quantitatively.

- We differentiate mixed methodology from mixed methods. We use *mixed methodology* as a scholarly field of developing, studying and discussing the issues and procedures for integrating the qualitative and quantitative approaches to research. We use *mixed methods* as the process of integrating the qualitative and quantitative approaches and procedures in a study to answer the research questions, as well as the specific strategies and procedures that are used (see Teddlie & Tashakkori, in press, for more details).

- Although the integration may occur at any stage of a research project, we believe that true mixed methods designs have clearly articulated mixed research questions, necessitating the integration of qualitative and quantitative approaches in *all* stages of the study. Strands of a study might have research questions that are qualitative or quantitative in approach. However, an overarching question, involving the integration of subquestions must drive every mixed methods study.

- Throughout the chapter, we make every effort to differentiate between purpose (agenda or reason motivating you to conduct a study), question (the professional or theoretical issue troubling you that needs an answer or solution), data (the information you need to answer your research question), data collection methods (how you collect the information you need for answering your research question), results (the outcome of summarizing and analyzing your collected data), inferences (the credible conclusions you make on the basis of the results), and policy/practice recommendations (credible suggestions you can make for policy and professional practice on the basis of your inferences).

Widespread popularity of mixed methods may partly be attributed to its empowerment of the investigators to go beyond the qualitative-quantitative divide. Flexibility to use both the qualitative and quantitative approaches and methods allows the applied researcher or evaluator to answer his or her research questions in the most effective manner. The evidence for this popularity comes from a variety of sources: Several texts have now been written in the area (e.g., Brewer & Hunter, 1989, 2006; Creswell, 2003; Creswell & Plano Clark, 2007; Greene, 2007; Greene & Caracelli, 1997, 2003; Newman & Benz, 1998; Tashakkori & Teddlie, 1998, 2003a).

There are a number of dynamic ongoing debates within the mixed methods field over issues such as basic definitions, research designs, and how to draw inferences. There are literally thousands of references to mixed methods research on the Internet. There also is a new journal devoted to the field (*Journal of Mixed Methods Research*). This chapter summarizes our thinking regarding the current status of mixed methods research in a variety of areas.

Definition and Utility of Mixed Methods

Probably, the most frequently asked question about mixed methods concerns its definition. Sandelowski (2003) summarizes the issue of definition very eloquently by suggesting that collecting and analyzing two types of data in a project should not be called mixed methods:

> In one kind of mixed methods study, qualitative and quantitative entities are in mixed company with each other, while in the other kind, they are actually blended. In the first kind of mixed methods study, entities are associated with or linked to each other but retain their essential characters; metaphorically, apple juice and orange juice both are used, but they are never mixed together to produce a new kind of fruit juice. (p. 326)

Referring to the characteristics of research articles that are labeled *mixed methods,* Tashakkori and Creswell (2007) concluded that these published articles

> are considered "mixed" because they utilize qualitative and quantitative approaches in one or more of the following ways:
>
> - two types of research questions (with qualitative and quantitative approaches);
> - the manner in which the research questions are developed (emergent vs. pre-planned);
> - two types of sampling procedures (e.g., probability and purposive, Teddlie & Yu, 2007);
> - two types of data collection procedures (e.g., focus groups and surveys);
> - two types of data (e.g., numerical and textual);
> - two types of data analysis (statistical and thematic); and
> - two types of conclusions (emic and etic representations, "objective" and "subjective," etc.).
>
> As an effort to be as inclusive as possible, we have broadly defined mixed methods here as research in which the investigator collects and analyzes data, integrates the findings, and draws inferences using both qualitative and quantitative approaches or methods in a single study or a program of inquiry. A key concept in this definition is integration. (pp. 3–7)

Earlier we (Tashakkori & Teddlie, 1998, 2003b) tried to distinguish between studies that are mixed in the methods only (i.e., mixed in data collection and analysis only, without serious integration) and those that are mixed in all stages of the study. Incorporating contemporary developments and clarifications, we recently (Teddlie & Tashakkori, 2006) have differentiated mixed methods designs into *quasi-mixed* (predominantly quantitative or qualitative approach in questions, two types of data, no serious integration) and *mixed methods* (two types of data or analysis, integrated in all stages).

Unfortunately, the reasons for using mixed methods are not always explicitly delineated and/or recognized by authors. Some of the reasons that have been identified by scholars include complementarity, completeness, developmental, expansion, corroboration/confirmation, compensation, and diversity (see Table 9.1 for details). Although in the past, triangulation was often assumed to be the most frequent purpose for conducting mixed methods, other purposes are more notable today. For example, mixed methods are particularly adept at identifying diverse results across different data sets. Then, researchers attempt to reconcile the diversity, which is a strength of the mixed methods approach.

The quality of a mixed methods study directly depends on the degree to which it meets the purpose for which the mixing of approaches was deemed necessary in that study. For example, if the main purpose for using mixed methods is for completeness, a good mixed methods study must provide a more complete understanding of the phenomenon under study than its qualitative and quantitative strands do separately. This might be called the *utilization quality* (or *pragmatic quality*) of mixed methods inferences; that is, inferences that are made at the end of a study are good only if they address the intended purpose for mixing.

Table 9.1 Purposes for Mixed Methods, as Often Stated by Researchers

Purpose	*Description*
Complementarity	Mixed methods are used to gain complementary views about the same phenomenon or relationship. Research questions for the two strands of the mixed study address related aspects of the same phenomenon.
Completeness	Mixed methods designs are used to make sure a complete picture of the phenomenon is obtained. The full picture is more meaningful than each of the components.
Developmental	Questions of one strand emerge from the inferences of a previous one (sequential mixed methods), or one strand provides hypotheses to be tested in the next one.
Expansion	Mixed methods are used to expand or explain the understanding obtained in a previous strand of a study.
Corroboration/ Confirmation	Mixed methods are used to assess the credibility of inferences obtained from one approach (strand). There usually are exploratory *and* explanatory/confirmatory questions.
Compensation	Mixed methods enable the researcher to compensate for the weaknesses of one approach by using the other. For example, errors in one type of data would be reduced by the other (Johnson & Turner, 2003).
Diversity	Mixed methods are used with the hope of obtaining divergent pictures of the same phenomenon. These divergent findings would ideally be compared and contrasted (pitted against each other, Greene & Caracelli, 2003).

SOURCES: This table is constructed on the basis of Greene, Caracelli, and Graham (1989), Patton (2002), Tashakkori and Teddlie (2003a), Creswell (2005), and Rossman and Wilson (1985).

The utilization quality of mixed methods also depends on the design of the mixed methods study. For parallel mixed methods, the purpose of mixing must be known from the start. For sequential mixed methods, the purpose might be known from the start, or it might emerge from the inferences of the first strand. For example, unexpected or ambiguous results from a quantitative study might necessitate the collection and analysis of in-depth qualitative data in a new strand of the study.

Mixed Methods Designs

There are a variety of typologies for mixed methods designs in the literature. These designs have been differentiated by scholars on the basis of various criteria (e.g., Creswell & Plano Clark, 2007; Greene & Caracelli, 1997; Johnson & Onwuegbuzie, 2004; Morgan, 1998; Morse 1991, 2003) including the following:

- Number of strands or phases in the mixed methods study
- Type of implementation process

- Stage of integration of approaches
- Priority of methodological approaches
- Purpose or function of the study
- Theoretical or paradigmatic perspective

Recently, we (Teddlie & Tashakkori, 2006) have categorized mixed designs into five families: sequential, parallel, conversion, multilevel, and fully integrated. This classification is based on three key dimensions: (1) number of strands in the research design, (2) type of implementation process, and (3) stage of integration (i.e., collecting and analyzing two types of data to answer predominantly qualitative or quantitative questions vs. integration in all stages of research to answer mixed questions). We do not use the other three criteria noted above in our typology, which focuses on the methodological components of research designs.

The first dimension in our typology is the number of strands or phases in the design. A strand of a research design is a phase of a study that includes three stages: the conceptualization stage, the experiential stage (methodological/analytical), and the inferential stage. A monostrand design employs only a single phase and it encompasses all the stages from conceptualization through inference, while a multistrand design employs more than one phase, each encompassing all the stages from conceptualization through inference.

The second dimension of our typology is the type of implementation process: *parallel, sequential,* and *conversion.* Parallel and sequential designs have been employed by numerous authors writing in the mixed methods tradition. In parallel mixed designs, the strands of a study occur in a synchronous manner (even though the data for one strand might be collected with some time lag), while in sequential designs they occur in chronological order with one strand emerging from the other. Conversion designs are a unique feature of mixed methods research and include the transformation of one type of data to another, to be reanalyzed accordingly. Conversion may be in the form of *quantitizing*[1] (converting qualitative data into numerical codes that can be reanalyzed statistically) or *qualitizing* (in which quantitative data are transformed into data that can be reanalyzed qualitatively).

The third dimension of our typology is the stage of integration of the qualitative and quantitative approaches. The most dynamic and innovative of the mixed methods designs are mixed across stages. However, various scholars have identified mixed studies in which two types of data are collected and analyzed to answer a predominantly qualitative or quantitative type of research question. We call these studies *quasi-mixed designs,* because there is no serious integration across the qualitative and quantitative approaches.

Monostrand conversion designs (also known as the simple conversion design) are used in single-strand studies in which research questions are answered through an analysis of transformed data (i.e., quantitized or qualitized data). These studies are mixed because they switch approach in the methods phase of the study, when the data that were originally collected are converted into the other form. Monostrand conversion designs may be planned before the study actually occurs, but many applications of this design occur serendipitously as a study unfolds. For instance, a researcher may determine that there are emerging patterns in the information gleaned from narrative interview data that can be converted into numerical form and then analyzed statistically, thereby allowing for a more thorough analysis of the data.

The monostrand conversion design has been used extensively in both the quantitative and qualitative traditions, without being recognized as "mixed" (see, e.g., Hunter & Brewer, 2003; Maxwell & Loomis, 2003; Waszak & Sines, 2003). An explicit example of quantitizing data in the mixed methods research literature is Sandelowski, Harris, and Holditch-Davis (1991) transformation of interview data into a frequency distribution that compared the "numbers of couples having and not having an amniocentesis with the number of physicians encouraging or not encouraging them to have the procedure," which was then analyzed statistically to determine the "relationship between physician encouragement and couple decision to have an amniocentesis" (Sandelowski, 2003, p. 327).

Multistrand mixed methods designs are more complex, containing at least two research strands. Mixing of the qualitative and quantitative approaches may occur both within and across all stages of the study. Five types of these designs, which we consider to be the most valuable are parallel mixed designs, sequential mixed designs, conversion mixed designs, multilevel, and fully integrated mixed designs. These five types of designs are families, since there may be several permutations of members of these families based on other design criteria.

Parallel mixed designs are designs in which there are at least two interconnected strands: one with qualitative questions and data collection and analysis techniques and the other with quantitative questions and data collection and analysis techniques. Data may be collected simultaneously or with some time lag (for this reason, we prefer the term *parallel*, as compared with concurrent). Analysis is performed independently in each strand, although one might also influence the other. Inferences made on the basis of the results from each strand are integrated to form meta-inferences at the end of the study. Using parallel mixed designs enables the researchers to answer exploratory (frequently, but not always, qualitative) and confirmatory (frequently, but not always, quantitative) questions.

Lopez and Tashakkori (2006) provide an example of a parallel mixed study of the effects of two types of bilingual education programs on attitudes and academic achievement of fifth-grade students. The quantitative strand of the study included standardized achievement tests in various academic subjects, as well as measured linguistic competence in English and Spanish. Also, a Likert-type scale was used to measure self-perceptions and self-beliefs in relation to bilingualism. The qualitative strand consisted of interviews with a random sample of 32 students in the two programs. Each set of data was analyzed independently, and conclusions were drawn. The findings of the two studies were integrated by (a) comparing and contrasting the conclusions and (b) by trying to construct a more comprehensive understanding of how the two programs affected the children.

Sequential mixed designs are designs in which there are at least two strands that occur chronologically (QUAN → QUAL or QUAL → QUAN). The conclusions that are made on the basis of the results of the first strand lead to formulation of questions, data collection, and data analysis for the next strand. The final inferences are based on the results of both strands of the study. The second strand of the study is conducted either to confirm/disconfirm the inferences of the first strand or to provide further explanation for findings from the first strand. Although the second strand of the study might emerge as a response to the unexpected and/or inexplicable results of the first strand, it is also possible to plan the two strands in advance.

An example of a sequential QUAL → QUAN mixed design comes from the consumer marketing literature (Hausman, 2000). The first part of the study was exploratory in nature using semistructured interviews to examine several questions related to impulse buying. Interview results were then used to generate a series of hypotheses. Trained interviewers conducted 60 interviews with consumers, and the resultant data were analyzed using grounded theory techniques. Based on these analyses, a series of five hypotheses were developed and tested using a 75-item questionnaire generated for the purposes of this study. A final sample of 272 consumers completed the questionnaire. Hypothesis testing involved both correlational and analysis of variance techniques.

The conversion mixed design is a multistrand parallel design in which mixing of qualitative and quantitative approaches occurs in all components/stages, with data transformed (qualitized or quantitized) and analyzed both qualitatively and quantitatively (Teddlie & Tashakkori, 2006). In these designs, one type of data (e.g., qualitative) is gathered and is analyzed accordingly (qualitatively) and then transformed and analyzed using the other methodological approach. The Witcher, Onwuegbuzie, Collins, Filer, and Wiedmaier (2003) study is an example of such a design. In this study, the researchers gathered qualitative data from 912 undergraduate/graduate students regarding their perceptions of the characteristics of effective college teachers. A qualitative thematic analysis revealed nine characteristics of effective college teachers, including student centeredness and enthusiasm about teaching. A series of binary codes (1, 0) were assigned to each student for each effective teaching characteristic. These quantitized data were subjected to a series of analyses that enabled the researchers to statistically associate each of the nine themes of effective college teaching with four demographic variables (gender, race, undergraduate/graduate status, preservice status). The researchers were able to connect students with certain demographic characteristics with preferences for certain effective teaching characteristics.

In a multilevel mixed design, mixing occurs as QUAN and QUAL data from different levels of analysis are analyzed and integrated to answer aspects of the same or related questions. These designs are described in more detail in the sampling section below.

The fully integrated mixed design, takes advantage of both a parallel and a sequential process in which mixing of qualitative and quantitative approaches occurs in an interactive (i.e., dynamic, reciprocal, interdependent, iterative) manner at all stages of the study. At each stage, information from one approach (e.g., qualitative) affects the formulation of the other approach (e.g., quantitative) (Teddlie & Tashakkori, 2006).

It should be evident to the reader that in the multistrand designs, one approach/strand might only be a small part of the overall study (what Creswell & Plano Clark, 2007, call "embedded designs"). For example, parallel with (or immediately following) an extended qualitative study, limited quantitative survey data might be collected and analyzed, to provide insights about a larger respondent group than the qualitative study included. Despite the larger sample size, such a survey study does not provide much more insight on the phenomenon than the original qualitative study. However, it would provide information regarding the degree of transferability of the results to the large group/population.

Sampling in Mixed Methods Studies

Sampling involves selecting units of analysis (e.g., people, groups, artifacts, settings) in a manner that maximizes the researcher's ability to answer research questions that are set forth in a study (Tashakkori & Teddlie, 2003b, p. 715). Mixed methods sampling techniques involve the selection of units or cases for a research study using both probability sampling and purposive sampling, to maximize inference quality and transferability (Collins et al., 2007). Three types of mixed methods sampling are introduced in this section: sequential mixed methods sampling, parallel mixed methods sampling, and multilevel mixed methods sampling. Before discussing these three, we need a brief review of traditional (monomethod) sampling techniques.

Traditional probability sampling techniques involve selecting specific units or cases randomly so that the probability of inclusion for every population member is "determinable" (Teddlie & Yu, 2007). There are three basic types of probability sampling:

- *Simple random sampling* occurs when each sampling unit in a clearly defined population has an equal chance of being included in the sample.
- *Stratified sampling* occurs when the researcher divides the population into subgroups (or strata) such that each unit belongs to a single stratum and then selects units from those strata.
- *Cluster sampling* occurs when the sampling unit is not an individual but a group (cluster) that occurs naturally in the population such as neighborhoods or classrooms.

Traditional purposive sampling techniques involve selecting certain units or cases based on a specific purpose or research question rather than randomly. Researchers using purposive sampling techniques want to generate much detail from a few cases, to maximize the possibility of answering the research questions. There are three broad categories of purposive sampling techniques (plus a category that involves multiple purposive techniques), each of which encompasses several specific types of strategies:

- *Sampling to achieve representativeness or comparability* is used when the researcher wants to (1) select a purposive sample that represents a broader group of cases as closely as possible or (2) set up comparisons among different types of cases.

- *Sampling special or unique cases* are employed when the individual case itself, or a specific group of cases, is a major focus of the investigation.

- *Sequential sampling* uses the gradual selection principle of sampling when (1) the goal of the research project is the generation of theory (or broadly defined themes) or (2) the sample evolves of its own accord as data are being collected. Gradual selection is the sequential selection of units or cases based on their relevance to the research questions, not their representativeness (e.g., Flick, 1998).

A purposive sample is typically (but not always) designed to pick a small number of cases that will yield the most information about a particular phenomenon, while a probability sample is planned to select a larger number of cases that are collectively representative of the population of interest. There is a classic methodological tradeoff involved in the sample size difference between the two techniques: purposive sampling leads to greater depth of information from a smaller number of carefully selected cases, while probability sampling leads to greater breadth of information from a larger number of units selected to be representative of the population (e.g., Patton, 2002).

Purposive sampling can occur before or during data collection, and it often occurs *both* before and during data collection. Probability sampling is preplanned and does not change during data collection, unless serious methodological problems arise, which often undermines or terminates the study. Purposive sampling relies heavily on the expert judgments of the researcher(s), while probability sampling is often based on preestablished mathematical formulas. Sampling frames may be formal (typically mathematically determined) or informal. When using informal sampling frames, the researcher determines a dimension of interest, visualizes a distribution of cases on that dimension, and then selects the cases of interest to him/her.

Mixed methods studies use both purposive and probability sampling techniques. The dichotomy between probability and purposive becomes a continuum when mixed methods sampling is added as a third type of sampling strategy technique. Table 9.2 presents the characteristics of mixed methods sampling techniques, which are combinations of (or intermediate points between) the quantitative and qualitative points of view.

Mixed sampling strategies may employ any of the probability and purposive techniques discussed earlier in this chapter. The researcher's ability to creatively combine these techniques in answering a study's questions is one of the defining characteristics of mixed methods research. Combining the two orientations to sampling allows the researcher to generate complementary databases that include information that has both depth and breadth regarding the phenomenon under study.

When drawing a mixed sample for multiple research strands, researchers necessarily use both formal and informal sampling frames. The first decision to be made in developing a mixed sampling strategy concerns *what* is to be sampled. In fact, there are three general types of units that can be sampled: cases, materials, and other elements in the social situation. The mixed methods researcher should consider all three data sources in drawing his/her sample and how they relate to the study's research questions.

We define mixed methods sampling as involving the selection of units of analysis for a study through both probability and purposive sampling strategies. As with all mixed methods techniques, the particular sampling strategy for any given study

Table 9.2 Characteristics of Mixed Methods Sampling Strategies

Dimension of Contrast	*Mixed Methods Sampling*
Purpose of sampling	Generate a sample that will address research questions
Transferability/generalizability	Simultaneous attention to transferability to/across population(s) and transferability across settings, modalities, and time periods
Sampling techniques	Both probability and purposive, within and across strands
Rationale for selecting cases/units	Simultaneous attention across the strands, to representativeness *and* potential for finding answers to research questions
Sample size	Multiple sample within and across strands, with equal or different sample sizes
Depth/breadth of information per case/unit	Focus on both depth and breadth of information, both within and across the strands
When the sample is selected	Preplanned sampling design while allowing for the emergence of other samples during the study
Sampling frame	Both formal and informal frames are used

is determined by the dictates of the research questions. There are four types of mixed methods sampling: basic mixed sampling strategies, sequential mixed sampling, parallel mixed sampling, and multilevel mixed sampling (Teddlie & Yu, 2007).

The basic mixed methods sampling strategies include stratified purposive sampling and purposive random sampling. These strategies are also identified as purposive sampling techniques (e.g., Patton, 2002), yet by definition they include a component of probability sampling (stratified, random). We will not discuss these techniques here since they are widely described elsewhere.

Sequential and parallel mixed methods sampling follow from the design types described above. Sequential mixed methods sampling involves the selection of units of analysis for a study through the sequential use of probability and purposive sampling strategies (QUAN → QUAL) or vice versa (QUAL → QUAN). Parallel mixed methods sampling involves the selection of units of analysis for a study through the parallel, or simultaneous, use of both probability and purposive sampling strategies. One type of sampling procedure does not set the stage for the other in parallel mixed methods sampling studies; instead, both probability and purposive sampling procedures are used simultaneously.

Multilevel mixed methods sampling is a general sampling strategy in which probability and purposive sampling techniques are used at different levels (e.g., student, class, school, district) (Tashakkori & Teddlie, 2003b, p. 712). This sampling strategy is common in contexts or settings in which different units of analysis are "nested" within one another, such as schools, hospitals, and various bureaucracies (Collins et al., 2007).

In sequential mixed methods sampling, the results from the first strand typically inform the methods (e.g., sample, instrumentation) employed in the second strand. In many QUAN → QUAL studies, the qualitative strand uses a subsample of the quantitative sample. One example of this comes from the work of Hancock, Calnan, and Manley (1999), in a study of perceptions and experiences of residents concerning private/public dental service in the United Kingdom. In the quantitative portion of the study, the researchers conducted a postal survey that involved both cluster and random sampling: (1) the researchers selected 13 wards out of 365 in a county in southern England using cluster sampling, and (2) they randomly selected one out of every 28 residents in those wards resulting in an accessible population of 2,747 individuals, from which they received 1,506 responses (55%).

The questionnaires included five items measuring satisfaction with dental care, which they labeled *the DentSat scores*. The researchers next selected their sample for the qualitative strand of the study using intensity and homogeneous sampling: (1) 20 individuals were selected who had high DentSat scores (upper 10% of scores) through intensity sampling; (2) 20 individuals were selected who had low DentSat scores (lower 10% of scores) through intensity sampling; and (3) 10 individuals were selected who had not received dental care in the past 5 years, but also did not have full dentures, using homogeneous sampling. This type of sampling is often used in mixed methods designs that involve extreme groups analysis. A good example of this sampling and data analysis (called *Group-Case Method* or GCM) may be found in Teddlie, Tashakkori, and Johnson (2008).

Parasnis, Samar, and Fischer's (2005) study provides an example of parallel mixed methods sampling. Their study was conducted on a college campus where there were a large number of deaf students (around 1,200). Selected students were sent surveys that included closed-ended and open-ended items; therefore, data for the quantitative and qualitative strands were gathered simultaneously. Data analysis from each strand informed the analysis of the other.

The mixed methods sampling procedure included both purposive and probability sampling techniques. First, all the individuals in the sample were deaf college students (homogeneous sampling). The research team had separate sampling procedures for selecting racial/ethnic minority deaf students and for selecting Caucasian deaf students. There were a relatively large number of Caucasian deaf students on campus, and a randomly selected number of them were sent surveys through regular mail and e-mail. Since there were a much smaller number of racial/ethnic minority deaf students, the purposive sampling technique known as complete collection was used (Teddlie & Yu, 2007). In this technique, all members of a population of interest are selected that meet some special criterion. Altogether, the research team distributed 500 surveys and received a total of 189 responses, 32 of which were eliminated because they were foreign students. Of the remaining 157 respondents, 81 were from racial/ethnic minority groups (African Americans, Asians, Hispanics), and 76 were Caucasians. The combination of purposive and probability sampling techniques in this parallel mixed methods study yielded a sample that allowed interesting comparisons between the two racial subgroups on a variety of issues, such as their perception of the social psychological climate on campus.

Multilevel mixed methods sampling techniques are common in educational systems or other organizations in which different units of analysis are "nested within one another." In studies of these nested organizations, researchers are often interested in answering questions related to two or more levels or units of analysis. Multilevel sampling examples from educational settings may involve up to five to six levels. An example of a multilevel sampling strategy is the *Prospects* study of Title I (Puma et al., 1997), which was a federally funded program for high-poverty schools that target children with low achievement. The complex multilevel sampling strategy for this congressionally mandated study involved sampling at six different levels ranging from region of country to the individual student level (25,000-plus students). The researchers in this study gathered a mix of quantitative and qualitative data across the six levels of sampling over a 5-year time period that involved three student cohorts.

The sampling strategies that were employed across the six levels of the *Prospects* study include complete collection, stratified sampling, stratified purposive sampling, intensity sampling, homogeneous sampling, and sampling politically important cases. Interesting details on the complex sampling strategy used in *Prospects* can be found in the original research syntheses (e.g., Puma et al., 1997) and later syntheses (e.g., Kemper, Stringfield, & Teddlie, 2003). Other examples of mixed methods sampling may also be found in Teddlie et al.'s (2008) discussion of participatory mixed methods studies.

Data Collection in Mixed Methods Research

Mixed data collection includes the gathering of both quantitative and qualitative data in a single study using either (1) within-strategy mixed data collection involving the gathering of both qualitative and quantitative data using the same data collection strategy (e.g., observation) or (2) between-strategies mixed data collection that involves the gathering of both qualitative and quantitative data using more than one data collection strategy (e.g., observation and interviews). We describe basic data collection techniques in this section and how they can be combined in mixed methods studies. These techniques include observations, interviews, focus group interviews, questionnaires, unobtrusive measures, and tests (e.g., Johnson & Turner, 2003; Teddlie & Tashakkori, in press). Due to space limitation, only a sample of the possible combinations of mixed methods data collection is presented here.

Within-Strategies Mixed Methods Data Collection

Observation is the oldest data collection technique in the social and behavioral sciences. It may be defined as the recording of units of interaction occurring in a defined social setting based on visual examination/inspection of that setting (e.g., Denzin, 1989; Flick, 1998). Observations may be recorded in two manners: (1) they may be recorded as a "running narrative," which means that the observer takes extensive field notes recording as many of the interactions as possible in written form, or (2) they may be recorded using instruments with a prespecified structured format, including numeric measurement scales.

The first type of observation protocol is known as an *unstructured (open-ended) observation instrument* and may simply involve the use of (1) blank sheets of paper or scripting forms or (2) a series of prompts, which guide the observer in terms of what to watch for and how to record it. Narrative data result from the collection of information from these open-ended instruments.

The second type of observation protocol is known as a *structured (closed-ended) observation instrument* and consists of items accompanied by different predesigned or precoded responses. These standardized coding instruments present the observer with a series of behavioral indicators, and the observer is supposed to select the most appropriate precoded response to describe those behaviors. Numeric data result from the coding of these instruments.

Many mixed methods studies employ both structured and unstructured observational instruments, either sequentially or in a parallel manner. The area of study known as teacher effectiveness research has been in existence for a number of years and has generated numerous instruments designed to assess how effective teachers are in elementary/secondary classrooms (e.g., Brophy & Good, 1986; Teddlie & Meza, 1999). These instruments range from the unstructured, qualitatively oriented end of the continuum to the structured, quantitatively oriented end.

Interviews are also capable of generating both qualitative and quantitative data in a mixed methods study. An interview is a research strategy that involves one person (the interviewer) asking questions of another person (the interviewee). The questions may be open-ended (generating qualitative data) or closed-ended (generating quantitative data) or both (generating mixed methods data). This latter type also includes funnel-sequenced interviews that start from general questions/topics and are gradually directed to focus on more specific emerging or preplanned issues (Tashakkori & Teddlie, 1998). Interviews are a powerful method of data collection, because they entail one-to-one interaction between the researcher and the individuals he or she is studying.

Open-ended interviews are usually nondirective and very general ("tell me about your school"). Structured interviews are usually closed-ended ("which one of the following would you say describes the food in the school cafeteria" *very good*, *good*, *bad*, or *very bad*?). Open-ended interviews generate in-depth information, which may lead to reconceptualization of the issues under study. Open-ended interviews are often used in the initial research on topics about which little is known. This is very important in research in areas involving cross-cultural and multicultural issues, when the psychological repertoire of a population is not known.

Some interview studies employ both open-ended and closed-ended formats. For example, Brannen (2005) presented an example of mixed methods interview research, including an explicit rationale for including both quantitative and qualitative items on her interview protocol. The research was longitudinal in design and was conducted during a 6-year period in the 1980s (Brannen & Moss, 1991). The topic of the study was mothers and their return to work after maternity leave. As the study evolved, the researchers became more interested in the qualitative nature of the mothers' experiences. The original highly structured interview protocol changed accordingly as described by Brannen (2005):

> The result was an interview schedule which combined structured questions (the responses to which were categorized according to predefined codes) with open-ended questions giving scope for probing (responses were transcribed and analyzed qualitatively). We remained committed to collecting the structured data originally promised but required the interviewers to collect such data while seeming to adopt a flexible, in-depth mode of interviewing. (p. 179)

The resulting data generated by the open-ended and closed-ended items represented "the experiences of the mothers in all their complexity and ambiguity" (Brannen, 2005, p. 180).

Focus group interviews are another source of data for mixed methods studies (see Stewart, Shamdasani, & Rook, Chapter 18, this volume). While primarily considered a group interviewing technique, observations of shifts of opinion among group members are considered a major part of focus group data collection and analysis. Krueger and Casey (2000) defined a focus group study as "a carefully planned series of discussions designed to obtain perceptions on a defined area of interest in a permissive, non-threatening environment" (p. 5).

Most researchers writing about focus groups consider them to be a qualitative technique, since (1) they are considered to be a combination of interviewing and observation, both of which are presented as qualitative data collection techniques in many texts and (2) focus group questions are (typically) open-ended, thereby generating narrative data. However, focus group studies often yield mixed data. This outcome from focus groups is more common than described in the traditional focus group literature and is gaining popularity among researchers.

An example of a study employing focus groups to collect mixed methods data was reported by Henwood and Pidgeon (2001) in the environmental psychology literature. In this study, researchers conducted "community" focus groups in Wales in which the topic of conversation was the importance, significance, and value of trees to people. The focus group had a seven-step protocol, which involved open discussions, exercises, and individual rankings of eight issues both for the participants individually and for the country of Wales. While the data were primarily QUAL, the rankings provided interesting information on the importance that participants placed on issues related to the value of trees in Wales from wildlife habitat to commercial-economic.

Questionnaires also may yield both qualitative and quantitative data. When questionnaires are used in a study, the researcher is employing a research strategy in which participants self-report their attitudes, beliefs, and feelings toward some topic. Questionnaire studies have traditionally involved paper-and-pencil methods for data collection, but personal computers have led to the Internet becoming a popular venue for data collection. The items in a questionnaire may be closed-ended, open-ended, or both (also see Fowler & Cosenza, Chapter 12, this volume).

A good example of the use of questionnaires in mixed methods research comes from the Parasnis et al. (2005) study of deaf students described earlier in the sampling section of this chapter. Selected students were sent questionnaires that included 32 closed-ended (5-point Likert-type scales) and three open-ended items.

The two types of data were gathered and analyzed simultaneously, and the analysis of data from each strand informed the analysis of the other. The closed-ended items addressed a variety of issues, including comparisons between the two campuses where the information was gathered, the advantages of diversity, the institutional commitment to diversity, the inclusion of diversity in the curriculum, and so forth. The open-ended items asked the following questions:

- Has anything happened to make you feel *comfortable* on the NTID/RIT (National Technical Institute for the Deaf/Rochester Institute of Technology) campus (related to race relations and diversity)? Please describe what happened.
- Has anything happened to make you feel *uncomfortable* on the NTID/RIT campus (related to race relations and diversity)? Please describe what happened.
- Do you have any comments about the experiences of deaf ethnic minority students on this campus? Please describe. (Parasnis et al., 2005, p. 54)

Unobtrusive measures are research techniques that allow investigators to examine aspects of a social phenomenon without interfering with or changing that phenomenon (e.g., Lee, 2000; Webb, Campbell, Schwartz, & Sechrest, 1966, 2000). Unobtrusive measures are considered to be nonreactive, because they are hidden within the context of the social setting under study; therefore, individuals being observed will not react to their being observed.

A typology of unobtrusive measures includes a wide variety of techniques, organized around two categories: artifacts and covert or nonreactive observations. Due to space limitations, we only consider artifacts in this chapter. Artifacts include archival records and physical trace evidence archival records include written public and private records, archived databases from research studies conducted previously, and information stored in various nonwritten formats (e.g., audiotapes, photographs, videotapes).

Physical trace evidence includes accretion and erosion measures, which provide the physical evidence for "crime scene investigations" within the social sciences. *Accretion measures* are concerned with the deposit of materials, while *erosion measures* consider the selective wear on materials.

A recent example of a study using mixed methods data generated from unobtrusive measures comes from a study of the potential impact of Hurricane Katrina on future housing patterns in New Orleans (Logan, 2006). The unobtrusive quantitative data came from the numeric census data in the affected neighborhoods broken down by numbers of black and white residents, which were initially published in 2000 and then updated periodically. The unobtrusive qualitative data came from a variety of sources, including (1) categorical estimates of devastation (e.g., Federal Emergency Management Agency [FEMA] estimates of degree of damage to dwellings), (2) maps of New Orleans illustrating the degree of the flooding, and (3) photographs taken by the researcher. This research example provides strong evidence for the power of unobtrusive measures to generate socially meaningful research. Obviously, this type of data may also be quantitized and reanalyzed to provide a better understanding of the phenomenon under investigation.

Between-Strategies Mixed Methods Data Collection

Between-strategies mixed methods data collection refers to research in which qualitative and quantitative data are gathered using multiple modes of collection (e.g., interview, observation, focus group). This use of different data collection strategies has also been called intermethod mixing (Johnson & Turner, 2003) or data triangulation/methodological triangulation (e.g., Denzin, 1989; Patton, 2002). Between-strategies mixed methods data collection may be associated with any of the sequential or parallel research designs presented earlier. The following section includes a few examples from the many types of between-strategies mixed methods data collection techniques.

Using structured (quantitative) questionnaires together with open-ended (qualitative) items is a popular technique in the literature. This combination allows for the strengths of each strategy to be combined in a complementary manner with the strengths of the other (e.g., Johnson & Turner, 2003). Both strategies are good for measuring attitudes and other constructs of interest. Quantitative questionnaires can be used to inexpensively generate large numbers of responses that produce information across a broad range of topics. Data gathered using qualitative interviews are based on a relatively small number of participants, who generate in-depth information in response to queries and probes from the interview protocol about particular areas of interest.

An example comes from a study of child welfare administrators' responses to increased demands for services provided by their agencies (Regehr, Chau, Leslie, & Howe, 2001). These researchers first administered a set of questionnaires, including a measure of perceived stress, to a sample of agency supervisors/managers and then conducted semistructured interviews that focused on stress on the job. Data collection included 47 completed questionnaires and 8 interviews. Results indicated that about one half of the managers/supervisors fell in the high or severe range of post-traumatic symptoms on the scale measuring stress. A thematic summary of the data gathered from the eight follow-up interviews indicated that new child welfare reform regulations resulted in increased workload, increased accountability, and the introduction of new staff. These factors led to stress on the administrators, which then led to coping mechanisms and support, which then led to either resilience or "giving up." The sequential quantitative and qualitative data from this study were highly complementary, since one component quantitatively confirmed the high levels of stress among the administrators and the other qualitatively interpreted the effects of that stress on the lives of those administrators.

Structured quantitative observation together with qualitative interviews is another commonly occurring mixed data collection strategy. For example, in educational research, researchers observe teachers using closed-ended protocols, such as the Virgilio Teacher Behavior Inventory (Teddlie, Virgilio, & Oescher, 1990). This protocol presents an observer with a series of 38 behavioral indicators (e.g., the teacher uses time during class transitions effectively), and the observer selects the most appropriate precoded response on 5-point Likert-type scales to describe those behaviors. Numeric data results from the coding of these instruments, which describe teachers' behavior in great detail.

Researchers then interview the same teachers whom they observed, asking questions about the topic of interest, which may evolve somewhat on the basis of the quantitative results. For instance, if the average scores for the teachers at a school were low on measures of classroom management, then researchers might ask open-ended questions regarding the teachers' perceptions of orderliness in their classrooms, why the disorder was occurring, and what could be done to improve classroom management. The combination of quantitative and qualitative data resulting from this research strategy is very informative, especially for educators wanting to improve classroom teaching practices.

Another mixed methods data collection strategy is to use focus groups together with structured or unstructured interviews. The Nieto, Mendez, and Carrasquilla (1999) study of attitudes and practices toward malaria control in Colombia is an example of this combination:

- The study included five focus groups that were formed to discuss a wide range of issues related to generic health problems and malaria in particular.
- The focus group results were subsequently employed by the investigators to construct a questionnaire with closed-ended items.
- Interviews were conducted to determine a baseline regarding the knowledge and practices of the general population based on a probability sample of 1,380 households.

The findings from the qualitative and quantitative components were congruent, as noted by Nieto et al. (1999): "The information obtained by the two methods was comparable on knowledge of symptoms, causes and ways of malaria transmission, and prevention practices like the use of bednets or provision of health services" (p. 608).

Using quantitative unobtrusive measures together with qualitative interviews is another commonly occurring mixed methods combination, especially in the evaluation literature. In these studies, researchers mix quantitative information that they have gathered from unobtrusive data sources (e.g., archival records, physical trace data) together with qualitative interview data from participants. In sequential studies, the qualitative interview questions may be aimed at trying to understand the results from the quantitative data generated by the unobtrusive measures.

An example of this combination of strategies comes from Detlor (2003) writing in the information systems literature. His research questions concerned how individuals working in organizations search and use information from Internet-based information systems. There were two primary sources of information in this study: Web tracking of participants' Internet use, followed by one-on-one interviews with the participants. Web tracking "consisted of the use of history files and custom-developed software installed on participants' computers that ran transparently whenever a participant's web browser was used during a two-week monitoring period" (Detlor, 2003, p. 123).

The tracking software recorded a large amount of unobtrusive data on the participants' Web actions, including the sites visited and the frequency of Web page visits made by the participants. Log tables indicating extended or frequent visits to particular Web sites were used to pinpoint "significant episodes" of information seeking.

One-on-one qualitative interviews were used to discuss these "significant episodes" in enough detail so that the researcher could understand why the Internet-based information systems were used and the degree to which the participants were successful in resolving their information needs. The mixed methods data collected allowed the researcher to describe an iterative cycle of "information needs-seeking-use activities" that the participants employed in their Internet environment.

The examples in this section of the chapter only present a fraction of the numerous ways that mixed data collection occurs. Numerous other examples may be found, especially in literature from the applied social and behavioral sciences. The reader could also browse the pages of the *Journal of Mixed Methods Research* for examples across disciplines.

Data Analysis in Mixed Methods Research

Using a combination of qualitative and quantitative data collections strategies, as described above, provides the mixed methods researcher with rich data sets including both narrative and numerical data. There are three obvious steps in the analysis of such data: (1) narrative data are analyzed using qualitative thematic data analysis techniques, (2) numeric data are analyzed statistically (descriptive or inferential), and (3) some of the results may be converted from one type to another and reanalyzed using a new approach.

Analyses Strategies for Qualitative Data

Narrative data are usually prepared for analysis by converting raw material (e.g., field notes, documents, audiotapes) into partially processed data (e.g., write-ups, transcripts), which are then coded and subjected to a particular analysis scheme (e.g., Huberman & Miles, 1994). These analysis schemes may be differentiated by whether the themes or categories emerged during the analysis (emergent themes) or were established a priori (predetermined themes). While inductive logic and grounded theory are essential components of qualitative data analysis, there are research areas where predetermined themes are viable due to the large amount of previous research and accumulated knowledge.

The essence of qualitative data analysis of any type is the development of a typology of categories or themes that summarize a mass of narrative data. While several different types of qualitative data analysis strategies exist, we will briefly focus on only three in this overview due to space limitations: latent content analysis, constant comparative analysis, and the developmental research sequence.

Latent Content Analysis. The distinction between the manifest and latent content of a document refers to the difference between the surface meaning of a text and the underlying meaning of that narrative. For example, one could count the number of violent acts (defined *a priori*) that occur during a television program and make conclusions concerning the degree of manifest violence that was demonstrated in the program. To truly understand the underlying latent content of the violence within a specific program, however, the "context" (e.g., Manning & Cullum-Swan, 1994) within which the program occurred would have to be analyzed. In this case, that context would be the narrative line or plot of the program. A television program with several violent scenes, yet with an underlying theme of trust or concern among the characters, might generate a latent content analysis very different from its manifest content analysis.

Constant Comparative Analysis. The constant comparative analytical scheme was first developed by Glaser and Strauss (1967) and then refined by Lincoln and Guba (1985). This analytical scheme involves two general processes: (1) *Unitizing*—breaking the text into units of information that will serve as the basis for defining categories and (2) *Categorizing*—bringing together into provisional categories those units that relate to the same content, devising rules that describe category properties, and rendering each category set internally consistent and the entire set mutually exclusive. The entire categorizing process involves 10 steps, some of which are iterative (Lincoln & Guba, 1985, pp. 347–351). The constant comparative analysis constitutes the first step in the process of grounded theory, open coding, which is then followed by axial and selective coding (e.g., Strauss & Corbin, 1998).

Developmental Research Sequence. The developmental research sequence of James Spradley is one of the most complex schemes for determining the themes associated with what he called a "cultural scene." The 12-step process for analyzing both interview (Spradley, 1979) and observational data (Spradley, 1980) involves three stages of data gathering (using descriptive, structural, and contrast questions) and three stages of data analysis (domain, taxonomic, componential). Each successive stage of data gathering and analysis results in a more comprehensive understanding of the phenomenon under study (see Teddlie & Tashakkori, in press, for a further summary).

Spradley (1979, p. 157) explicitly defined two of the major principles used in qualitative data analysis: the similarity principle and the contrast principle. The similarity principle states that the meaning of a symbol can be discovered by finding out how it is similar to other symbols. The contrast principle states that the meaning of a symbol can be discovered by finding out how it is different from other symbols.

Analyses Strategies for Quantitative Data

Analysis of numeric data may be in two broad forms. One is to summarize the data into meaningful forms/indicators that are easy to understand, compare, and communicate. These indicators are called descriptive statistics. The second general category consists of techniques for estimating population parameters, testing hypotheses, or making predictions. These techniques are called inferential statistics. All tests of statistical significance are examples of this type of quantitative data analysis.

Descriptive methods include presentations of results through simple statistics and graphic displays. The most commonly used methods of descriptive data analysis and presentation are (a) measures of central tendency, (b) measures of relative standing, and (c) measures of association/relationship between variables.

Descriptive statistics are not sufficient for estimation and testing hypotheses. Data analysis methods for testing hypotheses are based on estimations of how much error is involved in obtaining a difference between groups, or a relationship between variables. Inferential statistical analysis, involving significance tests, provides information regarding the possibility that the results happened "just by chance and random error" versus their occurrence due to some fundamentally true relationship that exists between variables. If the results (e.g., differences between means) are statistically significant, then the researcher concludes that they did not occur solely by chance. The basic assumption in such hypothesis testing is that any apparent relationship between variables (or difference between groups) might, in fact, be due to random fluctuations in measurement of the variables or in the individuals who are observed. Inferential statistics are methods of estimating the degree of such chance variation. In addition, these methods of data analysis provide information regarding the magnitude of the effect or the relationship.

Mixed Methods Data Analysis

There are several ways that quantitative and qualitative data analyses are used in mixed methods research. In many mixed methods studies, each type of data (narrative or numerical) are analyzed separately, using the techniques summarized above. In these studies, integration (mixing) occurs *after* the results of the two strands are interpreted in the meta-analysis phase of the study. Therefore, the quantitative and qualitative data and/or data analyses are not mixed.

On the other hand, in a smaller number of mixed methods studies, one type of data (e.g., narrative) is transformed to another (e.g., numbers) after it was initially analyzed, and then the transformed data are analyzed again, using appropriate techniques. Two aspects of this type of transformation are (a) converting qualitative information into numerical codes that can be statistically analyzed and (b) converting quantitative data into narratives that can be analyzed qualitatively. As indicated in the design section earlier in this chapter, we refer to the first type of transformation method as *quantitizing techniques* and the transformed data as *quantitized data.* The second method is referred to as *qualitizing techniques* and the transformed data as *qualitized data.*

Such transformation and reanalysis can provide additional understanding of the phenomenon under investigation by (a) confirming/expanding the inferences derived from one method of data analysis (e.g., qualitative) through a secondary analysis of the *same* data with a different approach (e.g., quantitative), (b) sequentially using the results obtained through one approach (e.g., classification of individuals into groups through qualitative analysis) as a starting point for the analysis of *other* data with the alternative approach (e.g., *statistically* compare the groups that were identified by qualitative observations), or (c) using the results of one analysis approach (e.g., initial interviews and/or content analysis of texts) as a starting point for designing further steps (e.g., instrument development) or collecting new data using another approach. For example, many survey questionnaires are constructed after an initial qualitative study in the appropriate population.

Examples of Data Analysis in Mixed Methods Studies

The following section presents eight examples of different types of data analysis using mixed methods. Mixed methods data analysis is an area that requires more development at this time (e.g., Onwuegbuzie & Teddlie, 2003), because very few writers have provided typologies of mixed methods data analysis techniques together with examples.

1. *Parallel mixed analysis,* also known as triangulation of data sources, parallel analysis of qualitative and quantitative data is probably the most widely used mixed data analysis strategy in the social and behavioral sciences. Many investigators collect a combination of qualitative *and* quantitative data in their studies. In laboratory experiments, the participants are interviewed at the end (postexperimental interview) to determine the type of interpretations and perceptions they had that could have affected their responses. Observation of the participants during the experiment is also a source of data in experiments. While the obtained quantitative data are analyzed through statistical procedures, the interview and observational data are (or can be) analyzed through content analysis.

In survey research, there often is a combination of open-ended and closed-ended response options. These closed-ended responses are analyzed statistically, and the open-ended responses are content analyzed. In highly unstructured *qualitative surveys* and *field studies* (e.g., Babbie, 2003), although the bulk of data is qualitative and is analyzed accordingly, there are variables that are (or can be) analyzed quantitatively. The simplest form of such quantitative analysis is to calculate descriptive statistics for the appropriate variables (see, e.g., Gall, Gall, & Borg, 2006).

Similar types of parallel data collection/analysis might be found in most other types of research. It is a hallmark of much educational research in which quantitative data (e.g., tests, formal measures of teachers' classroom behaviors) are collected and analyzed concurrently with qualitative data (e.g., informal school observations, principal and faculty interviews).

2. *Analysis of the same qualitative data with two methods* involves the transformation of the qualitative data to a numerical form. Earlier, we referred to this transformation as *quantitizing* the qualitative data. *Quantitizing* might include a simple frequency count of certain themes, responses, behaviors, or events. On the other hand, it may consist of more complex ratings of the strength or intensity of these events, behaviors, or expressions. Depending on the type of transformation, different quantitative techniques might be used for their analysis.

3. *Analysis of the same quantitative data with two methods* involves the transformation of the quantitative data to qualitative categories or narrative. Earlier, we referred to this transformation as *qualitizing* the quantitative data. An example of such transformation is found in the Hooper (1994) study regarding the effects of language-art tasks in multicultural classrooms. In that study, children's responses to quantitative items on an interview form were analyzed both quantitatively and qualitatively, with the latter resulting in categories indicating students' interest level.

4. *Forming groups of people/settings* on the basis of qualitative data/observations, and then comparing the groups on quantitative data (sequential QUAL → QUAN analysis). Following Caracelli and Greene (1993), we call this *typology development.* In typology development, individuals are first classified into different types. These groups are then statistically compared with each other on *other* available quantitative (or quantitized) data.

For example, teachers might be categorized into effective and ineffective groups on the basis of field notes taken during observations. The two groups of teachers might then be compared on quantitative variables/measures, such as their responses to survey instruments or their students' performance on tests. Comparisons might be performed through univariate or multivariate analysis of variance or covariance, discriminant function analysis, or other statistical techniques. The result of the discriminant function analysis, for example, is the identification of variables that "discriminate" the two groups, along with some statistical indicators that show which of these variables discriminates the groups from each other the best.

5. *Forming groups of attributes/themes* through content analysis followed by confirmatory statistical analysis of quantitative data that are collected (or are available). As an example, constant comparative analysis is first used to construct emergent themes from the qualitative data. Categories of themes, variables, or situations that "fit together" (and are distinctly different from other categories) are formed (*construct identification*). In the next step, the available (or subsequently collected) quantitative data are statistically analyzed to either confirm or expand the inferences obtained from the initial qualitative analysis (*construct validation*).

An example is the classification of teachers' statements (obtained from focus groups) into themes that represent different aspects of a "good principal." The emergent themes or categories are indicators of subconstructs that are parts of the general construct of "principal effectiveness." These categories are formed on the basis of similarities (and/or differences) between teachers' perceptions and beliefs. Survey instruments may then be constructed that include these groups of themes and are administered to a group of teachers. The obtained quantitative data may then be factor analyzed to determine the degree of agreement with the initial qualitative categories.

6. *Forming groups of people/settings* on the initial basis of quantitative data and then comparing the groups on subsequently collected or available qualitative data (i.e., sequential QUAN → QUAL) is similar to the previously discussed sequential QUAL → QUAN analysis. A widely used example is the qualitative follow-up of individuals/units that were initially identified on the basis of their *residual scores* from multiple regression, or *covariate-adjusted scores* from analysis of covariance. Detailed qualitative data are then collected on these individuals/units in a search for possible factors that led to their initial high (or low) quantitative scores. The qualitative data are either analyzed through content analysis (or they could be converted to quantitative data for further statistical analysis). An example of this sequence of analyses involves the initial classification of schools into effective and ineffective categories on the basis of standardized tests using regression residuals

(e.g., Kochan, Tashakkori, & Teddlie, 1996). These two types of schools were then observed and compared with each other to explore possible differences between them on other dimensions such as school climate.

7. *Forming categories of attributes/themes* through quantitative analysis, and then confirming these categories with the qualitative analysis of other data, is similar to the *construct identification* and *construct validation* procedures described previously. In this strategy, the objective is to first identify the components of a construct (subconstructs) through factor analysis of quantitative data and then to collect qualitative data to validate the categories, or to expand on the information that is available regarding these subconstructs. An example of such a type of mixed data analysis might involve the initial classification of dimensions of teachers' perceptions of school climate through factor analysis of survey data completed by a sample of faculties. Observational and/or other types of data (e.g., focus group interviews) might then be used to confirm the existence of such dimensions and/or to explore the degree to which these different dimensions are present in everyday interactions.

Caracelli and Greene (1993) discuss another application of this type of analysis. Unlike the above examples, in this application the objective is *not* to confirm or expand the results of construct validation efforts. Instead, the objective is to develop an initial framework for the qualitative/categorical analysis that follows as the next step. For example, factor analytic results might be used as a starting point for the *constant comparative analysis* defined earlier in this chapter. The categories of events/observations that are obtained through factor analysis might then be used for coding the initial qualitative data in the subsequent constant comparative analysis.

8. *Using inherently mixed data analysis techniques.* Inherently mixed data analysis techniques are those that provide two types of outputs: qualitative and quantitative. Social network analysis is an example of one such technique. In social network analysis, the investigator obtains both graphic (qualitative) "snapshots" of communication networks *and* numeric indicators of various aspects of communication patterns. Another example is the output from computerized data analysis packages for qualitative research, such as *Atlas-ti* and others. These programs usually provide two types of results, one consisting of qualitative themes and the other, numeric indicators that may be analyzed statistically.

Making Inferences in Mixed Methods Research

Inferences are conclusions and interpretations that are made on the basis of collected data in a study. As such, they must be distinguished from the data that produced them. Unfortunately, few scholars have tried to distinguish between these two. Among those who have recognized a clear need for distinguishing inferences from the evidence they are based on are Tashakkori and Teddlie (1998, 2003a), Newman and Benz (1998), and King, Keohane, and Verba (1994).

The term *inference* has been used to denote both a process and an outcome (see Miller, 2003, for a full discussion). As a process, making inferences consists of a set of steps that a researcher follows to create meaning out of a relatively large amount of collected information. As an outcome, inference is a conclusion made on the basis of obtained data. Such a conclusion may or may not be acceptable to other scholars and is subject to evaluation by the community of scholars and/or consumers of research. For example, an inference may be evaluated in terms of the degree to which it is consistent with the theories and the state of knowledge. Or, on the other hand, one might ask how good the conclusion is in terms of its relevance and usefulness to policymakers.

Making inferences in mixed methods involves integrating (comparing, contrasting, incorporating, etc.) the findings of the qualitative and quantitative strands of a study. Such integration is not the same in parallel and in sequential or conversion designs. In parallel mixed methods designs, two separate but related answers to the research questions are obtained, one from each strand of the study. The investigator must make meta-inferences by integrating the two sets of inferences that are gleaned from the two strands of the study. As we will discuss below, integration and its adequacy is directly related to the goal of the study and the purpose of using a mixed methods design.

In sequential and conversion designs, one strand emerges either as a response to the inferences of the previous one or provides an opportunity to conduct the next strand. For example, the conclusions gleaned from one strand might be controversial, incomplete, or highly unexpected. This leads to the need to conduct a second strand, in order to obtain more in-depth understanding of such findings. Alternatively, one strand might provide an opportunity for the next one by providing a framework for sampling (see examples of typology formation discussed above) or lead to procedures for data collection (e.g., instrument development in one strand, to be used in data collection for the next). Although there is a temporal sequence of making inferences, and the two sets of inferences might seem independent, in a mixed methods design (as compared with quasi-mixed designs), the inferences of each of the two (or more) strands must be incorporated into a meta-inference.

Quality Audits in Mixed Methods Research

There is a long-standing controversy over the issue of quality using the term *validity* and its types. The problem is in the myriad terms used in qualitative and quantitative research, all referring to some aspect of quality in a research project, be it the quality of collected information, the research question, the methods of data analysis, or the utilization of findings for policy. Investigators have used research validity, design validity, legitimacy, trustworthiness, credibility, and their numerous (and often inconsistent or conflicting) subtypes in qualitative and quantitative research. In the following sections, we discuss some of the quality issues in mixed methods research. We should mention that this coverage is not exhaustive (i.e., it does not cover all aspects/components of a research process).

Quality of Questions in Mixed Methods Research

In an earlier section of this chapter, we discussed the importance of keeping the purpose of mixing and research questions in mind when assessing the quality of mixed methods research. This is an issue of the match between purposes, research questions, and the final inferences. Obviously, the quality of the research questions has a direct affect on the overall quality of mixed methods research. Little has been written about the quality of research questions in qualitative, quantitative, and mixed methods research. A good mixed methods question incorporates two subquestions (qualitative and quantitative). The distinction between the qualitative and quantitative types of questions is arbitrary, since all research questions are on a continuum between these two (Teddlie, Tashakkori, & Johnson, 2008). Research questions might also be differentiated on other dimensions, such as exploratory-explanatory, holistic-molecular, or the degree of value orientation. These differentiations do not always match the quantitative-qualitative distinction. For example, exploratory questions are found in both qualitative and quantitative research; therefore, there is not always a match between confirmatory-exploratory and quantitative-qualitative.

Quality of Data in Mixed Methods Research

It is obvious that high-quality data are a necessary (but not sufficient) requirement for high-quality answers to research questions. The famous GIGO (garbage in, garbage out!) principle in research design is a simple expression of such a necessity. Data quality in mixed methods directly depends on data quality issues in the qualitative and quantitative strands of a mixed methods study. With one exception, *the quality of data in mixed methods is determined by standards of quality in the qualitative and quantitative strands.* In other words, if the qualitative and quantitative data are credible, then the mixed methods study has data quality. There is one exception to the italicized comment in the previous sentence: the quality of qualitizing or quantitizing efforts in a conversion mixed design (see above). In such designs, the transformed data are analyzed again, using an alternative approach. For example, already content-analyzed qualitative data are quantitized and are analyzed again, using statistical procedures. The quality of the transformations adds an additional condition, over and above the quality of the initial data coming from a strand of the mixed methods study. This, of course, might be considered an attribute of data analysis techniques (analytic adequacy, see the last section of this chapter).

A problem facing mixed methods researchers is that they have to use two different sets of standards for assessing the quality of their data: one for qualitative and one for quantitative. Quantitative researchers evaluate the quality of their data in terms of validity (whether or not the data represent the constructs they are assumed to capture) and reliability (accuracy of the quality or quantity of the constructs). Although qualitative researchers are also concerned about both concepts (representation and accuracy) in one form or another, they assess the quality of their observations in terms of the degree to which they accurately reconstruct the realities of the participants in the study. Furthermore, issues of data quality are highly entangled in the quality of analysis (i.e., the investigator's reconstructions of relationships and events). *Credibility* is a qualitative term used for both reputational and accuracy quality.

The terms and examples used in this section are associated with the quality of data, while the next section concerns quality of design and inference.

Quality of Design and Inferences in Mixed Methods Research

Despite the increasing utilization of mixed methods in social and behavioral research, there is a dearth of systematic literature on the quality of inferences in such studies. From one point of view, scholars have considered mixed methods as a vehicle for improving the quality of inferences that are potentially obtainable from either the qualitative or quantitative stands of a study. From another point of view, some scholars have expressed concern that mixed methods are potentially susceptible to weak inferences, given the difficulty of implementing two diverse types of designs/procedures for answering the same research question (or two closely related aspects of a single question).

This second point of view questions the feasibility of implementing mixed methods designs with acceptable quality to enable the investigators' strong and credible inferences. For a mixed methods researcher, the crucial stage of the study is to integrate (e.g., compare and contrast, infuse, modify one on the basis of another) the two sets of inferences that are generated within the two strands of the study. Obviously, sound inferences from a study are only possible if there is a strong and appropriate design that is implemented with quality. In such a study, research questions dictate what research design and procedures are needed for answering them. If the procedures are not implemented with quality and rigor, the quality of obtained inferences will be uncertain. In the following section, we discuss these two aspects under the topics of design quality and interpretive rigor.

Design quality refers to the degree to which the investigator has used the most appropriate procedures for answering the research question(s) and implemented them effectively. This is equally applicable to both qualitative and quantitative strands. Some of the questions asked about the quality of the research design and its implementation are

- *Suitability* (also known as translation fidelity, Krathwohl, 2004): Was the method of study appropriate for answering the research question(s)? In other words, were the research questions of the study adequately and appropriately translated into elements of the design (e.g., sampling, data collection) that could potentially answer the research questions? Obviously, different research designs are needed depending on the type of questions and research purposes that any given study has (see Newman, Ridenour, Newman, & DeMarco, 2003).

- *Adequacy/Fidelity:* Were the components of the design (e.g., sampling, data collection) implemented adequately? In experimental designs, implementation fidelity refers to the degree to which experimental procedures were strong enough (and were credible to the participants) to create the expected effect.

- *Within Design Consistency:* Did the components of the design fit together in a seamless and cohesive manner? Inconsistencies might happen if the data collection procedures (e.g., interview, focus group questions) are not compatible with the sampling process (do not match respondents' level of education, or language ability, etc.).

- *Analytic Adequacy:* Are the data analysis techniques appropriate and adequate for answering the research questions?

- *Interpretive Rigor:* It is the degree to which credible interpretations have been made on the basis of obtained results (e.g., Lincoln & Guba, 2000; Tashakkori & Teddlie, 2003b). In order to assess such rigor, and improve the quality of inferences, one has to meet five criteria described in the following section:

- *Interpretive Consistency:* Does each conclusion closely follow the findings? Also, do multiple conclusions based on the same results agree with each other? There are at least two indicators of this. First is the type of inference consistent with the type of evidence. For example, causal inferences that are made on the basis of correlational data in some quantitative research are clearly problematic. Second is the level of intensity that is reported consistent with the magnitude of the events or the effects that were found.

- *Theoretical Consistency* (also known as explanation credibility, Krathwohl, 2004): Is each inference (explanation for the results or for relationships) consistent with current theories and empirical finding of other researchers?

- *Interpretive Agreement:* Would other scholars reach the same conclusions on the basis of the results from the study? If the research approach or purpose places value on the perceptions or interpretations of participants, do the conclusions agree with their interpretations? Both in quantitative and qualitative research, a standard of quality has been the degree to which peers, other scholars, or the scientific community agree with the manner in which conclusions are drawn. In both qualitative and quantitative research, disagreement between scholars is an indication that other plausible interpretations of the same results exist.

- *Interpretive Distinctiveness:* Is each conclusion distinctively different from other plausible conclusions regarding the same results? In other words, is each conclusion clearly different and more defensible than other plausible conclusions that were eliminated by the investigator? In order to meet this condition, the investigator must be clearly able to refute or eliminate the other possible interpretations of the results. Attention to alternative plausible explanations of the results is not unique to quantitative research. Although qualitative researchers are expected to use an "emic" perspective in their interpretations, one of the criticisms of qualitative research has revolved around the gap between the investigator's construction of reality and meaning, and that of their informants. In ethnography, reflective analysis is used as a process of identifying and analyzing one's biases, to make sure the interpretations reflect the "truth" rather than purely emerge as a result of one's personal biases (e.g., Creswell, 2005).

- *Integrative Efficacy:* The degree to which inferences made in each strand of a mixed methods study are effectively integrated into a theoretically consistent meta-inference. In our discussion above, all criteria of quality are applicable both to each strand (qualitative, quantitative) *and* to the meta-inferences that emerge when the inferences of the two or more strands are integrated. Integrative efficacy, in contrast, is unique to meta-inferences in mixed methods (does not apply to qualitative or quantitative strands separately). It addresses the degree to which a mixed methods researcher adequately integrates the findings, conclusions, and policy recommendations gleaned from each of the two strands. A strong mixed methods inference (meta-inference) clearly links the inferences that are made from each strand of the study, evaluates the possible similarities (consistencies) and differences (inconsistencies) across various components, and provides explicit and credible explanations and implications for these variations and similarities. Also, a credible meta-inference clearly demonstrates how the process of linking/integrating the qualitative and quantitative approaches provided a more credible and complete understanding of the phenomenon than otherwise would be made possible in a monoapproach qualitative or quantitative study.

In mixed methods studies, integration does not necessarily mean creating a single understanding on the basis of the results. We are using the term *integration* as a mixed methods term that denotes making meaningful conclusions on the basis of consistent or inconsistent results. The term incorporates elaboration, complementarity, completeness, contrast, comparison and so forth. For mixed methods research, the *consistency* between two sets of inferences derived from qualitative and quantitative strands have been widely considered as an indicator of quality. However, some scholars have also cautioned against a simple interpretation of *inconsistency* (see Erzberger & Prein, 1997; Perlesz & Lindsay, 2003). Obtaining two alternative or complementary meanings is often considered one of the major advantages of mixed methods (see Tashakkori & Teddlie, 2008).

Inconsistency might be a diagnostic tool for detecting possible problems in data collection and analysis, or the inferences derived from the results of one strand or the other. If refocusing does not reveal any problems in the two sets of inferences, then the next step would be to evaluate the degree to which lack of consistency might indicate that the two sets are revealing two different aspects of the same phenomenon (complementarity). Not reaching a plausible explanation for the inconsistency, the next step would be to explore the possibility that one set of inferences provides the conditions for the applicability of the other (for detailed examples, see Perlesz & Lindsay, 2003). If none of these steps provide a meaningful justification for the apparent inconsistency, the inconsistency might be an indicator of the fact that there are two plausible but different answers to the question (i.e., two different but equally plausible realities exist).

Transferability of Inferences in Mixed Methods Research

Transferability is a term that comes from qualitative research and refers to the generalizability of results from that type of research (Lincoln & Guba, 1985). We use the term *transferability* to also include the concept of external validity from the quantitative research literature. Transferability is relative in that any high-quality inference is applicable to *some* condition, context, cultural group, organization, or individuals other than the one studied.

The degree of transferability depends on the similarity between those studied ("sending" conditions, contexts, entities, individuals) and the ones that the findings are being transferred to ("receiving" conditions, contexts, groups, etc.). Determining the degree of similarity is often beyond the scope of the investigator's knowledge and resources. Although it is up to the consumer of research to assess such a degree of similarity, it is necessary for the researcher to facilitate such a decision by providing full description of the study and its context, and to employ a research design that maximizes transferability to other settings.

Although authors often regard sampling adequacy as the main determinant of the degree of transferability, in truth it also highly depends on design quality and interpretive vigor. Inadequate implementation of the design components or inadequate interpretation of the findings would limit the transferability of the inferences (i.e., noncredible inferences do not hold in any context or group).

If a finding is not transferable to *any* other context, phenomenon, or group, it is of little value to scholars and professionals other than the researcher. Therefore, you are strongly encouraged to think of maximizing the possible transferability of your findings by maximizing the representativeness of your (purposive or probability) sample (of people, observations, entities, etc.), and providing rich descriptions of your study (procedures, data collection, etc.), and its context.

Summary

Mixed methods designs are used with increasing frequency across disciplines. Among the reasons for such utilization, researchers and program evaluators point to the necessity of using all possible approaches/methods (qualitative and quantitative) for answering their questions. We presented a brief overview of some of the issues in such utilization and also presented summaries of possible ways for conducting integrated research. Obviously, the main starting point for conducting such research is the purpose and research question, which in turn shapes your ideas about the type of design you might need to reach your objectives. The design you identify as the most appropriate for answering your research questions (e.g., sequential, parallel, conversion, multilevel, and fully integrated) would also shape your sampling and data collection procedures, steps for data analysis, and ultimately your inferences and policy/practice recommendations/decisions. We believe that the most important part of any study is when you make final inferences and make policy/practice recommendations on the basis of your findings. Therefore, we introduced the concept of *inference quality* and *inference transferability* as two categories of audits/assessments about your overall research.

Discussion Questions

1. Briefly summarize three sampling procedures in integrated research.

2. What are the similarities and differences between a sequential and a parallel mixed methods design? Provide an example for each.

3. Explain the reasons why Teddlie and Tashakkori (2006) have found it necessary to distinguish between mixed methods and quasi-mixed-methods research designs.

4. A concern among some researchers is that if mixed methods are used, they might find inconsistency between the findings of the qualitative and quantitative strands. Explain why mixed methods researchers consider inconsistency potentially valuable for understanding the phenomenon under investigation.

5. Explain the reason(s) why the authors of this chapter do not consider classification of integrated research design on the basis of priority (of qualitative and qualitative approaches) useful.

6. Define/explain *inference quality* and *inference transferability*. Why have the authors of this chapter proposed these terms?

Exercises

1. Mixed methods are appropriate for certain research questions but not others (see, e.g., Creswell & Tashakkori, 2007). Generate four or five examples of research questions for which a mixed methods design/approach would make sense. For each, also write at least one question for each strand (qualitative/quantitative).

2. For each question in Exercise 1 above, briefly write a short justification as to why a qualitative or quantitative approach is not enough for answering the research question.

3. Think about the mixed methods questions that you generated above. What mixed methods design is necessary/appropriate for answering each? Write a short description for a possible study that can potentially answer each research question. In your description, include brief sections for sampling design, data collection procedures, and possible data analysis steps.

4. Give an example of a conversion mixed methods design. Why is it potentially more useful than a single quantitative or qualitative project?

5. Describe the steps you will take if you find variation (difference, inconsistency) between the inferences drawn from qualitative and quantitative strands of a mixed methods study.

Note

1. *Quantitizing* (e.g., Miles & Huberman, 1994) and *qualitizing* (e.g., Tashakkori & Teddlie, 1998) are terms that are part of the mixed methodologists' lexicon. They are employed by almost everyone working in the field (e.g., Sandelowski, 2003).

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CHAPTER 10

Organizational Diagnosis

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What Is Organizational Diagnosis and How Is It Used?[1]

Organizational diagnosis is the use of conceptual models and applied research methods to assess an organization's current state and discover ways to solve problems, meet challenges, or enhance performance. When in-house or external consultants, applied researchers, or managers engage in diagnosis, they draw on ideas and techniques from a diverse range of disciplines within behavioral science and related fields—including psychology, sociology, management, and organization studies. Diagnosis helps decision makers and their advisers develop workable proposals for organizational change and improvement. Without careful diagnosis, decision makers may waste effort by failing to attack the root causes of problems (Senge, 1994). Hence, diagnosis can contribute to managerial decision making, just as it can provide a solid foundation for recommendations by organizational and management consultants.

Here is an example of a diagnostic project that I conducted:

> The head of training in a national health maintenance organization (HMO) received a request from the director of one of its member organizations—here called Contemporary Health Facility (CHF)—for an ambitious program that would train CHF employees to undertake a major organizational transformation. The transformation proposed by the director would radically redefine the goals and mission of CHF and alter its patient mix, personnel, size, structure, and its relations with other health care organizations. The director of CHF was worried that his nursing staff and administrative employees would oppose the far-reaching changes he envisioned. Unconvinced that the training program

was justified, the head of training in the HMO reached an agreement with the CHF director to ask an independent consultant to assess the situation. After discussions between the consultant, the head of training, and the top managers at CHF, all parties agreed to broaden the study goals to include assessment of the feasibility of the proposed transformation and the staff's readiness for the change.

Over a period of 3 weeks, the consultant conducted in-depth interviews with CHF's 3 top managers and 7 senior staff members. He also conducted focus-group interviews with 12 lower-level staff members, made site visits, and examined data on CHF's personnel, patient characteristics, and administration. The consultant analyzed and presented these data within the context of a guiding model of preconditions for strategic organizational change. This model drew concepts from research on open systems, organizational politics, and leadership for organizational transformation. The major diagnostic finding was that the transformation was both desirable and feasible; but accomplishing it would be risky and difficult. In his report and oral feedback to the CHF management and the HMO's director of training, the consultant conveyed these conclusions and some of the findings on which they were based. Moreover, the consultant recommended steps that the director of CHF could take to overcome opposition and build support for the proposed transformation of CHF and suggested ways of implementing the transformation. The report also recommended ways to improve organizational climate, enhance staffing procedures, and improve other aspects of organizational effectiveness *with or without* implementing the program to transform CHF.

As the CHF case suggests, diagnosis involves more than just gathering valid data. A successful diagnostic study must provide its clients with data, analyses, and recommendations that are useful as well as valid. To meet these dual standards, the diagnostic practitioner must fill the requirements of three key facets of diagnosis—*process, modeling,* and *methods*—and assure good alignments among all three. After a brief introduction to types of diagnostic studies and a comparison to other forms of applied research, this chapter introduces each of these three facets. Space limits prevent exploration of the many delicate interactions among them. These can best be learned by example—for instance, case studies and descriptions of actual consulting projects—and through mentored experience in conducting a diagnosis.

Types of Diagnostic Studies

Diagnosis can contribute to organization development projects (OD) and to business-oriented change management. OD, which includes action research and planned change, involves systematic applications of behavioral science to the planned development and reinforcement of strategies, structures, and processes that lead to organizational effectiveness (Cummings & Worley, 2001, p. 1; Waclawski & Church, 2002). Business-oriented change projects aim more explicitly than OD at improving a firm's economic performance and its competitive advantage and rely more on techniques drawn from business, engineering, and other technical fields (Beer & Nobria, 2000). Change management consultants can use diagnosis to help clients decide what changes in organizational features are likely to promote desired outcomes, how ready members are for these changes, and how managers can best implement changes and assure their sustainability. Unfortunately, many ambitious change projects that could benefit from careful diagnosis do not make much use of it (Harrison, 2004; Harrison & Shirom, 1999).

In either OD or business-oriented change management, diagnosis can form the core of a free-standing study or serve as an early stage in a consultant-guided change initiative. In free-standing diagnoses, as occurred at CHF, the practitioner contracts with clients about the nature of the study, designs it, gathers and analyzes data, provides written and oral feedback, and makes recommendations. Then the organization's executives are left to decide what actions, if any, to take in response to the diagnostic report. When diagnosis forms a stage in a consultant-guided intervention, the consultants take part in decision making and action planning (Kolb & Frohman, 1970; Waclawski & Church, 2002). Moreover, they may lead or facilitate implementation of steps designed to foster improvements (interventions) and may provide feedback on them.

Members of an organization can also conduct a self-diagnosis without the help of internal or external consultants. To engage in constructive self-diagnosis, members of the diagnostic team require skills in teamwork, data gathering, analysis, and feedback, along with openness to self-analysis and criticism.

Comparisons to Other Types of Applied Organizational Research

Another way of understanding diagnosis is to contrast it to other forms of applied organizational research. Investigations of programs or entire organizations by external agencies or commissions (e.g., Gormley & Weimer, 1999) do not usually involve organizational diagnosis; they do not create client-consultant relations of the sort described above nor do they rely mainly on behavioral science methods and models. In turn, diagnosis does not refer to applied research projects that assess specific programs (e.g., prevention of work accidents) or that help decision makers decide how to allocate funds (e.g., training vs. safety devices) (Freeman, Dynes, Rossi, & Whyte, 1983; Lusthaus et al., 2002; Majchrzak, 1984). These studies usually have a narrower research focus than diagnosis.

Diagnosis has more in common with evaluation research (Rossi, Lipsey, & Freeman, 1999). Like diagnosis, evaluation is practically oriented and may focus on effectiveness. But diagnostic studies usually examine a broader spectrum of indicators of organizational effectiveness than do summative evaluations, which assess program impacts or program efficiency. Diagnostic studies also differ from most formative evaluations, which monitor program implementation. Most diagnostic studies examine a broader range of organizational features, whereas formative evaluations usually concentrate on the extent to which a project was conducted according to plan. An additional difference is that diagnoses are often conducted on more restricted budgets, within shorter time frames, and rely on less extensive forms of data gathering and analysis.

Despite these differences, many of the models used in diagnosis can contribute to strategy assessments and program evaluations (Harrison & Shirom, 1999), and diagnostic practitioners can benefit from the extensive literature on evaluation techniques and processes (e.g., Patton, 1999; Rossi et al. 1999; Wholey, Harty, & Newcomer, 2004). Practitioners of diagnosis can also incorporate concepts and methods from strategic assessments of intraorganizational factors shaping performance and strategic advantage (Duncan, Ginter, & Swayne, 1998; Kaplan & Norton, 1996).

Process

Phases in Diagnosis

To provide genuinely useful findings and recommendations, consultants need to create and maintain cooperative and constructive relations with clients. Moreover, to ensure that diagnosis yields valid and useful results, practitioners of diagnosis must successfully negotiate their relations with other members of the focal organization as their study moves through a set of analytically distinct phases (Nadler, 1977). These phases can overlap in practice, and their sequence may vary. As the following description shows, diagnostic tasks, models, and methods shift within and between phases, as do relations between consultants, clients, and other members of the client organization:

- *Entry:* Clients and consultants explore expectations for the study; client presents problems and challenges; consultant assesses likelihood of cooperation with various types of research and probable receptiveness to feedback; consultant makes a preliminary reconnaissance of organizational problems and strengths.
- *Contracting:* Consultants and clients negotiate and agree on the nature of the diagnosis and client-consultant relations.
- *Study design:* Methods, measurement procedures, sampling, analysis, and administrative procedures are planned.
- *Data gathering:* Data are gathered through interviews, observations, questionnaires, analysis of secondary data, group discussions, and workshops.
- *Analysis:* Consultants analyze the data and summarize findings; consultants (and sometimes clients) interpret them and prepare for feedback.
- *Feedback:* Consultants present findings to clients and other members of the client organization. Feedback may include explicit recommendations or more general findings to stimulate discussion, decision making, and action planning.

Critical Process Issues

The relations that develop between practitioners and members of a client organization can greatly affect the outcomes of an organizational diagnosis, just as they affect other aspects of consultation (Block, 2000; Turner, 1982). Clients and practitioners should try to define their expectations early in the project. Nonetheless, as occurred in the CHF case, they will often need to redefine their relations and objectives during the course of the diagnosis to deal with issues that were neglected during initial contracting or arose subsequently. To manage the consulting relation successfully, practitioners need to handle the following key process issues (Nadler, 1977; Van de Ven & Ferry, 1980, pp. 22–51) in ways that promote cooperation between themselves and members of the client organization:

- *Purpose:* What are the goals of the study, how are they defined, and how can the outcomes of the study be evaluated? What issues, challenges, and problems are to be studied?
- *Design:* How will members of the organization be affected by the study design and methods (e.g., organizational features to be studied, units and individuals included in data gathering, and types of data collection techniques)?
- *Support and cooperation:* Who sponsors and supports the study and what resources will the client organization contribute? What are the attitudes of other members of the organization and of external stakeholders toward the study?
- *Participation:* What role will members of the organization play in planning the study, collecting data, interpreting them, and reacting to them?
- *Feedback:* When, how, and in what format will feedback be given? Who will receive feedback on the study, and what uses will they make of the data?

Modeling

The success of a diagnosis depends greatly on the ways that practitioners handle the analytic tasks of deciding what to study, framing and defining diagnostic problems, choosing criteria for assessing organizational effectiveness, analyzing data to identify conditions that promote or block effectiveness, organizing findings for feedback, and providing feedback. Behavioral science models and broader-orienting metaphors (Morgan, 1996) and frames (Bolman & Deal, 2003) can help practitioners handle these tasks.

Many practitioners use models developed by experienced consultants and applied researchers to guide their investigations (see Harrison, 2005, appendix B; Harrison & Shirom, 1999). These models specify organizational features that have proved critical in the past. Standardized models also help large consulting practices maintain consistency across projects. Unfortunately, work with available models runs the risks of generating a lot of hard-to-interpret data that fail to address challenges and problems that are critical to clients and do not reflect distinctive features of the client organization. To avoid these drawbacks, consultants often tailor standardized models to fit the client organization and its circumstances.

Developing Grounded Models

Another way of addressing these issues is to develop grounded models that emerge during initial study of the organization and focus more directly on client concerns and challenges facing them. For example, in "sharp-image diagnosis," (Harrison & Shirom, 1999), the practitioner uses one or more theoretical frames as orienting devices and then develops a model that specifies the forces affecting the problems or challenges presented by clients. This model also guides feedback.

Figure 10.1 shows the main steps in applying the sharp-image approach to developing a diagnostic model. In the CHF case, the diagnosis drew on two theoretical frames. The first applied open systems concepts to the analysis of strategic organizational change (Tichy, 1983). This frame guided analysis of the core challenge facing CHF—developing an appropriate strategy for revitalizing the organization and helping it cope with external challenges. Second, a political frame (Harrison, 2005, pp. 95–104; Harrison & Shirom, 1999, chap. 5; Tichy, 1983) guided analysis of the ability of CHF's director to mobilize support for the proposed transformation and overcome opposition among staff members. For the feedback stage, elements from both frames were combined into a single model that directed attention to findings and issues of greatest importance for action planning.

As they examine diagnostic issues and data, practitioners often frame issues differently than clients. For example, in the CHF case, the director of CHF originally defined the problem as one of resistance to change, whereas the director of training at the HMO phrased the original diagnostic problem in terms of assessing the need for the proposed training program. The consultant reframed the study task by dividing it in two: (1) assessing feasibility of accomplishing the proposed organizational transformation and (2) discovering steps that CHF management and the HMO could take to facilitate the transformation. This redefinition of the diagnostic task thus included an image of the organization's desired state that fit both client expectations and social science knowledge about organizational effectiveness. Moreover, this reformulation helped specify the issues that should be studied in depth and suggested ways in which the clients could deal with the problem that initially concerned them. The consultant's recommendations took into account which possible solutions to problems were more likely to be accepted and could be successfully implemented by the clients.

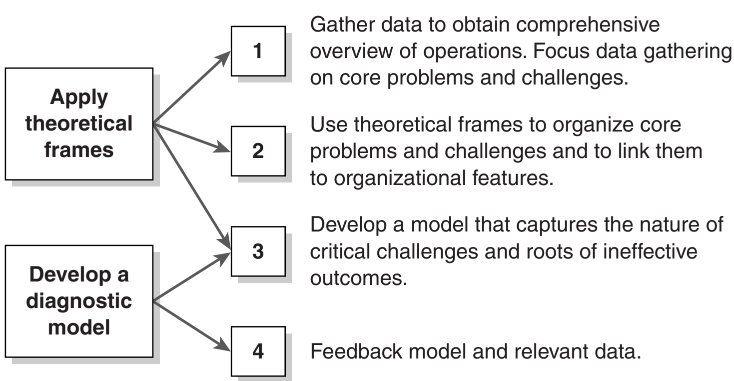

Figure 10.1 Sharp-Image Diagnosis

SOURCE: From *Organizational Diagnosis and Assessment* by M. Harrison and A. Shirom, 1999, p. 19, fig. 1.1). Reprinted with pemission of SAGE.

Choosing Effectiveness Criteria

To decide how well an organization or unit is operating, practitioners and their clients need clearly defined criteria of effectiveness. Organizational effectiveness is multidimensional and hard to measure. For example, the effectiveness of health care providers can be assessed in terms of very divergent criteria, each of which poses measurement challenges (Institute of Medicine, 2001) These criteria include the cost of care, the degree to which care is appropriate (i.e., based on scientific knowledge and avoids overuse and underuse), its safety, timeliness, equity, and patient-centeredness. As is often the case in publicly contested areas, stakeholders—including government agencies, payers, groups of providers, patient organizations, and consumer groups—assign divergent levels of importance to these criteria.

In selecting effectiveness criteria, practitioners of diagnosis and their clients face choices about five topics, which are listed here from the most general to the most specific:

1. Assessment approach: basing criteria on one or more of the following: organizational objectives and the quantity and quality of outputs; internal system states, such as work flows and organizational climate; external system relations—such as competitive position and market share; or satisfaction of multiple stakeholders (e.g., customers, funders or payers, community groups, employees, owners, regulators)
2. Domains: sets of conceptually related criteria, such as those for service quality or innovativeness
3. Criteria: specifications of domains, for example, types of innovativeness (technological vs. administrative)
4. Operational definitions and measures, for example, ways to measure technological innovativeness
5. Standards for analysis and evaluation, for example, time periods, absolute versus relative standards, and comparison groups

As is the case in any research project, the research design, the measures, and the findings in a diagnostic study will depend greatly on the choices made about each of these five facets of effectiveness (for further discussion and illustrations see Harrison, 2005; Harrison & Shirom, 1999). Let us now turn to examples of broad and focused models which are useful in diagnosis.

Diagnosing System Fits

The open systems approach gave rise to a broad model (or frame) that can guide the diagnosis of entire industries or sectors, sets of organizations, individual organizations, divisions, or units within organizations (Cummings & Worley, 2001; Harrison & Shirom, 1999; Nadler & Tushman, 1980; Senge, 1994). Open systems research calls attention to ways that developments in one part of a system or at one level (e.g., total organization, divisions, departments, units, and work groups) affect one another. In like manner, open systems studies examine exchanges between a focal organization or unit and its organizational environments and interdependencies among system subcomponents—including the focal organization's culture and subcultures, inputs (resources), behavior and processes (both intended and emergent), technologies, structures, and outputs.

There are many specifications of the open systems model that can contribute to diagnosis (Harrison & Shirom, 1999). One useful approach examines fits among system features. This approach is based on research showing that good fit among system parts, levels, or subcomponents contributes to several dimensions of organizational effectiveness.[2] Good fit (or alignment) occurs when elements within a system reinforce one another, rather than disrupting one another's operations. Organizational units, system components, or functions fit poorly if their activities erode or cancel each other; or if exchanges between units or components harm performance (e.g., by leading to avoidable losses of time, money, or energy).

Common signs of ineffectiveness–such as rapid turnover of personnel, high levels of conflict, low efficiency, and poor quality—are often symptoms of poor system fit. The following case (adapted from Beckhard & Harris, 1975) illustrates how poor fit between managerial processes (goal formation and leadership) and reward systems (structures and processes) at the divisional level can harm motivation and lead to unintended consequences:

> The head of a major corporate division at Advance Incorporated was frustrated by his subordinates' lack of motivation to work with him in planning for the future of the business and their lack of attention to helping subordinates developing their managerial potential. Repeated exhortations about these matters produced few results, although the division managers agreed that change was desirable. Diagnosis quickly uncovered the primary barrier to changing the division managers' behavior: there were no meaningful rewards for engaging in planning or management development and no punishments for not doing so. Moreover, managers were directly accountable for short-term profits in their divisions. If they failed to show a profit, they would be fired on the spot. (p. 52)

Figure 10.2 provides a schematic summary of the steps required to diagnose fits. When starting from presented problems and challenges, practitioners hunt for related, underlying conditions—such as the reward contingencies in the Advance Inc. firm—that may be causing ineffectiveness. By reporting these underlying conditions, the practitioner may help clients solve the original problems, reduce other signs of ineffectiveness, and enhance overall organizational effectiveness.

For example, a practitioner who encounters complaints about tasks being neglected or handled poorly can examine links between structure and two critical processes—decision making and communication. Responsibility charting—a procedure used in many large organizations (JISC Infonet, n.d.)—provides one way to clarify these links. First, during interviews or workshops, the practitioner asks group members to list key tasks or decision areas. In a project group, these might include budgeting, scheduling, allocating personnel, and changing design specifications of a

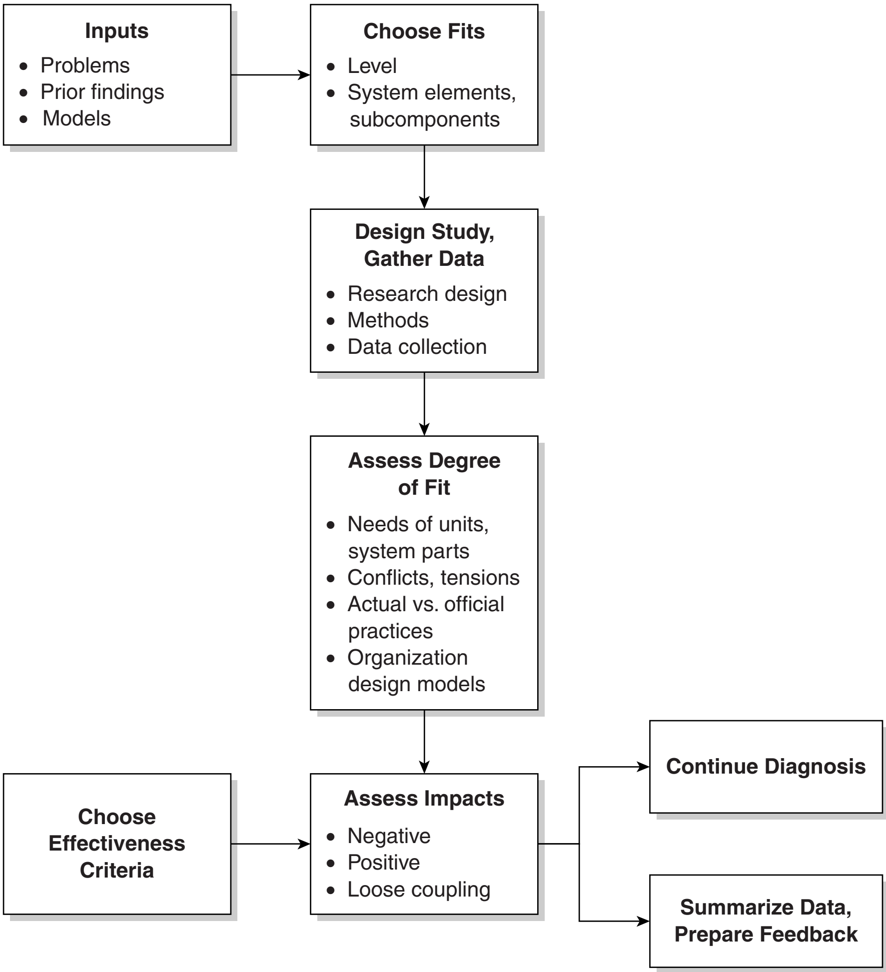

Figure 10.2 Diagnosing System Fits

SOURCE: From *Diagnosing Organizations* by M. Harrison, 2004, p. 80, fig. 4.1. Reprinted with pemission of SAGE.

product. Second, each member is asked to list the positions that will be involved in these areas (e.g., project director, general manager, laboratory manager); indicate who is assigned responsibility for performing tasks; and note who is supposed to approve the work, be consulted, and be informed. The data usually reveal ambiguities relating to one or more task areas. Consultants can use these data as feedback to stimulate efforts to redefine responsibilities and clarify relations. Feedback can also lead clients and consultants to evaluate fundamental organizational features, such as delegation of authority, coordination mechanisms, and the division of labor. For instance, discussion of approval procedures for work scheduling might reveal that many minor scheduling changes are needed and that scheduling would operate more smoothly if middle-level managers received authority to make such minor changes and inform the project head afterward.

One practical way to assess fits is to examine the compatibility of requirements, needs, or procedures in different units or system parts. Fits among units are weak if the work of one unit is disrupted because of inadequate inputs from another unit or poor synchronization with other units. For example, hospital emergency departments sometimes experience overcrowding because inpatient units encounter difficulties discharging patients and the housekeeping unit does not quickly prepare beds after patients vacate them (Urgent Matters, 2006).

A second way to assess fits is to investigate whether participants feel subject to conflicting expectations or pressures and check whether these conflicts are the result of poor fit. In the Advance, Inc. case, for example, a department manager might have complained during an interview, "My boss wants me to work on management development, but if I do, I'll be in hot water when he goes over my quarterly sales results!" The practitioner would then check whether other managers made similar comments and whether rewards were closely tied to quarterly performance, while ignoring management development activities.

Diagnosing Group Performance

To simplify diagnosis and intervention and enhance their impact, consultants and researchers have developed many focused models for diagnosing recurring organizational problems and challenges. For example, Hackman and his colleagues (1987, 1991) developed an Action Model for Group Task Performance, which examines organizational and group conditions that can serve as change levers for improving the task performance of work groups.[3] These conditions can serve well, both as focal points for diagnosis and as building blocks in the design of new work groups.

The model is summarized in Figure 10.3. As shown at the center of the figure, group performance (as defined by outputs) requires combining sufficient joint effort, adequate skills and knowledge, and a task-performance strategy that fits the work and the organization in which the work is done. Assessment of how well groups handle these critical processes can provide valuable diagnostic information. However, interventions are more likely to enhance group performance when they target *conditions that facilitate handling of critical group processes,* instead of trying to change the processes themselves. Each of the potentially facilitating conditions shown in Figure 10.3 identifies likely causes of ineffective group processes and outcomes and provides potential levers for intervention to improve group functioning and task performance (see also Hackman, 2002). First are conditions relating to the *organizational context* within which the group operates. Higher management can promote performance by defining challenging, yet specific *goals* for group performance. Performance is enhanced when management delegates much authority for deciding how to attain these goals to the team itself. Organizational *reward systems* promote performance by focusing on group actions and outcomes—rather than individual performance—and by recognizing and reinforcing good performance. The organization's *information system* can provide access to data and forecasts that help members formulate their tasks and their performance strategies and provide feedback on performance. Informal and formal *training systems* can contribute to performance by providing members with the necessary skills and knowledge in advance of task activity and in response to members' needs.

Second, *group design and culture can facilitate or hinder group processes and performance.* The most critical *task conditions* for groups include defining clear tasks, setting challenging objectives, assigning shared responsibility, and specifying accountability for task performance. In addition, it is important that groups be as small as possible, since larger groups encounter more coordination problems. *Compositional features* that contribute to performance focus include clear boundaries, inclusion of members possessing the needed skills and knowledge, including interpersonal skills, and creation of a good mix of members in terms of training and experience. This mix ensures cross-fertilization and creativity, while avoiding insurmountable divergences of opinion and working styles. Finally, groups are more successful when they possess clear and strong *norms* that regulate behavior and insure coordinated action. It is also important that these norms encourage members to act proactively and learn from their experiences.

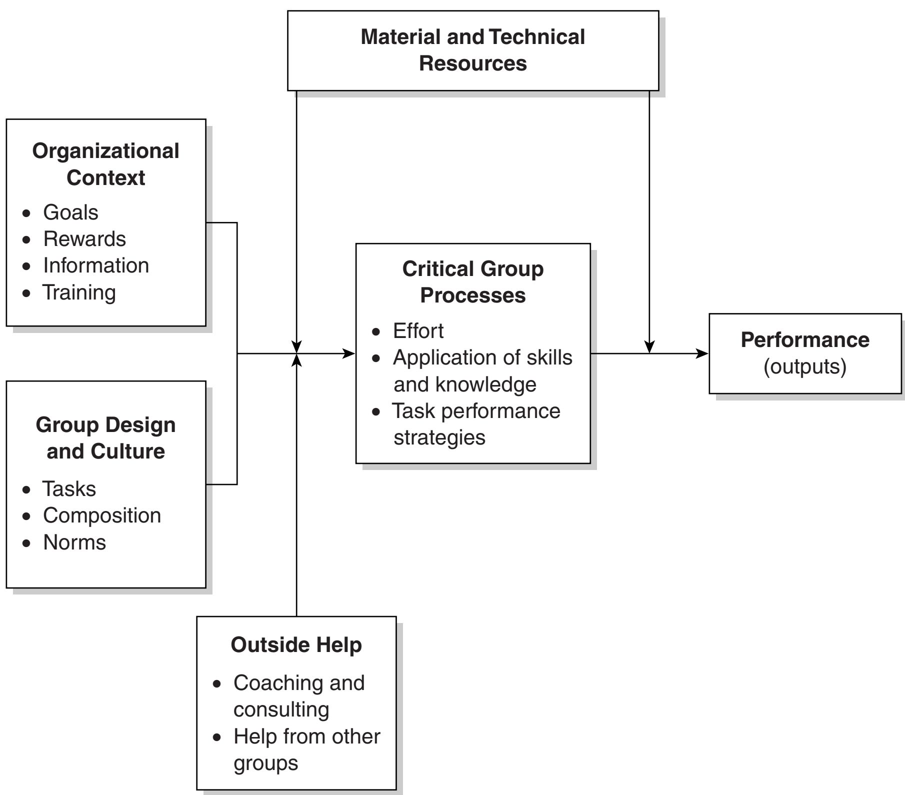

Figure 10.3 Action Model for Group Task Performance

SOURCE: From *Diagnosing Organizations* by M. Harrison, 2004, p. 62, fig. 3.2. Reprinted with pemission of SAGE.

The third set of facilitating conditions refer to access to *outside help,* such as coaching and consulting received by members. Like team leaders, external *coaches and consultants* can help members anticipate or resolve critical coordination problems and learn to collaborate effectively. Coaches can also help build commitment to the group and its task. Leaders and coaches facilitate performance when they help members decide how best to use participants' skills and knowledge, learn from one another, and learn from other groups. Leaders or coaches also help groups avoid performance strategies that are likely to fail and can help group members think creatively about new ways to handle their tasks.

Fourth, groups need access to appropriate material and technical *resources.* Without the needed equipment, funds, or raw material, group outputs will be inferior, even if the group members perform well on all the process criteria. Serious resource constraints and acute shortages can lead to frustration and even turnover among potential high performers and can erode a group's long-term performance capacity. Resource availability is particularly critical in groups that are undergoing structural change or learning new techniques for handling their tasks. Managers responsible for introducing change sometimes expect performance to improve immediately without investing in the necessary processes of learning, training, and experimentation that occur during change. By singling out material and technical resources as critical variables that intervene between group processes and performance, the Action Model reminds managers and consultants to pay attention to seemingly mundane issues, as well as examining the subtler questions of the availability of needed human resources, knowledge, and information.

Drawing on the Action Model, diagnostic studies can examine whether current conditions in each of these four areas lead to ineffective or effective performance. For example, based substantially on Hackman's model, Denison, Hart, and Kahn (1996) developed and validated a set of diagnostic questionnaire items for members of cross-functional teams. These items ask respondents to report the degree to which their team enjoys supportive facilitating conditions, handles team processes effectively, and obtains desired outcomes. Hackman and his colleagues also developed an instrument that measures concepts in the Action Model, along with those developed subsequently (Hackman, 2002). This instrument (see www.leadingteams.org/ToolsOnWeb/TDS-Guide.pdf) also assesses how well team members work together and their levels of motivation and satisfaction.

Another way to use the Action Model in diagnosis is to follow the problem-oriented, sharp-image logic shown in Figure 10.1. The diagnosis would start with troubling performance problems and then trace these signs of ineffectiveness back to difficulties in handling one or more critical group processes. Then these difficulties could be followed back to the other elements in the model, such as group design and organizational context, which can hinder or facilitate group processes. For instance, a consultant or manager might trace problems of low quality in an industrial work group back to a critical process such as pursuit of an inappropriate quality enhancement strategy. If the quality enhancement strategy is inappropriate, then the solution lies in redesigning the group's task (a facilitating condition) so as to include appropriate quality assurance techniques. Suppose, on the other hand, that the group had chosen an appropriate strategy for quality enhancement, but team members lacked the skills and knowledge needed to implement the strategy. In that case, the solutions lie in changing other conditions, such as coaching for skill use and development, training programs, or procedures for selecting team members.

Although the Action Model provides useful starting points for diagnosis, it may not adequately reflect a group's distinctive challenges and conditions. The distinctive challenge for air traffic controllers, for example, is reliability, whereas a repertoire theater group faces problems of maintaining spontaneity and artistic vigor night after night. In a similar fashion, groups and entire organizations face divergent challenges at different periods in their life cycles (Harrison & Shirom, 1999, pp. 299–324). Nor does the Action Model pay much attention to important "soft" aspects of group interaction, such as mutual expectations and understandings. A further limitation is the model's heavy stress on measurable outputs, which could lead analysts and clients to pay less attention than needed to other dimensions of effectiveness and ineffectiveness. Finally, the Action Model builds in strong assumptions about the likely indicators and causes of ineffectiveness and the best ways to intervene to enhance group performance. Hence, the model may discourage users from attending directly to client concerns and from identifying causes and possible solutions that reflect the organization's distinctive features and the contingencies affecting it.

Methods

Besides assuring valid findings, diagnosis requires identifying readily changeable factors affecting clients' problems. The data-gathering methods should help practitioners uncover these actionable solutions. The methods should also contribute to constructive relations between consultants and members of the client organization and enhance the chances that members of the client organization will regard the findings as valid and useful.

Choosing Methods

To provide valid results, practitioners should employ the most rigorous methods possible within the practical constraints imposed by the assignment. Rigorous methods—which need not be quantitative—follow accepted standards of scientific inquiry (King, Keohane, & Verba, 1994). They have a high probability of producing results that are valid and reliable (i.e., replicable by other trained investigators; Trochim, 2001). Nonrigorous approaches can yield valid results, but these cannot be externally evaluated or replicated. In assessing the validity of their diagnoses, practitioners need to be aware of the risk of false-positive results that might lead them to recommend steps that are unjustified, and even harmful, to the client organization (Rossi & Whyte, 1983).

To achieve replicability, practitioners can use structured data-gathering and measurement techniques, such as fixed-choice questionnaires (Faletta & Combs, 2002; Harrison, 2005, chap. 3, appendix B) or structured observations (Harrison, 2005, appendix C; Weick, 1985). Unfortunately, it is very hard to structure techniques for assessing many complex but important phenomena, such as the degree to which managers accurately interpret environmental developments.

To produce valid and reliable results, investigators must often sort out conflicting opinions and perspectives about the organization and construct an independent assessment. The quest for an independent viewpoint and scientific rigor should not, however, prevent investigators from treating the plurality of interests and perspectives within a focal organization as a significant organizational feature in its own right (Ramirez & Bartunek, 1989).

Whatever techniques practitioners use in diagnosis, it is best to avoid methodological overkill when only a rough estimate of the extent of a particular phenomenon is needed. In the Advance Inc. case, for example, the investigators needed to determine whether division heads were frustrated and dissatisfied and needed to find the sources of the managers' feelings. The practitioners did not need to specify the precise degree of managerial dissatisfaction, as they might have done in an academic research study.

Consultants need to consider the implications of their methods for the consulting process and the analytic issues at hand, as well as weighing strictly practical and methodological considerations. Thus, consultants might prefer to use less rigorous methods, such as discussions of organizational conditions in workshop settings (Biech, 2004; Harrison, 2005, chap. 5), because these methods can enhance the commitment of participants to the diagnostic study and its findings. Or they might prefer observations to interviews, so as not to encourage people to expect that the consultation would address the many concerns raised during interviews.

The methods chosen and the ways in which data are presented to clients also need to fit the culture of the client organization. In a high-technology firm, for example, people may regard qualitative research as impressionistic and unscientific. On the other hand, volunteers at a hospice might view standardized questionnaires and quantitative analysis as insensitive to their feelings and experiences.

Research Design

Three types of nonexperimental designs seem most appropriate for diagnosis. The first involves gathering data on important criteria that allow for comparisons between units, between entire organizations, or over a period of time (Glanz & Dailey, 1992; Harrison & Shirom, 1999, pp. 217–221). Comparisons may focus on criteria such as client satisfaction, organizational climate (e.g., perceptions of peer and subordinate-supervisor relations, identification with unit and organizational goals), personnel turnover, costs, and sales. Sometimes, practitioners can analyze available records or make repeated measurements to trace changes in key variables across time for each unit or for an entire set of related units.

The second design uses multivariate analysis of data to isolate the causes or predictors of variables linked to a particular organizational problem, such as work quality or employee turnover, or to some desirable outcome, such as product innovation or customer satisfaction. This design is less common in diagnosis than in academic research because of practical constraints during diagnosis on extensive and lengthy data collection and analysis. The third design uses qualitative field techniques to construct a portrait of the operations of a small organization or subunit (e.g., the executive team) and obtain in-depth data on subtle, hard-to-measure features that may be lost or distorted in closed-ended inquiries. Among such features are emergent practices, members' perceptions and assumptions, behind-the-scenes interactions, and work styles. In such qualitative studies, investigators use data-gathering techniques and inductive forms of inference such as those used in nonapplied qualitative research (Denzin & Lincoln, 2000; Miles & Huberman, 1994; Yin, 2002). However, to assure quick feedback and reduce costs, diagnostic studies usually seek less ethnographic detail than nonapplied qualitative research and use less rigorous forms of recording and analyzing field data. These less rigorous qualitative methods can yield helpful insights, but they are also more likely to yield biased or superficial interpretations of complex phenomena.

Data Collection

Table 10.1 surveys and assesses data collection techniques frequently used in diagnosis.[4] No single method for gathering and analyzing data can suit every diagnostic problem and situation, just as there is no universal model for guiding diagnostic analysis or one ideal procedure for managing the diagnostic process. By using several methods to gather and analyze their data, practitioners can compensate for many of the drawbacks associated with relying on a single method. They also need to choose methods that fit the diagnostic problems and contribute to cooperative, productive consulting relations. Let us consider two of the most popular data collection techniques in greater depth.

Structured Instruments

Self-administered questionnaires provide the least expensive way of eliciting attitudes, perceptions, beliefs, and reports of behavior from many people. Questionnaires can be administered in person, by mail, by telephone, or over the Internet (Miller & Salkind, 2002; Stanton & Rogelberg, 2001). Aggregations of individual responses can also provide a substitute for behavioral measures of group and organizational phenomena. Although questionnaires typically use fixed-choice answers, a few open-ended questions can be included to give respondents an opportunity to express themselves. Responses to such open-ended questions are often informative, but difficult to code. Questionnaires composed of items drawn from previous research studies and standardized organizational surveys can be prepared and administered rapidly, since there is less need to develop and pretest the instrument. By including standard measures, consultants may also be able to compare the responses obtained in the client organization with results from other organizations in which the same instrument was used.

Over the past few years, many standardized organizational survey instruments have been developed that can be used in diagnostic studies (see Harrison, 2005, appendix B). Focused instruments cover particular areas that are often of concern, including team functioning (e.g., the instruments discussed above measuring aspects of the Action Model for Group Task Performance), human resources practices, organizational climate and culture, leadership, and communication patterns. Broad instruments include scales or entire subsections that cover these topics and others of recurring interest. Classic examples include the well-documented, Michigan

Table 10.1 Comparison of Methods for Gathering Diagnostic Data

Method	*Advantages*	*Disadvantages*
Questionnaires		
Self-administered schedules, fixed choices	Easy to quantify and summarize; quickest and cheapest way to gather new data rigorously, neutral and objective; useful for large samples, repeat measures, comparisons among units or to norms; standardized instruments contain pretested items, reflect diagnostic models, good for studying attitudes	Hard to obtain data on structure, behavior; little information on how contexts shape behavior; not suited for subtle or sensitive issues; impersonal; risks: nonresponse, biased or invalid answers, over reliance on standard measures and models
Interviews		
Open-ended questions based on fixed schedule or interview guide	Can cover many topics; modifiable before or during interview; can convey empathy, build trust; rich data, allows understanding of respondents' viewpoints and perceptions	Expensive, hard to administer to large samples; respondent bias and socially desirable responses; noncomparable responses; hard to analyze responses to open-ended questions; modification of interviews to fit respondents reduces rigor
Observations		
Structured or open-ended observation of people, work settings	Data are independent of people's self-presentation and biases; data on situational, contextual effects; rich data on hard-to-measure topics (e.g., emergent behavior, culture); data yield new insights, hypotheses	Constraints on access to data; costly, time-consuming; observer bias, low reliability; may affect behavior of those observed; hard to analyze and report; less rigorous, may seem unscientific
Available records and data		
Use of documents, reports, files, statistical records, unobtrusive measures	Nonreactive; often quantifiable; repeated measures show change; organization's members can help analyze data; credibility of familiar measures (e.g., customer complaints, staff turnover); often cheaper and faster than gathering new data; independent sources; data on total organization, environments, industries	Access, retrieval, analysis problems can raise costs; validity, credibility of some sources and derived measures can be low; need to analyze data in context; limited information on many topics (e.g., emergent behavior)
Workshops, group discussions		
Discussions on group processes, culture, environment, challenges strategy; directed by consultant of manager; simulations, exercises	Useful data on complex, subtle process; interaction stimulates creativity, teamwork, planning; data available for immediate analysis and feedback; members share in diagnosis; self-diagnosis possible; consultant can build trust, empathy	Biases due to group processes, history, leader's influence (e.g., boss stifles dissent); requires high levels of trust and cooperation in group; impressionistic, nonrigorous; may yield superficial, biased results, unsubstantiated decisions

SOURCE: From *Diagnosing Organizations* by M. Harrison, 2004, table 1.1. Reprinted with pemission of SAGE.

Organizational Assessment Questionnaire (MOAQ; Cammann, Fichman, Jenkins, & Kelsh, 1983) and the related instruments in the Michigan Quality of Work Program (Seashore, Lawler, Mirvis, & Cammann, 1983). These instruments were often used in research and served as models for many subsequent instruments. MOAQ includes seven modules that cover individual performance—based on self-reported effort at work—and quality of work life outcomes—including job satisfaction. Also included in measures of individual responses to the job are intentions and opportunities to leave the organization or job. Other scales cover characteristics of jobs, roles and tasks; identification with work and the organization; adequacy of training and skills; perceived determinants of pay and importance of various types of rewards; and several facets of supervisory behavior. There are also measures of some group characteristics and processes—including diversity, goal clarity, cohesiveness, involvement in decision making, fragmentation, and openness of communications.

The Organizational Assessment Survey (OAS; Muldrow, Schay, & Buckley, 2002; available at www.opm.gov/employ/html/org_asse.asp) is another useful broad instrument, which is in the public domain. It was developed by the United States Office of Personnel Management to provide government agencies with a standardized tool for assessing organizational strengths and weaknesses, planning training and change programs, and making comparisons across time and among agencies (benchmarking). The survey has been used by many federal agencies and some states. It covers employee perceptions in 17 areas of organizational climate—including rewards, training, innovation, consumer orientation, teamwork, communication, performance, supervision, and diversity. It can be administered over the Internet or in a paper-and-pencil version.

To create a more comprehensive diagnostic instrument, practitioners can supplement data instruments based on individual perceptions with more behavioral data on working conditions and outputs. Data can also be gathered on additional facets of group performance, such as output quantity and quality, goal attainment, innovativeness, efficiency, morale, and reputation for excellence. The Organizational Assessment Inventory (OAI; Van de Ven & Ferry, 1980) contains scales in these areas as well as measures of group diversity and group processes, including conflict management and normative pressures. The instrument has been used in many basic and applied research investigations. Substantial evidence has accumulated for the reliability, construct validity, and predictive validity of scales constructed from OAI items (Gresov, 1989; Van de Ven & Chu, 1989; Van de Ven & Walker, 1984). Structural features assessed by OAI include control systems, job standardization, role relations, work and unit interdependencies, work flows, and authority distribution. OAI contains separate questionnaires for supervisors and group members so that comparisons of their attitudes and reports can be made. Other instruments within OAI assess divisional (interdepartmental) and organization-level phenomena.

Additional factors, such as group or organizational norms and culture, can be assessed with the aid of standardized research instruments (Rousseau, 1990). Such instruments are reviewed in source books and in review articles such as Ashkenasy, Wilderom, and Peterson (2000); Harrison (2005, appendix B), and Kraut (1996). In addition, academic journals and publications contain many instruments for specific types of organizations (e.g., Lester & Bishop, 2001; Scott, Mannion, Davies, & Marshall, 2003).

To obtain data on group-level phenomena from questionnaires such as MOAQ, OAS, and OAI, the responses from members of a particular work group or administrative unit are averaged to create group scores. For these averages to be meaningful and useful in analysis and feedback, the questionnaires must specify clearly which work groups and supervisors are referred to.

Instruments such as MOAQ, OAS, and OAI contain ready-to-use scales that usually produce valid and reliable measures for many organizational settings. In keeping with current research and organizational theory, these instruments reflect the assumption that there is no one best way to organize groups or organizations. Instead, the optimal combination of system traits is assumed to depend on many variables, including environmental conditions, tasks, technology, personnel, history, and size of the organization.

Despite their appeal, standardized diagnostic instruments also have weaknesses and drawbacks. First, they may give practitioners a false sense of confidence that all the factors relevant to a particular client organization have been covered adequately. Second, standard questions are necessarily abstract; hence, they may not be fully applicable to a particular organization or situation. For example, a typical questionnaire item in MOAQ asks respondents to indicate their degree of agreement with the statement, "My supervisor encourages subordinates to participate in making important decisions" (Cammann et al., 1983, p. 108). But the responses to this general statement may mask the fact that the supervisor encourages participation in decisions in one area, such as work scheduling, while making decisions alone in other areas, such as budgeting. To obtain data on such situational variations, investigators must determine the situations across which there may be broad variations and write questions about these situations (e.g., Moch, Cammann, & Cooke, 1983, pp. 199–200).

Third, as in any questionnaire, even apparently simple questions may contain concepts or phrases that may be understood in different ways. For instance, when reacting to the statement, "I get to do a number of different things on my job" (Cammann et al., 1983, p. 94), one person might see diversity in physical actions (e.g., snipping vs. scraping) or in minor changes in the tools needed for the job, whereas another would consider all those operations as "doing the same thing." Fourth, questionnaires are especially vulnerable to biases stemming from the respondent's desire to give socially acceptable answers or to avoid sensitive issues. There may also be tendencies to give artificially consistent responses (Salancik & Pfeffer, 1977; but compare Stone, 1992). Some instruments include questions designed to detect or minimize biases, whereas others may invite response bias by phrasing all questions in a single direction.

In designing samples, practitioners consider the attitudes of group members toward the study and the uses to which the data will be put, as well as strictly methodological considerations. Standardized diagnostic instruments are often administered to all members of a unit undergoing diagnosis so as to make the study findings more relevant and believable to all people who will receive feedback. Interviews can also be conducted with small units or organizations. Alternatively, practitioners may use a purposive sample for interviews, so as to include people holding key positions and a cross-section of those likely to have divergent perspectives and experiences.

To reach large numbers of people, self-administered questionnaires can be distributed to samples of members selected through probability sampling (Trochim, 2001). Probability samples can also be used to gather secondary data, such as absenteeism rates from large data sets.

Practitioners should try to sample critical situations and processes as well as individuals. For example, the characteristic ways in which conflicts are handled, the practitioner would look for typical or representative conflict episodes as well as questioning a representative cross-section of group members.

Semistructured Interviews

Semistructured interviews provide practitioners with opportunities to develop rapport with members of the organization and learn about critical areas that are not readily assessed through standardized questionnaires. These include organizational processes, basic assumptions and beliefs, and critical organization-level phenomena—such as management control processes, relations to clients, and business strategies. In the exploratory stage of a diagnosis, practitioners often conduct orientation interviews (Harrison, 2005, appendix A) with people who occupy leadership positions and perform crucial functions within the focal unit or organization. These interviews provide data on how the focal unit is organized and operates, as well as the respondent's view of the major challenges or problems facing it. Topics often include background on the interviewee, the unit's main products and services, controls and coordinating mechanisms, relations to other units—including broader organizational units such as divisions or corporate headquarters, relations to the external environment (markets, stakeholders, suppliers, and regulators), management structures, processes, and culture.

In seeking information about groups, divisions, or entire organizations, investigators need to pose questions that fit the positions and organizational level of respondents. For example, department heads may provide basic information on department regulations, history, and working relations with other departments; their subordinates may have little knowledge in such areas. In contrast, subordinates sometimes know better than their boss how work is actually done.

Interviews can also be structured around a particular topical area. When practitioners lack detailed knowledge of operations in a particular area or want to allow their interviews to be responsive to issues that arise during the interview, they may construct an interview guide, rather than prepare detailed questions in advance. The guide lists topics to be investigated and then allows the interviewer to frame questions about each topic that reflect the distinctive circumstances of the client organization; the guide also provides opportunities to take into account previous answers. Interview guides thus ensure coverage of major topics while allowing flexibility. But interview guides have lower reliability than standardized questionnaires, because they allow for more variation between interviews and among interviewers.

Using interview guides also requires more interviewer skill than does the use of more structured schedules.

Here is an illustration of the major headings for an interview guide that aims to assess relations between an organization and its external environments (see Harrison, 2005, chap. 5 on Environmental Relations Assessment):

1. Key external conditions in markets or fields
2. Main outside organizations, types of relations (ties, competition vs. cooperation, resource dependence)
3. Main units, people who handle external contacts
4. Current management of problems, demands, opportunities
5. Effectiveness of current actions—including specification of effectiveness criteria
6. Ways to improve current environmental management

Each major heading in the guide is broken down into subheadings to cover specific topics. For example, Items 4 and 5 could be specified as follows (with phrases in parentheses serving as interviewer guidelines):

4. Current management of problems, demands, opportunities
   - 4.1. Specific actions—describe in detail. What is/was done, by whom? (Interviewer: Look for internal adjustments, interventions in environment; incremental vs. strategic actions.)
   - 4.2. Other actions (e.g., Did your group make any other attempts to moderate these pressures/deflect these criticisms/anticipate such developments?) (Interviewer: Look for anticipatory vs. reactive moves.)

5. Effectiveness of current actions
   - 5.1. External impact of actions
     - 5.1.1. Impacts on external actors and conditions (e.g., How did *x* react to the steps you took?)
     - 5.1.2. Effectiveness (Interviewer: Apply effectiveness criteria suggested by respondent, e.g., Did these steps improve sales revenues?)
   - 5.2. Internal organizational impacts. (Interviewer: Probe for felt effects of one-time or recurring reliance on these responses; whether they produced desired results, how successful they seemed to respondent, and meaning of success for him or her.)
   - 5.3. Changes in tactics and impacts. Were similar problems handled in the same way in the past? What happened after changes in tactics? (Interviewer: Probe for shifts in tactics, stance toward environment, variations in impacts.)

Naturally, when practitioners use an interview guide, they prepare for the possibility that the answers will range across the topics listed in the guide. During the interview, they record the responses in the order given. Afterward, they can reorganize them according to the topics in the guide.

Interview and questionnaire studies are often subject to bias because respondents seek to present themselves in a favorable light or withhold information, such as negative descriptions of supervisors that might be used against them. By conducting interviews with members from different backgrounds and locations within a unit and by listening carefully to their accounts of important issues, investigators can become aware of members' distinct perspectives and viewpoints. For example, department heads might characterize their organization as dealing honestly and directly with employee grievances, while subordinates complain that their grievances are ignored or minimized by management. The people interviewed may be unaware of such a diversity of viewpoints or intolerant of the feelings and perceptions of others. In such cases, consultants can summarize the various viewpoints during feedback to stimulate communication and encourage people to respect diverse perspectives and opinions. In other instances, consultants can simply take note of divergent viewpoints and avoid giving undue weight to one particular interpretation when formulating their own descriptions and analyses.

By building relations of trust with group members, consultants can sometimes overcome people's reluctance to reveal sensitive information during interviews. Practitioners may also gain the trust of one or more members of an organization who know a lot about organizational affairs but are somewhat detached from them.[5] Assistants to high-level managers, for example, often have a broad view of their organization and may be more comfortable providing such information than are the managers. When such well-placed individuals trust consultants, they may provide useful information about sensitive subjects, such as the degree of influence of managers who officially have the same level of authority, or staff members' past reactions to risk-taking behavior.

The processes of gathering and reporting diagnostic information can pose tricky ethical and professional issues. These and other ethical issues facing diagnostic practitioners and other types of consultants deserve advance consideration (see American Psychological Association, 1992; Harrison, 2005, chap. 6).

Conclusion

Successful diagnosis requires practitioners to deal with three distinct challenges and to strike a good balance in their tactics for handling each. The *process* challenge requires constructive management of interactions with clients and other organizational stakeholders. The *methodological* challenge calls for using rigorous and valid techniques for gathering, summarizing, and analyzing data within the constraints imposed by the consulting assignment. The *analytic* challenge involves using research-based models to identify sources of effectiveness and ineffectiveness, discover routes toward organizational improvement, and frame feedback.

Despite their usefulness, the models, techniques, and methods reviewed here and those presented in the literature on diagnosis, applied research, and consultation cannot serve as step-by-step guides to diagnosis. Nor can they be used like equations into which bits of data are inserted to produce a completed assessment. No such recipes for diagnosis or action planning exist, and none is likely to be discovered. Instead, most models and methodological techniques work best as frames and guides that help both experienced and novice practitioners sort out what is going on within an organization. Because models and methods focus attention on particular system levels or types of phenomena, they may distract attention from other important organizational features. Only by combining frames and methods can practitioners deal with the multifaceted nature of organizational problems and challenges (Harrison & Shirom, 1999).

Anyone who undertakes a diagnosis, thus, faces many choices about which models and methods to use and how to manage the consulting process. In most cases, each alternative has some advantages and some drawbacks. Emerging relations between clients and practitioners and practical considerations, such as the accessibility of data, shape choices among alternatives.

To engage in diagnosis is to undertake a difficult but exciting and rewarding task—to use methods and models from the behavioral and organization sciences to help people find out what is going on in their organization and why, while engaged in a complex, changing web of relations; to find a way of serving clients who may be ambivalent about receiving help and deal with people who may be dead set against the project; to sort among project constraints and a tangle of compelling obligations, values, and professional standards (see Harrison, 2005, chap. 6). Readers who want to develop their ability to handle these challenges should seek firsthand, supervised experiences in diagnosis and consulting processes, along with advanced training in organizational analysis and research methods.

Exercises

1. Describe a planned change project with which you are familiar. Report how the consultants and main clients dealt with the Critical Process Issues discussed on page 321—the purpose of the diagnosis, its design, sources of support and cooperation, participation, and feedback. Explain how the diagnosis and the change project as a whole were affected by the consultants' handling of these Critical Process Issues. If you are not familiar with an actual change project, propose one for an organization you know well, explain how the consultant should address each of the Critical Process Issues, and justify your choices.

2. Describe a team or work group that you know well. Explain how you could gather diagnostic data about this team that would cover each of the factors highlighted in the discussion of the Action Model for Group Task Performance.

3. Use one of the standardized diagnostic instruments discussed in this article or another standardized instrument (questionnaire) to survey at least seven

members of a team or organizational unit. Then construct an interview guide for one or two of the topical areas covered by the questionnaire. Conduct three semi-structured interviews with team members using this guide. Summarize the findings obtained with each instrument and compare them. Discuss the advantages and disadvantages of each data-gathering method.

Notes

1. Portions of this chapter are drawn from Harrison (2005) and Harrison and Shirom (1999). See those sources for more detailed discussions and further references on the methods, techniques, and models reviewed here; those sources and the references cited in them provide many additional tools and diagnostic approaches besides those presented in this chapter.

2. Some effective organizations develop structures and practices that appear to be poorly aligned with one another. For example, managers in large organizations can use new information technologies to closely oversee the practices and performance of subordinate units or people, while also granting subordinates substantial decision authority and operating autonomy. See Harrison (2005, p. 91) for further discussion of combinations of opposing design principles.

3. This presentation of the model reflects both the work of Hackman and his colleagues and a modification and critique in Harrison and Shirom (1999, pp. 166–173).

4. See Harrison (2005) for additional discussion and references on data collection techniques.

5. In anthropological studies, such individuals are called informants, a term that cannot be used in diagnosis because of its negative connotations.

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CHAPTER 11

Research Synthesis and Meta-Analysis

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As the volume of primary research across all fields of social science continues to grow at rapid rates, research synthesis has become more important today than at any other time in history. With the development of meta-analysis, a set of procedures for summarizing the quantitative results from multiple studies, the rigor, systematicity, and transparency of research syntheses was greatly improved. However, a number of developments, including the creation of the Cochrane Collaboration and Campbell Collaboration, have heightened the profile of meta-analysis in recent years. Furthermore, recent advancements in analytic strategies, including the use of a random effects model of error, the development of meta-regression, and improved methods for dealing with missing data and data censoring, have enhanced the popularity, efficiency, and trustworthiness of meta-analyses.

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We begin this chapter with a brief history of meta-analysis and research synthesis. We then describe the different stages of a rigorous research synthesis. Next, we outline a set of generally useful meta-analytic techniques and follow this with a discussion of some of the difficult decisions that research synthesists face in carrying out a meta-analysis. We conclude by addressing some broader issues concerning criteria for evaluating the quality of knowledge syntheses in general and meta-analyses in particular.

A general theme of the chapter is that social scientists who are conducting research syntheses need to think about what distinguishes a good synthesis from a bad synthesis. This kind of effort is crucial for assessing the value of existing research syntheses and for promoting high-quality research synthesis in the future.

A Brief History of Research Synthesis and Meta-Analysis

In 1904, Karl Pearson published what is believed to be the first meta-analysis. Having been asked to synthesize the evidence on a vaccine against typhoid, Pearson gathered data from 11 relevant studies, and for each study, he calculated a recently developed statistic called the correlation coefficient. He averaged these measures of the treatment's effect across two groups of studies distinguished by the nature of their outcome variable. Based on the average correlations, Pearson concluded that other vaccines were more effective.

In 1932, Ronald Fisher, in his classic text *Statistical Methods for Research Workers*, noted, "It sometimes happens that although few or [no statistical tests] can be claimed individually as significant, yet the aggregate gives an impression that the probabilities are lower than would have been obtained by chance" (p. 99). Fisher then presented a technique for combining the *p* values that came from statistically independent tests of the same hypothesis. His work would be followed by more than a dozen papers published prior to 1960 on the same topic (see Olkin, 1990).

This early development of procedures for statistically combining results of independent studies went largely unused. However, beginning in the 1960s, with the tremendous growth in social scientific research and increasing interest in its social policy implications, these methods began to gain widespread use (Chalmers, Hedges, & Cooper, 2002). By the mid-1970s, when Robert Rosenthal and Donald Rubin undertook a synthesis of research studying the effects of interpersonal expectations on behavior, they found 345 studies that pertained to their hypothesis (Rosenthal & Rubin, 1978). Almost simultaneously, Gene Glass and Mary Lee Smith were conducting a synthesis of the relation between class size and academic achievement (Glass & Smith, 1979). They found 725 estimates of the relation, based on data from nearly 900,000 students. Smith and Glass (1977) also gathered assessments of the effectiveness of psychotherapy; this literature revealed 833 tests of the treatment. Likewise, John Hunter and Frank Schmidt uncovered 866 comparisons of the differential validity of employment tests for black and white workers (Hunter, Schmidt, & Hunter, 1979).

Each of these research teams realized that for some topic areas, prodigious amounts of empirical evidence had been amassed on why people act and feel the way they do and on the effectiveness of psychological, social, educational, and medical interventions. These researchers concluded that the traditional research synthesis simply would not suffice. Largely independently, the three research teams rediscovered and reinvented Pearson's and Fisher's solutions to their problem.

In discussing his solution, Glass (1976) coined the term *meta-analysis* to stand for "the statistical analysis of a large collection of analysis results from individual studies for purposes of integrating the findings" (p. 3). Shortly thereafter, other proponents of meta-analysis demonstrated that traditional synthesis procedures led to inaccurate or imprecise characterizations of the literature, even when the size of the literature was relatively small (Cooper, 1979; Cooper & Rosenthal, 1980).

The first half of the 1980s witnessed the appearance of five books devoted primarily to meta-analytic methods. The first, by Glass, McGaw, and Smith (1981) presented meta-analysis as a new application of analysis of variance and multiple regression procedures, with effect sizes treated as the dependent variable. In 1982, Hunter, Schmidt, and Jackson introduced meta-analytic procedures that focused on (a) comparing the observed variation in study outcomes to that expected by chance and (b) correcting observed correlations and their variance for known sources of bias (e.g., sampling errors, range restrictions, unreliability of measurements).

Rosenthal (1984) presented a compendium of meta-analytic methods covering, among other topics, the combining of significance levels, effect size estimation, and the analysis of variation in effect sizes. Rosenthal's procedures for testing moderators of effect size estimates were not based on traditional inferential statistics, but on a new set of techniques involving assumptions tailored specifically for the analysis of study outcomes.

Another text that appeared in 1984 also helped elevate research synthesis to a more rigorous level. Light and Pillemer (1984) focused on the use of research synthesis to help decision making in the social policy domain. Their approach placed special emphasis on the importance of meshing both numbers and narrative for the effective interpretation and communication of synthesis results.

Finally, in 1985, with the publication of *Statistical Methods for Meta-Analysis*, Hedges and Olkin helped to elevate the quantitative synthesis of research to an independent specialty within the statistical sciences. This book, summarizing and expanding nearly a decade of programmatic developments by the authors, not only covered the widest array of meta-analytic procedures but also established their legitimacy by presenting rigorous statistical proofs.

Meta-analysis did not go uncriticized. Some critics opposed quantitative synthesis, using arguments similar to those used to oppose primary data analysis (Barber, 1978; Mansfield & Bussey, 1977). Others linked meta-analysis with more general synthesis procedures that are inappropriate, but not necessarily related to the use of statistics in synthesis. We address several of these issues later in this chapter.

Since the mid-1980s, several other books have appeared on meta-analysis. Some of these treat the topic generally (e.g., Cooper, 1998; Hunter & Schmidt, 2004; Lipsey & Wilson, 2001), some treat it from the perspective of particular research design conceptualizations (e.g., Eddy, Hassleblad, & Schachter, 1992; Mullen, 1989), some are tied to particular software packages (e.g., Johnson, 1993; Wang & Bushman, 1999), and some look to the future of research synthesis as a scientific endeavor (e.g., Cook et al., 1992; Wachter & Straf, 1990).

During and after the years that the works mentioned above were appearing, literally thousands of meta-analyses were published. In 1994, the first edition of *Handbook of Research Synthesis* was published (Cooper & Hedges, 1994). Through the 1990s, the use of meta-analysis spread from psychology and education (see Hunt, 1997, for a history of these efforts) through many disciplines, especially social policy analysis and the medical sciences (see Chalmers, Hedges, & Cooper, 2002, for a history of meta-analysis in medicine). One of the most notable events in medicine was the establishment of the U.K. Cochrane Center in 1992. The Center was meant to facilitate the creation of an international network to prepare and maintain systematic synthesis of the effects of interventions across the spectrum of health care practices. At the end of 1993, an international network of individuals, called the Cochrane Collaboration (www.cochrane.org/index.htm), emerged from this initiative (Bero & Rennie, 1995; Chalmers, 1993). By 2006, the Cochrane Collaboration was an internationally renowned initiative with 11,000 people contributing to its work, in more than 90 countries. The Cochrane Collaboration is now the leading producer of research syntheses in health care and is considered by many to be the gold standard for determining the effectiveness of different health care interventions. Its library of systematic synthesis numbers in the thousands. In 2000, an initiative called the Campbell Collaboration (www.campbellcollaboration.org) was begun with similar objectives for the domain of social policy analysis, focusing initially on policies concerning education, social welfare, and crime and justice.

Research Synthesis as a Scientific Process

Several early attempts that framed the integrative research synthesis in the terms of a scientific process occurred independent of the meta-analysis movement. In 1971, Feldman published an article titled "Using the Work of Others: Some Observations on Reviewing and Integrating," in which he wrote, "Systematically reviewing and integrating . . . the literature of a field may be considered a type of research in its own right—one using a characteristic set of research techniques and methods" (p. 86).

In the same year, Light and Smith (1971) presented a "cluster approach" to research synthesis that was meant to redress some of the deficiencies in the existing strategies. They argued that if treated properly, the variation in outcomes among related studies could be a valuable source of information, rather than a source of consternation, as it appeared to be when treated with traditional synthesis methods.

Three years later, Taveggia (1974) struck a complementary theme:

> A methodological principle overlooked by [synthesists] . . . is that research results are probabilistic . . . they may have occurred simply by chance. It also follows that, if a large enough number of researches has been done on a particular topic, chance alone dictates that studies will exist that report inconsistent and contradictory findings! Thus, what appears to be contradictory may simply be the positive and negative details of a distribution of findings. (pp. 397–398)

Taveggia described six common problems in literature syntheses; selecting research; retrieving, indexing, and coding studies; analyzing the comparability of findings; accumulating comparable findings; analyzing the resulting distributions; and reporting the results.

Two articles that appeared in the *Synthesis of Educational Research* in the early 1980s brought the meta-analytic and synthesis-as-research perspectives together. First, Jackson (1980) proposed six synthesis tasks "analogous to those performed during primary research" (p. 441). Jackson portrayed the limitations of meta-analysis as well as its strengths. His article employed a sample of 36 synthesis articles from prestigious social science periodicals to examine the methods used in syntheses of empirical research. His conclusion was that "relatively little thought has been given to the methods for doing integrative reviews" (p. 459).

Cooper (1982) took the analogy between research synthesis and primary research to its logical conclusion. He presented a five-stage model of the integrative synthesis that viewed research synthesis as a data-gathering exercise and, as such, applied to it criteria similar to those employed to judge primary research. Similar to primary research, a research synthesis involves problem formulation, data collection (the literature search), data evaluation, data analysis and interpretation (the meta-analysis), and public presentation. For each stage, Cooper codified the research question asked, its primary function in the synthesis, and the procedural differences that might cause variation in synthesis' conclusions. In addition, Cooper applied the notion of threats to inferential validity—introduced by Campbell and Stanley (1966; expanded by Cook and Campbell, 1979, and further refined in Shadish, Cook, & Campbell, 2002) for evaluating the utility of primary research designs—to research synthesis. He identified numerous threats to validity associated with synthesis procedures that might undermine the trustworthiness of a research synthesis' findings. He also suggested that other threats might exist and that any particular synthesis' validity could be threatened by consistent deficiencies in the set of studies that formed its database.

Table 11.1 presents Cooper's (1982) conceptualization of the research synthesis process. In the next section, we describe briefly the critical decisions that characterize each stage.

The Stages of Research Synthesis

The Problem Formulation Stage

During the problem formulation stage, research synthesists must (a) define the variables of interest both conceptually and operationally and (b) clearly state the relationship of interest.

Conceptual definitions describe qualities of the variables that are independent of time and space but can be used to distinguish relevant from irrelevant events (Shoemaker, Tankard, & Lasorsa, 2004). The first source of variation in synthesis conclusions enters during this concept identification. Two synthesists using an identical label for an abstract concept can employ different definitions or levels of

Table 11.1 Research Synthesis Conceptualized as a Research Process

	Stage of Research				
Stage Characteristics	*Problem Formulation*	*Data Collection*	*Data Evaluation*	*Analysis and Interpretation*	*Public Presentation*
Research question asked	What evidence should be included in the review?	What procedures should be used to find relevant evidence?	What retrieved evidence should be included in the review?	What procedures should be used to make inferences about the literature as a whole?	What information should be included in the review report?
Primary function in review	Constructing definitions that distinguish relevant from irrelevant studies	Determining which sources of potentially relevant studies to examine	Applying criteria to separate "valid" from "invalid" studies	Synthesizing valid retrieved studies	Applying editorial criteria to separate important from unimportant information
Procedural differences that create variation in review conclusions	1. Differences in included operational definitions 2. Differences in operational detail	Differences in the research contained in sources of information	1. Differences in quality criteria 2. Differences in the influence of nonquality criteria	Differences in rules of inference	Differences in guidelines for editorial judgment
Sources of potential invalidity in review conclusions	1. Narrow concepts might make review conclusions less definitive and robust. 2. Superficial operational detail might obscure interacting variables.	1. Accessed studies might be qualitatively different from the target population of studies. 2. People sampled in accessible studies might be different from target population.	1. Nonquality factors might cause improper weighting of study. 2. Omissions in study reports might make conclusions unreliable.	1. Rules for distinguishing patterns from noise might be inappropriate. 2. Review-based evidence might be used to infer causality.	1. Omission of review procedures might make conclusions irreproducible. 2. Omissions of review findings and study procedures might make conclusions obsolete.

SOURCE: From *Synthesizing Research: A Guide for Literature Synthesis,* 3rd ed., by H. M. Cooper, 1998. Reprinted with pemission of SAGE.

abstraction. That is, conceptual definitions can differ in breadth, or in the number of events to which they refer. Let's take as an example the concept of homework. One synthesist may consider as homework only assignments meant to have students practice what they have learned in class, whereas another may include assignments to visit museums or to watch certain television programs. In such a case, the second synthesist employs a broader conception of homework, and this synthesis will likely contain more research than will the first.

As in primary research, in order to relate concepts to concrete events, the variables of interest in a research synthesis also must be operationally defined. An operational definition provides a description of the characteristics of observable events that are used to determine whether the event represents an occurrence of the conceptual variable. Synthesists can also vary in the way operations are treated *after* the relevant research has been retrieved. Thus, synthesists who employ identical conceptual definitions of homework and who include the same set of studies can still reach decidedly different conclusions if one synthesist retrieved more information about the features of studies and recognized a relation between a study feature and outcome that the other synthesist did not test. One synthesist might discover that the outcomes of homework studies depended on whether textbook or teacher-developed tests were used to assess impact, whereas another synthesist never even coded studies based on this feature of the outcome measure.

Each difference in how a problem is formulated introduces a potential threat to the trustworthiness of a synthesis' conclusions. First, synthesists who focus on very narrow conceptualizations provide little information about how many different contexts a finding applies to. Therefore, synthesists who employ broad conceptual definitions can *potentially* produce more valid conclusions than ones using narrow definitions. However, broad definitions can lead to the erroneous conclusion that research results are insensitive to variations in a study's context. We can assume, therefore, that synthesists who examine more operational details within their broader constructs will produce more trustworthy conclusions. These synthesists present more information about contextual variations that do and do not influence the synthesis outcome.

The Literature Search Stage

The decisions a synthesist makes during the literature search determine the nature of studies that will ultimately form the basis for conclusions. Identifying populations for research syntheses is complicated by the fact that syntheses involve two targets. First, a synthesist wants the findings to reflect the results of *all previous research* on the problem. The synthesist can exert some control over whether this goal is achieved through their choice of information sources. Second, the synthesist hopes that the included studies will allow generalizations to the *individuals or other units that interest researchers in the topic area.* The synthesist's influence is constrained at this point by the types of individuals or units who were sampled by the primary researchers. Thus, a synthesis of the homework research first should include as many of the previous studies as the synthesist can find, and it is hoped that these studies will include all the types of students for whom homework is a relevant issue.

Some discrepancies in synthesis conclusions are created by differences in the sources synthesists use to retrieve studies, such as journal networks, reference databases, listservs, and personal communications. The studies available through different sources are often different from one another. The first concern with the literature search is that the synthesis may not include, and probably will not include, all studies pertinent to the topic of interest. Synthesists who have used the broadest sources of information are most likely to retrieve a set of results that resembles the entire population of previous research. However, methodologists do differ in their opinions about how exhaustive a literature search needs to be, especially as it pertains to the inclusion of unpublished research. We take up this debate in the following sections.

The second concern that arises during the literature search is that the participants or other units in the retrieved studies may not represent all units in the target population. For instance, it may be that little or no research has been conducted that examines the effects of homework on first- or second-grade students. The synthesist cannot be faulted for the existence of this gap *if* the retrieval procedures used were exhaustive. However, synthesists who qualify conclusions with information about the kinds of units missing or overrepresented in studies probably run less risk of making overly broad generalizations.

The Data Evaluation Stage

After the literature is collected, the synthesist makes critical judgments about the quality of individual studies. Each study is examined to determine whether it is contaminated by factors irrelevant to the problem under consideration. Then, trained personnel use standardized coding procedures to extract the desired information from research reports.

Differences in syntheses are created by differences in synthesists' criteria for evaluating the quality of research. Just how this evaluation ought to proceed is another source of disagreement among researchers that we will address more fully below. Relatedly, variation in conclusions is created when factors other than research quality affect synthesists' decisions, for example, the reputation or institution of the primary researchers, or the research findings. The use of any criteria other than methodological quality ought to be considered a threat to the validity of a research synthesis (e.g., Mahoney, 1977).

A second threat to trustworthiness during research evaluation is completely beyond the control of the synthesist. This threat involves incomplete reporting by primary researchers. If the synthesist must estimate or omit what happened in these studies, wider confidence intervals must be placed around synthesis conclusions. We will examine some solutions to the problem of missing data below.

A third threat to the validity of conclusions drawn from a synthesis can result because synthesists are not immune to making mistakes themselves in coding information from reports. To address this issue, it is recommended that two or more synthesists code either all or a subset of studies in the synthesis. The extent to which study information has been reliably extracted from research reports can then be assessed by performing some sort of reliability assessment. This involves employing procedures akin to those used in assessing interjudge reliability in other research domains (e.g., Lipsey & Wilson, 2001; Orwin, 1994). A second strategy involves having two synthesists independently code all the studies in the synthesis. Disagreements then may be resolved in conference or by a third reader. This procedure raises the effective reliability of codes to very high levels.

The Analysis and Interpretation Stage

During analysis and interpretation, the separate research reports collected by the synthesist are integrated into a unified statement about the research problem. It is at this stage that the synthesist must decide whether or not to use meta-analysis. Synthesis conclusions can differ because synthesists employ different analytic interpretation techniques. A systematic relation that cannot be distinguished from noise under one set of rules may be discernible under another set.

One source of concern during the analysis and interpretation of studies involves the rules of inference employed by the synthesist. In nonquantitative syntheses, it is difficult to gauge the appropriateness of inference rules because they are not very often made explicit. For meta-analyses, the suppositions of statistical tests are generally known, and some statistical biases can be removed. Regardless of the strategy used for analysis and interpretation, the possibility always exists that the synthesist has used an invalid rule for inferring a characteristic of the target population. For this reason, the number of primary studies available, the degree of statistical detail presented in research reports, and the frequency of methodological replications need to be assessed before determining whether to perform a meta-analysis.

Meta-analysis should be the default option when the goal of a synthesis is to summarize a research literature for purposes of making a general statement about the support for, or size of, a relationship between variables. However, there are some instances in which the use of meta-analysis might be less appropriate, or perhaps completely unnecessary. First, meta-analysis is improper if the goal of the synthesis is to critically appraise a research literature study-by-study or to identify particular studies central to a field. Second, meta-analysis may be inappropriate in cases where conceptual and methodological approaches to research on a topic have changed over time. Third, under certain conditions meta-analysis might not lead to the kinds of generalizations the synthesist wishes to make. Under these circumstances, the synthesist might convincingly establish the generalization of a finding using conceptual and theoretical bridges rather than statistical ones. Finally, even if the synthesist wishes to summate statistical results across studies on the same topic, the studies might have been conducted using decidedly different methodologies, participants, and outcome measures. In such cases, statistical combinations might mask important differences in research findings. In these instances, it may make the most sense not to use meta-analysis, or to conduct several discrete meta-analyses within the same synthesis. Regardless of the technique used to analyze and integrate the results of individual studies, all research synthesists should provide justification for their methods and ensure that the synthesis techniques employed are transparent to the reader.

A second concern involves the misinterpretation of synthesis-based evidence as supporting statements about causality. For example, it might be that a study finding a larger-than-normal effect of homework on achievement was conducted at an upper-income school. However, it might also be the case, known or unknown to the synthesist, that this study used unusually long homework assignments. The synthesist cannot discern, therefore, which characteristic of the study, if either, produced the larger effect. Thus, when different study characteristics are found associated with the effects of a treatment, the synthesist should recommend that future researchers examine these factors within a single experiment.

The Public Presentation Stage

Finally, the production of a document describing the synthesis is a task with important implications for the accumulation of knowledge. Two threats to validity accompany report writing. First, the omission of details about how the synthesis was conducted reduces the possibility that others can replicate the conclusions. The second threat involves the omission of evidence that others find important. A synthesis will quickly become obsolete if it does not address the variables and relations that are (or will be) important to an area.

The Elements of Meta-Analysis

Suppose a research synthesist is interested in whether fear-arousing advertisements can be used to persuade adolescents that smoking is bad. Suppose further that the (hypothetical) synthesist is able to locate eight studies, each of which examined the question of interest. Of these, six studies reported nonsignificant differences between attitudes of adolescents exposed and not exposed to fear-arousing ads and two reported significant differences indicating less favorable attitudes held by adolescent viewers. One was significant at $p < .05$ and one at $p < .02$ (both two-tailed). Can the synthesist reject the null hypothesis that the ads had no effect?

There are multiple methods a research synthesist could employ to answer this question. First, the synthesist could cull through the eight reports, isolate those studies that present results counter to their own position, discard these disconfirming studies due to methodological limitations, and present the remaining supportive studies as presenting the truth of the matter. Such a research synthesis would be viewed with extreme skepticism. It would contribute little to answering the question.

The Vote Count

As an alternative procedure, the synthesist could take each report and place it into one of the three piles: statistically significant findings that indicate that ads were effective, statistically significant findings that indicate that the ads created more positive attitudes toward smoking (in this case, the pile would have no studies), and nonsignificant findings that do not permit rejection of the hypothesis that the fear-arousing ads had no effect. The synthesist then would declare the largest pile the winner. In our example, the null hypothesis wins.

This vote count of significant findings has much intuitive appeal and has been used quite often. However, the strategy is unacceptably conservative. The problem is that chance alone should produce only about 5% of all reports falsely indicating that viewing the ads created more negative attitudes toward smoking. Therefore, depending on the number of studies, 10% or less of positive and statistically significant findings might indicate a real difference due to the ads. However, the vote-counting strategy requires that a minimum 34% of findings be positive and statistically significant before the hypothesis is ruled a winner. Thus, the vote counting of significant findings could, and often does, lead to the suggested abandonment of hypotheses (and effective treatment programs) when, in fact, no such conclusion is warranted.

Hedges and Olkin (1980) describe a different way to perform vote counts in research synthesis. This procedure involves (a) counting the number of positive and negative results, regardless of significance, and (b) applying the sign test to determine if one direction appears in the literature more often than would be expected by chance. This vote-count method has the advantage of using all studies but suffers because it does not weight a study's contribution by its sample size. Thus, a study with 100 participants is given weight equal to a study with 1,000 participants. This is a potential problem because large samples are likely to provide more precise answers to questions. Therefore, results from larger samples should be given more weight. Furthermore, the revealed magnitude of the hypothesized relation (or impact of the treatment under evaluation) in each study is not considered—a study showing a small positive attitude change is given equal weight to a study showing a large negative attitude change. Still, the vote count of directional findings can be an informative complement to other meta-analytic procedures and can even be used to generate an effect size estimate (see Bushman, 1994; Hedges & Olkin, 1985).

Estimating Effect Sizes

While vote-counting addresses the question of whether or not an effect exists; it gives no information about whether that effect is large or small, important or trivial. Therefore, the question of greatest importance is often not "Do fear-arousing ads create more negative attitudes toward smoking in adolescents, yes or no?" Instead, the question should be "How much of an effect do fear-arousing ads have?" The answer might be zero or it might be either a positive or a negative value. Furthermore, the synthesist is likely interested in what factors influence the effect of fear-arousing ads. Given these new questions, the synthesist would turn to the calculation of average effect sizes.

Cohen (1988) has defined an effect size as "the degree to which the phenomenon is present in the population, or the degree to which the null hypothesis is false" (pp. 9–10). In meta-analysis, effect sizes are (a) calculated for the outcomes of studies (or sometimes comparisons within studies), (b) averaged across studies to estimate general magnitudes of effect, and (c) compared between studies to discover if variations in study outcomes exist and, if so, what features of studies might account for them.

Although numerous estimates of effect size are available, three dominate the literature. The first, called the *d*-index by Cohen (1988; also see Hedges & Olkin, 1985; Rosenthal, 1994), is a scale-free measure of the separation between two group means. Calculating the *d*-index for any study involves dividing the difference between the two group means by either their average standard deviation or the standard deviation of the control group. For example, Cooper, Robinson, and Patall (2006) examined the difference in academic achievement of students who did and did not do homework. Across five studies that manipulated the presence of homework, the average *d*-index was 0.60 favoring the homework doers. Thus, the average academic achievement of students who did homework was 0.60 standard deviations above the average score of students who did not.

Figure 11.1 presents the *d*-indices associated with three hypothetical studies. In Figure 11.1a, the fear-arousing ad has no effect on adolescents' reported attitudes toward smoking, thus $d = 0$. In Figure 11.1b, the average adolescent viewing the ad has an attitude score that is four tenths of a standard deviation more negative than the average adolescent viewing control ads. Here, $d = 0.40$. In Figure 11.1c, $d = 0.85$, indicating an even greater separation between the two group means.

In many instances, synthesists will find that primary researchers do not report the means and standard deviations of the separate groups. For such cases, meta-analysts can use one of a number of computational formulas that do not require means and standard deviations. The interested reader may refer to Rosenthal (1994) or Lipsey and Wilson's (2001) for listings of algebraically equivalent formulas that can be used to compute an effect size from various statistical information.

Another effect size metric is the *r*-index, or the Pearson product-moment correlation coefficient. Typically, it is used to measure the degree of linear relation between two variables. The correlation coefficient is familiar to most researchers and is most appropriate when describing the relationship between two continuous variables. For example, Cooper and colleagues (2006) found 32 studies that described the correlations between the time a student spent on homework and a measure of academic achievement. The average correlation for the 32 studies was $r = 0.24$, suggesting that more time spent on homework is related to greater academic achievement.

The third effect size metric is the odds ratio. The odds ratio is applicable when both variables are dichotomous and findings are presented as frequencies or proportions. This measure of effect is used most in medical sciences, in which the researcher is often interested in the effect of a treatment on mortality or the appearance or disappearance of disease. It also appears frequently in studies of educational interventions when the outcome of interest is drop-out or retention rates or criminal justice studies where the outcome is recidivism. For example, if the synthesist was interested in whether exposure to fear-arousing ads led adolescents to continue or quit smoking, then an odds ratio would be an appropriate effect size metric. First, the odds of smoking must be determined for each condition, when participants are exposed to fear-arousing advertisements versus control advertisements. Then, the ratio of the odds for being exposed to fear-arousing advertisements over control advertisements is then calculated as the ratio of the odds.

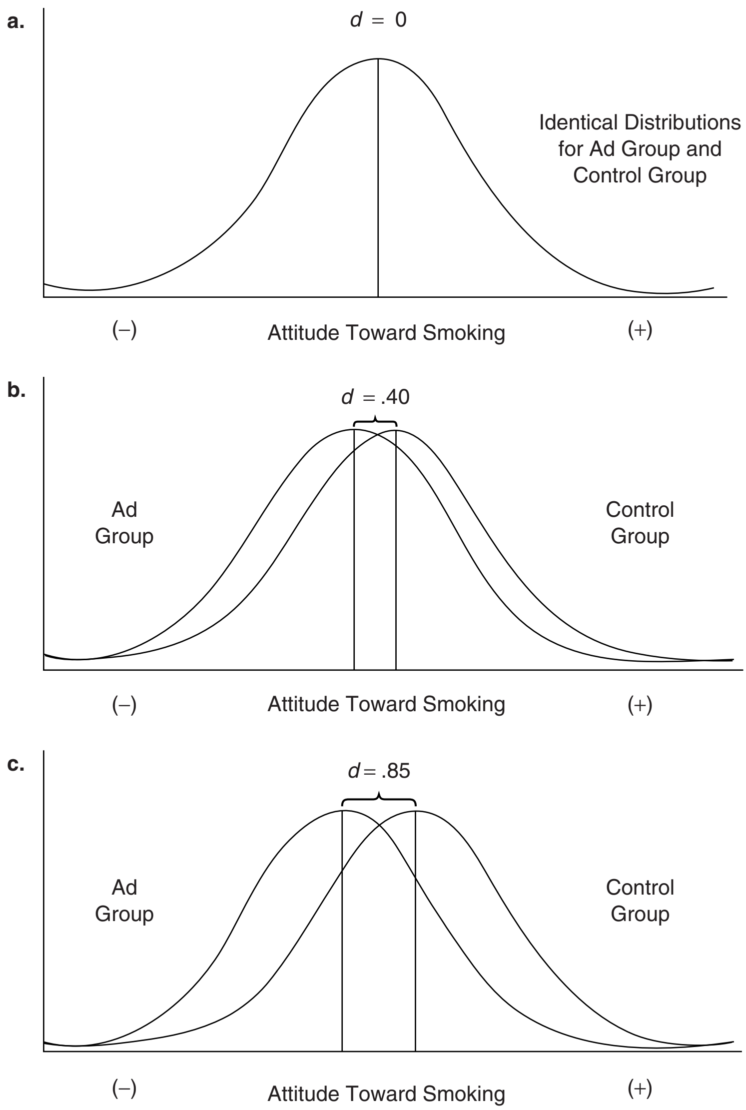

Figure 11.1 Three Relations Between Fear-Arousing Ads and Attitudes Toward Smoking Expressed by the *d*-Index

Averaging Effect Sizes and Measuring Dispersion

The most pivotal outcomes of meta-analyses are the average effect sizes and measures of dispersion that accompany them. State-of-the-art meta-analytic procedures call for the weighting of effect sizes when they are averaged across studies. In the weighted procedure, each independent effect size is first multiplied by the inverse of its variance and the sum of these products is then divided by the sum of the inverses.

The weighting procedure is generally preferred because it gives greater weight to effect sizes based on larger samples, and larger samples give more precise population estimates. Confidence intervals are then calculated to test the null hypothesis that the difference between two means, or the size of a correlation or odds ratio, is zero (Hedges, Cooper, & Bushman, 1992). Going back to the meta-analysis conducted by Cooper and colleagues (2006) looking at the effect of homework on academic achievement, the average *d*-index was 0.60 favoring homework doers, with a 95% confidence interval of 0.38 to 0.82. This confidence interval suggests that the effect of homework on achievement was significantly different from zero. Hedges and Olkin (1985), Shadish and Haddock (1994), and Lipsey and Wilson (2001) provide procedures for calculating the appropriate weights and confidence intervals.

In addition to the confidence interval as a measure of dispersion, meta-analysts usually carry out *homogeneity analyses*. Homogeneity analyses allow the meta-analyst to explore whether effect sizes vary from one study to the next. A homogeneity analysis compares the amount of variance in an observed set of effect sizes with the amount of variance that would be expected by sampling error alone and provides calculation of how probable it is that the variance exhibited by the effect sizes would be observed if only sampling error was making them different. If there is greater variation in effects than would be expected by chance, then the meta-analyst can begin the process of examining moderators of comparison outcomes. For example, in Cooper and colleagues meta-analysis on the effect of homework, the test of homogeneity revealed that the effect sizes were not significant, suggesting that the meta-analyst cannot reject the hypothesis that the effects from different studies are estimating the same underlying population value. In the case in which the observed variance is not significantly different from that expected by sampling error alone, many statisticians advise that meta-analysts stop the analysis there and not look for moderators. After all, chance is the most parsimonious explanation for the variation in effect sizes. Others suggest that meta-analysts may search for moderators in the absence of a statistically significant homogeneity analysis if there are good theoretical or practical reasons for doing so.

An alternative approach to examining if effect sizes vary across studies also compares the observed variation in obtained effect sizes with the variation expected due to sampling error, that is, the expected variance in effect sizes given that all observed effects are estimating the same underlying population value (Hunter & Schmidt, 2004). However, a formal statistical test of the difference between these two values is typically not carried out. Rather, meta-analysts adopt a critical value for the ratio of observed-to-expected variance to use as a means for rejecting the null hypothesis. In this approach, meta-analysts might also adjust effect sizes to account for methodological artifacts such as sampling error, range restrictions, or unreliability of measurements. This method has been applied most often in the areas of industrial and organizational psychology.

Moderator Analyses

Another advantage of performing a statistical integration of research is that it allows synthesists to test hypotheses about why the outcomes of studies differ. To continue with the fear-arousing ad example, the synthesist might calculate average *d*-indexes for subsets of studies, deciding that he or she wants different estimates based on certain characteristics of the data. For example, the synthesist might want to compare separate estimates for studies that use different outcomes, distinguishing between those that measured likelihood of smoking and those that measured attitude toward smoking. Or, the synthesist might wish to compare the average effect sizes for different media formats, distinguishing print from video advertisements. Or, the synthesist might want to look at whether advertisements are differentially effective for males and females.

The ability to ask these questions about variables that moderate effects reveals one of the major contributions of research synthesis. Specifically, even if no individual study has compared different outcomes, media, or adolescent sexes, by comparing results across studies the synthesist can get a first hint about whether these variables would be important to look at in future research and/or as guides to policy.

Without the aid of statistics, the synthesist simply examines the differences in outcomes across studies, groups them informally by study features, and decides (based on an "interocular inference test") whether the feature is a significant predictor of variation in outcomes. At best, this method is imprecise. At worst, it leads to incorrect inferences. In contrast, meta-analysis provides a formal means for testing whether different features of studies explain variation in their outcomes. After calculating the average effect sizes for different subgroups of studies, the synthesist can statistically test whether these factors are reliably associated with different magnitudes of effect also using homogeneity analyses. As previously suggested, homogeneity analysis allows meta-analysts to test whether sampling error alone accounts for variation in effect sizes or whether features of studies, samples, treatment designs, or outcome measures also play a role. This test is analogous to conducting an analysis of variance, in that a significant homogeneity statistic indicates that at least one group mean differs from the others. It is relatively simple to carry out a homogeneity analysis; formulas are described in Cooper (1998), Cooper and Hedges (1994), Hedges and Olkin (1985), and Lipsey and Wilson (2001).

An alternative strategy for examining whether particular characteristics of studies are related to the sizes of the treatment effect is meta-regression. Unlike the strategy previously discussed, meta-regression allows the meta-analyst to explore the relationship between continuous, as well as categorical, characteristics and effect size, and allows the effects of multiple factors to be investigated simultaneously (Thompson & Higgins, 2002). In our example, imagine that our studies ranged in the duration of exposure to fear-arousing ads. One option would be to group studies into several distinct categories of duration of exposure to fear-arousing ads and continue with subgroup moderator analyses as previously discussed. However, an alternative would be to employ meta-regression, leaving this characteristic continuous. The interested reader may refer to Thompson and Higgins (2002) or Higgins and Thompson (2004) for a full discussion of this method.

In sum, a generic meta-analysis might contain three or four separate sets of statistics: (a) a frequency analysis of positive and negative results, (b) estimates of average effect sizes with confidence intervals, (c) homogeneity analyses to assess dispersion and examine study features that might influence study outcomes, and possibly, (d) regression coefficients if meta-regression is used to examine the relationship between continuous study characteristics and effect size. The need for vote counts diminish as the body of literature grows or if the synthesist provides confidence intervals around effect size estimates.

Difficult Decisions in Research Synthesis

When conducting primary research, investigators encounter decision points at which they have multiple choices about how to proceed. The same is true when conducting research syntheses. Some of these decisions will be easy to make, with choices being dictated by topic area considerations and the nature of the research base. Other decisions will be less clear. Six choice points have been generally perplexing for research synthesists. One occurs during data collection, two during data evaluation, and three during data analysis. These involve (a) how exhaustive the literature search should be, (b) what rules should be used for including or excluding studies from syntheses, (c) how to handle data missing from research reports, (d) how to determine whether separate tests of hypotheses are actually independent of one another, (e) how to decide what model of error underlies the generation of study outcomes, and (f) how to synthesize slopes from multiple regression.

Publish or Perish

Research synthesists disagree about how exhaustive a literature search needs to be. Some synthesists go to great lengths to locate as much relevant material as possible; others are less thorough. Typically, disagreement centers on the importance of including unpublished research in syntheses.

Those in favor of limiting syntheses to only published material argue that publication is an important screening device for maintaining quality control. Because published research has been reviewed for quality, it provides the best evidence available. Also, the inclusion of unpublished material typically does not change the conclusions drawn by synthesists. Therefore, the studies found in unpublished sources do not warrant the additional time and effort needed to obtain them.

Those who argue that research should not be judged based on publication status give three rationales. First, they dispute the claim that published research and unpublished research yield similar results; statistically significant results are more likely to be published (Begg, 1994). That is, studies revealing smaller effects may be systematically censured from the published literature, making relationships appear stronger than if all estimates were retrieved (Rothstein, Sutton, & Borenstein, 2005). Lipsey and Wilson (1993) compared the magnitudes of effects reported in published versus unpublished studies contained in 92 different research syntheses. They reported that the impacts of interventions in unpublished research were, on average, one third smaller than published effects.

Second, even if publication status does relate to the quality of research, there will still be much overlap in the quality of published and unpublished studies. Superior studies sometimes are not submitted or are turned down for publication for other reasons. Inferior studies sometimes find their way into print. Application of the "publish or perish" rule may lead to the omission of numerous high-quality studies and will not ensure that only high-quality studies are included in the synthesis.

And finally, in a meta-analysis, both the reliability of effect size estimates, expressed through the size of confidence intervals, and tests for effect size moderators will depend on the amount of available data. Therefore, synthesists may unnecessarily impede their ability to make confident statistical inferences by excluding unpublished studies (Rothstein et al., 2005).

Consequently, it is accepted practice that rigorous research syntheses should always access multiple channels to retrieve studies and operate with the goal of obtaining all relevant research (Cooper, 1998; Lipsey & Wilson, 2001), regardless of whether or where it was published. If the synthesis includes only published research it must be accompanied by a convincing justification.

Judging the Quality of Primary Studies

Another area of controversy in meta-analysis is related to the publication issue. All research synthesists agree that the quality of a study should dictate how heavily it is weighted when inferences are drawn about a research literature. However, there is disagreement about whether studies should be excluded from syntheses entirely if they are flawed.

Proponents of excluding flawed studies often employ the "garbage in, garbage out" axiom (Eysenck, 1978). They argue that amassing numerous flawed studies cannot replace the need for better-designed ones. Others argue that synthesists should employ the principle of best evidence used in law. This principle argues that "the same evidence that would be essential in one case might be disregarded in another because in the second case there is better evidence available" (Slavin, 1986, p. 6). Thus, a synthesist evaluates the entire literature and then bases decisions on only those studies that are highest in quality, even if these are not ideal.

Opponents of excluding studies contend that flawed studies can, in fact, accumulate to valid inferences. This might happen if the studies do not share the same design flaws but do come to the same result. Furthermore, global decisions about what makes a study good or bad are fraught with difficulty. There is ample evidence that even the most sophisticated researchers can disagree about the dimensions that define quality and how these dimensions apply to particular studies (see Valentine & Cooper, 2005). And finally, opponents of exclusion contend that the effect of research design on study outcomes is an empirical question. Rather than leaving studies out based on disputable, global judgments of rigor, synthesists can examine the operational details of studies empirically for their relation to outcomes. That is, synthesists code study features are known to vary with the strength of inferences they permit (e.g., research design, sampling frame, measurement reliability), and determine if those features covary with effect sizes uncovered by different studies (Berlin & Rennie, 1999; Jüni, Witschi, Bloch, & Egger, 1999). Then, if studies with more desirable features produce results different from other studies, inferences about the literature can be adjusted accordingly (Lipsey & Wilson, 2001).

How to Handle Missing Data

Missing data constitute one of the most frustrating problems faced by research synthesists. Missing data can take two forms. First, the synthesist may miss entire research reports that are pertinent to the topic or that he or she knows about but cannot retrieve. The above discussion of publication bias is relevant to this issue. Second, there may be data missing from the reports themselves. Within a report, missing information might include (a) the magnitude of the effect size (because it is not reported and not enough information is given for the meta-analysts to calculate it) and/or (b) important study characteristics that might be tested as moderators of study outcomes. When data are missing, not only is the size of the sample gathered for the research synthesis reduced but the representativeness of the sample and the validity of the results are compromised, regardless of the quality of the meta-analysis in all other respects (Rothstein et al., 2005).

There are a number of strategies that meta-analysts can use to deal with missing data and data censoring. Rothstein et al. (2005) provide an in-depth treatment of numerous approaches. A number of graphical and statistical tests can be used to assess the possible presence of missing data and data censoring, and the implications of this threat to the validity of the conclusions drawn from the meta-analysis. Techniques include regression methods such as the rank correlation test (Begg & Mazumdar, 1994) and Egger's Test (Egger, Davey Smith, Schneider, & Minder, 1997), funnel plots (Light & Pillemer, 1984), as well as the Trim-and-Fill method (Duval & Tweedie, 2000a, 2000b). Furthermore, strategies for handling missing data within reports have been proposed. Some are simple. These include (a) omitting the cases with data missing from a given analysis or from the meta-analysis entirely, (b) assuming that missing values are equivalent to a very conservative estimate, such as zero, or (c) replacing missing values with the mean value calculated from available cases for that variable. Alternatively, missing data points can be estimated using single-value imputation procedures. More complex methods using multiple imputation procedures (Rubin, 1987) involve the employment of maximum likelihood models, though these methods are not widely used in meta-analysis. Details of these procedures are given by Pigott (1994). Regardless of which method is employed, meta-analysts are obligated to discuss the possibility and impact of missing entire reports and data censoring on the conclusion of the meta-analysis, how much data were missing within reports included in the synthesis, how they handled the situation, and why they chose the methods they did. Furthermore, it is becoming increasingly common practice for meta-analysts with much missing data to conduct their analyses using more than one strategy and determining whether their findings are robust across different missing data assumptions (see Greenhouse & Iyengar, 1994).

Finally, prospective registration and prospective meta-analysis have been recently proposed as two strategies which, if widely adopted, would decrease the occurrence of missing data and minimize publication bias (Berlin & Ghersi, 2005). Prospective registration entails registering a study on its inception when the researcher receives ethical or funding approval, allowing both the description of the study as well as eventual results to be publicly available. This would create an unbiased compilation of studies for subsequent meta-analyses and allow the synthesist to obtain information and results about studies regardless of the significance of their findings or publication status. In prospective meta-analysis, studies are identified and determined to be eligible before the results of any of the studies are known. Prospective meta-analysis may be accomplished when multiple groups of investigators agree to combine their findings on completion. Furthermore, the comparability of research included in the meta-analysis is improved when investigators also decide prospectively to employ the same methods and assessment instruments across studies. Because the studies and specific analyses to be included in the meta-analysis are determined prior to any single study being conducted, missing data and data censoring is virtually eliminated.

Identifying Independent Hypothesis Tests

Meta-analysts must make decisions concerning how to handle multiple effect sizes coming from the same study. These effect sizes may share method variance that make them nonindependent data points. The problem this creates is that the assumption that effects are independent underlies the meta-analysis procedures described above.

Sometimes, a single study can contain multiple estimates of the same relation because (a) more than one measure of the same construct is used and the measures are analyzed separately or (b) results are reported separately for different samples of people. Taken a step further, synthesists also might conclude that the separate but related studies in the same report, or multiple reports from the same laboratory, are not independent.

Meta-analysts employ multiple approaches to handling nonindependent tests. Some treat each effect size as independent, regardless of the number that come from the same study. The strength of this technique is that it does not lose any of the within-study information regarding potential moderators. However, this strategy violates the assumption that the estimates are independent. Furthermore, the results of studies will not be weighted equally in any overall conclusion about results. Rather, studies will contribute to the overall effect in relation to the number of statistical tests contained in it.

Others use the study as the unit of analysis. In this strategy, a mean or median result is calculated to represent the study. This strategy ensures that the assumption of independence is not violated and that each study contributes equally to the overall effect. However, some within-study information may be lost in this approach.

Sophisticated statistical models also have been suggested as a solution to the problem of dependent effect size estimates (Gleser & Olkin, 1994; Raudenbush, Becker, & Kalaian, 1988) but due to their complexity they are still rarely found in practice.

Other meta-analysts suggest a shifting unit (Cooper, 1998). Here, each study is allowed to contribute as many effects as there are categories in the given analysis, but effects within any category are averaged. For example, if a study on whether fear-arousing advertisements promotes change in smoking behavior by adolescents used two different measures, one attitudinal and one behavioral, two separate d-indexes would be calculated. In the shifting unit of analysis approach, for estimating the overall relation between exposure to fear-arousing ads and smoking, statistical independence would be maintained by averaging these two d-indexes prior to entry into the analysis, so that the study only contributes one effect size. However, in an analysis that examined the effect of measurement characteristics, attitudinal or behavioral, on smoking outcomes, this sample would contribute one estimate to each category in the moderator analysis. This shifting unit of analysis approach retains as much data as possible from each study while holding to a minimum any violations of the assumption that data points are independent.

Models of Error

Another aspect of conducting a meta-analysis that has recently received considerable attention involves the decision about whether a fixed-effects or random-effects model of error underlies the generation of study outcomes. In a fixed-effects model, all studies are assumed to be drawn from a common population. As such, variance in effect sizes is assumed to reflect only sampling error, that is, error solely due to participant differences. However, sometimes other features of studies can be viewed as random influences. For example, studies that look at the impact of fear-arousing advertisements on smoking might vary in the length of exposure to ads or in how the ads are introduced to participants. In this case, it may be most appropriate to consider advertisements as randomly sampled from all fear-arousing advertisements. That is, in a random-effect analysis, study-level variance is assumed to be present as an additional source of random influence.

The question meta-analysts must ask is whether the effect sizes in their data set are affected by a large number of these study-level random influences. If it is the case that the meta-analysts suspect a large number of these additional sources of random error, then a random-effects model is most appropriate to take these sources of variance into account. If the meta-analysts suspects that the data are most likely little affected by other sources of random variance, then a fixed-effects model can be applied. Alternatively, Hedges and Vevea (1998) state that fixed-effect models of error are most appropriate when the goal of the research is “to make inferences only about the effect size parameters in the set of studies that are observed (or a set of studies identical to the observed studies except for uncertainty associated with the sampling of subjects)” (p. 3). A further consideration is that in the search for moderators, fixed-effect models may seriously underestimate error variance and random-effects models may seriously overestimate error variance when their assumptions are violated (Overton, 1998).

In view of these competing sets of concerns, the meta-analysts might consider applying both models (e.g., Cooper et al., 2006). Specifically, all analyses could be conducted twice, once employing fixed-effect assumptions and once using random-effect assumptions. Differences in results based on which set of assumptions is used can be incorporated into the interpretation and discussion of findings.

Calculating random-effects estimates of the mean effect size, confidence intervals, and homogeneity statistics are complex and involve a two-stage process. As such, the interested reader should refer to Hedges and Olkin (1985), Raudenbush (1994), and Lipsey and Wilson (2001) for a full discussion of random-effects computation. In addition, several statistical packages have recently been developed specifically for meta-analysis that allow the meta-analysts to easily conduct analyses using both fixed-effects and random-effects assumptions (e.g., Borenstein, Hedges, Higgins, & Rothstein, 2005).

Combining Slopes From Multiple Regressions

Up to this point, the procedures for combining and comparing study results have generally assumed that the measure of effect is a mean difference, correlation, or odds ratio. However, regression analysis is a commonly used technique in the social sciences, particularly for nonexperimental studies. Like the standardized mean difference or correlation coefficient, the regression coefficient, b, or the standardized regression coefficient, β, are also measures of effect size. β will typically be used in meta-analyses because, like the d-index and r-index, it standardizes effect size estimates when different measures are used in different studies. β represents the standardized score change in a predictor variable, controlling for all other predictors, given one unit change in the criterion variable.

Syntheses of regression analyses are difficult to conduct for a variety of reasons. First, models using multiple regression generally differ from study to study. Each study may include different predictors in the regression model and therefore, the slope for the predictor of interest will represent a different partial relationship in each study (Wu & Becker, 2004). Second, the scale of the predictor of interest and outcome may vary across studies. This problem can be overcome by using β, the fully standardized estimate of the slope for a particular predictor. "Half-standardizing" is an alternative way to create similar slopes when only outcomes are dissimilar (Greenwald, Hedges, & Laine, 1996).

If slopes are independently and identically distributed, we can apply standard methods for meta-analysis. Slopes will be identically distributed across studies when the outcome and predictor of interest are measured in a similar fashion, the other predictors in the model are the same across studies, and when predictor and outcome scores are similarly distributed (Becker, 2005). However, it is rare that data sets meet the assumption of being identically and independently distributed. Typically, measures differ across studies and regression models are diverse in terms of which additional variables are included in them. And, because few studies provide descriptive statistics on the variables measured and included in the regression model, it remains difficult to assess whether the assumption that scores are distributed similarly across studies has been met. Given the current limitations, a common method for summarizing the results of regression analyses has been to use a vote-count strategy (see, e.g., Hanushek, 1989 or Cooper et al., 2006). What remains clear is that techniques for synthesizing results from multiple regression analyses need to be more extensively developed and studied.

Judging the Quality of Research Syntheses and Meta-Analyses

Given the potential value and increased dependence on research syntheses for assisting the development of effective explanations for behavior and behavioral interventions, an important question concerns how to distinguish good from bad syntheses. Throughout this chapter, we have suggested points of contention at which decisions the synthesist makes may affect the validity of conclusions drawn from the synthesis. The model of integrative synthesis as scientific research presented in Table 11.1 provides general guidelines for judging the quality of research syntheses. At each stage, explicit questions about synthesis methods that relate to quality are posed: (a) Do the operations appearing in the literature fit the synthesists' abstract definition? (b) Is enough attention paid to the methodological details of the primary studies? (c) Was the literature search thorough? (d) Were primary studies evaluated using explicit and consistent rules? (e) Were valid procedures used to combine the results of independent studies? Matt and Cook (1994) have expanded on this approach to assessing the validity of research synthesis conclusions. For example, Matt and Cook (1994) also suggest that the possibility that the meta-analyst has used an invalid rule for inferring a characteristic of the target population is another threat to the validity of meta-analytic conclusions. In addition, the validity of results might be threatened because of the probabilistic nature of statistical findings. First, as in primary research, the meta-analyst might conduct many statistical tests without adjusting for "synthesis-wise" error rates. Second, because of gaps in the literature, a meta-analyst might discover so few tests of a particular hypothesis that the statistical power of the meta-analysis is low. Shadish, Cook, and Campbell (2002) have expanded Matt and Cook's compendium of threats even further.

In sum, social research methodologists need to continue to identify and systematize criteria for the evaluation of meta-analyses. This effort should guide and facilitate the generation of high-quality research syntheses in the future. As the role of syntheses in our acquisition of knowledge expands, the ability to distinguish good from bad syntheses becomes more critical.

Discussion Questions

1. What is the primary impetus for adoption of meta-analysis in the social sciences?

2. Name several channels by which to search for relevant literature. What are the strengths, weaknesses, and cost-effectiveness of each?

3. Briefly review the key components of a meta-analysis. Discuss any potential threats to validity that may occur as a result of decisions the synthesist makes at the data analysis stage.

4. What criteria are most crucial to consider when evaluating the quality of primary research?

5. What criteria are most crucial to consider when evaluating the quality of a research synthesis?

Exercises

1. Identify a conceptual variable and list the operational definitions associated with it that are known to you now.

2. List the keywords that you would use to search for articles relevant to your conceptual variable in electronic reference databases. Use them to identify other related terms in the thesauri of at least two reference databases. What did you learn about your concepts from the new keywords you discovered? Did the keywords differ for the different reference databases and if so, how?

3. Find several reports that describe research relevant to your topic. How many new operational definitions did you find? Evaluate these with regard to their correspondence to the conceptual variable.

4. Read two research syntheses. Outline what the authors report on each of the following: (a) how the literature search was conducted, (b) what rules were used to decide if studies were relevant to the hypothesis, and (c) what rules were used to decide if cumulative relations existed. Was there any information that the synthesists did not report that would be needed to fully evaluate the quality of the research syntheses?

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PART III

Practical Data Collection

In this section, we move from the concept of research design to a diversity of approaches to collecting, managing, and analyzing data. The first chapter in Part III concentrates on the art of designing good survey questions. Too often, the actual wording of survey questions is overlooked. Fowler and Cosenza, building on a wealth of survey experience, provide valuable information on how to ask questions in Chapter 12. They place the design of questions within a total survey design framework that includes sampling, data collection techniques, interviewer training, and question construction.

The authors offer four characteristics of a good question to help guide question construction. They anticipate some of the question design challenges a researcher might face and provide a number of tips and suggestions for tackling them. One of the more exciting features of the second edition is the inclusion of more than 30 examples of how to avoid the many pitfalls in designing a survey. Fowler and Cosenza also discuss practical decisions that need to be made, such as how many response categories to use in writing questions. Finally, because techniques alone cannot guarantee good questions, the authors discuss three empirical approaches to producing better surveys including focus groups (see Stewart, Shamdasani, & Rook, Chapter 18), cognitive testing, and field pretesting.

A new chapter (Chapter 13) to this second edition is on collecting data on the Internet by Best and Harrison. This is an innovative technology gaining widespread use that will grow as the world gets more connected. Using the Internet offers some special challenges that other data collection approaches do not share. The authors detail how sampling can be accomplished on the Internet. The bottom line is that it is impractical to attempt representative sampling of the general population because of the still limited and biased access to the Internet. However, representative surveys within organizations known to have practically 100% access, such as universities, are feasible.

The authors spend most of the chapter reviewing the details that must be attended to if an Internet survey is going to be successful. Each of these issues may seem small, but together they can make a difference in determining the success of the survey. Studies are reviewed that discuss such seemingly mundane things ranging from how many items should appear on a page to how to write instructions. Best and Harrison provide several different approaches to contacting potential respondents that do not depend on the Internet. There is a wealth of practical advice that researchers using the Internet would be wise to follow.

In Chapter 14, another new addition to this second edition of the *Handbook*, Kane and Trochim provide an overview of concept mapping as it has been or can be used in applied social research. Concept mapping is a structured method for developing maps of theories and ideas, typically generated through a group process, that can be tied to practice. The technique is designed to help a group solve a problem, express a need, or design a plan or intervention.

The authors describe the specific steps that a researcher needs to understand to implement the methodology from the start of a project to the use of the results. The steps include (1) preparing for the process, by identifying a focus and selecting individuals to participate in the process; (2) generating the ideas and statements through a structured brainstorming process; (3) structuring the statements, so that they are sorted and rated, with participant-related information attached to allow comparison of ratings of one subgroup of participants to another; (4) analysis of the statements, integrating both qualitative input and quantitative analysis (including sort aggregation, multidimensional scaling, and hierarchical cluster analysis) that results in the development of concept maps and accompanying reports; (5) interpreting the maps, with involvement of the stakeholders involved in the idea generation now involved in the interpretation of the concept maps, and (6) utilization of the results, either in theory building, program development, and measurement and evaluation. The authors illustrate the method with detailed examples and outline some future areas of likely development in the methodology.

Mangione and Van Ness in Chapter 15 review the principles involved in conducting mail surveys. They start with a discussion of when mail surveys may be the most appropriate data collection method, providing a list of the advantages of mail surveys as well as the situations in which the mail survey is the best method to use. One of the most commonly discussed weaknesses of mail surveys is poor response rates. The authors provide several excellent suggestions for how to improve response rates, including the type of letter to send, the use of return postage, and how to preserve confidentiality. They provide "tip boxes" as well as an extensive discussion on how to remind people to return the survey and the use of incentives, and they offer some surprising conclusions about how the length of a survey affects the return rate. They also make several other practical suggestions with regard to such critical aspects as managing the survey process and how to improve the physical appearance of the survey instrument. Finally, the authors remind us that mail surveys need to follow good practice with regard to the wording of questions and sampling procedures as discussed in the prior chapters by Fowler and Cosenza (Chapter 12) and Henry (Chapter 3).

The third major survey approach involves the use of telephone interviews. In Chapter 16, Lavrakas provides a comprehensive overview of the design and implementation of telephone surveys. Similar to several other contributors to this *Handbook*, Lavrakas takes a total survey approach. This orientation recognizes that all aspects of research are interdependent and that a weakness in any one area will affect the quality of the data collected. For example, the researcher may have done an excellent job in selecting the sample and constructing the interview, but if the interviewers are not properly trained and supervised, the data may not be of sufficient quality. Lavrakas takes the reader through the entire process of conducting telephone surveys, from sample selection to interviewer supervision, and shows how each of these steps is critical to the quality of the data collected.

The next two chapters focus on two qualitative data collection approaches common in applied social research. Ethnography, as Fetterman defines it in Chapter 17, is the art and science of describing a group or culture. He presents an overview of the concepts, methods, equipment, analysis, writing, and ethics involved in conducting ethnographic research. Like other contributors to this volume, Fetterman highlights the need for organization in research but also notes the reality that much of what happens during the research will be unplanned and iterative. In ethnographic research in particular, the ethnographer is a human instrument who is often collecting and analyzing data simultaneously.

Fetterman has made several important changes to this chapter from the first edition. First, in addition to updating the relevant literature that supports the method, he has provided considerable material on several new tools that can be used by ethnographers as extensions of the human instrument. In particular, more detail is provided on the use of qualitative database programs that can allow for the development of emergent themes and help the ethnographer organize the data collected. Tools also described are those that help ethnographers communicate with colleagues and key actors in the field such as Internet telephony and videoconferencing and those that help the ethnographer more efficiently and completely collect and organize data in the field, such as digital voice recorders, digital camcorders, cinema and digital videos, and personal digital assistants. Finally, supplementing the exercises and discussion questions that all authors have provided are pictures that offer visual images to illustrate ethnographic concepts.

In Chapter 18, Stewart, Shamdasani, and Rook describe the collection of information from focus groups, a technique that is being used with increasing frequency in applied research. A focus group is a group of 8 to 12 persons who meet for a session of approximately 2 hours to discuss the topic presented to them by the researcher. Although typically used in the early stages of research projects to help frame the focus of an effort or to formulate a more structured set of survey questions, focus groups have also been used in hypothesis testing.

Stewart and colleagues describe the history of focus group research, when focus groups are most useful to use in applied research, and their advantages and limitations relative to other data collection methods. This chapter should help readers decide whether the focus group approach would be useful for answering their research questions. The authors outline the steps to designing, conducting, and analyzing a focus group, including framing its purpose, selecting the participants, developing the interview guide, conducting the group, and analyzing and interpreting the data. Opportunities offered by new technology both in analyzing the data and in conducting "virtual groups," groups that cannot be brought to one location, are described. The authors provide the important reminder that, regardless of the technology used either in the analysis or conduct of the focus group, validity is not ensured and needs to be addressed throughout the focus group process.

CHAPTER 12

Design and Evaluation of Survey Questions

Floyd J. Fowler Jr.

Carol Cosenza

The quality of data from a survey depends on the size and representativeness of the sample from which data are collected; the techniques used for collecting the data; the quality of the interviewing, if interviewers are used; and the extent to which the questions are good measures. Methodologists have a concept that they call total survey design (e.g., Fowler, 2002; Groves et al., 2004; Weisberg, 2005). By that, they refer to the perspective of looking at all sources of error, not just a single source, when making survey design decisions. The quality of data from a survey is no better than the worst aspect of the methodology.

When Sudman and Bradburn (1974) looked at sources of error in surveys, they concluded that perhaps the major source of error in survey estimates was the design of survey questions. When Fowler and Mangione (1990) looked at strategies for reducing interviewer effects on data, they, too, concluded that question design was one of the most important roads to minimizing interview effects on data. Moreover, improving the design and evaluation of survey questions is one of the least expensive components of the survey process. Compared with significantly increasing the size of a sample, or even the efforts required to improve response rates significantly, improving questions is very cost-effective. Thus, from the perspective of total survey design, investing in the design and evaluation of questions is a best buy, one of the endeavors that is most likely to yield results in the form of better, more error-free data.

What Is a Good Question?

A good question is one that produces answers that are reliable and valid measures of something we want to describe. Reliability is used here in the classic psychometric sense of the extent to which answers are consistent: When the state of what is being described is consistent, the answers are consistent as well (Nunnally, 1978). Validity, in turn, is the extent to which answers correspond to some hypothetical "true value" of what we are trying to describe or measure (Cronbach & Meehl, 1955).

There are four basic characteristics of questions and answers that are fundamental to a good measurement process:

1. Questions need to be consistently understood.
2. Respondents need to have access to the information required to answer the question.
3. The way in which respondents are asked to answer the question must provide an appropriate way to report what they have to say.
4. Respondents must be willing to provide the answers called for in the question.

A critical part of the science of survey research is the empirical evaluation of survey questions. Like measurement in all sciences, the quality of measurement in survey research varies. Good science entails attempting to minimize error and taking steps to measure the remaining error so that we know how good our data are and we can continue to improve our methods.

There are two types of question evaluation: those aimed at evaluating how well questions meet the four standards above, which can be thought of as process standards, and those aimed at assessing the validity of answers that result. In order to assess the extent to which questions meet process standards, we can take a number of possible steps. These include (a) systematic question review; (b) cognitive interviews, in which people's comprehension of questions and how they go about answering questions is probed and evaluated; and (c) field pretests under realistic conditions. Each of these activities has strengths and limitations in terms of the kinds of information they provide about questions. However, in the past decade, there has been growing appreciation of the importance of evaluating questions before using them in a research project, and a great deal has been learned about how to use these techniques to provide systematic information about questions (see, e.g., Presser et al., 2004).

The evaluation of validity usually occurs after data have been collected and entails specific analyses aimed at producing evidence that the answers are measuring what they were intended to measure.

We begin this chapter by describing what we know about how to design survey questions. The discussion is separated by whether the focus is on measuring objective facts or subjective states of respondents, such as knowledge, opinions, or feelings. The latter part of the chapter is devoted to the objective evaluation of survey questions. The overall goal in this chapter is to describe how to design survey questions that will be good measures.

Question Objectives

One of the hardest tasks for methodologists is to induce researchers, people who want to collect data, to define their objectives. The difference between a question objective and the question itself is a critical distinction. The objective defines the kind of information that is needed. Designing the particular question or questions to achieve the objective is an entirely different step. In fact, this chapter is basically about the process of going from a question objective to a set of words, a question, the answers to which will achieve that objective.

Sometimes the distance between the objective and the question is short:

Objective: Age

Possible Example 1a: How old were you on your last birthday?

Possible Example 1b: On what date were you born?

The answer to either of these questions probably will meet this question objective most of the time. An ambiguity might be whether age is required to the exact year, or whether broad categories, or a rounded number, will suffice. Example 1a produces more ages rounded to 0 or 5. Example 1b may be less sensitive to answer than Example 1a for some people, because it does not require that the respondent explicitly state an age. There also may be some difference between the questions in how likely people are to err in their answers due to recall or miscalculations. However, the relationship between the objective and the information asked for in the questions is close, and the two questions yield similar results.

Objective: Ethnic background

Possible Example 2a: Do you consider yourself to be black, white, Asian, American Indian, something else, or some combination?

Possible Example 2b: In what country were you born?

Possible Example 2c: Most people in this country think of themselves as American. However, in addition, is there any particular racial, ethnic, or nationality group that you consider yourself to be part of?

Each of these three questions has been used as a measure of ethnicity. However, the results are very different. Which question is best depends on the way the analyst plans to use the results and what is to be measured. The first question measures race, but it does not take into account national or cultural issues. The most common measures in the United States include at least one additional question that identifies those of Hispanic background. However, Hispanic is not a race; it cuts across race, as there are black, white, and Indian Hispanics. Example 2a also has a perceptual component for all those respondents who have some degree of racial mixture in their backgrounds, so that two people with the same racial backgrounds could answer the question differently.

In contrast, Example 2b—country of origin—is a good question in that it is easy to answer, and the answer is unambiguous. However, if one is interested in measuring cultural influences, Example 2b may be too limited. There are many people who were born in the United States for whom the effects of some ethnic or national culture is a critical part of their background. To capture such people, one might want to know where their parents were born or even grandparents. Also, even for those whose families came to the United States in recent generations, there are differences in how important those influences are. Some families live in areas with others with the same backgrounds and encourage their children to marry those with similar backgrounds, while others strive to become fully integrated into United States society and lose the trappings of their cultural pasts. If one wants to measure the presence of a cultural identity or potential influence, perhaps a question such as Example 2c would be the best measure of the desired construct.

A good question objective has to be more specific than simply "ethnic background." More broadly, a question objective can be defined only within the context of an analysis plan, a clear view of how the information will be used to meet a set of overall research objectives. Measuring ethnic background can be a way to measure the norms and expectations with which one was raised, language experience, the potential for having friends and social support, and how one is likely to be treated by the broader society. It is necessary to be explicit about the question objectives and how answers will be used in an analysis in order to choose a question.

It is good practice to produce a detailed list of question objectives and an analysis plan that outlines how the data will be used before designing a survey instrument.

Questions to Gather Factual Data

Consistent Understanding

One basic part of having people accurately report factual or objective information is ensuring that all respondents have the same understanding of what is to be reported, so that the researcher is sure that the same definitions have been used across all respondents. If respondents do not consistently understand what a question is asking for, the resulting answers are unlikely to be accurate measures. This is one of the most difficult tasks for the designer of survey questions, and failure to do it properly is a major source of error in survey research.

While it is probably not possible to write a question that everyone will understand in exactly the same way, there are certain characteristics of questions that make them more likely to be misunderstood. Questions that include jargon, unfamiliar or technical terms or phrases that are abstract are more likely to be understood inconsistently. Vocabulary choices when writing a question should take into account the sampled population. While it may be acceptable to use phrases such as "coinsurance," "low-income subsidy," and "IEP or 504 Plan" in surveys for certain populations (such as medical insurance benefit specialists, housing advocates, or special education teachers), for most people in a general population survey, these phrases are examples of unfamiliar or technical terms that will not be understood.

When a question includes words or phrases that a respondent cannot define, there are several things that the respondents can do—they can try to guess what the question is asking and answer the question anyway, they can skip the question and not answer it at all, or they can just choose an answer at random. All these options are detrimental to the reliability of the data. It is the responsibility of the researcher to provide the respondent with all the information needed to answer a question—including definitions or examples of words that may not be universally understood.

Sometimes, question ambiguity arises from using a common abstract word or phrase without a definition. When that happens, it is easy for respondents to wrongly assume that they know what the question means.

Example 3: Do you own a car?

The meaning of this question is unclear. It could be about access to transportation (e.g., trying to find out whether the respondent has a way to get to work or doctor appointments). Or it could be about material wealth (owning vs. leasing a vehicle). What if someone "owns" the car but someone else has it or drives it all the time? And what about trucks, SUVs, and motorcycles? Are they included here? Once the researcher knows what the goal is, a clearer question can be written.

Alternative 3: Do you have access to a car or other vehicle you can use every day to get to work?

Proper question design means making certain that the researcher and all respondents are using the same definitions when classifying people or counting events. In general, researchers have tended to solve the problem by giving the respondents a definition to use and then asking the respondents to do the classification work.

Example 4: A health provider is anyone you would see for health care. In the last 12 months, not counting the times you needed health care right away, did you make any appointments with a doctor or other health provider for health care?

Without a definition, respondents were confused about who should be included. Do nurse practitioners count? What about chiropractors? With the definition—"anyone you would see for health care"—the respondent has been given some guidance. However, sometimes the concept a researcher wants to measure is very complex—for example, income.

Example 5: What is your income?

The problem with this question is that there are numerous issues about how to calculate income. Among them are whether income is current or for some period of time in the past, whether it is only income earned from salaries and wages or includes income from other sources, and whether it is only the person's own income that is at issue or includes income of others in which the respondent might share.

Alternative 5: Next we need to get an estimate of the total income for you and family members living with you during 2008. When you calculate income, we

> would like you to include what you and other family members living with you made from jobs and also any income that you or other family members may have had from other sources, such as rents, welfare payments, social security, pensions, or even interest from stocks, bonds, or savings. So including income from all sources, before deductions for taxes, for you and for family members living with you, how much was your total family income in 2008?

This is a very complicated definition, but it would be necessary because what the researcher wants to measure is a very complicated concept. However, even this complex definition avoids, or fails to address, some important issues. For example, what does the respondent do if household composition at the time of the interview is different from how it was at the beginning of 2008?

When the rules for counting events are quite complex, providing a comprehensive, complex definition probably is not the right answer. At the extreme, respondents may end up more confused and the results may actually be worse than if definitions were not provided.

A different approach is probably needed. One approach is to add extra questions to cover commonly omitted kinds of events.

> *Example 6:* In the last 12 months, how many times have you seen or talked with a doctor?

It has been found that receiving advice over the telephone from a physician, seeing nurses or assistants who work for a physician, and receiving services from physicians who are not always thought of as "medical doctors" (such as psychiatrists) often are left out. One solution is to ask a general question and then ask some follow-up questions:

> *Example 6a:* Other then the visits to doctors that you just mentioned, how many times in the last 12 months have you gotten medical advice from a physician over the telephone?

> *Example 6b:* Other than what you've already mentioned, how many times in the last 12 months have you gotten medical services from a psychiatrist?

Using multiple questions to cover all aspects of what is to be reported, rather than trying to pack everything into a single definition, can be an effective way to simplify the reporting tasks for respondents. It is one of the easiest ways to make sure that commonly omitted types of events are included in the total count that is obtained.

In addition to being able to understand the vocabulary used in a question, it is also important for respondents to understand for what time period they should be answering. For any question that could reasonably be expected to vary from day to day, week to week, or month to month, the researcher should include a time frame or reference period. Without a time frame, it is left up to the respondent whether to answer about today, last week, or some longer period.

Example 7: How often do you ski—more than once a week, about once a week, two to four times a month, or less often than that?

If this question is asked in the winter, the same person could answer differently than if it was asked in the summer. By not including any specific reference period, respondents must choose on their own what time periods to think about. If they choose to think about the last 30 days, the answer will likely be different than if they think about the entire year. By allowing respondents to make their own choices about a time frame, answers can vary for that reason alone and the data reliability is reduced. In addition, this question also assumes a pattern of regularity that may not always be the case. The alternative question below fixes both of these problems.

Alternative 7: In the last 12 months, on about how many days did you ski?

Questions that require multiple cognitive steps: Answering questions is a cognitively complex task that is all the more challenging when researchers design questions that require several cognitive steps to arrive at an answer. When there are multiple concepts asked about in a single question, the respondent must decide on his or her own how to handle the separate pieces.

Example 8: In the last 6 months, how often did you buy a newspaper at a newsstand—always, sometimes, rarely, never?

This question requires at least three cognitive steps. First, respondents have to decide whether they have bought any newspapers in the last 6 months. Then, they have to figure out how many times they bought a newspaper at a newsstand. Then, they have to figure out the ratio of newspapers bought at a newsstand to newspapers bought elsewhere and decide which of the adjectival responses best describes their situation. A better way to ask these kinds of complex questions is to ask about each part separately.

Alternative 8a: In the last 30 days, about how many newspapers have you bought?

Alternative 8b: (if any) And about how many of those newspapers did you buy at a newsstand?

In a survey, how respondents understand a question is influenced not just by the words in the question itself, but also by the other questions around it. Context matters.

Example 9a: Do you belong to a gym or health club?

Example 9b: In the last 7 days, how often have you exercised?

Although there is nothing in Example 9b that mentions a gym or health club, it would not be surprising if a respondent felt that it was asking about exercising at a gym or health club. The simplest solution would be to ask the questions in reverse order, asking about exercising first, so question 9a would not be part of the context for 9b. Another alternative would be to add a phrase to the second question asking respondents to think about all the different places that they might have exercised.

Another example of the influence that context can have on answers is given below.

Example 10a: The next questions refer to the joints in your body. Please do not include the back or neck. During the past 30 days, have you had symptoms of pain, aching, or stiffness in or around a joint?

Example 10b:

1. Have you ever been told by a doctor or other health professional that you have some form of arthritis, rheumatoid arthritis, gout, lupus, or fibromyalgia?
2. During the past 30 days, have you had symptoms of pain, aching, or stiffness in or around a joint?

Example 10a above asks first about pain and then asks about a diagnosis. In a recent study comparing the two examples, 58.8% of the people who answered Example 10a answered that they had joint pain while 49.4% who answered Example 10b said that they had joint pain. One possible explanation might be that asking about the long list of medical conditions first gives the respondent the sense that the questions are asking only about significant or major pain. In Example 10a, with no previous mention of medical diagnosis, people may be more likely to report less significant pain.

There are other characteristics of questions that can lead to inconsistent understanding by respondents. Good survey questions about factual data should be about what people know and can answer. Since behavior is largely determined by situations, asking hypothetical questions about what respondents might do in the future is less of a factual question and more of a guess or opinion. In general, people are not good at predicting what they will do in circumstances that they have not yet encountered. Since it has yet to happen, respondents have to fill in what they imagine might happen. However, the more experience a respondent has with similar situations, the more likely meaningful answers can be provided. When questions are truly hypothetical, asking about situations or things with which the respondent has little or no experience, answers are unlikely to be meaningful. Moreover, because people have to fill in their own assumptions about what the situation will be, each respondent is likely to be answering a different question.

Another potential problem is multibarreled questions. If a question asks about more than one issue (e.g., "Do you want to be rich and famous?" or "Are you unhappy and overworked?"), respondents are faced with the task of deciding, on their own, what to do if the answer to each "barrel" is different ("I am not overworked, but I am unhappy"). To the extent that they decide differently, respondents are answering different questions.

Knowing and Remembering

Once a question has been designed so that all respondents understand what is wanted, the next issue is whether or not respondents have the information needed to answer the question. There are three possible sources of problems:

1. The respondent may never have had the information needed to answer the question.
2. The respondent may once have known the information, but may have difficulty recalling it.
3. For questions that require reporting events that occurred in a specific time period, respondents may recall that the events occurred, but have difficulty accurately placing them in the time frame called for in the question.

Lack of Knowledge

Sometimes, respondents simply do not have the information needed to answer a question. One critical part of the preliminary work a researcher must do in designing a survey instrument is to find out whether or not questions have been included to which some respondents do not know the answers. The limit of survey research is what people are able and willing to report. If a researcher wants to find out something that is not commonly known by respondents, the researcher must find another way to get the information.

Example 11: In the last 6 months, how often did you feel that your personal doctor had all the information needed to correctly diagnose and treat your health problems?

In addition to the fact that this is a double-barreled question (diagnosis and treatment may not be a unified concept), in general, patients and consumers of health care are not able to answer about the technical quality of physicians. While they can provide important information about their interactions and experiences with doctors, this is not an area of expertise for them. Furthermore, patients do not have a basis for knowing what information their doctors have (Fowler, 1997).

Sometimes, respondents have experiences or information related to a question but do not have the information in the form that the researcher wants it. A good example is a medical diagnosis. There is considerable evidence showing a lack of correspondence between the conditions patients say they have and the conditions recorded in medical records (Edwards et al., 1994; Jabine, 1987). At least part of this mismatch results from patients not being told, or not remembering, how to name their conditions. Having some information—but not in the form the researchers want—is also a common problem in nutrition studies. For example, asking how many 4-ounce servings of chicken someone ate last week is not only cognitively complex, but a person who knows exactly what was eaten (one drumstick, one thigh) may have no idea how to report quantity by weight.

Asking about other people: Sometimes, the problem of asking people questions to which they do not know the answers is one of respondent selection rather than question design. Many surveys ask a specific member of a household to report information about other household members or about the household as a whole. When such designs are chosen, a critical issue is whether or not the information required (such as insurance or employment status) is known to the person who will be doing the reporting.

In other situations, researchers make conscious decisions to ask a proxy respondent for information, rather than talking to the person of interest. For example, it is common to ask parents about their children and to ask family members to report on experiences of nursing home residents. However, in situations such as these, researchers need to be careful about the questions that they ask and the assumptions that they make. Parents could answer factual questions about the grade their child is in or how their child gets to school, but they are not the best reporters of whether their child is happy in school or how many cigarettes the child smokes. Family members of nursing home residents can report on what type of room the resident lives in and, of course, on their own experiences of visiting the nursing home, but they will most likely not be able to reliably answer how often a call light is answered quickly when help is requested or how day and night staffs compare.

There is a large literature comparing self-reporting with proxy reporting (Cannell, Marquis, & Laurent, 1977; Clarridge & Massagli, 1989; Moore, 1988; O'Muircheartaigh, 1991; Rodgers & Herzog, 1989; Tourangeau, Rips, & Rasinski, 2000). Across all topics, usually self-respondents are better reporters than proxy respondents.

Recall

Memory researchers tell us that few things, once directly experienced, are forgotten completely. The readiness with which information and experiences can be retrieved follows some fairly well-developed principles (Cannell, Marquis, et al., 1977; Eisenhower, Mathiowetz, & Morganstein, 1991; Tourangeau et al., 2000):

- The more recent the event, the more likely it is to be recalled.
- The greater the impact or current salience of the event, the more likely it is to be recalled.
- The more consistent an event was with the way the respondent thinks about things, the more likely it is to be recalled.

If the researcher wants information about very small events that had minimal impact, it follows that the reference period should be quite short. For example, when researchers want reporting about dietary intake or soft drink consumption, it has been found that even a 24-hour recall period can produce deterioration and reporting error due to recall. When people are asked to report their behavior over 1 or 2 weeks, they resort to giving estimates of their average or typical behavior, rather than trying to remember (Blair & Burton, 1987).

So if a researcher wants accurate information about something such as how many glasses of water someone drank, having respondents report for a very short period, such as a day, is probably the only way to get reasonably accurate answers (Smith, 1991). However, if a researcher is asking about events that probably had a greater impact in someone's life, a longer time period could be asked about.

Table 12.1 is from a study in which people were sampled based on having had a hospital stay. The survey asked respondents about recent hospital stays; then the researchers compared the survey responses with the actual hospital records. The table reports the percentages of known hospitalizations that were and were not reported. The shorter the stay in the hospital and the greater the time period between the discharge and the interview, the less likely the respondent was to report the hospital stay. More than 30% of the 1-day hospitalizations 40 weeks before the interview were not reported at all, while only 5% of the longer stays within 20 weeks were not mentioned. This table shows that the more important the event (such as a long hospital stay), the more likely it was to be reported—both in the immediate and recent past.

Table 12.1 Recorded Duration of Hospitalizations and Percentages of Discharges Not Reported in Interviews, by Time Elapsed

	Duration of Hospital Stay		
	1 Day	*2–4 Days*	*5 or More Days*
Time Elapsed	*Percentage of Discharge Not Reported*		
1–20 weeks	21	5	5
21–40 weeks	27	11	7
41–52 weeks	32	34	22

SOURCE: Summary of Studies (Cannell, Marquis, et al., 1977).

Researchers have explored strategies for improving the quality of the recall performance of respondents. One example is decomposing the question and asking several questions about smaller parts. Asking multiple questions improves the probability that an event will be recalled and reported (Cannell, Marquis, et al., 1977; Sudman & Bradburn, 1982).

Example 12: During the past 30 days, how many times have you used oils to cook food or added oils to foods like salad, pasta, or bread?

The respondent is being asked to count the number of times an unimportant task (using oil) was done over a 30-day period. Although the question offers numerous examples, it still is very complex. By splitting up the question, the respondent will be better able to focus on all the different ways oil can be used and thereby improve recall.

Alternative 12a: The next few questions are about oils used with food. You should include things like vegetable oil or olive oil, but not butter or margarine. During the past 30 days, how many times have you used oils to cook food?

Alternative 12b: During the past 30 days, how many times have you added oil to salads, such as oil and vinegar?

Alternative 12c: During the past 30 days, how many times have you added oils to other foods like pasta or bread?

In a recent study in which these series of questions were compared, the average number of times someone reported using oil was 11.9 times when asked Example 12 and 16.6 times when asked the three alternative questions.

Another strategy for increasing recall is stimulating associations likely to be tied to what the respondent is supposed to report. Activating the cognitive and intellectual network in which a memory is likely to be embedded is likely to improve recall as well (Eisenhower et al., 1991).

Example 13: In the last 12 months, to how many organizations did you volunteer your time?

There are many different things that could count as volunteering. To help remind the respondent of all the different things that could be included, a researcher could provide some cues that might stimulate memories. This could be done by asking additional questions or adding an introduction to the question.

Alternative 13: There are many ways that people volunteer their time—they could help at a church or school, help provide meals to the homeless during the holidays, or participate in a charity walk or other event. In the last 12 months, for how many organizations did you volunteer your time?

There are limits to what people are able to recall. If a question requires information that most people cannot recall easily, the data will almost certainly suffer. However, even when the recall task is comparatively simple for most people, if getting an accurate count is important, asking multiple questions and developing questions that trigger associations that may aid recall are both effective strategies for improving the quality of the data.

Placing Events in Time

Many of the issues discussed above could reflect an interrelationship between recalling an event at all and placing it in time. If a survey is to be used to estimate the annual number of hospitalizations for a particular sample, people are asked what essentially is a two-part question: (1) Have you been in the hospital recently and (2) how many times were you in the hospital in precisely the last 12 months?

There are two approaches researchers use to try to improve how well respondents place events in time:

1. Stimulating recall activities on the part of respondents to help them place events in time.
2. Designing data collection procedures that generate boundaries for reporting periods.

In order to improve the ability of respondents to place events in time, the simplest step is to help clarify the time frame. For example, rather than "In the last 6 months, did you go to any museums?" the question could add the actual month: "In the last 6 months, that is, since March, did you go to any museums?" This simple addition could help focus the respondent on exactly what is being asked about. For in-person surveys, showing respondents a calendar with the reference period outlined may be helpful. In addition, respondents can be asked to recall what was going on and what kinds of things were happening in their lives at the time of the boundary of the reporting period. Filling in any life events, such as birthdays or jobs, can help make the dates on the calendar more meaningful (Belli, 1998; Sudman, Finn, & Lannon, 1984).

A very different approach to improving the reporting of events in a time period is to create an actual boundary for respondents by conducting two or more interviews. During the initial interview, respondents are asked about events and situations that happened during some time period before the interview. In the subsequent interview, they are then asked about what has happened between the time of the initial interview and the time of the second interview. This method is used in several national surveys, including the National Crime Victimization Survey (NCVS, formally called the National Crime Survey; Groves et al., 2004). For example, the NCVS surveys are done every 6 months, and respondents are asked about any crimes they were a victim of "in the last 6 months." To prevent telescoping of events (talking about events that happened outside the 6 months time frame), respondents' answers are compared with their answers in the prior survey, and duplicate events are eliminated. Obviously, such reinterview designs are much more expensive to implement than are one-time surveys. However, when accurate reporting of events in time is very important, they provide a strategy that improves the quality of data.

The Form of the Answer

It is important that a question specifies the form the answer is supposed to take and that the form of the answer fits the answer the respondent has to give. For all questions, the response task must be clear. The respondent should know from the question the terms and the units they should use. This is especially problematic for open-ended questions, as seen in the two examples below.

Example 14: How long have you been working there?

This could be answered in many ways—"since I was 18," "13 years," and "a long time." All of these could be correct answers to the question "how long" something has been going on. By not providing the respondents information about what unit of measure to use, the researcher may be left with data that are not comparable. The alternative version below provides enough detail for the respondent to know how to answer.

Alternative 14: For how many years have you been working here?

Example 15 is another example where the response task is unclear.

Example 15: How do you usually go to work?

Again, with no direction from the researcher, a respondent could answer with mode of transportation (I drive), directions (I take the interstate), or even with whom they go to work (with my neighbor).

For closed-ended questions, there are several additional issues. Just as questions should not involve multiple cognitive tasks, neither should answer categories. The answer choices should be clear and should not combine multiple concepts.

Example 16: In the last week, did you drink coffee at breakfast—Yes, always; Yes, sometimes; or No?

The researcher correctly worried that this question is not a simple yes or no question. However, rather than altering the question, the response task was changed to combine the yes/no task with a frequency question (which is not explicitly asked). The cognitive complexity of the question can be reduced by either asking two questions or simply changing it to a frequency.

Alternative 16: In the last week, on how many days did you drink coffee at breakfast—every day, some days, no days?

Response tasks for closed-ended questions must always be mutually exclusive and exhaustive. That means that for any given situation, there should be only one answer that fits and there should be an answer for every situation.

Example 17: What is your current work situation—working full-time, working part-time, student, retired, homemaker, or something else?

The answer categories provided in this question are not mutually exclusive. It is possible that someone is working full- or part-time and is also a student. Or someone could be retired and a homemaker. Thus, respondents could legitimately put themselves into more than one category. Respondents (and interviewers) must decide how to handle this situation—mark more than one answer, choose one or the other, skip the question, or write something in the margins. When there is a possibility of people being in more than one category, it is sometimes better to ask a series of yes/no questions to describe the respondent's situation.

Closed-ended response tasks also need to be exhaustive—every situation must be taken into account in the answer choices available. Frequency scales, especially those that use a number of times per unit of measurement (see example below), are notorious for not being exhaustive.

Example 18: How often have you attended a sporting event—several times a week, about once a week, about once a month, a few times a year, about once a year?

In addition to the fact that this question is assuming a regularity that may not exist, this is also an example of answer choices that are not exhaustive. There is no answer to describe the situation of someone who goes to sporting events every other week, every other month, or less than once a year. With no place that exactly describes their situation, respondents will be left on their own to figure out the closest fit—and respondents who have the same answer to give will answer differently from one another.

As discussed earlier, often the answer to a question varies over time. When a question makes the assumption of regularity, respondents who vary will have trouble.

Example 19: In the last 30 days, were you able to climb a flight of stairs with no difficulty, with some difficulty, or were you not able to climb stairs at all?

This question imposes an assumption: that the respondent's situation was stable for 30 days. For a study of patients with AIDS, we found that questions in this form did not fit the answers of respondents, because their symptoms (and ability to climb stairs) varied widely from day to day (Cleary et al., 1993).

Reducing the Effect of Social Desirability on Answers

Studies of response accuracy suggest that there is a tendency among respondents to distort answers in ways that will make them look better or will avoid making them look bad. Locander, Sudman, and Bradburn (1976) found that convictions for drunken driving and experiences with bankruptcy were reported very poorly in surveys. Clearly, such events are significant enough that they are unlikely to have been forgotten; the explanation for poor reporting must be that people are reluctant to report such events about themselves. However, the effects of social desirability are much more pervasive than such extreme examples.

For example, when Cannell, Marquis, et al. (1977) coded the reasons for hospitalization by the likelihood that the condition leading to the hospitalization might be embarrassing or life threatening, they found that the hospitalizations associated with the most threatening conditions were significantly less likely to be reported in a health survey. Distortion can also produce overreporting. Anderson, Silver, and Abramson (1988) found notable overreporting of voting in elections.

Although social desirability has been used as a blanket term for these phenomena, there are probably several different forces operating to produce the response effects described above. First, there is no doubt some tendency for respondents to want to make themselves look good and avoid looking bad. In addition, sometimes surveys ask questions to which the answers could actually pose a threat to respondents. When surveys ask about illegal drug use, about drinking alcohol to excess, about the number of sexual partners people have had, the answers, if revealed, could expose respondents to divorce proceedings, loss of jobs, or even criminal prosecution. When the answer to a survey question poses such a risk for respondents, it is easy to understand why respondents might prefer to distort their answers rather than take a chance on giving accurate answers, even if the risk of improper disclosure is deemed to be small. Third, in a related but slightly different way, response distortion may come about because the literally accurate answer is not the way the respondent wants to think about himself or herself. When respondents distort answers about not drinking to excess or voting behavior, it may have as much to do with respondents' managing their own self-images as with their managing the images that others have of them.

It is fundamental to understand that the problem is not "sensitive questions," but "sensitive answers." Questions tend to be categorized as sensitive if a "yes" answer is likely to be judged by society as undesirable behavior. However, for those for whom the answer is "no," questions about any particular behavior are not sensitive. Questions about drug use or drunken driving are not sensitive for people who do not use drugs or drive after drinking.

It is also important to remember that people vary in what they consider to be sensitive. For example, asking whether or not a person has a library card apparently is a fairly sensitive question; some people interpret a "no" answer as indicating something negative about themselves (Parry & Crossley, 1950). Library card ownership is considerably overreported.

Thinking broadly about the reasons for distorting answers leads to the notion that the whole interview experience should be set up in such a way as to minimize the forces on respondents to distort answers.

With respect to data collection procedures, constructive steps to reduce the effects of these forces on answers include the following:

- Ensure and communicate to respondents that their answers will be confidential.
- Emphasize through the introduction and in other ways the importance of the accuracy of answers (Cannell, Oksenberg, & Converse, 1977).
- Use self-administration rather than interviewer administration, or have respondents enter their answers directly into a computer (Aquilino & Losciuto, 1990; Brener et al., 2006; Turner et al., 1998; Turner, Lessler, & Gfroerer, 1992).

In designing the questions themselves, constructive steps include the following:

- Explain the purposes of questions so that respondents can see why they are appropriate.
- Frame questions, and take care in wording, to reduce the extent to which respondents will perceive that particular answers will be interpreted in a negative or inaccurate light.

These steps are likely to improve the quality of reporting in every area of a survey, not just those deemed to be particularly sensitive. Researchers never know when a question may cause a respondent some embarrassment or unease. A survey instrument should be designed to minimize the extent to which such feelings will affect answers to any question asked.

Questions to Measure Subjective States

A distinctive feature of the measurement of subjective states is that there are, in fact, no right or wrong answers to questions. "Rightness" implies the possibility of an objective standard against which to evaluate answers. Although we can assess the consistency of answers with other information, there is no direct way to know about people's subjective states independent of what they tell us.

This does not mean that there are no standards for questions designed to measure subjective states. The standards are basically the same as for questions about factual things: Questions should be understood consistently by all respondents so they are all answering the same question, they should usually cover topics with which most respondents are familiar, and the response task, the way respondents are asked to answer the questions, should be one that respondents can use consistently and that provides meaningful information about what they have to say.

By far, the largest number of survey questions ask about respondents' perceptions or feelings about themselves, others, or ideas. The basic task for the respondent on most questions in this category is to place answers on a continuum. Such questions all have the same basic framework, which consists of three components: (a) what is to be rated, (b) what dimension or continuum the rated object is to be placed on, and (c) the characteristics of the continuum that are offered to the respondent.

Defining What Is to Be Rated

As with all survey questions, when researchers are designing questions to measure subjective states, it is important that they keep in mind that all respondents should be answering the same question.

> *Example 20:* In general, do you think government officials care about your interests a lot, some, only a little, or not at all?

"Government officials" are a very heterogeneous lot, and which government officials a respondent has in mind may affect how he or she answers the question. For example, people consistently rate local governments as more responsive than state and federal governments. Elected officials may not be rated the same as persons who have been appointed to positions in the executive branches of government. To the extent that people's answers vary based on the ways they interpret questions, a new source of error is introduced, and the answers will provide less than the best information on what the researchers are trying to measure.

> *Example 21:* Do you consider crime to be a big problem, some problem, or no problem at all?

Crime is also a heterogeneous category. Can people lump white-collar crime, drug dealing, and armed robbery into a single integrated whole? It would not be surprising for respondents to this question to key on different aspects of crime. Moreover, this particular question does not specify a locus for the problem: the neighborhood, the city, the state, the nation. The perspectives people take will affect their answers. People generally rate the crime problems in their own neighborhoods as less severe than average. To the extent that what is being rated can be specified more clearly, so that respondents do not vary in their interpretations of what they are rating, measurement will be better.

Seemingly small differences in wording can have big effects on answers (Schuman & Presser, 1981). Careful attention to wording is one key to good questions design.

The Response Task

Researchers have designed numerous strategies for evoking answers from respondents. The most common task for respondents is some variation of putting the object of the answer on a continuum.

The Direct Rating Task

Table 12.2 shows three different forms of a continuum with rankings from positive to negative. Such a continuum can be described to respondents in numerous ways, and there are numerous ways that respondents can be asked to assign answers to positions on the continuum.

Example 21a: Overall, how would rate your health—excellent, very good, good, fair, or poor?

Example 21b: Consider a scale from 0 to 10, where 10 represents the best your health can be, where 0 represents the worst your health can be, and the numbers in between represent health states in between. What number would you give your health today?

Example 21c: Overall, would you say you are in good health?

These three questions all ask the same thing; they differ only in the ways in which the respondents are asked to use the continuum.

When the goal is to have respondents place themselves or something else along a continuum, the researcher must make choices about the characteristics of the scale or response task to be offered to respondents. Two key issues include (a) how many categories to offer and (b) whether to use scales defined by numbers or by adjectives. In general, the goal of any rating task is to provide the researcher with as much information as possible about where respondents stand compared with others. Consider a continuum from positive to negative and the results of a question such as the following:

Example 23: In general, would you rate the job performance of the President as good or not so good?

Table 12.2 Some Examples of Forms for an Evaluative Continuum

Excellent			Very Good			Good		Fair		Poor
10	9	8	7	6	5	4	3	2	1	0
Best										Worst
			Good					Not Good		

Such a question divides respondents into two groups. That means that the information coming from this question is not very refined. Respondents who answer "good" are more positive than the people who say "not so good," but there is no information about the relative feelings of all the people who answer "good," even though there may be quite a bit of variation among them in the degree of positiveness that they feel about the President's job performance.

There is another issue as well: the distribution of answers. In the above example, suppose most of the respondents answered the question in a particular way; for example, suppose 90% said that the President is doing a "good" job. In that case, the value of the question is particularly minimal. The question gives meaningful information only for about 10% of the population, the 10% who responded "not good." For the 90% of the population that answered "good," absolutely nothing was learned about where they stand compared with others who gave the same answer.

This analysis suggests that there are two general principles for thinking about optimal categories for a response task. First, to the extent that valid information can be obtained, more categories are better than fewer categories. Second, generally speaking, an optimal set of categories along a continuum will maximize the extent to which people are distributed across the response categories.

Given these considerations, is there any limit to the number of categories that are useful? Is it always better to have more categories? There are at least two limiting factors to the principle that using more categories produces better measurement. First, there appear to be real limits to the extent to which people can use scales to provide meaningful information. Although the optimal number of categories on a scale may vary, in part with the dimension and in part based on the distribution of people or items rated, most studies have shown that little new valid information is provided by response tasks that provide more than 10 categories (Andrews, 1984). Beyond that, people seem not to provide new information; the variation that is added seems to be mainly a reflection of the different ways that people use the scales. In fact, five to seven categories are probably as many categories as most respondents can use meaningfully for most rating tasks.

A second issue has to do with ease of administration. If the survey instrument is being self-administered (with respondents reading the questions to themselves) or administered by an in-person interviewer (who can hand respondents a list of the response categories), long lists of scale points do not pose any particular problem. However, when surveys are done on the telephone, it is necessary for respondents to retain all the response options as the interviewer reads them in order to answer the question. There clearly are limits to peoples' abilities to retain complex lists of categories.

When long, complex scales are presented by telephone, sometimes it is found that this produces biases simply because respondents cannot remember the categories well. For example, there is some tendency for respondents to remember the first or the last categories better than some of those in the middle (Schwartz & Hippler, 1991). When questions are to be used on the telephone, researchers often prefer to use scales with only three or four response categories in order to ease the response task and ensure that respondents are aware of all the response alternatives when they answer questions.

Another decision is whether to use numerical or adjectival labels. The principal argument in favor of adjectival scales is that all the points are more consistently calibrated by the use of words. The other side of the story is that it is difficult to think up adjectives for more than 5 or 6 points along most continua. When researchers have tried, some of the adjectival descriptions have sounded very close or similar to one another. It is virtually impossible to find a list of adjectives that will define a 10-point scale.

A related advantage of numbers is that a numerical 10-point scale is easy to remember and use. Thus, when doing telephone interviews, whereas it may be difficult to teach respondents five or six adjectives, it is comparatively easy to define a 10-point scale numerically. Hence, using scales defined by numbers can increase the reliability of a rating task performed on the telephone, if numerous response alternatives are to be provided. Moreover, it may increase the comparability of measurement of subjective ratings across modes of data collection.

Finally, a problem in international research and increasingly in research in the United States is how to get consistent measurement of subjective states for different cultural groups. In particular, when scales are defined adjectivally, it has been found that it is virtually impossible to have exact translations across languages. Adjectival scaling tasks across languages are not comparable. Although it has not been documented, it seems reasonable that numerical scales could improve the comparability of data collected across languages.

Using an Indirect Rating Task

The tasks discussed in the preceding section were all geared to having respondents place something (themselves, their views, or their evaluations of something else) on a rating scale or order items on a scale. A large part of the survey research enterprise is focused on measuring people's responses to various ideas, analyses, or proposals. The content of such questions is as vast as the imagination of the survey research community. The common form of such questions is something such as the following:

Example 24: Higher taxes generally hurt the rich and benefit the poor. Do you agree or disagree?

An important distinction to be made in thinking about questions such as these is the nature of the task confronting the respondent. In the examples given previously, respondents were asked to place themselves or others on some defined continuum. For example, they would be asked to rate their own health on a scale from excellent to poor or they would be asked to rate the job that they thought that the President of the United States was doing from good to poor. The task posed by Example 24, however, is somewhat different. Instead of being asked to place some object on a defined continuum, the respondent is asked to rate the distance between his or her own views or preferences and the idea expressed in the question.

One principle issue is the same for all questions: it is important that what is being rated be unambiguous and understood consistently by all those answering the questions. It is very common to find multiple dimensions underlying questions posed in the agree-disagree format, or variations thereon. The following are examples cited by Robinson and Shaver (1973) that have this characteristic.

Would you strongly agree, agree, disagree, or strongly disagree with the following statements:

Example 25: America is getting so far away from the true American way of life that force may be necessary to restore it.

Three issues: How far America is from the true American way, whether or not the true American way should be restored, and whether or not force may be needed (or desirable) to restore it.

Example 26: There is little use writing public officials because they often aren't really interested in the problems of the average man.

Two issues: The value of writing to officials and how interested officials are in the problems of the average man.

With respect to both of these questions, it is not possible to define what an "agree" or "disagree" answer actually means.

There are three common problems with questions in the agree-disagree form—or related question forms such as the oppose-favor form. First, many questions in this form do not produce interpretable answers, either because they are not on a clearly defined place on a continuum or because they reflect more than one dimension. Those problems can be solved through careful question design. However, two other problems—that these questions usually sort people into only two groups (agree or disagree) and that they often are cognitively complex—are more generic to the question form.

The most important limitation of such questions, however, is that the question form itself introduces error into the measurement process that is unnecessary. Essentially the same question can be answered in either a direct or indirect way. Examples 27a and 27b illustrate the indirect and direct approaches to asking the same question.

Example 27a: Consider the statement, Federal income taxes should be reduced. Would you say you completely agree, generally agree, neither agree nor disagree, generally disagree, or strongly disagree with that statement?

Example 27b: How do you feel about the level of federal income taxes—would you say they should be much higher, a little higher, about as they are now, a little lower, or much lower?

For Example 27b, the respondent directly puts where he or she wants taxes to be on a continuum from much higher to much lower. In Example 27a, the respondent is asked to report on the distance between his or her own views and the position stated in the question stem. Figure 12.1 is a pictorial representation of the Example 27a task. The steps include the following:

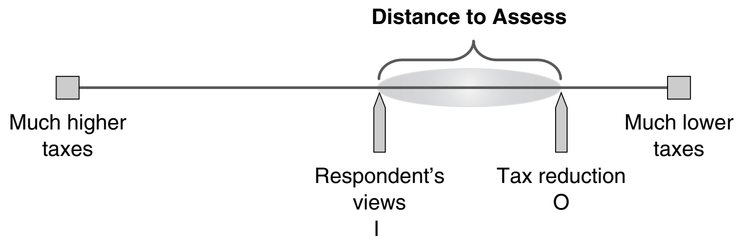

Figure 12.1 Cognitive Processes of Indirect Rating Task Visualized

1. Figuring out where on the continuum one's own views are (I in figure). This has to be done to answer either Example 27a or 27b. In addition, however, to answer Example 27a, the respondent must
2. Evaluate the distance between one's view and the position stated in the question stem (O in the figure)
3. Decide whether the distance should be considered "agreement" or "disagreement"; in other words, one has to code the distance into the agree-disagree categories.

To the extent that two respondents have differences of opinion about how close their views need to be to the stated position (O) in order to be considered agreement, they could give different answers for that reason alone, even if their views on income taxes are the same. In essence, indirect ratings introduce an additional source of potential error into the measurement process. This can be denoted as follows:

$$X = t + e_d + e_i,$$

where X is the answer, t is the true value or score that we want the respondent to report, e_d is the error related to the way the respondent performs the direct rating task of locating his or her own views on the oppose-favor continuum, and e_i is the error related to the way the respondent performs the process of coding the distance from his or her answer to the point stated in the question stem into the agree-disagree format.

It is fairly obvious that indirect ratings are cognitively more complicated than direct ratings. They also introduce a second task of coding the distance between the stimulus and the respondents' views that will be done differently from respondent to respondent and, hence, introduce an additional source of measurement error into the answer. We think researchers will almost always be better served by using direct ratings and avoiding agree-disagree and related question forms.

Rank Ordering

There are occasions when researchers want respondents to compare objects on some dimension.

Example 28: Which candidate do you prefer?

Example 29: What do you consider to be the most important problem facing the city?

Example 30: Here are some factors some people consider when deciding where to live. Which is most important to you?

Proximity to work

Quality of schools

Parks

Safety

Access to shopping

Example 31: I'm going to read you a list of candidates. I want you to tell me whom you consider to be the most liberal . . .

The basic question objectives can all be met through one of four tasks for respondents:

Task 1. Respondents can be given a list of options and asked to rank order them from top to bottom on some continuum.

Task 2. Respondents can be given a list of options and asked to name the most (second most, third most, and so on) extreme on the rating dimension.

Task 3. Respondents can be asked to make a series of paired comparisons, ordering two options at a time.

Task 4. Respondents can be given a list and asked to rate each one using some scale (rather than just putting them in order or picking one or more of the most extreme).

If there is a short list of options, Task 1 is not hard to do. However, as the list becomes longer, the task is harder, soon becoming impossible on the telephone, when respondents cannot see all the options. Task 2 is easier than Task 1 when the list is long (or even when the list is short). Often researchers are satisfied to know which are the one or two most important, rather than having a complete rank ordering. In that case, Task 2 is attractive. Psychometricians often like the paired comparison approach of Task 3, in which each alternative is compared with every other, one pair at a time. However, it is such a time-consuming and cumbersome way to create an ordered list that it is seldom used in general surveys.

Best of all may be Task 4. This task is probably easiest of all for respondents, regardless of data collection mode. Moreover, the rank ordering tasks (Tasks 1 through 3) do not provide any information about where the items are located on the rating continuum. They could all be perceived as very high or very low—the rank order provides no information. Task 4 provides information about where the items are located on the rating scale. Although there can be ties, so ordering is not known perfectly, usually an aggregate order will result as well. For all these reasons, very often a series of ratings, rather than a rank order task, is the best way to achieve these objectives.

Narrative Answers

When the goal is to place answers on a continuum, allowing people to answer in their own words will not do. Consider a question such as the following:

Example 32: How are you feeling today?

People can answer in all kinds of ways: Some will say "fine," some will say "great," some will say "not bad." If one were trying to order such comments, some ordinal properties would be clear. Those who say "terrible" would obviously be placed at a different point on a continuum from those who say "great." However, there is no way to order responses such as "not bad," "pretty good," "good enough," or "satisfactory."

In contrast, when the purpose of a question is to identify priorities or preferences among various items, there is a choice to be made between the following two approaches:

Example 33a: What do you consider to be the most important problem facing your local city government today?

Example 33b: The following is a list of some of the problems that are facing your local city government. Which do you consider to be most important?

Crime

Tax rates

Schools

Trash collection

The open-ended approach has several advantages. It does not limit answers to those the researcher thought of, so there is opportunity to learn the unexpected. It also requires no visual aids, so it works on the telephone. On the other hand, the diversity of answers may make the results hard to analyze. The more focused the question and the clearer the kind of answer desired, the more analyzable the answers. Moreover, Schuman and Presser (1981) found that the answers are probably more reliable and valid when a list is provided than when the question is asked in open form.

If the list of possible answers is not known or is very long, the open form may be the right approach. Although computer-assisted interviewing creates great pressure to use only fixed-response questions, respondents like to answer some questions in their own words. The measurement result may not be as easy to work with, but asking some questions to be answered in narrative form may be justified for that reason alone. However, if good measurement is the goal and the alternatives can be specified, providing respondents with a list and having them choose is usually the best.

The Relativity of Answers About Subjective States

The answers to questions about subjective states are always relative; they are never absolute. The kinds of statements that are justified based on answers to these kinds of questions are comparative. It is appropriate to say that Group A reports more positive feelings than Group B. It is appropriate to say that the population reports more positive feelings now than it did a year ago. It is not appropriate (at least not without some careful caveats) to say that people gave the president a positive rating, that they are satisfied with their schools, or that by and large they think that their health is good.

One of the most common abuses of survey measurement is treating data collected using measures of subjective states, which are designed to produce ordinal measures, as if they had produced data with absolute meaning. When statements are made such as, "Most people favor gun control," "Most people oppose abortion," and "Most people support the President," these statements should be viewed askance. All that happened in any of these cases was that a majority of respondents picked response alternatives to a particular question that the researcher chose to interpret as favorable or positive. That same group of people could be presented with different stimuli that apparently address the same topic that would produce different distributions and support very different statements. For example, Rasinski (1989) showed that many more people were willing to increase spending on people "with low incomes" than would increase spending on people on "welfare." Schuman and Presser (1981) found that nearly half the population would support "not allowing" communists to speak in public but only 20% would "forbid" it. The distribution of answers to questions depends critically on the details of the wording, and reporting the results in absolute, rather than relative, terms is not appropriate.

The Role of Language and Mode of Data Collection in Question Design

Survey questions can be asked by interviewers, either by telephone or in person. Respondents can be asked to fill out paper questionnaires or enter answers into a computer. It is very common for surveys to be conducted in more than one mode of data collection. It is also very common for surveys to be conducted in more than one language. There are many implications of mode of data collection for the design of questions, and this chapter will not address most of them. Dillman (2007) and Groves et al. (2004) are two good places to look for more information on those issues. However, we wanted to point out one very important principle that will have an important effect on data quality.

If a survey is going to be administered in more than one mode or more than one language, the researcher wants the results to be as comparable as possible. To that end, the questions that are asked should be identical. For that to happen, the survey should be designed from the beginning to be used in more than one language and/or mode. Designing a survey for one mode or one language and then trying to adapt it to another language or mode is the wrong way to proceed.

With respect to language, there are some words that translate much better than others. For example, the categories "excellent, good, fair, poor" are frequently used in English surveys. However, "poor" and particularly "fair" do not translate easily into other languages. If a researcher is thinking about how precisely questions can be translated when the initial questions are being written, choices can be made that will greatly increase the comparability of the questions across languages (Harkness, van de Vijver, & Mohler, 2007).

The same is true for mode of data collection issues (Dillman, 2007). If a survey is going to be administered in person or in a self-administered questionnaire, it is possible to ask respondents to choose from a long list of answers. However, if that same survey is going to be used on the telephone, respondents will not be able to remember more than a few answer options. In a self-administered survey, questions do not have to include all the words. The question and the answers can be combined by the respondent to understand what is wanted. However, a telephone interviewer must have a complete script so that the words that are read give the respondent all the information needed to know what is being asked and how to answer.

If the researcher thinks about the way questions will work in multiple languages or modes from the beginning, problems of comparability can be minimized. However, attempts to adapt surveys designed for a single mode or language to other modes or languages almost always produce major problems of comparability.

Presurvey Evaluation of Questions

Before a question is asked in a full-scale survey, testing should be done to find out if respondents can understand it, if they can perform the tasks that it requires, and if the interviewers can and will read it as worded.

There are three main kinds of presurvey question evaluation activities: using checklists to systematically review questions, conducting cognitive interviews, and field pretesting (replicating to a reasonable extent procedures to be used in a proposed survey).

Systematic Question Review

Sometimes, a problem with a question can be detected in a question just by reading it. Formal question reviews usually consist of checklists of question characteristics that are indicative of potential problems (Lessler & Forsyth, 1996). Willis and Lessler's (1999) Question Appraisal System (QAS) has an 8-step process that looks at everything, including the readability and clarity of the question, whether the question contains unstated assumptions or is inherently sensitive or biased, the knowledge and recall skills needed to answer the question, and characteristics of the response categories. The Question Understanding Aid (QUAID) is a computerized version of a QAS (Graesser, Cai, Louwerse, & Daniel, 2006). A computer program analyzes the wording of question, comparing it to a set of programmed algorithms—checking for uncommon words, vague terms, complex syntax, and the number of clauses in the question.

Often question appraisal checklists require the appraiser to make some sort of judgment. Whether the question is hard to understand or asks for information that a person may not have are all based on the impression of the appraiser. In Table 12.3, we present a systematic appraisal form that requires minimal judgment from the appraiser. Questions that are identified as having the characteristics listed in Table 12.3 can be rewritten or revised to make them better questions before any testing occurs. In some cases, if a particular question has a pedigree or no suitable alternative way of asking the question can be found, the appraisal can flag questions and issues for subsequent testing.

Cognitive Testing of Questions

The ultimate goal of all question evaluation is to determine if the answers given by respondents represent what the researcher wants to measure. In order to assess this, a researcher has to find out what respondents are thinking when they are trying to answer questions. Cognitive testing is a method of question evaluation that allows a researcher to know how the cognitive tasks posed by a question are being handled. These tasks—comprehension of the question, retrieval of information, and formation of the answer—can all be asked about, observed, and evaluated in a cognitive interview.

A conference on the Cognitive Aspects of Survey Methodology became the basis for much work in this field (Jabine, Straf, & Tanur, 1984). By using the methodology of cognitive psychologists, survey researchers have been able to better evaluate (at another level) questions.

Many federal and academic institutions (including the National Center for Health Statistics, the Census Bureau, and Research Triangle Institute) have been instrumental in defining and documenting this emerging field (DeMaio & Landreth, 2004; DeMaio & Rothgeb, 1996; Lessler & Tourangeau, 1989; Willis, 2005; Willis, DeMaio, & Harris-Kojetin, 1999).

Cognitive interviewing is often done using small numbers of individuals (usually 5 to 20 respondents) in several iterative rounds. After each round, the surveys are reviewed, the questions and protocol are modified as needed, and additional interviews are completed (Willis, 2005). Respondents are sometimes brought into a special setting in which interviews can be recorded and observed; hence, these are often referred to as "laboratory interviews." The priority is to find out how respondents understand questions and perform the response tasks; there is no particular

Table 12.3 Systematic Instrument Appraisal List

Comprehension Issues

1. Does the question have a reference period (time)? This applies to any question for which the answer could reasonably be expected to vary from day to day, week to week, or month to month.
2. Is the question hypothetical?
3. Are there multiple questions being asked in a single question? (Is the question multibarreled)?
4. Does the question include an abstract noun that is not defined?

Retrieval of Information

5. Is the question cognitively complex? Does the question require multiple calculations in order to answer the question?

Formation of Answer

6. Does the question contain assumptions about the respondent's situation, or the way the respondent thinks about things, that are not necessarily true but that are critical to answering the question?
7. Does the question make the response task clear to the respondent; that is, is it clear what kind of answer is required, and at what level of detail, in order to to meet the question objectives?
8. (If fixed-response question) Are the answer categories mutually exclusive and exhaustive?
9. Does the question give respondents a task other than a direct rating to provide information about where something (an idea, experience, person, or institution) is seen to lie on some continuum?

Usability Concerns

10. (If interviewer-administered question) Is the question fully scripted, including when and how to use any optional text?
11. Does the question end with a question? (Are definitions and introductory phrases at the beginning of the question?)
12. Are there appropriate "skip" instructions so that respondents are asked to answer only those questions that apply to them?
13. Are the response tasks that respondents are supposed to use appropriate to the question that is asked?

effort to replicate the data collection procedures to be used in the full-scale survey. The basic protocol involves reading questions to respondents (or having them read the questions themselves), having respondents answer the questions, and then having a specially trained interviewer use some strategy to find out what was going on in the respondents' minds during the question and answer process.

There are three common procedures for trying to monitor the cognitive processes of the respondent who is answering questions: "think-aloud" interviews; asking probe or follow-up questions after each question or short series of questions; and going through the questions twice, first having respondents answer them in the usual way, then returning to the questions and having a discussion with respondents about the response tasks.

Although the process of cognitive interviewing varies among research organizations, there are several goals of cognitive testing that are consistent:

1. Finding out the extent to which the understanding of questions is consistent from respondent to respondent and consistent with what the researchers intend.
2. Assessing the ability of respondents to retrieve the information needed to answer questions.
3. Assessing the ability of respondents to form answers based on the information available to them.
4. Assessing how well the answers that the respondents give reflect what they have to report.

As Willis (2005) describes, cognitive interviews are advisory in nature. Like systematic question appraisals, cognitive interviews do not improve questions by themselves. They provide qualitative information by identifying possible sources of problems. It falls to the question designer to take the information learned from the cognitive interviews and craft a better question.

Field Pretesting

A field pretest generally replicates procedures that will be used in the survey itself. The pretest should provide information about the usability of the proposed survey instrument for respondents and, if they are used, interviewers. If it is an interviewer-administered instrument, it should also provide information about how well the instrument facilitates a standardized question and answer process.

If a survey is self-administered, there are two approaches that can be used. Individuals or groups can be invited to a central location to fill out the instrument, then be debriefed about the experience. For a mail survey, a small mail pilot study can be undertaken. Feedback from respondents about usability and individual questions can come either from a series of debriefing questions at the end of the instrument itself, or, much better, from an interviewer-administered debriefing after the questionnaire has been filled out.

If a survey is interviewer-administered, one source of information is debriefing the interviewers who conducted the pretest interviews. There has been a prototype of a traditional field pretest for interviewer-administered surveys. When a survey instrument is in near final form, experienced interviewers conduct 15 to 35 interviews with people similar to those who will be respondents in the planned survey. Data collection procedures are designed to be similar to those to be used in the planned survey, except that the people interviewed are likely to be chosen on the basis of convenience and availability, rather than according to some probability sampling strategy. Question evaluation from such a survey mainly comes from interviewers (Converse & Presser, 1986).

One limitation of traditional field pretests is that, by themselves, they do not provide much information about question comprehension or response difficulty (Presser, 1989). The technique of systematically coding interviewer and respondent behavior during the pretest interview helps fill that gap (Fowler & Cannell, 1996; Oksenberg, Cannell, & Kalton, 1991).

The basic technique of behavior coding is straightforward. Pretest interviews are tape-recorded. For telephone interviews, it is important to inform respondents explicitly that the interview is being taped and to get their permission for that, in order not to break any laws. It has been well established that respondents seldom decline to have interviews tape-recorded if the idea is properly presented (Fowler & Mangione, 1990). The recordings are then listened to and the interviewer-respondent interactions are coded based on a set of criteria.

The rationale behind coding the behavior in pretest interviews is as follows: When a survey interview is going perfectly, the interviewer will read the question exactly as written once, after which the respondent will give an answer that meets the question objectives. When there are deviations from this—such as the respondent asking for clarification or the interviewer needing to repeat the question to get an adequate answer—it may be an indication of a question problem. The more often deviations occur, the more likely it is that there is a problem with the question.

It turns out that questions have reliable, predictable effects on the behavior of respondents and interviewers. In one study, the same survey instrument was pretested by two different survey organizations. The results of the behavior coding of the pretests were then compared, question by question. It was found that the rates at which three key behaviors occurred—reading questions exactly as worded, respondent requests for clarification, and respondents providing inadequate answers to questions—were highly and significantly correlated between the two pretests. Thus, regardless of who does the interviewing, the same questions were likely to produce misread questions, requests for clarification, and inadequate answers (Fowler & Cannell, 1996).

The product of the behavior coding is a simple distribution for each question. From the coding, the rate at which each of the behaviors occurred across all the pretest interviews is tabulated. The strengths of behavior coding results are that they are objective, systematic, replicable, and quantitative. Interviewers cannot have a real quantitative sense for how often they encounter respondents who have difficulty with questions. Indeed, interviewers are not even very good at identifying questions that they do not read exactly as written. Hence behavior coding adds considerably to the information researchers have about their questions.

The quantifiable nature of the results provides perspective by allowing comparison of how frequently problems occur across questions and across surveys. It also constitutes more credible evidence for researchers of the presence of a problem with a question. When interviewers say that they think respondents are having difficulty with a question, it is hard for researchers to know how much weight to give that perception. When the behavior coding shows that 25% of respondents asked for clarification before they answered a question, the evidence is clear that something should be done.

Other Presurvey Tools

There are many other tools and techniques to test different aspects of a survey instrument or a study protocol. *Focus groups* help define topics and research questions. *Usability testing* helps the researcher understand if the respondent or interviewer can navigate through the instrument correctly. With the growth of computer-based applications, this has become increasingly important in survey research. *Split ballot tests* provide a way to find out how wording changes affect distributions of answers (Fowler, 2004). Respondents are randomized to answer alternative versions of the same question and then response distributions are compared to see whether or not the changes affected the results. While such tests are not appropriate for all surveys, they are appropriate for large surveys, or surveys that are likely to be repeated.

Summary

A sensible protocol for the development of a survey instrument prior to virtually any substantial survey would include all the steps outlined above: systematic question review, cognitive interviewing, and field pretests with behavior coding. Moreover, in the ideal situation, at least two field pretests would be done, the second to make sure the problems identified in the first field pretest have been solved.

Arguments against this kind of question evaluation usually focus on time and money. Certainly the elapsed calendar time for the question design process will be longer if the researcher includes cognitive interviews than if he or she does not; however, these processes can be carried out in a few weeks. The time implications of question testing have less to do with the amount of time it takes to gather information about the questions than with the time it takes to design new and better questions when problems are found. For almost any survey, experience shows that each of these steps yields information that will enable the researcher to design better questions.

In recent years, there has been increased attention given to the evaluation of survey questions from the cognitive and interactional perspectives. The basic idea is that before a question is asked in a full-scale survey, testing should be done to find out if respondents can understand it, if they can perform the tasks that it requires, and if the interviewers can and will read it as worded.

Evaluating the Validity of Questions

The end result of good design should be a set of questions that produce answers that are valid measures of what we are trying to measure. Validity is evaluated by studying patterns of association. If one is measuring an objective fact, it may be possible to compare answers with some kind of gold standard. For example, Cannell, Marquis, et al. (1977) compared survey reports with data from hospital records. When comparing survey responses with records is not a possibility, as is common with respect to questions about facts and always the case for measures of subjective states, then validity is assessed by studying the relationship between answers to a question and the answers to other questions. If answers are good measures of their intended constructs, there should be a set of predictable relationships. For example, a good measure of health status should have predictable relationships to the amount of medical care a person receives, the number of days of work that are missed, and how able a person is to perform difficult physical tasks. Stewart and Ware (1992) provide a kind of prototype for how to systematically develop and validate measures of important health concepts. McDowell (2006) provides a compendium on the evidence for the reliability and validity of many of the measures related to health research. In the process, he describes the steps that researchers do (and sometimes do not) take to psychometrically evaluate their measures.

Validation studies are highly desirable, but they are not done routinely. Ideally, they should be done with the population in which they are being used. On occasion, measures are referred to as if being "validated" were some absolute state, such as beatification. Validity is the degree of correspondence between a measure and what is measured. Measures that can serve some purposes well are not necessarily good for other purposes. For example, some measurements that work well for group averages and to assess group effects are quite inadequate at an individual level (Ware, 1987). Validation studies for one population may not generalize to others (Kulka et al., 1989).

The challenges are of two sorts. First, we need to continue to encourage researchers to evaluate the validity of their measurement procedures routinely from a variety of perspectives. Second, we particularly need to develop clear standards for what validation means for particular analytic purposes.

Conclusion

To return to the topic of total survey design, no matter how big and representative the sample, no matter now much money is spent on data collection and what the response rate is, the quality of the resulting data from a survey will be no better than the questions that are asked. Although we can certainly hope that the number and specificity of principles for good question design will grow with time, the principles outlined in this chapter constitute a good, systematic core of guidelines for writing good questions. In addition, whereas the development of evaluative procedures will also evolve with time, cognitive testing, good field pretests, and appropriate validating analyses provide scientific, replicable, and quantified standards by which the success of question design efforts can be measured.

A final word is in order about standards for survey questions. In fact, there are four kinds of standards for survey questions:

1. Are they measuring the right thing—that is, what is needed for an analysis?
2. Do they meet cognitive standards?
3. Do they meet psychometric standards?
4. Do they meet usability standards?

The first three kinds of standards have been the primary focus of this chapter. The fourth refers to the fact that questions also have to work in the mode in which they are used. If a survey is interviewer administered, an interview schedule is also a protocol for an interaction. It has been shown that the quality of measurement can be compromised by the way the questions affect the way interviewers and respondents interact (Mangione, Fowler, & Louis, 1992; Schaeffer, 1991; Suchman & Jordan, 1990). If the survey is being done by mail or via the Internet, the questions also must be demonstrated to be able to be used comfortably by respondents. Indeed, with no interviewer to help, it is particularly important that the questions delivered in those modes be easy for respondents to manage.

A tension is created because these standards are not necessarily positively related, and in fact they can work against each other. For example, the easiest questions from a cognitive perspective may be weak psychometrically. One reason for weak survey questions is that researchers tend to one standard while neglecting the others (Fowler, 2001). A real challenge is to design questions that meet all four of these kinds of standards.

That said, certainly the most important challenge is to induce researchers to evaluate questions routinely. Unfortunately, there is a long history of researchers designing questions in haphazard ways that do not meet adequate standards and have not even been well evaluated. Moreover, we have a large body of social and medical science, collected over the past 50 years, that includes some very bad questions. The case for holding on to the questions that have been used in the past, in order to track change or to compare new results with those from old studies, is not without merit. However, a scientific enterprise is probably ill served by repeated use of poor measures, no matter how rich their tradition. In the long run, science will be best served by the use of survey questions that have been carefully and systematically evaluated and that meet the standards enunciated in this chapter.

Discussion Questions

1. What is the difference in the meaning of "validity" of answers to questions designed to measure objective facts (e.g., number of visits to doctors, cigarette smoking, eye color) and those designed to measure subjective states (e.g., happiness, feelings about political figures, interest in the arts)? What are the implications for how one would evaluate the validity of answers to questions?

2. If the answers to public opinion poll questions are always relative, as is averred in the chapter, what meaningful statements can be made about poll results? Are they worth anything?

3. The authors maintain that the cognitive tasks involved in agree-disagree questions, and other indirect ratings, are more complicated and make answers more difficult to interpret than when similar questions are asked in a direct rating form? There probably are more questions asked in the agree-disagree format in surveys than any other question form. Why do you think that is? What do you think of the authors' contention that agree-disagree questions should be avoided?

4. When asking people to do ratings, which kind of rating scale seems better: those that use numbers, such as 0 to 10, or those that use adjectives, such as excellent to poor? What are the pros and cons of each? Which provide a better way for people to say what they have to say?

5. How important is it to give respondents a time frame for questions? What are some of the kinds of questions for which a time frame is essential? Are there any kinds of questions for which a time frame is not important?

6. Why would someone want to ask about a respondent's income in a survey? What are some of the constructs for which income might be a measure? What are some examples of analysis questions for which a measure of income might be helpful? Depending on the hypotheses to be tested, what are the implications for what measure of income one might want to use?

7. Survey questions either ask respondents to choose from a set of provided answer categories or ask them to respond in their own words. If one was trying to describe how people felt about a government official or about their "significant others," which would be a better kind of question to ask? What are the pros and cons of each approach?

Exercises

1. Take a set of questions that have been used in professional surveys and cognitively test them with two or three people. Ask the questions, then probe until you understand how people understood the questions and whether or not their answers were good measures of what the questions are designed to measure. Write a critical evaluation of the questions as measures, based on your results. The Behavioral Risk Factor Social Survey, conducted by the Center for Disease Control, is a good source of questions on various aspects of health and health-related behavior. Questions used can be accessed at www.cdc.gov/brfss.

2. Write three questions in an agree-disagree form. Then design three questions in a direct rating form that measure the same constructs.

3. Write three questions that include a noun that could be interpreted in more than one way. Examples used in the chapter include "crime," "income," "car," and "political leaders," but you can use your own vague nouns. Then, for each, write another question in which you explain, define or clarify the term so that everyone will understand the question in the same way.

4. Use the standards outlined in Table 12.3 to critically evaluate the following questions. Refer to the numbers in the table in your answers.

   a. How often have you been feeling stressed—always, usually, sometimes, rarely, or never?
   b. Where did you live before you moved here?
   c. Given the crime rate where you live, how likely are you to move somewhere else in the next year or two—very likely, fairly likely, or not likely at all?

   d. When you go to the movies, how often do you have popcorn—very often, fairly often, not very often, or not at all?
   e. If an interviewer contacted you about being in a survey about using drugs and alcohol, do you think you would agree to be interviewed?
   f. How often do you have at least one alcoholic beverage to drink—every day, a couple of times a week, once a week, once a month, or less often?
   g. Are you married, living with a partner, divorced, separated, widowed, or have you never married?

5. Write questions to measure three of the following constructs:
   a. Age
   b. Weight
   c. Number of offspring
   d. Sexual orientation
   e. Physical fitness
   f. Mood
   g. Political conservatism
   h. Religiosity
   i. Soft drink consumption
   j. Music preferences

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CHAPTER 13

Internet Survey Methods

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Internet survey methods refer to surveys completed by respondents either by e-mail or over the World Wide Web (www). Decisions about the methods used to conduct surveys online hinge on whether the objective is to collect information for an existing sample or to recruit participants for study as well. When deployed as a mode of data collection, Web surveys have strengths and weaknesses much like any other data collection mode, such as telephone interviewing, in-person interviewing, or the use of self-administered questionnaires (Best & Krueger, 2004). In contrast to the Internet, however, most other data collection modes can be incorporated with well-developed sampling methods to provide survey data that are projectable to general populations, at least in developed countries. Until an exceptionally high proportion of the U.S. population has Web access, general Internet sampling methodologies will not be able to provide sufficient coverage to be used as serious alternatives for scientific sampling of general populations. Moreover, the technologies involved in generating scientific Web surveys of individuals are in their infancy and may never be fully developed.

In this chapter, we discuss the options available to researchers using the Internet for survey research and the implications of choosing them. We begin by detailing the process of drawing samples of research subjects on the Internet. Then, we consider the process of administering online instruments.

Drawing Samples on the Internet

At the outset, researchers must determine how members of the sample frame will be chosen. There are two basic approaches to sampling—probabilistic and nonprobabilistic—each serving different objectives. If the purpose of the study is to make inferences to or predictions about the target population, then a probabilistic sampling method is required. However, if the study is only intended to describe the group of individuals under observation for the purpose of theory building or illustration, then nonprobabilistic sampling methods can be used. The Internet can easily accommodate nonprobabilistic sampling methods; however, the coverage of the Internet and the accessibility of its users limit the circumstances in which probabilistic survey methods can be employed.

Probabilistic Sampling on the Internet

When the Internet is used as the medium for drawing samples, the range of possible populations that can be studied in a scientific manner is limited to groups where all members have access to the Internet. Although debates persist on how to define the Internet population (access to vs. use of the medium, household connectivity vs. personal connectivity, etc.), most studies estimate that the Internet has only penetrated between two thirds and three quarters of the U.S. population. Moreover, the backgrounds of those with Internet connections differ significantly from those without. U.S. Internet users are more likely to be young, white, married, higher educated, and wealthier compared with the general population (Fallows, 2005; Fox, 2005; Fox & Livingston, 2007). Even as millions of new individuals secure Internet connections annually, it is likely to be years before all members of the general population have sufficient access to the Internet to make the Internet appropriate as the sole data collection mode for a survey.

The sample frames used in traditional scientific surveys of general populations are often two-staged selection processes that initially select a household and then randomly select individuals who reside in that household. Households typically possess a fixed address assigned to a particular individual or group. For instance, most individuals reside in households with known addresses that can be reached by telephone numbers that can be incorporated into sampling methodologies designed to include all telephone households in specific geographies.

In contrast, the Internet is primarily arranged around its services and their associated content, rather than the clients of those services. Individual computers are often assigned a temporary IP address when they connect to the Internet from a pool of available addresses managed by their ISP, OSP, or local area network. For example, the IP address of a subscriber to an Internet provider such as a local cable or DSL company varies each time the subscriber logs on to the Internet because the pool of addresses administered by the service is far smaller than the pool of subscribers. Consequently, specific computers or their users cannot be identified or located in advance. The specific procedures for isolating individuals on the Internet depend on the nature of the service.

Internet surveys can be effectively employed in scientific surveys of specialized populations or groups who are all likely to have access to the Internet. For example, researchers may wish to study members of an organization. In these cases, a researcher considering an Internet study needs to consider the percentage of the population that is likely to have Web access. If the percentage is exceedingly large, then an appropriate scientific sampling strategy might be developed. On the other hand, if many members of the target population for a study do not typically have Internet access, then the ability to generalize all members of this group will be limited. For example, a survey of university faculty, at an institution where Internet access is universal, might be appropriately accomplished over the Internet. However, an Internet survey designed to study the characteristics of homeless persons would be futile and inappropriate because many such persons do not have access to the Internet in any meaningful way.

In samples of special populations of specific persons, researchers typically use some type of directory or database as a sample frame. Researchers can limit target populations to those possessing Internet access and for whom they can acquire a complete list of e-mail addresses, such as the many private organizations, public bureaucracies, trade associations, and schools that produce comprehensive e-mail directories of individuals affiliated with these institutions. If an available database or directory contains a valid e-mail address for most, or all, persons in the target population, or if such an e-mail address can be added from secondary sources, then the database can be very functional as a sample frame.

Nonprobabilistic Sampling on the Internet

Because nonprobabilistic sampling methods draw samples arbitrarily without a specific probability structure in mind, the Internet is exceptionally well suited for drawing nonprobabilistic samples. The Internet can easily, quickly, and inexpensively access an enormous subject pool. Potential participants can exhibit a broad range of traits due to the reach of the medium. Alternatively, the Internet can isolate groups of people exhibiting particular interests or characteristics by making use of the countless content-driven sites available on the Internet.

To get to these potential subjects, researchers can also purchase banner ads on Web sites or search engines to capture large numbers of persons to Web surveys. These "opt-in" panels represent a means of targeting Web users who are prescreened to have—or at least to have reported to someone that they might have—a particularly rare characteristic. Regardless of which approach is taken, it is important to remember that they are not scientific samples and consequently, cannot be used to make generalizations to greater populations using the assumptions of traditional probability sampling.

Inferring Nonprobabilistic Internet Samples to General Populations

While most scholars acknowledge the threats to generalizability posed by nonprobability samples, some believe measures can be taken to reduce or eliminate them. To this end, researchers have adopted techniques designed to minimize the impact that the distinctive characteristics may have on sample statistics. The methods most widely applied to Internet samples are poststratification weighting and propensity scoring. However, without accurate and reliable information about the individuals who are unreachable with Internet sampling methods, such efforts offer no greater assurances than if they were not adopted at all.

Poststratification weighting attempts to obtain more accurate population estimates by weighting respondents by the incidence of known characteristics in the target population. These population proportions are typically drawn from highly reliable estimates produced by federal statistical agencies. In the case of Internet samples, researchers often adjust samples drawn from the population of Internet users to the demographic characteristics of the population as a whole. These measures implicitly assume that any differences between a sample of Internet users and the general public are due to differences between these two populations on the demographic characteristics used in the weighting.

An alternative method used by researchers to improve the generalizability of online survey results is propensity scoring (Lee, 2004, 2006a; Schonlau et al., 2004). While poststratification weighting attempts to account for sample differences based on demographic characteristics, propensity scoring attempts to account for differences based on factors that might relate to an individual's propensity to respond to a survey. The propensity scoring approach estimates the likelihood of each participant being in a sample based on a set of covariates that would predict such recruitment (Rosenbaum, 1995) and then weights the responses for each individual by their score. In the case of Internet samples, this usually involves adjusting online results to match the results derived from a more representative sampling technique such as those produced by random-digit-dialing telephone recruitment.

Regardless of which approach is undertaken, there are no assurances that the threats from coverage error will be eliminated (Lee, 2006b). Both methods make two questionable assumptions (Mitofsky, 1999). They assume that the variables used for adjustment are the only variables related to the variables of interest. This is unlikely considering that each method relies on a limited set of demographics (Best & Krueger, 2002). Experiences, beliefs, or attitudes, for example, could underlie the variables of interest. If the weighting variables are flawed, not only will the quality of the measures fail to improve, but they may actually worsen.

Moreover, poststratification weighting and propensity scoring assume that respondents within Internet samples generate opinions in the same manner as those not in the sample. If the causal mechanisms generating the variables of interest do vary from online respondents to the population, then efforts to improve the representativeness of Internet samples will be undermined (Best, Krueger, Hubbard, & Smith, 2001). In other words, it is not simply the relationship between the variables of interest that matters, but the relationship between the variables of interest within the Internet sample and those outside the Internet sample. Poststratification improves the accuracy of estimates only if the relationship between the weighting variables and the variables of interest is equivalent between those who are in the online sample and those who are not. Similarly, propensity scoring reasons that the multivariate model uncovered in the probability samples holds in the Internet sample. If the relative importance of the weighting variables is adjusted without correcting for the differences in the relationships, then the distribution of preferences in the Internet sample is likely to remain at odds with the distribution of preferences in the population. Prior research, unfortunately, suggests that the decision-making processes of Internet respondents are likely to vary systematically from others in the U.S. populace across an array of issues. Internet users obtain information from different sources than nonusers (Fox, 2005; Stempel, Hargrove, & Bernt, 2000), participate in different social activities (Fox, 2005; Pew Internet and American Life, 2000), and socialize in different ways (Boase, Horrigan, Wellman, & Rainie, 2006; Nie & Erbring, 2000). The only way to reliably estimate differences in the relationships is to draw a probability sample of the U.S. population, which, of course, is not currently possible in the online environment (Mitofsky, 1999).

Although scientific surveys of many populations may be difficult or impossible to conduct solely over the Internet, Internet data collection is increasingly used in conjunction with other methodologies to enhance or improve the ability to easily and effectively contact individuals (Dillman, 2007). These multimode surveys often begin with a sampling method that selects individuals or establishments through specific addresses or telephone numbers. Rather than using a single method of collecting data, however, multimode surveys employ multiple methodologies, either to improve levels of survey response or to optimize the advantages of different data collection strategies.

Alternatively, representative samples of adults can be obtained by traditional communication modes and then outfitted with the equipment necessary to receive and/or respond to online instruments (Huggins & Eyerman, 2001). For example, the company, Knowledge Networks recruits households through random-digit-dial telephone calling and then equips them with free connections to the Internet and hardware (a WebTV unit) to use it. In exchange for the Internet access they receive, each household member must regularly participate in online instruments transmitted directly to their unit.

Implementing Contacting Procedures

Once researchers have decided on what type of sample they intend to pursue, they must develop procedures for contacting individuals and soliciting participation. Scholars can procure research participants by e-mailing list-based samples or soliciting visitors to Web sites (Couper, 2000). Additionally, in cases where a multimode approach is used, recipients can be offered a Web link through a traditional medium, such as a letter or through a human interviewer. Each approach is best suited for generating a particular type of sample, possessing certain advantages and disadvantages compared with the others.

E-mailing List-Based Samples. Research participants can be recruited by e-mail. Although it is not currently possible to generate a list of the entire population of e-mail users, there are subsets of e-mail users whose addresses are compiled and catalogued by various organizations for internal or external purposes. Researchers can acquire e-mail lists in a variety of ways. Many organizations maintain private lists of e-mail addresses of affiliated personnel. For instance, most colleges and universities, government institutions, and large companies maintain internal mail systems that provide an individual e-mail address for each member of their organization. These private databases, in most cases, not only provide universal coverage of these closed populations but are kept current, thereby eliminating the problem of nonworking addresses. Acquiring such lists, though, is often difficult. Private organizations are often keenly sensitive to potential privacy concerns of individual members, unwilling to part with such lists except under special circumstances.

Before contacting members of an e-mail sample, researchers should be mindful of the threat posed by spam, or unsolicited bulk commercial e-mail. In recent years, spam has proliferated, increasing disdain and frustration among the online community and prompting it to develop countermeasures (Fallows, 2007; Grimes, 2006). Many servers, organizations, and individual users actively employ software to stop the spread of spam. These programs filter bulk mailings or e-mails from unfamiliar sources, returning suspicious transmissions to senders as undeliverable. Distinguishing legitimate research studies from spam is one of the biggest challenges facing online researchers. Many Internet users fail to distinguish between requests for research participation and other solicitations. There is a fine line between promoting research and pushing products, and researchers need to be as direct and forthcoming as possible about the purposes, goals, and benefits of their projects. Researchers should be sensitive to the way e-mails may be perceived and interpreted by users, home organizations, and host servers. They should seek permission from administrators whenever possible, particularly when relying on organizational directories or transmitting e-mails to a single server. And they should construct e-mails that emphasize the legitimacy of the study.

After securing a list of e-mail addresses, recruitment e-mails can be sent. Researchers must construct messages that overcome the cloud of suspicion that hangs over unsolicited e-mail and leads many users to ignore or delete such messages. This places a premium on constructing an appealing e-mail heading.

An e-mail heading is the virtual equivalent of addressing information appearing on a postal envelope. It generally contains six text-entry fields: (1) a "from" field for the e-mail address of the sender, (2) a "to" field for the e-mail address of the recipient, (3) a "subject" field for the focal point of the e-mail, (4) a "cc" field for the e-mail addresses of users receiving a disclosed "carbon copy" of the e-mail, (5) a "bcc" field for the e-mail addresses of users receiving a "blind carbon copy" of the e-mail, and (6) an "attachment" field for any files to be added to the e-mail. Researchers must be careful about what information is entered on these lines, since e-mail users frequently base their decisions to open a message on it.

The "from" field should contain an e-mail address that will establish the legitimacy of the sender in the eyes of the respondent. If a private list is being used, someone inside the organization should be included as the sender, if possible. Only as a last resort should researchers include their own e-mail, and preferably, it should include a recognizable suffix identifying their home institution (Smith & Kiniorski, 2003). The "to" field should be limited to a single recipient. Although most e-mail software permits multiple names to be placed in the "to" field of an e-mail heading, the appearance of multiple names in the "to" field often impersonalizes the message and increases the chance of the note being filtered as spam. Personalizing the e-mail invitation, by including the full name of the respondent, has been found to dramatically increase cooperation (Heerwegh, 2005; Heerwegh & Loosveldt, 2006a, 2006b). The "subject" field should contain a brief, precise phrase or sentence inviting users to participate in a research study. Researchers should avoid words commonly used to market products, such as "free," "money," or "offer." As a result, incentives are best not mentioned in the subject field. Focus instead should be on legitimizing the e-mail, either by referencing the researcher's home institution or the objectives of the study. The "cc" and "bcc" fields should remain blank. The carbon copy and blind carbon copy fields enable identical e-mails to be transmitted to multiple e-mail addresses simultaneously. Although these features offer an efficient approach to transmitting bulk e-mail, they also are likely to trigger spam filters. Researchers are well-advised to use e-mail software that can be configured to send e-mails one-at-a-time. Last, the attachment field should be used prudently. Some users will not open e-mails containing attachments for fear of acquiring computer viruses. Therefore, the line should remain blank, if possible. Attachments containing the instrument, audio files, video streams, or any other extraneous materials, should never be affixed to the message.

The body of the message should disclose the objectives, procedures, expectations, and authors of the study, as well as how the individual's name and e-mail address were obtained. These messages should be brief and crafted as warily as the header. As with subject fields, researchers should avoid using words and phrases commonly found in product advertisements. Intrinsic appeals are less likely to be flagged than extrinsic appeals. Requests for survey participation should include a hyperlink to a Web page hosting the instrument. However, researchers cannot rely solely on the hyperlink to direct potential respondents to the data collection instrument. Some subjects will not be connected to the Internet when they open their e-mail, or their e-mail programs will not be configured to process the hyperlink. Therefore, the URL address of the Web site as well as instructions regarding how to import it into a browser should be included in the e-mail as well.

Soliciting Visitors to Web Sites. Researchers can also be recruited by soliciting visitors to a Web site. Online solicitations are most often used to recruit large, diverse, nonprobabilistic samples. Millions of Internet users surf the contents of the Web daily. By posting advertisements on frequented Web pages or popular search engines, researchers can invite a variety of visitors to take part in their studies. Interested parties simply click through the advertisement and are immediately directed to the research site, where they can be formally recruited and, if receptive, directed to the appropriate version of the instrument to complete. Such Web advertisements are the virtual equivalent of recruitment posters, with the advantage of providing potential subjects immediate access to research materials.

In a small number of cases, usually limited to cases where a researcher is interested in inferring their research to a survey of visitors to a specific Web page, survey solicitations on a Web can result in scientific samples. For example, a corporation or organization that is attempting to better design their Web page might seek a scientific survey of Web visitors. In this case, the sample frame—visitors who view a Web page—corresponds exactly, or almost exactly, to the target population of a survey, making scientific sampling from a full sample frame possible.

Online advertisements come in two forms: embedded and intercept advertisements. Embedded advertisements are displayed as part of a Web page. In contrast, intercept advertisements appear in a separate browser window from the one being used to retrieve a particular Web page. Whereas embedded advertisements are part of the page being retrieved, not interfering with its contents, intercept ads obstruct the content of the requested page if they are to be read. There are numerous types of intercept ads; pop-up ads appear over the requested page, floating ads move across the content of the page, and pop-under ads appear under the page. Interstitial ads appear before the browser brings up the requested page, while hijack ads divert the user from the requested page entirely, redirecting the user to the new browser window instead. Regardless of type, they too can take on any size, appear anywhere in the viewing frame, and feature text, graphics, or animation.

Intercept advertisements have traditionally enjoyed a higher degree of forced exposure than embedded advertisements (Comley, 2000). Whereas embedded advertisements can be ignored, intercept advertisements must be either opened or closed before the requested Web page can be viewed. This requirement not only ensures that all intercept advertisements will be seen, but that all viewers must consciously decide whether to participate or not. The primary disadvantage to using intercept advertisements is the increasing use of software capable of blocking intercept advertisements from overriding the requested Web page. Since embedded advertisements are part of the requested page, such programs cannot filter them. In either case, click-through rates are usually extremely low.

In recent years, a growing number of studies have assessed factors underlying effective Web advertisements, offering guidelines for the construction of successful research invitations. Banner advertisements that use an intrinsic appeal, such as "Contribute to an important study," have been found to be more effective than those with an external appeal, such as "Win valuable prizes" (Tuten, Bosnjak, & Bandilla, 1999). Consumers who are exposed to more colorful, image-laden Web sites rather than monotone, simple Web sites, were more likely to browse, engage in more unplanned purchasing, and seek out more stimulating products (Menon & Kahn, 2002). Advertisements with stationary black backgrounds have been found to have significantly more positive effects on judgments of the advertisement and purchasing intention than advertisements with blinking phrases and moving images (Stevenson, Bruner, & Kumar, 2000). Researchers who attempt to solicit participants through Web recruitment must be careful, however, to design ads that are neutral with respect to the underlying goals of the survey. If certain types of people are more likely than others to respond to a particular Web ad, selection bias can result.

Obtaining a high click-through rate, though, is not the same as securing research participants. Once Web users have clicked through the advertisement, they still must be formally recruited. A Web page must be constructed that informs visitors of the objectives, expectations, and procedures of the study in a manner that appeals for their participation. Generally, Web designers should focus on creating pages that are as basic as possible, with limited graphics or images. This ensures that visitors, regardless of the specifications of their browser, connection speed, or hardware will be able to view the page as researchers intended.

Regardless of the approach adopted, the likelihood that Web surfers will happenchance across these pages is slim. Most Web surfers are not looking to be participants in various research studies; hence, they are unlikely to look for them in search engines or visit Web sites of professional organizations. Research sites will have to be promoted by word-of-mouth or in offline publications. Such approaches, though, may defeat the purpose of Web recruitment, resulting in participants who could have been secured without Internet-based initiatives altogether.

Non–Web Recruitment Approaches. Some researchers recruit participants to Web-administered surveys through more traditional means of recruitment. In many cases, this occurs when the Web component of a survey is being used as a part of a multimode survey. For example, a traditional mail survey might offer a respondent the option of completing the survey online, as opposed to mailing a paper form. In this case, the researcher is best served if the Web link is clearly marked in a conspicuous place on the top of the survey, or if it stands out in bold lettering, colored print, or some other conspicuous way. Researchers who want to encourage this method of contact might also include a special postcard-sized enclosure that contains a clearly identifiable link to the survey.

Telephone interviews can also be supplemented with Web data collection methods in some cases. In this case, interviewers can read a Web link to provide instructions to a respondent on how to access a Web page containing the survey. In some cases, providing verbal instructions on a telephone answering device can enable respondents to complete a survey over the Web and increases response and participation rates to surveys.

In any case, the researcher is well served if they are able to provide the prospective respondent with a Web link that is easy to remember. If possible, researchers might secure a Web domain name specifically identifying the study. Alternatively, a researcher can direct a respondent to an easily identifiable Web page for the survey and then provide a clear link to the appropriate instrument.

Researchers who use traditional sampling and contact methods to contact participants to Web surveys need to keep several items in mind. Foremost, researchers need to be mindful that traditional probability sampling methods for Web surveys will only yield probabilistic results if they are applied to populations that entirely have Internet access. For example, a general population mail survey is bound to include many individuals who do not have Internet access. In addition to what is likely to be an unacceptably low cooperation rate, these surveys would also suffer from bias resulting from the inapplicability of the Internet survey method to non-Internet households.

Additionally, in an era where response rates to most surveys are declining rapidly, placing potentially burdensome requirements on respondents to a survey is not well-advised. Although some respondents might prefer to complete a survey over the Web rather than on paper, other respondents might be moved to complete a survey immediately on opening the mail. Thus, Web links to surveys using traditional sampling methods are best used only as a supplement to a traditional survey methodology, or in cases where respondents are likely to be so extremely motivated, to take a survey that a high response rate is assured if data collection is conducted in any mode.

Finally, any use of mixed-mode survey processes faces the possibility of mode effects, or differential responses to survey questions based on the method of completing the survey. Research concerning the nature and extent of mode effects between Internet surveys and other methodologies is complex (see Dillman, 2007, pp. 453–461). Although multimode surveys hold great future promise, researchers need to be mindful of potential difficulties comparing or combining data between modes.

Administering Instruments on the Internet

Once a sample has been drawn either using online or offline methods, researchers must administer a research instrument to them; in this respect, the Internet offers an array of opportunities. Online instruments can accommodate a variety of presentation styles, question formats, and response options. Unfortunately, users access the Internet with a variety of hardware, software, and connection equipment, any of which can alter the appearance and functioning of the instrument. Thus, researchers must administer instruments that are not only capable of addressing the hypotheses under investigation but can be presented and delivered in a uniform, yet usable, manner to each participant.

Regardless of the method in which subjects are contacted to complete an Internet survey, most surveys are posted on a Web page, which visitors can view and complete using a Web browser. Web postings possess considerable design flexibility, enabling researchers to integrate multimedia and interactive elements in visually appealing formats.

Researchers seeking to create Web surveys have a variety of software options. Web-based survey instruments can be designed from scratch using a number of user-friendly HTML editor programs that can walk researchers through the process. Programs specifically designed to create online research instruments have also emerged in recent years. As long as users stick to the templates provided, these programs can be useful, time-saving devices. However, they often lack the flexibility necessary to adapt to the specific needs of the researcher. Similar to these programs, some services integrate Web survey creation with online data collection and analysis services. Finally, many programming systems designed to be used for other types of computer-assisted survey interviewing such as telephone or face-to-face interviews have been adapted for creating Web surveys. In addition to being easy to integrate into multimode data collection strategies, many of these systems offer a nice balance between the flexibility to design custom protocols and the ease of using the types of design elements typically needed in advanced survey research.

Item Delivery

Researchers must initially determine the method for delivering individual items to subjects. Researchers can either display items on a single static Web page or disseminate them over multiple interactive Web pages. Each approach has advantages and disadvantages that researchers should weigh before making a selection.

Static delivery displays the entire instrument on a single Web page. Subjects can view all the questions simultaneously without having to access a new page. They can scroll from item to item either forward or backward through the instrument without limitation. They transmit responses to the server on one occasion, after they click a "submit responses" button at the end of the instrument. They are the electronic equivalent of a pencil-and-paper instrument.

Static Web instruments are easy to implement. They can be programmed straightforwardly with HTML forms. Client-side coding can be added without jeopardizing the integrity of the instrument. Conditional branching, whereby subjects are routed to different question sequences depending on their responses, can be implemented in single-screen instruments with either manual or automatic scrolling. Manual scrolling requires subjects to manipulate arrow keys with their fingers or an elevator button with a mouse either up or down to move to the appropriate question. Automatic scrolling, in contrast, positions the next appropriate question at the top of the browser window either involuntarily after the screening question has been answered or after a hyperlink has been clicked. In either case, though, respondents can easily move forward or backward through questions, even those not intended for them.

Static single-page delivery possesses a number of pros and cons. The programming simplicity of static delivery minimizes download times and is compatible with a wide variety of browsers, particularly outdated ones. Subjects can view the entire instrument, enabling them to monitor their progress, understand the context of the questions, and review and/or change previous answers. However, static delivery cannot accommodate randomized questions or validation checks. It loses subjects' responses if they do not properly submit the instrument or their Internet connection fails. And it is more susceptible to cross-contamination among questions, since respondents can see the items as related, thereby increasing the correlation among them.

Interactive multiple-page delivery is the alternative to a static single-page design. Interactive delivery displays items one at a time or in blocks on a single page that must be processed before users are provided with the next page of items. Subjects can be prevented from skipping pages or returning to prior ones. After each item is completed, responses are transmitted to the host server by clicking-through a "next question" or "forward" button affixed to the bottom of the page.

Interactive Web instruments are more difficult to implement than static Web pages, typically requiring programming to derive full benefits. However, the options available to researchers are often of great benefit. Interactive delivery enables automatic skipping and conditional branching. Question ordering and response options can be randomized, and item response times can be measured.

Regardless of which features are employed, researchers opting for interactive delivery must make two key decisions. They must determine the number of items to be included on each page. Some observers have suggested that researchers are better served by grouping related items together on the same screen. Couper, Traugott, and Lamias (2001) compared two different forms of interactive screen delivery: one item per screen and multiple, related items per screen. They found that the condition showing multiple-items per screen took less time to complete and generated less item nonresponse, though they also produced greater interitem correlations among questions. Peytchev and colleagues (2006) also compared two different implementations of the same Web survey: one with multiple screens and the other with a single long scrolling page. The overall completion time for the long scrolling page was less than that for the survey version with multiple screens, although overall levels of survey break-off rates were not different.

Researchers must determine whether respondents should be required to answer questions on one page before the next page appears or whether they can proceed without doing so. Although forcing responses can eliminate item nonresponse, it does so in an overbearing fashion that may lead subjects to drop out of the instrument. Researchers are better served by incorporating pop-up screens or conditional pages designed to inform respondents when they fail to answer questions and encourage them to reconsider (DeRouvray & Couper, 2002).

Interactive delivery has many advantages. It ensures greater uniformity and control of response conditions, reducing question order effects. It permits the analysis of dropouts through the inspection of their partially completed instruments. Prompts can be introduced after any page that is completed incorrectly or is left blank. And subjects who wish to pause and resume the instrument at a later time can do so from the point where they left off if a "stop temporarily" or "quit for now" button is included on a page. On the flip side, interactive delivery requires many more interactions with the host server, increasing download times and the possibility of connection failures. Navigation can be more challenging. And respondents prevented from inspecting the entire instrument simultaneously may lose track of the context of various questions.

In most cases, the design of the instrument dictates the choice between static or interactive delivery. For example, interactive delivery is the optimal choice if complex question ordering is a priority, whereas static displays are better suited for shorter instruments targeting technologically varied populations.

Response Style

Another important decision that researchers must make when formatting an instrument is the type of items to be asked. Questions can be asked open ended or closed ended. Open-ended questions enable subjects to answer in their own words, whereas closed-ended questions force subjects to choose from a predetermined set of responses. Researchers should recognize that open-ended questions, though beneficial for some analyses, require more effort from online subjects and often induce unit and item nonresponse (Knapp & Heidingsfelder, 1999).

Open-ended questions are straightforward to implement. Researchers simply insert a text-input field below the question for typed entry. These text-input fields may be programmed to limit responses to a fixed number of characters or accept as much text as desired. In either case, researchers must determine the initial size of the field that subjects confront. Prior research has found that longer entry fields elicit less nonresponse, lengthier responses, and more explicit answers than shorter ones (Couper, 2000; Couper et al., 2001; Fuchs & Couper, 2001). However, there is also evidence that longer entry fields are more prone to receiving invalid entries from subjects than shorter entry fields (Couper et al., 2001). These findings suggest that researchers should pretest the length of entry fields, spacing them according to what is expected to be the typical response.

Closed-ended questions pose challenges to researchers as well. Researchers can choose from text-input fields, pull-down menus, click tags, or slider bars. Each can be adapted to solicit single or multiple responses.

Text input fields are designated boxes on a Web page where subjects can indicate their preference by typing a character, usually an "X" or a numerical value. They can be programmed to accept single or multiple responses, enable options to be rank-ordered, and even compute running totals. The downside is that they require time and effort for subjects to complete and programming skills from researchers to ensure that they actually prevent invalid responses.

Another response format available on the Web is pull-down menus (or drop-boxes). Pull-down menus conceal the list of response options, save for a default category, until the subjects click on the menu with their cursor. Subjects indicate their preferences by clicking again on the appropriate response category. Researchers can program pull-down menus to accept multiple responses. Since respondents only see one response category until clicking the menu, researchers must be careful to set the default response option blank on one of the standard response categories to ensure that they can determine whether subjects do respond. Pull-down menus have the advantage of taking up little space on the screen, making an instrument appear shorter to subjects. Unfortunately, the two-step process that must be completed to respond both decreases usability and increases the time necessary to complete the instrument (Dillman, 2007). Respondents answering questions in this method have also been found to be more likely to select choices toward the top of the list, have higher nonresponse rates, and be more likely to inadvertently select unintended answers when using certain types of mice (Couper, Tourangeau, Conrad, & Crawford, 2004; Healey, 2007). Thus, although drop-down boxes are common features of Web surveys and forms, researchers are generally well-advised to avoid these answer formats.

The Web also enables researchers to collect responses to closed-ended questions with click tags. In this format, subjects respond by maneuvering their cursor over the input tag of their preferred choice and clicking their mouse. Click tags can be radio buttons or check boxes. Radio buttons are circular click tags that appear filled when selected. Radio buttons allow one and only one choice from the predetermined categories, thereby preventing multiple responses. In contrast, check boxes are square tags that display a checkmark when selected. Check boxes accommodate as many responses as desired by the individual taking the instrument. Click tags are easy to understand and fast to employ, but take up considerable space, and require hand-eye coordination to use efficiently.

Finally, the Web offers the opportunity to introduce slider bars. Slider bars align response options along a track containing a pointer or bar that can be moved back-and-forth. Subjects slide the bar until it aligns with the preferred response. Sliders are a particularly attractive option for questions with rating scales because they offer the sense of a continuum (Arnau, Thompson, & Cook, 2001). They can also be designed to permit more response options than their counterparts, while occupying no more space. However, sliders may not appear identically across all browsers, and it is difficult to differentiate preferences from nonresponse when the default position is left untouched. Moreover, respondents may be less likely to continue with a survey when receiving a slider-bar question than more common formats (Watson, Lissitz, & Rudner, 2006).

No consensus has emerged concerning the effectiveness of different closed-end formats. A series of studies have demonstrated that radio buttons produce faster completion times. Otherwise, choices on closed-ended response formats are best guided by the nature of the question, space considerations, and technical capabilities of the sample.

Researchers must also decide how to handle instances where subjects do not know the answer or do not care to convey it. Some researchers are tempted to not provide an option to respondents in an effort to increase the proportion of substantive responses. This is not only more likely to increase measurement error from subjects who feel compelled to respond but also prevents researchers from differentiating among subjects who leave the question blank. Researchers are better served by including "don't know" and "decline to answer" options after the response categories. Although this will generate fewer substantive responses, this effect can be diminished in multiple-page instruments, where pop-up screens prompting respondent to reconsider such responses can be added (DeRouvray & Couper, 2002).

Alignment

Researchers must also decide how to align or position items on subjects' computer screens. Item placement like display configurations can be fluid or fixed. Fluid layout enables items to expand or contract to fit various display configurations. Since this can obviously change the appearance of text for different subjects, fluid layouts should be avoided. Instead, researchers should implement fixed layouts, where items are positioned to originate from a particular part of the screen.

There are several different alignment decisions that must be made when formatting instruments. First, researchers must determine the horizontal positioning of questions. Although questions can be left justified, centered, or right justified on a computer screen, researchers should only employ left justification, as it is both consistent with user expectations and easier to follow as subjects move from top to bottom.

Second, researchers must determine the alignment of response options for closed-ended questions. Responses can be positioned either vertically (one below another) or horizontally (one after another) under the questions. Vertical positioning is less prone to alignment problems from technical variation, but it takes up more space, extending the physical length of the instrument. Conversely, horizontal positioning saves space but may extend past users' screen configurations requiring horizontal scrolling. For example, horizontal positioning is more appropriate when response options are intended to convey the sense of a continuum; whereas vertical positioning is more suitable when there are an extensive number of response options. In either approach, researchers should remain consistent throughout the instrument to avoid confusion.

Researchers choosing to vertically align response categories must also determine their arrangement relative to the questions that precede them. They must decide whether to place response options in a single column or in multiple columns. Although there is no evidence supporting one approach over another, Couper (2001) did find that users tend to gravitate toward the top half and leftmost options in columned categories. Moreover, researcher must determine whether to left, center, or right justify columns. Left justification is more familiar, centered is more visually appealing, and right justification is closer to the arrow keys used for navigation. Experimentally manipulating left-justified and right-justified response options, Bowker and Dillman (2000), however, found no statistical or substantive differences between users' preferences or their performance.

Researchers employing horizontal alignment also have the option of grouping related items that employ the same set of responses into matrices. Matrices structure items so that each row corresponds to a particular question and each column matches up with a particular response option. Matrices save considerable space (since subjects do not need to be reintroduced to the response options after each question), yield faster completion times (Couper et al., 2001), and yet do not produce higher interitem correlations (Bell, Mangione, & Kahn, 2001; Couper et al., 2001).

The final alignment decision to which researchers must attend is the spacing among questions and between response categories. Stand-alone questions should contain the equivalent of two text lines of blank space between them, while questions within a matrix should contain the equivalent of one text line of blank space between them. This preserves physical space on the screen without disorienting readers with textual density. Response categories should be spaced equally. Couper, Traugott, and Lamias (2001) varied the spacing of response categories, randomly assigning subjects to a question with equally spaced response categories, a question where the end points were wider than the middle categories, and a question where the categories widened as they became more distant from the midpoint. They found that subjects were increasingly drawn to the endpoints as the spacing of response options became more varied, thereby increasing the mean deviation from the midpoints. If columns or matrices are being employed, researchers need to ensure that column widths are set to be equal rather than letting them be defined by the length of the text.

Length

Last, researchers must decide on the length of the instrument. The length of an instrument can be measured in one of two ways, either as the number of items administered or the time it takes to complete them. Studies suggest that the length of online instruments can have detrimental effects on response and dropout rates. This not only reduces the number of cases available for analysis but can also increase bias in the data if the response and dropout rates correlate with the variables of interest.

Research suggests that the length of online instruments is correlated with dropout. Dropout occurs when subjects fail to complete an instrument after they have begun, leaving the remaining questions unanswered. Though researchers need to ensure that their instruments provide a sufficient number of variables for appropriate analysis, they also need to be mindful that each additional question appears to increase the odds that subjects will fail to complete the instrument (Galesic, 2006). Crawford, Couper, and Lamias (2001) offer some evidence that by disclosing the length of the instrument to subjects before they begin may serve to lessen the impact of length on response and dropout rates.

Providing Instructions

Instructions to participants must be formulated so that all targeted respondents clearly understand how to complete and submit the data collection instrument. Nonexistent or poorly worded instructions may induce subjects to perform tasks incorrectly, skip particular portions of the instrument, or fail to participate altogether. Considering the variation in sophistication and technologies employed by different users, researchers should not assume that any task is too simple or obvious unless they know otherwise. Instructions should be designed to inform all targeted participants—regardless of their education, technical skills, or online experience—how to perform each function.

Instructions should be helpful without being distracting. They should take on a distinct, consistent appearance—either decorated (e.g., in bold, italics, or color), sized larger/smaller, contained in parentheses, or aligned differently from the other items or stimuli—throughout the instrument (Dillman, 2007). The key is that respondents recognize them for what they are. In this way, more Web-savvy respondents can move quickly past them, while novice users will be drawn toward them. Instructions should also only be as wordy as necessary. Extensive instructions have the danger of disrupting the continuity of instruments or deterring subjects from completing them (Vehovar, Lozar Manfreda, & Batagelj, 2000).

Instruments should open with an introductory statement. Although the introduction should be brief, it should touch on several different issues. It should describe the objectives of the instrument, providing some context for the study. It should summarize the tasks facing subjects and disclose the expected time needed to complete them (Crawford et al., 2001). It should explain to subjects how to configure the screen for ideal viewing (e.g., maximizing the viewing window or changing color palettes) as well as answer the opening question.

Subjects need to be instructed how to operate each type of response format that they confront. Every format possesses certain aspects that are not obvious to those who have never used them before. Instructions for open-ended questions should mention where to provide responses, what characters are preferable, limits on response lengths, and whether boxes extend automatically. Closed-ended questions using text entry fields should explain what types of characters are permissible and where they should be placed. Pull-down menus, if used, should describe how to access hidden categories and how to provide responses. Instructions for click tags should inform subjects that radio buttons require clicking an alternative button to erase responses, while check boxes require a second click on them. And, slider bars should inform subjects how the mouse is used to move the pointer. Researchers also need to consider where to place instructions and how often to provide them. Subjects may forget a long series of instructions given at the outset; instead, instructions should be placed at the points where they are first used. For example, if the first check box comes 30 questions into an instrument, researchers should locate the instructions at or near the 30th question when these instructions become relevant. Determining how often to repeat instructions, if at all, is a bit more challenging. Redundant instructions increase the size of your instrument and may exacerbate long download times or distract more advanced users. Therefore, single-page instruments should only repeat instructions if the chain of questions employing a particular response format is broken. For multiple-page instruments, instructions should be included once on each page, since respondents are usually not permitted to visit previous pages, and even if they were, it would be difficult and burdensome for them to orient themselves to their location.

Instructions must not only describe how to complete various tasks but how to navigate through an instrument. Researchers delivering instruments in a single Web page should inform subjects how to use scrolling bars, whereas those employing multiple-page instruments should describe how to operate applicable action buttons. If subjects are provided with an option to quit, they should be instructed how to resume later and what they can expect when they do so. Instruments that employ conditional branching on single-page instruments must ensure that subjects can skip to the appropriate items. Researchers employing nonautomated delivery should include skip instructions to the right of the item where subjects can easily see them after reading the response option, while those employing single-screen instruments with automated skips should forewarn subjects of the movement to reduce their disorientation after it occurs.

Balancing the need for and threats from extensive instructions can be challenging. Researchers posting their instruments on the Web, though, possess several tools to make this effort easier. They can inset hyperlinks or pop-ups to provide more detailed instruction without lengthening or disrupting the continuity of the instrument. These should not entirely replace embedded instructions, since subjects may miss or ignore them. Instead, they should be used to provide greater depth for complicated explanations or information that is not applicable to everyone. Hyperlinks or pop-ups should be set off from, but adjacent to, related instructions with a clear, concise remark, such as "For further details, click here," with the words programmed to activate the hyperlink or pop-up. Researchers are also well served by affording respondents with opportunities to relay comments or questions to the researcher.

Collecting Submissions

After inducing subjects to complete the instrument, researchers must provide the means by which they can submit their responses. The approaches are somewhat different depending on whether the survey uses a static Web page or an interactive Web page. Each has several advantages and disadvantages.

Researchers administrating static Web instruments must instruct subjects that they can return them by clicking a "submit" button included at the end of the instrument. They should program the button to both transmit the instrument and click-through respondents to a corresponding page notifying them that the instrument has been successfully transmitted and thanking them for their cooperation. Submitted instruments are then e-mailed to the researcher's workstation unbeknownst to subjects.

This approach has several benefits. The instrument can be easily programmed to transform closed-ended responses into a preassigned numerical format and automatically imported into a database, saving researchers considerable time and effort. Moreover, the submission mechanism does not directly expose any personal identifying information, thereby accentuating the perception of anonymity. The flipside, though, is that subjects unfamiliar with Web transmissions can easily lose their responses by inadvertently closing their browser, instead of clicking the "submit" button. Moreover, if the connection fails or the transmission is corrupted, the entire set of responses vanishes as well.

Researchers administering interactive Web instruments can instruct subjects to submit either completed pages or the entire instrument. Since interactive Web instruments are usually administered over a series of pages, submission procedures are typically designed to appear as continuation buttons. When subjects click "continue" or "next page" buttons affixed to the bottom of the page, responses are transmitted directly to the corresponding Web server, without invoking e-mail. After arriving, responses are automatically compiled for each subject and then added to a preformatted database.

Interactive instrument submission possesses several advantages over their counterparts. It loses little existing data when Internet connections fail or subjects abandon the instrument. By avoiding e-mail altogether, it is likely to induce stronger perceptions of anonymity. And, it places far fewer demands on researchers' workstations. Unfortunately, these benefits come at a price. Interactive submissions are more expensive to manage than their counterparts and require more advanced programming skills to implement.

Conclusion

The Internet is an exciting and increasingly popular method for collecting survey and other sorts of data. Compared with other data collection modes, the Internet often has a relatively low marginal cost for conducting interviews, particularly when large samples are desired. Equally important, the Internet offers a way of incorporating experiments and visual stimuli into self-administered surveys. In many cases, experimental researchers find the Internet to be an efficient way of conducting studies among populations that are far broader and more representative than those typically found in a psychology lab.

At the same time, though, the Internet faces shortcomings that researchers need to be aware of. In cases where scientific samples of the general population are not important, or where probabilistic samples of Web users can be generated, the Internet can serve as an optimal data collection tool. The Internet is often used successfully in studies employing multimode data collection approaches that provide respondents with different options for completing survey questionnaires. However, Internet studies are usually seen as inadequate when used for estimating population parameters for groups that might not all have Internet access or who might not be easily identified or included in scientifically developed Internet sampling frames. Consequently, when considering Internet data collection, researchers need to think very carefully about the goals of their study. In particular, researchers need to be especially careful in specifying and considering the relationship between the target population of their study and the available sampling methods. Though Internet data collection offers great promise, it also has limitations that can make an otherwise useful data collection method inappropriate for some studies.

Discussion Questions

1. For what types of samples would the Internet be an appropriate tool to recruit subjects? For what types of samples would the Internet be inappropriate?

2. What methods are available for contacting individuals and soliciting participation? What are the advantages and disadvantages of each?

3. What two approaches can be taken to question delivery on the Web? What considerations should researchers weigh before making a decision?

4. What response styles are available to researchers designing questions for Internet surveys? What types of questions are best suited for each? How should they be aligned on a Web page?

5. Why are instructions so important for Web surveys? What conventions should be adopted to ensure that respondents understand how to complete Web surveys correctly?

6. What options are available for collecting survey submissions? What are the strengths and weaknesses of each option?

Exercises

1. Design an e-mail to send to prospective subjects inviting them to participate in a Web survey. Make sure to describe how you will construct the heading and the body of the message.

2. Construct a 20-item Web survey. Detail how you will approach the following considerations:

   a. What method will you use to deliver the questions to subjects?
   b. What response style will you use with each question?
   c. How will you align each question on the page?
   d. What instructions will you use to explain how the survey should be completed?
   e. How will the submissions be collected?

3. Design an Internet sample

   a. Who, specifically, do you want to target?
   b. What percentage of these people are likely to have Web access? How can you find this out?
   c. How are you going to develop your sample frame? What specific lists or sources will you use to develop your sample frame? How will you get access to these lists or sources; will you need permission?
   d. How well does your sample frame cover the population that you are intending to study? If the sample frame does not cover the entire population, how might the people you exclude be different from those who are in the list? Is this a problem for your data?

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CHAPTER 14

Concept Mapping for Applied Social Research

Mary Kane

William M. Trochim

The vitality of thought is in adventure.
Ideas won't keep. Something must be done about them.

—Alfred North Whitehead

Applied social research is fundamentally the adventure of connecting ideas with reality, research with practice. When we set out on adventures, it's useful to have a map of the territory we hope to traverse. The social psychologist Karl Weick tells this famous (or infamous) anecdote (Basbøll & Graham, 2006) on the importance of maps:

> This incident, related by the Hungarian Nobel Laureate Albert Szent-Gyorgyi and preserved in a poem by (Holub, 1977), happened during military maneuvers in Switzerland. The young lieutenant of a small Hungarian detachment in the Alps sent a reconnaissance unit out into the icy wilderness. It began to snow immediately, snowed for two days, and the unit did not return. The lieutenant suffered, fearing that he had dispatched his own people to death. But the third day the unit came back. Where had they been? How had they made their way? Yes, they said, we considered ourselves lost and waited for the end. And then one of us found a map in his pocket. That calmed us down.

> We pitched camp, lasted out the snowstorm, and then with the map we discovered our bearings. And here we are. The lieutenant borrowed this remarkable map and had a good look at it. He discovered to his astonishment that it was not a map of the Alps but of the Pyrenees. (Weick, 1995, p. 54)

This chapter is about developing maps, not of geographical territory, but of theories and ideas. It describes a structured applied social research methodology that can be used to connect theory to observation and research to practice.

Concept mapping is a method for designing and populating conceptual models, to inform, confirm, or revise a testable theory. Recognizing that a wide range of thought and practice are associated with the term *concept mapping*, we concentrate on one particular approach as having special relevance for applied social research and that is especially appropriate for this Handbook. This approach has strong roots in several important traditions in applied social research, and the method has broad utility in many applied social research contexts.

In this chapter, we place concept mapping within the more general context of structured conceptualization methods; we then describe the specific steps in implementing this methodology, from initiation of a project to utilization of results. We consider the variety of ways this concept mapping approach has been or could be used in applied social research. Finally, we discuss some of the current related issues in, and how it might evolve in the near term.

Conceptual Models and Applied Social Research: Theory to Practice

Kurt Lewin's (1951) aphorism that "there is nothing so practical as a good theory" is a fitting entreé to the role of conceptual models in applied social research. It reminds us that social research is constructed on conceptual frames that underlie every hypothesis, observation, analysis, or conclusion. The challenge for social scientists is to establish the connection from theory to practice. Our particular interest is in how concept mapping enables this connection.

The constructs of theory, concept, and model are central in social research and embedded in concept mapping. We develop theories, define concepts, and create models in order to respond to, investigate, or anticipate complex social issues. A *theory* might be defined as a proposed explanation or description of some phenomenon. There are a great variety of types of theories; just within the domain of science, diverse exemplars—the theory of evolution or relativity theory, for example—advance debate, discovery, or innovation. *Concepts* are the elements or components of theories. Simply, a concept is any abstract idea. Concepts are nested within other concepts, constituting a hierarchical, interrelated, and complementary structure of meaning. A conceptual *model* is a representation of a theory or some portion of it. It can be verbal, mathematical, graphical, or symbolic. A conceptual model shows how the concepts in a theory are related to each other, to their empirical manifestations, or to the theories they are intended to support.

A conceptual model, as the visual summary of a theory of abstract ideas, provides a view of how a person or group thinks the world operates in the context being considered. The model acts as a reference point, a device against which one can anticipate change or juxtapose empirical observations. In the simplest sense, discrepancies between the model and what we observe or have evidence for suggest that the model may need to be revised, that our observations may be inaccurate, or both. Theories and conceptual models are the everyday indispensable components of applied social research. Often implicit, frequently fluid, theories and models help define programs and interventions, identify causal variables, support decisions, or enable construct definitions and measurements. We cannot conduct a randomized experiment without even the simplest theory (e.g., if *x*, then *y*) or at least an operational sense of the key concepts (such as the meaning of *x* and *y*).

Given the prominence of theory development and conceptual models in applied social research, it seems logical that we would develop ways to generate and improve theories and models, and it's somewhat surprising to see how little work addresses this issue. While there is considerable folklore and mythology about the source of theories and how they are constructed (e.g., Einstein's thought experiments and the Watson and Crick's smoke helixes), these tend to be scientist-centric notions of where theory comes from. While much theory certainly does originate from scientists, there is no scientific basis for considering scientists the sole or even best source of ideas for theories—especially in social research contexts where the phenomena of interest are directly accessible to anyone.

Structured Conceptualization. Concepts characterize and define theory; and provide the elements of the models we develop to represent that theory. Models—visual, verbal, numeric—allow us to describe interrelationships of the concepts that comprise or inform the theory. While we use concepts routinely and perhaps subconsciously in everyday thought, we can approach them systematically, consciously, and collectively as well. The term *structured conceptualization* refers to this systematic effort to develop models and theories. Trochim and Linton (1986) developed a general model of structured conceptualization based on three major components:

- *Process steps:* To conceptualize, individuals or groups *generate* ideas that make up a conceptual domain; *structure* the domain by specifying how those ideas are related and *represent* the domain in words, pictures, or mathematical symbolic notation. These are represented simply as *G, S,* and *R.*

- *Perspectives:* There are multiple perspectives from which one accomplishes each process step (or combinations of steps), divided into three broad categories: from the point of view of an *individual,* a *group,* or a formalized predetermined *algorithm.* The letters *i, g,* and *a* represent these perspectives.

- *Representational forms:* Any conceptualization can be represented in one or more forms: *verbal* as in lists or text descriptions; *pictorial* such as maps, or *mathematical* or other symbolic notation. The letters *V, P, and M* represent these forms.

The structured conceptualization model's components can be combined (Figure 14.1) to represent a diverse range of conceptualization types, and they can be distinguished using a simple summary notational form.

In the figure, perspectives are shown in brackets to signify that any given process step can be done by one or more of them.

An example illustrates: In everyday thought we typically do not distinguish conceptualization stages methodologically. When we think about grocery shopping, for instance, we typically think about the items we want to purchase and how they are organized in the store, and, essentially simultaneously and unconsciously, make a grocery list. In structured conceptualization terminology, we might represent this as

$$(GSR)_{\mathrm{i}} \rightarrow V.$$

This model indicates that all process activities (G, S, and R) are accomplished essentially as one process step (thus enclosed within common parentheses), from

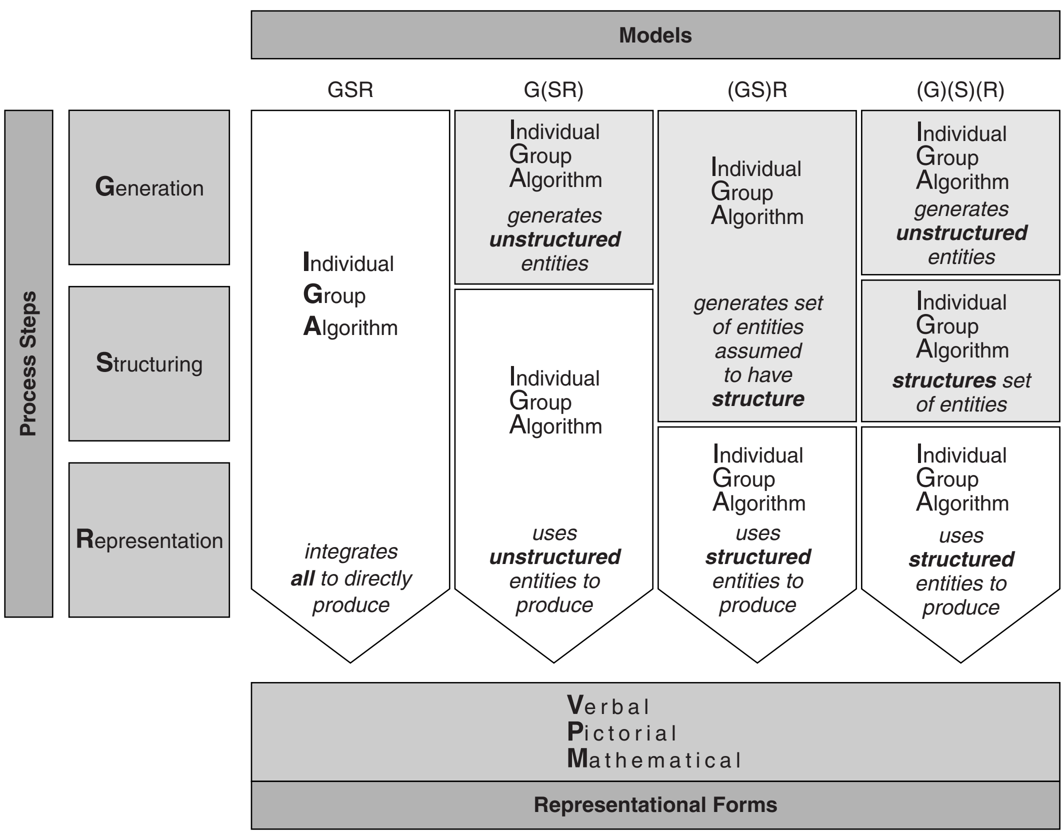


Figure 14.1 A General Model of Structured Conceptualization

an individual's perspective (the subscript i), and yielded a verbal representation (a list) of the conceptual territory of interest.

This everyday type of conceptualization would barely be considered structured and constitutes one end of the structured conceptualization spectrum. Other processes might yield different modes of conceptualization. For example, if a group (g) of people brainstorm (i.e., generate, *G*) a list of ideas and then, in a separate simultaneous step, organize (i.e., structure, *S*) and represent (*R*) them in (verbal, *V*) lists of related ideas, we might depict the process as

$$(G)_{\mathrm{g}}(SR)_{\mathrm{g}} \rightarrow V.$$

Some common examples might be in outline development or the development of an organization's operational structure.

Concept Mapping

The structured conceptualization model enables a more formal description of the central focus of this chapter: Concept mapping is a collaborative and algorithmic-structured conceptualization process that results in a visual representation of ideas and their interrelationships. In notational form, any process that generates a pictorial (*P*) representation could be described as concept mapping. While the notion of relating concepts to each other is as old as thought itself, the idea that the result might be represented visually is a relatively modern phenomenon.

Included under the broad rubric of concept mapping are approaches such as "idea maps" (Armbruster & Anderson, 1982, 1984), "mind maps" (Buzan & Buzan, 1993), "mental maps" (Dillon, Richardson, & McKnight, 1993), "cognitive maps" (Axelrod, 1976), and a host of literatures related to how to generate such structures, including lateral thinking (DeBono, 1971, 1973), brainstorming (Adams, 1979), and brainwriting (Hiltz & Turoff, 1978; Rothwell & Kazanas, 1989). In social science and education, a variety of different "concept mapping" approaches represent several traditions and methods. Many are individual learning, organizing, or writing methodologies.

In contrast, collaborative group concept mapping methods are explicitly designed to collect input about ideas from several or many individuals, identify how they organize the interrelationships among the ideas, and represent their group thinking pictorially or graphically. These approaches are highly structured, with each process step performed as a distinct activity.

Concept mapping as discussed in the remainder of this chapter is of this form:

$$(G)_{\mathrm{g}}(S)_{\mathrm{g}}(R)_{\mathrm{a}} \rightarrow P.$$

The generation of the domain of ideas takes place first, typically (although not necessarily) through some form of group brainstorming. Individuals contribute to the delineation of ideas, so it is notated with a subscripted 'g.' Structuring of the ideas is a distinct second step (within its own parentheses); usually by having each of the individuals in a group sort the ideas. The product of this step is also group based, even though each individual separately sorts the ideas before aggregation. An algorithm is used to compute the map, in this case, a sequence of multivariate statistical analyses as described later. The result is a map (*P*), which the participants discuss and interpret. This type of conceptualization method is both a child and a parent of applied social research methods: Its analytical tools and group processes are rooted in social research; and its integrated process is frequently used as a methodology in applied social research that generates and explores conceptual structures of group thinking.

Group concept mapping was developed in the early 1980s (Kane & Trochim, 2006; Trochim & Linton, 1986), and had its foundations in a variety of applied social research and organizational behavior traditions, including

- group process and facilitation methods, such as brainstorming (Adams, 1979; Osborn, 1948) and Delphi Methodology (Carroll & Wish, 1975);
- psychometrics and scale construction (Shepard, Romney, & Nerlove, 1972), especially thematic sorting and categorization (Coxon, 1999);
- qualitative and mixed methods (Greene & Caracelli, 1997); and
- multivariate statistics, such as multidimensional scaling (MDS; Carroll & Wish, 1975; Davison, 1983; Kruskal & Wish, 1978; Shepard et al., 1972) and cluster analysis (Anderberg, 1973; Everitt, 1980).

Group concept mapping draws on both qualitative and quantitative social research and analysis processes to simplify the complex task of including multiple stakeholder input to build an acceptable conceptual framework or group-authored concept map. In general, concept mapping has the following principles:

- *Concept mapping actively values individual knowledge* or articulation of content. A specific source—usually a person with knowledge or experience of relevance to the issue at hand—has specific input, knowledge, or observations that are valid as contributions to the conceptual picture of an issue. Concept mapping aggregates individual knowledge across knowledge sources—represented by those who contribute input.

- *Concept mapping provides rules* for building, or, often, recognizing, emergent relationships of meaning among the concepts. Concept mapping guides the process by which the participant operates to link the individual input of ideas and observations.

- *Concept mapping constructs a knowledge or conceptual model* from the participants' specific units of input. Combining input via the application of simple rules, the researcher creates an emergent framework, often a unique representation of issues that have not been combined in such a way before.

- *Concept mapping supports the inclusion of often disparate units of existing knowledge* in a unified conceptual framework. Individuals involved in group concept mapping each bring specific requirements and perspectives to the issue at hand.

- Fundamentally, *concept mapping facilitates the identification of common themes to enable theory development, decision making, action, or assessment.*

- *Concept mapping encourages application in the participants' context,* whether it is to understand the elements of the theory at hand, to enhance understanding and planning in an organizational setting, or to design a system for research or evaluation based on community-articulated requirements.

The emergent map that is developed through the engagement of multiple stakeholders and the application of rigorous processes and tools, is a product that, in applied social research, would be difficult to arrive at through other more traditional means.

In practical terms, concept mapping helps a group solve a problem, articulate a group need or desire, author a plan, or develop a program or intervention. A researcher might consider concept mapping an appropriate methodology when a group has unique experience that can inform theory or represents a range of opinions that are not easily reconciled in traditional group conceptualization modes; when the power differential in a group has the effect of reducing contributions of thought from certain quarters; or when the desired outcome of a group's thinking is not well articulated. Concept mapping is an especially applicable methodology for research or evaluation in organizations or communities where there is a history or culture of community participation in decision making and planning.

The Process: Simple Rules and Practical Steps for Concept Mapping

Concept mapping is designed to be useful in many different social research contexts, so it is flexible, transferable, and scalable as a research method. The researcher and his or her community or stakeholder group have flexibility in scheduling the activities required to participate. The time required to conduct an entire concept mapping exercise, as described later in the steps in the text, can vary greatly. A small, targeted initiative may require 6 hours over the course of a group retreat. On the other side of the spectrum, hundreds of individuals, dispersed over time and location, can participate by designing the process in phases that are scheduled to occur in sequence over weeks or months. The method provides scalability as well: It can involve as few as 10 participants or incorporate input from hundreds or even thousands of stakeholders.

The concept mapping process incorporates both participatory activities and rigorous statistical analysis. The simple rules for concept mapping (Figure 14.2) are few in number and well tested in applied social research (Kane & Trochim, 2006).

Note the correspondence of the three central steps in Figure 14.2 with the process steps in the structured conceptualization model. Generating the ideas corresponds with the generation (*G*) step and structuring the statements corresponds directly with the structuring (*S*) step. The concept mapping analysis is another way of describing the representation (*R*) step which, in this case, is a set of analytic algorithms (a).

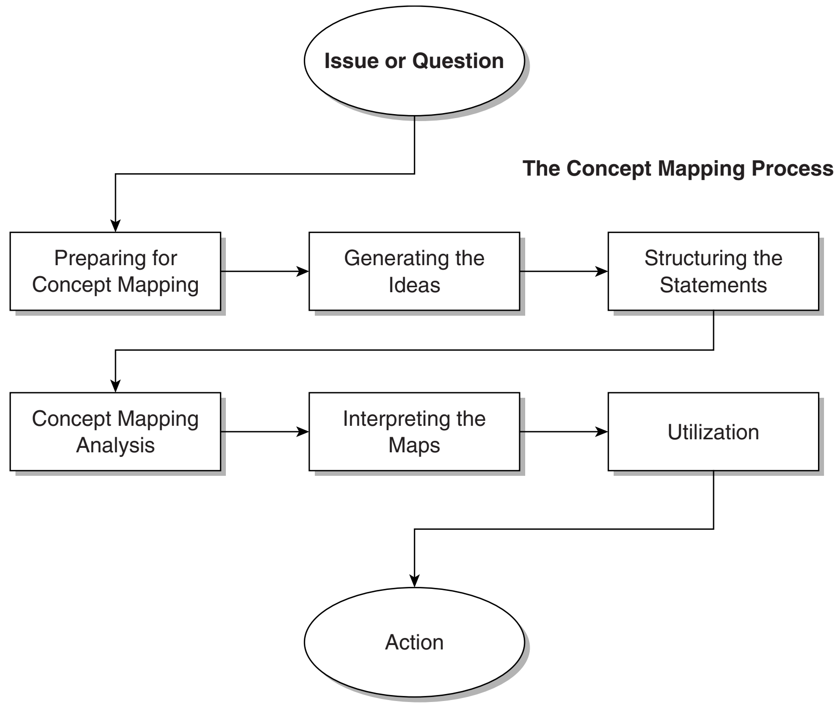


Figure 14.2 Overview of the Concept Mapping Process

SOURCE: From *Concept Mapping for Planning and Evaluation* by M. Kane and W. M. Trochim, 2006. Reprinted with permission of SAGE.

Steps in Concept Mapping

Figure 14.2 indicates that the start of the concept mapping process is the "issue or question." To illustrate the steps, we will use an example of developing a comprehensive statewide cancer control plan that was undertaken by the Delaware Advisory Council on Cancer Incidence and Mortality (Delaware Advisory Council on Cancer Incidence and Mortality, 2004).[1] Established in 2001 as an Advisory Council to serve the governor and legislature, the group was made permanent in 2002, and used concept mapping to develop the conceptual framework for its work. We reference and present the specifics of the initiative where appropriate to illustrate the process.

Step 1: Preparing for Concept Mapping

In preparing to conduct a concept mapping research initiative, investigators identify the two critical sources for the knowledge that will emerge: the *focus* and the *participants in the study*. Unlike many survey techniques, the concept mapping approach makes use of one general focus to elicit a wide range of responses that address the conceptual domain of interest. The focus is typically determined through discussions with stakeholders or research advisers, to ensure that it reflects the desired outcome of the research. Phrased either as a sentence completion prompt or as a directive, the focus asks for specific statements or expressions of need or interest from the participants. The following are examples of focus prompts, for a variety of concept mapping projects:

- "A specific issue that affects the mental health of women and girls is . . ."
- "In order to improve community services to vulnerable new residents in a city, the community clinic system should . . ."
- "We will know that our after-school program is a success when . . ."

Equally important is the selection of participants or respondents, who will provide the appropriate depth and breadth of ideas in response to the prompt. The number of participants is driven by the need of the research. In some cases, a formal sampling plan will be appropriate and necessary. In other research, an opportunistic participant pool is identified due to the strong connection or ability to contribute to the issue at hand—a "community of interest" is formed that is the knowledge base for the project.

The Delaware Example

In the Delaware Cancer Consortium, the Advisory Council identified this focus prompt: A specific issue that needs to be addressed in comprehensive cancer control planning in our state is . . .

Participants in idea generation numbered about 300 from the 500 invited; and included citizens of Delaware, cancer survivors, people related to those affected by cancer, policymakers, public health advocates, and researchers.

SOURCE: Delaware Department of Health and Social Services, Division of Public Health.

Step 2: Generating the Ideas

Typically, brainstorming in response to the focus prompt is used to develop the input for the conceptual framework. In face-to-face meetings or using Web-based tools, participants are encouraged to generate as many statements as possible. Participants create statements using a structured brainstorming process (Coxon, 1999; Osborn, 1948) guided by the specific focus prompt that limits the types of statements that are acceptable and helps ensure that their grammatical structure is similar. Brainstorming is by no means the only method that can be used. Researchers may consider extracting statements from existing key documents, interviews, or lists to form the statement set that responds to the focus prompt; or conducting targeted focus group sessions to ensure that certain stakeholder perspectives are heard.

As a postgeneration step, it is often useful for the researcher to conduct an idea synthesis, to create a rationalized set of ideas from the group, reducing the final number to around 100, by eliminating redundancies and items not relevant to the focus of the research. In some cases, a formal content analysis (Krippendorf, 2004) is used to synthesize the statement set. The Delaware initiative included idea synthesis.

In the Delaware Cancer Consortium, the number of statements generated was more than 500; after idea synthesis the number representing the conceptual domain was 118.

SOURCE: Delaware Department of Health and Social Services, Division of Public Health.

Step 3: Structuring the Statements

Having developed a unique set of statements that define the current territory of the issue at hand, the participants next perform three simple tasks: unstructured pile sorting of the statements, statement ratings (often of value or opinion), and responses to brief demographic or characteristic questions.

In unstructured pile sorting, each individual arranges the statements in piles or groups in a way that "makes sense" (Coxon, 1999; Rosenberg & Kim, 1975; Weller & Romney, 1988). Specific guidelines direct the participants to sort each item with those most related to it in meaning. Guidelines also state that there cannot be (a) the same number of groups as ideas; (b) one group consisting of all items; or (c) a "miscellaneous" group (any item thought to be unique is ideally put in its own separate pile). Weller and Romney (1988) point out that unstructured sorting (in their terms, the pile sort method) is appropriate in this context because it can accommodate a larger number of items than other common data collection methods. At the conclusion of the sorting activity, each sorter will have placed two statements together if he or she thought that those two statements shared some common elements. The sorter will have placed statements in different groups if they did not seem to be related to each other. The result will be a unique classification of similar and dissimilar statements for each sorter.

For the DCC Cancer plan, 32 individuals conducted individual pile sorts of the final statements, and the data were used as the foundation for the development of the concept map.

The project asked participants to *rate* (or provide value observations on) *importance* and *feasibility*. A total of 93 participants provided ratings on importance, and 80 provided ratings on feasibility.

Participants were asked to provide nonidentifying information in response to the following characteristics:

- County of residence
- Relationship to cancer control
- Type of organization

SOURCE: Delaware Department of Health and Social Services, Division of Public Health.

Ratings may take different forms and ask a range of questions, and are collected on each statement from each stakeholder participant. Although a standard Likert-like response scale is most common, ratings in concept mapping can consist of 0 to 1 (*no–yes*) ratings to 0 to 100 (percentage) ratings or may not even be "ratings"—they can consist of measurement of each statement on virtually any characteristic. Ratings are collected to enable the researcher to observe value or opinion differences on the specific ideas, from the participating stakeholders. It is often useful to collect and compare multiple ratings, such as the importance and feasibility of each statement. Because participants are unlikely to brainstorm statements that are actually unimportant to the focus, we typically emphasize that an importance rating should be considered a *relative* judgment of the importance of each item compared with all the other items brainstormed. Similarly, a feasibility rating would request a judgment of *relative* feasibility of each statement compared with the others in the set.

Demographics or respondent characteristics, combined with ratings information, provide the researcher rich opportunities to compare the ratings of one subgroup of participants to another. The researcher typically requests nonidentifying information that will make it possible to classify participants into subgroups for such detailed analysis. Respondent characteristics can be customized to ensure that relevant distinctions are captured. They may be personal, social, or organizational characteristics, depending on the purpose and the setting of the research.

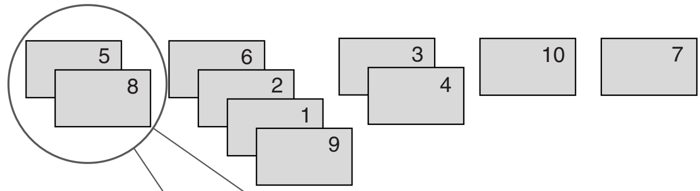
Step 4: Concept Mapping Analysis

Thus far the research methodology of concept mapping parallels that of many other qualitative group research and organizational management approaches. Soliciting a set of responses to a query, organizing and value rating the responses, and collecting participant-related information are routine processes. The data itself are uncomplicated and in common formats at this time. It is simply *N* sets of statement sorts, where *N* is the number of sorters, with their individual views of the interrelationship of the statements. The ratings data are simple numeric representations of the value or observation for each statement for each rater. Respondent question feedback is self-reported data on a small set of items.

The analysis step in concept mapping is inherently a mixed methods approach that integrates the qualitative input and quantitative analysis, and enables creation of concept maps and accompanying reports. The basic analysis[2] consists of the sequence of sort aggregation, MDS, and hierarchical cluster analysis.

The first step in the analysis involves transforming each participant's sort into quantitative information that can then be meaningfully aggregated for analysis. The analytic challenge is to reconcile the fact that each stakeholder likely has a different number and arrangement of sort piles, and the analysis requires the combination of the data across participants. The solution is to place each person's sort into the same-sized square matrix that consists of as many rows and columns as there are statements. Figure 14.3 illustrates a sort matrix for the simple example of a single participant and a 10-statement sort. The participant sorted the 10 statements into 5 piles or groups. Other participants may have had more or fewer groups, but it is assumed that all sorted the same number of statements—in this example, 10. Constructing a 10×10 matrix, or table of numbers, provides the necessary structure for aggregation across participants. For each individual, the table is binary,

Sort Cards for 10 Statements for One Person



Binary Square Similarity Matrix for One Person

	1	2	3	4	5	6	7	8	9	10
1	1	1	0	0	0	1	0	0	1	0
2	1	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	0	0
4	0	0	1	1	0	0	0	0	0	0
5	0	0	0	0	1	0	0	1	0	0
6	1	1	0	0	0	1	0	0	1	0
7	0	0	0	0	0	0	1	0	0	0
8	0	0	0	0	1	0	0	1	0	0
9	1	1	0	0	0	1	0	0	1	0
10	0	0	0	0	0	0	0	0	0	1

Figure 14.3 Transforming Sort Data Into a Binary Square Similarity Matrix

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consisting only of 0s and 1s. If two statements were placed together in a pile by the individual, their corresponding row and column numbers would contain a 1. If they weren't placed together, their joint row-column value would hold a 0. Because a statement is always sorted into the same pile as itself, the diagonal of the matrix always consists of 1s. The matrix is symmetric because, for example, if Statement 5 is sorted with Statement 8, it must always be the case that Statement 8 is sorted with Statement 5. Thus, the concept mapping analysis begins with construction from the sort information of an $N \times N$ (where N = the number of statements) binary, symmetric matrix of similarities, X_{ij}. For any two items i and j, a 1 is placed in X_{ij} if the two items were placed in the same pile by the participant, otherwise a 0 is entered (Weller & Romney, 1988, p. 22).

This creates a common data structure that is the same size for all participants, permitting aggregation across participants' input. Figure 14.4 shows how this might look when aggregating sort results from *5 participants* who each sorted the same *10-statement* set. The figure illustrates that, in effect, the individual binary matrices are "stacked" on top of each other and added. Thus, any cell in this aggregate matrix could take integer values between 0 and 5 (i.e., the number of people who sorted the statements); the value indicates the *number* of people who placed the i, j pair in the same pile. The total $N \times N$ similarity matrix, T_{ij} was obtained by summing across the individual X_{ij} matrices.

This total similarity matrix T_{ij} is the input for nonmetric MDS analysis with a two-dimensional solution. The solution is limited to two dimensions for ease of use, as recommended by Kruskal and Wish (1978).

The analysis yields a two-dimensional (x, y) configuration of the set of statements based on the criterion that statements piled together by more people are

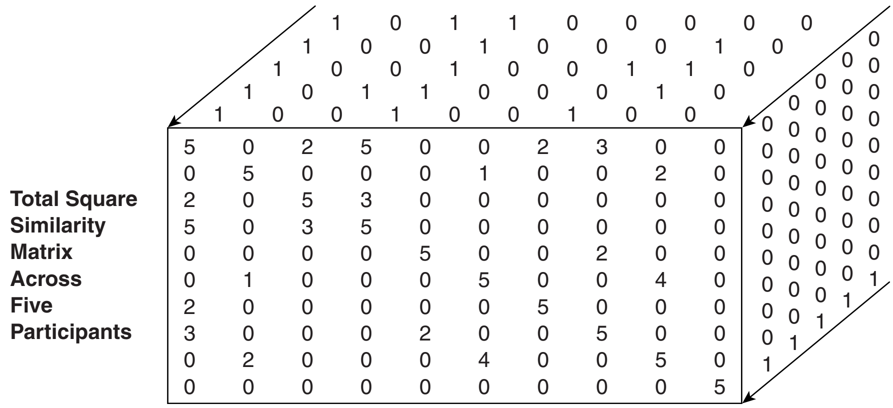

Figure 14.4 Aggregating Sort Data Across Five Participants Into the Total Square Similarity Matrix for a 10-Statement Map

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located closer to each other in two-dimensional space while those piled together less frequently are further apart. There are numerous mathematical descriptions of the MDS process (Davison, 1983; Kruskal & Wish, 1978); a visual, nonmathematical explanation follows to provide insight for social researchers whose work requires explanation of this analysis to others.

Here, we use a hypothetical example to illustrate the analysis. In the example, 80 statements are assumed to be generated by 10 participants. The first 10 statements from this example are given in Table 14.1.

Table 14.1 The First 10 Brainstormed Statements (of 80) From a Hypothetical Example Concept Mapping Process on Organizational Development and Sustainability

1. Advertise the organization's image rather than just specific programs
2. Establish a "quality circle" team approach for program employees
3. Improve employee medical benefits
4. Improve communication among employees
5. Friendlier program managers
6. Reduce unnecessary reports, memos, meetings
7. Improve cleanliness of offices and program locations
8. Computerize communication mailing lists
9. Allow employees flex-time options
10. Conduct program effectiveness analysis for all major current programs

Figure 14.5 shows an excerpt of the aggregate 80 × 80 sort matrix that shows the results for the first 10 statements. Each cell shows how many of the 10 participants sorted each statement with each other statement. The maximum number in each cell is necessarily 10, since that is the total number of "sorters." The minimum number in each cell is 0, since that is the lowest possible number of sorters connecting a specific statement to another statement.

Expanding this example to the entire data set, the data we have are 10 individual sorts of the 80 statements. MDS takes a square matrix of similarities[3] for a set of items/objects, such as the one above, as input and produces a map[4] as output.

The map shown in Figure 14.6 represents statements and their relationship to each other, and highlights the first 10 statements. The cells in the table in Figure 14.5 indicate, for example, that 8 out of 10 people sorted Statements 5 and 7 together; and these statements are consequently located next to each other on the bottom of the map. Similarly, 8 out of 10 people sorted Statements 3 and 9 together and they are located next to each other on the top of the map. On the other hand, none of the participants sorted Statement 3 with either 5 or 7 or Statement 9 with either 5 or 7. Statements 3 and 9 on the top are located far away from 5 and 7 on the bottom. Interstatement relationships in the similarity matrix are translated by MDS into distances on the map.

How does MDS take the aggregate sort matrix and produce the two-dimensional point map? The following simple illustration of MDS is not an exact explanation for how the statistical algorithm works, but rather provides a visual metaphor that suggests what the formula is grappling with. We find this example useful for students and nonstatisticians who are interested in the analysis. For an exact

Statement	1	2	3	4	5	6	7	8	9	10
1	10	0	0	0	0	0	0	0	0	5
2	0	10	0	2	2	1	1	0	0	0
3	0	0	10	1	0	0	0	0	8	0
4	0	2	1	10	1	2	0	1	2	0
5	0	2	0	1	10	0	8	0	0	0
6	0	1	0	2	0	10	0	4	0	0
7	0	1	0	0	8	0	10	0	0	0
8	0	0	0	1	0	4	0	10	0	0
9	0	0	8	2	0	4	0	0	10	0
10	5	0	0	0	0	0	0	0	0	10

Figure 14.5 Excerpt of the Aggregate Similarity Matrix Showing Results for the First 10 Statements for 10 Participants in an 80 × 80 Similarity Matrix

SOURCE: From *The Concept System Training Manual,* 2007. Reprinted with permission of Concept Systems, Inc. www.ConceptSystems.com

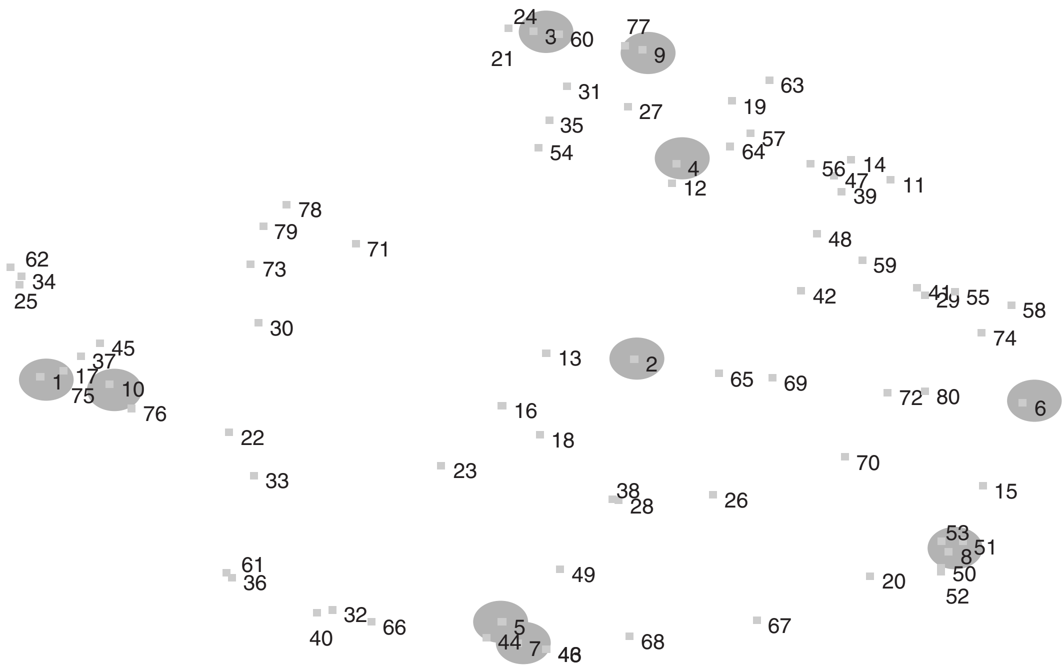

Figure 14.6 Final Map of 80 Statements as Sorted by 10 Participants, With the First 10 Statements Highlighted

SOURCE: From *Concept Mapping for Planning and Evaluation* by M. Kane and W. M. Trochim, 2006. Reprinted with permission of SAGE.

description of the statistical computations, consult one of the texts on MDS (Davison, 1983; Kruskal & Wish, 1978).

Using an even smaller example, imagine a mapping project that consists of three statements that have been sorted by five participants. A similarity matrix might be like the one represented in the upper left in Figure 14.7.

Possible matrix values range from 0 (a pair of statements sorted together by *none* of the participants) to 5 (a pair of statements sorted together by *all* participants). We might imagine how MDS constructs a map by creating one by hand. First, we place Statement 1 somewhere arbitrarily on a page, as shown in Figure 14.7. We then need to place Statement 2 in relation to Statement 1, using the data in the matrix. Before we can do this, we have to set an arbitrary measurement scale that shows distance between points on the map in units based on sorting. This scale is indicated by the concentric circles surrounding Statement 1. Since there are only five sorters, we use five equally spaced concentric circles from Statement 1's location. The matrix indicates that only one person sorted the Statements 1 and 2 together. Using this scale, Statement 2 can be placed anywhere on the fourth circle away from Statement 1. We choose a location on this circle in the upper right of the map, in Figure 14.7.

Next, we place Statement 3 in relation to both Statements 1 and 2, by arraying concentric circles from Statement 1 *and* Statement 2 using the same distance scale

Similarity Matrix

	1	2	3
1	5	1	2
2	1	5	0
3	2	0	5

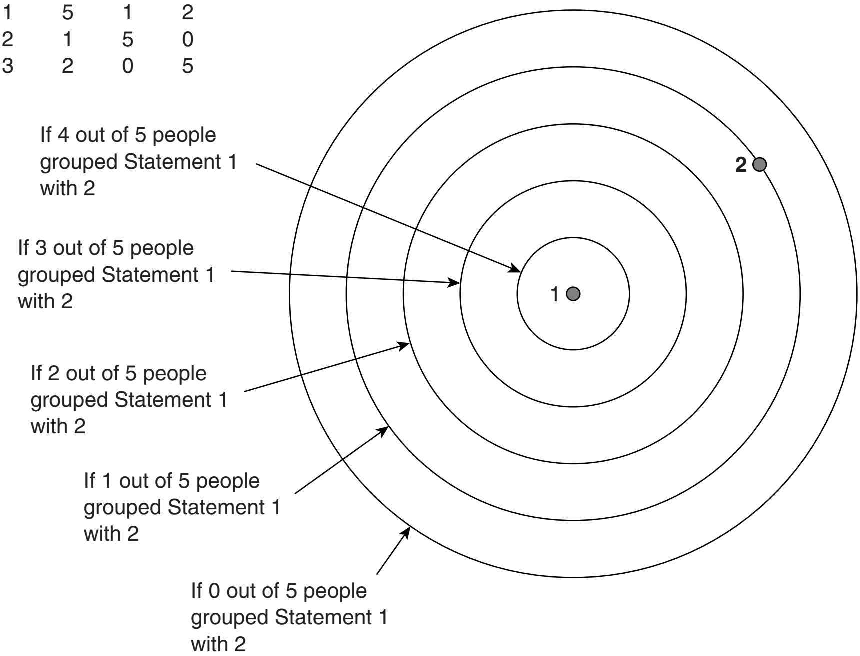

Figure 14.7 Similarity Matrix for Three Items, and Theoretical Distance of Item 2 From Item 1, Based on Sorters' Input

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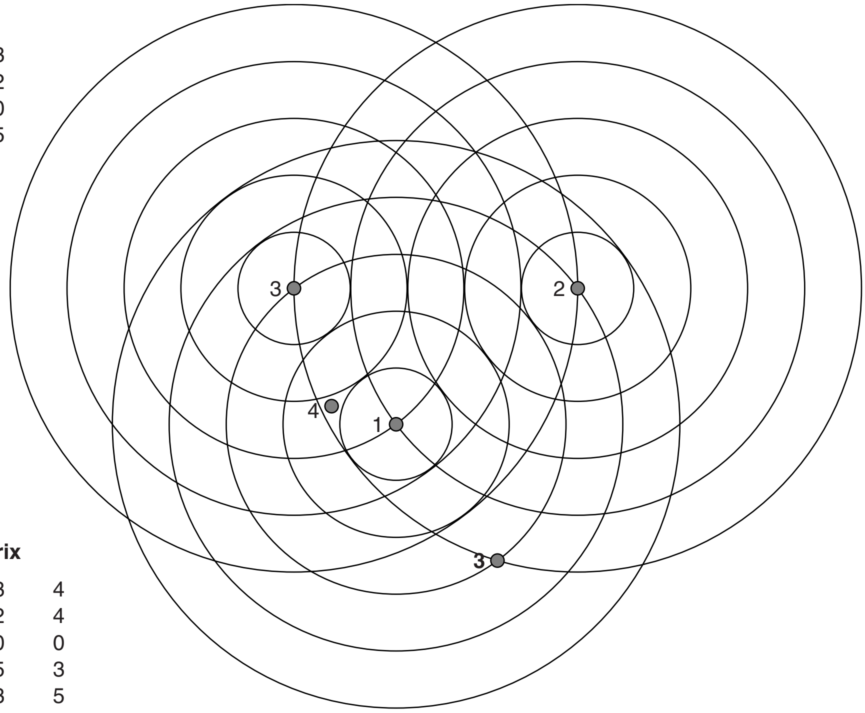
as indicated in Figure 14.8. Two people sorted Statements 1 and 3, and none sorted 2 and 3. To place Statement 3, we locate the position that is *simultaneously* three circles from Statement 1 and five from Statement 2. Figure 14.8 shows that there are two equally accurate locations for Statement 3. We arbitrarily select the one on the upper left (the highlighted Statement 3).

With only three points in two dimensions, it is always possible to place the points exactly in two dimensions. However, the process gets more complicated with a fourth statement to add to the project. In the lower left of Figure 14.8, we see the same hypothetical similarity matrix for the five sorters, but with a fourth statement added. To place this fourth point on the map, we need to locate its distance simultaneously from each of the other three statements. The concentric circles in Figure 14.8 show the possibilities. The best location is an intersection that is simultaneously one unit away from Statement 1, five units away from Statement 2, and two units away from Statement 3. But note that the required concentric circles do not have such an intersection point. In two dimensions, the best we can do is to locate Statement 4 as closely as possible to the intersection, as shown in Figure 14.8.

Several important insights emerge from this simple visual description of what MDS is grappling with. MDS does not know directions, so that when we place Point 2 in

Similarity Matrix

	1	2	3
1	5	1	2
2	1	5	0
3	2	0	5



Similarity Matrix

	1	2	3	4
1	5	1	2	4
2	1	5	0	0
3	2	0	5	3
4	4	0	3	5

Figure 14.8 Constructing an MDS-like Point Plot for the Hypothetical Case of Four Statements and Five Sorters

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relation to Point 1, it is equally accurate to place it anywhere on the fourth circle from Point 1. This means that when you look at a concept map generated by MDS, direction on the map is arbitrary. One can flip a concept map horizontally or vertically and/or rotate it clockwise or counterclockwise any amount *and this would have no effect on the distances among the points.* The simple exercise shows that MDS yields a relational picture and is indifferent to directional orientation.

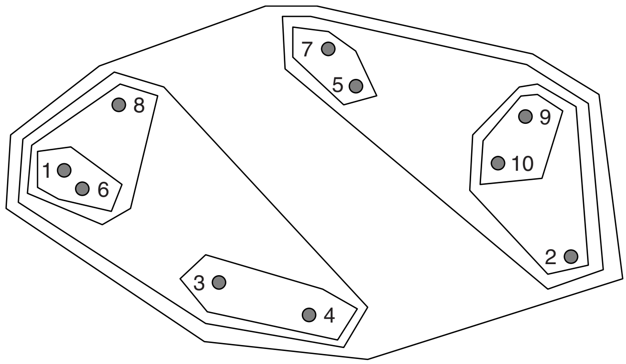
In our three-point example, there is always a two-dimensional solution that represents the table exactly, with no error. When we move to larger numbers of points this is no longer the case; it is not likely that the visual result will represent the similarities perfectly. In MDS, we estimate the overall degree of correspondence between the input (i.e., the similarity matrix) and the output (i.e., distances between points on the map) using a value called the *Stress Value.* A *lower* stress value indicates a *better* fit; higher stress means the fit is less exact. In general, lower stress is preferred.

The normative range for judging stress values in a particular study should be determined from comparisons with similar types of data collected under similar circumstances. A study of the reliability of concept mapping (Trochim, 1993) reported that the average stress value across 33 concept map projects was .285 with a range from .155 to .352. While in general the stress value indicates how well the two-dimensional map fits the sort data, it is not clear that maps with lower stress are more interpretable or useful than ones with considerably higher stress. The Stress Value is sensitive to even the smallest distance discrepancies on the map. Interpretation of micromeasurements of distances among points is typically neither necessary nor useful. Slight variances between the input and the placement can contribute to higher stress without diminishing the general map result or its interpretability.

The discussion thus far shows how the concept mapping analysis uses aggregate sorting results and MDS to produce the basic "point map" that is the foundation for all other maps. While this is a useful result in itself, it is helpful to be able to view a concept map at different levels of detail. The point map generated by MDS is a fairly detailed map, especially when it contains as many as a hundred points. To arrive at a higher-level view of the map, a procedure known as *hierarchical cluster analysis* (Anderberg, 1973; Everitt, 1980) is used. The input to the cluster analysis is the point map, specifically, the x, y values for all the points, or units of input, on the MDS map. Using the MDS configuration as input to the cluster analysis forces the cluster analysis to *partition* the MDS configuration into nonoverlapping clusters in two-dimensional space. Mathematicians do not agree on what constitutes a cluster mathematically, so several algorithms exist for conducting cluster analysis, each of them likely to yield different results. In group concept mapping, we typically conduct hierarchical cluster analysis using Ward's algorithm (Everitt, 1980) as the basis for defining a cluster. Ward's algorithm has the advantage of being especially appropriate with the type of distance data that comes from the MDS analysis. The hierarchical cluster analysis uses the point map data to construct a "tree" that at one extreme represents all points together (in the trunk of the tree) and at another represents all points as individual end points of the "branches." Cluster analysis approaches can be classified as either divisive (i.e., top down) or agglomerative (i.e., bottom up). Ward's algorithm is an agglomerative approach.

Figure 14.9 illustrates how agglomerative hierarchical cluster analysis is related to an MDS point map. Returning to the 10-statement example, the top of the figure shows a 10-statement point map. The bottom shows the cluster analysis tree. Each statement is the end-point of a branch. The tree shows, moving from top to bottom, how statements are agglomerated and eventually combined onto a single trunk—a one-cluster solution. To illustrate, it is visually apparent that Statements 1 and 6 are closer to each other than any other pair of statements on the map. In the cluster tree, they are the first two branches that are merged. The next closest pair is Statements 5 and 7, and they are grouped next. The merge table on the bottom left of Figure 14.9 shows which statements (or previously formed clusters of statements) are combined at each number of clusters. By taking horizontal slices at different heights of the tree, one can look at different numbers of clusters. For instance, for a five-cluster solution, we look at the horizontal slice at the 5 "Number of Clusters" level in the cluster tree to see that the following statements would be grouped in clusters: (1, 6, 8) (3, 4) (7, 5) (9, 10) (2).

The resulting graphic representation partitions the universe of statements—ideas, issues, or articulations of knowledge—into groups or "clusters" that appropriately represent the map content on a higher conceptual level. The process of selecting the appropriate level of detail, or concept, is typically driven by the needs of the group in the study or the research. In brief, the process relies on qualitative review of a range of cluster solutions, from fairly granular—in the case of a map



Merge	Points Merged
1	1 + 6
2	5 + 7
3	9 + 10
4	(1 + 6) + 8
5	3 + 4
6	2 + (9 + 10)
7	((1 + 6) + 8)) + (3 + 4)
8	(5 + 7) + ((2 + (9 + 10))
9	(((1 + 6) + 8)) + (3 + 4)) + (5 + 7) + ((2 + (9 + 10))

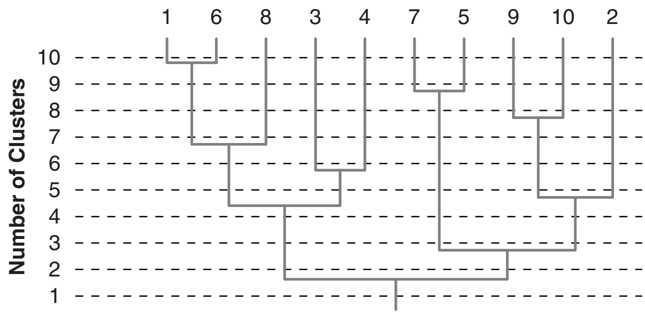


Figure 14.9 Agglomerative Cluster Analysis for a 10-Statement Map

SOURCE: From *Concept Mapping for Planning and Evaluation* by M. Kane and W. M. Trochim, 2006. Reprinted with permission of SAGE.

with 80 statements, we may begin by looking at around 20 clusters—to broadly conceptual—perhaps as few as 4 or 5. The procedure typically followed is to examine an initial cluster solution that was the maximum thought desirable for interpretation in this context. Successively lower cluster solutions are examined, with a judgment made at each level as to whether the merger seems substantively reasonable or desirable for the purpose at hand. If deciding in Figure 14.9 whether we prefer a five- or four-cluster solution, our focus is whether we prefer what is merged when we move from the horizontal slice at 5 to 4; or, whether we prefer Statements 9, 10, and 2 to be grouped together or separate into two clusters of (9, 10) and (2). The suitability of different cluster solutions is examined and results in a decision of a specific cluster solution for the project.

Figure 14.10 illustrates a final labeled cluster map of 8 clusters of the 80 statements in the example we are using. Cluster names may come from the results of a "sort pile" analysis conducted during the data aggregation phase; or from a group interpretation and labeling of each cluster's contents after the map is completed.

Step 5: Interpreting the Maps

Because the social research approach that concept mapping supports best is participatory and stakeholder driven, it usually makes sense to involve the stakeholders

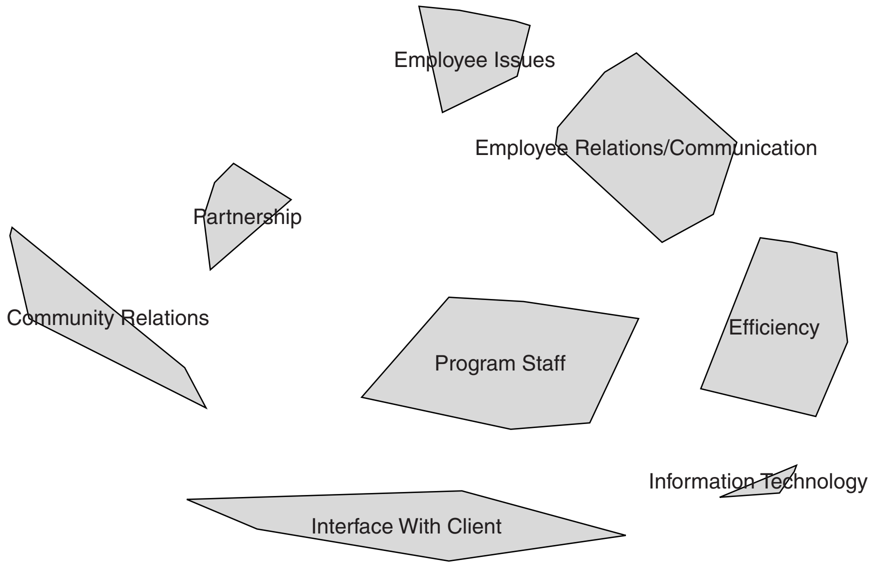


Figure 14.10 Eight-Cluster Concept Map

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directly in the interpreting and understanding the results. Inclusion of participants in attributing relevance to the maps develops joint authorship, greater richness of result, and higher likelihood that the research will yield value both for the researcher and the community of interest. Tools for the interpretation include the following:

- Basic concept maps—point maps and cluster maps—that form the foundation for further analysis.
- Pattern matches, that use the value ratings gathered at the structuring phase (Step 3) to show consensus, or differences of opinion or judgment between groups, at the cluster level.
- Bivariate value plots, called *go zones*, that use the value rating data on a statement-by-statement level within each cluster.

Ratings data from Step 3 can be used by the researcher to describe the ratings of all participants or a subgroup, to compare across subgroups; or to compare all participants across different dimensions, such as importance, feasibility, or potential impact. Figure 14.11 represents a cluster *ratings* map, which illustrates the range of, in this case, importance levels that the participants as a whole associate with each conceptual cluster on the map.

The overall values related to each concept provide rich feedback to the researcher and community of interest. Here, the group would likely notice that the "northeast ridge" indicates high importance associated with the concepts of

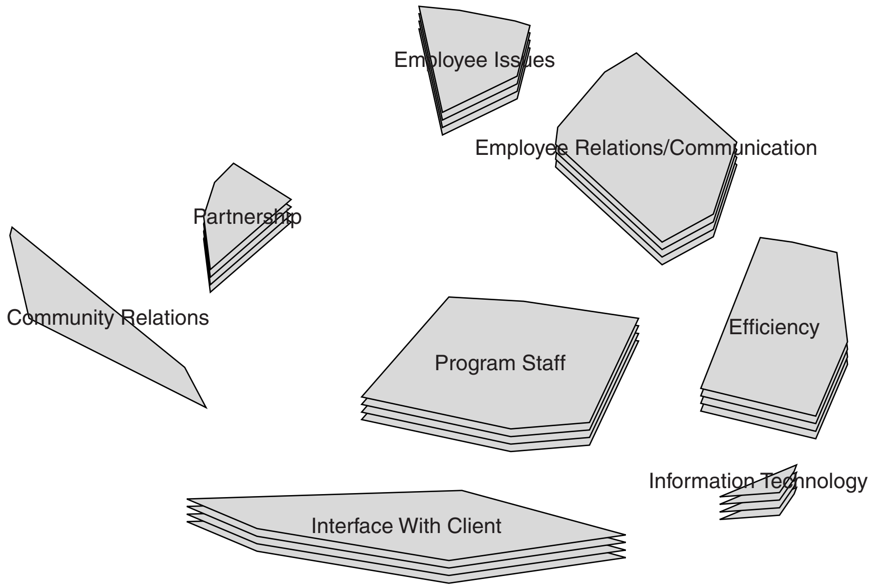

Figure 14.11 Cluster Rating Map

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employee issues, employee relations, and efficiency. Some might note the complementary relationship among those concepts, as a "region" of interest for planning. In contrast, the "west coast" is relatively less important to the organization, according to those who participated.

More detailed rating comparison tools are also typically used. Pattern matching (Trochim, 1985, 1989b) is used to explore consensus across different stakeholders or stakeholder groups. Pattern matching is both a statistical and a graphic analysis. Graphically, a pattern match is portrayed using a "ladder graph" that consists of two vertical axes (one for each "pattern") as shown in Figure 14.12. The vertical axes are joined by lines that indicate the average values for each cluster on the concept map for any variable and group specified. Statistically, the two patterns are compared with a Pearson Product Moment correlation that is displayed at the bottom of the ladder graph. The graphic is derived from the ratings data taken on each statement from each participant and the demographic information collected at the same time. The analysis segments the stakeholders by self-identified group; it also averages the value ratings of the statements within each cluster (as on the cluster rating map) and aligns them on a vertical number line for each subgroup of interest. Connecting Cluster A on the left side with Cluster A on the right side shows us graphically the relative importance of the opinions between Groups 1 and 2. Figure 14.12 represents a pattern match that compares, in this case, managers and staff opinions of importance on each of the concepts or clusters.

This pattern match is a cluster-level representation of the average values of the statements in each cluster, and how they compare for two subgroups. The cluster

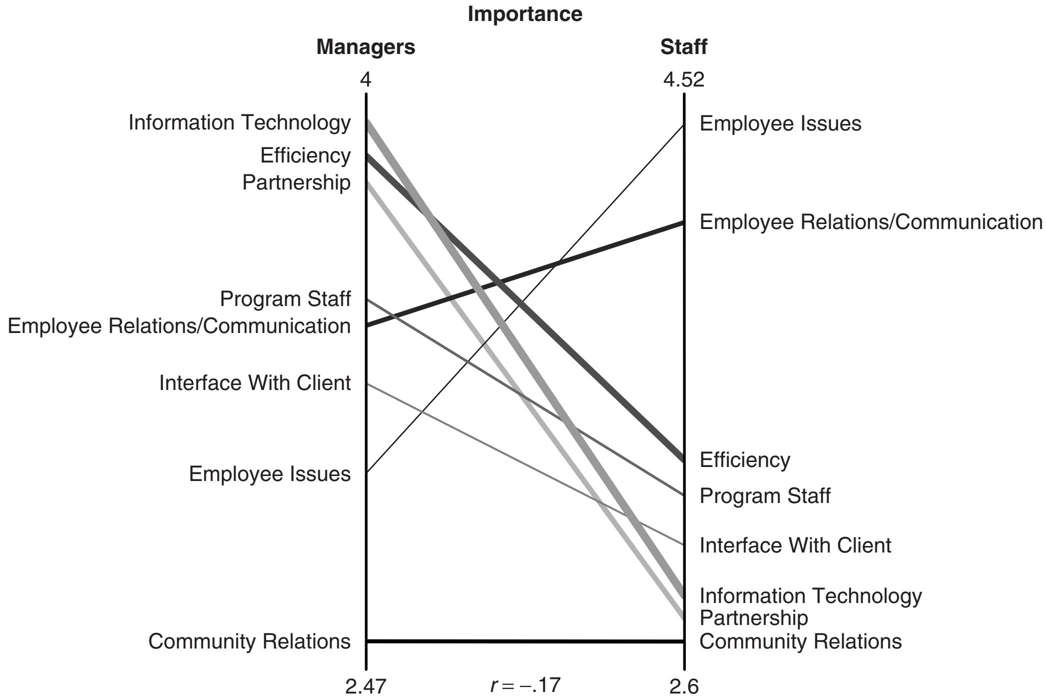

Figure 14.12 Pattern Match Comparing Managers and Staff on Importance Ratings by Cluster

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names from the concept map are arrayed according to their average rating, in comparison with the other clusters. In this pattern match, managers and staff agree that "community relations" is not as important as the other concepts associated with organizational planning (which is the focus of this project). On the other hand, the conceptual areas that managers feel are important are rated relatively low by the staff and vice versa. As a planning tool, pattern matches can point to elements that require attention before decisions are made, as in this case.

The next step in exploring concept map data allows the researcher to come full circle, back to the level of specific statements or issues in the domain. The researcher can develop bivariate value plots, also labeled *go zones*, that show the average rating values of each statement in relation to the other statements in its conceptual cluster. An example is shown in Figure 14.13. The horizontal axis shows the importance rating for managers; the vertical shows for staff. Statements are displayed with their identifying number. The plot is divided into quadrants based on the average for each axis. The upper-right quadrant indicates the statements that are rated above average on importance by both managers and staff. The plot takes its name from this quadrant, which is sometimes called the *go zone*, to indicate that these are the first issues one might typically "go" to in thinking about action planning, because they are the ones both managers and staff agree are important. The participants review these plots and use them as the basis for an initial discussion about action. Such plots can be valuable to planners, researchers, and evaluators in agencies or organizations because they enable one to identify issues that are high value by agreement.

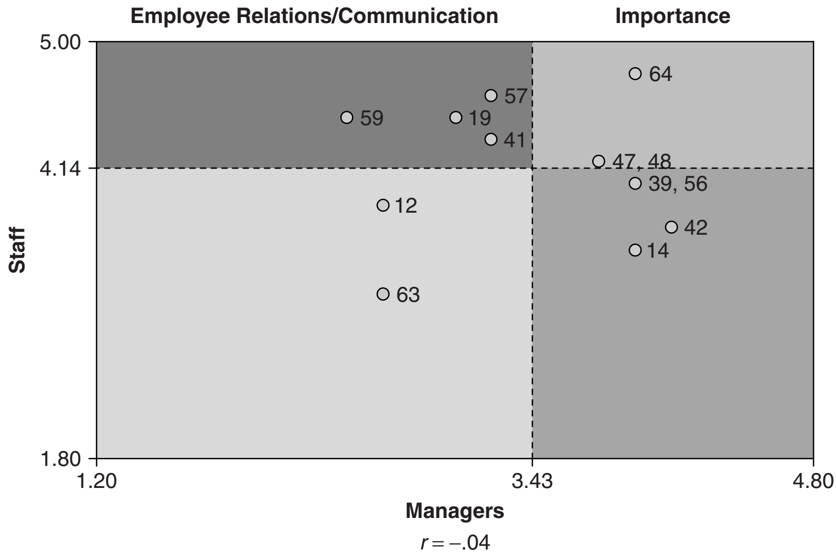

Figure 14.13 Go Zone Bivariate Plot for Cluster Employee Relations/Communication

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Step 6: Utilization

The last step in the process is utilization. The critical issue here is ensuring that the work undertaken to construct the rich conceptual framework, both by the researcher and by the community of interest, is used as the foundation for whatever application is desired. The next section describes a variety of applications in social science research, in both use (theory building, program development, measurement and evaluation) and area of study or research (health, mental health, education, etc.).

Looking back at our Delaware Advisory Council on Cancer Incidence and Mortality example gives us some insight into how a larger-scale project might evolve. Figure 14.14 shows the cluster map of 118 statements, which was authored, in effect, by the sorts of 32 individuals. The key concepts for programming and innovation focus are the ring of clusters surrounding the central cluster; and the central cluster itself represents management and oversight as part of the overall plan. Figure 14.15 is an example of one of the DCC cluster's Go Zones, indicating in the top-right quadrant the items of highest importance and highest feasibility according to participants. This provided the consortium with specific recommendations for action to address the focus of the initiative; the Go Zone for each cluster was queried, interpreted, and used to inform the plan and set milestones for each topic.

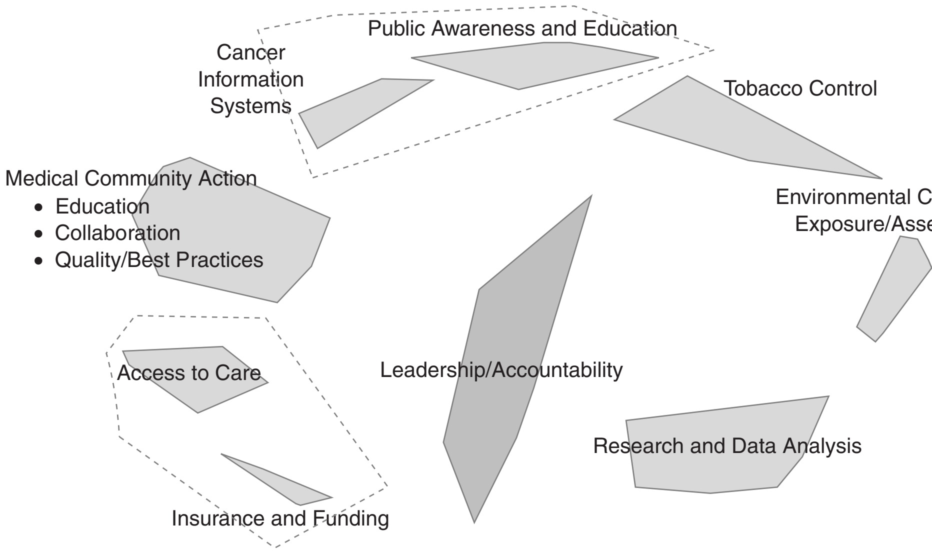
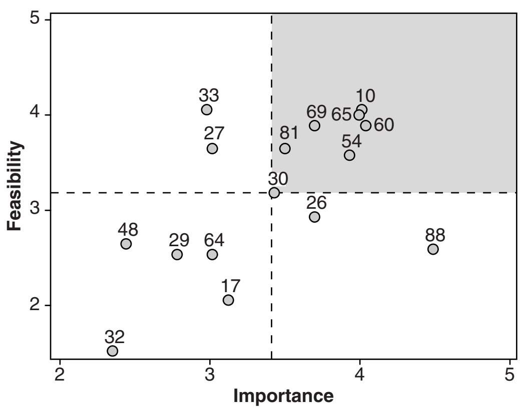


Figure 14.14 Final Concept Map of the Delaware Advisory Council on Cancer Incidence and Mortality

SOURCE: Delaware Department of Health and Social Services, Division of Public Health.



Standardize and centralize the collection and dissemination of data on cancer incidence in DE, including occupational history, demographics, and exposures. Budgetary priority. (10)

Find out why energetic, robust children get cancer. (17)

Increase high quality causative factor and clinical trial research. (26)

Look at whether the high cancer rate is related to an aging population. (27)

Use miscarriage/birth defect case data to look at the effects of carcinogenic exposure. (29)

Determine how many deaths from the 6 leading cancers in DE were avoidable. (30)

Determine the effect of the Cancer Society on recidivism rates in DE. (32)

Consider the potential years of life lost to certain types of cancer when setting state priorities. (33)

Conduct a study to determine the impact of the use and abuse of alcohol. (48)

Address the high incidence and death rate for African Americans. (54)

Standardize specific, accurate reporting of the cause of death for cancer patients. (60)

Examine how the high incidence of HIV and other sexually transmitted diseases impacts the cancer rate (sarcomas, cervical) (64)

Use registry data to create a geographic map of cancer incidence (by type), including the risk factors and causes associated with each area. (65)

Focus on survival rates due to early detection and advancements in treatment. (69)

Research cancer incidence and cure rates for children (18 yrs. and younger). (81)

Determine why DE cancer rate is so high. (88)

Figure 14.15 An Example of a Bivariate "Go Zone" Plot for Cluster "Research and Data Analysis" in the Delaware Advisory Council on Cancer Incidence and Mortality Project

SOURCE: Delaware Department of Health and Social Services, Division of Public Health.

Using Concept Mapping in Applied Social Research

The range of application for concept mapping in social research is broad. Here, we describe the application of concept mapping in theory development, planning, and implementation of social programs and interventions, measurement or scale construction, and the evaluation of social programs.

Theory Development

Because concept mapping is a structured methodology for identifying what a group of people think about some topic, it is hardly surprising that one of its major uses in applied social research contexts has been for exploring or developing theories or models that can subsequently be assessed empirically. Over the past several decades, there has been a broad recognition in applied social research that articulation of program theory (Bickman, 1986; Chen & Rossi, 1990) is critical to the understanding of causal relationships between interventions and outcomes (Chen & Rossi, 1983, 1984; Trochim, 1985, 1989b).

Concept mapping has often been employed to explore the theory or meaning of some construct or area from a multistakeholder perspective. It has been used to explore multistakeholder perspectives on primary health care services (Southern Young, Dunt, Appleby, & Batterham, 2002); how patients cope with illness and with the health care system (DeRidder, Richardson, Severens, & Malsch, 1997); the needs of children in pediatric hospice and palliative care (Donnelly, Huff, Lindsey, McMahon, & Schumacher, 2005); how stakeholders perceive services in mental health (Johnsen, Biegel, & Shafran, 2000); the barriers to racial or ethnic minority application and competition for NIH (National Institutes of Health) research funding (Shavers et al., 2005); the problems that persons with traumatic brain injury face (J. P. Donnelly, K. Z. Donnelly, & Grohman, 2005; K. Z. Donnelly, J. P. Donnelly, & Grohman, 2000); what is meant by systems thinking in public health (Trochim, Cabrera, Milstein, Gallagher, & Leischow, 2006); staff's views of a supported employment program for persons with severe mental illness (Trochim, Cook, & Setze, 1994); gender differences in perceptions of sexual harassment in the workplace (Hurt, Weiner, Russell, & Mannen, 1999); what clients perceive as helpful in counseling (Paulson, Truscott, & Stuart, 1999); factors that affect psychiatric hospitalization (Dumont, 1993); quality of life (Boevink, Wolf, van Nieuwenhuizen, & Schene, 1995; van Nieuwenhuizen, Schene, Koester, & Huxley, 2001); quality of care (VanderWaal, Casparie, & Lako, 1996); the differences in perceptions between student employees and recreational sports administrators about student employee work in a recreational sports setting (Miller & Grayson, 2006); student perceptions of issues in their lives as students (Trochim, 1989a); and in a project designed to see what clients experience as helpful in counseling (Paulson et al., 1999). In many of these projects, formal models or theories were not the explicit goal, even though they sometimes resulted from the process.

Concept mapping has been used explicitly for more formal development of a theory, model, or framework. Some examples include the development of theories or models of multiconstruct issues such as general practice in health (Batterham et al., 2002); depression in college students (Daughtry & Kunkel, 1993); tobacco industry tactics to undermine tobacco control (Trochim, Stillman, Clark, & Schmitt, 2003); women's perceptions of intimate partner violence experiences (Burke et al., 2005); and group conflict in organizations (Jackson & Trochim, 2002). Researchers sometimes focus on a specific construct or concept such as complementary and alternative medicine (Baldwin, Kroesen, Trochim, & Bell, 2004); feminism (Linton, 1989a, 1989b); caring in nursing (Valentine, 1989); and the construct of listening (Witkin & Trochim, 1997). In applied studies, researchers engage stakeholders to validate or extend theories and models, as in the study regarding the challenges faced by foster parents (Brown & Calder, 1999).

Planning Programs and Social Interventions

The planning and implementation of social programs or applied social research interventions is often a complex endeavor that involves coordinating disparate groups of stakeholders who often have different motivations and operate under differing incentives. Concept mapping provides a structured method in such contexts for groups to work together to understand what they are trying to do. A key function that the concept mapping approach enables with such groups is the organic creation of a commonly authored framework, thus enabling disparate views their place in the model.

Researchers have used concept mapping in planning programs and social interventions in areas such as the development of new technology products (Cousins & MacDonald, 1998); integration of computer technology into education (Abrahams, 2004; Keith, 1989); intensive family-based in-home services (Mannes, 1989); the development of a family support program (Rosas, 2005); and university student services (Gurowitz, Trochim, & Kramer, 1988). It has also been used to address broad, multifaceted planning problems, including public health priorities for end-of-life initiatives (Rao et al., 2005); barriers to African American families' involvement in treatment of their family members for mental illness (Biegel, Johnsen, & Shafran, 1997); and to accomplish large-scale community-based public health planning (Trochim, Milstein, Wood, Jackson, & Pressler, 2004).

A major issue in applied social research is the degree to which successful interventions get translated or adapted when they move from highly controlled research contexts to the world of practice. Concept mapping has helped examine how larger systematic factors influence the adaptation of telemedicine technology in child abuse examination settings (Pammer et al., 2001) and to help ensure fidelity of model transfer (Shern, Trochim, & Lacomb, 1995), for example.

Measurement Development and Scaling

The basic steps in concept mapping—the generation of a large set of ideas, collection of judgments about their relationships and representation based on their metric coordinates—are similar to the general steps in many scaling methods. It should not be surprising, then, that the method is useful in the development and revision of measures and in the assessment of their validity. In this vein, concept mapping has been used to develop frameworks for instruments or measures of psychosocial preferences to enable individualizing care of older persons (Carpenter, Van Haitsma, Ruckdeschel, & Lawton, 2000); quality of care in the treatment of chronic disease (VanderWaal et al., 1996); a questionnaire to evaluate a Big Brother, Big Sister program for youth (Galvin, 1989); and a general practitioner integration index in health care (Southern et al., 2002).

As useful as the approach has proven in measurement development, it can also support revision and expansion of the definition of what should be measured, as in the case of the Lancashire Quality of Life Profile (LQoLP). Stakeholders identified what was important for the quality of life of long-term care-dependent psychiatric patients (van Nieuwenhuizen et al., 2001). Nine distinct conceptual domains emerged in the mapping, only six of which were addressed in existing measures of quality of life. The LQoLP was revised to add items for the other domains. The revised instrument demonstrated strong reliability and validity in subsequent testing. Similarly, concept mapping was used to refine the Revised Children's Manifest Anxiety Scale (White & Farrell, 2001), one of the most widely used self-report measures of children's anxiety.

The methodology also makes possible new and more sensitive methods for assessing construct validity in developing new measures. Because concept mapping is based on a scaling of similarities among theoretical ideas, the map provides a continuous interval-level measure of the expected similarity among concepts. This pattern of distances on a map can be compared with the pattern of observed intercorrelations among corresponding items on a measure to obtain a more sensitive estimate of construct validity (Davis, 1989; Trochim, 1989b). For example, in a study that assessed the construct validity of a questionnaire with this pattern-matching method found a .76 overall construct validity coefficient (Marquart, 1989) relating map distances with observed interitem correlations.

Evaluation of Social Programs

Theories need to be examined against empirical reality. We need feedback about programs and interventions to assess how they perform and what works or not. Measures are central to this assessment function. All these—theories, programs, and measures—come together in the applied social research process of evaluation. Concept mapping plays an important role in helping evaluators and evaluation stakeholders articulate their implicit theories, program activities and outcomes, and potential measures. Concept mapping has been used as the basis for generating a program theory or logic model for evaluating children's mental health programs (Yampolskaya, Nesman, Hernandez, & Koch, 2004); a family support program (Rosas, 2005); education and training programs in business and industry (McLinden & Trochim, 1998a, 1998b; Michalski & Cousins, 2000); a drop-in center for youth (Mercier et al., 2000); and a Big Brother, Big Sister program for youth (Galvin, 1989).

Concept mapping's methodology has also been used to identify outcome criteria for evaluation (Trochim, 1996). For example, it was used to incorporate the views of a large and diverse group of stakeholders throughout the United States to develop a logic model that was used as the basis for evaluation of the Centers for Disease Control and Prevention Research Centers Program (Anderson et al., 2006).

In a similar manner, it was used to develop a logic model of transdisciplinary research in science (Stokols et al., 2003) that provided a framework for a preliminary evaluation of the Transdisciplinary Tobacco User Research Center initiative of the National Cancer Institute.

The analytic and reporting side of evaluation can also use concept mapping, such as to display and contrast results from statistical tests from multiple groups on multiple measures, and showed promise as a mechanism for addressing situations where there may be low statistical power (Caracelli, 1989).

Issues in Concept Mapping

In the approximately two decades since its inception, concept mapping has become an established applied social research methodology with an impressive technical literature and broad range of applications. The concept mapping method is continuously being developed and adapted and is influenced considerably by other changes in the applied social research environment. We conclude this chapter by considering briefly some of the major areas of likely development in concept mapping in the near future in methodology, group process, meta-analysis, and technology issues.

In terms of *methodology,* it is likely that we will see advances and variations in how data are collected and statistically analyzed. Concept mapping has typically relied on a simple unstructured sort as the primary data collection approach. Although easy to conduct, this method captures a simple categorical depiction of the conceptual structure implicit in a person's thought. Another approach would use a paired comparison of each statement to each other, a prohibitively time-consuming process for a map with more than a few statements (Weller & Romney, 1988). However, other procedures may emerge that are both feasible to accomplish and richer in information about statement interrelationship. For example, Cooksy (1989) demonstrated the use of hierarchical outlining or graphing methods for data collection. In addition to getting more precise data from each participant, and therefore also yielding a more informed group map, some of these methods may also be precise enough to enable development of a map for an individual.

One of the most important advances in statistical analysis over the past two decades has been the development of new methods for appropriately handling the hierarchical structure of much social research data. These approaches, variously called multilevel, hierarchical, or mixed effects models (Luke, 2004; Raudenbush & Bryk, 2002; Singer, 1998; Snijders & Bosker, 1999) are especially applicable for the analysis of concept mapping rating data. These data are inherently hierarchical with statements nested within cluster and persons nested within demographic groups (e.g., gender or organizational role). It is now feasible to construct models for rating data in a concept mapping project that assess whether there are statistically significant differences between average cluster ratings, between the ratings for multiple groups of participants, whether there are changes in ratings over time, and even to assess the pattern-matching hypothesis of whether expected outcomes on a set of measures (operationalized by statement or cluster) are statistically related to observed outcomes (Trochim, 1985, 1989b). These mixed effects models enable testing of both the overall differences and, if significant effects are found, of multiple comparisons to identify specific differences, in a fashion directly analogous to multiple comparison tests in analysis of variance frameworks.

Concept mapping is a pattern-oriented method. One of the most intriguing issues is whether patterns can be used in applied social research to help address problems of noise or variation in measures, that is, to improve statistical power. Consider a common situation in applied social research where we might conduct many similarly structured analyses. For instance, using a multi-item scale, we estimate significant change from before to after an intervention is in place on an item level. Comparing on an item level, we might be tempted to conclude that the intervention was not significant. But if it were possible to rank order or scale the expected outcomes of the set of tests, we would be able to correlate the expected outcomes with the observed test values. It is possible that this correlation is statistically significant—even though none of the individual tests was. If it is easier to detect significant patterns in situations of low statistical power than to detect point-specific predictions (Trochim, 1989b), this approach may provide support for an expanded application of pattern matching. This pattern-matching approach was taken in a test of the effects of a psychiatric rehabilitation program for supported employment (Trochim & Cook, 1992). Here, the intriguing finding was that there was a significant negative correlation between theoretical expectations and observed change scores (estimated through *t* tests), leading program staff and researchers to rethink the theory of the program. In a similar manner, it may be that by overlaying statistical results onto a concept map, we could detect patterns of similarity among treatment effect estimates not detectable from the estimates themselves (Caracelli, 1989). This notion has yet to be thoroughly investigated but continues to have significant potential.

Although concept mapping has traditionally been viewed as a method for developing conceptual frameworks, theories, or constructs, it opens up new possibilities as an analysis approach for qualitative data. Many qualitative methods use procedures that are analogous to the steps in concept mapping or that could easily be coupled with it. For example, if one had transcribed interview text, typical qualitative analysis would involve the identification of key themes and the organization of these themes into broader frameworks or rubrics. Concept mapping suggests a way to do this collaboratively as a type of participatory qualitative analysis where a group of individual interviewees could be directly involved in the collective thematic analysis of their own interview data through sorting and rating of a common set of excerpted statements from each of their interviews. Such approaches are already being explored in the use of concept mapping for the participatory analysis of short open-ended questions on surveys (Burke et al., 2005; Jackson & Trochim, 2002) and in the conduct of community-based participatory research projects (Trochim et al., 2004).

Group process issues related to concept mapping benefit from ongoing attention and research. At the very beginning of a project, for example, developing the focus statement is one of the most critical tasks, but no clear standard method exists to accomplish this. A structured method for developing and pilot testing alternative focus statements for the context at hand would be useful (Mercer, 1992).

Embedded in the concept mapping analysis process is the selection of the number of clusters on the map. Essentially and necessarily situational, it is nevertheless currently the purview of the researcher/facilitator. A promising approach, based on the same principles that guide concept mapping generally, is to develop a structured method for participants to follow to arrive at a consensus regarding cluster-level selection. We might profitably use a variation of Delphi Methodology (Carroll & Wish, 1975; Linstone & Turoff, 1975) in iterative rounds where participants are first asked to review a range of cluster solutions (as the facilitator currently does), decide on the number of clusters they prefer, share their results and reasons why, and then repeat the cycle until either a consensus solution evolves or they recognize that there are different views on potential use of the maps.

Concept mapping as a methodology is sufficiently mature to require cross-project assessments and syntheses to help researchers understand how the concept mapping process works in practice and to develop benchmarks for subsequent concept mapping projects. The wealth of completed concept mapping projects that exist in a range of substantive areas and constructed for a broad range of uses can be used to investigate a range of methodological and substantive questions, which include the following:

- What are the different types of focus statements that have been used?
- How many statements are typically brainstormed in a concept mapping project?
- What's the typical person-to-statement brainstorming ratio?
- How many piles do people typically sort statements into?
- What is the length of time for participation in each step of the mapping process?
- What is the typical distribution of rating variables?
- How many clusters do maps typically have, and how much does this vary from project to project?
- What is the distribution of pattern-matching or go-zone correlations?

Having answers to such questions can provide ranges or descriptions of "typical" projects that can help guide concept mapping practitioners and can suggest ways the concept mapping methodology might be improved or extended. This analysis is underway and will be reported within the next few years.

The revolution in *computing technology* since the development of concept mapping was impossible to fully anticipate. The evolution of the Internet in particular has enormous implications for concept mapping as it does for the rest of applied social research. Methods for collecting concept mapping, brainstorming, sorting, and rating information asynchronously over the Web are already available and relatively easy to use with any recent version Web browser, making it possible for hundreds or even thousands of people to collaborate on projects worldwide. We expect that in the next decade, this capability will be extended to the analysis, display, and interpretation of results as well, and that in the future an entire mapping process from inception to utilization will be able to be accomplished with virtual groups.

This points to what may be the most important eventual evolution of concept mapping: the development of complex and adaptive mapping. Currently, the focus prompt sets the direction of the project and the brainstormed set of statements is the universe of available data on that issue. But evolving computer power and the availability of continuously networked Web-based participant groups suggest a different approach, where focus statements evolve over time and new statements may be created as previous ones decline in relevance, based on algorithms that use participant input. This kind of dynamic modeling would enable maps to evolve in real time and change as our understanding of the problem changes. This would almost certainly require the evolution of new statistical and analytic methods, and new data structures, integrating the principles of MDS but using different algorithms than those currently used. Where the current method assumes a fixed number of sorters who process a fixed set of statements, this dynamic alternative would assume that different people would organize different subsets of statements where the overlap enables broader common data structures to be estimated and to emerge. The future may focus on "meta-mapping" that knits together multiple existing maps or shards of maps. Newer and more dynamic analytic methods and data structures will make this more feasible, suggesting the possibility that concept mapping might be a foundation for a broader, more integrated, and continuously adaptive mapping of a more general semantic space. Expanding the definitions of theory, concept, and model with which we began this chapter is both a process, and a result, of the continued exploration of concept mapping in applied social research.

Discussion Questions

1. What are the characteristics of a research context or research question that would benefit from using concept mapping as described in this chapter?
2. What are the advantages and disadvantages of using concept mapping for theory development?
3. What are the similarities and differences between concept mapping and community-based participatory research?
4. Consider a social research question that you have addressed or worked on recently—one that requires or would benefit from the development of a common conceptual framework involving several or many individual points of view.
   - What methods did you use to develop that framework?
   - How are they similar to the concept mapping method described here? How are they different?
   - What does the concept mapping approach allow you to accomplish as a social researcher that the methods you selected do not?
   - What do the methods you used allow you to accomplish that the concept mapping approach does not?

Exercises

Exercise 1: Focus, Brainstorming, and Sorting

This exercise is designed for classroom or workshop use. The objectives are as follows:

- To communicate the importance of the focus prompt
- To increase understanding of the difference between traditional focus group brainstorming and brainstorming for concept mapping
- To support understanding of the fundamental data unit of the individual sort

Focus

1. Instruct the group to identify and agree on a social issue or a problem in their context that requires group input to address. An example might be improving student housing or increasing sustainability of a specific social program.

2. Draft the focus prompt using the following structure:

- A specific (thing, issue, element, need) we need to (do, investigate, identify, solve) in order to (accomplish the goal of the project) is . . .
- Discuss and get agreement on the wording, for the purposes of the exercise.

Brainstorm

1. Identify a "facilitator" from the group.

2. Instruct the facilitator that the brainstorm is a *focused* response to the prompt described above. Basic rules are as follows: all input to the focus is acceptable; no editing of others' input except for clarity and understanding. Also, redundancy is acceptable at the brainstorming stage. Items not related to the focus should be recorded so as not to sidetrack the topic, but still capture the input. Statements should be short and contain only one main idea in response to the prompt.

3. The facilitator will state the focus prompt and ask for input. Another participant will write these statements on a white board, or type on a computer, so that the group can see the statements.

4. After 12 specific statements are generated, the facilitator can end the session.

Sorting

The sorting routine that each stakeholder conducts is the key to the data input and analysis.

1. Provide each student with a number of blank index cards or small slips of paper to correspond to the number of statements.

2. Instruct each to write the statements down and number them exactly as they were numbered in the brainstorming session. The card should contain the statement and the statement number (in parentheses).

3. Each person will conduct a "sort" of the 12 statements, according to the following rules:

   - The focus is how similar in meaning statements are, in relation to the others.
   - There is no specific number of clusters that is better than any other.
   - A statement must be put in one group—it cannot be put in two at the same time.
   - An individual statement may be considered its own "group"; do not put all statements in one pile.
   - Do not create a "miscellaneous" pile or sort things according to importance; this is a meaning sort, not a sort for value.

After giving the group 5 to 7 minutes to complete the sort, ask for a show of hands on the following questions:

- How many ended up with 10 piles, how many with 9, how many with 8, etc.? Ask the group for observations: Where did most people end up on the sort spectrum? What was the range of sort numbers?
- Reinforce that all sort results are valid, provided they follow the simple rules described.

Exercise 2: Analysis: Organizing the Sorting Results

This exercise is designed for classroom or workshop use. It is intended to follow the previous exercise. The objectives are

- To show the linkage between the individual sort data and the similarity matrix
- To enhance understanding of the unit of interest in MDS as applied in concept mapping

Using the sort piles that the participants developed in the previous exercise, they will create a matrix of similarities. This matrix is the data source for the MDS analysis.

Instruct the participants to look at their sorts and finalize them if needed.

1. Provide the following instructions:

   - Take out a piece of paper and draw a grid that has as many columns and rows as the number of statements—it will be 12, according to Exercise 1's instructions.
   - Write 1 through 12 down the left side of the grid or matrix and across the top of the matrix. The matrix should look like this. The marks in the columns are explained below.

Statement	1	2	3	4	5	6	7	8	9	10	11	12
1	/											
2		/										
3			/					/		/		
4				/								
5					/							
6						/						
7							/					
8			/					/		/		
9									/			
10			/					/		/		
11											/	
12												/

- Put a slash in the diagonal boxes in the middle of the matrix. Each statement is always sorted with itself.
- Pick up one of the piles that you have sorted the statements into and notice which statements are sorted together in that pile by their identifying number.
- Look at the first pair of statements (e.g., 3 and 8). On the matrix, put a slash in the boxes that represent where 3 and 8 intersect. There will be two of them.
- Look at what else is sorting with Statement 3 (e.g., 10). Put a slash in the boxes that represent where 3 and 10 intersect. There will be two.
- Notice that if you put 3 and 8 together and 3 and 10 together, then 8 and 10 are also together. Put a slash in the two boxes that represent where 8 and 10 intersect.
- Continue in this way until all statements that were placed together are recorded.

2. Check to see if the group is finished after 7 or 8 minutes or has reached a point of understanding.

3. Draw the students' attention to the fact that they have each created a binary square.

4. Instruct each person to work with another person and combine their sorts. Instruct as follows:

   - Get together with the person next to you. Decide which sort matrix you'll use as the "base" for combining your sorts. It doesn't matter which.

- Review the piles that you each had: How many did you each have?
- Look at the two matrices side by side. Do they look the same?
- Using the "base" matrix, transfer the information from the second matrix to combine them on one sheet.

5. This should take about 5 to 7 minutes. Ask

- How complicated was that to do? What would a database of 80 statements with, say, 25 sorters, look like? Would it be feasible to do?
- This is a manual exercise to show the construction of the database that the analysis is built on. This should help to illustrate that the data unit of interest is not the person who sorted, or even the piles of sorted items, but rather, the relationship of one idea to every other idea in the set.

Notes

1. The authors wish to thank the Division of Public Health, Delaware Department of Health and Social Services, for permission to reproduce the information related to the Delaware Cancer Consortium.

2. The analysis can be accomplished in most standard statistical packages such as SAS or SPSS. Some programming in these statistical packages would typically be required to get the data into the appropriate form and to sequence the analytic steps appropriately. Alternatively, the entire sequence of analytic steps have already been integrated along with data entry and graphics output of maps, pattern matches, and go-zone graphs into the Concept System software that is available from Concept Systems Incorporated (http://www.conceptsystems.com).

3. The term *(dis)similarity* is used in the MDS literature to indicate that the data can consist of either dissimilarities or similarities. In concept mapping, the data are always the square symmetric similarity matrix that is generated from the sorting data, so this discussion only considers similarity input.

4. The "map" is the distribution of points that represent the location of objects in *N*-dimensional space. In concept mapping, the objects are the brainstormed (or otherwise generated) statements and the map that MDS produces is the point map in two dimensions.

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CHAPTER 15

Mail Surveys

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When Is a Mail Survey the Right Choice?

While there are challenges to overcome and pitfalls lurking for the inexperienced researcher, a mail survey can be a very appropriate and efficient way of gathering high-quality information. How can you decide whether a mailed survey is the appropriate data collection strategy for your research question? This chapter is a short treatise on how to make that decision and how to conduct mail surveys.

In comparison with telephone or in-person interviews, mail surveys have many advantages.

- They are relatively inexpensive.
- They allow for large numbers of respondents to be surveyed in a relatively short period even if they are widely distributed geographically.
- They allow respondents to take their time in answering and to look up information if they need to.
- They give privacy in responding.
- They allow respondents to answer questions at times that are convenient to them.
- They allow respondents to see the context of a series of questions.
- They insulate respondents from the expectations of an interviewer.

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Keep in mind that some of these may not be advantageous in your own area of research. However, a mail survey can be an especially good choice when (a) you have limited human resources to help you conduct your study, (b) your questions can be written in a closed-ended style, (c) your research sample has a moderate to high interest or investment in the topic, and (d) your list of research objectives is modest in length.

Many key steps to conceptualizing research questions, developing questionnaires, avoiding errors, and ensuring quality are the same for mailed surveys as for those administered by other means. Rather than repeating the excellent guidance provided elsewhere, we focus, instead, on the core elements and considerations that are unique to the mail survey process (e.g., cover letters, questionnaire graphic design and instructions, procedures for encouraging returns) and on those aspects where the mailed format itself could increase the risk for error.

Keep Quality in Mind to Reduce Errors

As we describe the broad process of developing, conducting, and managing mail survey research, keep this fundamental principle in mind: Quality is not an average of the efforts that you put into a project; rather, it is achieved only by designing quality into all its phases and component parts. If you cut corners in one area to do an excellent job in another, the final product may have significant quality problems. The concept of optimizing your efforts across all areas has been referred to as "total survey design" (Biemer, Groves, Lyberg, Mathiowetz, & Sudman, 1991).

Once you have made the decision to conduct your research by mail, choose methods and procedures at each phase of the project with an eye toward reducing the likelihood that errors will affect the quality of your data. In particular, try to reduce four types of errors: sample selection bias, item response error, item nonresponse error, and nonresponse error.

By drawing a sample from a list that is incomplete or deficient in a significant way, you may end up with inaccurate survey results; this is called "sample selection bias." For example, your mailing list may be out of date—people may have left the area, new people may have arrived, but the list does not represent these recent changes. Clearly, sampling from outdated lists will produce outdated samples. However, problems with lists can be much more subtle. For example, you might choose to use the wrong list. If you want to find out why people may or may not use a neighborhood health center, you might be tempted to draw a sample from patient files—some who have visited recently and others who have not. However, the real question being asked is, "Why do people who live in the neighborhood use the health center and why do others who live there not use it?" Thus, the most appropriate sample is one of neighborhood residents, not of health center clients. Making sure that your list fits the population you want to study is a critical point on the road to gathering quality data. Having chosen the correct list, a reliable way to ensure that your sample is not biased is to use a method of random selection to draw a sample from the population in which you are interested. Methods of selecting samples and a fuller discussion of the issues mentioned above are found in the chapter by Henry (Chapter 3, this volume).

A second broad problem area comes from *response error,* for example, respondents misunderstanding the wording of the questions as presented. A central tenet of quantitative survey research is that all respondents should understand each question in the same way so that they are able to provide answers to each from the same frame of reference. This simply stated goal can be vexing to achieve. Two general rules will help you write good questions: (a) make them clear and (b) keep them simple—do not go beyond what is reasonable to expect people to understand or remember. There are many new tools available to help you reach your goal of creating valid questions, but still it takes effort to get there. For an excellent summary of the major issues you need to address to avoid response error, see Fowler and Cosenza (Chapter 12, this volume).

A third problem area is *item nonresponse error,* the failure of respondents to answer individual questions. Respondents may leave questions blank or accidentally skip over items. They may not follow instructions and then fill out answers incorrectly. They may write marginal comments that cannot be equated with your printed answer categories. If this happens often enough, the data that remain may be biased. With a mail survey, respondents do not have the benefit of an interviewer who is able to make clarifications or point the way though various skip patterns. We will discuss some design and content considerations that should help respondents to fill out your questionnaire properly.

Finally, the most challenging pitfall, for the mail survey researcher involves *nonresponse error,* the biased nature of the responding sample. It does not matter how accurately and randomly you draw a sample if returns come mainly from people who are biased in a particular way. Unfortunately, it can be difficult to determine whether a responding sample is biased. Thus, the standard safeguard is to aim to achieve a high response rate so that nonresponders would have to be very different from responders to affect your overall estimates for the population (Etter & Perneger, 1997). The next section of this chapter outlines in broad strokes the capacity of nonresponse error to wreak havoc on your data quality and offers two proven strategies for avoiding this common and potentially fatal problem. Later in the chapter, we detail additional strategies to ensure that every component of your mail survey project is carried out with an eye toward maximizing response rates.

Nonresponse Error

Nonresponse error is the bias that results when you do not get returns from 100% of your sample. Nonresponse errors distort your picture of the population and create problems for your study in two ways. First, if those who do not respond hold different views or behave differently from the majority of people, your study will incorrectly report the population average. It will also drastically underreport the number of people who feel as the nonresponders do. How far off the mark you are depends on how big the nonresponse is and how different the nonresponders are from responders (Armstrong & Overton, 1977; Barnette, 1950; Baur, 1947; Blair, 1964; Blumberg, Fuller, & Hare, 1974; Brennan & Hoek, 1992; Campbell, 1949; Champion & Sear, 1969; Clausen & Ford, 1947; Cox, Anderson, & Fulcher, 1974; Daniel, 1975; Dillman, 1978; Donald, 1960; Eichner & Habermehl, 1981; Filion, 1975; Gannon, Northern, & Carroll, 1971; Gough & Hall, 1977; Jones & Lang, 1980; Larson & Catton, 1959; Newman, 1962; Ognibene, 1970; Reuss, 1943; Suchman & McCandless, 1940). Second, even if nonresponders are not that different, low response rates give the appearance of a poor-quality study and undermine confidence in its results. The study becomes less useful or less influential simply because it does not have the trappings of quality.

Nonresponse error poses a particular risk for mail surveys, in that it is *so very easy* for recipients not to respond. It is not as if they have to close the door in someone's face, or even hang up the phone on a persistent interviewer; all they have to do is toss the survey questionnaire into the wastebasket. In addition, some recipients of mail surveys who are interested and have good intentions to participate become nonresponders simply because they never get around to filling out the questionnaire. Unfortunately, in many studies very little can be discerned about the nonresponders, and we are thus left with uncertainty about the quality of the data. By obtaining a very high return, you reduce the likelihood that the nonresponders will have an impact on the validity of your population estimates, even if the nonresponders are different.

What is considered a high response rate? Certainly, a rate of return in excess of 85% is viewed as excellent. With such a rate, it would take a highly unusual set of circumstances to throw off your results by very much. Response rates in the 70% to 85% range are viewed as very good. While rates in the 60% to 70% range are considered acceptable, at this level you should begin to feel uneasy about the characteristics of nonresponders. Response rates between 50% and 60% are barely acceptable; at this level, you really need some additional information that can contribute to confidence about the quality of your data. Response rates below 50% are not scientifically acceptable—after all, at this level, the majority of the sample is not represented in the results.

In addition to striving for high response rates, it is always useful to try to obtain information about the nonresponders, so that you can compare them with responders. Sometimes this information is available from the list that you originally sampled. For instance, city lists that are used to confirm eligibility for voter registration have each person's age; gender (not listed explicitly, but you can usually figure it out from the first name); occupation (in broad categories); precinct or voting district; whether the person is registered to vote or not and, if registered, party affiliation. By keeping track of who from your original sample has and has not responded, you can compare the characteristics of one group to the other.

It turns out that there are some common traits of nonresponders (Baur, 1947; Campbell, 1949; Gannon et al., 1971; Gelb, 1975; Goodstadt, Chung, Kronitz, & Cook, 1977; Ognibene, 1970; Peterson, 1975; Robins, 1963; Suchman, 1962); we can get a picture of some common traits of nonresponders. Compared with responders, they tend to be less educated; they also tend to be elderly, unmarried and male; or they have some characteristic that makes them seem less relevant to the study (e.g., abstainers for a drinking study, nondrivers for a traffic safety study, or lower income people for a study about mortgages).

Although a variety of response rates are reported in the literature, it is safe to assume that some of the worst response rates never are published. If you were to simply stuff questionnaires in envelopes and mail them to people asking them to fill them out, it would be common to see response rates in the 20% range, though it would not be surprising to see them in the 5% range too. This is much lower than the rate of 70% or higher that can inspire confidence in the data. So the fundamental question is, how can you achieve the highest possible response rates?

Sending Reminders

The key technique for producing high response rates is the use of reminders (Denton, Tsai, & Chevrette, 1988; Diamantopoulos & Schlegelmilch, 1996; De Rada, 2005; Dillman, 1978; Dillman et al., 1974; Eckland, 1965; Edwards et al., 2002; Erdogan & Baker, 2002; Etzel & Walker, 1974; Evangelista, Albaum, & Poon, 1999; Filion, 1976; Ford & Zeisel, 1949; Fox, Robinson, & Boardley, 1998; Furse, Stewart, & Rados, 1981; House, Gerber, & McMichael, 1977; Jones & Lang, 1980; Kanso, 2000; Kanuk & Berenson, 1975; Kephart & Bressler, 1958; Linsky, 1975; Scott, 1961; Yammarino, Skinner, & Childers, 1991). Even under the best of circumstances, you will not achieve acceptable levels of return if you send no reminders. In fact, it is important to send out several, and it is imperative to pay attention to their timing.

As you track the daily returns, an interesting pattern becomes apparent. For the first few days after questionnaires have been mailed, you will receive nothing. This makes sense because it takes time for the surveys to be delivered, it takes a short period for respondents to fill them out, and then it takes a day or two for the respondents to mail them back (actually this can be a day or two longer with business-reply returns). About 5 to 7 days after the initial mailing, you will receive a few returns; then in the next few days you receive many more, with more coming in each day than the day before. Around the 10th day after the mailing, returns will start to level off, and around the 14th day they will drop off precipitously.

An abrupt reduction in returns is a signal that whatever motivational influence your initial letter had is now fading. Those who have not returned the questionnaire by now are going to begin to forget about doing it, or the survey is going to get buried on their desks. At this point in the return pattern—about the 14th day—you want to have your first reminder arrive.

The initial pattern repeats itself after you send out the first reminder. After a few days of inactivity, a burst of returns with more coming in each day will be followed by a precipitous decline at about 14 days. Another interesting feature of this pattern is that whatever return rate you got in the first wave (e.g., 40%), you will get about half that number in the second wave (e.g., 20%), and so on for each succeeding wave. Aiming for at least a 75% return rate, you should plan for at least four mailings—the initial mailing and three reminders. Each of these mailings should be spaced about 2 weeks apart. This will result in a pattern of returns something like this:

$$40\% + 20\% + 10\% + 5\% = 75\%.$$

Thus, your total mailing period will take about 8 to 9 weeks, leaving some time after your last reminder for the final returns to come in. A final point of interest about this pattern is that the rate of returns and the number of reminders is unrelated to the total size of your sample: Follow the same procedures whether your sample size is 200 or 20,000. The only impact of scale is that you need more staff to put together the mailings in each round.

Sending reminders more frequently than every 2 weeks does not speed up the returns—it merely wastes time and money reminding people who were going to respond anyway. Conversely, spacing out two or three reminders over a longer period than at 2-week intervals (hoping to save money on postage, for example) is not as effective in producing a good return rate. Your reminder sequence will not build momentum among the nonresponders, because the time lapse is so long that they would have forgotten about the survey. In this case, each reminder must start all over again to motivate people to participate.

Tip

If budget constraints are causing you to consider reducing the number of reminders or not doing any at all, there is a middle ground. From the sample sent to the initial mailing, select a random subset to receive the full sequence of reminders. You will be able to compare their results with those received from the group with fewer reminders to gauge the extent of bias.

Should each reminder be a repeat of the first mailing? We recommend sending a complete package (respondent letter, questionnaire, and return envelope) only in the first and third mailings. In the second and fourth mailings, send a reminder postcard or letter.

The series of letters you use for each of the four mailings should focus on slightly different issues. The first mailing should be the most thorough, covering all the issues. In the second, be gentle and friendly—for example, "Just a reminder in case you have not yet sent in your questionnaire. We would really like to hear from you." In the third mailing, emphasize the confidentiality of responses and the importance of getting a good return so that all points of view are represented. Make note of the fact that you are including another copy of the questionnaire in case the recipient misplaced the first one you sent. The fourth mailing should be a "last call." Consider setting a specific deadline and encourage the recipient to send in the questionnaire so that his or her point of view can be represented.

Who should receive reminders? When using a procedure that promises confidentiality, you are able to track returned questionnaires through their code numbers. In that case, send reminders only to those who have not yet responded. This saves money on postage, printing, and supplies and keeps respondents from being annoyed (or confused) by reminders after they have already sent in their surveys.

When using a procedure that promises anonymity, the process for sending reminders is a little more complicated. Since you do not know who in your sample has returned their questionnaires, you must use one of two alternate strategies. The first is to send reminders to everyone and explain that, because the returns are anonymous, you do not know who has and who has not responded; thus you are sending reminders to everyone. Always include a line that says, "If you have already sent in your questionnaire, thank you very much." This strategy has disadvantages in that (a) it wastes postage, supplies, and resources; (b) it irritates respondents to receive reminders when they have already returned their questionnaires; (c) it dilutes your message by apologizing to people who have already returned their questionnaires and not focusing exclusively on those who have yet to respond; and, furthermore, (d) reminders sent to all respondents may confuse some, lead them to worry that their surveys got lost in the mail and prompt them to fill out new ones. With no way of knowing which surveys might be duplicates, you cannot remove them from your returns.

A technique that one might call the *reminder postcard strategy* can sidestep these concerns. This method maintains complete anonymity for the respondents' returned questionnaires while letting you know who has and has not returned the questionnaire. Thus, reminders need be sent only to those who have yet to respond. When using the reminder postcard strategy, enclose in the original mailing of the questionnaire a postage-paid return postcard imprinted with either an identification code or the recipient's name (or both). Be sure that the questionnaire itself bears no identification. In the survey instructions, state explicitly that returning the postcard lets you know that they do not need any more reminders and ask that they mail the postcard back *separately* from the questionnaire. By using this procedure, you know who has returned the questionnaire without having to put any identifying information on the questionnaire itself.

Some may worry that respondents might simply return the postcard and not the questionnaire. That would certainly be a problem, but in our experience that has not been the case. More questionnaires than postcards are returned. Some respondents forget to mail their postcards, some lose them, and a small number (e.g., 5% or so) purposely do not return them as a way to ensure their anonymity. Thankfully, there are only a few who take this last route; otherwise the method would not achieve its intended purpose of providing information about who has responded while maintaining respondent anonymity.

Eliminating the Element of Surprise

An interesting addition to the use of reminders is the strategy of contacting respondents by mail or phone in advance of sending the survey. In effect, you are giving a "heads up" that they have been selected to be in a survey, and they should watch for its arrival within the next week or two. Prenotification can offer the benefit of shortening the interval between the first mailing of the survey and the last reminder. You can "gain" as much as 2 weeks on your return schedule by prenotifying while you are wrapping up the process of printing the survey, stamping envelopes, and assembling the mailing. This "reminder done ahead of time" has generally been found to be equivalent to one follow-up reminder (Allen, Schewe, & Wijk, 1980; Brunner & Carroll, 1969; Edwards et al., 2002; Ford, 1967; Furse et al., 1981; Heaton, 1965; Jolson, 1977; Kanso, 2000; Kerin & Peterson, 1977; Myers & Haug, 1969; Parsons & Medford, 1972; Schegelmilch & Diamantopoulos, 1991; Stafford, 1966; Taylor & Lynn, 1998; Walker & Burdick, 1977; Wynn & McDaniel, 1985; Yammarino et al., 1991).

Providing Incentives

A second technique that is another powerful tool to increase response rates is to provide incentives (Gendall, Hoek, & Brennan, 1998; Helgeson, Voss, & Terpening, 2002; Jobber & O'Reilly, 1998; Kanso, 2000). The logic of offering an incentive is simple: explicitly raise the stakes by offering something in return for filling out the questionnaire. However, research findings hold some surprises regarding options for providing incentives to respondents. The challenge is to figure out what reward to give and when to give it.

It makes sense to provide the incentive after respondents return their questionnaires. They would be informed in the initial letter that this is the proposition, and more respondents would be motivated to participate because of the promise of this reward. Clearly, respondents would have to value what is being offered, or there would be no motivational benefit. One disadvantage with this strategy is that respondents receive delayed gratification; they receive their rewards as many as several weeks later, after they have demonstrated their "good" behavior.

A second option is to offer the reward in advance, including it with the initial mailing in anticipation of the respondent's participation. The advantage here is that the impact is immediate; the respondent receives the benefit right away. We should not underestimate the motivational power of the implied contract: "They gave me this reward, so if I don't do my part by filling out the questionnaire, I will not be living up to my end of the bargain." The disadvantage here (both financially and morally) is that some people receive the reward but do not deserve it because they do not return the surveys anyway. Because of this problem, one goal in using this technique is to figure out the lowest value of the reward you need to give in order to achieve the effects that you want.

Using money as an incentive is very successful. A variety of studies and literature reviews have shown that offering monetary incentives tends to improve response rates (Duncan, 1979; Edwards et al., 2002; Fox et al., 1998; Heberlein & Baumgartner, 1978; Hopkins & Gullickson, 1992; Kanuk & Berenson, 1975; Leung, Ho, Chan, Johnston, & Wong, 2002; Linsky, 1975; Scott, 1961; Yammarino et al., 1991; Yu & Cooper, 1983). What is also clear from this research is that prepaid monetary incentives are more effective than promised monetary rewards (Blumberg et al., 1974; Edwards et al., 2002; Hancock, 1940; Saunders, Jobber, & Mitchell, 2006; Schewe & Cournoyer, 1976; Warriner, Goyer, Gjertsen, Hohner, & McSpurren, 1996; Wotruba, 1996). There have been many studies that show some impact of promised monetary rewards compared with no rewards, but there are many more examples of studies that have shown even better results for prepaid rewards compared with promised rewards (Edwards et al., 2002; Yu & Cooper, 1983).

These studies find that not only do prepaid rewards have substantial impact, but that it does not seem to take a very large reward to stimulate an improved response rate. Many studies reported in the literature show the benefits of providing just 25 and 50 cents. However, many of these studies were conducted 20 to 30 years ago. It seems reasonable to extrapolate the findings from these studies to the "current" value of the dollar. Hopkins and Gullickson (1992) conducted a review and equated these values to 1990 dollars, and still showed improvements for values less than 50 cents. A more recent study showed that both $2 and $5 generated respectable and similar response rates (Shaw, Beebe, Jensen, & Adlis, 2001).

The question of whether increased benefit accrues for increasing dollar amounts is harder to answer definitively. Much of the research to test alternate amounts has not tended to use sums more than $1; therefore the number of studies we have available to make generalizations about larger-sized incentives is relatively small. In their review, Hopkins and Gullickson (1992) did find an increasing percentage of returns over "no incentive" control methods for greater incentive values, but their top group was designated as $2 or more and included only eight studies. A more recent study by Edwards, Cooper, Roberts, and Frost, (2005) showed steady increase in response rates for amounts up to $5. In addition, our experience with a recent nonexperimental study dealing with alcohol use and work included one work site where we used a $5 prepaid incentive; the resulting response rate was 82%.

Understanding the meaning of the reward to the respondent helps interpret these findings about larger ($5) and smaller incentives ($1–$2). With small amounts of money, people clearly do *not* interpret the reward as fair market exchange for their time. Even a $1 reward for filling out a 20-minute questionnaire works out to only a $3 per hour rate of pay. Therefore, people must view the reward in another light; one idea is that it represents to the respondent a token of good faith or a "trust builder" (Dillman, 1978). The respondent feels that the research staff are nice to show their appreciation by giving the incentive and therefore feels motivated to reciprocate by filling out the questionnaire.

There have not been many studies reporting on the provision of larger-sized rewards, but it looks as though response rates tend to be higher for these than for lesser amounts (Hopkins & Gullickson, 1992; Martinson et al., 2000; Yu & Cooper, 1983). In particular, higher incentive amounts are reported in the literature for surveys conducted with persons in professional occupations, particularly doctors. Incentive amounts from $20 to $50 have been used (Godwin, 1979). In these circumstances, higher response rates are obtained with higher rewards (Berry & Kanouse, 1987; James & Bolstein, 1992; Jobber, Saunders, & Mitchell, 2004).

Another monetary incentive technique is the use of a "lottery" prize. This technique falls within the "promised reward" category, but has a twist. Respondents are offered a "chance" to win a "big" prize, although they also have, of course, a chance of getting nothing. Again, research on this variation is limited, so definitive generalizations about its effectiveness are not possible (Gajraj, Faria, & Dickinson, 1990; Hopkins & Gullickson, 1992; Leung et al., 2002; Lorenzi, Friedmann, & Paolillo, 1988; Martinson et al., 2000). The logic behind this idea is that the chance of hitting big will be such an inducement that respondents will fill out their surveys to qualify. This technique also works well if you are trying to encourage respondents to mail in their surveys by a particular deadline.

Of course, to give out the lottery prize incentives, respondents cannot remain anonymous. To conduct a drawing and give out prizes, you need to know the name and address (and possibly a phone number) associated with each returned survey. This lack of anonymity may be counterproductive in some circumstances. The postcard mechanism discussed above provides a solution to this dilemma. To be eligible, respondents would need to return their postcards. Enterprising respondents could realize that all they really have to do to be eligible is to turn in their postcards. No one would be able to tell if they had actually sent in their questionnaires. It would seem that the more attractive the "prize," the more motivation there might be to cheat. However, respondents do not seem to do that. In our recent experience with this technique in 12 different work sites across the country, we never received more postcards than questionnaires, even though we were offering three $250 prize drawings at each work site!

Nonmonetary rewards have also been shown to act as incentives (Brennan, 1958; Bright & Smith, 2002; Dommeyer, 1985; Edwards et al., 2002; Furse & Stewart, 1982; Hansen, 1980; Hubbard & Little, 1988; Jobber & O'Reilly, 1996; Nederhof, 1983). The logic of giving a "gift" is similar to that of giving a token amount of money. The idea is to acknowledge respondents' efforts and thank them for their participation. A wide range of nonmonetary incentives can be used, including ballpoint pens, cups, gift certificates, postage stamps, and movie tickets. As with money, the incentive can be framed as a prepaid "thank you" gift or a promised "reward" sent after the survey is returned (Brennan, 1958; Pucel, Nelson, & Wheeler, 1971; White, Carney, & Kolar, 2005).

Although few research studies report on response rate differences between prepaid and promised gifts, one would assume that the effectiveness would follow the same pattern as with monetary rewards—prepaid gifts would probably have a better effect (Kalafatis & Madden, 1995). Similarly, there has not been much research done to see what the trends are with more valuable gifts. To some extent, the concept of value can be less obvious with many types of gifts with monetary rewards. Also, it is possible that a gift's perceived value may exceed its actual cost. Some respondents may not be aware of how much particular gifts cost, or perhaps you can get a discount for buying in bulk. For example, movie passes that cost about $6 to $7 each in a quantity purchase can be redeemed for movies that may cost up to $10.

Incentives that relate to a survey's topic or that are of interest to a wide range of respondents may also increase response rates. One of us recently had the opportunity to be part of a survey study in which respondents were asked to fill out a short questionnaire concerning their nutritional intake. The researchers also needed respondents to include clippings from their toenails. As an incentive, respondents were told that when they returned the survey they would receive detailed nutritional analyses of their own diets based on their reports and the requested clippings. *Returns were more than 70% with only one reminder.*

Another variation on the "gift" incentive is to offer a contribution to charity in the respondents' names if surveys are returned (Dickinson & Faria, 1995; Robertson & Bellenger, 1978; Warriner et al., 1996). This technique can be used on an individual or group basis. The individual strategy would be to contribute a certain amount (say $5) to a charity for each survey returned. Obviously, the perceived value of the charity might have some impact on the effectiveness of this incentive. Specific charities can be designated or you can allow respondents to choose from among a few offerings, or you can ask them to write in their own suggestions. The group strategy would provide a significant payment to a charity if the sample as a whole provides a certain number or percentage of returns (e.g., a 70% return rate). Our recent work site study included two sites in which we used the group strategy, with a $750 contribution to a local charity. We achieved response rates of 68% and 78%.

Reminders or Incentives: Which to Choose?

If you had to choose, which is the more effective technique—reminders or incentives? The question can be answered from the perspective of final response rates, cost-effectiveness, and speed of returns. James and Bolstein (1990) conducted a study that offers some insight into this issue. They conducted an experiment using different amounts of incentives (none, 25 cents, 50 cents, $1, and $2) and tracked response rates at the end of each of their four mailings of a four-page questionnaire. The highest rates of returns resulted from the use of both methods in combination—four mailings and a $2 prepaid incentive. This strategy was also the most expensive. Good return rates (albeit a little lower than for the combination method) were also obtained through the use of two mailings and a $2 incentive and from four mailings with no incentive. The no-incentive strategy was slightly less expensive than the incentive strategy, but of course it took more time for the additional waves of mailings to be administered.

To summarize, if a shorter data collection period is more important than keeping costs low, then using incentives may allow you to save some time; if money is the limiting factor, then planning for multiple mailings with no incentives may be the best. However, if a high response rate is your major goal, you should use multiple mailings and incentives together (Larson & Chow, 2003).

Basic Mail Survey Planning Considerations

There are several procedural decisions that, early on, can ensure that a mail study is carried out well and that recipients are more likely to respond. These are described in the following sections.

Preserving Confidentiality/Anonymity

If respondents believe that their answers will be kept confidential, rather than being attributed to them directly, they will be more likely to return a survey (Boek & Lade, 1963; Bradt, 1955; Childers & Skinner, 1985; Cox et al., 1974; Fuller, 1974; Futrell & Hise, 1982; Futrell & Swan, 1977; Kerin & Peterson, 1977; McDaniel & Jackson, 1981; Pearlin, 1961; Rosen, 1960; Wildman, 1977). There are a number of straightforward safeguards for maintaining confidentiality. First, never write respondent names or addresses directly on the questionnaires. Instead, use code numbers on the surveys and maintain a separate list of names and addresses with their corresponding code numbers. Keep the list out of the view of people who are not on the research team. Second, when the questionnaires come back, do not leave them lying around for curious eyes to peruse. Store returns in file cabinets, preferably locked when you are not present; lock your office when you are not there. Third, do not talk to colleagues, friends, or family about the responses you receive on individual questionnaires. Fourth, do not present the data, in reports or papers, in such a way that readers are able to figure out who individual respondents are. Sometimes, this means describing individuals with characteristics somewhat different from those they really have, and sometimes it means not presenting information on very small subsets of respondents. For example, in a company report, you would not present data on a group of three vice presidents by saying, "Two-thirds of the senior management group reported thinking about changing jobs in the next year." Data for organizations such as companies, schools, and hospitals should be presented without the names of the organizations unless there has been a prior specific agreement that this would be done.

Tip

For both community-based surveys and institutional-based surveys, any publicity you can garner that includes support from leaders (mayor, plant manager, company doctor, or union leader) will reassure people who are concerned that the study will not make a difference or is not strictly confidential..

There is an important distinction between procedures that maintain anonymity compared with those that only maintain confidentiality. For confidentiality, you know who filled out which questionnaire, but you promise not to divulge that information to anyone outside the research team. For anonymity, no one—not even you—knows which questionnaire belongs to which person. You can achieve this by not writing the code numbers on the questionnaires before they are sent out. That way there is absolutely no link between the returned questionnaires and any sample list you have.

A reasonable bet would be that studies offering true anonymity (no identification numbers on the questionnaires) compared with those offering only confidentiality (a promise of no disclosure) would produce better response rates. However, studies have not clearly demonstrated such an advantage (Andreasen, 1970; Boek & Lade, 1963; Bradt, 1955; Kalafatis & Blankson, 1996; Mason, Dressel, & Bain, 1961; Pearlin, 1961; Rosen, 1960; Scott, 1961). Perhaps this is too technical a distinction for respondents to understand. Perhaps they assume that because you knew their address, you can somehow find them again if you want to. There is also the cynical interpretation: "They could figure out who I am by putting together several demographic characteristics, so their promise of anonymity is really not much more than a promise of confidentiality." However, a study by Jobber and O'Reilly (1998) showed that anonymity improved response rates when sensitive questions were asked. Of course, many surveys are rather innocuous, and respondents do not care if people know who they are and what they think on these topics. It is probably best to provide anonymity if you can, as no one has shown that promising anonymity produces worse response rates. Even when the data are anonymous, you still must follow the other procedures described above to maintain confidentiality—that is, you should not leave questionnaires lying around for curious eyes to view and you should not report data for small, identifiable, groups of respondents.

Supplying Return Postage

It almost goes without saying that if you are asking a respondent to do you the favor of participating, enclosing a return envelope, already addressed to you with return postage affixed is the least you can do. Perhaps because this is so obvious, few studies have explicitly tested whether the enclosure enhances returns. Those that have been done certainly confirm this point (Armstrong & Lusk, 1987; Blumberg et al., 1974; Ferris, 1951; Harris & Guffey, 1978; Jobber & O'Reilly, 1998; Kanso, 2000; McCrohan & Lowe, 1981; Price, 1950; Yammarino et al., 1991).

The issue of what types of postage to affix to the return envelope has, however, received a lot of attention by researchers. The alternatives are to use some kind of business-reply franking or affix stamps to the return envelopes. The advantage of the business-reply method is that you get charged only for questionnaires that are actually returned. The post office does charge a fee for setting up this service, and it adds a charge that can range from 6 to 65 cents per returned questionnaire. Having a large volume of returns (more than 900) or using automation-compatible mail pieces can, however, keep these costs low. Be sure to factor in these charges when comparing the costs of alternate postage mechanisms. A major disadvantage of this return postage choice is that it appears more impersonal than the alternative.

Putting actual stamps on return envelopes rather than using a business-reply franking, seems to produce a small increase in return rates (Brook, 1978; Edwards et al., 2002; Jones & Linda, 1978; Kimball, 1961; Watson, 1965). The reason for this is that respondents often do not want to "waste" stamps by not returning their questionnaires and yet they are not crass enough to peel the stamps off and use them for their own purposes. Some studies have also shown that using attractive commemorative-type stamps has a slight advantage over the use of regular stamps (Henley, 1976; Jones & Linda, 1978; Martin & McConnell, 1970). A disadvantage of this approach is its cost. Not only must you pay for stamps that ultimately never get used but it also costs personnel time and money to purchase the stamps and affix them to all the envelopes.

Considering the Effect of Outgoing Postage

When considering the alternatives for postage that you affix to the outgoing envelope, the most common choices are stamps or metered postage. A few studies have shown a slight advantage of using stamps—in particular, commemorative stamps—on outgoing envelopes (Blumenfeld, 1973; Dillman, 1972; Edwards et al., 2002; Hopkins & Podolak, 1983; Kernan, 1971; McCrohan & Lowe, 1981; Peterson, 1975; Vocino, 1977). The explanation for this difference is that respondents are less likely to assume a mailing as "junk mail" if there is a stamp on the envelope, and so they are more likely to actually open the envelope. The only disadvantage of using stamps, as noted previously, is the extra cost of staff time to affix them on the envelopes.

First-class indicia can also be considered for outgoing postage. This is similar to the business-reply franking, except that it is used for outgoing first-class mail. You set up a prepaid account with the postal service and print your account number and a first-class designation on your outgoing envelopes (illustration at right). The postal service keeps track of your mailings and deducts the postage amounts from your account. This is the least labor-intensive method of sending out your questionnaires, but it probably suffers somewhat from the same problem as metered mail in that it may be confused with junk mail.

First Class Mail *U.S. Postage* *PAID* *Boston, MA* *Permit No. 108*

Another alternative to consider for outgoing postage is to use premium postage/shipping for mailings, such as special delivery or next-day delivery services. The research shows that there is some advantage to using this type of postage, but the costs are so substantial that many consider it prohibitive (Clausen & Ford, 1947; Hager, Wilson, Pollak, & Rooney, 2003; Kephart & Bressler, 1958). When special postage is used, it is most often for final reminders. At least at this stage of the process you are mailing only to part of your sample and therefore the cost impact is less. However, a study by Schmidt, Calantone, Griffin, & Montoya-Weiss (2005) showed no extra benefit for certified mail over first-class mail when used as a third reminder.

The Mail Survey Package

In a best-case scenario, recipients will pull your survey package from the letterbox, tear open the envelope with curiosity, read your letter with growing enthusiasm, complete the questionnaire without hesitation (or confusion), and drop it in the mail, safely sealed in the stamped and addressed return envelope that you have thoughtfully provided. In this section, we provide suggestions that will help the recipient move smoothly through this process.

Composing an Engaging Respondent Letter

The enclosed respondent letter in a survey packet plays a critical role for producing good response rates, since this is usually the only means of communication between the study team and the respondents. Because most mail surveys arrive at a potential respondent's doorstep out of the blue, the cover letter has to do all the work of describing the study purposes, explaining the general procedures to be followed, and motivating the recipient to participate (Andreasen, 1970; Champion & Sear, 1969; Hornik, 1981; Houston & Nevin, 1977; Simon, 1967). To produce a letter that is "just right," several characteristics and elements are critical:

- Keep the letter short with the text confined to one side of one page and printed on professionally produced letterhead. This makes it clear who has sent the survey and what institution is supporting the research. Recipients are more likely to respond to surveys that they consider important or prestigious (Baldauf, Reisinger, & Moncrief, 1999; Doob, Freedman, & Carlsmith, 1973; Houston & Nevin, 1977;

Jones & Lang, 1980; Jones & Linda, 1978; Kanso, 2000; Peterson, 1975; Roeher, 1963; Watson, 1965). For example, they are more likely to respond to surveys that are sponsored by government agencies or well-known universities (Houston & Nevin, 1977; Jones & Lang, 1980; Jones & Linda, 1978; Peterson, 1975). Also, when the cover letter is on university or government agency letterhead, recipients may be less concerned that the survey is a ploy to send them a credit card or sell them insurance. Taking further advantage of the institutional affiliation, do not refer to the study name alone (e.g., the Healthy Family Study); instead, include the name of the university or research institution as well (e.g., the Famous and Well-Regarded University's Healthy Family Study).

- Start with a first sentence that captures attention and encourages the recipient to read the rest of the letter. For example, in a study of police officers concerning gambling enforcement policies, we started our letter with, "We would like the benefit of your professional experience and 10 minutes of your time!" For a corporate study of alcohol policies, we started with, "Many people are concerned about alcohol abuse in the workplace."

- Describe why this study is important and how the information may be used. Respondents want to participate in activities that they think are useful and that relate to their lives in some specific way.

- Explain who is being asked to participate in the survey and how you got this person's name and address.

- Discuss whether this survey is confidential or anonymous, and describe exactly how privacy will be achieved.

- Make it clear that that participation in the study is voluntary, but emphasize the importance of the recipient's participation. If an incentive is to be provided, be sure to describe it as a good-will gesture, not as a ploy to coerce participation.

- Tell the recipient how to get in touch if he or she has questions. Include the name of a contact person and a phone number—perhaps even a toll-free number or the instruction to "call collect."

- Show how the respondent can return the questionnaire to you, pointing out the return envelope and noting that it is stamped and preaddressed, for example.

- Use clear language, and keep the needs of your audience in mind when choosing font and type size, reading and language level, and layout and reproduction quality.

Offering Clear Guidance

The clarity of the instructions that come with the questionnaire also strongly influences response rates. It is not surprising that forms with complicated, confusing, or incorrect instructions create frustration for respondents. The results of this frustration can be either completion errors or failure to return the questionnaire altogether.

In addition to making instructions clear and short, efforts should be made to make them clearly visible. Various formatting aids such as boldface type, instructions enclosed in boxes, and arrows directing the respondent to the next question can supplement written directions and can help respondents to comply.

Formatting Questions and Their Response Categories

Decide on uniform question and answer formats that you will maintain throughout the document. For example, each question should have a question number, either applied sequentially throughout the whole survey or numbered sequentially within sections (e.g., A1, A2, B1, B2, and so on). By numbering each question, you are providing a mechanism that helps the respondent move through the questionnaire efficiently. Sometimes researchers are tempted to leave follow-up, probing questions unnumbered, but this can be a shortsighted strategy, because many respondents accidentally skip unnumbered questions.

All similar questions should be formatted in the same way. For example, multiple choice questions might be formatted in boldface type with the response categories in regular (nonboldface) type. This style facilitates the respondent's ability to scan from one response category to another and from one question to another. Furthermore, response categories should have an established style. You might display all the categories vertically in one column or horizontally in one row. If possible, avoid doubling up the categories into two columns or two rows, because this creates an ambiguous sequence for reading the categories. If you have a sequence of questions that use the same response categories, they should all be lined up vertically on the page with the response options arrayed horizontally.

Though virtually all your questions should be written in a closed-ended style, you may be including one or two which allow open-ended responses. Questions seeking a one- or two-word response should be followed by a short response line to help the respondent frame the answer succinctly. Similarly, if you are including open-ended probes (e.g., "Do you have any suggestions on how we could improve our services?"), provide enough blank space for the respondent to give a complete, thoughtful, and legible response. Be aware, however, that many respondents (as many as 75%) may just leave blank questions that require a sentence or two written response.

Keeping Aesthetics in Mind

Paying attention to aesthetic issues can improve your response rates as well as the quality of your data (Blumenfeld, 1973; Dillman, 1978; Ford, 1968). Your response rates will be higher because a visually pleasing questionnaire is more likely to be considered important and competently prepared. Neat and stylishly presented response alternatives and instructional messages make it easier for respondents to comply in a correct fashion (Edwards et al., 2002; LaGarce & Washburn, 1995).

First and foremost, the pages of your questionnaire should be balanced. Page layout standards should include uniform top, bottom, left and right margin widths, consistent page numbering format and placement, consistent indentation between question numbers and question text, and consistent line spacing between question and their groups of response categories and between questions themselves. On occasion, a sequence of questions might create the need for a page break three quarters of the way down because the next question will not fit in the remaining space. It is important to make such a page look complete by increasing the space between items so that the page balances with the one that it faces. The same principle applies to the use of pages with a two-column format. Make sure that both columns are equally "filled." In achieving balance, it is better to have more white space around questions than to produce an overly crammed or squeezed look. This is an area where you will benefit by having someone with a good eye for graphic detail periodically review the layout of your questionnaire as it takes shape.

Tip

Although research on the impact of colored paper stock (pastel please!) is not strong, it certainly helps those with messy desks to locate the survey when they finally decide to complete it.

Choosing Type Style and Size

Choose one or two familiar and clean-looking type fonts (e.g., Helvetica for headings and Times New Roman for text) rather than novelty or script-like ones. Busy or hard-to-read type styles can be annoying for many readers, and for those with lower literacy skills, they may be a barrier to comprehension or to participation altogether. A second key issue is type size. Choose a type size that is large enough to be read easily. It stands to reason that a questionnaire produced entirely in an 8 point font will not be an easy read. If your study focuses on elderly respondents, you may want to use a type size that is larger than normal.

In a sequence where items are very similar except for one phrase, you may find it helpful to visually emphasize the changing phrase. Consider formatting it in a boldface or italics style or increase the type size. This will help the respondent to pick up the difference in the questions on the first reading. Do not, however, go overboard with this mechanism. If you use it in every question, readers will tend to tune out its subtle message. We suggest that you avoid underlining altogether; it is less easy to read and is an artifact from the days when our typewriters offered fewer options for adding emphasis.

Deciding on the Physical Dimensions of the Questionnaire

Research has suggested that questionnaires be produced on sheets of paper that are smaller than 8.5 by 11 inches, to give the appearance of a "small" task. In general, there is nothing wrong with this notion if it does not conflict with other formatting and production priorities. For instance, the type size should not be made too small to compensate for the smaller page. Also, if the smaller page size generates significantly more pages, then it is not clear that there is a net benefit. If a commercial printer is involved, you can easily specify any dimension; however, if you are relying on your office printer or copy machine, using odd size paper may not be worth its trouble. Finally, smaller-sized questionnaires raise the issue of the size of the envelope. A small questionnaire rattling around in an envelope made to hold standard 8.5-by-11-inch paper may not make a polished first impression.

Printing pages back-to-back cuts the number of sheets needed in half, resulting in a questionnaire that looks "less weighty," which may help response rates. We recommend this style provided that the paper is of sufficient weight to keep the print from "bleeding through" to the other side. It also lowers postage costs and blunts criticism from environmentally conscious respondents.

Another style feature that will have a direct influence on the number of pages in your questionnaire is the use of a two-column, newspaper-type format. Many questions that have relatively short response categories (e.g., yes/no, agree/disagree) can be easily placed in a two-column format. The questions themselves may take up a few extra lines, but the response categories take up no more space. Using this technique can reduce the number of pages in your questionnaire by anywhere from 25% to 50%.

With a commercial print vendor, it is a simple task to commission a multipage questionnaire in a booklet format. However, with a printer/copy machine enabled for booklet/two-sided printing, you too can produce a polished-looking booklet. For an 8.5-by-11-inch finished product, use 17-by-11-inch paper (with the pages set up in the right order to flow in the correct sequence when the pages are folded), fold each set in the middle and staple into the fold. You may need to purchase a long-arm stapler if your copier does not have this feature, called saddle stitching. Keep in mind that the total number of pages in your finished booklet must be divisible by four, though you can "cheat" if need be by placing the overall instructions on the "cover" page and leaving the back page blank (good for inviting additional comments).

As a last step to ensure that your instructions, formatting, and overall layout actually do make the questionnaire easy to follow and to fill out, see if a few of your detail-oriented colleagues can complete the survey correctly. Incorporate their feedback if appropriate, and then ask a few volunteers who are not part of the survey research world to do the same.

Other Strategies for Reducing Nonresponse Error

Other techniques beyond reminders and incentives and those noted above have been shown to improve response rates. Some have been associated with consistent improvements, whereas some have shown improvements only in some circumstances.

Length of the Questionnaire

It almost goes without saying that you are likely to get a better response rate with a shorter questionnaire than with a longer one (Smith, Olah, Hansen, & Cumbo, 2003). That said, there are no clear demarcation points for length. It is not the case that a 12-page questionnaire will generate a decent response rate but a 13-page questionnaire will not. There has been a fair amount of research on this issue, but the results are muddled because of several confounding factors (Berdie, 1973; Burchell & Marsh, 1992; Champion & Sear, 1969; Childers & Ferrell, 1979; Lockhart, 1991; Mason et al., 1961; Roscoe, Lang, & Sheth, 1975; Scott, 1961).

Part of the explanation for these contradictory findings is the different meanings of "length" of a questionnaire used in the research study. Is length determined by the number of questions, the number of pages, or some combination of the two? For example, 30 questions on three pages may seem different from 30 questions on six pages. Another confound is that different-length questionnaires may be perceived differently by respondents in terms of interest levels or in terms of importance. Longer questionnaires may actually be seen as more interesting or more important because they can impart a fuller picture of a topic than a more cursory version. Even within one methodological study to test the effects of varying questionnaire length, it is hard to "hold constant" other factors that may play a role in response rates. Many studies that try hard to control for these issues wind up comparing different-length questionnaires that are actually not so different. For example, Adams and Gale (1982) compared surveys with one page versus three pages versus five pages. They found no difference in response rates between one- and three-page surveys but did find a lower response rate for five-page surveys.

Another limitation on drawing conclusions from findings on a series of studies are differences in topics covered, samples, reminder procedures, and so on. In an ambitious review covering 98 methodological studies, Heberlein and Baumgartner (1978) were unable to document any zero-order correlation between length measures and overall responses. However, a more recent review by Edwards et al. (2005) of randomized clinical trials did find a significant effect for length of the survey.

What is clear from this research is that length by itself is not the sole determining factor driving response rates. Whatever the length of a questionnaire, other design factors can influence whether a good response rate is obtained or not. However, in general, it makes sense that shorter questionnaires will on average do better than substantially longer versions. To put this statement in its proper context, however, our recent work site study—which included reminders and incentives—used a 24-page survey and generated an average response rate of 71% across all 16 work sites.

The real challenge for the researcher is to design a questionnaire that *efficiently* asks about all the elements that are important to the study. In particular, steer clear of questions that seem off the topic or that are overly redundant. Avoid long sequences of questions that try to measure very minor differences in issues: For example, it is not a good idea to first ask about the length of time the respondent had to wait in a doctor's waiting room; then, ask how long he or she had to wait in the examining room before the doctor came in; later ask how long the wait was overall; and finally, prolonging the agony, ask how satisfied the respondent was with the waiting time (Helgeson et al., 2002).

The Personal Approach

Trying to personalize the respondent letter may improve response rates. This can be achieved by putting the respondent's name in the salutation (as opposed to a more generic greeting, such as "Dear Boston resident") or through the use of personally signed letters. However, neither procedure has consistently shown benefits for response rates (Andreasen, 1970; Carpenter, 1975; Dillman & Frey, 1974; Edwards et al., 2002; Frazier & Bird, 1958; Gendall, 2005; Houston & Jefferson, 1975; Kawash & Aleamoni, 1971; Kerin & Peterson, 1977; Kimball, 1961; Rucker, Hughes, Thompson, Harrison, & Vanderllp, 1984; Simon, 1967; Weilbacher & Walsh, 1952). Some authors have commented that personalizing the letters may have just the opposite effect of reducing response rates, because it calls attention to the fact that the researcher knows the respondent's name.

Giving a Deadline

By giving respondents a deadline they will try harder to return the questionnaire, rather than putting it aside, meaning to get to it later and then forgetting it. The use of a deadline gets a little complicated, however, when you are also using reminders. It is not a good idea to set 2 weeks from now as the deadline for responding, and then send the respondent a reminder at that time saying, "Please respond—we are giving you 2 more weeks." On the other hand, giving a deadline of 8 weeks in the future hardly serves a motivating purpose.

Research, however, does not show any particular advantage in final response rates by using deadlines. What the research does show is that the returns come in a little faster (Edwards et al., 2002; Futrell & Hise, 1982; Henley, 1976; Kanuk & Berenson, 1975; Linsky, 1975; Nevin & Ford, 1976; Roberts, McCrory, & Forthofer, 1978; Vocino, 1977). Consider using soft deadlines that also incorporate the information about subsequent reminders. For instance, "Please try to respond within the next week, so we will not have to send you any reminders" (Green, 1996).

Managing the Survey Development and Implementation Process

Even though the mail survey process removes the need to manage a staff of interviewers who need to be hired, trained, and supervised, it is still important to manage the mail survey process. Two areas of management require particular attention: the design of a realistic schedule and the incorporation of a quality control system.

The Schedule

Preparing a written schedule will help more effectively manage the mail survey process. The schedule allows you to appreciate how the various parts of the mail survey study must fit together like a jigsaw puzzle for the project to roll out in a timely fashion. By having a schedule, you can anticipate milestones and their inherent challenges so that you are not overly rushed to get particular steps accomplished.

In developing a schedule, you will find that several independent processes must, at various points, merge to create a high-quality mail survey study. These include the sampling process, the development of the questionnaire, the development of supplemental materials, the print production of the questionnaire, envelopes and other collateral materials, the data collection period, and the coding and data entry process.

There are different approaches to constructing a schedule. Some like to start at the end—when the results are due—and work backward toward the start date. This assumes that the time window to conduct the study is fixed and that time is to be allocated among the various phases of the study within this defined period. As study components are allotted portions of time, those with hard and fast time requirements are entered into the schedule first. For example, we know that the data collection period is predetermined after you decide how many reminders you are going to send out and exactly how long you are going to wait between reminders. If you choose to have an original mailing and then three reminders with 2 weeks between mailings, then the data collection process will take 8 to 9 weeks.

How long it will take a printer to produce copies of your surveys can be readily identified. This time period runs from the day you supply the final copy through checks of the print proof to the day the surveys are delivered to you.

By using these landmarks to build your schedule, you can then allocate the remaining time to the remaining components and phases of the study. Inevitably, in some phases you become pressed for time. You may discover that you have only 2 weeks to develop your questionnaire, or you may find that you have only a 3-week period after the last questionnaire arrives to code, analyze, and write up your report. One way to deal with this time crunch is to overlap various functions in your schedule as illustrated in Figure 15.1. For instance, you can construct your sample while you are developing your questionnaire, or you can begin the coding and data entry process even while questionnaires are still coming in.

An alternate way to construct a schedule is to start at the beginning of the project and allocate time to various phases based on your estimates of the time you will need. Again, for some phases you can be relatively certain about how much time will be required; for others you will need to make some educated guesses. With more experience, you will become more proficient at estimating how long each phase may require.

It can be very anxiety provoking to see a project getting off a “carved in stone” schedule. Instead, consider the schedule to be a dynamic list, understanding that when something changes, it may affect many subsequent dates that you have outlined and therefore require adjustment. For instance, the printer might not come through with the questionnaire when promised, or a particular section of troublesome questions may need to be pretested one more time. As you put the schedule together, allocate some time within it for unexpected crises or slippage.

Quality Controls

Checking the quality of work from your mailing team is also important. There are many areas to consider. Everything that is word processed must be sent through a spell check program with every change reviewed before acceptance. All materials must be carefully proofread before they are sent to the printer. Special attention should be paid to contact information and telephone numbers, return addresses

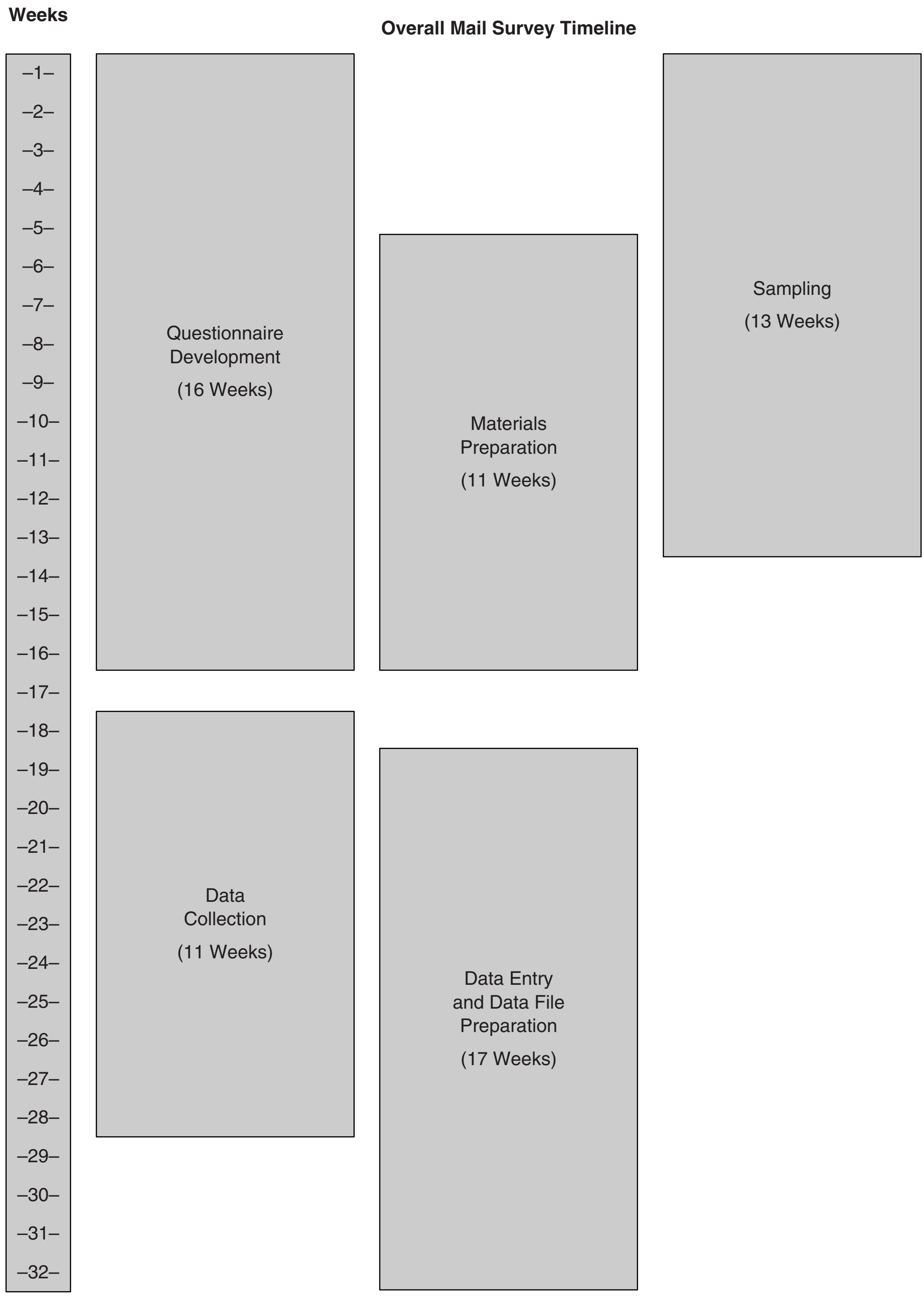

Figure 15.1 A General Model of Structured Conceptualization

SOURCE: Adapted from *Mail Surveys: Improving the Quality,* by T. W. Mangione, 1995. With permission of SAGE.

and postal indicia, and consistency in punctuation and layout. Ideally, proofreading should be done by at least two individuals: Someone who is familiar with the project and someone who is not involved in the study on a day-to-day basis. Above all, be extremely careful about last-minute changes; sometimes in the rush to revise something, new errors are created.

The core tasks for bringing out a mail survey are "stuffing" the envelopes and putting on mailing labels. Often, this phase has relatively simple steps: insert a letter, a numbered questionnaire and stamped return envelope in an envelope, put a mailing label on the envelope, seal it, affix postage, and mail. However, even with a straightforward process, things can, and do, go wrong. Someone can forget to insert a cover letter or may incorrectly number or forget to number a questionnaire. The wrong labels could go with the wrong questionnaires or they might be put on crooked. The postage could be insufficient or missing; the envelopes sent without being sealed or with the seal not firmly glued. Assume that if something can go wrong, it will go wrong on occasion. All these nightmares have happened at one point or another on our projects, even though we were trying to be diligent.

If the study is more complex then even more things can go wrong. As the mailing process requires more steps and more people to carry them out, there is much more room for things to go wrong. One way to help ensure the ultimate quality of your product is to analyze the work flow of the questionnaire mailing assembly process. As you do so, think about mistakes that could be made, then design processes in a way that minimizes the potential for mistakes and maximizes your ability to monitor the work of others.

Surveys in Cyberspace

Many of us think back to the years before the dawn of the Information Age and fondly recall the quaint ways we reviewed the literature (went to the library), wrote proposals (stocked up on correction fluid), bought lunch (slipped out for pizza), and stayed in touch with Aunt Martha (rummaged for a stamp). Electronic technologies have transformed our lives to the extent that these and so many other activities now can be accomplished from the comfort of our offices. When carried out electronically, each of these involves completing forms or questionnaires of one kind or another. Furthermore, each transaction is dispatched into the ether with the sender's faith that information or a product will be returned or that key details will be securely recorded in a database.

It is astonishing to consider how these new developments in technology will make things easier as we go forward. This is especially true as we peer over the leading edge of electronic survey development and administration. With programmers and Web design professionals on our team, we can use e-mail to contact potential subjects, explain the study, invite them to participate, and automatically send reminders. We can embed within that invitation and later reminders a direct link to the Web site where the survey can be accessed. We can attach a unique code to the invitation so that respondents can complete the survey only once. Or, if the research is related to something the respondent "registers" for online, we can help the respondent choose a unique user identification and password.

Our questionnaire can be delivered instantaneously with no postal costs; large numbers of recipients can complete it and submit their responses on the spot (Tse, 1998). Even more exciting to research assistants everywhere, we can "help" respondents fill out the survey correctly by automatically "skipping" to the next appropriate question; by insisting (nicely) that some responses are required; by ensuring that "check one only" instructions are never violated; and by making sure answers are logically consistent (e.g., not allowing someone to say they were born in the current year or that they started smoking before they were born).

At some point in the future, nearly all mail surveys may be conducted electronically rather than by "snail mail." However, that future has not quite arrived. While good programming can reduce item response and item nonresponse errors to a great extent, surveys administered electronically are at least as vulnerable to sample bias and nonresponse error as are their hard-copy cousins (Tse et al., 1995). While the promise is great, the reality for the moment is that an e-mail/Web survey is a sensible choice in some fairly limited circumstances (Dillman & Bowker, 2001).

Of course, each mail survey must be sent somewhere, and so potential subjects' mailing addresses must be known to the research team. When using e-mail to invite participants to the study, one must have a current list of e-mail addresses. For the general public, reliable lists of e-mail addresses do not exist. Right now, it is possible to obtain e-mail lists for affiliates of a particular institution (e.g., a school or company) if the institution is interested in collecting the data and willing to make them available to the research team. However, even for institution-based surveys, some recipients may not use the organization's e-mail system but use alternative e-mail systems; some may not check their e-mails very often or at all. This can be especially true when people have multiple affiliations and use one e-mail system as their primary one and never or hardly ever check the others.

Some researchers try to overcome the problem of sample bias by disseminating notification of the study by standard mail and including the address of the Web site where the survey instrument can be found in the cover letter. This is not a bad solution by any means; however, it does presume that all who fall in the sample have ready access to the Internet. (Many households do of course, but not all by a long stretch (Ranchhod & Zhou, 2001).

However, even if your study design solves the sample access problems, the traditional, major problem with mail surveys, nonresponse error, is waiting in the wings (Couper, 2001; Dillman & Bowker, 2001; Kaplowitz, Hadlock, & Levine, 2004; Sills & Song, 2002; Tourangeau, 2004; Tse, 1998). E-mail and Web-based surveys make it difficult to implement two critical procedures discussed earlier that ensure good response rates—reminders and up-front incentives. The reminder problem is twofold. First, will the respondent even open the e-mail to read the reminder? As more and more spam saturates the Internet, many of us purposely ignore e-mails that do not come from familiar sources. Plus many Internet providers or institutions use sophisticated spam detectors and filters to block the delivery of "blast" e-mails (the equivalent of bulk postal mail) or those sent from "unknown" sources.

Second, how often should we send our reminders? Since e-mails are easy to produce (no envelopes to stuff!) and cost nothing to send, it can be tempting to send too many, thereby making the respondent feel harassed. Plus, sending several reminders to people who may have already completed the questionnaire will make them very angry indeed. We suggest sending reminders only to those who have not yet responded. Since there is no need to take into account the time it takes for the postal service to deliver your letter and return a completed questionnaire, timing e-mail reminders at intervals of about one week seems appropriate.

It is very difficult to deliver "token" up-front incentives by electronic means and, therefore, most Web-based surveys, if they offer incentives at all, are likely to frame them as a promised reward (where the respondent may need to provide a mailing address—another sticking point, perhaps). E-mailed gift certificates available from many online merchants can be presented up-front, but this attractive option comes with a high price tag: minimum denominations range from $5 to $15 or more, a potential budget breaker. Of late, the Internet has become a distribution medium for traditional "Cents-" and "Dollars-Off" coupons which the customers can print and redeem online or at specific retail locations. However, safeguards against coupon manipulation (counterfeiting, changing the value or expiration date), coupon reuse, unauthorized use, and customer privacy will have to be standardized and widely available before these systems can be trusted to distribute incentives. One solution that currently exists is to make the first contact with a potential responder via regular mail; then, of course, you can include the up-front reward in the mailer.

A final point of concern that may contribute to non-response error is that many people are justifiably concerned about how personal information relayed electronically is—and often is not—safeguarded. For example, promises of confidentiality can be viewed with suspicion because of the relative ease of forwarding information via e-mail (e.g., to a person's supervisor).Your pledge to maintain respondents' anonymity or confidentiality must be buttressed by reliable security controls over both electronic and human resources to protect data from hackers, viruses and threats to privacy. Be sure to detail these measures and policies, but realize that your descriptions may be too technical, may be ignored, or simply may not ease some skeptical subjects' concerns.

In sum, the adoption of electronic methods for survey administration holds great promise, but we are not there yet. When the technology does arrive in full force, however, all of the issues we have discussed about improving the quality of "mail" surveys will still be relevant.

Summary

We have focused in turn on the various components and phases of the mail survey process and have tried to give you an in-depth understanding of the unique issues, potential hazards, and procedures to follow. However, in the real world, we rarely have the luxury of conducting an "ideal" project—one where time and money are no object and where quality can be maximized at each decision point. Instead, each project is a series of trade-offs and balancing efforts that we perform in an effort to produce an optimum combination of decisions that produce the best-quality result.

This process of trying to achieve an optimum balance is called a total survey design approach. To review quickly, all mail surveys should include the following basic elements:

- An engaging respondent letter
- Return postage on a return envelope
- At a minimum, a guarantee of confidentiality, with anonymity even better

To ensure good response rates, use one or both of the following:

- Reminders—up to 3 reminders spaced at 2-week intervals
- Prepaid incentives—usually a small amount of money ($1 to $10)

In addition, to maximize response rates, use as many of these procedures as possible:

- Keep your questionnaire modest in length.
- Work extra hard to make your instructions clear and the questionnaire's visual presentation attractive.
- Use an attractive commemorative stamp on the outgoing envelope and/or the return envelope.
- Prenotify respondents of your survey.
- Use letterhead that identifies your institutional sponsorship.
- Personalize the salutation or the signature.
- Mention a soft deadline in your respondent letter.

The thrill of opening your mailbox to an avalanche of returns will be exceeded only by the satisfaction of knowing that you are conducting high-quality research and are making an important contribution to your field of interest. Good luck to you!

Discussion Questions

1. Discuss types of studies or populations for which a mailed, self-administered survey would *not* be the best choice.

2. What are the differences in manpower levels and the variations in those levels for a mailed survey compared with a telephone survey?

3. Discuss how a data collection effort would proceed if it combined in-person contact and mailed, self-administered surveys.

4. Discuss how a data collection effort would proceed if it combined the telephone and the mails.

5. What kinds of question formats do *not* work well in mailed surveys, and how would you get around this problem if you really wanted data collected in that manner?

6. In addition to a good respondent letter, a postage-paid return envelope, reminders, and an up-front incentive, what would be your choice for two other mechanisms to improve the quality of the survey effort?

7. Name a few groups in which an e-mail/Web-based survey might be expected to succeed.

8. What are the fatal problems with conducting e-mail/Web-based surveys in the general population? What are some of the ways to overcome such problems?

Exercises

Exercise 1

Describe your data collection procedures and develop a detailed timeline for a 15-page mail survey project with a sample size of 2,000 people from a funder that will not allow monetary incentives to be used but still demands a high response rate. Include in the timeline questionnaire development, sampling details of the data collection process, data entry, and data analysis.

Exercise 2

How would your procedures and timeline change (if at all) if the funder for the survey described in Exercise 1, would allow a $10 monetary incentive to be used?

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CHAPTER 16

Methods for Sampling and Interviewing in Telephone Surveys

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When and Why Use a Telephone Survey?

Telephone survey methods have undergone serious methodological development only in the past 30 years. Prior to that time, the penetration (coverage) of households with telephones in the United States, Europe, and elsewhere had been too low to justify use of the telephone as a representative survey sampling mode. However, by the 1980s, household telephone coverage in the United States well exceeded 90%, and telephone surveying had become commonplace. Nonetheless, even as of 2008, there remained low-income geographic areas—both inner city and rural—in which telephone coverage in the United States was below 90%. In Europe, telephone coverage has increased to 97% of all households, with two thirds having both a wired (fixed) line and mobile service (IPSOS-INRA, 2004).

Why did telephone surveying gain prominence as a means of providing accurate measures on various topics of interest? Since the late 1980s, the telephone has been the sampling and data collection mode of preference for gathering survey data of the public in the United States. This occurred for three primary reasons: (1) the data gathered via well-conceived and well-executed telephone surveys was shown to be reliable and valid (see de Leeuw & van der Zouwen, 1988; Groves, 1989); (2) telephone survey data could be gathered much more quickly than in-person or mail survey data; and (3) telephone surveys were far less costly than in-person surveying and fairly close in cost to high quality mail surveys that achieved similar response rates.

However, a series of telecommunication-related behavioral trends and government policies in the United States since the mid-1990s and declining response rates have begun to call into question whether telephone surveys of the American public will remain representative in the coming decades. In particular, the growing movement from landline (wired) telephone service to cell phone (wireless) service is threatening the validity of traditional telephone surveying in the United States; (as of 2008, it is estimated that less than 80% of households could be reached via landline telephone, and this proportion is expected to continue to drop during the next 5 years). These telecomm factors are not likely to be of concern in Europe and other countries, raising the future prospect of distinct national differences in how the telephone can be used for representative sampling of the public.

The Advantages and Disadvantages of the Telephone Survey Mode

Advantages

Although many fail to recognize or acknowledge it, an important advantage of telephone surveying over other modes of survey data collection is the opportunity it provides for quality control over the entire data collection process. This includes sampling, respondent selection, administering a questionnaire, and data entry. It is this *quality control advantage* that recommended the telephone as the preferred mode for surveying in the past three decades, providing there were no overriding concerns that rule against it. Unfortunately, too often, researchers have not instituted the quality control procedures that make this potential advantage a reality (see Steve, Burks, Lavrakas, Brown, & Hoover, 2008).

A second major advantage is its cost-efficiency. Telephone surveys can collect data far more efficiently than in-person interviewing. Groves (1989) estimated that individual questionnaire items administered via telephone take 10% to 20% less time than the same items administered in person. And although telephone surveys are typically more expensive than mail and Web surveys, their potential advantages for addressing total survey error (TSE) factors often outweigh this cost disadvantage.

A third major advantage is the speed with which data can be gathered and processed. In less than a week, a group of skilled interviewers can gather high-quality opinion data via telephone that might take a month or more using in-person interviews. An even longer period often would be needed to conduct a high-quality mail survey on the same topic with the same sample size, given the necessity of follow-up mailings to increase typically low response rates to the first mailing. High-quality survey data could not be gathered via mail or in-person surveys, within this time frame, for the same costs as the telephone survey, and researchers could not be confident about data accuracy gathered via a Web survey because the Internet—unlike the telephone—at present, cannot be used to gather data from a fully representative sample of the citizenry without a great deal of additional effort and cost to initially recruit the sample via another mode of contact, such as telephone and/or in-person recruitment.

Disadvantages

A major disadvantage of telephone surveys—even when well executed—is the limitations they place on the complexity and length of the interview. Unlike the dynamics of face-to-face interviewing, the average respondent often finds it tiresome to be kept on the telephone for longer than 20 minutes, especially when the topic does not interest her or him. In contrast, personal interviewers do not seem to notice much respondent fatigue even with interviews that last 30 minutes or longer. Mail and Web surveys also do not suffer as much from this disadvantage as those questionnaires often can be completed at a respondent's leisure over multiple sessions. Similarly, complicated questions, especially those that require the respondent to see or read something, heretofore have been impossible to display via the telephone. With the advent of video telecommunication technology via the Web and telephones, this limitation should diminish.

Other traditional concerns about telephone surveys include potential coverage error that may occur. For example, not everyone in the United States lives in a household with telephone service, and among those who do, not every demographic group is equally willing or can be reached for interviewing via telephone. According to the most recently available Federal Communications Commission statistics, in 2004 approximately 6% of the U.S. public lived in a home without any telephone—with Arizona (8%), Arkansas (11%), the District of Columbia (8%), Georgia (9%), Illinois (10%), Indiana (8%), Kentucky (9%), Louisiana (9%), Mississippi (10%), New Mexico (9%), Oklahoma (9%), and Texas (8%) having the highest rates of noncoverage. In contrast, regional coverage in European Union countries was not as problematic, with only Portugal at 90% coverage and Belgium at 94% coverage, having more than 5 in 100 households without a telephone line (IPSOS-INRA, 2004).

Furthermore, currently there are no scientifically accepted ways to incorporate cell phone and Voice-Over-Internet (VoIP) telephone numbers into the traditional sampling methods used to survey the U.S. public via telephone (see Brick et al., 2007; Brick, Dipko, Presser, Tucker, & Yuan, 2006; www.nielsenmedia.com/cellphone summit/cellphone.html). By the end of 2007, an estimated 20% of U.S. households had only cell phone coverage (see Blumberg, Luke, & Cynamon, 2006; Tucker, Brick, & Meekins, 2007). Thus, landline telephone surveys in the United States are at a disadvantage in reaching certain segments of the general population such as renters and adults younger than 25 years of age. For other countries, these problems do not exist because the business model used to charge their customers does not hamper respondents' willingness to be interviewed on their wireless phone—as it often does in the United States—nor are there as many restrictive federal telecommunications policies that currently hamper survey researchers in the United States from surveying respondents reached on a cell phone.

In addition, since the advent of *number portability*[1] in the United States in 2004, researchers can no longer be certain *where* (in a geographical sense) a respondent has been reached when contacted on a telephone. Depending on the extent to which the people continue to exercise their right to *port* their telephone number(s)—and in 2005, approximately 3 million already had done so—and depending on how far they move from the original geographic area in which they were assigned their phone number, telephone surveys may suffer the considerable burden of having to conduct explicit geographic screening of respondents to determine whether the respondent lives within the geopolitical area being surveyed (see Lavrakas, 2004). If this were not done, then serious Errors of Commission (false positives) and Errors of Omission (false negatives) may result from interviewing respondents who are geographically ineligible for the survey. Furthermore, geographic screening would lead to increase in nonresponse. These problems do not exist for researchers outside the United States.

Total Survey Error Perspective

In addition to considerations of sampling error, survey researcher should attend to the potential effects of coverage error, nonresponse error, and measurement error. Together, all these potential sources of variance and bias constitute TSE (Groves, 1989; Lavrakas, 1996). Thus, researchers should consider each element of TSE separately when planning, implementing, and interpreting a telephone survey. Concern about a survey's total error will lead the researcher to deploy methods to (a) reduce the likely sources of error and/or (b) measure the nature and size of potential errors. Ultimately, it remains the researcher's responsibility to allocate the resources available to conduct the survey so as to achieve the best-quality data possible within the finite budget. Often, this requires many difficult cost-benefit tradeoffs, such as whether to use more resources to hire and train high-quality interviewers or, instead, make additional callbacks to the hardest-to-reach respondents, or to deploy a "refusal conversion" process, since a researcher never will have enough resources to address all potential sources of survey error.

Noncoverage

As it applies to telephone surveys, noncoverage is the "gap" that often exists between the sampling frame (the set of telephone numbers from which a sample is drawn) and the larger population the survey is meant to represent. To the extent the group "covered" by the sampling frame differs in nonignorable ways on variables of interest from the group not included in the sampling frame, the survey will have coverage biases. For example, all household telephone surveys in the United States using random-digit dialing (RDD) landline sampling frames miss households and persons without telephones and persons with only cell phone service. Thus, RDD landline surveys have the potential for coverage error if researchers infer findings to the general public about issues that are correlated with whether or not someone can be surveyed via a landline telephone. Worldwide, not having a telephone is related to very low income, low education, rural residency, younger ages of household heads, and minority racial status. In the United States, having only wireless phone service is related to many of these same demographic factors and to being a renter. Thus, there will be some level of nonnegligible coverage errors in many telephone surveys that sample only households with wired telephone service.

Another source of potential coverage error is multiple-line households. Approximately one in six households with wired lines in the United States as of 2007 had more than one line, whereas more than half of households in many European countries have multiple lines, when considering both wired lines and mobile lines. Whenever an RDD or a list-based frame of household telephone numbers is used, residences with more than one telephone number have a greater probability of being sampled than those with only one number. Thus, researchers conducting a survey via telephone should ask respondents about the number of different telephone numbers in the household on which the respondent could have been reached and then take this into account when conducting post hoc statistical weighting adjustments for unequal probabilities of selection caused by the multiple phone lines within certain households.

Nonresponse

Nonresponse error in a telephone survey occurs when people who are sampled, but not interviewed, differ as a group in some nonnegligible way from those who are successfully interviewed. Nonresponse in telephone surveys is due primarily to: (a) failure to contact sampled respondents, (b) sampled respondents who refuse to participate, and (c) sampled respondents who have language or health problems. Since the early 1990s, response rates in telephone surveys of the United States and European publics have noticeably and continuously declined each year, albeit slowly (see Curtin, Presser, & Singer, 2005; de Heer, 1999). This is due to a combination of the public's increasing unwillingness to participate in telephone surveys because of busy lifestyles, the problems caused by telemarketers and the public's behavioral responses to avoid such nuisance calls, and the increase in telecommunications system challenges to reaching a sampled respondent within a fixed length field period, especially within the United States.

In the United States, the implementation of the Do Not Call List (DNCL) in October 2003, appears to have significantly reduced the telemarketing nuisance call problem, but it is too soon to know with confidence what long-run effect this will have on response rates in legitimate telephone surveys. Some evidence to date is promising in that those listed on the DNCL appear more likely to participate when subsequently sampled for a telephone survey than those who are not. But other findings are troubling, in that a large minority of the U.S. public would like to have the DNCL restrictions extended to opinion polls and other types of research surveys (Lavrakas, 2004).

One of the most effective ways to counter nonresponse in a telephone survey is to make an advance contact via mail with the sampled household before contacting them via telephone (see de Leeuw, Joop, Korendijk, Mulders, & Callegaro, 2005). The most effective type of advance-mailed contact is a polite, informative, and persuasive letter that is accompanied by a token cash incentive. Lavrakas and Shuttles (2004) reported experimental findings in very large national surveys of gains in RDD response rates of 10 percentage points with as little as $2 mailed in advance of phone contact. Of course, this advance mail treatment requires the ability to match sampled telephone numbers with accurate mailing addresses, which is possible in approximately 60% to 70% of the time for many U.S. RDD samples if researchers use multiple vendors for the matching process.

Special training for interviewers is a different approach to reducing the problem of refusals in telephone surveys. Groves and others (e.g., Groves & McGonagle, 2001; Shuttles, Welch, Hoover, & Lavrakas, 2002) have made advances using carefully controlled experiments in testing a theory-based “Refusal Avoidance” interviewer training curriculum that includes the following:

(a) focus groups with top performing interviewers that identify the actual verbiage they hear from refusers and then map persuasive replies that these interviewers use to try to convert reluctant respondents to each reason for refusing;

(b) communication discourse techniques for extending the time that reluctant respondents stay on the telephone before hanging up on the interviewer, for example, posing a conversational question back to the respondent to engage her or him in a two-way dialogue;[2] and

(c) correctly and rapidly identifying the reasons why the respondent is refusing and delivering relevant persuasive verbiage to counter them.

The results of these experiments have been mixed with some studies showing upwards of a 10 percentage point gain in cooperation by those interviewers receiving this training and other studies showing no effects whatsoever.

In terms of reducing nonresponse associated with noncontacts in telephone surveys, the basic technique is to make many callbacks, scheduled at various times of the day and days of the week, over as long a field period as possible. That is, the more callbacks made and the longer the field period, the higher will be the contact rate in RDD surveys, all other factors being equal. This is problematical for many surveys, especially those conducted for news purposes because newsworthiness often exists only for a brief moment in time. In these instances, the only choices a researcher faces is to exercise care in considering the effect of noncontact-related nonresponse and to weight the data by gathering information in the survey about the propensity of the respondent to be at home over a longer field period (e.g., the past week), with those least likely to be at home being assigned weights greater than 1.0 and those most likely to be at home being assigned weights less than 1.0.

In considering how to handle callbacks during any finite field period, not all RDD telephone numbers merit equal calling effort since many of them are nonworking or otherwise nonresidential, yet are not reliably detected as such by autodialers or live interviewers. In the United States, this is due in part to the inconsistent manner in which local telephone companies handle such nonresidential numbers. Using data from several extremely large national RDD surveys, Stec, Lavrakas, and Shuttles (2005) reported that U.S. telephone numbers that have a repeated Busy-Signal outcome (>5 times) or a repeated Ring-No-Answer outcome (>10 times) are very unlikely to ever produce an interview with as many as 30 call attempts. On the other hand, when encountering a residential answering machine, persistence often appears to pay off, regardless of how many times such an outcome results (Piazza, 1993). Leaving messages on answering machines is generally thought to be a good practice to increase subsequent contact rates, but the literature is inconclusive on the issues of what should be said in the message and when, and how often, such messages should be left. Leaving too many messages is assumed to be more harmful than helpful in eventually gaining cooperation from a household, but exactly how many is "too many" remains uncertain.

With the growth of the Caller ID and Privacy Manager technologies, it is becoming harder to get people to pick up their telephone when they receive a call from an unknown source. Tuckel and O'Neill (2002) and the Pew Research Center (2004) reported that more than half of U.S. households have Caller ID capability. Leverage-saliency theory (Groves, Singer, & Corning, 2000) would suggest that depending on what information is displayed about the caller on the household's telephone equipment, the response propensity to answer the incoming call will be affected either positively or negatively. Trussell and Lavrakas (2005) reported the results of two very large national experiments with RDD samples in which displaying the name "Nielsen Ratings" (a generally well-known and positively valued brand in the United States) raised telephone survey response rates by more than 2 percentage points, although these gains were not due solely to increasing the contact rate. But other results in these experiments suggested that caution should be exercised in displaying something on Caller ID too many times in a field period, if a telephone survey is using a large number of callbacks (e.g., >10). Callegaro, McCutcheon, and Ludwig (2006) also found mixed results with Caller ID. Depending on the target population, in some cases, a Caller ID display lowered the response rate, whereas with an RDD of the general population, sending out the name of the survey organization on Caller ID increased the response rate by 3 percentage points.

Measurement

Not all data that interviewers record during an interview are accurate measures of the attitudes, behaviors, and demographics of interest. These inaccuracies, in the forms of both bias and variance, may be due to errors associated with (a) the questionnaire and/or (b) the interviewers and/or (c) the respondents (see Biemer, Groves, Lyberg, Mathiowetz, & Sudman, 1991). In thinking about these potential sources of measurement error, the researchers should consider ways that the nature and size of such errors might be measured so that the researcher can consider post hoc adjustments to the "raw data" gathered from respondents by interviewers. The best way to base such adjustments on sound empirical evidence is to build experiments into the telephone questionnaire. This is especially important whenever a researcher is using questions that have not been used in previous surveys, and thus, their wording is not validated by solid experience. In this case, a researcher should use an experimental design to test different wordings, even if only a small part of the sample is exposed to alternative wordings.

Cost-Benefit Trade-Offs

Every telephone survey should be viewed as an endeavor with a finite set of resources available. The challenge faced by the researchers is to deploy those resources in the most cost-beneficial way possible, so as to maximize the quality of the data that are gathered. The TSE perspective can guide researchers through a series of choices (trade-offs) that often pit what they know or assume about one source of potential error against what they know or assume about another source of potential error. For novice researchers, these considerations can seem forbidding or even overwhelming. When faced with all the potential threats to a survey's validity, some may throw up their hands and question the value of the entire survey enterprise. To do so, however, is to fail to remember that highly accurate surveys are routinely conducted by researchers who exercise the necessary care.

This chapter serves as an introduction to these considerations as they apply to telephone surveys. This discussion of TSE is meant to alert future researchers to the many challenges they face in conducting telephone surveys that will be accurate enough for the purposes for which they are intended. The message to the novice should be clear: Planning, implementing, and interpreting a survey that is likely to be accurate is a methodical and time-consuming process, but one well worth the effort.

Computer-Assisted Telephone Interviewing (CATI)

Traditionally, telephone surveys were conducted by interviewers asking questions read from paper questionnaires and then recording the answers on the questionnaires with pencils. By the late 1990s, this approach was almost entirely replaced by CATI whereby the interviewer is guided through the introductory script and questionnaire by a computer program. The CATI software is also used to control the sample of telephone numbers used during the field period (see Hansen, 2008).

In CATI, human interviewers work at computer workstations that control the administration of the questionnaire and most often control the sampling process. CATI software can control the distribution of the *sampling pool* (the set of telephone numbers dialed by interviewers in a given survey),[3] even to the point of dialing the appropriate telephone number for a "ready" interviewer, as in the case of predictive dialers. CATI has the potential to provide many types of statistics on interviewer productivity to aid supervisory staff (see Tarnai & Moore, 2008). In presenting the questionnaire to the interviewer, CATI makes complicated skip patterns (question-sequence contingencies) very easy to use, and can randomly order sequences of items and incorporate previous answers into the wording of subsequent items. Of course, CATI also provides for simultaneous entry of the answers into a computer database.

Despite what was expected of CATI in its early years, it has not been found generally to lower survey costs or to reduce the length of the field period (see Lavrakas, 1991, 1996), because with CATI, the interviewer typically has less control over the speed at which the questionnaire is administered. CATI is not a panacea, but rather a tool that, when properly implemented on appropriate studies, has the potential to improve the quality of resulting data by reducing TSE and/or more readily, producing data that allow a researcher to conduct post hoc investigations of possible error sources.

Proper implementation of CATI calls for much more than merely purchasing computers, other hardware, and software. It also requires a proper channeling of the physical and social environment within a survey facility (see Hansen, 2008; Kelly, Link, Petty, Hobson, & Cagney, 2008). Ideally, the use of CATI should be based on a survey organization's desire to reduce TSE. CATI offers great promise for those concerned with minimizing TSE, but it should never be viewed as a "technological fix" that replaces the need for intensive human quality control procedures. Just the opposite is true: When properly implemented, CATI allows for an increase in the quality control that humans can impose on the telephone survey process.

Steps and Considerations in Conducting a Telephone Survey

Anyone planning a telephone survey should develop a detailed administrative plan that lays out all the tasks that must be accomplished and identifies the personnel to be involved in each task (see Frey, 1989; Hansen, 2008; Kelly et al., 2008; Lyberg, 1988; Steve et al., 2008; Tarnai & Moore, 2008). The following are the steps an opinion researcher typically needs to perform in order to conduct a high-quality telephone survey:

1. Decide on a sampling design, including identification of the sampling frame from which sampled units will be selected and the method of respondent selection within a sampling unit, if the sampling unit is not also the sampling element (discussed later in more detail). In many telephone surveys, some variation of RDD sampling and some variation of the "last birthday" within-unit respondent selection method will be best to use (see Gaziano, 2005).

2. Choose a method to generate or select the set of telephone numbers that will be used in sampling (sampling pool) from the sampling frame. Create the sampling pool and divide it randomly into replicates to help control the allocation of the numbers that will be dialed during the field period. This is often done by purchasing the sampling pool from one of many commercial vendors.

3. Decide on the length, in days, of the field period, and the "calling rules" that will be used to reach a proper final disposition for all telephone numbers in the sampling pool that are dialed within the field period. Also, decide at what hours of each day and on which days of the week calling will occur. For the calling rules, decide on the maximum number of call attempts per telephone number, how much time should be allowed to elapse before recalling a busy number, and whether or not refusal conversions will be performed. In terms of refusal conversions, decide how much time should elapse before redialing the number while recognizing that

"best practice" is to allow as many days as possible to pass, within the finite constraints of the field period, before redialing the refusing number (discussed later in more detail).

4. Produce a call-record for each telephone number that will be used to track and control its call history during the field period. Most CATI systems that control the processing of a sample have such a feature built in. The information in these call-records—sometimes referred to as "paradata"—is very informative for interviewers to review before making each callback to help prepare themselves for the recontact. The more detailed the information recorded by the previous interviewers who contacted the household, the more prepared an interviewer will be for any subsequent contacts within the household.

5. As the sampling design is being selected, develop and format a draft questionnaire keeping in mind how much time, on average, the questionnaire can afford to take to complete, given the available resources and the purpose and needs of the survey project.

6. Develop a draft introduction and respondent selection sequence and draft "fallback statements" (persuaders) for use by interviewers to help tailor their introduction and help gain cooperation from reluctant sampled respondents (Lavrakas, 1993).

7. Decide whether advance contact will be made with sampled respondents, such as an advance letter, and, if so, whether an incentive will be included in the advance mailing.

8. Pilot test and revise survey procedures and the questionnaire. Pilot testing of all materials and procedures is an extremely important part of any high-quality telephone survey; an adequate pilot test often can be accomplished with as few as 20 to 30 "practice" interviews. As part of the pilot stage, the researcher should hold a debriefing session with the interviewers who participated, the project management team, and (ideally) the survey sponsor, to identify any changes that are needed before the sampling scheme and the respondent selection procedures are finalized, and before final versions of the questionnaire and other survey materials are printed or programmed into CATI.

9. Program the script (introduction, respondent selection method, and questionnaire) into CATI (see House & Nicholls, 1988) or print them onto paper.

10. Hire interviewers and supervisors, and schedule interviewer training and the data collection sessions. When doing a survey in more than one language, it is best from a data accuracy standpoint and response rate standpoint to have interviewers interview in only one language. It is best to use native speakers of a language rather than using bilingual speakers whose primary language is not the one in which they will interview exclusively. The value of this approach is that native speakers also will share cultural similarities with many respondents who speak that language and, thus, will be able to gain cooperation more readily and probe unclear answers more effectively.

11. Train interviewers and supervisors (see Tarnai & Moore, 2008). When doing a survey in more than one language, each group of interviewers should have supervisory personnel whose primary language matches the language they will use to conduct interviews.

12. Conduct fully supervised interviews. Decide what portion, if any, of the interviewing will be monitored (see Steve et al., 2008) and whether any respondents will be called back to validate the completed interviews (see Lavrakas, 1993).

13. Edit/code completed questionnaires. If coding open-end verbatims, devise coding categories, train coders, and monitor their reliability.

14. Assign weights (if any) to correct for unequal probability of selection (e.g., for multiple-telephone-line households; the number of adults in a household; the proportion of time in the past year that the household did not have telephone service), and for deviations in sample demographic statistics (gender, age, race, education, etc.) from known population parameters. In the latter case, adjustments for education are likely to be the most important because most telephone surveys of the general public vastly oversample those with high educational attainment and vastly undersample those with low educational attainment, and many behaviors and attitudes are highly correlated with educational attainment.

15. Perform data analyses and report preparation.

An additional design consideration in any telephone survey should be an explicit decision about whether experiments will be built into the study. When planning their surveys, far too few researchers take advantage of the power of true experiments to test causal relationships in the data being gathered—which often can be done at essentially no additional cost. The common ways that experiments can be used in telephone surveys is to test the effects of (a) various question wording or ordering sequences, (b) different introductory scripts and/or respondent selection methods, and (c) incentives and other treatments to raise response propensities, improve the demographic characteristics of the unweighted sample, improve data quality, and/or reduce nonresponse bias.

Sampling Frames in Telephone Surveys

Prior to choosing a sampling frame, the researcher must choose between the use of a probability sample and the use of a nonprobability sample. As Henry (1990) notes, the great advantage of probability samples is that "the bias and likely error stemming from [their use] can be rigorously examined and estimated; [but this] is not the case for nonprobability samples" (p. 32). As such, only probability samples permit the portion of TSE that is due to sampling variance to be rigorously quantified, as only a probability sample provides every element in the sampling frame a known nonzero chance of selection.

Once these decisions are made, the researcher must make a number of other sampling design decisions. These include explicit identification of the following:

1. *The population of inference,* that is, the group, the setting, and the time to which the findings must generalize: For many telephone surveys, this will be the entire adult population within a specific geopolitical area. For example, in the United States, this might be the entire nation; the 48 contiguous states, some region of the nation (e.g., the South or the West), a state, a large metropolitan area, a county or a combination of counties, a city, a precinct, or even a smaller neighborhood area. Another key consideration in choosing the population of inference is the implications such a decision has on the language(s) in which the survey will be conducted.

2. *The target population,* that is, the finite population that is purportedly surveyed.

3. *The sampling frame,* often in list form, that will operationalize the target population.

In most instances in which the U.S. general public within a geopolitical area is being surveyed, including rare subgroups within the general population, a researcher will need to use an RDD frame.[4] In contrast, in many European countries, RDD sampling is not always necessary to reach a representative sample of the public, as unlike in the United States, nearly all residences have listed telephone numbers (Kuusela, 2003; Taylor, 2003). In these instances, a directory may exist that can be used as the sampling frame. When sampling elites or members of special interest groups via a telephone survey, RDD essentially is never the preferred frame because it is highly inefficient in reaching these types of respondents. Instead, a list frame (e.g., the membership of a professional organization) needs to be acquired (or built) that well covers the target population of interest.

When the RDD frame was first embraced, the Mitofsky-Waksberg approach became the standard methodology, but this proved to be difficult and costly to implement accurately and was rather inefficient. Subsequently, many approaches to list-assisted RDD sampling were devised that were more easily administered, much more efficient, and thus less costly in reaching sampled respondents (Brick, Waksberg, Kulp, & Starer, 1995; Tucker, Lepkowski, & Piekarski, 2002). Nowadays, there are several reputable commercial vendors that supply accurate, efficient, and reasonably priced list-assisted RDD sampling pools to survey the public in just about any geographical area in the United States and in many other countries as well. Thus, it is unusual for a researcher to engage in the manual approach to generate an RDD sampling pool for the target population (see Lavrakas, 1987, 1993). For those conducting cross-national telephone surveys, the work of Kish (1994) and Gabler and Hader (2001) is recommended for guidance in building sampling frames and probability sampling designs that best represent the respective target population in each country.

Reaching the Cell-Phone-Only Population

When surveying the general U.S. public, it is very important to make an explicit decision about whether or not known cell phone telephone exchanges will be included in the sampling frame. This is an extremely thorny issue for which Best

Practices have not as yet been identified by the survey industry, but issues that must be balanced are as follows:

(a) the extent to which those who can be reached only by cell phone have different attitudes and behaviors from those who can be reached via a traditional landline (see Blumberg & Luke, 2007; Callegaro & Poggio, 2004; Keeter, Kennedy, Clark, Tompson, & Mokrzycki, 2007; Vehovar, Belak, Batagelj, & Cikic, 2004);

(b) the size of the final sample of cell phone respondents with whom interviews must be completed and whether this will be restricted to cell phone only respondents or not (see Kennedy, 2007; Lavrakas et al., 2008);

(c) how wireless phone and wired phone exchanges will be mixed in the sampling pool and how respondents reached via a wired line versus those reached via a wireless phone will be weighted at the analysis stage (see Brick, Edwards, & Lee, 2007; Kennedy, 2007; Lavrakas et al., 2008; Link, Battaglia, Frankel, Osborn, & Mokdad, 2007);

(d) how long a questionnaire is reasonable to use with someone reached on their cell phone (see Brick et al., 2007);

(e) the considerable greater costs that sampling U.S. cell phone numbers require, in part, because of the restrictions placed by federal and state regulations on the use of automatic dialing technologies when calling cell phone numbers (see Lavrakas, Shuttles, Steeh, & Fienberg, 2007; Lavrakas et al., 2008); and

(f) how respondents reached on a cell phone will be incented, have their safety protected, and how the accuracy of the responses that they provide will be maximized (see Lavrakas et al., 2007, 2008).

Furthermore, as of 2008, there exists no data in the United States on the percentage of households that are cell phone only at the state, county, or city level (see Lavrakas et al., 2007). As such, researchers who are conducting a subnational telephone survey can at best make only an informed "guesstimate" about the proportion of a mixed landline and cell phone sample should come from each frame and how to weight and integrate the data that are gathered from each type of sample.

Size of the Sampling Pool

A general rule is that the shorter the field period for a telephone survey, the larger the sampling pool of telephone numbers needs to be. Shorter field periods lead to lower response rates, all other things being equal. Thus, for example, surveys that strive to complete 1,000 interviews over a weekend will need a much larger sampling pool than surveys striving to complete the same number of interviews during a field period of a month or longer.

Within-Unit Respondent Selection/Screening Techniques

Some persons unfamiliar with valid telephone survey methods mistakenly assume that the person who initially answers the telephone is always the one who is interviewed. This is almost never the case with any survey designed to gather a representative within-unit sample of the general population. For example, although males and females are born at a near 50:50 rate, the adult population in most urban communities is closer to a 55:45 female/male split. A survey that strives to conduct interviews with a representative sample of an area's adult population must rely on a systematic respondent selection procedure to achieve a valid female/male balance, in part because, on average, a female is more likely than a male to answer the telephone when an interviewer calls. Thus, always interviewing the first person who answers the telephone would lead to an oversampling of females.

Obviously, when sampling is done from a list and the respondent is known by name, respondent selection requires merely that the interviewer ask to speak with that person. But in many instances with list sampling, and with all RDD sampling, the interviewer will not know the name of the person within the household who should be interviewed, unless this has been learned in a previous contact with the household. Therefore, a survey designed to gather estimates of person-level population parameters (as opposed to household-level measures) must employ a systematic selection technique to maximize external validity by lessening the chance of within-unit noncoverage error.

As a rule, interviewers should neither be allowed to interview the first person who answers the telephone nor be allowed to interview anyone who is merely willing to be surveyed. Instead, the interviewer should select one designated respondent in a systematic and unbiased fashion from among all possible eligible respondents within the unit, who meet the the survey's demographic/experiential definition of a respondent (e.g., an adult who is 18 years of age or older).

Respondents can be selected within a sampling unit using a true probability sampling scheme—one that gives every possibly eligible respondent a known and nonzero chance of selection—although researchers will not always need, nor necessarily want, to employ such an approach. For the purposes of most surveys, it is acceptable to use a procedure that systematically balances selection along the lines of both gender and age. Because most sampling units (e.g., households) are quite homogeneous on many other demographic characteristics (e.g., race, education, religion), random sampling of units should provide adequate coverage of the population on these other demographic factors.

During the past 30 years, most of the techniques that have been commonly employed for respondent selection were devised to be minimally intrusive about gathering personal information at the start of the interviewer's contact with the household, while attempting to provide a demographically balanced sample of respondents across an entire survey. Because asking for "sensitive" information before adequate trust has been developed by the interviewer can seriously increase telephone survey refusals, and thus nonresponse, researchers have tried to strike a somewhat difficult balance in their respondent selection techniques, between avoiding coverage error and avoiding nonresponse error.

The Kish Method of Random Selection

The most rigorous respondent selection method that is the accepted standard for in-person interviews, was developed by Kish (1949, 1965). This method can also be used in telephone surveys that require as complete a representation as possible of all eligibles from within sampling units. The Kish method minimizes noncoverage within sampling units compared with other less rigorous selection methods; although, due to its intrusive nature, it may increase refusal/nonresponse rates, especially when used by unskilled interviewers. Of note, whether it will increase nonresponse bias is unknown.

In the Kish selection method, immediately after reading the introductory spiel, the interviewer identifies all eligibles within the sampling unit. In most cases, this means determining all persons living in the household who meet the survey's age criterion. Some researchers prefer to have interviewers identify eligibles in terms of the familial relationships within the household, whereas others have interviewers ask for eligibles' first names or their initials. Either way, it is typical for the interviewer to begin by identifying the household head(s) and then follow by listing other eligibles. After the interviewer has made certain that all eligibles are accounted for, he or she asks for and records the age of each person listed. The interviewer then pauses, briefly, to check that the age of each person listed meets the age requirements of the survey, eliminating any who do not meet the requirements from further consideration. Then the interviewer rank orders all eligibles according to the following traditional rule: oldest male numbered 1, next-oldest male (if there is one) numbered 2, and so on through all males listed, then followed by oldest female, next-oldest female, and so forth. The interviewer next consults one of several versions of a "selection table" to determine which one person should be interviewed for that household. Of course, with Kish, as with any selection method, if there is only one eligible person in the household, then that person automatically becomes the selected respondent. Used across an entire survey, Kish comes very close to providing a truly random within-unit selection of all possible eligible respondents in the units that are sampled.

Birthday Methods for Respondent Selection

In the past three decades, a different and much more streamlined approach for yielding, in theory, a probability selection of respondents within sampling units has been explored (see Gaziano, 2005; Lavrakas, Harpuder, & Stasny, 2000). In using these "birthday" methods, the interviewer asks either for the eligible person within the sampling unit whose birthday was "most recent" (i.e., who had the "last" birthday) or for the eligible who will have the "next birthday." Due to their nonintrusive nature and the heterogeneous within-unit sample they produce, birthday selection methods have been widely embraced in telephone surveys by academic, public sector, and private sector researchers. Because birthday selection methods are neither intrusive nor time-consuming and are easy for interviewers to administer, their appeal is great. In particular, it is thought that nonresponse is lessened by such an easy, nonintrusive approach to respondent selection.

After reading the introductory spiel, the interviewer using a birthday selection method asks for a respondent with wording such as the following: "For this survey, I'd like to speak with the person in your household, 18 years of age or older, who had the last birthday." There is evidence that the birthday methods lead to the correct eligible being interviewed in most, but not all, cases. Evidence also suggests that some of these errors are not random across a sample (see Lavrakas et al., 2000). As such, a prudent strategy when using the birthday method is to randomly assign sampled households to either the "next" or "last" birthday, as it is reasoned that the errors that occur with each technique will balance out across the two.

Hybrid Approach for Respondent Selection

Rizzo, Brick, and Park (2004) reported on a new method for respondent selection that has a solid scientific foundation and a great deal of commonsensical appeal. The new method essentially treats households as one of three types, based on the number of eligibles in the household. The method requires that CATI be used and begins by determining how many eligible persons reside in a household. For example, for surveys in which the only eligibility criterion is being 18 years of age or older—which is the case for the majority of telephone surveys—in order to start the within-unit selection process, all that needs to be asked of the adult household member who is first contacted, is a question along the following lines: *Including yourself, how many people aged 18 or older currently live in this household?*

If there is only one eligible person, then the interview commences with the person already being spoken to. If there are two eligible persons, then a random process is used to alert the interviewer to either proceed to interview the person already being spoken to or to ask for the other eligible person. If three or more eligible people reside in the household, then another selection process, such as the birthday method, is used to choose only one of them to interview.

Approximately 15% of U.S. households have three or more adults residing in them. As such, this within-unit procedure is noninvasive for the vast majority of sampled households. It provides a relatively nonintrusive approach that has not been found to increase nonresponse, provided that skilled interviewers are deploying it, and it entirely eliminates respondent-related error in choosing the "wrong" person in the households with one or two eligibles.

Other Criteria for Respondent Selection

Whenever a telephone survey requires only a certain type of respondent (e.g., women between the ages of 30 and 49 who are college graduates), the researcher will need to employ other respondent selection (or screening) methods. Some surveys, for example, require interviews only with heads of households, or taxpayers, or registered voters. For other surveys, researchers may need to select people who live within a relatively small geographic boundary, or some unique subsample of the general population. For more information about how to sample within households for "head of household," "likely voters," and other subsets of the adult population, including those who live within certain small area boundaries, see Lavrakas (1993, pp. 116–120).

Regardless of the respondent selection method the researcher chooses, the method should be pretested along with the questionnaire in any pilot study that is implemented. This will provide the researcher with a chance to look for evidence that the method and its interaction with the group of interviewers being used and respondents being sampled may be contributing errors of omission or commission.

In sum, respondent selection is a nonissue for any telephone survey in which respondents are sampled by name. However, in a survey that does not sample people by name, the researcher's purpose in using a systematic within-unit respondent selection procedure is to choose one and only one person from within each sampling unit in an unbiased fashion—one that will not contribute to possible coverage error.

Sampling and Coverage in Mixed-Mode Surveys

The past two decades have seen a growing interest in surveys that combine two or more modes of sampling and data collection. The appeal of combining different survey modes—personal, telephone, mail and/or Internet—follows the reasoning that TSE may be reduced if the limitations of one mode are offset by the strengths of another (see Dillman & Tarnai, 1988).

For the researcher who is planning how a sampling pool will be generated, use of a mixed-mode approach may require additional time and resources to assemble the sampling frame(s) but in the end may bring the payoff of much higher coverage of the population. For example, a survey of a specific community that does not conform well with the geographic boundaries of the telephone prefixes that reach the area might employ an address-based sampling frame with multimode data collection approach (see Link, Battaglia, Frankel, Osborn, & Mokdad, 2008). The sampling pool could be a list of all addresses at which the USPS makes deliveries (USPS Delivery Sequence File) within the geography of interest. The researcher then can have these addresses matched with telephone numbers where such a match is possible. For those addresses with a telephone number, the households can be contacted and interviewed via telephone. For those addresses without a telephone number, a mail questionnaire can be sent. For those households in which neither of these modes leads to successful data collection, an in-person interviewer could be sent to the household address. Furthermore, because each of the households in the telephone-matched part sampling pool has a listed address, advance letters to reduce initial nonresponse could be mailed to inform residents that they have been selected for the survey. (Of note, regardless of whether a household is sampled via telephone, mail, or in-person, the Internet can be used as a mode for respondents to complete the questionnaire; see Steve et al., 2007.)

All in all, it remains the responsibility of the researcher to determine how best to use the survey's fixed budget so as to balance the possibility of reducing coverage error by using a mixed-mode approach versus the cost of doing so—that is, whether the potential for gains is "justified when costs and other error considerations are taken into account" (Lepkowski, 1988, p. 98).

Telephone Survey Introductions

The introduction that the interviewer reads on making contact with a potential respondent is critical for the success of the survey. A poorly worded introductory spiel will lead to many refusals and can increase nonresponse to a point that entirely invalidates the survey. There are differing opinions among survey professionals regarding how much information should be given in the introductory spiel, and the research literature does not provide a definitive answer (see Groves & Lyberg, 1988, pp. 202–210). I side with those who believe that the introduction should be reasonably brief, so that the respondent can be actively engaged via the start of the questionnaire. Exceptions to this rule exist, as in cases where introductions must contain instructions regarding how the questionnaire is organized or about unusual types of questions. Furthermore, although the content of the spiel is important, how well interviewers deploy and tailor it is even more important.

I recommend that an introductory spiel contain enough information to reduce, as much as possible, any apprehension and uncertainty on the part of the person answering the telephone who hears that a stranger is calling to conduct a survey interview (cf. Frey, 1989, pp. 125–137). In other words, the credibility of the interviewer (and thus the survey effort) must be established as soon as possible, and it is the purpose of the introduction to do this. At the same time, experience demonstrates that it is easier to get someone's full cooperation once he or she begins the questionnaire—somewhat like the "foot-in-the-door" technique. Thus, logic suggests that the longer the introduction and the more a potential respondent must listen without active involvement, the greater the chance that he or she will lose interest before questioning even begins (cf. Dillman, Gallegos, & Frey, 1976; Burks, Camayd-Freixas, Lavrakas, & Bennett, 2007).

To this latter point, no matter how long or short the total introduction is, it is advisable to not have interviewers ramble on during the introduction without allowing the respondent to become engaged in the "conversation." Experience suggests that this can lead to abrupt hang-ups during the first 5 to 20 seconds of contact made by an interviewer. Instead, the interviewer should tailor the beginning of the introduction in a conversational manner that elicits responses from the respondent every 5 to 10 seconds.

Although survey researchers may differ in the exact ways they prefer to word introductions, it is strongly recommended that all telephone survey introductions include the following information, consistent with the disclosure guidelines of the American Association for Public Opinion Research (1991) and the National Council on Public Polls (Gawiser & Witt, 1992):

(a) identification of the interviewer (i.e., her or his actual first name at a minimum), the interviewer's affiliation, and the survey's sponsor;

(b) a brief explanation of the purpose of the survey and its sampling area (or target population);

(c) some "positively" worded phrase to encourage cooperation;

(d) verification of the telephone number dialed by the interviewer; and

(e) assurances that confidentiality will be maintained.

In most surveys it is unnecessary, and thus highly inadvisable, to devise an introductory spiel that contains a detailed explanation of what the survey is about, as this is likely to increase nonresponse. In doing so, it may also increase nonresponse error as it makes it more likely that those who are interested in the survey topic will continue and those not interested will not continue. For those respondents who want to know more about the survey before making a decision to participate, interviewers should be given an honest, standardized explanation to read or paraphrase.

There are some basic types of information-seeking exchanges that occasionally are initiated with interviewers by prospective respondents (see Lavrakas & Merkle, 1991). The word *occasionally* is important to keep in mind: If interviewers were asked these questions often, it would be wise to incorporate the information conveyed in the answers into the introductory spiel spoken to everyone. The types of information respondents sometimes ask for include

(a) the purpose of the survey and how the findings will be used;

(b) how the respondent's number was selected;

(c) more about the survey firm and/or sponsor than simply a name; and

(d) why the particular respondent selection method is being used.

For each of these questions, written fallback statements (or persuaders) should be provided to interviewers to enable them to give honest, standardized answers to respondents who ask them. The goal of these statements is to help interviewers convince potential respondents that the survey is a worthwhile (and harmless) endeavor; this should be kept in mind by the person who composes the statements. (For more details on telephone survey introductions and fallback statements, see Frey, 1989, pp. 125–137; Lavrakas, 1993, pp. 100–105.)

Refusals and Refusal Conversions

Currently, even in high-quality RDD surveys of the public, a majority of eligible households will end as either noncontacts or refusals, with the latter making up three quarters or more of a survey's nonresponse. For telephone surveys that use poorly skilled and/or poorly trained interviewers and/or that have poorly crafted introductory spiels, refusals can occur at two thirds or more of the households reached.

It is noteworthy that the vast majority of telephone survey refusals occur within the first 20 seconds of contact with a respondent and many occur within the first 10 seconds—that is, during the introduction, before the questionnaire has begun to be administered. Traditionally, good telephone surveys have invested many resources to reduce the number of refusals, in hopes that nonresponse error that might otherwise be associated with refusals will be markedly reduced. The most important of these are the resources spent on (a) developing an effective introductory spiel and (b) employing a skilled and well-trained group of interviewers.

Refusal Avoidance Training

The single factor that seems to differentiate the best of interviewers from those who are not so good is the ability to handle difficult respondents and outright refusals. This is one of the reasons, in most cases, interviewers should *not* be required to read an introductory spiel exactly as it is written, but should be allowed to convey the information accurately to the respondent in some of their own words (of course, this "tailoring" of the wording of the introduction by an interviewer is unacceptable when it comes to the reading of the actual survey items). The part of the interviewer's training that covers general expectations, therefore, should include a detailed discussion of the nature of refusals and explicit advice on how to be politely persuasive without being overly aggressive. Interested readers are encouraged to study Groves's (1989, pp. 215–236) review of the social science literature on persuasion and compliance as it relates to respondents' willingness to participate in surveys and interviewing strategies to reduce nonresponse. The work of Groves and his colleagues pertaining to leverage-saliency theory as a means to address nonresponse is also highly recommended (see Groves et al., 2000).

Based on three decades of experience with telephone interviewing, I believe that it is best to assume that all potential respondents need to be provided "incentives" for participating. Fortunately, with many respondents it seems to be enough incentive if they are told that they are being helpful by providing answers. For others, it appears to make them feel important to know that it is *their* opinions that are being sought. However, for approximately one half of all contacted respondents in surveys of both the general public and special populations, interviewers will have to work harder at "selling" the interview.

In these challenging cases, one option is to assume that the timing of the contact is wrong and to suggest calling back on another occasion. Interviewers might be trained to make a statement such as, "I'm sorry we've bothered you at what is apparently a bad time." Interviewers then must exercise discretion on a case-by-case basis concerning asking if there is a better time to call back, simply stating that a callback will be made, or not saying anything else. Another option is for the interviewer to politely "plead" with the potential respondent. When a telephone questionnaire is a relatively short one (i.e., it can be administered in 10 minutes or less), an interviewer can try to convince a reluctant respondent that it will not take very long. Another tactic for countering reluctance is to state that any question the respondent is uncomfortable answering may be left unanswered. Or interviewers can be trained to give several levels of assurance of both the legitimacy and importance of the survey through use of the survey's fallback statements. However, the simple provision of assurances, such as offering the respondent the name and phone number of the project director, often goes a long way toward alleviating the concerns of a reluctant respondent. As a last resort, the interviewer might consider reminding the respondent that by cooperating, the respondent is helping the interviewer earn a living (or, for the unpaid interviewer, the respondent is helping the interviewer fulfill her or his obligation). By personalizing the issue of cooperation, the interviewer is neither referring to an abstract incentive, such as "to help plan better social programs," nor appealing in the name of another party (the survey organization or sponsor).

In addition to training interviewers about *what* to say to minimize the refusals they experience, researchers should train them in *how* to say it—in terms of both attitude and voice. Collins, Sykes, Wilson, and Blackshaw (1988) found that less successful interviewers, when confronted with problems such as reluctant respondents, "showed a lack of confidence and a tendency to panic; they seemed unprepared for problems, gave in too easily, and failed to avoid 'deadends'" (p. 229). The confidence that successful interviewers feel is conveyed in the way they speak. Oksenberg and Cannell (1988) have reported that "dominance" appears to win out, with interviewers with low refusal rates being "generally more potent" (p. 268), rather than trying to be overly friendly, ingratiating, and/or nonthreatening. In terms of interviewers' voices, Oksenberg and Cannell found that those who spoke somewhat faster, louder, with greater confidence, and in a "falling" tone (declarative vs. interrogative) had the lowest refusal rates (cf. Groves, O'Hare, Gould-Smith, Benki, & Maher, 2008).

Refusal Conversions

Due in part to continuing difficulties in eliciting respondent cooperation over the past three decades, procedures have been developed and tested that are designed to lessen the potential problems refusals may cause (see Lyberg & Dean, 1992). One approach involves the use of a structured refusal report form (RRF) that the interviewer completes after encountering a refusal (see Lavrakas, 1993, pp. 78–81). This form can provide information that may help the sampling pool controller and interviewers in subsequent efforts to convert refusals—calling back at another time to try to convince a respondent to complete the interview after a refusal was previously encountered—and may help the researcher learn more about the size and nature of potential nonresponse error. If a researcher chooses to incorporate an RRF into the sampling process, it is not entirely obvious what information should be recorded. That is, even in the late-2000s, use of these forms has not received much attention in the survey methods literature. With this in mind, I urge interested readers to consider the following discussion of RRFs as suggestive and to follow the future literature on this topic.

Figure 16.1 is an example of an RRF used at my former university survey organizations. The interviewer completes the RRF *immediately after* encountering a refusal. Using the RRF shown in Figure 16.1, the interviewer would begin by recording "who" it was within the household that refused, although this is not always obvious and depends on information that the interviewer is able to glean prior to the termination of the call. The interview might also code some basic demographics about the person refusing, but only if the interviewer has some degree of certainty in doing so. Research suggests that interviewers can do this accurately in a majority of cases for gender, age, and race (Bauman, Merkle, &

Lavrakas, 1992; Lavrakas, Merkle, & Bauman, 1992). To the extent that this demographic information is accurate, the supervisor can use it to make decisions about which interviewers should attempt which subsequent refusal conversions. For example, my own experience and research suggests that an interviewer of the same race as the person who initially refused will have better success in converting a refusal. Furthermore, to the extent that respondent demographic characteristics correlate with survey measures, the researcher could investigate the effects of nonresponse by considering the demographic characteristics of the unconverted refusals; however, much more needs to be learned before the validity of this strategy is known. The interviewer can also rate the "severity" of the refusal, as shown in Figure 16.1, as well as add comments and answer other questions that may help explain the exact nature of the verbal exchange (if any) that transpired prior to the termination of the call. It also is recommended that households in which someone has told the interviewer at the initial refusal, "Don't call back!" or some such explicit comment not be recontacted.

No definitive evidence exists about the success rate of refusal-conversion attempts, although Groves and Lyberg (1988) placed it in the 25% to 40% range; my own experience leads me to put it in the 10% to 20% range nowadays. In making decisions about whether or not to attempt to convert refusals, the researcher is faced with this trade-off: the investment of resources to convert refusals so as to possibly decrease potential nonresponse error, versus the possible increase in other potential sources of survey error that otherwise might be reduced if those same resources were invested differently (e.g., paying more to have better-quality interviewers or refining the questionnaire more with additional pilot testing). Of note, Stec and Lavrakas (2007) reported that it is considerably more cost-efficient to gain completed interviews from converted refusals than from releasing new numbers from the sampling pool.

Measurement Issues in Telephone Surveys

Measurement issues in surveying include the effects of the questionnaire, the interviewers, the respondents, and the survey mode. However, this section on measurement issues in telephone surveys focuses almost entirely on the interviewer and on how a researcher can plan to minimize the potential error (bias and variance) that interviewers can contribute in telephone surveys.

As Groves (1989) noted, "Interviewers are the medium through which measurements are taken in [personal and telephone] surveys" (p. 404). This includes not only asking questions and recording responses but also processing the sample and securing respondent cooperation. Given the central role of interviewers, it is not surprising that they can add significant bias and variance to survey measures. However, there are many strategies for reducing interviewer-related error (see Fowler & Mangione, 1990, p. 9) that too often goes unused.

Interviewing is a part of the telephone survey process that, to date, has been much more a craft than a science. The quality of interviewing starts with the caliber of the persons recruited/hired to serve as interviewers, includes preinterviewing

User Supplied Title

Interviewer #: ____

1. Did the person who refused have the last (most recent) birthday?

Yes	1
No/Uncertain	2

2. Demographics of the person refusing:

GENDER		*AGE*		*RACE*	
Female	1	Child	0	Asian	1
Male	2	18–29 years	1	Black	2
Uncertain	9	30–59 years	2	Hispanic	3
		60 or older	3	White	4
		Uncertain	9	Uncertain	9

3. Reason for refusal:

____

____

4. Refusal strength:	VERY WEAK	1	2	3	4	5	6	7	VERY STRONG
Respondent attitude:	VERY POLITE	1	2	3	4	5	6	7	VERY RUDE
	NOT AT ALL ANGRY	1	2	3	4	5	6	7	VERY ANGRY

5. Did you tell the person:

	YES	*NO*
A. How he or she was sampled?	1	2
B. The nature/purpose of survey beyond the *standard* intro?	1	2
C. Confidentiality?	1	2
D. How the data would be used?	1	2
E. Verification with supervisor/sponsor?	1	2

6. What can you recommend, if anything, for gaining respondent/household cooperation if a conversion attempt were made?

____

____

____

____

Figure 16.1 Example of a Refusal Report Form training, and continues through supervisor monitoring and constant on-the-job training. As noted previously, I have long held the view that the great strength of the telephone survey method is its *potentially* large advantage over other modes of gathering survey data to reduce measurement error through centralized data collection (Lavrakas, 1987, 1993). Surprisingly, although many researchers appear to recognize the importance of a representative sampling pool, a low rate of nonresponse, and a well-constructed questionnaire, they often are lax in the control that they institute over the telephone interviewing process. Cost appears to be the primary reason for the lack of adequate attention given to rigorous control of interviewing in telephone surveys. Although it is expensive to institute strict and constant controls over telephone interviewers, in the absence of such a system, the researcher should be concerned that money spent on other parts of the survey enterprise (e.g., sampling) may be money wasted.

Interviewer Recruitment

A basic consideration regarding interviewers is whether they are paid for their work or unpaid, such as volunteers or students who do interviewing as part of their course work. When a telephone survey employs paid interviewers, there should be a greater likelihood of higher-quality interviewing, due to several factors. In situations in which interviewers are paid, the researchers can select carefully from among the most skilled individuals. With unpaid interviewers, researchers have much less control over who will *not* be allowed to interview. Paid interviewers are more likely to have an objective detachment from the survey's topic. In contrast, unpaid interviewers often have expectancies of the data; that is, volunteers by nature are often committed to an organization's purpose in conducting a survey and may hold preconceived notions of results, which can alter their behavior as interviewers and contribute bias to the data that they gather. Similarly, students who interview for academic credit often have an interest in the survey outcomes, especially if the survey is their class's own project.

Regardless of whether interviewers are paid or unpaid, I recommend that each interviewer be asked to enter into a written agreement with the researcher. This agreement should include a clause about not violating respondents' confidentiality. Also, the researcher must make it very clear to all prospective interviewers that telephone surveys normally require "standardized survey interviewing" (see Fowler & Mangione, 1990)—a highly structured and rather sterile style of asking questions. Standardized survey interviewing—as opposed to "conversational" interviewing (Schober & Conrad, 1997)—does not allow for creativity on the part of interviewers in the ordering or wording of particular questionnaire items or in deciding who can be interviewed. Furthermore, the researcher should inform all prospective telephone interviewers that constant monitoring will be conducted by supervisors, including listening to ongoing interviews (see Steve et al., 2008). The researcher's informing prospective interviewers of quality control features such as these in advance of making a final decision about their beginning to work will create realistic expectations. In the case of paid interviewers, it may discourage those who are not likely to conform to highly structured situations from applying. Good-quality telephone interviewers are best recruited through the use of a careful personnel screening procedure and the offer of a decent wage ($10-$12 per hour) to attract persons with ability and experience who might otherwise not be interested in telephone interviewing. Simply stated, the more the researcher pays interviewers, the more he or she can (and should) expect from them, in terms of both quality and quantity. (For more details about these matters, see Lavrakas, 1993, pp. 126–129.)

Survey administrators may be concerned with whether there are any demographic characteristics that are associated with high-quality interviewing—such as gender, age, or education—and whether they should take these characteristics into account in making hiring decisions. Within the perspective of wanting to avoid hiring practices that might be discriminatory, it should be noted that "other than good reading and writing skills and a reasonably pleasant personality, [there appear to be] no other credible selection criteria for distinguishing among potential interviewers" (see Fowler & Mangione, 1990, p. 140; Groves et al., 2008). Even in the case of strong regional accents, Bass and Totora (1988) report no interviewer-related effects. On the other hand, if the survey topic is related to interviewer demographics, there is consistent evidence that interviewer-respondent effects can and do occur that can increase TSE (see Fowler & Mangione, 1990, pp. 98–105). For example, a telephone survey about sexual harassment found that male respondents were twice as likely to report having sexually harassed someone at work if they were interviewed by a male versus a female (Lavrakas, 1992). In such cases, criteria used to hire and assign interviewers certainly should take into account the needs of the survey, but it should consider interviewer demographics in a nondiscriminatory manner.

Interviewer Training

The training of telephone survey interviewers, prior to the on-the-job training that they should constantly receive by working with their supervisors, has two distinct components: general training and project-specific training. New interviewers should receive general training to start their learning process. General training also should be repeated, or at least "refreshed," for experienced interviewers. Project-specific training is given to everyone, no matter what seniority or ability they have as interviewers.

The following issues should be addressed in the part of training that covers general practices and expectancies:

(a) What makes a good telephone interviewer, including behaviors related to processing the sampling pool, introducing the survey, selecting and securing the cooperation of the correct respondent, avoiding refusals by tailoring the introduction, and administering the questionnaire in a standardized fashion

(b) How the survey group's CATI system hardware and software works

(c) How interviewing is monitored, including an explication of standards for quality and quantity

(d) Ethical considerations in survey research

(e) The particulars of employment with the organization or person conducting the survey

All interviewers must be trained in the particulars of each new survey. Generally, this second, project-specific, part of training should be structured as follows:

(a) An explanation of the purpose of the survey

(b) A review of how the sampling pool was generated and how telephone numbers will be processed

(c) An explanation of the use of the introduction/selection sequence

(d) A review of fallback statements (persuaders) and practice in their use

(e) An explanation of the RRF

(f) A detailed item-by-item explanation of the questionnaire, including role-playing practice in its use

Fowler and Mangione (1990) suggest that prospective interviewers cannot be expected to behave acceptably as standardized survey interviewers with fewer than 20 to 30 hours of training. Researchers planning for interviewer training and the costs associated with it should take this into consideration. (For more suggestions on the general training telephone survey interviewers might receive, see Lavrakas, 1993, pp. 130–140; Tarnai & Moore, 2008)

Interviewer Supervision and Monitoring

The demands on supervisors in high-quality telephone survey operations are great. It is the responsibility of supervisors to ensure the integrity of sampling and quality of the data that are gathered. For these reasons, researchers should employ energetic and skilled persons in supervisory positions and should pay them accordingly. In general, considering both costs and data quality, an optimal ratio should be one supervisor for every 8 to 10 experienced interviewers (see Groves, 1989, pp. 61–62). Supervisors are responsible for maintaining the quality of the interviewing that occurs during their sessions, and interviewers should clearly perceive that their supervisors feel and display this responsibility.

Supervisors should themselves be trained to determine the levels at which interviewing-related problems occur (Cannell & Oksenberg, 1988). It may be that an interviewer has yet to receive adequate training and, therefore, is unfamiliar with proper techniques. Or it may be that the interviewer knows what to do, but not exactly how to operationalize it. Or the interviewer may know how something is supposed to be done, but lacks the skill/ability to do it properly. Unless the supervisor can judge accurately the level of the problem, she or he is not likely to be able to propose an effective solution to the interviewer. The rapport that supervisors develop with interviewers will affect the quality of data produced. To achieve a high level of quality, there must be constant verbal and/or written feedback from supervisors to interviewers, especially during the early part of a field period, when on-the-job training is critical.

Whenever possible, a telephone survey should use a centralized bank of telephones with equipment that allows the supervisor's telephone to monitor all interviewers' lines. There are special telephones that can be used to monitor an ongoing interview without the interviewer or respondent being aware of it. With CATI surveys, monitoring ongoing interviews often is a supervisor's primary responsibility. The use of a structured Interviewer Monitoring Form (IMF) is recommended (see Lavrakas, 1993, pp. 157–161; Steve et al., 2008). Supervisors need not listen to complete interviews, but rather they should systematically apportion their listening, a few minutes at a time, across all interviewers, concentrating more frequently and at longer intervals on less-experienced ones. All aspects of interviewer-respondent contact should be monitored, including the interviewer's use of the introduction, the respondent selection sequence, fallback statements, and administration of the questionnaire itself. An IMF can (a) aid the supervisor by providing documented on-the-job feedback to interviewers, (b) generate interviewer performance data for the field director, and (c) provide the researcher with a valuable type of data for investigating item-specific interviewer-related measurement error (see Cannell & Oksenberg, 1988; Groves, 1989, pp. 381–389).

In addition to noting whether or not interviewers are reading the items exactly as they are written, supervisors should pay special attention to the ways in which interviewers probe incomplete, ambiguous, or irrelevant responses, and to whether or not interviewers adequately repeat questions and define/clarity terms respondents may not understand in an unbiased fashion, if the latter is appropriate for the survey. Supervisors also need to pay close attention to anything interviewers may be saying or doing (verbally) that might reinforce certain response patterns that may bias answers. With many CATI systems, monitoring an ongoing interview includes being able to view the interviewer's use of the keyboard as it happens. Listening to ongoing interviewing and providing frequent feedback is especially important in the early stages of the field period and with new interviewers, and at these times extra supervisors may be needed.

Ethical Considerations, Telemarketing, and Pseudopolls

High-quality telephone surveys practice the principle of "informed consent." Respondents are informed, either explicitly or implicitly, that their participation is voluntary and that no harm will come to them regardless of whether they choose to participate or not. In addition to practicing these ethical standards, legitimate telephone surveys assure respondents that the answers they provide will be confidential; that is, no one other than the survey organization will know "who said what," unless respondents explicitly provide permission for their answers to be linked with their names.

Nowadays, many unethical survey practices are masquerading as legitimate surveys (see Traugott & Lavrakas, 2008). For example, there are so-called push-polls (political propagandizing disguised as legitimate polling, but using biased question wording solely to expose "respondents" to a highly partisan viewpoint), "FRUGing" (fund-raising under the guise of surveying), and "SUGing" (selling under the guise of surveying). With these telemarketing scams occurring, it is no wonder that many citizens hold negative (albeit uninformed) views of telephone surveying. Thus, all legitimate telephone researchers face the dual challenge of having to work to counter the negative effects of these pseudopolls and having to make certain that they do nothing inadvertent to compromise the integrity of ethical surveying.

Discussion Questions

1. What were the factors that led to telephone surveys becoming the "mode of preference" in the 1980s and 1990s for sampling and gathering data from the general population?

2. Currently, what are the major advantages and disadvantages in using telephone surveys to sample the general public? What about sampling populations other than the general public, such as members of a professional organization, students at a university, or members of a synagogue?

3. Why are there certain states in the United States that have relatively low coverage of residential telephone service, instead of coverage being essentially equal across all states? What effect does this have on telephone survey results in those states with low residential telephone coverage rates? What survey topics would be most biased by low coverage of telephone service?

4. What effects will number portability in the United States have on telephone surveys of the general population? What effects will the trend toward more U.S. residents using only a cell phone for their telephone service have on telephone surveys?

5. Why have telephone survey response rates dropped in the United States in the past decade? What direction will this trend likely go in the next decade? What implications does this have for the accuracy of telephone surveys used to measure the general population? What can be done to raise telephone survey response rates in the Unites States?

6. Discuss how the prior calling history on a given telephone number chosen for a telephone survey might affect future outcomes when calling back the same number as part of a telephone survey.

7. What are the advantages and disadvantages of using computer-assisted telephone interviewing (CATI) compared with a "paper and pencil" (PAPI) method? What are some circumstances when a telephone survey should be done using PAPI?

8. What reasoning should be considered in deciding whether or not initial refusals in a telephone survey should be called again and refusal conversions tried?

9. Discuss the considerations that need to be addressed in deciding what languages other than English should be used to conduct a telephone survey of the general population in the United States.

10. Explain the difference between the following concepts and illustrate those differences using a specific survey example: (a) population of inference, (b) target population, (c) sampling frame, (d) sampling pool, and (e) the final sample.

11. Explain when, if ever, a telephone directory can be used as the sampling frame for a survey of the general population in the United States.

12. Explain why a telephone survey of the general population should use some form of within-household respondent selection rather than merely interviewing the first person who answers the telephone. When, if ever, would it be acceptable to interview the first adult who answers the telephone?

13. Discuss the role of telephone surveying in mixed-mode surveys, both for the purpose of gaining cooperation from the sampled respondent and for gathering data from the sampled respondent.

Exercises

1. As a class, develop a short questionnaire that measures the type of telephone service(s) someone has in her or his home residence and the proportion of calls they receive and make on each type of service; include a few demographic questions at the end. Have all students complete 10 telephone interviews with other students at their college/university using the questionnaire. Have students write up and discuss their experiences as telephone interviewers.

2. Develop a set of "persuaders" for telephone interviewers to use with reluctant respondents who express the following: (a) How did you get my number? (b) How long is this going to take? (c) Why can't you pick someone else? (d) How do I know you won't sell the information that I give you? (e) My wife is busy, why can't I do the survey?

3. Create an introductory spiel for a telephone survey about seat belt usage that is being conducted by a university for a government agency in which adults will be sampled.

4. Create a sequence of introductory screener questions to determine if an eligible respondent resides within a household for a telephone survey that is sampling the opinions of African American adult females who have had at least some college education beyond being a high school graduate.

5. Create an Interviewer Monitoring Form for a telephone survey that includes measuring how well the interviewer administers the introduction and the questionnaire.

Notes

1. *Number portability* refers to an option that went into effect in November 2004 in the United States allowing people to transfer (port) their 10-digit telephone number to another geographic area when they moved and/or allowing them to keep the same number when they changed their telephone service from a landline to a cell phone or vice versa.

2. Starting in 2005, Linguist Dr. Erik Camayd-Freixas and researchers at the Nielsen Company began a series of "progressive involvement" experiments with training interviewers to use relatively brief sentences to "encourage" respondents reached on the telephone to become engaged in the conversation so as to counter the tendency of many respondents to hang up within the first few seconds after an interviewer has made contact (see Burks et al., 2007).

3. The concept of a sampling pool is not often addressed explicitly in the survey methods literature. A naive observer might assume, for example, that a telephone survey in which 1,000 persons were interviewed, actually sampled only those 1,000 persons and no others—but this is almost never the case, for many reasons, including the problem of nonresponse. Thus, a researcher is faced with the reality of often needing many times more telephone numbers for interviewers to process than the total number of interviews that the survey requires. Although most researchers refer to the set of telephone numbers that will be dialed as their sample and also use the word *sample* to refer to the final number of completed interviews achieved, Lavrakas (1987, 1993) proposed using the term *sampling pool* for the starting set of numbers to be dialed and the word *sample* for the final set of interviews that are achieved from the sampling pool.

4. First proposed by Cooper (1964), random-digit dialing, or RDD, comprises a group of probability sampling techniques that provide a nonzero chance of reaching any household with a telephone access line in a sampling area (assuming all exchanges/prefixes in the area are represented in the frame), regardless of whether its telephone number is published or listed. RDD does not provide an equal probability of reaching every telephone household in a sampling area because some households have more than one telephone number. For households with two or more numbers, postsampling adjustments (weighting) typically need to be made before the data are analyzed to correct for this unequal probability of selection; thus, data must be gathered via the questionnaire in RDD sampling about how many telephone numbers reach each household. Recent estimates are that, about two in five residential telephone numbers in the United States are unlisted. In theory, using RDD eliminates the potential problem of coverage error that might result from missing households with unlisted telephone numbers.

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CHAPTER 17

Ethnography

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Ethnography is about telling a credible, rigorous, and authentic story. Ethnography gives voice to people in their own local context, typically relying on verbatim quotations and a "thick" description of events. The story is told through the eyes of local people as they pursue their daily lives in their own communities. The ethnographer adopts a cultural lens to interpret observed behavior, ensuring that the behaviors are placed in a culturally relevant and meaningful context. The ethnographer is focused on the predictable, daily patterns of human thought and behavior. Ethnography is thus both a research method and a product, typically a written text.

Ethnographers are noted for their ability to keep an open mind about the groups or cultures they are studying. However, this quality does not imply any lack of rigor. The ethnographer enters the field with an open mind, not an empty head. Before asking the first question in the field, the ethnographer begins with a problem, a theory or model, a research design, specific data collection techniques, tools for analysis, and a specific writing style. The ethnographer also begins with biases and preconceived notions about how people behave and what they think—as do researchers in every field. Indeed, the choice of what problem, geographic area, or people to study is in itself biased. Biases serve both positive and negative functions. Controlled, biases can focus and limit the research effort. Uncontrolled, they can undermine the quality of ethnographic research. To mitigate the negative effects of bias, the ethnographer must first make specific biases explicit. A series of additional quality controls, such as triangulation, contextualization, and a nonjudgmental orientation, place a check on the negative influence of bias.

An open mind also allows the ethnographer to explore rich, untapped sources of data not mapped out in the research design. The ethnographic study allows multiple interpretations of reality and alternative interpretations of data throughout the study. As discussed, the ethnographer is interested in understanding and describing a social and cultural scene from the emic, or insider's perspective. The ethnographer is both storyteller and scientist; the closer the reader of an ethnography comes to understanding the native's point of view, the better the story and the better the science.

Overview

This chapter presents an overview of the steps involved in ethnographic work (see Fetterman, 1998, for additional detail). The process begins when the ethnographer selects a problem or topic and a theory or model to guide the study. The ethnographer simultaneously chooses whether to follow a basic or applied research approach to delineate and shape the effort. The research design then provides a basic set of instructions about what to do and where to go during the study. Fieldwork is the heart of the ethnographic research design. In the field, basic anthropological concepts, data collection methods and techniques, and analysis are the fundamental elements of "doing ethnography." Selection and use of various pieces of equipment—including the human instrument—facilitate the work. This process becomes product through analysis at various stages in ethnographic work—in field notes, memoranda, and interim reports, but most dramatically in the published report, article, or book.

This chapter presents the concepts, methods and techniques, equipment, analysis, writing, and ethics involved in ethnographic research. This approach highlights the utility of planning and organization in ethnographic work. The more organized the ethnographer, the easier his or her task of making sense of the mountains of data collected in the field. Sifting through notepads filled with illegible scrawl, listening to hours of digital voice recordings, labeling and organizing digital photographs and video, and conducting cross tabs and various data sorts in online surveys are much less daunting to the ethnographer who has taken an organized, carefully planned approach.

The reality, however, is that ethnographic work is not always orderly. It involves serendipity, creativity, being in the right place at the right or wrong time, a lot of hard work, and old-fashioned luck. Thus, although this discussion proceeds within the confines of an orderly structure, I have made a concerted effort to ensure that it conveys as well the unplanned, sometimes chaotic, and always intriguing character of ethnographic research.

Whereas in most research analysis follows data collection, in ethnographic research analysis and data collection begin simultaneously. An ethnographer is a human instrument and must discriminate among different types of data and analyze the relative worth of one path over another at every turn in fieldwork, well before any formalized analysis takes place. Clearly, ethnographic research involves all different levels of analysis. Analysis is an ongoing responsibility and joy from the first moment an ethnographer envisions a new project to the final stages of writing and reporting the findings.

Concepts

The most important concepts that guide ethnographers in their fieldwork include culture, a holistic perspective, contextualization, emic perspective and multiple realities, etic perspective, nonjudgmental orientation, inter- and intracultural diversity, structure and function, symbol and ritual, micro- and macrolevel studies, and operationalism.

Culture

Culture is the broadest ethnographic concept. Definitions of culture typically espouse either a materialist or an ideational perspective. The classic materialist interpretation of culture focuses on behavior. In this view, culture is the sum of a social group's observable patterns of behavior, customs, and way of life (see Harris, 1968, p. 16; Murphy & Margolis, 1995; Ross, 1980). The most popular ideational definition of culture is the cognitive definition. According to the cognitive approach, culture comprises the ideas, beliefs, and knowledge that characterize a particular group of people (Strauss & Quinn, 1997). This second—and currently most popular—definition specifically excludes behavior. Obviously, ethnographers need to know about both cultural behavior and cultural knowledge to describe a culture or subculture adequately. Although neither definition is sufficient, each offers the ethnographer a starting point and a perspective from which to approach the group under study.

Both material and ideational definitions are useful at different times in exploring fully how groups of people think and behave in their natural environments. However defined, the concept of culture helps the ethnographer search for a logical, cohesive pattern in the myriad, often ritualistic behaviors and ideas that characterize a group.

Anthropologists learn about the intricacies of a subgroup or community to describe it in all its richness and complexity. In the process of studying these details, they typically discover underlying forces that make the system tick. These cultural elements are values or beliefs that can unite or divide a group, but that are commonly shared focal points. An awareness of what role these abstract elements play in a given culture can give the researcher a clearer picture of how the culture works.

Many anthropologists consider cultural interpretation ethnography's primary contribution. Cultural interpretation involves the researcher's ability to describe what he or she has heard and seen within the framework of the social group's view of reality. A classic example of the interpretive contribution involves the wink and the blink. A mechanical difference between the two may not be evident. However, the cultural context of each movement, the relationship between individuals that each act suggests, and the contexts surrounding the two help define and differentiate these two significantly different behaviors. Anyone who has ever mistaken a blink for a wink is fully aware of the significance of cultural interpretation (see Fetterman, 1982, p. 24; Geertz, 1973, p. 6; Roberts, Byram, Barro, Jordan, & Street, 2001; Wolcott, 1980, pp. 57, 59).

Adopting a cultural interpretation is critical for classroom observation. For example, in an ethnographic study of an inner-city educational program, two students looking at each other's work might be interpreted as "cheating" in a conventional classroom. However, the accurate characterization is cooperation, given the philosophy of the school and the specific instructions provided by the teacher (see Figure 17.1).

Holistic Perspective and Contextualization

Ethnographers assume a holistic outlook in research to gain a comprehensive and complete picture of a social group. Ethnographers attempt to describe as much as possible about a culture or a social group. This description might include the group's history, religion, politics, economy, and environment. No study can capture an entire culture or group. The holistic orientation forces the fieldworker to see beyond an immediate cultural scene or event in a classroom, hospital room, city street, or plush offices in Washington, D.C., New York, or Chicago. Each scene exists within a multilayered and interrelated context.

Contextualizing data involves placing observations into a larger perspective. For example, in one of my studies of an alternative high school for dropouts, policymakers were contemplating terminating one dropout program because of its low attendance—approximately 60% to 70%. My reminder that the baseline with which to compare 60% to 70% attendance was zero attendance—these were students who systematically skipped school—helped the policymakers make a more informed

Figure 17.1 Classroom in Which the Accurate Description Is Sharing and Cooperating, Rather Than Cheating (far left two students)

decision about the program. In this case, contextualization ensured that the program would continue serving former dropouts (see Fetterman, 1987a).

In the same study, it was important to describe the inner-city environment in which the schools were located—an impoverished neighborhood in which pimping, prostitution, arson for hire, rape, and murder were commonplace (see Figure 17.2). This helped policymakers understand the power of certain elements in the community to distract students from their studies. This description also provided some insight into the often lucrative alternatives with which the school competed in attracting and retaining students. Contextualization helped provide a more accurate characterization of the school's degree of difficulty and helped prevent a common error—blaming the victim.

Emic and Etic Perspectives

The emic perspective—the insider's or native's perspective of reality—is at the heart of most ethnographic research. The insider's perception of reality is instrumental to understanding and accurately describing situations and behaviors. Native perceptions may not conform to an "objective" reality, but they help the fieldworker understand why members of the social group do what they do. In contrast to a priori assumptions about how systems work from a simple, linear, logical perspective—which might be completely off target—ethnography typically takes a phenomenologically oriented research approach.

An emic perspective compels the recognition and acceptance of multiple realities. Documenting multiple perspectives of reality in a given study is crucial to

Figure 17.2 New York Inner-City Neighborhood

an understanding of why people think and act in the different ways they do. Differing perceptions of reality can be useful clues to individuals' religious, economic, or political status and can help a researcher understand maladaptive behavior patterns.

An etic perspective is an external, social scientific perspective on reality. Some ethnographers are interested only in describing the emic view, without placing their data in an etic or scientific perspective. They stand at the ideational and phenomenological end of the ethnographic spectrum. Other ethnographers prefer to rely on etically derived data first, and consider emically derived data secondary in their analysis. They stand at the materialist and positivist philosophical end of the ethnographic spectrum. At one time, a conflict (ideational, typically emically oriented perspective) or the environment (materialist, often etically based perspective) consumed the field. Today, most ethnographers simply see emic and etic orientations as markers along a continuum of styles or different levels of analysis. Most ethnographers start collecting data from the emic perspective, then try to make sense of what they have collected in terms of both the native's view and their own scientific analysis. Just as thorough fieldwork requires an insightful and sensitive cultural interpretation combined with rigorous data collection techniques, so good ethnography requires both emic and etic perspectives.

Figure 17.3 Burnt-Out Building in the Inner City

A burnt-out building in the inner city across from the alternative school for dropouts provides an excellent example of why it is important to combine emic and etic perspectives (see Figure 17.3). From an initial etic perspective, it looks like there was a fire, possibly due to faulty electrical wiring. A few interviews with the students, and an alternative emic view is revealed. This was arson for hire. Some of the students are hired to "torch" a building after the landlord has increased the insurance coverage on the building. An interview with the local fire department (another emic view with considerable traditional authority) confirmed the students' emic view, adding a new insight into the alternative school's "competition" for the students' attention—particularly concerning alternative sources of activity and revenue. An etic view based on these emic views provides a more accurate depiction of what happened to the house and more to point, the social circumstances shaping what happened to the house (see Wolcott, 1999, p. 156).

Nonjudgmental Orientation and Inter- and Intracultural Diversity

A nonjudgmental orientation requires the ethnographer to suspend personal valuation of any given cultural practice. Maintaining a nonjudgmental orientation is similar to suspending disbelief while watching a movie or play, or reading a book—one accepts what may be an obviously illogical or unbelievable set of circumstances in order to allow the author to unravel a riveting story.

Intercultural diversity refers to the differences between two cultures, *intracultural diversity* to the differences between subcultures within a culture. Intercultural differences are reasonably easy to see. Compare the descriptions of two different cultures on a point-by-point basis—their political, religious, economic, kinship, ecological systems, and other pertinent dimensions. Intracultural differences, however, are more likely to go unnoticed.

These concepts place a check on our observations. They help the fieldworker see differences that may invalidate pat theories or hypotheses about observed events in the field. In some cases, these differences are systematic, patterned activities for a broad spectrum of the community, compelling the fieldworker to readjust the research focus; to throw away outdated and inappropriate theories, models, hypotheses, and assumptions; and to modify the vision of the finished puzzle. In other cases, the differences are idiosyncratic but useful in underscoring another, dominant pattern—the exception proves the rule. In most cases, however, such differences are instructive about a level or dimension of the community that had not received sufficient consideration.

Housing in the inner city provides an example of intracultural diversity. Most of the houses in the inner-city neighborhood we were studying were in disrepair, many were marked by graffiti by local gangs, and entire blocks were in rubble (see Figure 17.4). This was the "norm" concerning quality of housing in the neighborhood. However, there were families which were attempting to improve the quality of the neighborhood, and they "put their money where their mouth was" by painting and repairing their homes. They were, albeit, in the minority. However, they represented a special group with a symbolic message of hope in the community. This is an example of intracultural diversity. (For additional illustrations of intracultural diversity in qualitative research, see Fetterman, 1998; Marcus, 1998, p. 65.)

Structure and Function and Symbol and Ritual

Structure and function are traditional concepts that guide research in social organization. *Structure* here refers to the social structure or configuration of the group, such as the kinship or political structure. *Function* refers to the social relations among members of the group. Most groups have identifiable internal structures and established sets of social relationships that help regulate behavior.

Figure 17.4 Example of Intracultural Diversity in Terms of Housing in the Neighborhood

Ethnographers use the concepts of structure and function to guide their inquiry. They extract information from the group under study to construct a skeletal structure and then thread in the social functions—the muscle, flesh, and nerves that fill out the skeleton. A detailed understanding of the underlying structure of a system provides the ethnographer with a foundation on and frame within which to construct an ethnographic description.

In addition, ethnographers look for symbols that help them understand and describe a culture. Symbols are condensed expressions of meaning that evoke powerful feelings and thoughts. A cross or a menorah represents an entire religion, a swastika represents a movement, whether the original Nazi movement or one of the many neo-Nazi movements. A flag represents an entire country, evoking both patriotic fervor and epithets.

Symbols may signify historical influences in a community. For example, a Jewish star or Star of David (with Hebrew words carved into the stone) of a building marred by graffiti and broken glass, marks the historical presence of an orthodox Jewish community (see Figure 17.5). This symbol of the past provides some insight into the roots of current tensions between young African Americans in the community and older orthodox Jews (see Abramovitch & Galvin, 2001, p. 252).

Rituals are repeated patterns of symbolic behavior that play a part in both religious and secular life. Ethnographers see symbols and rituals as a form of cultural shorthand. Symbols open doors to initial understanding and crystallize critical cultural knowledge. Together, symbols and rituals help ethnographers make sense of observations by providing a framework in which to classify and categorize behavior (see Dolgin, Kemnitzer, & Schneider, 1977; Swatos, 1998, p. 505).

Figure 17.5 Yeshiva in the Inner City With Graffitti

Micro- or Macrolevel Studies and Operationalism

A microstudy is a close-up view, as if under a microscope, of a small social unit or an identifiable activity within the social unit. Typically, an ethnomethodologist or symbolic interactionist will conduct a microanalysis (see Denzin, 1989; Hinkel, 2004, p. 194). The areas of proxemics and kinesics in anthropology involve microstudies. Proxemics is the study of how the socially defined physical distance between people varies under differing social circumstances (Barfield, 1997; Birdwhistell, 1970). Kinesics is the study of body language (Birdwhistell, 1970; Psathas, 1995, p. 5). A macrostudy focuses on the large picture. In anthropology, the large picture can range from a single school to worldwide systems. The typical ethnography focuses on a community or specific sociocultural system. The selection of a micro- or macrolevel of study depends on what the researcher wants to know, and thus what theory the study involves and how the researcher has defined the problem under study.

Operationalism, simply, means defining one's terms and methods of measurement (Anderson, 1996, p. 19). In simple descriptive accounts, saying that "a few people said this and a few others said that" may not be problematic. However, establishing a significant relationship between facts and theory, or interpreting "the facts," requires greater specificity. Operationalism tests ethnographers and forces them to be honest with themselves. Instead of leaving conclusions to strong impressions, the fieldworker should quantify or identify the source of ethnographic insights whenever possible. Specifying how one arrives at one's conclusions gives other researchers something concrete to go on, something to prove or disprove.

In this section of the chapter, I have provided a discussion of some of the most important concepts in the profession, beginning with such global concepts as culture, a holistic orientation, and contextualization and gradually shifting to more narrow concepts—inter- and intracultural diversity, structure and function, symbol and ritual, and operationalism. In the next section, I detail the ethnographic methods and techniques that grow out of these concepts and allow the researcher to carry out the work of ethnography.

Methods and Techniques

The ethnographer is a human instrument. Ethnographic methods and techniques help guide the ethnographer through the wilderness of personal observation and to identify and classify accurately the bewildering variety of events and actions that form a social situation.

Fieldwork

Fieldwork is the hallmark of research for both sociologists and anthropologists. The method is essentially the same for both types of researchers—working with people for long periods of time in their natural setting. The ethnographer conducts research in the native environment to see people and their behavior given all the real-world incentives and constraints. This naturalist approach avoids the artificial response typical of controlled or laboratory conditions. Understanding the world—or some small fragment of it—requires studying it in all its wonder and complexity. The task is in many ways more difficult than laboratory study, but it can also be more rewarding (see Atkinson, 2002; McCall, 2006).

One of the benefits of fieldwork is that it provides a common sense perspective to data. For example, in a study of schools in the rural south, I received boxes of records indicating very low academic performance and high school attendance. This was counterintuitive and contrary to my experience working with schools in urban areas where students who received poor grades dropped out of school or were often truant or late. However, traveling to the school watching cotton, rice, and soy fields pass by, mile after mile, it became clear to me that the data made sense (see Figure 17.6). There was nothing else to do but show up to school. It was the only "social game" in town. As one student put it, "It (school) sure beat sittin' in the field, doing nothing, all by yourself."

The fieldworker uses a variety of methods and techniques to ensure the integrity of the data. These methods and techniques objectify and standardize the researcher's perceptions. Of course, the ethnographer must adapt each one of the methods and techniques discussed later to the local environment. Resource constraints and deadlines may also limit the length of time for data gathering in the field—exploring, cross-checking, and recording information.

Selection, Sampling, and Entry

The research questions shape the selection of a place and a people or program to study. The ideal site for investigation of the research problem is not always accessible. In that event, the researcher accepts and notes the limitations of the study from the onset. Ideally, the focus of the investigation shifts to match the site under study.

The next step is to decide how to sample members of the target population. Most ethnographers use the big-net approach conducive to participant observation—mixing and mingling with everyone they can at first. As the study progresses, the focus narrows to specific portions of the population under study. The big-net

Figure 17.6 Cotton Fields in the Arkansas Delta

approach ensures a wide-angle view of events before the microscopic study of specific interactions begins.

Ethnographers typically use informal strategies to begin fieldwork, such as starting wherever they can slip a foot in the door. (An introduction by a member is the ethnographer's best ticket into the community.) The most common technique is judgmental sampling; that is, ethnographers rely on their judgment to select the most appropriate members of the subculture or unit, based on the research question. Some experienced ethnographers use a rigorous randomized strategy to begin work—particularly when they already know a great deal about the culture or unit they are studying. However, using a highly structured randomized design without a basic understanding of the people under study may cause the researcher to narrow the focus prematurely, thus eliminating perhaps the very people or subjects relevant to the study. (See Weisner et al., 2001, for additional discussion about sampling.)

Participant Observation

Participant observation characterizes most ethnographic research and is crucial to effective fieldwork. Participant observation combines participation in the lives of the people under study with maintenance of a professional distance that allows adequate observation and recording of data.

Participant observation is immersion in a culture. Ideally, the ethnographer lives and works in the community for 6 months to a year or more, learning the language and seeing patterns of behavior over time. Long-term residence helps the researcher internalize the basic beliefs, fears, hopes, and expectations of the people under study. The simple, ritualistic behaviors of going to the market or to the well for water teach how people use their time and space, how they determine what is precious, sacred, and profane. The process may seem unsystematic; in the beginning, it is somewhat uncontrolled and haphazard. However, even in the early stages of fieldwork the ethnographer searches out experiences and events as they come to his or her attention. Participant observation sets the stage for more refined techniques—including projective techniques and questionnaires—and becomes more refined itself as the fieldworker understands more and more about the culture. Ideas and behaviors that were only a blur to the ethnographer on entering the community take on a sharper focus. Participant observation can also help clarify the results of more refined instruments by providing a baseline of meaning and a way to reenter the field to explore the context for those (often unexpected) results (DeWalt & DeWalt, 2002).

In applied settings, participant observation is often noncontinuous, spread out over an extended time. Often, contract research budgets or time schedules do not allow long periods of study—continuous or noncontinuous. In these situations, the researcher can apply ethnographic techniques to the study, but cannot conduct an ethnography (see Fetterman, 1988).

Interviewing

The interview is the ethnographer's most important data-gathering technique. Interviews explain and put into a larger context what the ethnographer sees and experiences. General interview types include structured, semistructured, informal, and retrospective interviews.

Formally structured and semistructured interviews are verbal approximations of a questionnaire with explicit research goals. These interviews generally serve comparative and representative purposes—comparing responses and putting them in the context of common group beliefs and themes. A structured or semistructured interview is most valuable when the fieldworker comprehends the fundamentals of a community from the "insider's" perspective. At this point, questions are more likely to conform to the native's perception of reality than to the researcher's (see Schensul, LeCompte, & Schensul, 1999).

Informal interviews are the most common in ethnographic work. They seem to be casual conversations, but where structured interviews have an explicit agenda, informal interviews have a specific but implicit research agenda. The researcher uses informal approaches to discover the categories of meaning in a culture. Informal interviews are useful throughout an ethnographic study for discovering what people think and how one person's perceptions compare with another's. Such comparisons help the fieldworker identify shared values in the community—values that inform behavior. Informal interviews are also useful for establishing and maintaining healthy rapport.

Retrospective interviews can be structured, semistructured, or informal. The ethnographer uses retrospective interviews to reconstruct the past, asking informants to recall personal historical information. This type of interview does not elicit the most accurate data. People forget or filter past events. In some cases, retrospective interviews are the only way to gather information about the past. In situations where the ethnographer already has an accurate understanding of the historical facts, retrospective interviews provide useful information about individuals.

All interviews share some generic kinds of questions. The most common types are survey or grand tour, detail or specific, and open- or closed-ended questions. Survey questions help identify significant topics to explore. Specific questions explore these topics in more detail. They determine similarities and differences in the ways people see the world. Open- and closed-ended questions help the ethnographer discover and confirm the participant's experiences and perceptions. (See sections on permission and institutional review boards presented later in this chapter.)

Survey Questions

A survey question—or what Spradley and McCurdy (1972) call a grand tour question—is designed to elicit a broad picture of the participant or native's world, to map the cultural terrain. Survey questions help the ethnographer define the boundaries of a study and plan wise use of resources. The participant's overview of the physical setting, universe of activities, and thoughts help focus and direct the investigation.

Once survey questions reveal a category of some significance to both fieldworker and native, specific questions about that category become most useful. The difference between a survey question and a specific or detailed question depends largely on context.

Specific questions probe further into established categories of meaning or activity. Whereas survey questions shape and inform a global understanding, specific questions refine and expand that understanding. Structural and attribute questions—subcategories of specific questions—are often the most appropriate approach to this level of inquiry. Structural and attribute questions are useful to the ethnographer in organizing and understanding of the native's view. Structural questions reveal the similarities that exist across the conceptual spectrum—in the native's head. (See Spradley & McCurdy, 1972, for additional information about the construction of taxonomic definitions. See also Clair, 2003.) Attribute questions—questions about the characteristics of a role or a structural element—ferret out the differences between conceptual categories. Typically, the interview will juxtapose structural with attribute questions. Information from a structural question might suggest a question about the differences among various newly identified categories.

Ethnographic research requires the fieldworker to move back and forth between survey and specific questions. Focusing in on one segment of a person's activities or worldview prematurely may drain all the ethnographer's resources before the investigation is half done. The fieldworker must maintain a delicate balance of questions throughout the study; in general, however, survey questions should predominate in the early stages of fieldwork, and more specific questions in the middle and final stages.

Open-Ended or Closed-Ended Questions

Ethnographers use both open- and closed-ended questions to pursue fieldwork. An open-ended question allows participants to interpret it. A typical open-ended question in the field is, "How are things going?" Closed-ended questions are useful in trying to quantify behavior patterns. For example, "How many times do you visit the city each month?" Ethnographers typically ask more open-ended questions during discovery phases of their research and more closed-ended questions during confirmational periods. The most important type of question to avoid is the stand-alone vague question.

Interviewing Protocols and Strategies

A protocol exists for all interviews—the product of the interviewer's and the participant's personalities and moods, the formality or informality of the setting, the stage of research, and an assortment of other conditions. The first element common to every protocol is the ethnographer's respect for the culture of the group under study. In an interview or any other interaction, ethnographers try to be sensitive to the group's cultural norms. This sensitivity manifests itself in apparel, language, and behavior. Second, an overarching guide in all interviews is respect for the person. An individual does the fieldworker a favor by giving up time to answer questions. Thus, the interview is not an excuse to interrogate an individual or criticize cultural practices. It is an opportunity to learn from the interviewee. Furthermore, the individual's time is precious: Both the industrial executive and the school janitor have work to do, and the ethnographer should plan initial interviews, whether formal or informal, around their work obligations and schedules. Later, the fieldworker becomes an integral part of the work. (See the permission section of this chapter for additional discussion.)

In formal settings—such as a school district—a highly formalized, ritualistic protocol is necessary to gain access to and to interview students and teachers. Structured interviews require a more structured protocol of introductions, permission, instructions, formal cues to mark major changes in the interview, closure, and possible follow-up communications.

Informal interviews require the same initial protocol. However, the researcher casually and implicitly communicates permission, instructions, cues, closure, and follow-up signals. Pleasantries and icebreakers are important in both informal interviews and formally structured interviews, but they differ in the degree of subtlety each interview type requires. Sensitivity to the appropriate protocol can enhance the interviewer's effectiveness.

Particular strategies or techniques can also enhance the quality of an interview. They include being a good listener, appreciating status differences, patience, respect, and engaging in the ebb and flow of conversation. The most effective strategy is, paradoxically, no strategy. Being natural is much more convincing than any performance (see Fetterman, 1998, for detail in this area).

Key Actor or Informant Interviewing

Some people are more articulate and culturally sensitive than others. These individuals make excellent key actors or informants. *Informant* is the traditional anthropological term; however, I use the term *key actor* to describe this individual, to avoid both the stigma of the term *informant* and its historical roots. In the social group under study, this individual is one of many actors, and may not be a central or even an indispensable community member. Yet this individual becomes a key actor in the theater of ethnographic research and plays a pivotal role, linking the fieldworker and the community.

Key actors can provide detailed historical data, knowledge about contemporary interpersonal relationships (including conflicts), and a wealth of information about the nuances of everyday life. Although the ethnographer tries to speak with as many people as possible, time is always a factor. Therefore, anthropologists have traditionally relied most heavily on one or two individuals in a given group.

Typically, the key actor will find many of the ethnographer's questions obvious or stupid. The fieldworker is asking about basic features of the culture—elementary knowledge to the key actor. However, such naive questions often lead to global explanations of how a culture works. Such responses point out the difference between the key actor and a respondent. The key actor generally answers questions in a comprehensive, albeit meandering, fashion. A respondent answers a question specifically, without explanations about the larger picture and conversational tangents, with all their richness and texture. Interviewing a respondent is usually a more efficient data collection strategy, but it is also less revealing and potentially less valid than discussions with a key actor.

Key actors require careful selection. They are rarely perfect representatives of the group. However, they are usually members of the mainstream—otherwise, they would not have access to up-to-date cultural information. Key actors may be cultural brokers, straddling two cultures. This position may give them a special vantage point and objectivity about their culture. They may also be informal or formal leaders in the community. Key actors come from all walks of life and all socioeconomic and age groups.

Key actor and ethnographer must share a bond of trust (see Figure 17.7). Respect on both sides is earned slowly. The ethnographer must take time to search out and spend time with these articulate individuals. The fieldworker learns to depend on the key actor's information—particularly as cross-checks with other sources prove it to be accurate and revealing. Sometimes key actors are initially selected simply because they and the ethnographer have personality similarities or mutual interests. Ethnographers establish long-term relationships with key actors who continually provide reliable and insightful information. Key actors can be extremely effective and efficient sources of data and analysis.

At the same time, the ethnographer must judge the key actor's information cautiously. Overreliance on a key actor can be dangerous. Every study requires multiple sources. In addition, the fieldworker must take care to ensure that key actors do not simply provide answers they think the fieldworker wants to hear. The ethnographer

Figure 17.7 Key Informant Interviewing by Dr. Fetterman

can check answers rather easily, but must stay on guard against such distortion and contamination. Another, subtler problem occurs when a key actor begins to adopt the ethnographer's theoretical and conceptual framework. The key actor may inadvertently begin to describe the culture in terms of this a priori construct, undermining the fieldwork and distorting the emic or insider's perspective. (For further discussion of the role of key informants, see Dobbert, 1982; Ellen, 1984; Freilick, 1970; Goetz & LeCompte, 1984; Pelto, 1970; Spradley, 1979; Taylor & Bogdan, 1984; Wolcott, 1999.)

Life Histories and Expressive-Autobiographical Interviews

Key actors often provide ethnographers with rich, detailed autobiographical descriptions. These life histories are usually quite personal; the individual is usually not completely representative of the group. However, how a key actor weaves a personal story tells much about the fabric of the social group. Personal description provides an integrated picture of the target culture.

Many of these oral histories are verifiable with additional work. However, in some instances, the life history may not be verifiable or even factually accurate. In these cases, the life history is still invaluable because the record captures an individual's perception of the past, providing a unique look at how the key actor thinks and how personal and cultural values shape his or her perception of the past. Together with observation and interviewing, taking life histories allows the ethnographer to assemble a massive amount of perceptual date with which to generate and answer basic cultural questions about the social group.

The life history approach is usually rewarding for both key actor and ethnographer. However, it is exceedingly time-consuming. Approximations of this approach, including expressive-autobiographical interviewing, are particularly valuable contributions to a study with resource limitations and time constraints. The expressive-autobiographical interview consists of a highly abbreviated chronological autobiography, interrupted at critical points with questions of concern to the researcher to narrow the scope almost immediately, for example, stress, puberty, marriage, employment, and so on (see Spindler & Spindler, 1970, p. 293; 1987, p. 25).

Lists and Forms

A number of techniques can stimulate the interviewer's recall and help organize the data. During a semistructured interview, the ethnographer may find a protocol or topical checklist useful. Printed or unobtrusively displayed on a laptop computer screen or a PDA (personal digital assistant), such a list usually contains the major topics and questions the ethnographer plans to cover during the interview. A checklist can be both a reminder and a mechanism to guide the interview when a more efficient approach is desirable. Similarly, after some experience in the field, the fieldworker can develop forms that facilitate data capture.

Checklists and forms help organize and discipline data collection and analysis. Their construction should rely on some knowledge from the field to ensure their appropriateness and usefulness. Checklists and forms also require consistent use. However, such lists and forms are not cast in stone; new topics emerge that merit exploration. New conceptualizations arise, and different forms are necessary for collection and analysis of the relevant data (see Carspecken, 1996, p. 29).

Questionnaires

Structured interviews are close approximations of questionnaires. Questionnaires represent perhaps the most formal and rigid form of exchange in the interviewing spectrum—the logical extension of an increasingly structured interview. However, questionnaires are qualitatively different from interviews because of the distance between the researcher and the respondent. Interviews have an interactive nature that questionnaires lack. In filling out a questionnaire, the respondent completes the researcher's form without any verbal exchange or clarification. Knowing whether or not the researcher and the respondent are on the same wavelength, sharing common assumptions and understandings about the questions, is difficult—perhaps impossible.

Misinterpretations and misrepresentations are common with questionnaires. Many people present idealized images of themselves on questionnaires, answering as they think they should to conform to a certain image. The researcher has no control over this type of response and no interpersonal cues to guide the interpretation of responses. Other problems include bias in the questions and poor return rates.

Despite these caveats, questionnaires are an excellent way for fieldworkers to tackle questions dealing with representativeness. They are the only realistic way of taking the pulses of hundreds or thousands of people. Anthropologists usually develop questionnaires to explore scientific concerns after they have a good grasp of how the larger pieces of the puzzle fit together. The questionnaire is a product of the ethnographer's knowledge about the system, and the researcher can adapt it to a specific topic or set of concerns. Ethnographers also use existing questionnaires to test hypotheses about specific conceptions and behaviors. However, the ethnographer must establish the relevance of a particular questionnaire to the target culture or subculture before administering it.

Online surveys and questionnaires provide an efficient way to document the views of large groups in a short period of time. The questions are posted on the Web, including *yes/no*, all that apply, open-ended, and 5-point Likert scale questions. Respondents are notified about the location of the survey on the Web (with a specific URL), enter their responses, and submit their survey online. The results are automatically calculated. The responses are often visually represented in a bar chart or similar graphic display as soon as the data are entered (see Figure 17.8). This saves the ethnographer from the initial mailing costs, time-consuming and expensive postal reminders, and the expense of data entry concerning all the submitted surveys. Ethnographers can help computer-phobic respondents or those who do not have access to computer or the Internet complete the survey and enter their responses in the same online data base if necessary (see Flick, 2006).

There are also many other ways to conduct surveys, ranging from PDAs to wireless polling devices. One of the benefits of wireless polling devices (where people use a hand-held instrument to record their answers and the results are immediately tabulated and visible) is the immediacy and transparency of the tool. Participants can see and share their responses in real time. The approach provides an excellent vehicle to launch focus group discussions. Individuals are also able to compare their answers with the group and (if comfortable) discuss their reasons for a specific response.

The credibility of survey findings (hard copy or online) depends on the response rate. Response rates refer to the percentage of people who complete a survey. There are many ways of increasing the response rate, ranging from keeping the survey short (reducing the respondent burden) to offering incentives. In general, the higher the response rate the better it is (see Fink, 2005, p. 6).

Projective Techniques

Projective techniques supplement and enhance fieldwork, they do not replace it. These techniques are employed by the ethnographer to elicit cultural and often psychological information from group members. Typically, the ethnographer holds an

11. Tobacco prevention & education programs are a good use of public money?			
Yes	████████████████████	883	98%
No	▌	14	2%
Total		897	100%

Figure 17.8 Computer Screen Snapshot of an Online Survey With Response Automatically Calculated

item up and asks the participant what it is. The researcher may have an idea about what the item represents, but that idea is less important than the participant's perception. The participant's responses usually reveal individual needs, fears, inclinations, and general worldview.

I typically share pictures and brief videos of the group I am working with while I am on site or in their community. In part, it is a natural form of reciprocity. However, it also yields important data. The pictures or videos elicit both confirming and unexpected comments. In one case, students yelled, "Idi Amin," when they saw the director's picture. This surprised me because I had only heard high praise about him before that. The reaction led me to understand another side or dimension to the director that made him successful—"caring but firm."

Projective techniques, however revealing, rarely stand alone. The researcher needs to set these techniques in a larger research context to understand the elicited response completely. Projective techniques can elicit cues that can lead to further inquiry or can be one of several sources of information to support an ongoing hypothesis. Only the ethnographer's imagination limits the number of possible projective techniques. However, the fieldworker should use only those tests that can be relevant to the local group and the study.

Additional Eliciting Devices

A variety of other tools are available with which the fieldworker can elicit the insider's classification and categorization of a target culture. Ethnographers ask participants to rank order people in their communities to understand the various social hierarchies. The semantic differential technique (Osgood, 1964) elicits an insider's rating of certain concepts. Cognitive mapping is also useful in eliciting the insider's perspective. Asking a student to map out his or her walk to school with various landmarks—for example, a route that identifies gang territories by block—provides insight into how that individual sees the world. As with projective techniques, the ethnographer requires some baseline knowledge of the community before he or she can design and use such techniques.

Unobtrusive Measures

I began this section on methods and techniques by stating that ethnographers are human instruments, dependent on all their senses for data collection and analysis. Most ethnographic methods are interactive: They involve dealing with people. The ethnographer attempts to be as unobtrusive as possible to minimize effects on the participant's behavior. However, data collection techniques—except for questionnaires—fundamentally depend on that human interaction.

A variety of other measures, however, do not require human interaction and can supplement interactive methods of data collection and analysis. These methods require only that the ethnographer keep eyes and ears open. Ranging from outcropping to folktales, these unobtrusive measures allow the ethnographer to draw social and cultural inferences from physical evidence (see Webb, Campbell, Schwartz, & Sechrest, 2000).

Outcroppings

Outcropping is a geological term referring to a portion of the bedrock that is visible on the surface—in other words, something that sticks out. Outcroppings in inner-city ethnographic research include skyscrapers, burned-out buildings, graffiti, the smell of urine on city streets, yard littered with garbage, a Rolls-Royce, and a syringe in the schoolyard. The researcher can quickly estimate the relative wealth or poverty of an area from these outcroppings. Initial inferences are possible without any human interaction. However, such cues by themselves can be misleading. A house with all the modern conveniences and luxuries imaginable can signal wealth or financial overextension verging on bankruptcy. The researcher must place each outcropping in a larger context. A broken syringe can have several meanings, depending on whether it lies on the floor of a doctor's office or in an elementary schoolyard late at night. On the walls of an inner-city school, the absence of graffiti is as important as its presence.

An expensive "white elephant" of a building takes on special significance when viewed within the confines of a township lacking rudimentary services and utilities. The outcropping hints at political patronage, poor planning, and/or misdirected resources. A South African woman standing in front of her modest home takes on greater meaning and significance, when situated within a larger squatter settlement in South Africa (see Figures 17.9 and 17.10). It becomes a political statement about the scope of poverty and injustice.

Changes in a physical setting over time can also be revealing. For example, an increase in the number of burned-out and empty buildings on a block indicates a decaying neighborhood. Conversely, an increase in the number of remodeled and

Figure 17.9 Woman in Squatter Settlement

Figure 17.10 Squatter Settlement in South Africa

revitalized houses may be indicative of gentrification, in which wealthy investors take over the neighborhood. The fieldworker must assess this abundant information with care, but should not ignore it or take it for granted.

Written and Electronic Information

In literate societies, written documents provide one of the most valuable and time-saving forms of data collection. In studies of office life, I have found past reports, memoranda, and personnel and payroll records invaluable. Mission statements and annual reports provide the organization's purpose or stated purpose and indicate the image the organization wishes to present to the outside world. Internal evaluation reports indicate areas of concern. Budgets tell a great deal about organizational values. Electronic mail is often less inhibited than general correspondence and thus quite revealing about office interrelationships, turf, and various power struggles. Proper use of this type of information can save the ethnographer years of work.

Proxemics and Kinesics

Proxemics is the analysis of socially defined distance between people, and kinesics focuses on body language (see Birdwhistell, 1970; Hall, 1974). In American culture, a salesperson speaking about a product while standing 2 inches away from a prospective buyer's face has probably intruded on the buyer's sense of private space. A skillful use of such intrusion may overwhelm the customer and make the sale, but it is more likely to turn the customer off.

Sensitivity to body language can also be instrumental in ethnographic research. A clenched fist, a student's head on a desk, a condescending superior's facial expression, a scowl, a blush, a student sitting at the edge of a chair with eyes fixed on the lecturer, and many other physical statements provide useful information to the observant fieldworker. In context, this information can generate hypotheses, partially confirm suspicions, and add another layer of understanding to fieldwork.

Folktales

Folktales are important to both literate and nonliterate societies. They crystallize an ethos or a way of being. Cultures often use folktales to transmit critical cultural values and lessons from one generation to the next. Folktales usually draw on familiar surroundings and on figures relevant to the local setting, but the stories themselves are facades. Beneath the thin veneer is another layer of meaning. This inner layer reveals the stories' underlying values. Stories provide ethnographers with insight into the secular and the sacred, the intellectual and the emotional life of a people.

All the methods and techniques discussed above are used together in ethnographic research. They reinforce one another. Like concepts, methods and techniques guide the ethnographer through the maze of human existence. Discovery and understanding are at the heart of this endeavor. The next section explores a wide range of useful devices that make the ethnographer's expedition through time and space more productive and pleasant.

Equipment

Notepads, computers, tape recorders, PDAs, cameras—all the tools of ethnography are merely extensions of the human instrument, aids to memory and vision. Yet these useful devices can facilitate the ethnographic mission by capturing the rich detail and flavor of the ethnographic experience and then helping organize and analyze these data. Ethnographic equipment ranges from simple paper and pen to high-tech laptop and mainframe computers, from tape recorders and cameras to digital camcorders. The proper equipment can make the ethnographer's sojourn in an alien culture more pleasant, safe, productive, and rewarding.

Pen and Paper

The most common tools ethnographers use are pen and paper. With these tools, the fieldworkers record notes from interviews during or after each session, sketches an area's physical layout, traces an organizational chart, and outlines informal social networks. Notepads can hold initial impressions, detailed conversations, and preliminary analyses. Most academics have had a great deal of experience with these simple tools, having taken extensive notes in classes. Note-taking skill is easily transferable to the field. Pen and paper have several advantages: ease of use, minimal expense, and unobtrusiveness. There are many occasions where it is disruptive, inappropriate, and/or dangerous to record notes, ranging from the observation of drug transactions in a playground to funerals. However, most ethnographers use their trained recall to record the information immediately after the event when necessary, typically using paper and pen. The drawbacks are obvious: The note-taking fieldworker cannot record every word and nuance in a social situation, has difficulty maintaining eye contact with other participants, and must expend a great deal of effort to record data legibly and in an organized manner. In addition, recall, even when trained, is typically more faulty than an immediate record of the event. Computer writing tablets emulate paper and pen, however, they also transcribe the written word almost immediately.

Laptop Computers

The laptop computer is a significant improvement over pen and notepad. Laptop computers are truly portable computers for use in the office, on a plane, or in the field. I often use the laptop in lieu of pen and paper during interviews (once I have established rapport and as long as it does not distance me from the person I am working with). In a technologically sophisticated setting, a laptop is rarely obtrusive or distracting if the fieldworker introduces the device casually and with consideration for the person and the situation. Laptop computers can save ethnographers time they can better spend thinking and analyzing. They greatly reduce the fieldworker's need to type up raw data interview notes every day, because the fieldworker enters these data into the computer only once, during or immediately after an interview. These notes can then be expanded and revised with ease. The files can be transferred from the laptop to a personal computer or mainframe with an external disk drive, appropriate software, and/or a high-speed modem or wireless connection. These files can then be merged with other field data, forming a highly organized (dated and cross-referenced), cumulative record of the fieldwork.

Laptops also provide the ethnographer with an opportunity to interact with participants at critical analytic moments. Ethnographers can share and revise notes, spreadsheets, and graphs with participants on the spot. I routinely ask participants to review my notes and memoranda as a way to improve the accuracy of my observations and to sensitize me to their concerns. We also produce bar charts and other graphic representations of the data together, providing an immediate cross-check on the preliminary analysis.

The laptop computer is not a panacea, but it is a real time-saver and is particularly useful in contract research. An ethnographer who conducts multisite research can carry a laptop to the sites and send files home via modem linkup or wireless with a home computer. Laptops also greatly facilitate communication from the field to the research center through interactive electronic mail systems. Laptops have drawbacks, of course, as any equipment does. The fieldworker must learn about the operating system and the programs. They must configure the computer properly with enough memory and storage. The ethnographer must also possess enough patience to work through bugs, viruses, slow-downs, and crashes. In addition, the fieldworker needs to take time to acquaint people with the device before thrusting it in front of them. Certain people will explicitly or implicitly prohibit the use of even a pen and notepad, never mind a laptop or other device. Also, the clatter of the keyboard can be distracting and obtrusive in certain situations. In most cases, however, a brief desensitization period will make people feel comfortable with the equipment. In fact, the laptop can be an icebreaker, helping the fieldworker to develop a strong rapport with people and at the same time inuring them to its presence. Given a careful introduction, laptops or any other useful pieces of equipment can greatly facilitate ethnographic work.

Desktop Computers

Many researchers use laptops to compose memos, reports, and articles to conduct interviews, and for general data collection, and then upload or send their files to a desktop computer. There are convenient tools to mechanically transfer files. However, an increasing number of researchers are skipping the transfer issue completely by using their laptop or notebook-type computers as their primary computers, because they are as powerful as the larger systems and are more convenient.

Database Software

Database programs enable the ethnographer to play a multitude of what-if games, to test a variety of hypotheses with the push of a button (and a few macros—strings of commands—assigned to that button). I have used a variety of database programs to test my perceptions of the frequency of certain behaviors, to test specific hypotheses, and to provide new insights into the data. NUDIST, Ethnograph, HyperQual, HyperResearch, AnSWR, EZ-Text, AskSam, Qualpro, and Atlas/ti are some programs that are well suited to ethnographic research (see Figure 17.11).

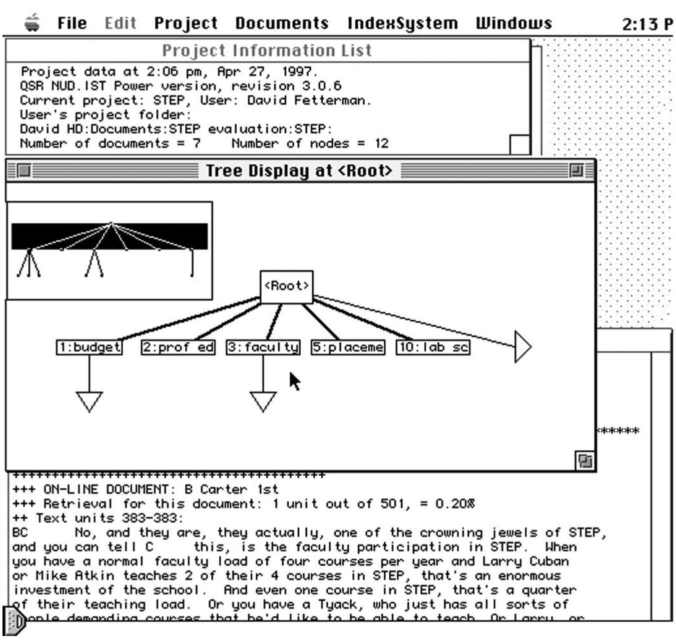

Figure 17.11 Computer Screen Snapshot of NUDIST Software

These database programs allow for the development of emergent themes. In addition, these tools help the ethnographer visualize and organize the data into "bins" or categories. FileMaker Pro and similar programs are less suitable for field notes, but are useful for more limited data sets and manipulation. Fixed fields do not allow for the addition of new fields that emerge along the way as the ethnographer learns more about the multidimensional nature of the topic and the field. (See Weitzman & Miles, 1995, for a detailed review of qualitative data analysis software. See also Hardy, 2004; O'Reilly, 2005.)

Internet Telephony

Internet telephone software, such as Skype and Jajah, enable people to speak with one another for free over the Internet. Ethnographers are increasingly using these tools to speak with colleagues and key actors in the field without long-distance charges. They are also a free or inexpensive way to maintain contact with community members as well.

Videoconferencing Technology

Videoconferencing technology allows geographically disparate parties to see and hear each other—around the globe. Free or inexpensive software programs, including iVisit, iChat, and CU-SeeMe, are available that allow videoconferencing online over the Internet, with no satellite or long-distance charges. With only this software and a small, relatively inexpensive digital camera plugged directly into a personal computer,[1] individuals can videoconference through their computer screens with any other similarly equipped users worldwide. I use videoconferencing to conduct follow-up interviews and observations at remote sites, after initially interviewing on-site and establishing rapport in person. I also use it to consult with colleagues and staff members on the ethnographic research team (see Fetterman, 1996, for additional details).

Videoconferencing was instrumental in a $15 million Hewlett-Packard funded Digital Divide project (Fetterman, 2004). The purpose of the project was to help people "bridge the digital divide," specifically establishing wireless communication within and outside the reservation. Videoconferencing facilitated communication throughout the project. In addition, digital photographs of videoconference exchanges between Native Americans in the Tribal Digital Village and ethnographers at Stanford University were used as evidence that the project was successful (see Figure 17.12).

Ethnographers have conducted fieldwork for generations without the benefit of laptop and desktop computers, printers, database software, and videoconferencing, and continue to conduct it without them. However, these tools are becoming indispensable in many disciplines, and few anthropologists conduct research without the use of some type of computer. Yet computers have limitations: They are only as good as the data the user enters. They still require the eyes and ears of the ethnographer to determine what to collect and how to record it, as well as how to interpret the data from a cultural perspective. (For further information about computing in ethnographic and qualitative research, see Best & Krueger, 2004; Brent, 1984; Conrad & Reinharz, 1984; Fischer, 1994; Friese, 2006; Podolefsky & McCarthy, 1983; Sproull & Sproull, 1982; Weitzman & Miles, 1995; also see Dow, 1987. My Web page provides a list of ethnographic resources on the Internet at www.stanford.edu/~davidf/ethnography.html.)

Digital Voice Recorders

Ethnographers attempt to immerse themselves in the field, working with people rather than devices. Tools that free the ethnographer from recording devices, whether pen and paper or laptop computers, are welcome. Tape recorders allow the

Figure 17.12 Videoconferencing Between Dr. Fetterman's Class/Research Team and Native Americans on Their Reservation

ethnographer to engage in lengthy informal and semistructured interviews without the distraction of manual recording devices. Tape recorders effectively capture long verbatim quotations, essential to good fieldwork, while the ethnographer maintains a natural conversational flow. Digital audio recordings can be analyzed over and over again. In all cases, however, the fieldworker should use the tape recorder judiciously and only with consent.

Tape recorders can inhibit some individuals from speaking freely during interviews. Some individuals may fear reprisals because their voices are identifiable on tape. The ethnographer must assure these people of the confidentiality of the data. Sometimes, easing into the use of tape recorders slowly can avoid unnecessary tension. I usually begin with pen and pad, and then ask if I can switch to the digital voice recorder simply because I cannot write fast enough to catch every word. I also stop the digital voice recorder whenever I touch on a topic that the interviewee thinks is too sensitive. A quick response to such requests highlights the ethnographer's sensitivity and integrity, and strengthens the bond between ethnographer and participant.

Digital recorders are useful icebreakers. On several occasions, I have recorded students' songs on the tape recorder and played the music back for them before asking them about the school under study. During group interviews, I typically ask students to pass the tape recorder around and introduce themselves on it as though they were celebrities. This approach often makes them eager to participate in the discussion and usually makes them comfortable with the machine. It also enables me to identify accurately each participants' words long after I have left the field.

Tape recorders do, however, have some hidden costs. Transcribing tapes is an extremely time-consuming and tedious task (even when they are digitally recorded and transferred to a computer). Listening to a tape takes as much time as making the original recording—hours of interview data require hours of listening. Transcribing tapes adds another dimension to the concept of time-consumption.[2] Typically, the fieldworker edits the tapes, transcribing only the most important sections. This keeps the ethnographer "close to the data," enabling the ethnographer to identify subtle themes and patterns that might be overlooked by a professional transcriber that is not familiar with the local community. However, a carefully selected professional transcriber can remove the pedestrian part of the process if funds are available (see Carspecken, 1996, p. 29; Robinson, 1994; Roper & Shapira, 2000).

Cameras

Cameras, particularly digital cameras, have a special role in ethnographic research. They can function as a can opener, providing rapid entry into a community or classroom (see Collier, 1967; Fetterman, 1980). They are a known commodity to most industrialized and many nonindustrialized groups. I use cameras to help establish an immediate familiarity with people. Cameras can create pictures useful in projective techniques or can be projective tools themselves. They are most useful, however, for documenting field observations.

Cameras document people, places, events, and settings over time. They enable the ethnographer to create a photographic record of specific behaviors. As Collier (1967) explains,

> Photography is a legitimate abstracting process in observation. It is one of the first steps in evidence refinement that turns raw circumstances into data that are manageable in research analysis. Photographs are precise records of material reality. They are also documents that can be filed and cross-filed as can verbal statements. Photographic evidence can be endlessly duplicated, enlarged or reduced in visual dimension, and fitted into many schemes or diagrams, and by scientific reading, into many statistical designs. (p. 5)

Photographs are mnemonic devices. During analysis and writing periods, photographs can bring a rush of detail that the fieldworker might not remember otherwise. By capturing cultural scenes and episodes on film at the beginning of a study—before he or she has a grasp of the situation—the ethnographer can use the pictures to interpret events retroactively, producing a rare second chance. Also, the camera often captures details that the human eye has missed. Although the camera is an extension of the subjective eye, it can be a more objective observer, less dependent on the fieldworkers' biases and expectations. A photographic record provides information that the fieldworker may not have noticed at the time. Photographs are also excellent educational tools, in the classroom, in a sponsor's conference room, or on a protected blog.

Computer software programs help organize digital photos and videos into "folders" based on themes or topics. Similarly, Web storage filing programs, such as Picasa and Dropshots, make it easy to organize and share photographs and digital videos with colleagues and the people you work with on the Internet. The same software can be used to "tell a story" by using these pictures to create digital slide shows and digital videos. I produce these kinds of videos for many of my projects and post them on blogs and Web pages. They help document a key event, share group projects with others who could not attend meetings, and give voice to community members who would not have otherwise been heard. They also serve as useful projective techniques, particularly as community members provide feedback on the video during the editing phase of video production.

The use of the camera or any photographic or audio recording mechanisms in fieldwork requires the subjects' permission. Some people are uncomfortable having their pictures taken; others cannot afford exposure. The ethnographer may enter the lives of people on their terms, but may not invade individual privacy. Photography is often perceived to be an intrusion. People are usually self-conscious about their self-presentation and concerned about how and where their pictures will be seen. An individual's verbal permission is usually sufficient to take a picture. However, written permission is necessary to publish or to display that picture in a public forum. Even with verbal and written permission in hand, the ethnographer must exercise judgment in choosing an appropriate display and suitable forum. Cameras, too, can be problematic. Inappropriate use of cameras can annoy and irritate people, undermining report and degrading the quality of the data. I typically use a pocket-sized digital camera that works under low-light conditions to minimize obtrusiveness. Cameras can also distort reality. A skillful photographer uses angles and shadows to exaggerate the size of a building or to shape the expression on a person's face. The same techniques can present a distorted picture of an individual's behavior. Photoshop and related software can easily modify and manipulate visual images. (See Aldridge, 1995; Becker, 1979, for an excellent discussion of photography and threats to validity. See also Pink, 2001, and the visual anthropology journal *Studies in Visual Communication.*)

Digital Camcorder

Digital camcorder recordings are extremely useful in ethnographic (and particularly microethnographic) studies. They are instrumental data collection tools when producing videos or digital vignettes of social situations. Camcorders can capture the ebb and flow of an activity or ritual. The three-dimensional movement brings the viewer closer to the natural movement and activity of the people you are describing. Raw digital video that is skillfully edited, much like a documentary, can tell a compelling and authentic story. Most digital cameras have a camcorder built into them, enabling the ethnographer to combine functions with a single device.

Ethnographers usually have a fraction of a second to reflect on a gesture or a person's posture or gait. Camcorders provide the observer with the ability to stop time. The ethnographer can tape a class and watch it over and over again, each time finding new layers of meaning or nonverbal signals from teacher to student, from student to teacher, and from student to student. Over time, visual and verbal patterns of communication become clear.

Camcorder equipment is essential to any microethnographic research effort. Gatekeeping procedures (Erickson, 1976) and the politics of the classroom (McDermott, 1974) are some elements of complex social situations that the fieldworker can capture on tape. However, the fieldworker must weigh the expense of the equipment and the time required to use it against the value of the information it will capture. Many ethnographic studies simply do not need fine-grained pictures of social reality. The equipment can be obtrusive, however, many fit in the palm of your hand. Even after participants have spent time with the ethnographer with and without the equipment, mugging and posing for the camera are not uncommon. The most significant hazard in using a camcorder is the risk of tunnel vision. Ideally, the ethnographer has studied the social group long enough to know what to focus on. The ethnographer may need months to develop a reasonably clear conception of specific behaviors before deciding to focus on them for a time. The camcorder can focus in on a certain type of behavior to the exclusion of almost everything else. Thus, the ethnographer may arrive at a very good understanding of a specific cultural mechanism but achieve little understanding of its real role in a particular environment.

In spite of the distinctions being made between visual media, the lines between them, especially digital photography and video, are becoming blurred. I often produce videos consisting of a combination of digital pictures and video recordings, with a voice track narrating the video and royalty free music in the background to convey a culturally appropriate and meaningful tone (see http://homepage.mac.com/profdavidf).

Cinema and Digital Videos

The use of cinema or movies in ethnographic research has been rare until recently. In ethnography, movies have typically presented finished pictures of cultural groups; they were not tools that researchers used to compose these pictures. Cost and the expertise needed to function as a filmmaker and editor were probably the primary reasons underlying this emphasis. However, with the advent of digital software such as iMovie, iPhoto, and Windows XP Movie Maker any ethnographer can produce videos and high-quality movies. Final Cut Pro is an even more advanced video software for professionals. Ethnographers can produce "draft" videos, used to test one's understanding of the social situation, much like a draft memoranda or a projective technique. They may be used to collaborate with community members, jointly videoing events and editing them together. This lends greater validity to the effort or finished product, because the key informant and ethnographer are making meaning together. Ethnographic films have rigorous requirements, ranging from actual time sequencing to authenticity of the events recorded. Heider (1976, pp. 46–117) has produced a scale of "ethnographicness" with which to judge ethnographic films (see also Lewis, 2004; Rouch & Feld, 2003).

Personal Digital Assistants

PDAs are pocket-sized tools designed to send and receive text messages, e-mail, and pictures. PDAs are also used to tell the time, record notes and to-do lists, search the Internet, find travel directions and maps, maintain calendars, synchronize files, and make telephone calls. Ethnographers increasingly rely on PDAs to organize and prioritize their schedules, communicate, and share preliminary insights and understandings while in the field. I routinely use my PDA to maintain my calendar and sync it with my office computer, verify directions while traveling, document my observations using the digital note pad and built-in camera (in an unobtrusive manner). I use my PDA routinely in the Stanford School of Medicine to document clinical teaching. I use the PDA to record observations and take photographs of clinical training activities (with permission). Then, I e-mail the notes and photographs to colleagues in the School of Medicine to verify my observations and interpretations. (See Masten & Plowman, 2003, for applications of digital ethnography to understanding consumers.)

The brief review of ethnographic equipment offered in this section is certainly not exhaustive. For example, many novel computer-aided design tools provide three-dimensional pictures of objects—an extremely useful tool for anthropologists working in space exploration. However, the tools discussed here are the ones that an ethnographer will most often use in the field. As aids to the ethnographer's own senses and abilities, they ease the difficult task of analysis, which is the subject of the next section of this chapter.

Analysis

Analysis is one of the most engaging features of ethnography. It begins at the moment a fieldworker selects a problem to study and ends with the last word in the report or ethnography. Ethnography involves many levels of analysis. Some are simple and informal; others require some statistical sophistication. Ethnographic analysis is iterative, building on ideas throughout the study. Analyzing data in the field enables the ethnographer to know precisely which methods to use next, as well as when and how to use them. Through analysis, the ethnographer tests hypotheses and perceptions to construct an accurate conceptual framework about what is happening in the social group under study. Analysis in ethnography is as much a test of the ethnographer as it is a test of the data.

Thinking

First and foremost, analysis is a test of the ethnographer's ability to think—to process information in a meaningful and useful manner. The ethnographer confronts a vast array of complex information and needs to make some sense of it all—piece by piece. The initial stage in analysis involves simple perception. However, even perception is selective. The ethnographer selects and isolates pieces of information from all the data in the field. The ethnographer's personal or idiosyncratic approach, together with an assortment of academic theories and models, focuses and limits the scope of inquiry. However, the field presents a vast amount of material, and in understanding day-to-day human interaction, elementary thinking skills are as important as ethnographic concepts and methods.

A focus on relevant, manageable topics is essential and is possible through the refinement of the unit of analysis. But then the fieldworker must probe those topics by comparing and contrasting data, trying to fit pieces of data into the bigger puzzle—all the while hypothesizing about the best fit and the best picture.

The ethnographer employs many useful techniques to make sense of the forests of data, from triangulation to the use of statistical software packages. All these techniques, however, require critical thinking skills—notably, the ability to synthesize and evaluate information—and a large dose of common sense.

Triangulation

Triangulation is basic in ethnographic research. It is at the heart of ethnographic validity, testing one source of information against another to strip away alternative explanations and prove a hypothesis. Typically, the ethnographer compares information sources to test the quality of the information (and the person sharing it), to understand more completely the part an actor plays in the social drama, and ultimately to put the whole situation into perspective.

I will typically ask how a student is doing in a particular study. I might also hear reports from the teacher about the student's performance. The student's parent might offer an insight into the student's performance as well. When these three separate sources converge and reinforce each other, I am more confident reporting that the student's performance has indeed improved. At least it helps me "rule out rival hypotheses" concerning the student's performance. (See Flick, Kardorff, & Steinke, 2004; Webb et al., 2000, for a detailed discussion of triangulation.)

Patterns

Ethnographers look for patterns of thought and behavior. Patterns are a form of ethnographic reliability. Ethnographers see patterns of thought and action repeat in various situations and among various players. Looking for patterns is a form of analysis. The ethnographer begins with a mass of undifferentiated ideas and behavior, and then collects pieces of information, comparing, contrasting, and sorting gross categories and minutiae until a discernible thought or behavior becomes identifiable. Next the ethnographer must listen and observe, and then compare his or her observations with this poorly defined model. Exceptions to the rule emerge, variations on a theme are detectable. These variants help circumscribe the activity and clarify its meaning. The process requires further sifting and sorting to make a match between categories. The theme or ritualistic activity finally emerges, consisting of a collection of such matches between the model (abstracted from reality) and the ongoing observed reality.

Any cultural group's patterns of thought and behavior are interwoven strands. As soon as the ethnographer finishes analyzing and identifying one pattern, another pattern emerges for analysis and identification. The fieldworker can then compare the two patterns. In practice, the ethnographer works simultaneously on many patterns. The level of understanding increases geometrically as the ethnographer moves up the conceptual ladder—mixing and matching patterns and building theory from the ground up. (See Glaser & Strauss, 1967, for a discussion of grounded theory.)

The observer can make preliminary inferences about the entire economic system by analyzing the behavior that is subsumed within the pattern, as well as the patterns themselves. Ethnographers acquire a deeper understanding of and appreciation for a culture as they weave each part of the ornate human tapestry together, by observing and analyzing the patterns of everyday life (see Davies, 1999, p. 146; Wolcott, 1999, p. 256).

Key Events

Key or focal events that the fieldworker can use to analyze an entire culture occur in every social group. Geertz (1973) eloquently used the cockfight to understand and portray Balinese life. Key events come in all shapes and sizes. Some tell more about a culture than others, but all provide a focus of analysis (see also Atkinson, 2002; Geertz, 1957).

Key events, like digital photographs or Quicktime videos, concretely convey a wealth of information. Some images are clear representations of social activity; others provide a tremendous amount of embedded meaning. Once the event is recorded, the ethnographer can enlarge or reduce any portion of the picture. A rudimentary knowledge of the social situation will enable the ethnographer to infer a great deal from key events. In many cases, the event is a metaphor for a way of life or a specific social value. Key events provide lenses through which to view a culture.

Key events are extraordinarily useful for analysis. Not only do they help the fieldworker understand a social group, but the fieldworker in turn can use them to explain the culture to others. The key event thus becomes a metaphor for the culture. Key events also illustrate how participation, observation, and analysis are inextricably bound together during fieldwork.

Maps, Flowcharts, Organizational Charts, and Matrices

Visual representations are useful tools in ethnographic research. Having to draw a map of the community tests an ethnographer's understanding of the area's physical layout. It can also help the ethnographer chart a course through the community. Flowcharts are useful in studies of production line operations. Mapping out what happens to a book in a research library, from the time it is received on the shipping dock to the time it is cataloged and available on the shelf, can provide a baseline of understanding about the system. We found that one library used to accept the books at the loading dock and then have them moved to the opposite end of the library instead of opening them up and processing them right by the dock. The simple act of creating a map or flowchart together made the inefficiency apparent.

Flowcharting a social welfare program is also common in evaluation. The analytic process of mapping the flow of activity and information can also serve as a vehicle to initiate additional discussions. Drawing organizational charts is a useful analytic tool. It tests the ethnographer's knowledge of the system, much as drawing a map or a flowchart does. Both formal and informal organizational hierarchies can be charted for comparison. In addition, organizational charts can measure changes over time, as people move in and out or up and down the hierarchy. Organizational charts clarify the structure and function of any institutional form of human organization.

Matrices provide a simple, systematic, graphic way to compare and contrast data. The researcher can compare and cross-reference categories of information to establish a picture of a range of behaviors or thought categories. Matrices also help the researcher identify emerging patterns in the data. (See Handwerker, 2001, p. 222; Miles & Huberman, 1994, for detailed presentation of the use of matrices in qualitative research.) Maps, flowcharts, organizational charts, and matrices all help crystallize and display consolidated information.

Content Analysis

Ethnographers analyze written and electronic data in much the same way they analyze observed behavior. They triangulate information within documents to test for internal consistency. They attempt to discover patterns within the text and seek key events recorded and memorialized in print.

Ethnographers may subject internal documents to special scrutiny to determine whether they are internally consistent with program philosophy. Reviews may also reveal significant patterns. It is often possible for the ethnographer to infer the significance of a concept from its frequency and context in the text (see Graneheim & Lundman, 2004; Krippendorff, 2004, p. 87; Neuendorf, 2001; Roberts, 1997; Stemler, 2001; Titscher, 2000, p. 224; Tuval-Mashiach, Zilber, & Lieblich, 1998).

Statistics

Ethnographers use nonparametric statistics more often than parametric statistics because they typically work with small samples. Parametric statistics require large samples for statistical significance. The use of nonparametric statistics is also more consistent with the needs and concerns of most anthropologists. Anthropologists typically work with nominal and ordinal scales. Nominal scales consist of discrete categories, such as sex and religion. Ordinal scales also provide discrete categories as well as a range of variation within each category—for example, reform, conservative, and orthodox variants within the category of Judaism. Ordinal scales do not determine the degree of difference between subcategories. The Guttman (1944) scale also known as cumulative scaling or scalogram (Trochim, 2006a) analysis is one example of an ordinal scale that is useful in ethnographic research.

The chi-square test and the Fisher exact probability test are popular nonparametric statistical tools in anthropology. However, all statistical formulas require that certain assumptions be met before the formulas may be applied to any situation. A disregard for these variables in the statistical equation is as dangerous as neglect of comparable assumptions in the human equation in conducting ethnographic fieldwork. Both errors result in distorted and misleading efforts at worst, and waste valuable time at best.

Ethnographers use parametric statistics when they have large samples and limited time and resources to conduct all the interviews. Survey and questionnaire work often requires sophisticated statistical tests of significance. *t* Tests are used to determine whether the means of two groups are statistically significant (Trochim, 2006b). Analysis of covariance design in regression analysis is another common test used when sample size permits in ethnography. Ethnographers also use the results of parametric statistics to test certain hypotheses, cross-check their own observations, and generally provide additional insight. (See Fetterman, 1998, for discussion of problems with statistics. See also Handwerker, 2001, p. 222, for examples ranging from factor analysis to logistic regression.)

Crystallization

Ethnographers crystallize their thoughts at various stages throughout an ethnographic endeavor. The crystallization may bring a mundane conclusion, a novel insight, or an earth-shattering epiphany. The crystallization is typically the result of a convergence of similarities that spontaneously strike the ethnographer as relevant or important to the study. Crystallization may be an exciting process or the result of painstaking, boring, methodical work. This research gestalt requires attention to all pertinent variables in an equation.

Every study has classic moments when everything falls into place. After months of thought and immersion in the culture, the ethnographer discovers that a special configuration gels. All the subtopics, miniexperiments, layers of triangulated effort, key events, and patterns of behavior form a coherent and often cogent picture of what is happening. One of the most exciting moments in ethnographic research is when an ethnographer discovers a counterintuitive conception of reality—a conception that defies common sense. Such moments make the long days and nights worthwhile.

Analysis has no single form or state in ethnography. Multiple analyses and forms of analyses are essential. Analysis takes place throughout any ethnographic endeavor, from the selection of the problem to the final stages of writing. Analysis is iterative and often cyclical in ethnography (see Atkinson, 2002, pp. 52, 384; Goetz & LeCompte, 1984; Hammersley & Atkinson, 1983; Taylor & Bogdan, 1984). The researcher builds a firm knowledge base in bits and pieces, asking questions, listening, probing, comparing and contrasting, synthesizing, and evaluating information. The ethnographer must run sophisticated tests on data long before leaving the field. However, a formal, identifiable stage of analysis does take place when the ethnographer physically leaves the field. Half the analysis at this stage involves additional triangulation, sifting for patterns, developing new matrices, and applying statistical tests to the data. The other half takes place during the final stage of writing an ethnography or an ethnographically informed report.

Writing

Ethnography requires good writing skills at every stage of the enterprise. Research proposals, field notes, memoranda, blogs, shared collaborative Web-based word processing and spreadsheet documents, interim reports, final reports, articles, and books are the tangible products of ethnographic work. The ethnographer can share these written works with participants to verify their accuracy and with colleagues for review and consideration. Ethnography offers many intangibles, through the media of participation and verbal communication. However, written products, unlike transitory conversations and interactions, withstand the test of time.

Writing good field notes is very different from writing a solid and illuminating ethnography or ethnographically informed report. Note taking is the rawest kind of writing. The note taker typically has an audience of one. Thus, although clarity, concision, and completeness are vital in note taking, style is not a primary consideration (see Emerson, Fretz, & Shaw, 1995).

Writing for an audience, however, means writing to that audience. Reports for academics, government bureaucrats, private and public industry officials, medical professionals, and various educational program sponsors require different formats, languages, and levels of abstraction. The brevity and emphasis on findings in a report written for a program-level audience might raise some academics' eyebrows and cause them to question the project's intellectual effort. Similarly, a refereed scholarly publication would frustrate program personnel, who would likely feel that the researcher is wasting their time with irrelevant concerns, time that they need to take care of business. In essence, both parties feel that the researcher is simply not in touch with their reality. These two audiences are both interested in the fieldwork and the researcher's conclusions, but have different needs and concerns. Good ethnographic work can usually produce information that is relevant to both parties.

This is possible when performance writing is used to guide ethnographic writing. Performance writing involves writing for an audience, caring about them, and hoping that your work will make a difference to them (Madison, 2005, p. 192). It is not unnecessarily complicated. It is relational in that it treats the reader like a gyroscope or a compass, in which the writer's words revolve around them. The skillful ethnographer will communicate effectively with all audiences—in part because the ethnographer cares about each audience—using the right smoke signals for the right tribe. However, it is not simply a matter of language. (See Fetterman, 1987b, for discussion of the ethnographer as rhetorician. See also Yin, 1994, for discussion of differing audiences in the presentation of a case study.)

Blogs and Web pages provide a powerful medium for writing progress reports, posting videos of key events, and capturing the spirit of the community you work with. They are tools to facilitate reciprocity, by posting reports, tools, and information the community values. Blogs and Web pages are also easily customized to multiple audiences, including scholarly audiences, program staff, and members of the community. These Web-based documents also are highly accessible. They provide an immediacy and transparency to ethnographic insights and understandings. They help solidify a sense of community between the ethnographer and the people they work with. Blogs and Web pages can be informal or scholarly, however, they are typically a form of writing that falls between field notes and final reports or articles (with formal articles and publications linked to the blog or Web page).

Writing is part of the analysis process as well as a means of communication (see also Hammersley & Atkinson, 1983). Writing clarifies thinking. In sitting down to put thoughts on paper, an individual must organize those thoughts and sort out specific ideas and relationships. Writing often reveals gaps in knowledge. If the researcher is still in the field when he or she discovers those gaps, the researcher needs to conduct additional interviews and observations of specific settings. If the ethnographer is a collaborative researcher, they might share Web-based word processing and spreadsheet documents with community members. This enables community members to edit and cowrite ethnographic insights and findings. This places a check on the ethnographer's interpretation and promotes collaboration (community building). I use an interactive spreadsheet, with an Arkansas tobacco prevention project, to manage incoming data concerning numbers of people who quit smoking and how this translates into dollars saved in terms of excess medical expenses. Data collection, for this project, is iterative and a collaborative experience. If the researcher has left the field, field notes, e-mails (including digital photographs), and telephone calls must suffice (unless they also use Web-based documents and share them with community members after leaving the field). Embryonic ideas often come to maturity during writing, as the ethnographer crystallizes months of thought on a particular topic. From conception—as a twinkle in the ethnographer's eye—to delivery in the final report, an ethnographic study progresses through written stages. (For additional discussions of ethnographic writing, see Fetterman, 1998; Madison, 2005; O'Reilly, 2005; Wolcott, 1990. See Van Maanen, 1988, for some of the rhetorical and narrative devices used in ethnographic work, including realist, confessional, and impressionist tales.)

Ethics

Ethnographers do not work in a vacuum, they work with people. They often pry into people's innermost secrets, sacred rites, achievements, and failures. In pursuing these personal sciences, ethnographers subscribe to a code of ethics that preserves the participants' rights, facilitates communication in the field, and leaves the door open for further research.

This code specifies first and foremost that the ethnographer do no harm to the people or the community under study. In seeking a logical path through the cultural wilds, the ethnographer is careful not to trample the feelings of natives or desecrate what the culture calls sacred. This respect for social environment ensures not only the rights of the people but also the integrity of the data and a productive, enduring relationship between the people and the researcher. Professionalism and a delicate step demonstrate the ethnographer's deep respect, admiration, and appreciation for the people's way of life. Noninvasive ethnography is not only good ethics but also good science (see American Anthropological Association, 1990, 1998; Rynkiewich

& Spradley, 1976; Weaver, 1973). Basic underlying ethical standards include the securing of permission (to protect individual privacy), honesty, trust (both implicit and explicit), and reciprocity (see Sieber, Chapter 4, this volume).

Permission

Ethnographers must formally or informally seek informed consent to conduct their work. In a school district, formal written requests are requisite. Often, the ethnographer's request is accompanied by a detailed account of the purpose and design of the study. Similarly, in most government agencies and private industry, the researcher must submit a formal request and receive written permission. The nature of the request and the consent changes according to the context of the study. For example, no formal structure exists for the researcher to communicate within a study of tramps. However, permission is still necessary to conduct a study. In this situation, the request may be as simple as the following embedded question to a tramp: "I am interested in learning about your life, and I would like to ask you a few questions, if that's all right with you." In this context, a detailed explanation of purpose and method might be counterproductive unless the individual asks for additional detail. (See the section on institutional review boards presented later in this chapter for more discussion on this topic.)

Honesty

Ethnographers must be candid about their task, explaining what they plan to study and how they plan to study it. In some cases detailed description is appropriate, and in others extremely general statements are best, according to the type of audience and the interest in the topic. Few individuals want to hear a detailed discussion of the theoretical and methodological bases of an ethnographer's work. However, the ethnographer should be ready throughout the study to present this information to any participant who requests it. Deceptive techniques are unnecessary and inappropriate in ethnographic research. Ethnographers need not disguise their efforts or use elaborate ploys to trick people into responding to specific stimuli.

Trust

Ethnographers need the trust of the people they work with to complete their task. An ethnographer who establishes a bond of trust will learn about the many layers of meaning in any community or program under study. The ethnographer builds this bond on a foundation of honesty, and communicates this trust verbally and nonverbally. He or she may speak simply and promise confidentiality as the need arises. Nonverbally, the ethnographer communicates this trust through self-presentation and general demeanor. Appropriate apparel, an open physical posture, handshakes, and other nonverbal cues can establish and maintain trust between an ethnographer and a participant.

Actions speak louder than words. An ethnographer's behavior in the field is usually his or her most effective means of cementing relationships and building trust. People like to talk, and ethnographers love to listen. As people learn that the ethnographer will respect and protect their conversations, they open up a little more each day in the belief that the researcher will not betray their trust. Trust can be an instant and spontaneous chemical reaction, but more often it is a long, steady process, like building a friendship.

Pseudonyms

Ethnographic descriptions are usually detailed and revealing. They probe beyond the facade of normal human interaction. Such descriptions can jeopardize individuals. One person may speak candidly about a neighbor's wild parties and mention calling the police to complain about them. Another individual may reveal the arbitrary and punitive behavior of a program director or principal. Each individual has provided invaluable information about how the system really works. However, the delicate web of interrelationships in a neighborhood, a school, or an office might be destroyed if the researcher reveals the source of this information. Similarly, individuals involved in illegal activity—ranging from handling venomous rattlesnakes in a religious ceremony to selling heroin in East Detroit in order to build a gang empire—have a legitimate concern about the repercussions of the researcher's disclosing their identities.

The use of pseudonyms is a simple way to disguise the identities of individuals and protect them from potential harm. Disguising the name of the village or program can also prevent the curious from descending on the community and disrupting the social fabric of its members' lives. Similarly, coding confidential data helps prevent them from falling into the wrong hands. However, there are limits to confidentiality in litigation.

Reciprocity

Ethnographers use a great deal of people's time, and they owe something in return. In some cases, ethnographers provide a service simply by lending a sympathetic ear to troubled individuals. In other situations, the ethnographer may offer time and expertise as barter—for example, teaching a participant English or math, milking cows and cleaning chicken coops, or helping a key actor set up a new computer and learn to use the software. Ethnographers also offer the results of their research in its final form as a type of reciprocity.

Some circumstances legitimate direct payment for services rendered, such as having participants help distribute questionnaires, hiring them as guides on expeditions, and soliciting various kinds of technical assistance. However, direct payment is not a highly recommended form of reciprocity. This approach often reinforces patterns of artificial dependence and fosters inappropriate expectations. Direct payment may also shape a person's responses or recommendations throughout a study. Reciprocity in some form is essential during fieldwork (and, in some cases, after the study is complete), but it should not become an obtrusive, contaminating, or unethical activity.

Guilty Knowledge and Dirty Hands

During the more advanced stages of fieldwork, the ethnographer is likely to encounter the problems of guilty knowledge and dirty hands. Guilty knowledge refers to confidential knowledge of illegal or illicit activities. Dirty hands refers to situations in which the ethnographer cannot emerge innocent of wrongdoing (see Fetterman, 1983, 1998; Klockars, 1977, 1979; Polsky, 1967). Ethics guide the first and last steps of an ethnography. Ethnographers stand at ethical crossroads throughout their research. The moments of ethical decision making are guided by codes of ethical practice and case examples of ethical dilemmas. However, many agree that ethical decision making should be handled situationally because of the complexity of each problem (British Sociological Association, 2001; Burgess, 1989; Goode, 1999; Goodwin, Pope, Mort, & Smith, 2003; Lee-Treweek, 2000; Punch, 1994; Riddell, 1989). This fact of ethnographic life sharpens the senses and ultimately refines and enhances the quality of the endeavor. (See Fetterman, 1998, for detailed discussion of the complexity of ethical decision making in ethnography.)

Institutional Review Boards (IRBs)

Ethnographic research is guided by the principles and standards briefly discussed above ranging from permission to reciprocity. These principles are used by ethnographers to guide and inform their ethnographic practice. In addition, ethnographic work supported by federal funding, is reviewed by IRB. Their approval is required before research can be conducted. A panel of researchers and administrators review ethnographic (and other) proposals to protect "human subjects" or those being researched from harm. (The need for the Board emerged as a result of research that did in fact damage individuals.) They require that research subjects have enough information to make an informed decision about their participation. In addition, they must be able to withdraw from the study at any time. Unnecessary risk to them must be eliminated. The benefits to society from the research must outweigh the risk. The IRB represents a significant hurdle for many ethnographers because the ethnographer does not always know what questions will be asked in the field. Many of the reviewers adopt a biomedical model instead of a sociological or anthropological model. Participants are often collaborators working together, rather than subject receiving a treatment. In spite of these legitimate concerns (Denzin, 2003; Madison, 2005, p. 118–119), it is possible and useful to draft well-written proposals with a detailed methods section, interview questions, formal survey questions, and informed consent forms acceptable to most IRBs. The proposal forces the ethnographer to think ahead and plan the effort (with the benefit of a "second set of eyes") in spite of the inevitable detours and diversions required in the field (see Sieber, Chapter 4, this volume).

Conclusion

This chapter has provided a brisk walk through the intellectual landscape of ethnography, leading the reader step by step through the ethnographic terrain, periodically stopping to smell the roses and contemplate the value of one concept or technique over another.

Each section of the chapter is built on the one before—as each step on a path follows the step before. Discussion about the selection of a problem or issue has been followed by a detailed discussion of guiding concepts. The ethnographer's next logical step is to become acquainted with the tools of the trade—the methods and techniques required to conduct ethnographic research and the equipment used to chisel out this scientific art form. A discussion of analysis in ethnographic research becomes more meaningful at this stage, once the preceding facets of ethnography have laid the foundation. Similarly, I have discussed the role of writing in the second-to-last section of this chapter because writing is one of the final stages in the process and because the meaning of writing in ethnography is amplified and made more illuminating by a series of discussions about what "doing ethnography" entails. Finally, ethics comes last because the complete ethnographic context is necessary to a meaningful discussion of this topic. Step by step, this chapter provides a path through the complex terrain of ethnographic work.

Exercises

I have presented three assignments that I have found useful in teaching ethnography at Stanford University.

The first is called "artifacts." It is designed to help students become aware of how knowledgeable and insightful they already are, relying primarily on their observational skills and common sense.

The assignment also highlights the limitations of observation and the need to ask questions and interview people to more accurately learn about what's going on.

The second assignment is designed to help students apply ethnographic concepts and techniques to their observations.

The third assignment is designed to provide students with an opportunity to apply ethnographic concepts and techniques to the art of interviewing.

Artifacts Assignment

Students are asked to bring in objects, pictures, and other relevant materials that they can share with a peer. These items should tell someone something about who they are.

Instructions to students:

1. You will be paired with another student in the classroom.

2. You will share your artifacts with them. However, you will not say anything about the items to them. In other words, you do not explain what the artifacts mean or say about your life. That's your partner's responsibility.

3. Your partner will share his/her artifacts with you at the same time. You will record what your partner's artifacts say about him/her but he/she will not help explain anything about the meaning of the artifacts.

4. Both of you will be taking notes on what you observe. Describe the items or artifacts and then briefly explain what the artifacts mean or tell you about the other person.

5. Then you will take turns explaining to your partner what you think the artifacts mean or say about your partner. Do not interrupt your peer. Let them complete their explanation. If you interrupt or correct them it will alter the rest of their explanation.

6. After your partner has completed his/her explanation, you can confirm and correct their story about you (based on the artifacts).

7. After completing the exercise, share
   a. how powerful the experience was,
   b. what you learned about your observational skills, and
   c. how important it is to be able to ask people what they think.

Observation Assignment

The second assignment involves observing a situation or event.

Student assignment:

1. We would like you to observe something for 15 to 20 minutes. Write it down (2–3 pages) and share it with us by posting it in the appropriate observation folder (in the virtual classroom).

2. Please read as many of your peer's postings as possible. Feel free to comment on them by posting messages in their folders.

3. Guidelines concerning the selection of a person or situation to observe.

4. You should pick a situation that allows you to observe individuals unobtrusively. We do not want you just staring at someone or making them feel uncomfortable for 15 to 20 minutes. However, this is an observation—not an interview, so observe and record your observations without interviewing the individual.

5. During this observation, we want you to use the ethnographic concepts and methods you have been reading about and hearing about in class. These tools should guide your observation. For example, you should be observing using concepts such as culture, holistic perspective, emic and etic perspective, nonjudgmental orientation, symbolism, and so on.

6. You should also be using methods such as participant or nonparticipant observation, outcroppings and relying on written information as is available.

7. You may want to read about field notes and thick description and verbatim quotations in Fetterman (1998, chap. 6) to assist you in this assignment. Remember detail is important in description. Concrete description is desired. You want to bring back a description of what you saw with enough detail that the reader feels like they were there or pretty close anyway.

8. We will discuss the assignment in class and provide a brief critique of each presentation about your observations.

Interview Assignment

The third assignment involves interviewing and critiquing the interview.

1. Your task is to conduct an informal interview with someone. Write it down (approximately, 2–3 pages and share it with us by posting it in the appropriate interview folder in the virtual classroom).
2. Please read as many of your peer's postings as possible. Feel free to comment on them by posting messages in their folders.
3. The interview should be guided by your readings and our discussions about chaps. 1, 2, 3, and 4 (Fetterman, 1998) with a focus on anthropological concepts and methods.
4. The interview should be nonjudgmental. This is not a "60 minutes" interrogation-type interview. It should be emically based. You are trying to learn about their perception of reality: who they are from their perspective.
5. You want detail. What do they look like? What are they doing? What do they say? You want verbatim quotations. Remember to use and build on your observational skills while conducting the interview.
6. What is the context/setting? Remember, permission is required to conduct an interview and to tape record it as well.
7. Post your assignment in the virtual classroom in the appropriate interview folder and be prepared to present your interview in class. We will provide a critique of the interview.

Notes

1. Many computers have built-in cameras to facilitate videoconferencing.
2. The accuracy of voice recognition software is improving. However, a fair amount of time is required to correct transcription errors. In addition, the software must be "trained" for each person's voice. This limits the software's utility for conducting interviews.

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CHAPTER 18

Group Depth Interviews

Focus Group Research

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Focus group research is among the most common research methods used by social scientists, marketers, policy analysts, health and social services professionals, political consultants, and other scientists and decision makers to gather information. Originally called "focussed" interviews, this technique came into vogue after World War II and has been a part of the social scientist's tool kit ever since. Focus groups emerged in behavioral science research as a distinctive member of the qualitative research family, which also includes individual depth interviewing, ethnographic participant observation, and projective methods, among others. Like its qualitative siblings, the popularity and status of focus groups among behavioral researchers has ebbed and flowed over the years, with distinctive patterns in particular fields. For example, in qualitative marketing studies, the use of focus groups has grown steadily since the 1970s; and today, business expenditures on focus groups are estimated to account for at least 80% of the $1.1 billion spent annually on qualitative research (Wellner, 2003).

In sociology, arguably the first field to embrace group research, qualitative research flourished through the 1950s, faded away in the 1960s and 1970s, and

Authors' Note: This chapter is an updated adaptation of Stewart, Shamdasani, and Rook (2007) and Shamdasani and Stewart (1992).

reemerged in the 1980s. Various patterns of focus group ascendance, decline, and revival characterize other fields, yet it is reasonable to conclude that focus group research has never enjoyed such widespread usage across an array of behavioral science disciplines and subfields as it does today. They are used by academic researchers, government policymakers, and business decision makers. Focus groups provide a rich and detailed set of data about perceptions, thoughts, feelings, and impressions of group members in the members' own words. They represent a remarkably flexible research tool, in that they can be adapted to obtain information about almost any topic in a wide array of settings and from very different types of individuals. Group discussions may be very general or very specific; they may be highly structured or quite unstructured. Visual stimuli, demonstrations, or other activities may be used within the context of a focus group to provide a basis for discussion. This flexibility makes the focus group a particularly useful tool and explains its popularity.

A focus group involves a group discussion of a topic that is the "focus" of the conversation. The contemporary focus group interview generally involves 8 to 12 individuals who discuss a particular topic under the direction of a professional moderator, who promotes interaction and assures that the discussion remains on the topic of interest. A typical focus group session will last from 1.5 to 2.5 hours. The most common purpose of a focus group interview is to stimulate an in-depth exploration of a topic about which little is known. Focus group research is uniquely suited for quickly identifying qualitative similarities and differences among people. Focus groups also provide an efficient means for determining the language people use when thinking and talking about specific issues and objects, and for suggesting a range of hypotheses about the topic of interest. Focus groups may be useful at virtually any point in a research program, but they are particularly useful for exploratory research when rather little is known about the phenomenon of interest. As a result, focus groups tend to be used very early in research projects and are often followed by other types of research that provide more quantifiable data from larger groups of respondents.

Focus groups have also been proven useful following analyses of large-scale, quantitative surveys. In this use, the focus group facilitates interpretation of quantitative results and adds depth to the responses obtained in the more structured survey. Focus groups also have a place as a confirmatory method that may be used for testing hypotheses. This application may arise when the researcher has strong reasons to believe that a hypothesis is correct, and where disconfirmation by even a small group would tend to result in rejection of the hypothesis.

Focus groups can produce quantitative data, but this is at odds with their nature and primary purpose, which is the collection of qualitative data. Focus groups, when properly designed and conducted, generate a rich body of data expressed in the respondents' own words and expressions. The degrees of freedom in participants' responses are high, unlike survey questionnaires that narrow responses to 5-point rating scales or other constrained response categories. In focus groups, participants can qualify their responses or identify important contingencies associated with their answers. Thus, responses have a certain ecological validity not found in traditional survey research. On the other hand, the data provided by focus groups may be idiosyncratic and unique to the group.

Although focus groups can be conducted in a variety of sites, ranging from homes to offices, they are typically held in commercial facilities designed especially for focus group interviewing. Such facilities provide one-way mirrors and viewing rooms where observers may unobtrusively observe an interview in progress. Focus group facilities may also include equipment for audio- or videotaping interviews and perhaps even small receivers for moderators to wear in their ears, so that observers may speak to them and thus provide input into interviews. In an age of online communication and videoconferencing, focus group facilities also tend to be equipped for "virtual" groups where the members may be broadly dispersed geographically and communicate through electronic media. Focus group facilities tend to be situated either in locations that are easy to get to, such as just off a major commuter traffic artery, or in places such as shopping malls, where people tend to gather naturally.

Today, focus groups are in use almost everywhere around the globe, but they are particularly important research tools in nations where survey research is difficult to conduct due to an unavailability of lists of representative customers, norms governing contact via telephone or mail, unreliable mail or telephone service, or language and literacy problems. In such settings, focus groups are often the only practical vehicle for collecting information, even when other methods might be more appropriate for the question at hand.

A variety of research needs lend themselves to the use of focus group interviews. Among the more common uses of focus groups are the following:

1. Obtaining general background information about a topic of interest;
2. Generating research hypotheses that can be submitted to further research and testing using more quantitative approaches;
3. Identifying similarities and differences among respondents with respect to specific behaviors, experiences, interests, perceptions, opinions, attitudes, or other characteristics;
4. Stimulating new ideas and creative concepts;
5. Diagnosing the potential for problems with a new program, service, or product;
6. Generating impressions of products, programs, services, institutions, or other objects of interest;
7. Learning how respondents talk about the phenomenon of interest (which may, in turn, facilitate the design of questionnaires, survey instruments, or other research tools that might be employed in more quantitative research); and
8. Interpreting previously obtained quantitative results.

Philosophical Perspectives on Focus Group Research

Focus groups are particularly well suited for exploratory research that addresses broad, "grand tour" questions about "why," "how," "when," "where," and "what kind." This is a distinctive advantage, because it is impossible to answer related quantitative questions efficiently—such as "how many," "how much," and "how often"—without first knowing, for example, "what kinds" to quantify. In this regard, qualitative and quantitative research complement one another, because the former helps identify important dimensions and aspects of phenomena, while the latter provides a means of assessing the frequency and/or magnitude of the types of phenomena discovered. Individual depth interviews also help answer these broad foundational questions. However, focus groups are often more efficient in terms of time and (sometimes) budgetary considerations in providing a quick overview of within-group differences, range of ideas, and so on. Furthermore, as with individual interviews, focus groups elicit a rich body of data expressed through respondents' own words and activities.

This begs the broader question of how focus groups differ from other scientific research tools—what purposes do they serve that are not served by other methods? The answer lies in the nature or character of the data generated by focus group interviews. Krippendorf (2004) distinguishes between two fundamental types of data: emic and etic. Emic data are data that arise in a natural or indigenous form. They are only minimally influenced by any structure imposed by the researcher or the research setting. Unobtrusive observation of computer users surfing the Web would be an example of emic data. Etic data, on the other hand, represent the researcher's imposed view of the situation. For example, the typical paper and pencil measure of attitude and beliefs impose the research view that certain attitudes and beliefs are important (or least important enough to draw attention to) and impose specific ways of responding.

Little of the research that is actually carried out can be described as completely etic or completely emic. Even the most structured type of research will be influenced to some extent by the idiosyncratic nature of the respondent and his or her environment. On the other hand, even the most natural of situations may not yield data that are completely emic, because the researcher must make decisions about what to attend to and what to ignore. Thus, it is perhaps more useful to think of a continuum of research, with some methods lying closer to the emic side of the continuum and some techniques lying closer to the etic side (see Table 18.1).

Focus groups, along with a few other techniques such as unstructured individual depth interviews, provide data that are closer to the emic side of the continuum

Table 18.1 The Emic and Etic Research Continuum

Etic Data ________				________ *Emic Data*
Experiments	Survey Research	Focus Groups	Individual Interviews	Ethnography

because they allow individuals to respond in their own words using their own categorizations and perceived associations. They are not completely void of structure, however, because the researcher does raise questions of one type or another and the artificial group setting also influences the character of data obtained from focus groups. Prototypic ethnographic research is probably the most emic due to its immersion in natural settings and bottom-up approach to data collection. Survey research and experimentation tend to produce data that are closer to the etic side of the continuum, because the response categories used by the respondent are generally prescribed by the researcher. These response categories may or may not be those with which the respondent is comfortable, though the respondent may still select an answer. And even when closed-ended survey questions are the only options available, some respondents elect to give answers in their own words, as most experienced survey researchers have discovered.

Neither emic nor etic data are inherently better or worse than the other; they simply differ. Both kinds of data have their place in social science research; they complement each other, each compensating for the limitations of the other. Indeed, one way to view social science research is as a process that moves from the emic to the etic and back, in a cycle. Phenomena that are not well understood are often first studied with tools that yield more emic data. As a particular phenomenon is better understood and greater theoretical and empirical structure is built around it, tools that yield more etic types of data tend to predominate. As knowledge accumulates, it often becomes apparent that the exploratory structure surrounding a given phenomenon is incomplete. This frequently leads to the need for data that are more emic, and the process continues. (Further discussion of the philosophical issues associated with the use of qualitative research and the complementarity of structured and unstructured approaches to social science research can be found in Bogdan & Biklen, 2006; Denzin & Lincoln, 2005; Marshall & Rothman, 2006; Maxwell, Chapter 7, this volume.)

Focus groups are widely used because they provide useful information and offer researchers a number of advantages. This information and the advantages of the technique come at a price, however. We review the relative advantages and limitations of focus group research below. We then present a discussion of the steps involved in the use and design of focus groups.

Advantages and Limitations of Focus Group Research

Advantages

Appropriate use of focus groups provides a number of advantages relative to other types of research as listed below:

1. Focus groups can collect data from a group of people much more quickly and at less cost than would be the case if each individual were interviewed separately.

They can also be assembled on much shorter notice than would be required for a more systematic, larger survey.

2. Focus groups allow researchers to interact directly with respondents. This provides opportunities for clarification and probing of responses as well as follow-up questions. Respondents can qualify responses or give contingent answers to questions. In addition, researchers can observe nonverbal responses, such as gestures, smiles, and frowns that may carry information that supplements and, on occasion, even contradicts, verbal responses.

3. The open-response format of focus groups provides researchers the opportunity to obtain large and rich amounts of data in the respondents' own words. Researchers can determine deeper levels of meaning, make important connections, and identify subtle nuances in expression and meaning.

4. Focus groups allow respondents to react to and build on the responses of other group members. This synergistic effect of the group setting may result in the production of data or ideas that might not have been uncovered in individual interviews.

5. Focus groups are very flexible. They can be used to examine a wide range of topics with a variety of individuals and in a variety of settings.

6. Focus groups may be one of the few research tools available for obtaining data from children or from individuals who are not particularly literate.

7. The results of focus group research are usually easy to understand. Researchers and decision makers can readily understand the verbal responses of most respondents. This is not always the case with more sophisticated survey research that employs complex statistical analyses.

8. Multiple individuals can view a focus group as it is conducted or review video- or audiotape of the group session. This provides a useful vehicle for creating a common understanding of an issue or problem. Such an understanding can be especially helpful for team building and for reducing conflict among decision makers.

Limitations

Although the focus group technique is a valuable research tool when used appropriately and offers a number of advantages, it is not a panacea for all research needs. It does have significant limitations, many of which are simply the negative sides of the advantages listed above:

1. The small numbers of respondents that participate in even several different focus groups and the convenient nature of most focus group recruiting practices significantly limit generalization to larger populations. Indeed, persons who are willing to travel to a locale to participate in a 1- to 2-hour group discussion may be quite different from the population of interest.

2. The interaction of respondents with one another and with the moderator has two potentially undesirable effects. First, the responses from members of the group

are not independent of one another; this restricts the generalizability of results. Second, the results obtained in a focus group may be biased by a very dominant or opinionated member. More reserved group members may be hesitant to talk.

3. The "live" and immediate nature of the interaction may lead a researcher or decision maker to place greater faith in the findings than is actually warranted. There is a certain credibility attached to the opinion of a live respondent that is often not present in statistical summaries.

4. The open-ended nature of responses obtained in focus groups often makes summarization and interpretation of results difficult. Statements by respondents are frequently characterized by qualifications and contingencies that make direct comparison of respondents' opinions difficult.

5. A moderator, especially one who is unskilled or inexperienced, may bias results by knowingly or unknowingly providing cues about what types of responses and answers are desirable.

Focus group research has been the subject of much controversy and criticism. Such criticism is generally associated with the view that focus group interviews do not yield "hard" data and the concern that group members may not be representative of a larger population because of both the small numbers and the idiosyncratic nature of the group discussion. Such criticisms are unfair, however. Although focus groups do have important limitations of which researchers should be aware, limitations are not unique to focus group research; all research tools in the social sciences have significant limitations. The key to using focus groups successfully in social science research is assuring that their use is consistent with the objectives and purpose of the research. It is also important to recognize and appreciate the philosophical underpinnings of focus group research.

There is a basis for criticizing focus group research that is poorly designed and applied to inappropriate research questions. These are problems with any type of research, but focus group research appears to have become especially prone to abuse and misapplication (Nelems, 2003). The abuse of the focus group research is, in large measure, a result of its apparent ease and low cost, relative to other tools for social science research. This is, of course, an illusion because a properly designed focus group, or a collection of focus groups addressing a common research question, is not any easier and cheaper than a survey or experimental design and, indeed, may be more difficult in some situations.

Designing, Conducting, and Analyzing Focus Group Research

As with any behavioral science research, methodological aspects and decisions are not merely minutiae that are relegated to a study's appendix. Rather, research design considerations are critical inputs that exert a strong influence on the quality and usefulness of the data collected. The key design elements of focus groups are generally not unique; in fact, they are common to other types of both qualitative and quantitative research. On the other hand, the communal nature of focus groups makes some research design issues loom large, particularly those related to the composition and likely interpersonal dynamics of group participants. The main design elements of focus group research and their attendant considerations are summarized in Table 18.2, and elaborated in the following discussion.

Research Purpose and Data

It is especially important that a researcher have a well-framed research purpose prior to initiating focus group research. This research question will guide the type of selection of respondents, the types of questions posed during the group session, and the types of analyses conducted following the group session. The exploratory nature of focus group research makes it tempting for researchers to use it as a substitute for constructing clear research questions. Such use of focus group research is likely to produce highly unsatisfactory results, however. Unfocused questions addressed to an inappropriate set of respondents by an ill-prepared moderator may not only fail to produce useful information, but actually mislead the researcher. Focus groups are designed to do exactly what the name implies—focus.

A focus group is not a freewheeling conversation among group members; it should have a clear focus and an identifiable agenda. In specifying the research purpose and key question(s), it is useful to consider how the data will be used and what decisions it will inform and guide. Will the data help marketers select the most promising new product concepts or advertising campaign; will it give government officials and health care professionals ideas for improving the lives of homeless people; or will the focus groups help gerontologists better understand the dimensions and causes of depression among the recently widowed? Despite the diversity of their research agenda, all would benefit from "researching backward" (Andreasen, 1985) to appreciate exactly how and for what purposes the focus group data will be used.

This step provides a focus for the next one—specifying the research constructs and measures; that is, questions. A clear statement of the research purpose helps identify what kinds of information are desirable and from whom it should be

Table 18.2 Focus Group Design Elements and Key Considerations

Design Element	*Key Considerations*
Research purpose	Data use/application constructs and measures
Data type and amount	Number and location of groups, type of questions
Sample	Group composition: homogeneity or heterogeneity, likely group dynamics
Interview guide	Mix of direct and indirect questions, number of questions
Group moderator	Role and responsibilities, required specialization
Analysis and interpretation	Qualitative or quantitative

obtained. A hypothetical focus group on consumers' perceptions about cars today, designed to help automobile manufacturers better respond to their concerns and aspirations, might help prioritize the study of consumers' (1) current car satisfaction, (2) attitudes and attributions about current gas prices, (3) awareness and evaluation of alternative fuel vehicles, and (4) next car purchase plans. Specifying these four key constructs or topics leads directly to crafting questions that tap the behavioral domain of each.

Clarity about the key research topics and questions helps address decisions concerning the amount and type of data that are required to answer the research questions. These issues have implications for the number of focus groups that are fielded and their geographic locations. The amount of data required will vary with the purpose of the research, the complexity of its design, magnitude of the focal issue or decision, and the extent to which conclusions can be reached easily and definitively. The type of data required is an important and relatively subtle issue. There is some tendency to assume that focus groups entirely comprise moderators' questions and participants' answers. While a Q&A format well serves the purposes of most focus groups, many research situations require alternative "questioning" tools. When focus group participants are likely to be unable to articulate answers—when a topic is sensitive or controversial, or when social desirability influences may distort responses—researchers may need to employ the less direct approaches that projective methods such as story telling, picture drawing, sentence completion, and psychodrama provide (Rook, 2007). In other situations, participants' body language may provide valuable data; for example, when toy manufacturers conduct focus groups with children to determine a particular toy's play value.

Group Composition

Once the researcher has generated a clear statement of the research purpose and key questions, he or she can move to the second stage of focus group research. As for a survey, it is important for the researcher to identify a sampling frame—that is, a list of people (households, organizations) the researcher has reason to believe is representative of the larger population of interest. The sampling frame is the operational definition of the population. The identification of a sound sampling frame is far more critical in large-scale survey research than it is for focus group research, however. Because it is generally inappropriate to generalize far beyond the members of focus groups, the sampling frame need only be a good approximation of the population of interest. Thus, if the research is concerned with middle-class parents of schoolchildren, a membership list for the local PTA might be an appropriate sampling frame.

Indeed, random samples, which are the rule in most survey research, are less frequently employed in focus group research. The reason for this is that the topics of some focus group discussions are topics that require special expertise, experience, or unique knowledge. For example, a random sample of the population of any given country would be unlikely to produce individuals who could talk knowledgeably about the direction of information technology over the next 50 years or persons who could discuss their feelings about having contracted AIDS. Thus, purposive sampling, in which respondents are purposely selected because they have certain characteristics, is often used in focus group research. Random sampling is also common in recruiting focus group participants, but it is important to recognize that the representativeness of any set of focus group participants is diminished by their participation in the group experience.

Unlike survey research, where data are obtained from respondents whose answers are independent of one another, the design of focus group research must also include consideration of the likely dynamics that will be produced by any particular combination of individuals (Carey & Smith, 1994). For example, the interaction among a group of 15-year-olds will be very different when their parents are a part of the group versus when they are alone. Similarly, men may respond differently in groups composed only of other men than the way they would in groups made up of a mixture of men and women. Furthermore, it may be unwise to include individuals whose socioeconomic circumstances are quite different. This idea is illustrated in the focus group application discussed below.

FOCUS GROUP APPLICATION: WHITE- AND BLUE-COLLAR BEER DRINKERS

Beer is a mass market consumer product whose enduring popularity extends across age, gender, and social class boundaries. However, this does not mean that the questions that beer marketers need answered are best addressed by focus groups comprising individuals who reflect beer's demographically diverse consumer base. For several years, a major American beer manufacturer fielded focus groups that were populated with young (aged 21–27) male beer drinkers from different occupational, educational, and income strata. The manufacturer's beverage products and their advertising are historically designed to appeal to a broad spectrum of male consumers, so it seemed logical to include everybody in the focus groups. Over time, it became clear that this was a bad idea. Although the respondents performed effectively in answering questions, social class stereotypes and mutual discomforts surfaced predictably, and sometimes interfered with the work at hand. The upscale, yuppie-type men chatted about their consuming exclusive imported brands in chic urban clubs, and they tended to wrinkle their noses when working class men described drinking a case of Budweiser on a fishing trip. Similarly, the blue-collar guys viewed the yuppie world as overprivileged and effete, and tended to dismiss their refined brand preferences for "sissy" beers. Often, the hostility was below the surface, but it was visible in respondents' mutually disapproving body language. Sometimes, it emerged in snide comments. The manufacturer eventually decided to conduct separate focus groups with white- and blue-collar beer drinkers, which took the battle of the social classes out of the equation.

Care also needs to be exercised in mixing groups across cultures. For example, in a 90-minute focus group session involving strangers, participants from more aggressive cultures are likely to dominate. Therefore, the safest strategy would be to avoid such mixing of participants from diverse cultures. Additionally, some topics and issues (e.g., sexual habits and contraception use) are perceived to be more personal and sensitive by members of some cultural groups than by others (Asians compared with Westerners, for instance). Thus, the moderators of focus groups investigating such sensitive topics need to exercise a great deal of tact and diplomacy, because members of some cultures are quite reserved and reluctant to discuss openly behaviors and issues that may lead to embarrassment or "loss of face." The general preference for homogeneous group composition has logical foundations, but one caveat should be mentioned. Many of the social science studies whose findings discouraged fielding demographically or culturally diverse focus groups were conducted years ago, and most have not been replicated recently. Arguably, Americans today would be more comfortable sitting among people who reflect the nation's demographic and cultural diversity than their parents or grandparents were.

A growing body of research has focused on the use of focus groups with various special populations. Such research has examined the unique issues that arise in the use of focus groups in developing nations (Folch-Lyon, de la Macorra, & Schearer, 1981; Fuller, Edwards, Vorakitphokatorn, & Sermsri, 1993; Knodel, 1995; Stewart & Shamdasani, 1992), with children (Hoppe, Wells, Morrison, Gillmore, & Wilsdon, 1995; Krueger & Casey, 2000; Vaughn, Schumm, & Singagub, 1996), and among low-income and minority populations (Jarrett, 1993; Magill, 1993). Although such populations do require some adaptation of technique, they have all been included successfully in focus group research.

There is no "best" mix of individuals in a focus group. Rather, the researcher needs to consider what group dynamic is most consistent with the research objectives. If the interaction of children and their parents is important for purposes of the research, then groups should be composed of parents and their children. On the other hand, if the focus of the research is on adolescents' perspectives on a topic, the presence of parents in the group may reduce the willingness of the adolescents to speak out and express their feelings. In the latter case, it would be more consistent with research objectives for the researcher to design groups that include only adolescents.

The interaction among members of a focus group adds a dimension to data collection that is not common in other forms of social science research. Because the results obtained from a group are the outcome of both the individuals in the group and the dynamics of the group interaction, it is common for focus group researchers to use several groups that differ with respect to composition. Indeed, it is uncommon for focus group research to use only a single group. More often, the research includes multiple groups composed of different types of individuals and different mixes of individuals. The specific number of groups that may be included in any research project is a function of the number of distinct *types of individuals* from which the researcher wishes to obtain data and the number of *mixtures of individuals* of interest to the researcher.

The Interview Guide

Although focus groups are relatively unstructured compared with the typical survey or other types of quantitative research, they are not completely without structure, as discussed earlier. The group's discussion needs to be guided and directed so that it remains focused on the topic of interest and the questions prepared. The moderator plays an important role in maintaining this focus, but an especially important tool for creating the agenda for a focus group discussion is the interview guide. The interview guide for a focus group discussion generally consists of a set of very general open-ended questions about the topic or issue of interest. It does not include all the questions that may be asked during the group discussion; rather, it serves to introduce broad areas for discussion and to assure that all the topics relevant to the research are included in the research. The interview guide is not a script for the discussion, nor should it be regarded as an immutable agenda. Rather, it is simply a guide, and it may be modified in response to the discussion and interaction among the respondents. As Grant McCracken (1988) astutely observes, the discussion guide "must not be allowed to destroy the elements of freedom and variability within the interview" (p. 25).

A typical interview guide for a 90-minute discussion ideally includes no more than 10 to 20 questions. Generally, questions of a more general nature are raised first, and more specific issues are raised later in the guide. This assures that background information, context, and broader issues are discussed before the group focuses on very specific issues. The use of very specific questions early in a discussion often results in a premature narrowing of the focus of the group and reduces the richness of the information that is obtained. An unfortunate trend in focus groups in marketing research is the tendency to ask too many questions. Veteran focus group moderator Naomi Henderson (2004) estimates that focus groups today include twice as many questions in the same amount of time as they did 15 years ago. This reflects both a poor understanding of the basic nature of qualitative research as well as a pragmatic desire to "get more" for each research dollar. It also tends to have a negative effect on a focus group's overall quality, the depth of responses, and the nature of participants' interactions. Groups with too many questions often devolve into surveys in disguise (Rook, 2003), as the following anecdote illustrates.

Focus Group Application: A Discussion Guide With No Discussion

An American cosmetic manufacturer had just hired an ad agency and sought its ideas for improving sales of lip gloss to preteen girls. The idea surfaced to conduct a "grand tour" focus group exploring girls' cosmetic usage behavior, their likes and dislikes of various cosmetic products, and their attitudes about different cosmetic brands. The agency recommended a gifted moderator who specialized in interviewing children, but the manufacturer declined any assistance in preparing the discussion guide. When they arrived at the focus group facility, the ad agency team members were surprised to see a large stack (approximately one half reams) of paper on the table in front of each girl's seat. "This should be interesting," whispered the agency's account executive. The moderator got the group off to a good start, inviting the girls to introduce themselves and share their current hobbies and interests. Things went downhill quickly. The discussion guide was almost entirely comprised of 15 to 20 evaluative ratings of 30 different lip gloss products. After laboring over their written evaluations of the first product concept, the girls were asked to explain why they liked or disliked it. Flavor issues loomed large: "I think blueberry is icky." At this point, the girls remained enthusiastic, but then it was back to the looming stack of paper concepts and ratings. They quickly appreciated how much work they had to get through and hunkered down in silence to complete the required forms. Much of the moderator's time was spent collecting and collating the completed materials. As the girls completed each consecutive concept, their discussion of likes and dislikes diminished to brief phrases, and some declined to comment at all, knowing how much paperwork remained. Each concept evaluation took about 10 minutes, during which the focus group observers watched the girls working away in silence. One new ad agency executive who had never attended a focus group asked an agency research staff member, "Is this what focus groups are like?" He was told "No, this is a group survey, unfortunately." He responded, "It's like watching people take the SAT."

Too many questions was only one problem with the lip gloss focus group. Given the ostensibly broad, exploratory purpose of the research, restricting its scope to evaluative ratings of alternative product concepts failed entirely to achieve the main objective. Also, given the age of the participants, other approaches to asking questions would have yielded richer data. For example, more playful and indirect questions, or actually trying (rather than reading about) different products would have generated greater enthusiasm and within-group interaction.

The Role of the Focus Group Moderator

The moderator is the key to assuring that the group discussion goes smoothly. The focus group moderator is generally a specialist who is well trained in group dynamics and interview skills. Depending on the intent of the research, the moderator may be more or less directive with respect to the discussion, and often is quite nondirective, letting the discussion flow naturally, as long as it remains on the topic of interest. Indeed, one of the strengths of focus group research is that it may be adapted to provide the most desirable level of focus and structure. If the researcher is interested in how parents have adapted to the child care requirements created by dual careers, the moderator can ask very general and nonspecific questions about the topic to determine the most salient issues on the minds of the participants. On the other hand, if the researcher is interested in parents' reactions to a very specific concept for child care, the moderator can provide specific information about the concept and ask very specific questions.

The amount of direction provided by the moderator influences the types and quality of the data obtained from the group. The moderator provides the agenda or structure for the discussion by virtue of his or her role in the group. When a moderator suggests a new topic for discussion by asking a new question, the group has a tendency to comply. This is important for assuring that all the topics of interest are covered in the time available. A group discussion might never cover particular topics or issues unless the moderator intervenes. On the other hand, the frequency and type of intervention by the moderator clearly affects the nature of the discussion. This raises the question of the most appropriate amount of structure for a given group. There is, of course, no best answer to this question, because the amount of structure and the directiveness of the moderator must be determined by the broader research agenda that gave rise to the focus group: the types of information sought, the specificity of the information required, and the way the information will be used.

There is also a balance that must be struck between what is important to members of the group and what is important to the researcher. Less structured groups will tend to pursue those issues and topics of greater importance, relevance, and interest to the group. This is perfectly appropriate if the objective of the researcher is to learn about the things that are most important to the group. Often, however, the researcher has rather specific information needs. Discussion of issues relevant to these needs may occur only when the moderator takes a more directive and structured approach. It is important for the researcher to remember that when this occurs, participants are discussing what is important to the researcher, not necessarily what they consider significant.

Analysis and Interpretation of Focus Group Research

The most common analyses of focus group results involve transcripts of the group interviews and discussion of the conclusions that can be drawn. There are occasions, however, when transcripts are unnecessary. When decisions must be made quickly and the conclusions of the research are rather straightforward, a brief summary may be all that is necessary and justifiable. In some cases, there may be time or budget constraints that prevent detailed analysis. In other cases, all interested parties and decision makers may be able to observe or participate in the groups, so there may be little need for detailed analyses or reports.

Apart from the few occasions when only short summaries of the focus group discussions are required, all analytic techniques for focus group data require transcription of the interviews as a first step. Transcription not only facilitates further analysis, it establishes a permanent written record of the interviews that can be shared with other interested parties. The amount of editing an analyst does on a transcribed interview is a matter of preference. Transcriptions are not always complete, and the moderator may want to fill in gaps and missing words, as well as correct spelling and typographical errors. There is a danger in this, of course, because the moderator's memory may be fallible or knowledge of what was said later in the course of the interview may color his or her memory of what happened earlier. Although editing may increase readability, it is important that the character of the respondents' comments be maintained, even if at times they use poor grammar or appear to be confused. Because one use of focus group interviewing is to learn how respondents think and talk about a particular issue, too much editing and cleaning of the transcript is undesirable. Too much editing and cleaning tends to censor ideas and information, often based on the analyst's preconceived ideas.

It should be noted, however, that the transcript does not reflect the entire character of the discussion. Nonverbal communication, gestures, and behavioral responses are not reflected in a transcript. Thus, the interviewer observer may wish to supplement the transcript with some additional observational data that were obtained during the interview, such as a videotape or notes by an observer. Such observational data may be quite useful, but they will be available only if their collection is planned in advance. Preplanning of the analyses of the data to be obtained from focus groups is as important as it is for any other type of research.

As with other types of research, the analysis and interpretation of focus group data require a great deal of judgment and care. Unfortunately, focus group research is easily abused and often inappropriately applied. A great deal of the skepticism about the value of focus groups probably arises from (a) the perception that focus group data are subjective and difficult to interpret and (b) the concern that focus group participants may not be representative of a larger population because of both the small numbers and the idiosyncratic nature of the group discussion.

The analysis and interpretation of focus group data can be as rigorous as the analysis and interpretation generated by any other method. Focus group data can even be quantified and submitted to sophisticated mathematical analyses, though the purpose of focus group interviews seldom requires this type of analysis. Indeed, there is no one best or correct approach to the analysis of focus group data. The nature of the analysis of focus group interview data should be determined by the research question and the purpose for which the data are collected. This, in turn, has implications for the validity of the findings generated from focus groups. Researchers should constantly be aware of the possible sources of bias at various stages of the focus group research process and take appropriate steps to deal with threats to the validity of the results.

A number of books and papers on focus group research have appeared in recent years (e.g., Fern, 2001; Greenbaum, 2000; Krueger & Casey, 2000; Morgan, 1997; Templeton, 1994). Although these publications are useful, their focus has tended to be more on the mechanics of the interviews themselves rather than on the analysis of the data generated in focus group sessions (see Stewart, Shamdasani, & Rook, 2007, for an exception). Where analysis is treated, the discussion is often limited to efforts to identify key themes in focus group sessions. Researchers interested in more sophisticated approaches have limited options. They can consult the rather voluminous literature on content analysis that exists outside the marketing domain, but this literature is not always readily accessible to researchers, particularly those outside academic settings. The more common approaches to content analysis are described below.

The Cut-and-Sort Technique

The cut-and-sort technique is a quick and cost-effective method for analyzing a transcript from a focus group discussion. This process may also be readily carried out on any computer with a word processing program. Regardless of whether scissors or a personal computer is employed, this method yields a set of sorted materials that provides the basis for the development of a summary report. Each topic is treated, in turn, with a brief introduction. The various pieces of interview transcription are used as supporting materials and incorporated within an interpretative analysis.

Although the cut-and-sort technique is useful, it tends to rely very heavily on the judgment of a single analyst. This analyst determines which segments of the transcript are important, develops a categorization system for the topics discussed by the group, selects representative statements regarding these topics from the transcript, and develops an interpretation of what it all means. There is obviously much opportunity for subjectivity and potential bias in this approach. Yet it shares many of the characteristics of more sophisticated and time-consuming approaches. It may be desirable to have two or more analysts independently code the focus group transcript. The use of multiple analysts provides an opportunity to assess the reliability of coding, at least with respect to major themes and issues. When determination of the reliability of more detailed types of codes is needed, more sophisticated content-analytic coding procedures are required.

Formal Content Analysis

Every effort to interpret a focus group represents analysis of content. Some efforts are more formal than others, however. There are rigorous approaches to the analysis of content, approaches that emphasize the reliability and replicability of observations and subsequent interpretation (Neuendorf, 2001). These approaches include a variety of specific methods and techniques that are collectively known as content analysis (Krippendorf, 2004). There are frequent occasions when the use of this more rigorous approach is appropriate for the analysis of data generated by focus groups. In addition, the literature on content analysis provides the foundation for computer-assisted approaches to the analysis of focus group data. Computer-assisted approaches to content analysis are increasingly being applied to focus group data because they maintain much of the rigor of traditional content analysis while greatly reducing the time and cost required to complete such analysis. It is important to note that in addition to verbal communication, there is a great deal of communication that takes place in a focus group discussion that is nonverbal and that is not captured in the written transcript. It is therefore desirable to videotape focus group sessions, so that the nonverbal behavior of participants can be recorded and coded. If videotaping is not possible, an observer may be used to record nonverbal behavior. By subjecting nonverbal communication to content analysis, the researcher can enhance the overall information content of the focus group research.

Janis (1965) defined content analysis as

> any technique (a) for the classification of the sign-vehicles (b) which relies solely upon the judgments (which theoretically may range from perceptual discrimination to sheer guesses) of an analyst or group of analysts as to which sign-vehicles fall into which categories, (c) provided that the analyst's judgments are regarded as the report of a scientific observer. (p. 55)

A sign-vehicle is anything that may carry meaning, though most often it is likely to be a word or set of words in the context of a focus group interview. Sign-vehicles may also include gestures, facial expressions, or any of a variety of other means of communication, however. Indeed, such nonverbal signs may carry a great deal of information and should not be overlooked as sources of information.

A substantial body of literature now exists on content analysis, including books by Krippendorf (2004), Neuendorf (2001), and West (2001). A number of specific instruments have been developed to facilitate content analysis, including the Message Measurement Inventory (Smith, 1978) and the Gottschalk-Gleser Content Analysis Scale (Gottschalk, Winget, & Gleser, 1969). The Message Measurement Inventory was originally designed for the analysis of communications in the mass media, such as television programming and newsmagazines. The Gottschalk-Gleser Content Analysis Scale, on the other hand, was designed for the analysis of interpersonal communication. Both scales have been adapted for other purposes, but they are generally representative of the types of formal content analysis scales that are in use.

Although content analysis is a specific type of research tool, it shares many features in common with certain types of research. The same stages of the research process are found in content analysis as are present in any research project (Krippendorf, 2004): data making, data reduction, inference, analysis, validation, testing for correspondence with other methods, and testing hypotheses regarding other data.

Data Making. Data used in content analysis include human speech, observations of behavior, and various forms of nonverbal communication. The speech itself may be recorded, and, if video cameras are available, at least some of the behavior and nonverbal communication may be permanently archived. Such data are highly unstructured, however, at least for the purposes of the researcher. Before the researcher can analyze the content of a focus group session, he or she must convert it into specific units of information. The particular organizing structure a researcher chooses will depend on the particular purpose of the research, but there are specific steps in the structuring process that are common to all applications. These steps are unitizing, sampling, and recording.

Unitizing involves defining the appropriate unit or level of analysis. It would be possible to consider each word spoken in a focus group session as a unit of analysis. Alternatively, the unit of analysis could be a sentence, a sequence of sentences, or a complete dialogue about a particular topic. Krippendorf (2004) suggests that in content analysis, there are three kinds of units that must be considered: sampling units, recording units, and context units. Sampling units are those parts of the larger whole that can be regarded as independent of each other. Sampling units tend to have physically identified boundaries. For example, sampling units may be defined as individual words, complete statements of an individual, or the totality of an exchange between two or more individuals.

Recording units tend to grow out of the descriptive system that is being employed. Generally, recording units are subsets of sampling units. For example, the set of words with emotional connotations would describe certain types of words and would be a subset of the total words used. Alternatively, individual statements of several group members may be recording units that make up a sampling unit that consists of all the interaction concerned with a particular topic or issue. In this latter case, the recording units might provide a means for describing those exchanges that are hostile, supportive, friendly, and so forth.

Context units provide a basis for interpreting a recording unit. They may be identical to recording units in some cases, whereas in other cases they may be quite independent. Context units are often defined in terms of the syntax or structure in which a recording unit occurs. For example, in marketing research, it is often useful to learn how frequently evaluative words are used in the context of describing particular products or services. Thus, context units provide a reference for the content of the recording units.

Sampling units, then, represent the way in which the broad structure of the information within the discussion is divided. Sampling units provide a way of organizing information that is related. Within these broader sampling units, the recording units represent specific statements and the context units represent the environment or context in which the statement occurs. The way in which these units are defined can have a significant influence on the interpretation of the content of a particular focus group discussion. These units can be defined in a number of different ways. The definition of the appropriate unit of analysis must be driven by both the purpose of the research and the ability of the researcher to achieve reliability in the coding system. The reliability of such coding systems must be determined empirically, and in many cases involves the use of measures of interrater agreement.

It is seldom practical to try to unitize all discussion that arises in a focus group. When multiple focus groups are carried out on the same general topic, complete unitization becomes even more difficult. For this reason, most content analyses of focus groups involve some sampling of the total group discussion for purposes of analysis. The analyst may seek to identify important themes and sample statements within themes, or use some other approach, such as examining statements made in response to particular types of questions, or at particular points in the conversation. Like other types of sampling, the intent of sampling in content analysis is to provide a representative subset of the larger population. It is relatively easy for a researcher to draw incorrect conclusions from a focus group if he or she does not take care to ensure representative sampling of the content of the group discussion. One can support almost any contention by taking a set of unrepresentative statements out of the context in which they were spoken. Thus, it is important for the analyst to devise a plan for sampling the total content of group discussions. The final stage of data making is the recording of the data in such a way so as to ensure their reliability and meaningfulness. The recording phase of content analysis is not simply the rewriting of a statement of one or more respondents. Rather, it is the use of the defined units of analysis to classify the content of the discussion into categories such that the meaning of the discussions is maintained and explicated. It is only after the researcher has accomplished this latter stage that he or she can claim to actually have data for purposes of analysis and interpretation.

The recording phase of content analysis requires the execution of an explicit set of recording instructions. These instructions represent the rules for assigning units (words, phrases, sentences, gestures, and so on) to categories. These instructions must address at least four different aspects of the recording process (Krippendorf, 2004):

1. The nature of the raw data from which the recording is to be done (transcript tape recording, film, and so on)

2. The characteristics of coders (recorders), including any special skills such as familiarity with the subject matter and scientific research
3. The training that coders will need to do the recording
4. The specific rules for placing units into categories

The specific rules referred to above are critical to the establishment of the reliability of the recording exercise and the entire data-making process. Furthermore, it is necessary that the researcher make these rules explicit and demonstrate that the rules produce reliable results when used by individuals other than those who developed them in the first place. Lorr and McNair (1966) question the practice of reporting high interrater reliability coefficients when they are based solely on the agreement of individuals who have worked closely together to develop a coding system. Rather, these researchers suggest that the minimum requirement for establishing the reliability of a coding system is a demonstration that judges using only the coding rules exhibit agreement.

Once a set of recording rules has been defined and demonstrated to produce reliable results, the researcher can complete the data-making process by applying the recording rules to the full content of the material of interest. Under ideal circumstances, recording will involve more than one judge, so that the coding of each specific unit can be examined for reliability and sources of disagreement can be identified and corrected. There is a difference between developing a generally reliable set of recording rules and assuring that an individual element in a transcript is reliably coded.

The assessment of the reliability of a coding system may be carried out in a variety of ways. As noted above, there is a difference between establishing that multiple recorders are in general agreement (manifest a high degree of interrater reliability) and establishing that a particular unit is reliably coded. The researcher must decide which approach is more useful for the given research question. It is safe to conclude that in most focus group projects, general rater reliability will be more important because the emphasis is on general themes in the group discussion rather than specific units.

Computation of a coefficient of agreement provides a quantitative index of the reliability of the recording system. There exists a substantial literature on coefficients of agreement. Treatment of this literature and issues related to the selection of a specific coefficient of agreement are beyond the scope of this chapter. Among the more common coefficients in use are kappa (Cohen, 1960), pi (Scott, 1955), and alpha (Krippendorf, 2004). All these coefficients correct the observed level of agreement (or disagreement) for the level that would be expected by chance alone. Krippendorf offers a useful discussion of reliability coefficients in content analysis, including procedures for use with more than two judges (see also Spiegelman, Terwilliger, & Fearing, 1953).

Data making tends to be the most time-consuming of all the stages in content analysis. It is also the stage that has received the greatest attention in the content analysis literature. The reason for this is that content analysis involves data making after observations have been obtained, rather than before. Content analysis uses the observations themselves to suggest what should be examined and submitted to further analysis, whereas many other types of research establish the specific domain of interest prior to observation. In survey research, much of the data making occurs prior to administration of the survey. Such data making involves identification of reasonable alternatives from which a respondent selects an answer. Thus, data making is a step in survey research, and all types of research, but it occurs prior to observation. In content analysis, data making occurs after observation.

Data Analysis. The recording or coding of individual units is not content analysis. It is merely the first stage in preparation for analysis. The specific types of analyses that might be used in a given application will depend on the purpose of the research. Virtually any analytic tool may be employed, ranging from simple descriptive analysis to more elaborate data reduction and multivariate associative techniques. Much of the content analysis work that occurs in the context of focus group data tends to be descriptive, but this need not be the case. Indeed, although focus group data tend to be regarded as qualitative, proper content analysis of the data can make them amenable to the most sophisticated quantitative analysis. This is well illustrated by development of computer-assisted methods for content analysis.

Computer-Assisted Content Analysis. Content analysts were quick to recognize the value of the computer as an analytic tool. The time-consuming and tedious task of data making can be greatly facilitated through use of the computer. Computers can be programmed to follow the data-making rules described earlier. The importance of assuring that these rules are well designed is made even clearer in the context of their use by a computer. In recent years, computer-assisted interpretation of focus group interviews has received attention and has built on the earlier foundations of research on content analysis.

The computer is capable of a great deal more than automation of search, find, and cut-and-paste activities. One problem with simple counting and sorting of words is that these procedures lose the contexts in which the words occur. For example, a simple count of the frequency with which emotionally charged words are used loses information about the objects of those emotional words. Because the meanings of words are frequently context dependent, it is useful to try to capture context. This is one reason that content analysts recommend the identification and coding of context units as a routine part of content analysis.

One computer-assisted approach to capturing the context as well as content of a passage of text is the key-word-in-context (KWIC) technique. In the KWIC approach, the computer is used to search for key words, which are then shown along with the text that surrounds them. The amount of text obtained on either side of the key word can be controlled by specification of the number of words or letters to be printed. One of the earliest computer programs for KWIC analyses was the General Inquirer (Stone, Dunphy, Smith, & Ogilvie, 1966), which is still in use today (the home page can be found at www.wjh.harvard.edu/~inquirer). The General Inquirer uses a theoretically derived dictionary for classifying words. A variety of similar systems have since been developed, and many use specially designed dictionaries for particular applications. Some of these programs are simply designated as KWIC, whereas others are named for particular applications for which KWIC may be used.

Among the more frequently cited software programs for content analysis are TEXTPACK (Mohler & Zuell, 1998), Concordance (Watt, 2004), Wordstat (Provalis Research, 2005), and TextQuest (Social Science Consulting, 2005). Software for text analysis is frequently reviewed in journals such as *Computers and the Humanities* and *Literary and Linguistic Computing*. Specialized dictionaries for use in conjunction with text analysis programs such as the General Inquirer and TEXTPACK are also available. Antworth and Valentine (1998) provide a brief introduction to several of these specialized programs and dictionaries.

Work on content analysis has also built on the research on artificial intelligence and in cognitive science. This more recent work recognizes that associations among words are often important determinants of meaning. Furthermore, meaning may be related to the frequency of association of certain words, the distance between associated words or concepts (often measured by the number of intervening words), and the number of different associations. The basic idea in this work is that the way people use language provides insights into the way people organize information, impressions, and feelings in memory and, thus, how they tend to think. The view that language provides insight into the way individuals think about the world has existed for many years.

The anthropologist Edward Sapir (1929) has noted that language plays a critical role in how people experience the world. Social psychologists have also long had an interest in the role language plays in the assignment of meaning and in adjustment to the environment (see e.g., Bruner, Goodnow, & Austin, 1956; Chomsky, 1965; Sherif & Sherif, 1969). In more recent years, the study of categorization has become a discipline in its own right and has benefited from research on naturalistic categories in anthropology, philosophy, and developmental psychology, and the work on modeling natural concepts that has occurred in the areas of semantic memory and artificial intelligence (see Hahn & Ramscar, 2001; Medin, Lynch, & Solomon, 2000, for a review of this literature).

Such research has been extended to the examination of focus groups. Building on theoretical work in the cognitive sciences (Anderson, 1983; Grunert, 1982), Grunert and Bader (1986) developed a computer-assisted procedure for analyzing the proximities of word associations. Their approach builds on prior work on content analysis as well. Indeed, the data-making phase of the approach uses the KWIC approach as an interactive tool for designing a customized dictionary of categories. The construction of a customized dictionary of categories is particularly important for the content analysis of focus groups because the range and specificity of topics that may be dealt with by focus group interviews is very broad, and no general purpose dictionary or set of codes and categories is likely to suit the needs of a researcher with a specific research application.

For example, to analyze focus group sessions designed to examine the way groups of respondents think and talk about computer workstations, the researcher will need to develop a dictionary of categories that refer specifically to the features of workstations, particular applications, and specific work environments. To analyze focus groups designed to examine the use of condoms among inner-city adolescents, it is likely that a dictionary of categories that includes the slang vernacular of the respondents will be required to capture the content of the discussion. Although the dictionaries developed for other applications may provide some helpful suggestions, the specificity of the language used by particular groups of respondents to discuss specific objects within given contexts almost always means that the focus group analyst will have to develop a customized categorization system.

Once the data-making phase is complete, the researcher can analyze the associative structure of the discussion content. He or she accomplishes this by counting the distances between various cognitive categories. Distance, or the proximity of two categories of content, is defined as the number of intervening constructs. Thus, two constructs that appear next to one another would have a distance of 1. To simplify computations, Grunert and Bader (1986) recommend examining categories that are at a maximum value of 10. This maximum value is then used as a reference point and distances are subtracted from it to obtain a numeric value that varies directly (rather than inversely) with intensity of association. This procedure yields a proximity value rather than a distance measure; that is, the higher scores represent closer associations among categories. Because most categories appear more than once, the measures of association are summed over all occurrences to obtain a total proximity score for each pair of constructs. These proximity data may then be used for further analysis.

Whether the amount of effort needed for further analysis is justified in focus group applications depends on a variety of factors—time and budget constraints, the nature of the research question, and the availability of a computer and the necessary software. The important point is that the level and detail of analysis of focus group data can be increased considerably through the use of the computer. At the same time, the computer can be an extremely useful tool for data reduction. It can also be used to uncover relationships that might otherwise go unnoticed. Thus, like most of the research tools in the social sciences, the focus group interview has benefited from the advent of the computer. Users of focus group interviews have also become increasingly facile in the use of the computer as an aid to the analysis, summarization, and interpretation of focus group data.

Virtual Focus Groups

Technology has made it possible to link people who are scattered across very broad geographic regions. This has made it possible to conduct interviews with highly specialized groups that might be difficult to assemble in a single location. The potential anonymity of virtual groups may also make participants more willing to participate when the topic is sensitive or potentially embarrassing. This latter advantage needs to be weighed against the prospect that group participants may not be who they represent themselves to be and the concern of some potential participants about sharing personal information with strangers in an electronic context. These latter issues are unlikely to be problems when respondents are prerecruited, identities verified, and topics are not of a sensitive nature. Such circumstances would be typical of focus groups used in many marketing research situations and interviews with professionals, but may be less typical in other applications of focus groups.

Use of virtual groups greatly expands the pool of potential participants and adds considerable flexibility to the process of scheduling an interview. Busy professionals and executives, who might otherwise be unavailable for a face-to-face meeting, can often be reached by means of information technologies. Virtual focus groups may be the only option for certain types of samples, but they are not without some costs relative to more traditional groups. The lack of face-to-face interaction often reduces the spontaneity of the group and eliminates the nonverbal communication that plays a key role in eliciting responses. Such nonverbal communication is often critical for determining when further questioning or probing will be useful, and it is often an important source of interplay among group members. Use of virtual groups tends to reduce the intimacy of the group as well, making group members less likely to be open and spontaneous.

The moderators' role is made more difficult, since it is harder to control the participants. Dominant participants are more difficult to quieten, and less active participants are more difficult to recognize. On the other hand, the moderator's task can be aided by electronic monitoring equipment that keeps an ongoing record of who has talked and for how long. A visual display can keep the names and frequency of participation of group members before the moderator. Thus, the moderator can draw out the quiet participant, just as in a more typical focus group.

Virtual groups can take several forms. Telephonic groups (essentially conference calls) have long been used by researchers but such groups are very awkward and it is difficult to manage any serious group interaction. Spontaneity is highly constrained in such groups. Real-time videoconferences have become a common means for conducting virtual groups in the last several years. Videoconferencing via telephone lines or the Internet can provide an opportunity for the moderator to see participants and for participants to see the moderator and other participants. The success of such groups critically depends on the reliability of the technology. It is always important that a technical expert be available during the group research.

Many research firms that specialize in focus group research now include virtual group capabilities as part of their facility offerings. Virtual groups conducted by videoconference are not a perfect substitute for on-site groups. The facial expressions and other behavior of group members may not be visible at all or may not be as visible as in face-to-face group encounters. Group interaction tends to be less spontaneous. Such groups are inevitably more expensive than more traditional on-site groups because of the cost of the technology, the need for a technician, and the cost of connect time.

Two other alternatives for conducting virtual groups involve the use of chat rooms and bulletin boards. Chat rooms involve real-time interaction among the moderator and group members. Bulletin boards are asynchronous, so questions can be posed and answers provided over some extended period of time. Such virtual groups can be very real social groups, but many people remain uncomfortable with such online sharing. It is also the case where the moderator and participants cannot see one another, so information that might be present in facial expressions, tone of voice, and other nonverbal behavior is lost.

Conclusion

With the advent of computer-assisted analysis and real-time, interactive electronic focus groups, the issue of validity in focus group research may, on the surface, seem to occupy a higher plane of importance and sophistication now that it is "technologically" more accessible. However, the use of computers alone does not ensure validity. Like other quantitative techniques, computer analysis of focus group results also suffers from the GIGO (garbage in, garbage out) problem. Therefore, it is worthwhile for social science researchers to take note of Brinberg and McGrath's (1985) succinct reminder that validity "is not a commodity that can be purchased with techniques . . . Rather validity is like integrity, character, or quality, to be assessed relative to purposes and circumstances" (p. 13).

In this regard, the validity of focus group findings should be assessed relative to the research objectives and circumstances that gave rise to the research. Furthermore, the issue of validity needs to be addressed throughout the focus group research process—from planning and data collection to data making, analysis, and interpretation. The execution of each step of this research process has the potential to influence the validity of focus group findings, either positively or negatively. Understanding the limitations and possible sources of bias at each stage of the focus group process will enable the researcher to take appropriate measures to deal objectively with threats to the integrity of the research results.

Discussion Questions

In this chapter, we have examined many facets of focus group research, including the appropriate role of such research in the social sciences, the design and conduct of focus group research, the interpretation of the results of focus group research, and the types of research questions to which focus group research should appropriately be applied. Focus group research is not just a group conversation; it is a complex research tool. You should carefully review this chapter before embarking on focus group research. The questions that follow will help you identify some of the critical issues and decisions associated with the use and conduct of focus group research.

1. For what types of research is the group depth interview (focus group) appropriate? For what types of research questions is a focus group inappropriate?

2. What are the differences between etic and emic research? How are these differences relevant to the use and conduct of focus group research?

3. What does it mean to say that a focus group produces a single observation rather than observations associated with each member of the group?

4. How does sampling differ in the context of focus groups as compared with survey research? What are the implications of these differences for the interpretation of the results of focus group research?

5. How does the composition of a focus group influence the results obtained? What are some of the social factors that can influence the interaction of focus group members?

6. What is the role of the moderator of a focus group? What are the characteristics of a good focus group moderator? Are there different styles for moderating groups that may be more or less appropriate for particular types of groups?

7. What is an interview guide? What is a "good" question for focus group research? What are the characteristics of good questions for use in a focus group?

8. How are probes and follow-up questions used in focus group research? What is the effect of using probes and follow-up questions on the generalizability of focus group research?

9. What types of "results" are produced by a focus group? How are such results summarized and interpreted?

10. What is content analysis? How might it be applied to the results obtained from focus group research?

11. Do you agree or disagree with the statement that focus groups should never be used for evaluative research? Why or why not?

12. List examples of the types of questions for which focus groups might be appropriate.

Exercises

1. Go online and do a search using the key words "focus group research." Find a report of a study that uses focus group research as the primary research method. Based on what you have learned from this chapter, critique the research. In developing your critique consider the following questions:

   a. What was the purpose of the research? How appropriate was focus group research for the research question(s) addressed in the research?
   b. How appropriate was the sample employed in the research? How generalizable are the results of the research?
   c. What types of questions were asked of the group(s)? Did these questions fully address the issues that motivated the research? What other questions might have been asked?
   d. How were the results of the group(s) analyzed? Was this analysis appropriate? What alternative analyses would you suggest instead of, or in addition to, what was reported in the paper?
   e. What was concluded as a result of conducting the group(s)? Do you agree or disagree with the conclusion(s)? What would be an appropriate follow-up to the research?

2. Pick a topic which you think is appropriate for investigation using a focus group. Such topics might include determinants of customer satisfaction with a

product or a service, factors that influence selection of a product or service, views related to a political candidate or social issue or other types of behavior or decision making. Design and conduct a small focus group on the topic. Be sure to carefully consider group composition, the types of questions used in the interview guide, and how you will analyze the results. Conduct the interview yourself. On completion of the group interview ask yourself: (a) What did you learn about the topic? and (b) what did you learn about the role of moderator?

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